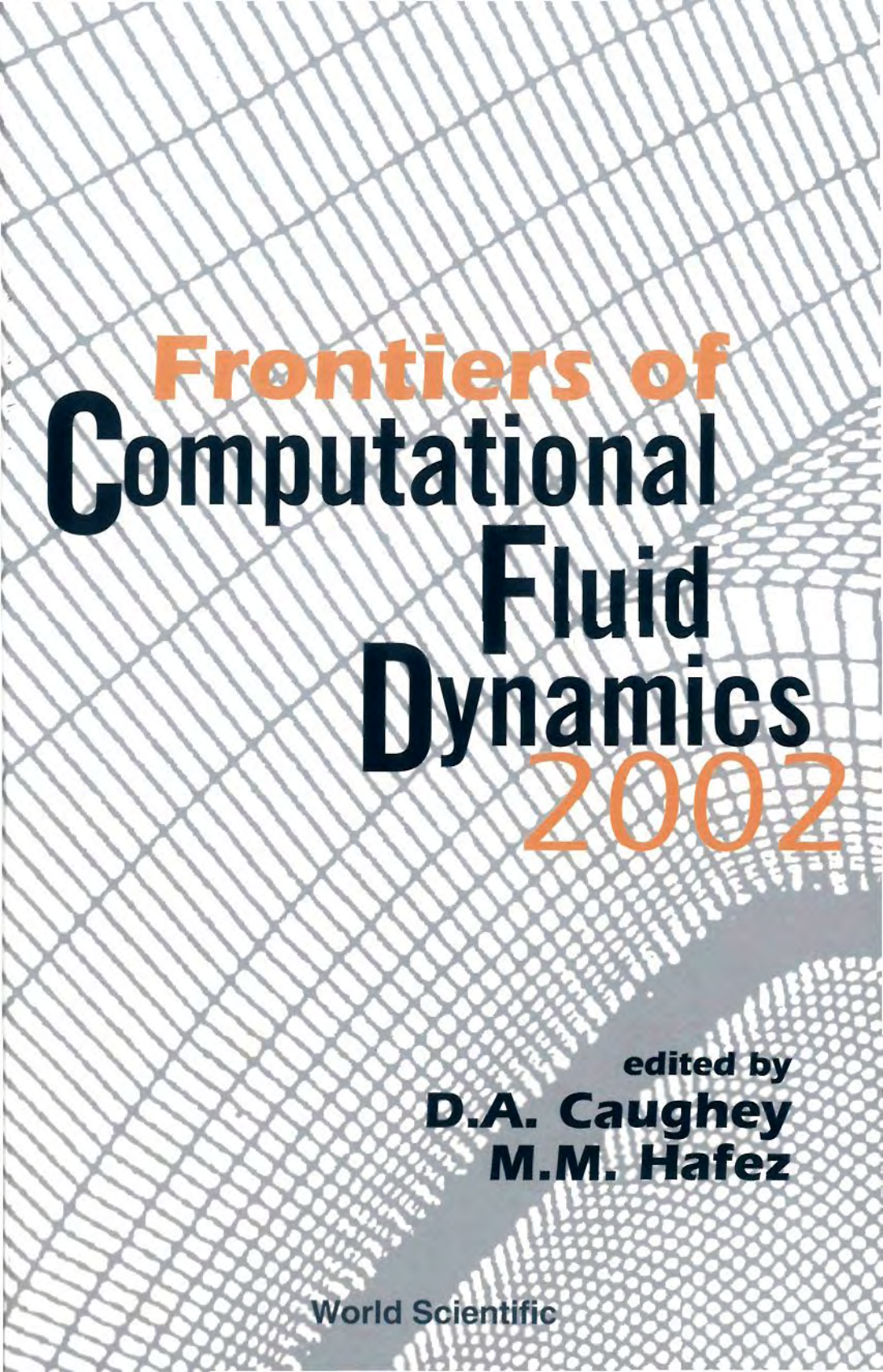




# **Frontiers of Computational Fluid Dynamics 2002**

**edited by  
D.A. Caughey  
M.M. Hafez**

**World Scientific**

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Computational  
Fluid  
Dynamics  
2002**

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**edited by**

**D.A. Caughey**

Cornell University

**M.M. Hafez**

University of California, Davis



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# Dedication

This volume consists of papers presented at a symposium honoring Robert W. MacCormack and recognizing his seminal contributions to the field of computational fluid dynamics (CFD) for more than three decades.

The symposium, entitled *Computing the Future III: Frontiers of Computational Fluid Dynamics*, was held in Half Moon Bay, California on June 26-28, 2000. The authors were selected from among internationally known researchers working in aerodynamics and CFD, where the impact of MacCormack's contributions have been so important. It is the pleasure of the authors and the editors to dedicate this book to Bob in recognition of the important role he has played in our technology and in our lives.

Bob MacCormack was born on February 21, 1940 in Brooklyn, New York. He was raised there and received his undergraduate education at Brooklyn College, majoring in physics and mathematics. He joined the NASA Ames Research Center in 1961, working initially in the Hypersonic Free Flight Branch. While at Ames he completed the M. Sc. degree in mathematics at Stanford University and, in 1971, moved to become Assistant Chief of the newly-formed Computational Fluid Dynamics Branch. He subsequently served as Senior Staff Scientist of the Thermo- and Gas- Dynamics Division at Ames, before beginning his academic career in 1981 as Professor in the Department of Aeronautics and Astronautics at the University of Washington in Seattle. He returned to the Bay Area in 1985 when he accepted the position of Professor in the Department of Aeronautics and Astronautics at Stanford.

Bob has delivered keynote lectures at international conferences in Italy, Japan, and the (former) Soviet Union, as well as in the United States. He has lectured in Short Courses on CFD at the von Karman Institute in Brussels, the Vikram Sarabhai Space Center in Trivandrum, India, National Cheng Kung University in Taiwan, Quinghua University in Beijing, Northwest Polytechnic University in Xi'an, and the China Aerodynamic Research and Development Center in Sichuan, as well as on numerous occasions in this country. He advises and consults with more than a dozen U. S. aerospace companies and

government agencies.

Bob's contributions to CFD have been recognized by a number of major awards. He received the NASA Ames Research Center H. Julian Allen Award in 1973, a NASA Medal for Exceptional Scientific Achievement in 1981, and was selected to deliver the Theodorsen Lecture in 2001. He has a long history of service to the American Institute of Aeronautics and Astronautics (AIAA), including membership on the Fluid Dynamics Technical Committee from 1977-79, and service as an Associate Editor of the AIAA Journal. He was elected a Fellow of the AIAA in 1988, and received that society's Fluid Dynamics Award in 1996. He has served as Associate Editor of the Journal of Computational Physics from 1970-76, and was a member of National Academy of Engineering Committees assessing the status and growth of CFD in 1982 and 1985, and of fluid mechanics in 1983-84. He has been a member of the National Academy of Engineering since 1992.

In the first chapter of this book, Bob's technical contributions will be discussed in more detail, particularly their impact on hypersonic aerodynamics and CFD in general. Virtually all of the attendees of the Symposium had a story to tell of their first experience with the "MacCormack Scheme." The second chapter reprints Bob's famous paper introducing the scheme, and the third chapter summarizes his interactions with several researchers in the CFD Branch at NASA Ames. The remaining chapters present topics of current interest written by leading experts in the field.

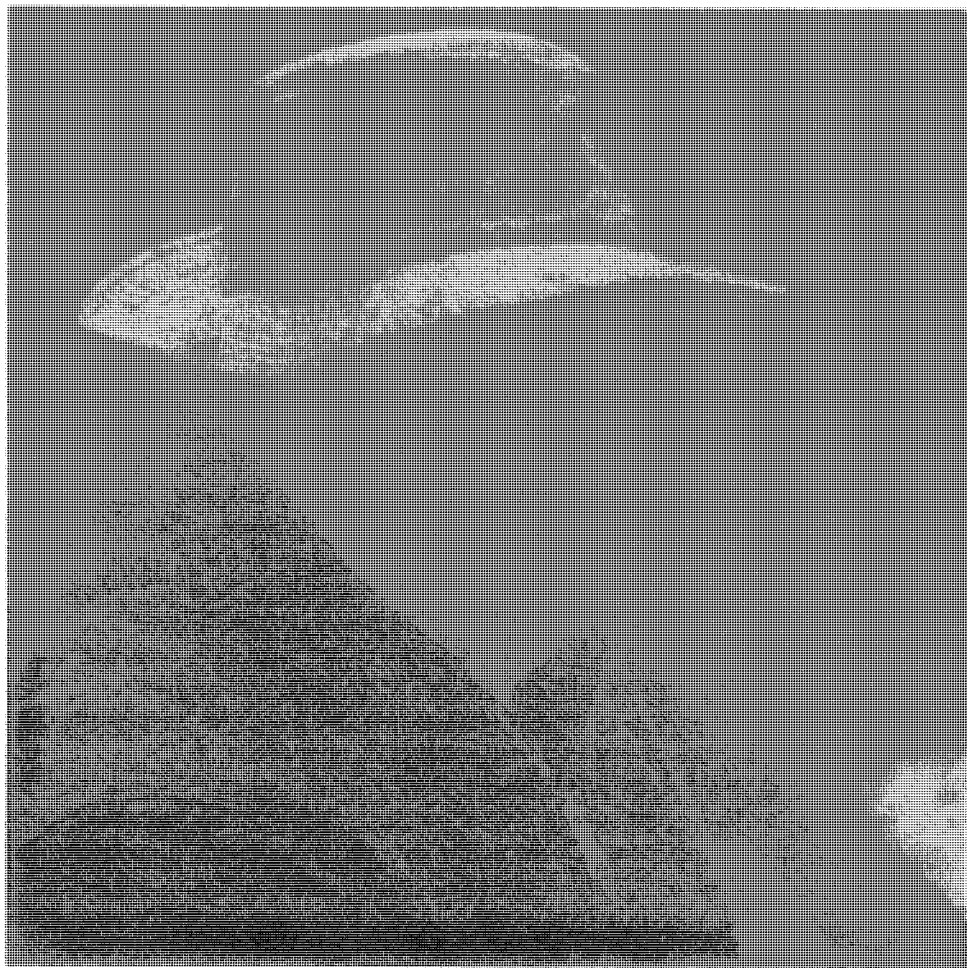
But Bob's contributions are not restricted to his technical ideas, his national leadership, the courses he has taught, or his supervision of many talented students at the University of Washington and Stanford. Bob is a gentleman in the truest sense of the word, and his grace and good humor have enriched all of those who have known him throughout the span of his remarkable career. A photograph of Bob, taken at the Symposium Banquet, is shown on the facing page.

The day after the Symposium, a number of attendees joined Bob for a day of salmon fishing in the Pacific Ocean off the California coast. A photograph of Bob, standing on the aft deck of the *New Captain Pete*, waiting for the next salmon to bite, is shown on the following page.



**Robert W. MacCormack**





Bob MacCormack on the deck of the *New Captain Pete*.

# Contents

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| <i>Dedication</i> . . . . .                                                    | v  |
| <b>1 Contributions of Robert W. MacCormack to Computational Fluid Dynamics</b> |    |
| <i>Caughey &amp; Hafez</i> . . . . .                                           | 1  |
| 1.1 Introduction . . . . .                                                     | 1  |
| 1.2 Overview . . . . .                                                         | 2  |
| 1.3 Technical Contributions . . . . .                                          | 3  |
| 1.4 Concluding Remarks . . . . .                                               | 17 |
| REFERENCES . . . . .                                                           | 18 |
| <b>2 The Effect of Viscosity in Hypervelocity Impact Cratering</b>             |    |
| <i>MacCormack</i> . . . . .                                                    | 27 |
| 2.1 Abstract . . . . .                                                         | 27 |
| 2.2 Introduction . . . . .                                                     | 27 |
| 2.3 The Numerical Method . . . . .                                             | 28 |
| 2.4 Numerical Calculations . . . . .                                           | 37 |
| 2.5 Concluding Remarks . . . . .                                               | 40 |
| REFERENCES . . . . .                                                           | 42 |
| <b>3 The MacCormack Method – Historical Perspective</b>                        |    |
| <i>Hung, Deiwert &amp; Inouye</i> . . . . .                                    | 45 |
| 3.1 Introduction . . . . .                                                     | 45 |
| 3.2 Evolution of the MacCormack Method . . . . .                               | 46 |
| 3.3 Applications . . . . .                                                     | 50 |
| 3.4 Closing Remarks . . . . .                                                  | 57 |
| REFERENCES . . . . .                                                           | 58 |

#### 4 General Framework for Achieving Textbook Multigrid Efficiency: One-Dimensional Euler Example

|                                                     |    |
|-----------------------------------------------------|----|
| <i>Thomas, Diskin, Brandt &amp; South</i> . . . . . | 61 |
| Abstract . . . . .                                  | 61 |
| Introduction . . . . .                              | 61 |
| General Framework . . . . .                         | 63 |
| Quasi-One-Dimensional Equations . . . . .           | 65 |
| Relaxation Schemes . . . . .                        | 66 |
| Distributed Relaxation . . . . .                    | 68 |
| Computational Results . . . . .                     | 71 |
| Transonic Flows . . . . .                           | 74 |
| Concluding Remarks . . . . .                        | 75 |
| REFERENCES . . . . .                                | 76 |
| Appendices . . . . .                                | 77 |
| I - Conservative Fluxes . . . . .                   | 77 |
| II - Distribution Matrices . . . . .                | 78 |
| III - Transonic Shock — ENO Differencing . . . . .  | 79 |

#### 5 Numerical Solutions of Cauchy-Riemann Equations for Two and Three Dimensional Flows

|                                                           |    |
|-----------------------------------------------------------|----|
| <i>Hafez &amp; Houseman</i> . . . . .                     | 81 |
| 5.1 Introduction . . . . .                                | 82 |
| 5.2 Governing Equations and Boundary Conditions . . . . . | 83 |
| 5.3 Numerical Methods . . . . .                           | 84 |
| 5.4 Numerical Results . . . . .                           | 85 |
| 5.5 Concluding Remarks . . . . .                          | 86 |
| 5.6 Appendix: Multigrid Convergence Results . . . . .     | 86 |
| REFERENCES . . . . .                                      | 86 |

#### 6 Efficient High-order Schemes on Non-uniform Meshes for Multi-Dimensional Compressible Flows

|                                                           |     |
|-----------------------------------------------------------|-----|
| <i>Lerat, Corre and Hanss</i> . . . . .                   | 89  |
| 6.1 Introduction . . . . .                                | 89  |
| 6.2 Euler solver on a regular Cartesian mesh . . . . .    | 90  |
| 6.3 Euler solver on an irregular Cartesian mesh . . . . . | 93  |
| 6.4 Navier-Stokes solver . . . . .                        | 98  |
| 6.5 Numerical experiments . . . . .                       | 100 |
| 6.6 Conclusion . . . . .                                  | 105 |
| REFERENCES . . . . .                                      | 105 |

#### 7 Future directions for computing compressible flows: higher-order centering vs multidimensional upwinding

|                                    |     |
|------------------------------------|-----|
| <i>Napolitano et al.</i> . . . . . | 113 |
|------------------------------------|-----|

|           |                                                                                                       |     |
|-----------|-------------------------------------------------------------------------------------------------------|-----|
| 7.1       | Introduction . . . . .                                                                                | 113 |
| 7.2       | High-order centred numerical method . . . . .                                                         | 115 |
| 7.3       | Fluctuation splitting method . . . . .                                                                | 116 |
| 7.4       | Results and Discussion . . . . .                                                                      | 118 |
| 7.5       | Conclusions . . . . .                                                                                 | 124 |
| 7.6       | Acknowledgements . . . . .                                                                            | 125 |
|           | REFERENCES . . . . .                                                                                  | 125 |
| <b>8</b>  | <b>Extension of Efficient Low Dissipation High Order Schemes<br/>for 3-D Curvilinear Moving Grids</b> |     |
|           | <i>Vinokur &amp; Yee</i> . . . . .                                                                    | 129 |
| 8.1       | Introduction . . . . .                                                                                | 130 |
| 8.2       | Formulation of Equations . . . . .                                                                    | 134 |
| 8.3       | Numerical Methods . . . . .                                                                           | 143 |
| 8.4       | Concluding Remarks . . . . .                                                                          | 155 |
|           | Acknowledgment . . . . .                                                                              | 156 |
|           | Appendix A: The Commutativity of a Class of Numerical Mixed<br>Partial Derivatives . . . . .          | 156 |
|           | Appendix B: Riemann Solver for Non-equilibrium Flow . . . . .                                         | 160 |
|           | REFERENCES . . . . .                                                                                  | 163 |
| <b>9</b>  | <b>Fourth Order Methods for the Stokes and Navier-Stokes<br/>Equations on Staggered Grids</b>         |     |
|           | <i>Gustafsson &amp; Nilsson</i> . . . . .                                                             | 165 |
| 9.1       | Introduction . . . . .                                                                                | 165 |
| 9.2       | The Steady Stokes Equations and Staggered Grids . . . . .                                             | 167 |
| 9.3       | A Fourth Order Method for the Stokes Equations . . . . .                                              | 171 |
| 9.4       | A Fourth Order Method for the Navier-Stokes Equations . . . . .                                       | 175 |
|           | REFERENCES . . . . .                                                                                  | 178 |
| <b>10</b> | <b>Scalable Parallel Implicit Multigrid Solution of Unsteady<br/>Incompressible Flows</b>             |     |
|           | <i>Pankajakshan et al</i> . . . . .                                                                   | 181 |
| 10.1      | Abstract . . . . .                                                                                    | 181 |
| 10.2      | Introduction . . . . .                                                                                | 182 |
| 10.3      | Basic Unsteady Flow Solver . . . . .                                                                  | 182 |
| 10.4      | Scalable Parallel Implicit Algorithm . . . . .                                                        | 185 |
| 10.5      | Parallel Performance Estimates and Scalability . . . . .                                              | 188 |
| 10.6      | Demonstration: Rudder-Induced Maneuvering Simulation . . . . .                                        | 193 |
| 10.7      | Acknowledgements . . . . .                                                                            | 195 |
|           | REFERENCES . . . . .                                                                                  | 195 |

**11 Application of Vorticity Confinement to the Prediction of the Flow over Complex Bodies**  
*Steinhoff* . . . . . 197  
11.1 Introduction . . . . . 198  
11.2 Conventional Eulerian Methods . . . . . 199  
11.3 Vorticity Confinement . . . . . 200  
11.4 Current Results . . . . . 206  
11.5 Conclusion . . . . . 213  
REFERENCES . . . . . 214

**12 Lattice Boltzmann Simulation of Incompressible Flows**  
*Satofuka & Ishikura* . . . . . 227  
12.1 Introduction . . . . . 227  
12.2 Lattice Boltzmann Method for Two-dimension . . . . . 228  
12.3 Two-dimensional Homogeneous Isotropic Turbulence . . . . . 231  
12.4 Two-dimensional Channel with Sudden Expansion . . . . . 233  
12.5 Lattice Boltzmann Method for Three-dimension . . . . . 235  
12.6 Three-dimensional Homogeneous Isotropic Turbulence . . . . . 236  
12.7 Three-dimensional Duct Flow . . . . . 237  
12.8 Parallelization . . . . . 239  
12.9 Conclusion . . . . . 240  
REFERENCES . . . . . 240

**13 Numerical Simulation of MHD Effects on Hypersonic Flow of a Weakly Ionized Gas in an Inlet**  
*Agarwal & Deb* . . . . . 243  
13.1 Abstract . . . . . 243  
13.2 Nomenclature . . . . . 244  
13.3 Introduction . . . . . 246  
13.4 Governing Equations of Electro-Magnetohydrodynamics . . . . . 247  
13.5 Governing Equations in Weak Conservation Law Form . . . . . 249  
13.6 Governing Equations in Generalized Coordinates . . . . . 252  
13.7 Numerical Method . . . . . 254  
13.8 Significant Parameters . . . . . 259  
13.9 Numerical Simulation of Supersonic Flow in an Inlet . . . . . 260  
13.10 Conclusions . . . . . 263  
13.11 Acknowledgements . . . . . 263  
REFERENCES . . . . . 263

**14 Progress in Computational Magneto-Aerodynamics**  
*Shang, Canupp & Gaitonde* . . . . . 273  
14.1 Introduction . . . . . 273  
14.2 Governing Equations . . . . . 275

|                                                                                                                               |     |
|-------------------------------------------------------------------------------------------------------------------------------|-----|
| 14.3 Numerical Procedures . . . . .                                                                                           | 277 |
| 14.4 Rankine-Hugoniot Jump Condition . . . . .                                                                                | 282 |
| 14.5 Ideal MHD Shock Tube Simulation . . . . .                                                                                | 285 |
| 14.6 Hypersonic MHD Blunt Body Simulation . . . . .                                                                           | 289 |
| 14.7 Concluding Remarks . . . . .                                                                                             | 293 |
| 14.8 Acknowledgments . . . . .                                                                                                | 294 |
| 14.9 References . . . . .                                                                                                     | 294 |
| <b>15 Development of 3D DRAGON Grid Method for Complex Geometry</b>                                                           |     |
| <i>Liou &amp; Zheng</i> . . . . .                                                                                             | 299 |
| 15.1 Introduction . . . . .                                                                                                   | 299 |
| 15.2 DRAGON Grid . . . . .                                                                                                    | 301 |
| 15.3 Three-Dimensional DRAGON Grid Generation . . . . .                                                                       | 303 |
| 15.4 Flow Solver . . . . .                                                                                                    | 308 |
| 15.5 Test Cases . . . . .                                                                                                     | 309 |
| 15.6 Concluding Remarks . . . . .                                                                                             | 312 |
| Acknowledgments . . . . .                                                                                                     | 313 |
| REFERENCES . . . . .                                                                                                          | 314 |
| <b>16 Application of Multi-Block, Patched Grid Topologies to Navier-Stokes Predictions of the Aerodynamics of Army Shells</b> |     |
| <i>Sturek &amp; Haroldsen</i> . . . . .                                                                                       | 319 |
| 16.1 Introduction . . . . .                                                                                                   | 319 |
| 16.2 Missile Configurations . . . . .                                                                                         | 320 |
| 16.3 Boundary/Initial Conditions . . . . .                                                                                    | 322 |
| 16.4 Performance/Convergence Criteria . . . . .                                                                               | 323 |
| 16.5 Results . . . . .                                                                                                        | 323 |
| 16.6 Concluding Remarks . . . . .                                                                                             | 324 |
| 16.7 Acknowledgements . . . . .                                                                                               | 324 |
| REFERENCES . . . . .                                                                                                          | 324 |
| <b>17 On Aerodynamic Prediction by Solution of the Reynolds-Averaged Navier-Stokes Equations</b>                              |     |
| <i>Hall</i> . . . . .                                                                                                         | 333 |
| 17.1 Introduction . . . . .                                                                                                   | 333 |
| 17.2 The RANS Scheme and the Menter Turbulence Model . . . . .                                                                | 336 |
| 17.3 RANS Results for the Menter Turbulence Model . . . . .                                                                   | 338 |
| 17.4 A modification to the Menter turbulence model . . . . .                                                                  | 341 |
| 17.5 Concluding Remarks . . . . .                                                                                             | 345 |
| REFERENCES . . . . .                                                                                                          | 346 |

**18 Advances in Algorithms for Computing Aerodynamic Flows**

*Zingg, De Rango & Pueyo* . . . . . 347

18.1 Introduction . . . . . 347

18.2 Newton-Krylov Algorithm . . . . . 349

18.3 Higher-Order Spatial Discretization . . . . . 356

18.4 Concluding Remarks . . . . . 366

Acknowledgements . . . . . 367

REFERENCES . . . . . 367

**19 Numerical Simulation of Hypersonic Boundary Layer Stability and Receptivity**

*Zhong, Whang & Ma* . . . . . 381

19.1 Introduction . . . . . 381

19.2 Governing Equations and Numerical Methods . . . . . 382

19.3 Results and Discussion . . . . . 383

19.4 Concluding Remarks . . . . . 395

REFERENCES . . . . . 396

**20 Time-Dependent Simulation of Incompressible Flow in a Turbopump using Overset Grid Approach**

*Kiris & Kwak* . . . . . 399

20.1 Introduction . . . . . 399

20.2 Numerical Method . . . . . 400

20.3 Approach and Computational Models . . . . . 402

20.4 Computed Results . . . . . 406

20.5 Summary . . . . . 413

20.6 Acknowledgements . . . . . 414

REFERENCES . . . . . 414

**21 Aspects of the Simulation of Vortex Flows over Delta Wings**

*Rizzi, Görtz & LeMoigne* . . . . . 415

21.1 Introduction . . . . . 415

21.2 Computational Method . . . . . 419

21.3 Test Cases and Grids . . . . . 421

21.4 Stationary-Wing Computations and Results . . . . . 426

21.5 Preliminary results for Pitching Delta . . . . . 434

21.6 Conclusions and Outlook . . . . . 438

21.7 Acknowledgments . . . . . 439

REFERENCES . . . . . 439

**22 Selected CFD Capabilities at DLR**

*Kordulla* . . . . . 443

|                                                                                       |     |
|---------------------------------------------------------------------------------------|-----|
| 22.1 Introduction . . . . .                                                           | 443 |
| 22.2 CFD Developments . . . . .                                                       | 444 |
| 22.3 Recent Applications . . . . .                                                    | 449 |
| 22.4 Where to go . . . . .                                                            | 454 |
| 22.5 Acknowledgements . . . . .                                                       | 455 |
| REFERENCES . . . . .                                                                  | 455 |
| <b>23 CFD Applications to Space Transportation Systems</b>                            |     |
| <i>Fujii</i> . . . . .                                                                | 459 |
| 23.1 Introduction . . . . .                                                           | 459 |
| 23.2 Numerical Method . . . . .                                                       | 460 |
| 23.3 Results and Discussion . . . . .                                                 | 460 |
| 23.4 Conclusions . . . . .                                                            | 471 |
| 23.5 Acknowledgement . . . . .                                                        | 472 |
| REFERENCES . . . . .                                                                  | 472 |
| <b>24 Multipoint Optimal Design of Supersonic Wings Using Evolutionary Algorithms</b> |     |
| <i>Obayashi, Takeguchi &amp; Sasaki</i> . . . . .                                     | 475 |
| 24.1 Introduction . . . . .                                                           | 475 |
| 24.2 Optimization Method . . . . .                                                    | 476 |
| 24.3 Formulation of the Present Optimization Problem . . . . .                        | 477 |
| 24.4 Optimization of a Supersonic Transport Wing . . . . .                            | 478 |
| 24.5 Conclusion . . . . .                                                             | 480 |
| REFERENCES . . . . .                                                                  | 481 |
| <b>25 Information Science – A New Frontier of CFD</b>                                 |     |
| <i>Oshima &amp; Oshima</i> . . . . .                                                  | 489 |
| 25.1 Out of Deterministic Systems Into Complex Systems . . . . .                      | 489 |
| 25.2 Computers vs Human Brain . . . . .                                               | 490 |
| 25.3 Information Science . . . . .                                                    | 491 |
| <b>26 Integration of CFD into Aerodynamics Education</b>                              |     |
| <i>Murman &amp; Rizzi</i> . . . . .                                                   | 493 |
| 26.1 Introduction . . . . .                                                           | 493 |
| 26.2 Changes from 1981 to 2000 . . . . .                                              | 494 |
| 26.3 Educational Considerations and Questions . . . . .                               | 497 |
| 26.4 Findings from an Informal Survey . . . . .                                       | 499 |
| 26.5 Examples of Integration . . . . .                                                | 503 |
| 26.6 Summary . . . . .                                                                | 505 |
| 26.7 Acknowledgements . . . . .                                                       | 506 |
| REFERENCES . . . . .                                                                  | 506 |



# Contributions of Robert W. MacCormack to Computational Fluid Dynamics

David A. Caughey<sup>1</sup> and Mohamed M. Hafez<sup>2</sup>

## 1.1 Introduction

Robert W. MacCormack has been a major force in the development of computational fluid dynamics (CFD) since the infancy of the field. He has made significant and seminal contributions to basic numerical methods for solving the equations of compressible fluid flow, including high-speed flows with non-equilibrium chemistry, and applied these methods to important fundamental problems, including shock-wave boundary layer interactions and supersonic flows on compression ramps, as well as more applied problems, including the flow past complete aerospace vehicles.

Most CFD researchers are familiar with Bob's highly efficient modification of the explicit Lax-Wendroff method, but many are unaware of the number of other important concepts that can be traced to Bob's papers. These include the finite volume method, the use of second- and fourth- difference numerical dissipation, his implicit scheme, the use of line relaxation techniques to iterate the compressible equations to steady state, and his modified approximate factorization scheme. He also used sub-iteration to eliminate (or minimize) splitting errors, and he advocated the introduction of numerical viscosity in a form similar to the natural viscosity to preserve the frame independence of the

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<sup>1</sup> Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, New York 14853-7501.

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Navier-Stokes equations, even when solved in arbitrary curvilinear coordinate systems. Many other ideas can be also found in his papers.

Bob applied his methods to many physical problems, including hypersonic laminar and turbulent flows and to flows with thermo-chemical non-equilibrium, as well as to supersonic and transonic flows. He studied the continuum limit and various forms of the Burnett equations and their stabilization. More recently, he worked on Magnetofluid Dynamics (and also worked in 1978 on nonlinear optics propagation using splitting and re-zoning techniques).

The purpose of this chapter is to describe these technical contributions in greater detail and, in the process, provide an historical overview of important developments in CFD.

## 1.2 Overview

The character of Bob MacCormack's contributions to the field of computational fluid dynamics can be seen by counting the occurrences of various words in the titles of his papers. A partial list of Bob's publications is included at the end of this chapter. The results of such a word search, applied to this list, are summarized in Table 1.

It is perhaps no surprise that the most commonly occurring word is *numerical*, appearing in more than 1/3 of the titles. It is more interesting to note that the word *computational* appears only 4 times, and the acronym *CFD* only twice (once in a personal review of the 25 years' progress in the field, since the development of the "MacCormack Method," presented at the 11th AIAA CFD Conference in Orlando, Florida in 1993). The practical focus of Bob's work is illustrated by the fact that the phrase *Navier-Stokes* appears 22 times, the word *viscous* appears 12 times, and *turbulent* (or *turbulence*) occurs 11 times, while *laminar* appears only 4 times. Further, the compound word *Three-Dimensional* appears 7 times while *Two-Dimensional* appears only once. It is noteworthy that the word *inviscid* appears only once and *Euler* is completely absent; a remarkable feat for a researcher actively developing numerical methods for compressible fluid flow problems during the decade of the 1980s. Finally, the focus of much of Bob's work is illustrated by the fact that the word *hypersonic* appears 19 times, while *compressible* and *shock* (or *shock wave*) appear a total of 33 times.

The word *hypervelocity* appears only three times, but is notable because it appears in what is likely Bob's most frequently cited paper, *The Effect of Viscosity in Hypervelocity Impact Cratering*<sup>3</sup>, which introduced the world to the *MacCormack scheme*.

---

<sup>3</sup> This classic paper is reproduced as Chapter 2 of this volume.

| Word              | Number of Titles |
|-------------------|------------------|
| Numerical         | 37               |
| Navier-Stokes     | 22               |
| Hypersonic        | 19               |
| Compressible      | 17               |
| Shock(-wave)      | 16               |
| Viscous           | 12               |
| Turbulen(t/ce)    | 11               |
| Boundary-Layer    | 9                |
| Three-Dimensional | 7                |
| Non-equilibrium   | 5                |
| Laminar           | 4                |
| Computational     | 4                |
| Hypervelocity     | 3                |
| CFD               | 2                |
| Inviscid          | 1                |
| Two-Dimensional   | 1                |
| Euler             | 0                |

**Table 1** Occurrences of selected words in the titles of nearly 100 papers by Robert W. MacCormack.

### 1.3 Technical Contributions

Bob's earliest work in the Hypersonic Free Flight Branch at NASA Ames was in the area of hypervelocity impact cratering. His report [3] summarizes experiments designed to resolve a controversy over whether the flash of visible radiation associated with hypervelocity impact required a gaseous atmosphere. These experiments were motivated by a proposal to determine the chemical composition of the lunar surface by spectroscopic analysis of the hypervelocity impact of a projectile into the surface of the dark side of the moon. Bob's analysis of the energies required for various mechanisms that might be responsible for the flash, showed that the dependence of the observed values of the onset rate of luminosity on the ambient pressure, were consistent with the interaction of high-speed ejecta with the atmosphere being the principal cause of the observed radiation. In his summary, Bob noted that for higher impact velocities, or for materials other than the aluminum and basalt rock of their tests, the cratering mechanism itself might be sufficient to produce observable radiation, although the editors don't know if the lunar experiment was ever performed.

Uncertainty over the physical mechanism responsible for observed power-law scalings of the ratios of penetration depth to projectile diameter and of final target momentum to initial projectile momentum motivated Bob to analyze the effect of viscosity on the hypervelocity impact phenomenon. It was, of course, necessary to solve the problem numerically, even for the axisymmetric case, and Bob's numerical solution of this problem required an efficient, second-order accurate scheme, so he developed the alternating backward-space predictor forward-space corrector version of the Lax-Wendroff scheme that has been synonymous with his name ever since [5]. In the paper Bob analyzes the stability of the scheme and conjectures that alternating between the order in which the backward/forward steps are applied will allow the full (one-dimensional) CFL limit on time step. This conjecture certainly has been verified again and again in numerous applications of the method, but has not been proved (at least, to the editors' knowledge). Bob describes the trade-off, in terms of efficiency, of his method, arguing that it would be effective only if the full CFL limit were attainable. Solutions of this historic problem, on grids containing  $32 \times 33$  mesh cells, required about 15 minutes CPU time on the IBM 7094. Bob carried out calculations for the impact of cylindrical aluminum projectiles (having  $\ell = d$ ) into aluminum targets (both, semi-infinite and thin sheets), using Sakharov's value for the viscosity of aluminum. The results showed that the total axial and radial momentum "... exhibit an effect ... consistent with ... measurements," and Bob went on to suggest specific experiments that "could confirm the importance of viscosity in hypervelocity impact."

After this research was completed, Bob apparently decided that fluid mechanics contained enough unsolved problems for a career. At about the same time, Dean Chapman saw the potential for the development of numerical methods in fluid dynamics and aerodynamics, and set up the CFD Branch at NASA Ames, which Bob was invited to join. His next paper applied his new numerical scheme to the problem of an oblique shock wave interacting with the laminar boundary layer on a flat plate. In order to attack problems involving boundary layers, he developed a spatially split version of his scheme having two advantages [6]. First, since each one-dimensional operator remains stable up to the full CFL limit, the earlier question about stability in this regard becomes moot. Second, the splitting allows multiple time-steps to be performed in the direction normal to the boundary (i.e., the boundary-layer "normal" direction) for each time step in the direction parallel to the boundary. The computations were performed on a multi-block mesh to allow finer resolution near the wall; the meshes contained  $34 \times 32$  and  $34 \times 22$  mesh points, respectively, and a typical computation requiring  $256\Delta t_x$  time steps required about 4 hours of CPU time on the IBM 360/67. Surface pressure and skin friction results for both un-separated and separated flow cases compared quite well with the widely-used experimental measurements

of Hakkinen *et al.*, especially when the mesh was refined in the recirculation zone for the separated flow case. It is interesting to note that one of the first presentations of “computer graphics” applied to CFD appeared in this paper: reproductions of streamline plots superimposed on boundary layer velocity profiles, generated from a cathode-ray display tube.

Bob becomes a proponent of spatial splitting in a paper with A. J. Paullay [7], presented at the AIAA Aerospace Sciences meeting in January 1972. The authors here suggest a positive correlation between the accuracy and efficiency of a numerical method, pointing out that an explicit scheme operating at its maximum allowable time step has all the data needed to advance the solution, with a minimum of extraneous data. The purpose of this paper was to demonstrate the advantages of operator splitting used earlier for the shock wave/laminar boundary layer interaction problem to more general fluid dynamics problems. The split explicit MacCormack scheme is applied to the inviscid equations of compressible flow to solve for the supersonic flow past symmetric diamond-shaped airfoils and double compression corners using simple, non-orthogonal, sheared meshes. They achieve results in excellent agreement with the exact (inviscid) solutions for these problems, demonstrating a reduction in computational time of more than a factor of two, relative to the unsplit method. The split method allows both 1) advancing the solution at the full one-dimensional CFL limit in each space dimension, and 2) advancing the solution in the direction of the smaller mesh spacing multiple time steps for each time step in the coarser direction, allowing a better matching of the numerical and physical domains of dependence.

In a subsequent paper [13] MacCormack and Paullay discuss the influence of the computational mesh on solution accuracy, introducing the concept of “mesh fitting” for the accurate treatment of shock waves. They introduce the concept of *weak solutions* to the CFD community, and introduce a *finite volume* form of the time-split, explicit MacCormack method – the first time this now standard class of approximation to the equations of motion is found in the literature. The authors use three problems to illustrate three different points. The linear wave (advection) equation is used to show that the MacCormack explicit method reproduces the exact solution at a Courant number of unity due, the authors argue, to the alignment of the space-time mesh with the solution for this value of Courant number. Second, the inviscid Burgers equation is used to show that, without corrective measures, the numerical scheme may capture (physically incorrect) expansion “shocks.” They provide two remedies for this problem; 1) a simple modification to the flux computation to ensure continuity of the velocity in the expansion region, and 2) the addition of a fourth-difference dissipation term (an element in the widely used blended second- and fourth-difference dissipation of Jameson, Schmidt, & Turkel). The authors note that it is dangerous to make these

modifications in regions where the solution is discontinuous because the additional truncation error may be large; this is consistent with the strategy of Jameson, Schmidt, & Turkel to use a nonlinear switch to turn off the fourth difference dissipation near shock waves. One significant difference between the strategies of MacCormack & Paullay and of Jameson, Schmidt, & Turkel should be noted: the former suggest adding the fourth difference terms only where they are needed to avoid expansion shocks while the latter suggest adding them everywhere *except* near discontinuities. Finally, the authors consider solutions of the Euler equations for several two-dimensional, supersonic flows, including flows past wedges, diamond airfoils, and a sphere. For these flows it is shown that the numerical error is reduced when the mesh is aligned with the shock position. This requires a solution-adaptive procedure when the shock position is unknown *a priori* – i.e., for the case of the sphere. A mesh position correction scheme is employed using the Rankine-Hugoniot conditions, but it should be emphasized that the shock jump relations still are *captured* by the numerical scheme (as opposed to being *fitted*). For the diamond airfoil example, fourth-difference dissipation is used to avoid the expansion shock that otherwise would emanate from the expansion corner of the body.

Bob next turned his attention to the more difficult case of the interaction of an oblique shock with a turbulent boundary layer in a series of papers with Barrett Baldwin [10, 11, 14]. The flow past a flat plate at  $M_\infty = 8.47$  was computed, with an oblique shock of strength  $\delta p/p_\infty = 83$  impinging on the boundary layer at a point corresponding to a Reynolds number  $Re_x = 22.5 \times 10^6$ . The spatially-split version of MacCormack’s explicit scheme was used, on a mesh now containing four regions of successively finer grids, with the finest grid adjacent to the wall. The fine mesh near the plate allowed the plane-normal factor to be advanced 96 time steps for each time step of the streamwise factor. This work introduced the idea of augmenting the numerical viscosity by a term proportional to

$$\frac{\delta_{xx}p}{\mu_{xx}p} \delta_{xx} \mathbf{w}$$

where  $p$  is the fluid pressure,  $\mathbf{w}$  is the vector of conserved variables, and  $\delta_{xx}$  and  $\mu_{xx}$  are 3-point differencing and averaging operators, respectively. The authors found it necessary to add this additional dissipation to stabilize the scheme for a case with such a strong shock wave. This term would, of course, later become an important element of the Jameson, Schmidt, Turkel blended 2nd/4th difference adaptive dissipation. The authors also introduced special treatment to achieve exponential accuracy in the viscous sublayer, across which the turbulence kinetic energy and dissipation rate vary by several orders of magnitude. Computed skin friction and heat transfer distributions along the plate were compared with experiment; agreement was fair – not

nearly so good as had been achieved earlier for the laminar case (surprise!).

Bob also worked with Art Rizzi to develop his method for spatial marching of supersonic flows in generalized coordinates [12]. This paper represents one of the earliest presentations of the inviscid equations of motion in generalized coordinates, describing the fluxes in terms of contravariant velocities. MacCormack's two-step, dimensionally-split explicit scheme is marched spatially for supersonic flows, on a body-fitted mesh that also is "fitted" to the shock wave to reduce oscillations there (see also [13]). Results of computations for  $M_\infty = 14.9$  flow past a blunted cone in helium and  $M_\infty = 21.7$  flow past a smooth three-dimensional body in air are presented.

A significant advance in implicit methods is described in MacCormack's paper presented at the AIAA 19th Aerospace Sciences Meeting [37]. The paper also is notable for containing the equation that perhaps best characterizes Bob's approach to CFD:

$$\begin{aligned} \{\text{NUMERICS}\} \delta U_{i,j}^{n+1} &= \{\text{PHYSICS}\} \\ U_{i,j}^{n+1} &= U_{i,j}^n + \delta U_{i,j}^{n+1} \end{aligned} \quad (1.1)$$

In other words, the right hand side of the equation that drives the solution updates should be an accurate local approximation to the equations governing the physics of the problem, while the responsibility of the left hand side is to propagate the locally determined solution changes globally in a stable manner to allow rapid convergence of the solution.

For the Navier-Stokes equations, written in the compact vector form

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0 \quad (1.2)$$

the right hand side of Eq. (1.2) becomes

$$\{\text{PHYSICS}\} = \Delta U_{i,j}^n = -\Delta t \left( \frac{\Delta \mathbf{F}}{\Delta x} + \frac{\Delta \mathbf{G}}{\Delta y} \right)_{i,j}^n \quad (1.3)$$

For steady problems, the  $\{\text{NUMERICS}\}$  in Eq. (1.2) can be interpreted as a preconditioning operator, while differentiation of the quasilinear form of Eq. (1.2) for general time-dependent problems gives

$$\frac{\partial(\partial \mathbf{U} / \partial t)}{\partial t} = - \frac{\partial \mathbf{A}(\partial \mathbf{U} / \partial t)}{\partial x} - \frac{\partial \mathbf{B}(\partial \mathbf{U} / \partial t)}{\partial y} \quad (1.4)$$

where  $\mathbf{A} = \partial \mathbf{F} / \partial \mathbf{U}$  and  $\mathbf{B} = \partial \mathbf{G} / \partial \mathbf{U}$  are the Jacobians of the flux vectors  $\mathbf{F}$  and  $\mathbf{G}$ , respectively. This equation describes how changes  $\Delta t(\partial \mathbf{U} / \partial t)$  in the solution should propagate throughout the domain. Implicit approximation of Eq. (1.4) yields,

$$\left( \mathbf{I} + \Delta t \frac{\partial \mathbf{A}}{\partial x} + \Delta t \frac{\partial \mathbf{B}}{\partial y} \right) \delta \mathbf{U}_{i,j}^{n+1} = \delta \mathbf{U}_{i,j}^n = \Delta \mathbf{U}_{i,j}^n \quad (1.5)$$

where the dots in the numerators of this equation indicate that the partial derivatives with respect to  $x$  and  $y$  also operate on the corrections  $\delta \mathbf{U}_{i,j}^{n+1}$ . In Eq. (1.5)  $\Delta t$  is assumed to be independent of  $x$  and  $y$ . These considerations suggest that an efficient implementation of the MacCormack predictor-corrector scheme can be written

$$\begin{aligned}
 \Delta \mathbf{U}_{i,j}^n &= -\Delta t \left( \frac{\Delta_+ \mathbf{F}_{i,j}^n}{\Delta x} + \frac{\Delta_+ \mathbf{G}_{i,j}^n}{\Delta y} \right) \\
 \left( \mathbf{I} - \Delta t \frac{\Delta_+ |\mathbf{A}| \cdot}{\Delta x} \right) \left( \mathbf{I} - \Delta t \frac{\Delta_+ |\mathbf{B}| \cdot}{\Delta y} \right) \delta \mathbf{U}_{i,j}^{n+1} &= \Delta \mathbf{U}_{i,j}^n \\
 \mathbf{U}_{i,j}^{n+1} &= \mathbf{U}_{i,j}^n + \delta \mathbf{U}_{i,j}^{n+1} \\
 \Delta \mathbf{U}_{i,j}^{n+1} &= -\Delta t \left( \frac{\Delta_- \mathbf{F}_{i,j}^{n+1}}{\Delta x} + \frac{\Delta_- \mathbf{G}_{i,j}^{n+1}}{\Delta y} \right) \\
 \left( \mathbf{I} + \Delta t \frac{\Delta_- |\mathbf{A}| \cdot}{\Delta x} \right) \left( \mathbf{I} + \Delta t \frac{\Delta_- |\mathbf{B}| \cdot}{\Delta y} \right) \delta \mathbf{U}_{i,j}^{n+1} &= \Delta \mathbf{U}_{i,j}^{n+1} \\
 \mathbf{U}_{i,j}^{n+1} &= \frac{1}{2} \left( \mathbf{U}_{i,j}^n + \mathbf{U}_{i,j}^{n+1} + \delta \mathbf{U}_{i,j}^{n+1} \right)
 \end{aligned} \tag{1.6}$$

where  $\Delta_+$  and  $\Delta_-$  are two-point forward and backward differences in the appropriate coordinate directions, respectively. The matrices  $|\mathbf{A}|$  and  $|\mathbf{B}|$  are matrices having nonnegative eigenvalues, computed from the corresponding Jacobian matrices in such a way that they are non-zero only when the local (explicit) CFL condition is violated. Thus, by virtue of the form of Eqs. (1.6) the implicitness of the scheme is incorporated as an approximate LU factorization, and the scheme can be marched spatially; by virtue of the construction of  $|\mathbf{A}|$  and  $|\mathbf{B}|$ , the scheme reduces to MacCormack's original explicit predictor-corrector Lax-Wendroff scheme when the local CFL condition is satisfied, and no effort is wasted on the local block inversions when they are not needed.<sup>4</sup>

Results for the turbulent boundary layer/shock interaction at a Reynolds number of  $3 \times 10^7$  produced virtually the same results as the earlier explicit scheme with the required CPU time reduced by more than a factor of 1,000. The scheme is only slightly more efficient than MacCormack's earlier explicit-implicit characteristic scheme [19, 22, 23], but is much easier to implement.

The implicit-explicit LU factored scheme was applied to the prediction of transonic flows past airfoils by Kordulla & MacCormack [39]. The finite-volume form of the explicit-implicit, predictor-corrector scheme was applied

<sup>4</sup> Note that without the dimension by dimension splitting in each step, the above arrangement becomes similar to point implicit symmetric Gauss-Seidel iteration for steady-state calculations.



to solve the Reynolds-Averaged Navier-Stokes equations on body-fitted grids. Several modifications were made to the basic numerical scheme. First, it was found advantageous to retain as much of the explicit contribution as possible in the final solution, and a CFL-based weighting of the implicit operator was introduced to provide as smooth a blending as possible between the explicit and implicit operators. Second, it was found necessary to add more dissipation for the more complicated airfoil problems; this dissipation is described as being "... third order small with the derivatives in the sweeping directions as coefficients." Third, the boundary condition at the solid wall is modified to cancel the flux there immediately (rather than carry this over to the corrector step as suggested in the original method).

Relative to conventional, fully-implicit (ADI) methods, the explicit-implicit, predictor-corrector scheme has the advantages of 1) requiring the solution only of bi-diagonal factors; 2) requiring the use only of (modified) Euler Jacobians, and 3) reverting to an explicit predictor-corrector scheme when the local CFL condition is satisfied. Computations for three different airfoils at Mach numbers in the range  $0.30 \leq M_\infty \leq 0.73$  and Reynolds numbers in the range  $4 \times 10^6 \leq Re_c \leq 6.5 \times 10^6$  show good agreement with other computations, and with experimental results when sufficiently fine grids are used (on the order of  $210 \times 60$  cells).

Gupta, Gnoffo, and MacCormack [42, 50] applied the new explicit-implicit method to the viscous shock layer on a blunt cone. The bow shock was again fitted, and the implicit operator was developed in the resulting body-fitted coordinate system. The results were shown to be relatively insensitive to Courant number, demonstrating the benefit of implicit methods as a convergence-acceleration technique for steady flow problems. Kneile and MacCormack [45] applied the explicit-implicit method to the Navier-Stokes equations for three-dimensional, internal flows. This work demonstrated the benefit of developing an implicit technique that could be implemented as an "add-on" to an existing explicit code. A version of Bob's explicit Euler code was (relatively) easily modified to include the viscous terms of the Navier-Stokes equations and the bi-diagonal implicit algorithm. Results are presented for several test cases, including a three-dimensional converging-diverging nozzle flow.

Bob investigated the use of multigrid to accelerate the convergence of solutions to the Navier-Stokes equations for steady flows in [47]. The method was based on his earlier explicit-implicit algorithm [37], applied in finite-volume form. The multigrid implementation was based on the Ni scheme (as implemented by Johnson for the Navier-Stokes equations). The method was applied to the laminar shock-induced separation problem, and resulted in a convergence rate speed-up of only about three (compared to the expected factor of  $8 \frac{1}{2}$ ). The results did demonstrate that the multigrid method was capable of greatly accelerating the rate of signal propagation in hyperbolic

problems, but realization of the full potential of multigrid would have to wait for further developments.

In [46], MacCormack extended the Gauss-Seidel line relaxation method developed for the flux-split, type-dependent difference scheme used by Chakravarthy for the Euler equations to the Navier-Stokes equations.

Noting that the estimates made by Dean R. Chapman for the computational resources required for the solution of the Navier-Stokes equations about a complete aircraft soon would be available, MacCormack [48] reviewed the problems associated with fitting a computational mesh about a complete aircraft configuration.<sup>5</sup> In this survey article, Bob argues that the Reynolds-Averaged Navier-Stokes equations soon would be solved with the same degree of accuracy as then current (1984) high Reynolds number calculations of flows past relatively simple configurations, such as two-dimensional airfoils and bodies of revolution. The article provides a complete recipe of the technologies needed to accomplish the goal of computing the viscous flow past a complete aircraft configuration, including (multi-block, structured) mesh generation and algorithms for solving the Reynolds-Averaged Navier-Stokes (RANS) equations, including developments in flux-vector splitting, the finite-volume formulation, implicit algorithms, and multigrid.

In the far-reaching survey paper [49] Bob goes beyond summarizing past work and emphasizes his line Gauss-Seidel implicit scheme. After doing a masterful job of placing the important ideas in historical context, he repeats the philosophy expressed in Eq. (1.2):

$$\begin{aligned}\{\text{NUMERICS}\} \delta U_{i,j}^{n+1} &= \{\text{PHYSICS}\} \\ U_{i,j}^{n+1} &= U_{i,j}^n + \delta U_{i,j}^{n+1}\end{aligned}$$

as motivation for the development of this scheme. He presents several example calculations, including the supersonic flow past a spherically-blunted cone and the transonic flow in a converging-diverging nozzle. He demonstrates that adequately-converged results for the Navier-Stokes equations can be obtained in about 10 iterations, but points out that the results are only first-order accurate and that important work remains to achieve comparable iterative efficiency with higher-order accuracy.

In [55] MacCormack, Chapman, and Göçken introduced new slip boundary conditions for the Navier-Stokes equations that reduce to those of Maxwell at small Knudsen numbers, and that yield the correct shear stress in the limiting case of free-molecule flow. Comparison of the skin friction and heat transfer rates computed for two-dimensional, hypersonic flow past a flat plate compare surprisingly well with experimental results and with results of Direct Simulation Monte Carlo calculations throughout the transitional flow regime,

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<sup>5</sup> It is interesting to note that these computational resources are currently available on high-end lap-top computers.

from continuum to free molecule flow when the new boundary conditions are used.

In [51] Viegas, MacCormack, and Rubesin discuss the prediction of turbulent flows in the trailing-edge region of circulation-control airfoils. Results of computations using two algebraic eddy viscosity models are presented.

In the first paper with his student Graham Candler [54], Bob extends his Gauss-Seidel method [49] to treat hypersonic flows past three-dimensional configurations. The method is fully implicit, using Gauss-Seidel line relaxation, uses flux-dependent differencing, and uses shock fitting for the bow wave. However, the method is fully conservative, allowing embedded cross flow shocks to be captured. The Reynolds-Averaged Navier-Stokes equations are solved, using the Baldwin-Lomax turbulence model to close the system. Solutions are presented for a biconic body and also for the X-24C-10D lifting body computed previously by Shang and Scherr.<sup>6</sup> In order to fit their computation within the memory limit of the Cray X-MP 48, MacCormack and Candler used a coarser grid (by a factor of 2 in the meridional direction), but obtained good agreement with the earlier solution, achieving a two order-of-magnitude reduction in computational time relative to the explicit method.

In [56, 57, 58, 63, 65] MacCormack and Candler develop and present their method for solving hypersonic flow problems, including the effects of finite rate chemistry and thermal non-equilibrium. Such flowfields are described by coupled, time-dependent, partial differential equations for the conservation of species, mass, mass-average momentum, the vibrational energies of each diatomic species, the electron energy, and the total mass-averaged energy. The solution procedure is fully implicit, coupling the fluid flow equations with the gas physics and chemistry relations. The Euler fluxes are approximated using flux splitting, while the viscous terms are central-differenced. The method preserves elements in the strong chemistry source terms, and the equations are solved using Gauss-Seidel line relaxation. The method requires only a few hundred time steps to solve axisymmetric flows past simple body shapes, and extension to more complex two-dimensional body geometries is expected to be straightforward. In [72] the method is extended to include electron number densities for weakly ionized flows. Electron densities computed for the hypersonic flow past a spherically blunted cone agree well with flight measurements over a range of altitudes.

In [59] and [62] Viegas, Rubesin and MacCormack describe their computer code for solving the flow past a circulation-control airfoil in a wind tunnel test section. After introducing the idea of code validation, results computed using variants of both the Baldwin-Lomax and the Jones-Launder turbulence

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<sup>6</sup> This computation, presented in AIAA Paper 85-1509 at the 23rd Aerospace Sciences Meeting in Reno, is broadly acknowledged to have been the first solution of the Navier-Stokes equations for a complete aerospace vehicle configuration.

models were compared with the extensive experimental data available for the low subsonic flow past a two-dimensional, circulation-control airfoil. Variants added to the turbulence models included a method of accounting for the history of the jet development and for the effects of streamwise curvature.

In [60] Candler and MacCormack summarize hypersonic research at Stanford University, highlighting recent results in the numerical simulation of radiating, reacting, and thermally excited flows, the investigation and numerical simulation of hypersonic shock wave physics, the extension of the continuum fluid dynamic equations to the transition regime between continuum and free-molecule flow, and the development of novel numerical algorithms for efficient particulate simulations of rarefied flow fields. It is a measure of Bob's ability to mentor his students (and other young researchers) that he encouraged his student Candler to be first author on this overview paper.

In [61] MacCormack reviews the difficulties of constructing efficient algorithms for three-dimensional flow. A number of candidates are analyzed and tested, with most found to have shortcomings. Nevertheless, Bob concludes there is promise that an efficient class of algorithms can be found between the severely time-step-size limited explicit or approximately-factored algorithms and those requiring the computationally intensive direct inversion of large sparse matrices. He spends most of his words and equations in this paper showing how factored algorithms do not necessarily follow the old saw that "extension to three dimensions is straightforward." Nevertheless, he provides a Gauss-Seidel algorithm that converges to the solution of a three-dimensional transonic cascade problem (admittedly a turbine nozzle, not a compressor blade passage) in about 50 iterations.

In [64] MacCormack and Göçken describe a thermochemical non-equilibrium formulation for hypersonic, transitional flows of air. The air is assumed to have five chemical species ( $N_2$ ,  $O_2$ ,  $NO$ ,  $N$ , and  $O$ ), and three temperatures corresponding to the translational, rotational, and vibrational modes of energy. Slip boundary conditions are introduced for both velocity and temperatures to extend the validity of the continuum formulation for low-density flows. Solutions for the rarefied, hypersonic flow past a 5-degree, spherically-blunted cone are compared with DSMC results to indicate the range of transitional Knudsen numbers for which the continuum results remain valid.

In [67] Bob discusses the impact of computational fluid dynamics on the design of fluid flow devices. He reviews his efficient numerical procedure for solving the Navier-Stokes equations in three dimensions, based on block tridiagonal inversion in two directions with Gauss-Seidel relaxation in the third direction [61], presenting results for the hypersonic (Mach 20) flow past a winged re-entry vehicle, computed on an inexpensive desk-top work station.

In [68] and [78] MacCormack and Wilson present the coupling of a fully-implicit finite-volume algorithm for two-dimensional axisymmetric flows to a

detailed hydrogen-air reaction mechanism (represented by 33 reactions among 13 species) to investigate supersonic combustion phenomena. They compare the results of numerical computations with ballistic-range shadowgraphs that exhibit two discontinuities as a blunt body passes through a premixed, stoichiometric mixture of hydrogen and air. They discuss the suitability of the numerical procedure for simulating these double-front phenomena, and examine the sensitivity of these flow fields to key reaction rates.

In [69] [81] MacCormack and his student Zhong use linearized stability analysis to develop a set of (stabilized) augmented Burnett equations. The new equations are solved for one-dimensional shock wave structures and two-dimensional flows past blunt leading edges. The stability of the conventional and augmented equation sets are tested numerically, confirming that the augmented equations are always stable and maintain the same accuracy as the conventional set. They show that at high altitudes the difference between solutions of the Burnett equations and the Navier-Stokes equations is significant, especially for parameters sensitive to flow field details, such as radiation. In [74] Zhong and MacCormack evaluate a number of models for surface slip boundary conditions for the augmented Burnett equations.

In [70] and [76] MacCormack and Conti merge Bob's implicit numerical method for the Navier-Stokes equations [49] with materials response technology for carbonaceous materials to yield two-dimensional, transient solutions for the coupled flow-materials problem. The vehicle surface temperature and heat shield ablation rate are computed, and the resulting change in vehicle shape is accounted for. Results of a test computation is presented for a typical ballistic re-entry vehicle, covering an altitude range from 43 kilometers to sea level. Also, in [77] MacCormack and Conti apply MacCormack's implicit method to the problem of laminar, axisymmetric near wakes with gas injection. The flow past a spherically-blunted 7-degree cone at Mach 22 is computed with the transient injection of cool inert gas into equilibrium air for two different locations of injection ports.

In [66, 75] MacCormack and Rostand apply the fully-implicit technique to the simulation of a nitrogen plasma in thermodynamic non-equilibrium. This requires the incorporation of state-of-the-art physical models, as well as MacCormack and Candler's numerical techniques. Results are compared with an arc-heated nitrogen plasma jet, with generally good agreement.

In [71] Moreau, Chapman, and MacCormack present a fully-implicit finite-volume algorithm for axisymmetric flows, including complete thermal and chemical non-equilibrium and a higher-order simplified Burnett stress tensor, coupled to an improved detailed non-equilibrium radiation code. A low-speed bow-shock ultraviolet flight experiment is used to benchmark the effect of rarefaction modeling on radiation at high altitudes. They demonstrate that inclusion of the rotational non-equilibrium and simplified Burnett terms does not improve the trend for the low-speed test, but does make a

difference at higher speeds. It is shown that the results are highly sensitive to radiation modeling, and that the maximum vibrational temperature and *NO* concentration are more critical than the maximum translational temperature for getting accurate radiation results.

Menon and MacCormack [73] applied the implicit, Gauss-Seidel line relaxation solver to the problem of supersonic mixing of air and helium in the region downstream of a rearward-facing step. Agreement with experimental results for the same case was only fair, doubtless degraded significantly by the rather simple (algebraic) turbulence models used.

Moreau, Laux, Chapman, and MacCormack [79] describe improvements to the *NEQAIR* computer code based on results of two experiments: a plasma torch experiment conducted at Stanford and measurements from the SDIO/IST Bow-Shock-Ultra-Violet missile flight. The computer code also was extended to handle any number of species and radiative bands in a gas whose thermodynamic state can be described by up to four temperatures. It provides greater efficiency for computing very fine spectra, and includes transport phenomena along the line of sight.

Moreau, Chapman and MacCormack [80] developed a quasi-one-dimensional flux-split, finite-volume computer code including additional rotational relaxation and separate vibrational modes. The code was used to compute the shock wave in a radiation experiment conducted by Sharma and Gillespie. The results demonstrated that the commonly used rotational model of Parker was inadequate to simulate the observed rotational temperature at peak radiation, and that a correction to the Parker model, introduced to account for the diffusional nature of the relaxation process, is able to recover the large initial difference.

Comeau, Chapman and MacCormack [82] study the shock interaction produced when an incident shock wave impinges on a blunt body, such as the engine inlet cowl lip of a hypersonic vehicle. The flux-vector split scheme of Steger and Warming is used to solve the Navier-Stokes equations for a perfect gas at altitudes ranging from continuum conditions to transitional flow conditions. The authors show that the interaction becomes fundamentally different as the fluid density is decreased, with its effect on the overheating problem correspondingly diminished. They find that the maximum stagnation point heating at the highest altitude is reached only when the incident shock misses the cowl lip entirely, and any interaction with the cowl bow shock that does occur takes place downstream (and, thus, has little effect on the conditions at the stagnation point).

In [83] Welder, Chapman, and MacCormack study alternative forms of the Burnett equations in which the inviscid, isentropic approximation for the material derivative, present in both the viscous stress and heat conduction expressions of the equations in their original form is replaced by the exact material derivative, and also using improved approximations based on the

Navier-Stokes (rather than the inviscid) equations. The various Burnett-order equations are studied to determine their stability to small wave length disturbances, and numerical accuracy for a one-dimensional shock structure. It is discovered that formulations that do not make some approximation to the material derivatives can lead to un-physical heat conduction. Two modified formulations are developed that greatly minimize this problem, while at the same time improving the accuracy of shock structure computations.

In [84] Bob summarizes a quarter century of CFD research in a very warm and personal way. He discusses developments in transonic potential flow computations, including those for the small-disturbance theory, as well as numerical approaches for the Euler and Navier-Stokes equations, and provides a number of enjoyable personal anecdotes along the way. He also provides his predictions about future work on computational grids, computer architectures, algorithms, and turbulence research.

In [85] Bob reviews progress of two decades of CFD research, and points the way to the issues that must be resolved for the field to become fully mature. Bob predicts that future decisions will be concerned with structured, multi-block grids versus unstructured grids, the modeling of turbulence versus direct simulation of turbulence phenomena, and indirect relaxation (or approximately factored) schemes versus direct solution procedures. In retrospect, these were highly accurate predictions, as most of these battles continue to be fought today.

In [86] Comeaux, Chapman, and MacCormack look at the entropy balance relation for the Burnett equations from two points of view: from classical thermodynamic theory using the Gibbs equation and the continuum conservation relations for mass, momentum, and energy; and from kinetic theory using Boltzmann's H-theorem in conjunction with the Chapman-Enskog expansion. They find that in both cases the irreversible entropy production is not positive semi-definite, in violation of the second law of thermodynamics. They also show that the two formulations are completely equivalent (to second order in the Knudsen number), indicating that the Gibbs equation is consistent with the Burnett equations (in contradiction to the results of earlier researchers who did not carry the derivation to its culmination). The inconsistency with the second law is proposed as a source of the numerical problems experienced by researchers attempting to solve the Burnett equations over the previous five decades.

In [87] Kao, von Ellenrieder, MacCormack, and Bershader study the interaction of a two-dimensional compressible vortex with a shock wave, both experimentally and numerically. The unsteady Navier-Stokes equations are solved using a second-order accurate, shock-capturing, total-variation diminishing (TVD) scheme, with results of the computations in good qualitative agreement with the physical experiment.

In [88] Moreau, Chapman, and MacCormack propose a new temperature

dependence expression for the Zeldovich reaction rate that accounts for the observed delayed formation of *NO* due to vibrational excitation. The new model of the exchange reaction brings a natural extension to the multi-temperature reaction rate model of Park for the dissociation reactions. A detailed analysis of the energy exchange mechanisms also emphasizes the better physical behavior of the extended Schwartz-Slawsky-Herzfeld (*SSH*) model (compared to simpler models). These simple modifications are expected to significantly improve the predictive capabilities of state-of-the-art detailed thermochemical non-equilibrium codes used to study low-density gas flows.

In [89] Melville and MacCormack present a methodology for designing optimal integration schemes for ordinary differential equations. Linear analysis is used to construct a generalized two-step, predictor-corrector method, and then to optimize it for hyperbolic and parabolic systems. For both cases, computational efficiency is improved over previous (standard) schemes, with no significant loss in accuracy. No extra memory is needed, but initialization is required.

The three-dimensional, compressible Euler equations were used by Melville and MacCormack [90] to study the unsteady behavior of the double helix mode of vortex breakdown. The convection of a longitudinal vortex through an adverse pressure gradient shows that the unsteady flow field is dominated by a single, spatially uniform frequency, associated with the rotation of the helical vortex structure.

In [91] Bob used efficient matrix decomposition to construct implicit algorithms. He first analyzed three strategies for solving the implicit matrix equations. The approximate *LU* decomposition via the Strongly Implicit Procedure (*SIP*) where the *LU* matrix is inverted by a forward elimination down the diagonal of the *L*-matrix, followed by a backward substitution up the diagonal of the *U*-matrix. During the forward elimination procedure,  $2N$  matrix elements of size  $4 \times 4$  for 2-D flows (and  $3N$  matrix elements of size  $5 \times 5$  for three dimensions) are calculated and stored, where  $N$  is the number of grid points. The approximate decomposition introduces an asymmetry into the calculation which can be minimized by reordering the matrix equation on alternate time steps (or by averaging the original asymmetric operators). Gauss Seidel Line Relaxation (*GSLR*) is another strategy which introduces a preferred direction, usually crossing through a boundary layer with a block tridiagonal inversion. "It is therefore unsuitable for domains with corners containing intersecting boundary layers, although it is usually exceptionally efficient otherwise."

The third strategy is Approximate Factorization (*AF*) of the differential equations. Both *GSLR* and *AF* have the advantage of inverting a matrix associated with "a line at a time" in two or three dimensions. (On the other hand, approximate *LU* decomposition or *SIP* can be used for totally unstructured grids).



After this analysis, Bob modified *SIP* by adding additional elements in the  $L$  and  $U$  matrices to eliminate the asymmetry in the calculations, in such a manner that the exact same elements of the original matrix are returned as before, but the matrix is now decomposed into three factors, one of them a block diagonal matrix (in three dimensions, there are five factors with two block diagonal matrices). This modified *SIP* is related to a special Approximate Factorization (*AF*) where only matrices associated with “a line at a time” are inverted.<sup>7</sup>

Bob also proposed to iterate on the splitting error (twice or three times) to improve the approximation, see [92].

In [92, 93, 97] Bob introduces his latest implicit scheme for solving the unsteady Euler or Navier-Stokes equations. The method is based on using (limited) iteration to dramatically reduce the factorization error of implicit schemes based on approximate factorization, and allows the convergence of solutions to within “engineering accuracy” in about 50 – 100 time steps for three test problems (including supersonic flow past a blunt body, and transonic and subsonic flows through a nozzle). The new method is compared to standard approximate-factorization schemes by MacCormack and Pulliam [94], with indications that the new procedure is about five times more efficient. Pulliam, MacCormack, and Venkateswaran examine the convergence characteristics of a number of implicit approximation schemes, including the DDADI scheme, in [96]. They also show the benefit of subiterations, and conclude that ADI and D3ADI with subiterations perform equally well, with D3ADI being possibly more robust. There are two ways to improve the performance of Approximate Factorization Schemes, either to cycle a parameter or to cycle grids. In [94], MacCormack and Pulliam used the new modified approximate factorization with two subiterations (AF2) combined with multigrid and obtained impressive results.

In [98, 99] MacCormack shows how the equations of magnetofluid dynamics can be modified to make the flux vectors homogeneous of degree one. This allows their solution in conservation form, and allows a modified Steger-Warming flux-vector splitting to be used.

## 1.4 Concluding Remarks

The preceding summary of his contributions makes clear the many original contributions that Bob MacCormack has made to computational fluid dynamics, and the enormous impact he has had on the development of CFD and its application to practical problems in engineering.

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<sup>7</sup> An alternative approach would be alternating direction symmetric Gauss Seidel Line Relaxation or Alternating Direction Zebra Relaxation, with alternating odd and even lines in each direction.

Bob was amongst the first to use directional splitting (backward differences in  $x$  and  $y$ , followed by forward differences in both directions), dimensional splitting or dimension by dimension factorization, and physical splitting (convection, or hyperbolic, and diffusion, or parabolic) as discussed in the paper by his colleagues at NASA Ames (Hung, Deiwert and Inouye) in this volume. Bob worked with many people, including, for example, H. Lomax, R. Warming, B. Baldwin, A. Paullay, M. Inouye, G. Deiwert, C. Hung, A. Rizzi, J. Viegas, M. Rubesin and T. Pulliam at NASA Ames; with J. Shang at the Air Force Research Laboratory at Wright Field; with W. Kordulla when he was an NRC senior research associate at Ames. At Stanford he worked with D. Chapman and D. Bershader, and he was advisor of many students, including T. Gocken, G. Candler, X. Zhong, S. Moreau, K. Comeaux, R. Melville, G. Wilson, C. Laux, W. Welder, P. Bourquin, C. Kao, K. von Ellenreider and many others. He visited the Institute for Computer Applications in Science and Engineering (ICASE) at NASA Langley frequently, and gave there the Theodorsen Lecture there in 2001.

The editors are pleased to have been able to bring together the researchers who have contributed to this volume to express our thanks to Bob, and to provide this summary of his technical contributions. As noted earlier, as impressive as these technical contributions have been, they represent only one dimension of Bob's impact; his personal presence and energy, and his willingness to help others, especially younger researchers, are particularly noteworthy. He is respected by everyone of this community in the U.S. and abroad. We wish Bob continued success for many years to come.

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# The Effect of Viscosity in Hypervelocity Impact Cratering

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## 2.1 Abstract

A numerical method, of second order in both time and space, for the solution of the time-dependent compressible Navier-Stokes equations is presented. Conditions for stability are discussed. The method has been applied to calculate an axisymmetric flow field produced by hypervelocity impact. Results are given for impacts of aluminum cylinders (having diameters of 0.16, 0.32, and 0.64 cm) into aluminum targets. Viscosities of zero and  $10^4$  poise were assumed. Both plates and semi-infinite targets are considered at an impact speed of 10 km/sec. It is concluded that viscous effects become increasingly important as projectile size diminishes and cannot be neglected during the initial stages of crater formation for projectiles smaller than 0.5 cm in diameter.

## 2.2 Introduction

Denardo [1, 2] in 1964, reported a deviation from simple linear scaling in the hypervelocity impact of aluminum spheres, of diameters 0.16, 0.32, 0.64, and 1.27 cm, into aluminum targets. Penetration and momentum transfer

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to the target decreased more rapidly than simple scaling rules would imply as the projectile size was reduced. More explicitly, the ratios of penetration to projectile diameter and target momentum after impact to projectile momentum varied as the  $1/18$  and  $1/6$  powers, respectively, of projectile diameter. Two possible sources of this phenomenon are: (a) a pure scale effect in the static strength of the materials, as shown by Kuhn and Figge [3]; (b) a rate-dependent stress effect, for example, viscosity. The former would act toward the end of crater formation; the latter during the earlier part of crater formation where high shear rates exist. The purpose of this study is to observe the effect of viscosity during the initial stage of cratering (defined to take place from the initiation of impact until static strength effects become significant). During this time, hypervelocity impact cratering may be described by the Navier-Stokes equations of fluid dynamics, whose solutions scale nonlinearly for viscosities different from zero. A numerical method is described to solve these time-dependent equations. The method is of second order in both time and space, and is thus more accurate than the methods derived from the Los Alamos Particle In Cell Code [4], which have been previously used [5, 6, 7] to calculate hypervelocity impact phenomena. The results of the computation of several impact cases are examined to determine the effect of viscosity.

## 2.3 The Numerical Method

### 2.3.1 Differential Equations

The Navier-Stokes equations of fluid dynamics, neglecting body forces and heat sources, may be written [8]

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} &= 0 \quad (\text{continuity equation}) \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} + \frac{\partial p}{\partial x_i} - \frac{\partial \sigma_{i,j}}{\partial x_j} &= 0 \quad (\text{momentum equation}) \\ \frac{\partial e}{\partial t} + \frac{\partial (e + p) u_j}{\partial x_j} - \frac{\partial (u_i \sigma_{i,j} - q_j)}{\partial x_j} &= 0 \quad (\text{energy equation}) \\ p = f(\epsilon, \rho) &= f\left(\frac{e}{\rho} - \frac{|\vec{u}|^2}{2}, \rho\right) \quad (\text{equation of state})\end{aligned}$$

where Einstein notation has been used (repeated indices are summed) and

$t$  = time

$\rho$  = the mass per unit volume

$u_i$  = component of velocity  $\vec{u}$  in the  $x_i$  direction

$\sigma_{i,j}$  = viscous stress  $= \lambda \delta_{i,j} \frac{\partial u_k}{\partial x_k} + \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$

where

$\delta_{i,j}$  = the Kronecker delta

$\mu, \lambda$  = first and second coefficients of viscosity, respectively

$e$  = total energy per unit volume

$$q_j = -k \frac{\partial T}{\partial x_j}$$

where  $k$  is the coefficient of heat conductivity, and the temperature  $T$  and specific internal energy  $\epsilon$  are related by an equation of form  $\epsilon = g(\rho, T)$ . Finally the pressure  $p$  is expressed as a function of  $\epsilon$  and  $p$ .

This set of equations may easily be expressed in other coordinate systems [9]. A two-step difference method has been devised to numerically solve the above set. For simplicity and also to be consistent with descriptions of other numerical techniques in the literature, the method will be illustrated in a two-dimensional cartesian coordinate system. The application to an axisymmetric system will be briefly commented on later. Also, for the analysis of numerical stability only the inviscid non-heat-conducting equations will be treated in detail. Both the viscous and heat conduction terms, physically and also numerically (if their magnitudes in the differenced equations are not too great), tend to damp out the high frequency components of the solution. These components are normally the ones which cause numerical instability. Thus, the approach here of choosing  $\mu$  and  $k$  equal to zero is conservative for the purposes of stability analysis. The equations in vector form are

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0 \quad (2.1)$$

where

$$U = \begin{pmatrix} \rho \\ m \\ n \\ e \end{pmatrix}, \quad F = \begin{pmatrix} m \\ m^2/\rho + p \\ mn/\rho \\ (e+p)m/\rho \end{pmatrix}, \quad G = \begin{pmatrix} n \\ mn/\rho \\ n^2/\rho + p \\ (e+p)n/\rho \end{pmatrix}$$

$$p = f(\epsilon, \rho) = f\left(\frac{e}{\rho} - \frac{m^2 + n^2}{2\rho^2}, \rho\right)$$

and  $m = \rho u$  and  $n = \rho v$  are the momenta per unit volume in the  $x$  and  $y$  directions, respectively.

### 2.3.2 Difference Equations

The two-step difference method to solve Eq. (2.1) is defined by

$$U_{j,k}^{\overline{n+1}} = U_{j,k}^n - \frac{\Delta t}{\Delta x} (F_{j+1,k}^n - F_{j,k}^n) - \frac{\Delta t}{\Delta y} (G_{j,k+1}^n - G_{j,k}^n) \quad (2.2)$$

$$U_{j,k}^{n+1} = \frac{1}{2} \left\{ U_{j,k}^n + U_{j,k}^{\overline{n+1}} - \frac{\Delta t}{\Delta x} (F_{j,k}^{\overline{n+1}} - F_{j-1,k}^{\overline{n+1}}) - \frac{\Delta t}{\Delta y} (G_{j,k}^{\overline{n+1}} - G_{j,k-1}^{\overline{n+1}}) \right\} \quad (2.3)$$

where  $F_{j,k}^n$  and  $G_{j,k}^n$  equal  $F(U_{j,k}^n)$  and  $G(U_{j,k}^n)$  respectively. The subscripts now and from here on refer to a spatial mesh of points  $(x_j, y_k)$  with spacing  $\Delta x$  and  $\Delta y$ , and the superscripts refer to times  $t = n\Delta t$ , where  $\Delta t$  is the time increment that the solution is advanced during each cycle of Eqs. (2.2,2.3). The method first obtains approximate values,  $\overline{U_{j,k}^{n+1}}$  at each point, by Eq. (2.2), which uses two forward differences to approximate the two spatial derivatives. The approximate solution is then used to calculate  $\overline{F_{j,k}^{n+1}}$  and  $\overline{G_{j,k}^{n+1}}$ , which are used in Eq.(2.3) with two backward differences to obtain the new accepted value  $\overline{U_{j,k}^{n+1}}$ . As in the Particle In Cell method, the flow field can be viewed as divided into cells with forces acting on cell faces and with mass, momentum, and energy being transported between cells. The set of differential equations is in conservation law form [10]. Similarly, examination of the set of difference equations Eqs. (2.2,2.3), shows that mass, momentum, and energy are conserved during the calculation; that is, the differenced quantities in Eqs. (2.2,2.3) at interior points of the mesh appear exactly twice during a sweep through the mesh, each time of opposite sign. Thus, the vector sum,  $\sum_{j,k} \overline{U_{j,k}^{n+1}}$  or  $\sum_{j,k} \overline{U_{j,k}^{n+1}}$  represents the net transport and stress effects at the boundaries.

### 2.3.3 Accuracy and Stability

The numerical stability of methods of this type, namely, those of Lax and Wendroff [10], cannot presently be completely analyzed in the general nonlinear form. The most successful attempt at analysis to date is to first linearize the set of differential Eqs. (2.1) and then to study the amplification of Fourier components of the solution by the difference method applied to the linearized set. The new set of equations is then

$$\frac{\partial U}{\partial t} + J_F \frac{\partial U}{\partial x} + J_G \frac{\partial U}{\partial y} = 0 \quad (2.4)$$

where  $J_F$  and  $J_G$  are the Jacobian matrices of  $F$  and  $G$  with respect to  $U$  and are considered to be constant.

$$J_F = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\frac{m^2}{\rho^2} + \frac{\partial p}{\partial \rho} & \frac{2m}{\rho} + \frac{\partial p}{\partial m} & \frac{\partial p}{\partial n} & \frac{\partial p}{\partial e} \\ -\frac{mn}{\rho^2} & \frac{n}{\rho} & \frac{m}{\rho} & 0 \\ -\frac{m}{\rho} \frac{e+p}{\rho} + \frac{m}{\rho} \frac{\partial p}{\partial \rho} & \frac{e+p}{\rho} + \frac{m}{\rho} \frac{\partial p}{\partial m} & \frac{m}{\rho} \frac{\partial p}{\partial n} & \frac{m}{\rho} + \frac{m}{\rho} \frac{\partial p}{\partial e} \end{pmatrix}$$

$$J_G = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{mn}{\rho} & \frac{n}{\rho} & \frac{m}{\rho} & 0 \\ -\frac{n^2}{\rho^2} + \frac{\partial p}{\partial \rho} & \frac{\partial p}{\partial m} & \frac{2n}{\rho} + \frac{\partial p}{\partial n} & \frac{\partial p}{\partial e} \\ -\frac{n}{\rho} \frac{e+p}{\rho} + \frac{n}{\rho} \frac{\partial p}{\partial \rho} & \frac{n}{\rho} \frac{\partial p}{\partial m} & \frac{e+p}{\rho} + \frac{n}{\rho} \frac{\partial p}{\partial n} & \frac{n}{\rho} + \frac{n}{\rho} \frac{\partial p}{\partial e} \end{pmatrix}$$

$$\begin{aligned}
\frac{\partial p}{\partial e} &= \frac{\partial p}{\partial \epsilon} \frac{\partial \epsilon}{\partial e} = \frac{1}{\rho} \frac{\partial p}{\partial \epsilon} \\
\frac{\partial p}{\partial m} &= \frac{\partial p}{\partial \epsilon} \frac{\partial \epsilon}{\partial m} = -\frac{m}{\rho^2} \frac{\partial p}{\partial \epsilon} \\
\frac{\partial p}{\partial n} &= \frac{\partial p}{\partial \epsilon} \frac{\partial \epsilon}{\partial n} = -\frac{n}{\rho^2} \frac{\partial p}{\partial \epsilon} \\
\frac{\partial p}{\partial \rho} &= \frac{\partial p}{\partial \rho} \Big|_{\epsilon} + \frac{\partial p}{\partial \epsilon} \frac{\partial \epsilon}{\partial \rho} = \frac{\partial p}{\partial \rho} \Big|_{\epsilon} + \left( -\frac{e}{\rho^2} + \frac{m^2 + n^2}{\rho^2} \right) \frac{\partial p}{\partial \epsilon}
\end{aligned}$$

The set of Eqs. (2.4) approximates Eq. (2.1) locally, and difference methods found unstable on it can be expected to be unstable in the general nonlinear case. Two conditions inherent in such an analysis are: (a) the boundary conditions have no effect on stability, and (b) the exact solution to Eq. (2.1) is smooth. The latter condition allows the matrices  $J_F$  and  $J_G$  to be treated as constant. The results of the analysis thus apply principally to regions of the flow away from the boundaries and in which there are no discontinuities, such as shock waves. In the next few paragraphs, this method of analysis will be used to obtain a bound for the maximum amplification of any Fourier component of the solution in regions which satisfy these conditions.

The amplification matrix of the present method applied to Eq. (2.4) for a single Fourier component of the solution,

$$W(t) \exp[i(k_1 x + k_2 y)]$$

becomes [11],

$$\begin{aligned}
G = I + & \frac{i \frac{\Delta t}{\Delta x} (J_F \sin \xi + J_G \sin \eta)}{-\frac{1}{2} \left( \frac{\Delta t}{\Delta x} \right)^2 \left( (1 - e^{-i\xi}) J_F + (1 - e^{-i\eta}) J_G \right)} \\
& \left( (1 - e^{i\xi}) J_F + (1 - e^{i\eta}) J_G \right)
\end{aligned} \tag{2.5}$$

where  $\xi = k_1 \Delta x$ ,  $\eta = k_2 \Delta y$ , and equal spacing is assumed ( $\Delta x = \Delta y$ ).

For  $\xi$  and  $\eta \ll 1$ , the approximations  $\sin \xi = \xi$ ,  $1 - \cos \xi = \xi^2/2$ , and  $1 + \cos \xi = 2 - \xi^2/2$  and similar ones for  $\eta$  may be used to show that

$$G = I + i \frac{\Delta t}{\Delta x} (J_F \xi + J_G \eta) - \frac{1}{2} \left( \frac{\Delta t}{\Delta x} \right)^2 (J_F \xi + J_G \eta)^2 + O(\xi^3, \eta^3)$$

Thus  $G = \exp \left[ i \frac{\Delta t}{\Delta x} (J_F \xi + J_G \eta) \right]$  modulo terms of third order in  $\xi$  and  $\eta$ . The exact solution of the differential Eq. (2.4) for the above Fourier component is

$$\exp[it(k_1 J_F + k_2 J_G)] \exp[i(k_1 x + k_2 y)] W(0)$$

Hence, the exact amplification of the solution from  $t = 0$  to  $t = \Delta t$  is

$$\exp[i\Delta t(k_1 J_F + k_2 J_G)] = \exp\left[i \frac{\Delta t}{\Delta x} (J_F \xi + J_G \eta)\right]$$

The difference between the two is of third order in  $\xi$  and  $\eta$ . Thus, the method defined by Eqs. (2.2,2.3) is of second order accuracy [10].

$G$  may also be written as

$$G = I + M - M^* - 2M^*M$$

where

$$M = \frac{1}{2} \frac{\Delta t}{\Delta x} ((1 - e^{i\xi}) J_F + (1 - e^{i\eta}) J_G)$$

and  $M^*$  is the complex conjugate of  $M$ .

Stability is assured if it can be shown that all of the eigenvalues  $\lambda_i$  of  $G$  satisfy the von Neumann condition  $|\lambda_i| \leq 1 + O(\Delta t)$  [11]. The eigenvalues of  $G$  are invariant under a similarity transformation. Using such a transformation  $S$ ,  $M$  may be written as

$$M = S \frac{1}{2} \frac{\Delta t}{\Delta x} ((1 - e^{i\xi}) A + (1 - e^{i\eta}) B) S^{-1}$$

where

$$A = \begin{pmatrix} u & c & 0 & 0 \\ c & u & 0 & 0 \\ 0 & 0 & u & 0 \\ 0 & 0 & 0 & u \end{pmatrix}$$

$$B = \begin{pmatrix} v & 0 & c & 0 \\ 0 & v & 0 & 0 \\ c & 0 & v & 0 \\ 0 & 0 & 0 & v \end{pmatrix}$$

$$S^{-1} = \begin{pmatrix} c & 0 & 0 & \beta/c \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -u & 1 & 0 & 0 \\ -v & 0 & 1 & 0 \\ -\alpha & -u & -v & 1 \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ u & 1 & 0 & 0 \\ v & 0 & 1 & 0 \\ u^2 + v^2 + \alpha & u & v & 1 \end{pmatrix} \begin{pmatrix} 1/c & 0 & 0 & -\beta/c^2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where

$$c^2 = \frac{\partial p}{\partial \rho} \Big|_{\epsilon} + \frac{p}{\rho^2} \frac{\partial p}{\partial \epsilon},$$



is the square of the local adiabatic speed of sound,

$$\alpha = \frac{e + p}{\rho} - u^2 - v^2$$

and

$$\beta = \frac{1}{\rho} \frac{\partial p}{\partial \epsilon}$$

Then  $G' = S^{-1}GS = I + M' - M'^* - 2M'^*M'$ , where  $M' = S^{-1}MS$  and is symmetric.

For any unit vector  $w$ , the inner product

$$\begin{aligned} (w, G'w) &= (w, w) + (w, M'w) - (w, M'^*w) - 2(w, M'^*M'w) \\ &= 1 - 2\|M'w\|^2 + (w, M'w) - (w, M'^*w) \end{aligned}$$

where the norm of  $M'w$  is denoted by  $\|M'w\| = (M'w, M'w)^{1/2}$ .

The quantity  $(w, M'w) - (w, M'^*w)$  is pure imaginary and since

$$|(w, M'w)| \leq \|w\|\|M'w\| = \|M'w\|$$

by the Cauchy-Schwartz inequality

$$|(w, G'w)|^2 \leq (1 - 2\|M'w\|^2)^2 + (2\|M'w\|)^2 \leq 1 + 4\|M'w\|^4$$

Gerschgorin's theorem [12] may be used to show

$$\|M'w\| \leq \frac{\Delta t}{\Delta x} (|u| + |v| + 2c)$$

Thus, if  $\nu = 4\frac{\Delta t^3}{\Delta x^4}(|u| + |v| + 2c)^4 \approx 1$  then  $|(w, G'w)| \leq 1 + \nu O(\Delta t)$ . If, in particular,  $w$  is any eigenvector of  $G'$ ,  $G'w = \lambda w$ . Then

$$|\lambda|^2 \leq 1 + \nu \Delta t$$

or

$$|\lambda| \leq 1 + O(\Delta t)$$

Hence, all the eigenvalues of  $G$  satisfy  $|\lambda_i| \leq 1 + O(\Delta t)$  if  $\nu$  is of the order of unity.

That is, if  $\nu$  is held constant as  $\Delta t, \Delta x \rightarrow 0$ , and Eqs. (2.2,2.3) are used to calculate the solution to time  $T$ , the number  $n$  of time steps is  $T/\Delta t$  and the amplification of any Fourier component is then less than

$$\sup_{\text{All } w, (w,w)=1} |(w, Gw)|^n \leq (1 + \nu \Delta t)^{n/2} \leq e^{(n/2)\nu \Delta t} = e^{\nu T/2}$$

The amplification of every component of the solution can thus be made arbitrarily small, in computing to a given time by suitably choosing  $\nu$ .

In one dimension (i.e.,  $U = \begin{pmatrix} \rho \\ m \\ e \end{pmatrix}$ ,  $F = \begin{pmatrix} m \\ m^2/\rho + p \\ (e + p)m/\rho \end{pmatrix}$ , and  $G = 0$ )

it is easily shown that the eigenvalues of the amplification matrix of the method are less than unity if  $\frac{\Delta t}{\Delta x}(|u| + c) \leq 1$ . This condition is the well-known Courant-Friedrichs-Lewy (C.F.L.) condition that often appears in fluid dynamics. This is the best bound that can be realized in numerical methods. The noncommutativity of the matrices  $A$  and  $B$  has presently prevented the calculation of the eigenvalues for two dimensions. The condition,  $\frac{\Delta t}{\Delta x}(|u| + |v| + 2c) \approx \Delta t^{1/4}$ , obtained from the derived bound is substantially more restrictive than the two-dimensional C.F.L. condition. Although it can be shown that the derived eigenvalue bound is not the least upper bound, it also can be shown that an eigenvalue does exist such that a more restrictive condition than the C.F.L. condition is required for stability. However, the method defined by Eqs. (2.2,2.3) is only one of four methods of second order accuracy of essentially the same form. For example, if instead of first using two forward spatial differences and then two backward differences, the reverse procedure could be followed, or one forward and one backward difference could be followed by corresponding backward and forward differences. The amplification matrix of each would have different eigenvalues and eigenvectors for the same Fourier component of the solution. Thus if the indices of the method defined by Eqs. (2.2,2.3) were permuted so that the four methods followed one another cyclically, a smaller amplification would be expected; that is, although the maximum eigenvalue in magnitude  $|\lambda_{\max}(G_i)|$  for each  $G_i$ , maximized on the set of all  $\xi$ ,  $\eta$ ,  $u$  and  $v$ , such that  $|u| + |v| \leq \text{constant}$  and  $c$  is constant would be the same, a single choice of  $\xi$ ,  $\eta$ ,  $u$  and  $v$  would not in general maximize all  $G_i$  (i.e.,  $|\lambda_{\max}(G_1 G_2 G_3 G_4)| \leq |\lambda_{\max}(G_i)|^4$ ,  $i = 1, 2, 3$  and  $4$ ). It is conjectured that  $|\lambda_{\max}(G_1 G_2 G_3 G_4)| \leq 1 + O(\Delta t)$  if  $\Delta t$  and  $\Delta x$  satisfy a condition close to the C.F.L. condition.

As previously stated, the addition of viscous and heat conduction terms does not disturb the numerical stability if their magnitudes are not too great (that is, if  $\Delta t$  and  $\Delta x$  are chosen so that  $\frac{\mu}{\rho} \frac{\Delta t}{\Delta x^2}$ ,  $k \frac{\Delta t}{\Delta x^2}$  are sufficiently less than one). Second-order accuracy will also be maintained if the terms are differenced so that their truncation error is also of second order. For example, the viscous term  $\frac{\partial \mu(\frac{\partial u}{\partial x})}{\partial x}$ , if differenced centrally,

$$\frac{\left( \frac{\mu_{i+1} + \mu_i}{2} \frac{u_{i+1} - u_i}{\Delta x} - \frac{\mu_i + \mu_{i-1}}{2} \frac{u_i - u_{i-1}}{\Delta x} \right)}{\Delta x}$$

or if  $\mu$  is constant

$$\mu \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}$$

will suffice for second-order accuracy.

The stability analysis is also essentially the same in axisymmetric cylindrical coordinates. For example, the set of equations corresponding to Eq. (2.4) is

$$\frac{\partial U}{\partial t} + J_F \frac{1}{r} \frac{\partial r U}{\partial r} + J_G \frac{\partial U}{\partial z} = J_H \frac{U}{r} \quad (2.6)$$

where  $r$  and  $z$  are the radial and axial coordinates,  $J_F$ ,  $J_G$  and  $U$  are the corresponding matrices and vector defined in these coordinates and  $J_H$  is the Jacobian of  $H$ ,  $H^T = (0, p, 0, 0)$ , with respect to  $U$ . In flow regions away from the axis,  $r \gg \Delta r$ , it can be shown that the effect of deleting the term  $J_H \frac{U}{r}$  from Eq. (2.6) causes a change in the eigenvalues of the associated amplification matrix of only the order of  $\Delta t$ . Thus, to analyze the stability of the difference method applied to Eq. (4) in regions away from the axis, the right hand side of Eq. (4) may be set to the zero vector. The Fourier component of the solution for this equation with the same wave numbers  $k_1$  and  $k_2$  as considered earlier is

$$\frac{1}{r} \exp[it(k_1 J_F + k_2 J_G)] \exp[i(k_1 r + k_2 z)]$$

Similarly, the corresponding component of the solution to the difference equations, where  $\frac{1}{r} \frac{\partial r U}{\partial r}$  is forward differenced as

$$\frac{1}{(i-1/2)\Delta r} \frac{(i+1/2)\Delta r U_{i+1,j} - (i-1/2)\Delta r U_{i,j}}{\Delta r}$$

and backward differenced as

$$\frac{1}{(i-1/2)\Delta r} \frac{(i-1/2)\Delta r U_{i,j} - (i-3/2)\Delta r U_{i-1,j}}{\Delta r}$$

is

$$\frac{1}{r} W(t) \exp[i(k_1 r + k_2 z)]$$

The amplification matrix for this component by Eq. (2.6), after differencing, is the same as obtained earlier, except that now  $x$  and  $y$  are replaced by  $r$  and  $z$ . Near the axis  $r = 0$ , the boundary conditions induced by axial symmetry (i.e.,  $u_{1,j} = -u_{-1,j}$ ,  $v_{1,j} = v_{-1,j}$ ,  $p_{1,j} = p_{-1,j}$ , etc.) are expected to influence stability, and the above linearized analysis is thus not sufficient. The numerical stability in this region has not yet been analyzed. Also, the nonzero component of  $H$ , occurring from the radial momentum equation, does not allow the equations to be expressed in divergence form [13]. Thus, the difference method applied to Eq. (2.6) rigorously conserves only mass, axial momentum and energy and not radial momentum as well. Again the second-order terms of the differential equations are not expected to disturb the numerical stability and if also differenced to second-order accuracy, the

method itself will be of second-order accuracy. For example, differencing the term  $\frac{1}{r} \frac{\partial r \mu (\partial u / \partial r)}{\partial r}$  by

$$\frac{i \Delta r \left( \frac{\mu_{i+1,j} + \mu_{i,j}}{2} \right) \left( \frac{u_{i+1,j} - u_{i,j}}{\Delta r} \right) - (i-1) \Delta r \left( \frac{\mu_{i,j} + \mu_{i-1,j}}{2} \right) \left( \frac{u_{i,j} - u_{i-1,j}}{\Delta r} \right)}{(i-1/2) \Delta r \cdot \Delta r}$$

will preserve accuracy.

The advantage of the described method in comparison to the Particle In Cell method is its second-order accuracy. The necessity of using a method of greater accuracy than first order in computing hypervelocity impact problems which include the effect of viscosity will be discussed in the Numerical Calculations. The advantages in comparison to others of second-order accuracy [10, 11, 13, 14] are: (a) The extension to any Eulerian coordinate system is straightforward; (b) The calculation to advance the solution at one point, for the inviscid difference Eqs. (2.2,2.3), requires knowledge of only seven neighboring points, rather than the usual nine; (c) If the mesh is swept row-wise ( $x$  direction) and the solution is modified only by the differences in the  $x$  direction, say,  $\frac{\Delta t}{\Delta x} (F_{j+1,k} - F_{j,k})$ , then for each  $j$  only  $F_{j+1,k}$  need be calculated since  $F_{j,k}$  is known from the previous calculation at "cell $_{j-1,k}$ ". Similarly, after completion of this sweep, the mesh is then swept column-wise to account for the difference terms in the  $y$  direction, again computing and saving the values of the transport, stress, and conduction terms at only one "cell face" for each  $k$ , and hence reducing the amount of computation significantly. This procedure could be followed to differing extents by other Lax-Wendroff methods, some requiring the values at two previous cell faces to be saved and others able to use again only parts of the calculation at each face. The disadvantage of the method is that the eigenvalues of the amplification matrix, as discussed above, are not known. If the restriction on  $\Delta t$  necessary to fulfill the von Neumann condition is severe, the efficiency gained by advantages (b) and (c) may be more than offset in some problems.

For the problems considered in this paper  $\Delta t$  was simply chosen to be the smaller of the two values  $\frac{1}{3} \frac{\Delta x}{v_p}$  and  $\frac{\rho_0}{8} \frac{\Delta x^2}{\mu}$ , where  $v_p$  is the projectile impact velocity and  $\rho_0$  is the initial density. With this choice and with no permutation of the indices of the method defined by Eqs. (2.2,2.3), no sign of numerical instability was observed. Each problem, with a computational mesh of  $32 \times 33$  cells, took about 130 time-steps to complete. The machine time was approximately 15 minutes on the IBM 7094.

## 2.4 Numerical Calculations

The method defined by Eqs. (2.2,2.3) was applied to solve the Navier-Stokes equation for a compressible, non-heat-conducting viscous fluid in cylindrical coordinates. The hydrostatic pressure was assumed equal to the average normal stress (i.e., the "second coefficient of viscosity" was set equal to  $-2/3$  the "first coefficient." See Ref. 15). The solutions of these equations do not scale linearly with characteristic size as do their inviscid counterparts. However if a solution for one characteristic size  $d$  and viscosity  $\mu$  is obtained, then all solutions of characteristic size and viscosity  $d'$  and  $\mu'$  such that  $\frac{\mu'}{d'} = \frac{\mu}{d}$  are known, all other parameters being kept equal. That is, time  $t$ , distance and viscosity scale as

$$\begin{aligned} t &\rightarrow st \\ d &\rightarrow sd \end{aligned}$$

and

$$\mu \rightarrow s\mu$$

where  $s$  is any real number.

Thus, the particular choice of  $\mu$  is not as important as the choice of the ratio  $\frac{\mu}{d}$ .

For all cases studied the projectile was an aluminum right circular cylinder of length equal to its diameter impacting an aluminum target at a velocity of 1 cm/ $\mu$ sec. The equation of state used in the calculations was that formulated by Tillotson [16] for aluminum. Sakharov [17] deduced from shock-wave experiments that the coefficient of viscosity  $\mu$  of an aluminum alloy (90% Al) at 0.3 Mb (megabar) was approximately 0.02 Mp (megapoise) and increased weakly, but did not exceed 0.1 Mp for shock pressures up to 1 Mb. For this paper a constant value of 0.01 Mp was assumed to be representative of the values of  $\mu$  during the compressive phases, from the initial impact at which the shock pressure was 1.54 Mb until the calculations ceased and the shock had attenuated to approximately an order of magnitude greater than the material strength of aluminum (2 or 3 kb). As previously stated, the particular choice of  $\mu$  is not as important as the ratio  $\frac{\mu}{d}$  and the results for the chosen value of  $\mu$  may be scaled to any other choice. During the calculations, regions of expansion were treated as inviscid flows. More explicitly, when  $\rho$  became less than  $\rho_0/1.1$ , where  $\rho_0$  is the initial density,  $\mu$  was set to zero.

The chosen value of  $\mu$  was, in general, of the same order numerical magnitude as the mesh spacing  $\Delta x$ . The magnitude of the viscous stress terms is then proportional to  $\Delta x$ , while that of the truncation error for the method of second-order accuracy, described in the last section, is proportional to  $\Delta x^2$ . Thus, if a method of only first-order accuracy were used, namely, the Particle In Cell Code, with the same mesh spacing, the viscous stress and truncation

error would be of the same order of magnitude. A mesh spacing, say,  $\Delta x \approx \mu^2$ , chosen to insure that the viscous stress is dominant in comparison with the truncation error is impractical ( $\Delta t \approx \frac{\Delta x^2}{\mu} \approx \mu^3$ ). Also, there is a danger that the stability of the Particle In Cell method would be destroyed by such a choice (i.e., the terms introduced by truncation in P.I.C. themselves act viscously). Therefore, because of the order of magnitude of the coefficient of viscosity of aluminum, a method of at least second-order accuracy is necessary.

The computational mesh was re-zoned  $\Delta x \rightarrow 2\Delta x$  and  $\Delta y \rightarrow 2\Delta y$  each time the target shock wave or ejecta approached the mesh boundaries.

At intervals during each calculation the total positive component of axial momentum  $Z_+$  and the total radial momentum  $R$  were determined. That is,

$$Z_+ = \sum_{\text{Cells with } u_{i,j} > 0} \rho_{i,j} u_{i,j} (\text{cell volume})_{i,j}$$

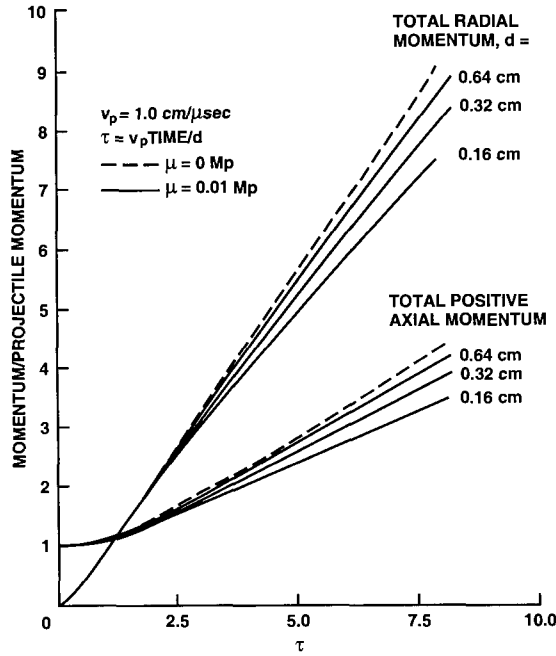
and

$$R = \sum_{\text{All cells}} \rho_{i,j} v_{i,j} (\text{cell volume})_{i,j}$$

The total negative component of axial momentum  $Z_-$  is, by conservation of momentum, equal to  $mv_p - Z_+$ , where  $mv_p$  is the projectile momentum. To be precise  $R$ , unlike  $Z_+$  and  $Z_-$ , is not a vector since the quantities  $\rho_{i,j} v_{i,j} (\text{volume of cell})_{i,j}$  have been summed algebraically. The vector sum would vanish because of the axial symmetry.

#### 2.4.1 Semi-Infinite Targets

The impact into thick targets of projectiles of diameters 0.16, 0.32, and 0.64 cm with  $\mu = 0.01$  Mp and, for comparison, with  $\mu = 0$ , was studied to determine if a momentum scale effect, comparable to that observed experimentally, could be caused by viscosity. The values for  $Z_+$  and  $R$  normalized by the initial projectile momentum are shown in Fig. 1 versus the nondimensional time  $\tau$ , where  $\tau = v_p \frac{t}{d}$ . The effect of viscosity is clearly shown here by each impact case having a distinct curve. It is also observed that at late times  $Z_+$  and  $R$  for each case increase nearly linearly. Fig. 2 shows the relationship of  $Z_+$  and  $R$  to  $d$  for  $\tau = 8$ , a time near the end of computation. Quantitatively, the scale effect is displayed here by the slopes of the curves, different from zero for both  $Z_+$  and  $R$ . The slope is seen to increase slightly for both curves as  $d$  decreases. For example, the slope of a straight line through the points of  $Z_+$  at  $d = 0.32$  cm and at 0.64 cm is 0.113, and that for  $d = 0.16$  cm and 0.32 cm is 0.146. Similarly, the corresponding slopes for the curve for  $R$  are 0.082 and 0.1126. These values are typical of those near the end of computation and do not appear to be changing appreciably. It is to be stressed that  $Z_+$  is not the same quantity measured experimentally as target momentum. The target during the

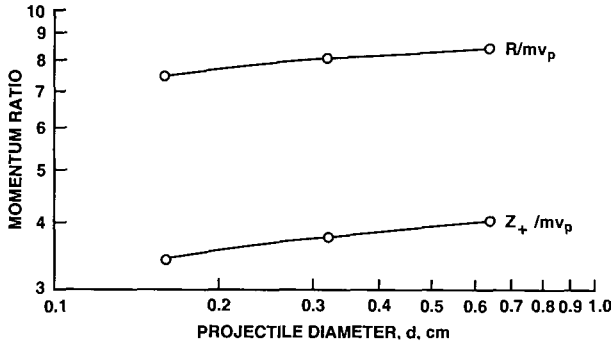


**Figure 1** Momentum versus  $\tau$  for  $d = 0.16, 0.32$ , and  $0.64$  cm

calculation was observed to contain large amounts of positive axial momentum near the axis and also appreciable amounts of negative axial momentum in the region forming the crater lip. The net effect would be the momentum of the target. Nevertheless the observed deviation from simple linear scaling in momentum in the numerical calculations would be expected to be reflected in the experimental measurements. The diameter exponents (slopes of the curves of Fig. 2) are somewhat lower than those found by experiment for spherical projectiles in approximately the same size range. The change in slope of both curves is an indication that the exponent of  $d$  depends on  $\mu/d$ , and a better correlation with experiment would be expected for a somewhat larger value of  $\mu$  at the same reported values of  $d$ . Also a greater deviation from simple linear scaling is to be expected in the momentum measurements of micrometeoroids than that of laboratory-sized projectiles.

#### 2.4.2 Finite Targets

The effect of viscosity in the impact of thin-sheet targets was also investigated. In each case the projectile diameter was  $0.16$  cm and the impact velocity was



**Figure 2** Momentum ratios of total positive axial momentum  $Z_+$  and total radial momentum  $R$  to projectile momentum  $mv_p$  versus projectile diameter  $d$  at nondimensional time  $\tau = 8$

1.0 cm/ $\mu$ sec. The momentum results for the impact of sheets of thickness  $th$  equal to 0.08, 0.16, and 0.24 cm are shown in Figs. 3(a) and (b). The most significant feature is the large attenuation of total positive axial momentum caused by viscosity in comparison with that of total radial momentum. In fact, for the cases of  $th/d \leq 1$  there is little or no reduction in  $R$ . The expected consequence of the greater attenuation in axial momentum, because of viscosity, is that the momentum of the spray, composed of both projectile and target material, moving through the impacted sheet, will be less intense and more divergent and thus will be less damaging to any subsequently impacted structure.

For finite targets, the impact process, because of the rapid attenuation of pressure caused by free surfaces, is expected to be dominated by the initial fluid dynamic stage. A finite-target scale effect found experimentally in spray-momentum measurements and in the solid angles in which the spray is distributed would add convincing evidence that the scale effect found in semi-infinite targets is caused by viscosity. Conversely, the absence of such an effect would lend credence to theory that the semi-infinite target scale effect is caused during the later, strength-dependent stages.

## 2.5 Concluding Remarks

1. Though it has not been shown conclusively that the scale effect found experimentally in semi-infinite targets is caused by viscosity, it has been shown that the total positive axial and radial momentum during the initial stages of cratering exhibit an effect, caused by expected levels of viscosity, consistent



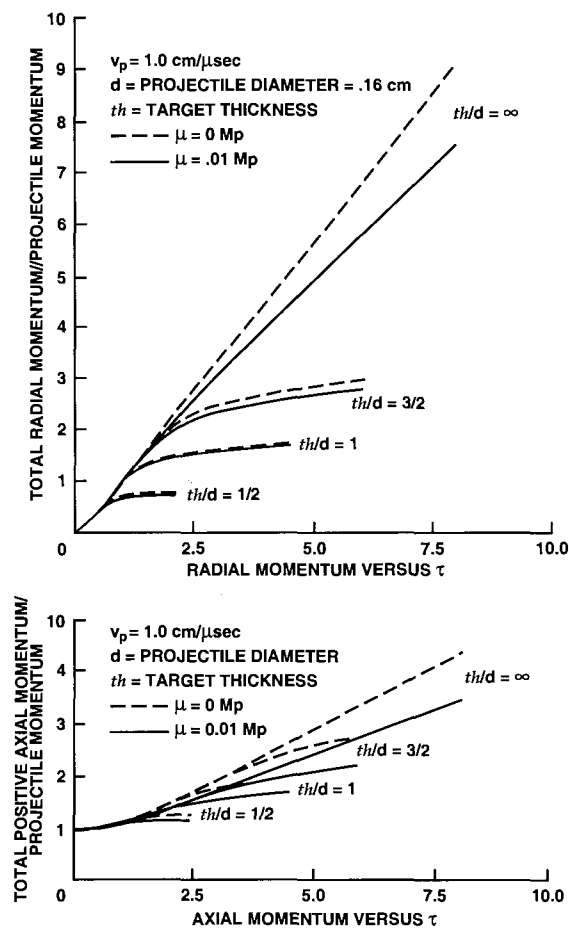


Figure 3 Momentum versus  $\tau$  for  $th/d = 1/2$ , 1, and  $3/2$

with experimental momentum measurements. It is expected that this effect will become increasingly important as projectile size diminishes.

2. Viscosity in thin targets is expected to reduce the momentum of the spray passing through the perforated target. An experimental study, in which the projectile-thin sheet geometry is unchanged as size is varied, could confirm the importance of viscosity in hypervelocity impact and provide an approach to the experimental evaluation of effective viscosity under the conditions of hypervelocity impact.

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# The MacCormack Method – Historical Perspective

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## 3.1 Introduction

Major advancements in computational fluid dynamics (CFD) have their roots in Brooklyn, New York, where Bob MacCormack was born on February 21, in the year of the Dragon, 1940. Bob attended public schools and graduated from Brooklyn College in 1961 with a Bachelor's degree in Mathematics and Physics. He answered President Kennedy's call to "send a man to the moon by the end of the decade" and decided to join NASA. Fortunately for Ames Research Center, he heeded an earlier call to go west rather than work for a NASA center close to his birthplace.

When Bob arrived at Ames, he was assigned to the Hypervelocity Ballistics Range Branch which became shortly thereafter, the Hypersonic Free-Flight Branch. His initial task was to study impact cratering using a light gas gun. In what must be the first recognition of his true talents, Branch Chief Tom Canning suggested that Bob study the problem numerically using an IBM 7094 computer, since the Branch was being charged for the Center supercomputer anyway. Bob's illustrious career in CFD began by learning Fortran IV in a weekend. In the meantime he earned a M. S. in mathematics from Stanford University in 1967.

Engineers knew how to solve viscous flow problems using only boundary

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layer theory. But Bob, who was trained as a mathematician and physicist, and preferred to think of himself as a research scientist, tackled the full Navier-Stokes equations that had been around for a century but had been solved for just a few simple problems. In an era when such research was encouraged, Bob succeeded in developing a method to solve the equations numerically for the impact of a body impinging on a surface. What followed was a succession of MacCormack methods that will be discussed in chronological order in this paper, followed by selected descriptions of application to various problems.

## 3.2 Evolution of the MacCormack Method

### 3.2.1 Mac-0

The original MacCormack method, which we will refer to as Mac-0, was developed over thirty years ago to solve the unsteady Navier-Stokes equations for impact cratering. The method was explicit, a two-step predictor/corrector and second order accurate in time and space. It had a stability limitation corresponding to the Courant-Friedrichs-Levy (CFL) condition equal to unity. The Mac-0 method was simpler than existing methods of the day. It did not require the calculation of Jacobian matrices, as in the Lax-Wendroff method, and it used a simple non-staggered grid, unlike Richtmyer's two-step version of the Lax-Wendroff method.

Mac-0 was introduced at the AIAA Hypervelocity Impact Conference held at Cincinnati, Ohio in April 1969. AIAA Paper 69-354 [1], entitled The Effect of Viscosity in Hypervelocity Impact Cratering, did not attract the attention of the aeronautics or fluid mechanics community at the time. That required solution of an aerodynamics problem.

### 3.2.2 Mac-1

The complex interaction that occurs when a shock wave impinges on a laminar boundary layer was the problem selected by Bob to demonstrate his method to the aeronautics and fluid mechanics community. Where others had tried and failed, he succeeded in solving numerically the compressible Navier-Stokes equations. Mac-1 was a simple modification to Mac-0.

The reasons for Bob's success were "obvious," as mathematicians always would say. First the method was simple. Second, it was split in time to become two one-dimensional operations and still maintain second-order accuracy. Third, it also split the computation domain into inner viscous and outer inviscid regions and used a different operating sequence to enhance the computational efficiency (as will be discussed below). Fourth, and least appreciated, a bias differencing technique was used for expansion regions and

a fourth order smoothing for oscillations were developed to “glue” the solution together to make the scheme stable for CFL conditions near unity.

Writing the conservative form of the Navier-Stokes equations as:

$$U_t + F_x + G_y = 0$$

and following the time-splitting concept of Strang [2], one-dimensional difference operators are defined as

$$\begin{aligned} L_x \text{ operator : } U_t + F_x &= 0 \\ L_y \text{ operator : } U_t + G_y &= 0 \end{aligned}$$

Each operator,  $L_x$  and  $L_y$ , is solved in a 2-step, predictor-corrector process as in the Mac-0 method. For a simple straight operation as

$$U(t + dt) = L_x L_y U(t)$$

the scheme is only first order accurate. However, if one symmetrizes the operating sequence as

$$U(t + 2dt) = L_y L_x L_x L_y U(t)$$

the scheme is second order accurate in both time and space. This allows one to have further variations such as

$$U(t + 2dt) = L_y(dt) L_x(2dt) L_y(dt) U(t)$$

with different stability time-steps dependent on each operator. This leads to the possibility of splitting the computational domain into inner viscous fine grid near the wall and outer inviscid coarse grid. The stable time step for the  $L_y$ -operator is very small compared to that of the  $L_x$ -operator in the fine grid, and, conversely, the stable time step in  $L_x$ -operator is smaller than that in the  $L_y$ -operator. Therefore one can enhance the computational efficiency by having a scheme such as

$$\begin{aligned} \text{Inner : } U(t + 2dt) &= L_y \dots L_y L_x(dt) L_x(dt) L_y \dots L_y U(t) \\ \text{Outer : } U(t + 2dt) &= L_x(dt) L_y(2dt) L_x(dt) U(t) \end{aligned}$$

This splitting in the computational domain avoids the disparity of different stable time steps in various flow regions, and allows the scheme to treat a fine resolution near the solid boundary. Since each operator is one-dimensional, the scheme was easily extended to three dimensions, as

$$U(t + 2dt) = L_z(dt) L_y(dt) L_x(2dt) L_y(dt) L_z(dt) U(t)$$

or other 3-D variations on this sequence. That was the beauty of the Mac-1 method and it really took off in the CFD world and became hugely successful. It was eventually used worldwide to solve a variety of problems.

A major contribution Bob made to CFD was his formulation using a control volume concept to achieve conservation law form. Instead of taking differences directly from the governing differential equations, he considered that the flow field was locally divided into controlled finite volumes (or cells) with forces acting on cell faces and with mass, momentum, and energy being transported between cells. The resulting set of governing equations was in conservation law form. This concept was eventually adopted and employed widely by the CFD community, and referred to as the “finite-volume” formulation.

A side note is worth mentioning here. Sutherland’s empirical formula was used to evaluate the molecular viscosity in his paper. Due to MacCormack’s success, from then on Sutherland’s formula was used everywhere for air, even to some applications where its validity is questionable. Sutherland himself could never dream of that his name would be cited so often and in so many papers because one man, Bob MacCormack, had taken it from NACA Report 1135.

Mac-1 was presented at the 2nd International Conference on Numerical Methods in Fluid Dynamics, (ICNMFD) held at Berkeley, California in September 1970. The paper, entitled Numerical Solution of the Interaction of a Shock Wave with a Laminar Boundary Layer [3], came at the right time, right place, and with the right title. It opened the door to solving the compressible Navier-Stokes equations – the governing equations of motion for fluid dynamics. It caught the attention of people in the aerospace industries who, by this time, were looking for ways to employ the computer to help them solve complicated flow problems. Even more important, it caught the attention of Dr. Dean Chapman, Chief of the Thermo- and Gas-dynamics Division at the NASA Ames Research Center. Chapman had foreseen the importance of numerical applications in fluid dynamics. Dr. Chapman was a “shock wave” man and most of his earlier research works were related to supersonic flow, shock waves, and thermo-physics. He was overjoyed to see that this kind of high-speed problem could be numerically simulated. As a result of this work, Bob was selected Assistant Branch Chief of the newly formed CFD Branch (1970), under the direction of Harvard Lomax in Dr. Chapman’s Division. He was permitted to continue the development of his method, training others to use the method and applying the method to problems of current interest. Bob was awarded the prestigious H. Julian Allen Award for the best Ames scientific paper in 1973 for this work.

### 3.2.3 Mac-2

As the importance of time stability in the inner viscous region increased (in order to permit the treatment of more complex geometries, higher Reynolds number, etc.), MacCormack pushed the idea of splitting a step further. In 1976 [4], two new operators had been developed for replacing the  $L_y$  operator



in the fine mesh calculation as

$$L_y(dt) \rightarrow L_{y_h}(dt)L_{y_p}(dt)$$

The operator  $L_{y_h}$  solves for the inviscid (hyperbolic) terms of  $G$ . It is explicit and uses characteristic relations to predict convection and pressure field. The operator  $L_{y_p}$  solves for the viscous (parabolic) terms in  $G$ . It is implicit and uses simple tridiagonal inversion. It is unconditionally stable. The operator sequence for all  $(i, j)$  in the fine mesh is

$$U(t + m dt) = \left\{ L_{y_h} \left( \frac{dt}{2m} \right) L_{y_p} \left( \frac{dt}{2m} \right) L_x \left( \frac{dt}{m} \right) L_{y_p} \left( \frac{dt}{2m} \right) L_{y_h} \left( \frac{dt}{2m} \right) \right\}^m U(t)$$

where  $m$  is a small integer, usually equal to 2 in value, possible because of the greatly relaxed stability requirements.

This approach, termed here Mac-2, was the ultimate use of time splitting. It not only split the equations into one-dimensional operations, it also split the one-dimensional operators based on convective and dissipative terms. Should Claude Navier or George Stokes see this formulation in heaven, they would not recognize the equations named after them.

In this way the method was speeded up substantially. However, the programming had become very complicated. Around the whole world, only a handful of people, closely associated with MacCormack, had ever used this method. Interestingly enough, the application of this method, even by only a few people, was very successful.

### 3.2.4 Mac-3 – An explicit-implicit scheme

Due to time-step restrictions in the explicit method, in late 1970s implicit methods were being developed to improve computational efficiency. The ideas of flux-splitting, upwind differencing, and total variational diminishing (TVD) were also under development. In 1981 [5] MacCormack incorporated a bidiagonal implicit procedure into the explicit predictor-corrector method. In the paper, MacCormack advocated a very important concept in the development of numerical method for steady state calculations. He suggested that a desirable form for a numerical method was, written in delta form, putting numerics on one-side, say left-hand side, and the accurate local approximation to the physical equations on the other side, say right-hand side, as residual. The responsibility of the left-hand side (numerics) was to convey the locally determined solution changes globally in a stable manner without violating the laws of physics. For numerical efficiency, the left-hand side should be as simple and straight forward as possible, and should drive the residual on the right-hand side to zero as fast as possible. The delta form was introduced earlier in Beam and Warming's implicit method. It was

MacCormack who made it so clear and easy for a CFDer to understand and follow.

The Navier-Stokes equations were still split into locally one-dimensional operators as before. The forward and backward differences were used in the locally linearized matrix for the flux terms in implicit operators. Forward differences for the predictor step and backward differences for the corrector resulted in simple bidiagonal matrix inversions for the implicit procedure. The combination of the two (predictor and corrector) scalar bidiagonal matrices produced, effectively, a diagonally dominant matrix operator.

With the addition of the implicit procedure, the method was theoretically stable for any time step. It required no scalar or block tridiagonal matrix inversions. Hence it was very efficient and had achieved a speed-up of about two orders of magnitude.

This is the zenith of “predictor-corrector” MacCormack method, termed here Mac-3. After this paper, Bob left Ames for the University of Washington to become a mentor of young students and continued to spread the seeds of CFD. By this time, with continuing advances in numerical methods and in computer capabilities, CFD had emerged as a branch of fluid dynamics. Various implicit schemes, coupled with flux-splitting, local time stepping had come along. And it was then that users began to solve Navier-Stokes equations for complex real geometries on a routine basis. The importance of the contributions made by MacCormack remain in every aspect of the development of subsequent numerical schemes.

### 3.3 Applications

#### 3.3.1 Inviscid

##### 3.3.1.1 *Mac-0, Three-Dimensional, Supersonic*

Even though the most important impact of the MacCormack method was in solving the Navier-Stokes equations, it was the early immediate applications to inviscid blunt body and supersonic Space-Shuttle solutions that drew first attention to his method. In the early 1970s, the Space Shuttle was the agency’s “space project.” Rizzi and Inouye [6] applied the method to supersonic flow over three-dimensional blunt bodies. Kutler [7] replaced his previous non-central scheme with the Mac-0 method to compute the inviscid flow over the Space Shuttle configuration, and later treated several other supersonic flow problems.

### *3.3.1.2 Mac-0, Transonic Flow, Subsonic Boundary Conditions*

Deiwert (1974) [8], while developing a code using the Mac-1 method to study transonic flow past airfoil configurations, first obtained results for inviscid transonic flow in air, Freon and cryogenic nitrogen over a biconvex airfoil shape. The increasing concern of performing wind tunnel tests at high Reynolds numbers had prompted the consideration of test gases other than air in order to increase gas density whilst maintaining manageable stagnation pressure levels. One way to increase the density is to use a test gas of high molecular weight, such as Freon 12. Another is by significantly lowering the gas temperature, such as by using cryogenic nitrogen. At that time the agency was developing plans to build a national transonic wind tunnel that would use cryogenic nitrogen as the test gas. In the former case (Freon) it is possible to consider the gas as ideal, yet with an isentropic exponent ( $\gamma$ ) considerably different from that of air. In the second case (cryogenic nitrogen) the gas does not always behave in an ideal manner, and regions of expansion may be critically near the two-phase region. One of the questions being asked was whether conditions could develop in the cryogenic transonic wind tunnel, that would result in liquefaction. A Van der Waals equation of state was used to describe the thermally and calorically imperfect cryogenic nitrogen, and simulations were performed for flow past the 18% biconvex circular arc at stream Mach numbers of 0.775 and 0.95.

The results of the study, which were presented at the 41st Semi-Annual Meeting of the Supersonic Wind Tunnel Association, Los Angeles, in March 1974, showed that liquefaction was not predicted under the conditions simulated, and that, in fact, the use of cryogenic nitrogen appeared viable. Additional information from the study quantified some of the differences in the isentropic exponent ( $\gamma$ ) of the different gases on the flowfield structure. At that time researchers were using the concept of “effective  $\gamma$ ” to “match” their inviscid solutions to experimentally observed results for lift and drag. This, along with “effective angle of attack,” were actually artificial ways to account for viscous displacement phenomena that occur in the real world experiments.

## **3.3.2 Viscous Transonic Flows**

### *3.3.2.1 Mac-1, Generalized Curvilinear Coordinates, Symmetric*

Deiwert (1974) [9] extended the basic MacCormack explicit method (Mac-1) with time splitting to treat nonorthogonal computational meshes of arbitrary configuration for application to viscous flow past bodies of general curvilinear shape. The objective was to “capture” the shock over an airfoil configuration and simulate its interaction with the boundary layer. Dr. Dean Chapman, who was one of the first to recognize the powerful potential of

the MacCormack method to practical applications of this sort, identified this particular important application. It was, in fact, Dr. Chapman who asked Dr. Deiwert to look into this problem area and even provided office space next to Bob's office to help facilitate the study. Being young and naïve at the time, and not realizing that such a complex application was unheard of, Deiwert readily agreed to take up the task and hence became one of Bob's first "students."

The configuration selected for this study was an 18% thick biconvex circular arc airfoil shape. Coordinate transformations were developed for differencing the viscous terms and a compressible turbulent transport model was implemented. The boundary conditions for this configuration are all subsonic. A transonic Mach number was identified that would produce a shock strong enough to induce flow separation at the foot of the shock. A companion experimental program was also initiated to acquire detailed data over such a configuration in the Ames High Reynolds Number Channel.

The results of the first study showing the viscous/inviscid interaction with shock induced separation were presented at the AIAA 7th Fluid and Plasma Dynamics Conference, Palo Alto, California in June 1974 and at the Fourth International Conference on Numerical Methods in Fluid Dynamics, in Boulder, Colorado in the same month [10]. A generalized transformation was developed to map the  $L_x$  and  $L_y$  operations onto a generalized nonorthogonal mesh in the viscous flow regions. A simple mixing length model was used to model viscous transport in the turbulent boundary layer.

Computing resources were considered quite large by the standards of those days. Using the Mac-1 method, solutions required from 2 to 10 hours on a CDC 7600 computer (the state of the art at the time). When the results were presented at the Fourth International Conference questions were raised from the audience about the computer requirements. When the answer was given that 2 to 10 hours on a CDC 7600 were required to reach a converged solution, the session chairman, Dr. Belotserkovskii, Director of the Computing Center, Academy of Science, Moscow, USSR, said to the speaker: "you must be very rich." Computing times are Reynolds number (and therefore, mesh resolution) dependent.

The code was subsequently vectorized, bringing the computer time to less than half. The code was also written and run on the experimental Illiac IV computer (a 64 processor parallel computer) on which computing times ranged from 0.6 to 3 hours per solution. These solutions were, in fact, the first published solutions obtained on the Illiac IV. The Illiac was operated at 11.5 MHz. Today the same computations could be performed in less time on a desk top or lap top computer which operate at several hundred MHz.

The transonic biconvex airfoil study was continued to develop and further assess algebraic turbulent transport models, including those proposed by Shang and Hankey and by Baldwin and Rose, applicable to separated flows. These results were compared with new experimental data obtained

by McDevitt and Levy in the Ames High Reynolds Number Channel and reported at the AIAA 8th Fluid and Plasma Dynamics, Hartford, CT, in June 1975 [11]. An unexpected result of the experimental study of transonic flow was the observation of a periodic unsteady flow in the aft region of the airfoil for a small select Mach number range (see Fig. 13, ref. 11). The imposed symmetry boundary condition in the computer code, and the computation time constraints with the Mac-1 method, precluded simulation of these phenomena.

### 3.3.2.2 *Mac-2, Lifting Airfoils, Adaptive Grid*

Bob's improvements to his method, the explicit/implicit concept (Mac-2), were implemented to increase the speed and computational efficiency of the code. Speed improvement of 95% was realized while still maintaining time accuracy at the inviscid time scale (Fifth International Conference on Numerical Methods in Fluid Dynamics, Enschede, The Netherlands, June 1976 [4].) This opened the way to remove the symmetry constraint, to consider lifting airfoil configurations, and to begin to address some of the unsteady issues, such as buffet and the unsteady phenomena observed in the High Reynolds Number Channel.

Deiwert's code was modified to treat the asymmetric behavior in the near wake region. A mesh adaption procedure was implemented, in a manner developed by Schiff, to follow the shear flow in the wake and to follow the shock wave, which moves in time in an unsteady flow. These enhancements greatly increased the capability of the code to simulate practical transonic airfoil flows. The first treatment of lifting airfoils was reported by Deiwert at the SQUID Workshop on Lifting Airfoils in February 1976 [12]. The significant advancement was the treatment of asymmetric airfoil shapes and the near wake region as well as the much improved computational efficiency resulting from using the Mac-2 method.

A particularly interesting configuration at the time was the supercritical configuration proposed by Richard Whitcomb. Supercritical airfoils were getting a lot of attention at the time. Particularly notable were the analyses of Garabedian and Korn. In these inviscid studies the concept of "effective angle of attack" was used to match computed results with experimental data. The results of a study by Deiwert [13] showed conclusively that proper account of viscous phenomena (i.e., boundary layer displacement effects) was necessary and sufficient to accurately simulate the performance of these airfoil shapes (see Figs. 2 and 4, ref. 13).

### 3.3.2.3 *Mac-2, Unsteady Transonic Flows*

Levy [14] used the revised Deiwert code with the latest MacCormack (Mac-2) method and the asymmetric wake treatment to study the unsteady processes observed in the experimental study of the 18% circular arc airfoil. Remarkable agreement was found between the computation and experiment, both in the amplitude and frequency of the unsteady process thus giving even more credibility to the power of the time accurate MacCormack method. Levy was eventually (1979) awarded the H. Julian Allen award for this study.

Subsequent studies by Levy and Bailey [15] and by Deiwert and Bailey [16] delved further into the applicability of unsteady flow simulation with the time-accurate MacCormack method and studied the buffet phenomenon and the phenomenon of aileron buzz. Application of such an approach was found to work remarkably well for flows in which there is a single dominant frequency with a time scale of the order of the inviscid time. In 1976 Deiwert performed studies in collaboration with Prof. Peter Bradshaw and developed a one-equation shear model, which was implemented in the code for near wake studies. The issue of dynamic grid adaption necessary to treat wake flows and unsteady flows ultimately led to the dynamic adaptive grid scheme developed by Nakahashi and Deiwert.

Horstman used Deiwert's code in his studies of trailing edge flows and studied a variety of turbulence transport models. Rose also used the code for several of his interactive flow studies. Comments from all that used this code were universal in agreement in that "the code was robust and always gave the correct answers." In fact, the MacCormack method was always extremely robust; the critical pacing item in the code was, and remains, the turbulent transport model.

## 3.3.3 **Viscous Supersonic Flows**

### 3.3.3.1 *Two-Dimensional Supersonic Flows*

After the presentation of Mac-1 in 1970, the challenge to solve the N-S equations started to pick up steam and activity charged forward. Parallel to Deiwert's viscous transonic flows efforts, in 1972 Baldwin took MacCormack's code and added the eddy viscosity term to solve the turbulent shock reflection problem, and presented an AIAA paper in summer 1974 in Palo Alto [17]. Hung joined the CFD Branch in as an NRC postdoctoral associate in 1973 and started to modify MacCormack's code to study laminar flow in a compression corner. Two other groups were also on the same trail. One was at Wright-Patterson AFB under Hankey, where Shang was studying the turbulent compression ramp. Another was an experimental group at Ames under Marvin (also in Chapman's Division) to develop well-documented experimental data for developing and validating turbulence transport models for high speed.

In addition to the transonic data studied by Deiwert, Levy and McDevitt, Horstman and Kussoy carried out experiments and Coakley carried out computations for an axisymmetric shock reflection problem at hypersonic speeds. These led to a fanfare at the 1975 AIAA Aerospace Sciences Meeting in Pasadena. The first four papers of the first session, AIAA paper 75-01 [18] by MacCormack and Baldwin, 75-02 [19] by Hung and MacCormack, 75-03 [20] by Shang and Hankey, and 75-04 [21] by Horstman, Kussoy, Coakley, Rubesin, and Marvin, all used MacCormack's method (Mac-1) to solve shock induced separation problems. From then on, theoretical study of shock-wave induced separation, or shock-wave/boundary layer interaction belonged to numerical simulation, and, for a long while, belonged to MacCormack's method.

As everyone realized, most flows of practical interest were turbulent and the pacing item for numerical simulation was the turbulence model. NASA-Ames invested substantial resources in the effort to develop these models. The results of Baldwin and of Coakley were reasonable, but not very good. Surprisingly, Shang and Hankey used a simple relaxation turbulence model and the results showed not only excellent agreement in the surface pressure and location of separation, but more dramatically, very good agreement of the density and velocity profiles at several upstream and downstream locations (see Figs. 9 and 10 of Ref. 20). Based on their results, one almost could claim that the turbulent flow problem was solved. That created a substantial interest at Ames and led to the formation of a small group to study "the relaxation" model. Unfortunately, no one could obtain good results similar to those obtained by Shang. A paper presented by Hung and MacCormack in summer 1976 was the result of one study of the relaxation model. The most common finding was that the relaxation length suggested by Shang was too large.

It was not until some time later that Hung reported two program "errors" in Shang's code. The first and most serious one was that, while searching for the boundary layer outer velocity for calculation of the displacement thickness needed by the Cebeci-Smith model, the maximum velocity would be obtained, instead of the intended edge velocity, due to the existence of the shock wave. A small difference in the edge velocity could result in a big difference in calculated displacement thickness which led to a big difference in its corresponding outer layer eddy viscosity. This "error" made the model work great for their problem. Another error had a minor effect on the calculation of eddy viscosity and is not described here. Let's just follow the rule of Jameson, (quoted from MacCormack's paper [22]) "In any program consisting in length of at least one box of IBM cards (modern translation: 2000 statements long) there is always a bug."

### 3.3.3.2 *Three Dimensional Supersonic Flows*

The development of Mac-2 made solutions to the 3-D Navier-Stokes equations feasible. Shang started with a 3-D compression corner simulation using the Mac-1 scheme for hypersonic laminar flow, and then switched to the Mac-2 scheme for turbulent flow. Hung applied the Mac-2 scheme to a 3-D compression corner for supersonic laminar and turbulent flows, to an axisymmetric body with a flare at angles of attack and various 3-D problems, such as the impingement of an oblique shock wave on a cylinder.

After the development of Mac-3 in 1981, Kordulla applied it to two-dimensional transonic airfoil flows, and Hung and Kordulla extended it to general 3-D geometries and applied it to a case of a blunt fin on a flat plate. They simulated the existence of a horseshoe vortex in front of the blunt fin (see Hung and Kordulla [23]), and obtained very good agreement with experimental data obtained by Bogdonoff's group at Princeton University. A simulation movie was shown by Hung and Buning [24] in 1984 at the AIAA Aerospace Sciences Meeting. From then on, for some period of time, movies of flow field simulation became very popular. At that time, Ames produced ten to twenty CFD movies a year and was almost like "Hollywood-North," and CFD jokingly stood for "Color Film Displays." One of the most successful users of the MacCormack method was Shang. He took advantage of computer architectures with the simplicity of Mac-0 and Mac-1, simulated many flow problems, 2-D and 3-D. He even applied the method to 2-D flow oscillations around a cylinder and 3-D unsteady flow over spike-tipped bodies. He was the first one to carry out a complete aircraft simulation, an X24C-10D calculation.

The MacCormack method has been employed to tackle many supersonic and hypersonic problems and conquer those complicated 3-D shock-wave and boundary-layer interactions for which the theoreticians never dreamed about and the experimentalists could only carry out very limited surface measurements. The agreement with experimental data for 3-D flows very often was much better than in 2-D cases. There were three important reasons. First, except for the computation time and data memory, 3-D problems are easier than 2-D problems. It is easier to get "good agreement" with experimental data in 3-D computation than in 2-D ones. The reason is obvious; most 3-D problems have one extra dimension for disturbance relief and hence are dominated by the inviscid mechanism which is easier to solve compared to the viscous mechanism. Knight and Horstman [25] showed in a 3-D swept shock calculation that, even with differences in eddy viscosity values up to a factor of fourteen in many places, two different calculations could still be in pretty good agreement. This does not mean that turbulence modeling is not important. It only means that 3-D problems have more room for error. Next, while shock waves and separation cause a lot of problems for theoreticians, conversely, supersonic and hypersonic problems are easier to solve numerically



than incompressible or transonic problems. Disturbances propagate in only one direction (downstream) and at a fast rate. The boundary conditions are easy to set up and the solution converges in a relatively short computation time. The last reason is that, even though compression shocks cause more engineering problems because their adverse pressure gradients can lead to boundary-layer separation and increased aerodynamic heating, compressive flows are easier to solve than expanding flows. An expansion has a chance to drain out a computation cell and leads to a negative density or pressure and makes the scheme unstable. Again, these fortunate reasons contribute to the success of application of the MacCormack method.

### 3.4 Closing Remarks

Looking back into history, it was the MacCormack method that was the torch bearer for solving frontier problems and paving the way for the growth of CFD. The method's success can be attributed not only to its power and robustness but also to Bob's willingness to help others implement the method to solve their problems.

With continuing advances in numerical methods and in computation power, CFD has become a separate branch of fluid dynamics. Using the computer as a tool, CFD is now not only able to simulate real industrial engineering problems, but moreover, is able to supplement the experimental and theoretical studies. It can be used to carry out research on its own merits, and development and further advancement in computational fluid dynamics can be considered as a separate field of physical sciences.

MacCormack's methods have played an important guiding role in the birth, growth and development of CFD. NASA recognized Bob's contributions in 1981 with the Medal for Exceptional Scientific Achievement. Bob left Ames for the academic world and has continued to spread the seeds of CFD, first at the University of Washington and now at Stanford University.

The authors have attempted to summarize the development of the MacCormack methods in the early years and to highlight some of the significant accomplishments made possible by the application of these methods. A comprehensive description of the methods and a complete documentation of their applications is beyond the scope of this paper. While every attempt has been made to assure historical and technical accuracy we recognize that some inaccuracies may exist; for this we apologize.

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# General Framework for Achieving Textbook Multigrid Efficiency: One-Dimensional Euler Example

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## Abstract

A general multigrid framework is discussed for obtaining textbook efficiency to solutions of the compressible Euler and Navier-Stokes equations in conservation law form. The general methodology relies on a distributed relaxation procedure to reduce errors in regular (smoothly varying) flow regions; separate and distinct treatments for each of the factors (elliptic and/or hyperbolic) are used to attain optimal reductions of errors. Near boundaries and discontinuities (shocks), additional local relaxations of the conservative equations are necessary. Example calculations are made for the quasi-one-dimensional Euler equations; the calculations illustrate the general procedure.

## Introduction

Computational fluid dynamics (CFD) has become an integral part of the aircraft design cycle because of the availability of faster computers with

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more memory and improved numerical algorithms and physical models. More impact is possible if reliable methods can be devised for off-design performance, generally associated with unsteady, separated, vortical flows with strong shock waves. Such computations demand significantly more computing resources than are currently available.

The current Reynolds-averaged Navier-Stokes (RANS) solvers with multigrid algorithms require on the order of 1500 residual evaluations to converge the lift and drag to one percent of their final values, even for relatively simple wing-body geometries at transonic cruise conditions. It is well known for elliptic problems that solutions can be attained optimally by using a full multigrid (FMG) process in far fewer (on the order of 2 to 4) residual evaluations. A multigrid method is defined by Brandt [1, 2, 3] as having textbook multigrid efficiency (TME) if the solutions to the governing system of equations are attained in a computational work that is a small (less than 10) multiple of the operation count in the discretized system of equations. Thus, operation count may be reduced by several orders of magnitude if TME can be attained for the RANS equations.

State-of-the-art multigrid methodologies for large-scale compressible flow applications use a block-matrix relaxation and/or a pseudo-time-dependent approach to solve the equations; significant improvements have been demonstrated with multigrid approaches, but the methods are not optimally convergent. The RANS equation sets are systems of coupled nonlinear equations which are not, even for subsonic Mach numbers, fully elliptic, but contain hyperbolic partitions. The distributed relaxation approach of Brandt [1, 2] decomposes the system of equations into separable, usually scalar, factors that can be treated with optimal methods. Several years ago, an investigation was started to extend this approach to large-scale applications; at that time, several TME demonstrations for incompressible simulations had been completed and Ta'asan had shown promising results for the subsonic Euler equations [13]. Progress has been shown in extending the methodology to viscous compressible flow applications [12] and to compressible Euler equations using a compact differencing scheme [15]. Further incompressible flow applications have been made, including complex geometries [14] and high-Reynolds-number viscous flow in two [17] and three dimensions [18]. Brandt has summarized the progress and remaining barriers in TME for the equations of fluid dynamics [3].

The purpose of this paper is to discuss the general framework expected to be required for large-scale compressible flow applications. The quasi-one-dimensional Euler equations are solved to illustrate the framework. Fully subsonic and supersonic applications, as well as transonic applications with a captured shock, are shown.

## General Framework

The viscous compressible equations for the time-dependent conservation of mass, momentum, and energy can be written as

$$\partial_t \mathbf{Q} + \mathbf{R} = 0, \quad (4.1)$$

where the conserved variables are  $\mathbf{Q} \equiv (\rho, \rho u, \rho v, \rho w, \rho E)^T$ , representing the density, momentum vector, and total energy per a unit volume, and  $\mathbf{R}(\mathbf{Q})$  is the spatial divergence of a vector function representing convection and viscous and heat transfer effects. In general, the simplest form of the differential equations corresponds to nonconservative equations expressed in primitive variables, here taken as the set composed of velocity, pressure, and internal energy,  $\mathbf{q} = (u, v, w, p, \epsilon)^T$ . These equations are found readily by transforming the time-dependent conservation equations to time-dependent primitive variable equations. Similarly, a set of nonconservative correction equations can be derived, with a right hand side vector composed of a combination of the conserved residual terms, given as

$$\mathbf{L} \delta \mathbf{q} = - \frac{\partial \mathbf{q}}{\partial \mathbf{Q}} \mathbf{R}. \quad (4.2)$$

In Eq. (4.2),  $\frac{\partial \mathbf{q}}{\partial \mathbf{Q}}$  is the Jacobian matrix of the transformation and the correction  $\delta \mathbf{q} \equiv \mathbf{q}^{n+1} - \mathbf{q}^n$ , where  $n$  is an iteration counter. For steady-state equations, the time derivative is dropped. At the discrete level, the right side of the correction equation (4.2) is a combination of conservative-discretization residuals while the left side is the principal linearization of the nonconservative operator.

Note this is not a Newton linearization; only the principal terms in a linearization of  $\mathbf{R}$  are retained. The principal terms are those terms that make a major contribution to the residual per a unit change in  $\mathbf{q}$ . The principal terms thus generally depend on the scale, or mesh size, of interest. For a *scalar* equation, the discretized highest derivative terms are principal on grids with small enough mesh size  $h$ . In deriving the principal linearization for high-Reynolds-number simulations, it is essential to consider both inviscid and viscous scales — the inviscid scales dominate over most of the flow field and the viscous scales are important in the thin viscous layers near bodies and in their wakes. Note that, for a discretized *system* of differential equations, such as  $\mathbf{R} = 0$ , the principal terms are those that contribute to the principal terms of the *determinant* of the matrix operator  $\frac{\partial \mathbf{R}}{\partial \mathbf{q}}$ . The coefficients of the principal terms in  $\mathbf{L}$  are evaluated from the current approximation.

The principal linearization is applied to the correction equation based on a nonconservative approximation. Thus, we expect the correction to be good away from discontinuities (shocks, slip lines) in the flow field. It is in these

regular (smoothly varying) flow regions that we apply distributed relaxation. The distributed relaxation method replaces  $\delta \mathbf{q}$  by  $\mathbf{M} \delta \mathbf{w}$  so that the resulting matrix  $\mathbf{L} \mathbf{M}$  becomes a diagonal or lower triangular matrix, as

$$\mathbf{L} \mathbf{M} \delta \mathbf{w} = - \frac{\partial \mathbf{q}}{\partial \mathbf{Q}} \mathbf{R}. \quad (4.3)$$

The diagonal elements of  $\mathbf{L} \mathbf{M}$  are composed ideally of the separable components of the determinant of the matrix  $\mathbf{L}$  and represent the elliptic or hyperbolic factors of the equation. Brandt termed the  $\delta \mathbf{w}$  variables as “ghost variables,” because they need not explicitly appear in the calculations.

The distributed relaxation approach yields fast convergence for both steady and unsteady simulations if the constituent scalar diagonal operators in  $\mathbf{L} \mathbf{M}$  are solved with fast methods. The approach can be applied to quite general equations; Brandt has derived a set of matrices  $\mathbf{M}$  that provide a convenient lower triangular form for the compressible and incompressible equations of fluid dynamics (including a variable equation of state) [2].

For the compressible Euler equations, the scalar factors constituting the main diagonal of  $\mathbf{L} \mathbf{M}$  are convection and full-potential operators. An efficient solver for the former can be based on downstream marching, with additional special procedures for recirculating flows [6, 7, 9]; the latter is a variable type operator, and its solution requires different procedures in subsonic, transonic, and supersonic regions. In deep subsonic regions, the full-potential operator is uniformly elliptic and therefore standard multigrid methods yield optimal efficiency. When the Mach number approaches unity, the operator becomes increasingly anisotropic and, because some smooth characteristic error components cannot be approximated adequately on coarse grids, classical multigrid methods severely degrade. The characteristic components are those components that are much smoother in the characteristic directions than in other directions [3, 10, 11]. In the deep supersonic regions, the full-potential operator is uniformly hyperbolic with the stream direction serving as the time-like direction. In this region, an efficient solver can be obtained with a downstream marching procedure. However, this procedure becomes problematic for the Mach number dropping towards unity, because the Courant number associated with the downstream marching procedure is large. Thus, a special procedure is required to provide an efficient solution for transonic regions. This local procedure [4, 5, 8] is based on piecewise semicoarsening and some rules for adding dissipation at the coarse-grid levels.

Boundaries introduce some additional complexity in distributed relaxation. The determinant of  $\mathbf{L} \mathbf{M}$  is usually higher order than the determinant of  $\mathbf{L}$ . Thus, as a set of new variables,  $\delta \mathbf{w}$  would generally need additional boundary conditions. In relaxation, because the ghost variables can be added in the external part of the domain, it is usually possible to determine suitable boundary conditions for  $\delta \mathbf{w}$  that satisfy the original boundary



conditions for the primitive variables. Examples are given by Thomas, Diskin and Brandt [17] in incompressible flow for entering and no-slip boundaries. However, to construct such a remedy may be difficult and/or time-consuming in general. In addition, enforcing these boundary conditions causes the relaxation equations to be coupled near the boundaries, not decoupled as they are in the interior of the domain.

Thus, near boundaries and discontinuities, the general approach [2, 3] is to relax the governing equations directly in terms of primitive variables. Several sweeps of robust (but possibly slowly converging) relaxation, such as Newton-Kacmarcz relaxation, can be made in this region. The additional sweeps will not affect the overall complexity because the number of boundary and/or discontinuity points is usually negligible in comparison with the number of interior points.

This general framework is used below to solve the quasi-one-dimensional Euler equations in fully subsonic, fully supersonic, and transonic (with and without shock) flow regimes. The regular flow regions are relaxed with distributed relaxation. Boundaries and shocks are treated by applying local relaxation — corresponding here to updates through a direct solution of an approximate-Newton linearization of the conservative equations.

In all cases, an FMG algorithm is used; at each level, the equations are solved with an FV(2,1) full approximation scheme (FAS) [1, 17] multigrid cycle. Six levels are used in the cycle wherever possible.

## Quasi-One-Dimensional Equations

The quasi-one-dimensional equations express the conservation of mass, momentum, and total energy as

$$\partial_t(\hat{\mathbf{Q}}) + \hat{\mathbf{R}} = 0, \quad (4.4)$$

where

$$\hat{\mathbf{Q}} \equiv \mathbf{Q}\sigma \equiv (\rho, \rho u, \rho E)^T \sigma, \quad (4.5)$$

$$\hat{\mathbf{R}} \equiv \partial_x(\mathbf{F}\sigma) + \mathbf{S}, \quad (4.6)$$

and  $\sigma = \sigma(x)$  is the area term. The flux  $\mathbf{F}$  and the source term  $\mathbf{S}$  are defined as

$$\mathbf{F} \equiv \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho u E + up \end{pmatrix}, \quad (4.7)$$

$$\mathbf{S} \equiv - \begin{pmatrix} 0 \\ p \\ 0 \end{pmatrix} \frac{d\sigma}{dx}. \quad (4.8)$$

The pressure  $p$ , internal (thermal) energy  $\epsilon$ , and sound speed  $c$  are related through the equation of state as

$$p = (\gamma - 1)\rho\epsilon, \quad (4.9)$$

$$\epsilon = E - u^2/2, \quad (4.10)$$

$$c^2 = \gamma p / \rho, \quad (4.11)$$

and  $\gamma$  is the ratio of specific heats.

A discrete conservative upwind-biased differencing approximation to  $\hat{\mathbf{R}}$  can be defined for first- or second-order accuracy as

$$\begin{aligned} \hat{\mathbf{R}}_j &= \frac{1}{h} [(\mathbf{F}\sigma)_{j+\frac{1}{2}} - (\mathbf{F}\sigma)_{j-\frac{1}{2}}] \\ &- (0, p_j, 0)^T [\sigma_{j+\frac{1}{2}} - \sigma_{j-\frac{1}{2}}]. \end{aligned} \quad (4.12)$$

Here, a finite-volume discretization is used, where subscripts  $j + \frac{1}{2}$  and  $j - \frac{1}{2}$  denote the right and left interfaces, respectively, of the cell centered at location  $j$  and the cell spacing in the  $x$ -direction is  $h$ . The flux-differencing splitting of Roe [20] is used to construct the interface flux  $\mathbf{F}_{j+\frac{1}{2}}$ ; pertinent details are described in appendix I. We consider only first- or second-order accuracy and do not differentiate between average and pointwise values of  $\mathbf{Q}$ . The area distribution is defined as  $\sigma(x) = 1 - 0.8x(1 - x)$ . For all of the results presented below, we overprescribe the boundary values from the exact solution of the differential problem at the cell centers that lie outside of the domain  $0 \leq x \leq 1$ .

## Relaxation Schemes

Several schemes are considered below for relaxation of the steady-state equations

$$\hat{\mathbf{R}}_j = 0. \quad (4.13)$$

The schemes are all written in delta form, so that the correction to the solution is solved at each iteration, denoted as  $\delta\mathbf{Q} \equiv \mathbf{Q}^{n+1} - \mathbf{Q}^n$ .

### Conservative Relaxation Scheme

The exact Newton linearization of the discrete conservative equations generally leads to an overly complicated linear system to solve. In practice, a quasi-Newton (approximate) linearization is used to reduce the algebraic complexity of the construction and/or the bandwidth of the resulting linear system. For the purpose of constructing a conservative relaxation scheme, an approximate but conservative linearization of the operator (4.12) can be written as

$$[\delta_x^- \mathbf{A}^+ + \delta_x^+ \mathbf{A}^-](\mathbf{Q}). \quad (4.14)$$

The operators  $\delta_x^-$  and  $\delta_x^+$  denote backward and forward first-order-accurate difference operators, respectively, and  $\mathbf{A}^+$  and  $\mathbf{A}^-$  denote the positive and negative eigenvalue contributions to the similarity matrices (see appendix I).

The conservative relaxation update  $\delta \mathbf{Q}$  is computed from

$$[\delta_x^- \mathbf{A}^+ + \delta_x^+ \mathbf{A}^-](\delta \mathbf{Q}) = -\frac{1}{\sigma_j} \hat{\mathbf{R}}_j. \quad (4.15)$$

The residuals in the right side of Eq. (4.15) are computed for the target discretization (4.13). The relaxation procedure requires a tridiagonal equation to be directly solved:

$$(-\mathbf{A}_{j-1}^+, |\mathbf{A}|_j, \mathbf{A}_{j+1}^-)(\delta \mathbf{Q}) = -\frac{h}{\sigma_j} \hat{\mathbf{R}}_j. \quad (4.16)$$

Barth [19] analyzed this relaxation with fixed-point analysis and showed the relaxation is nearly as good as a full Newton iteration. An underrelaxation parameter, e.g., a pseudo-time step, can be introduced in the relaxation scheme to improve robustness and stability. Extensive computations made during the course of this investigation verified that this relaxation method can be used throughout the domain, including shocks and boundaries. In the general framework described and used in this paper, this relaxation is used only locally to reduce residuals near boundaries and shocks and to solve the coarsest grid equation.

### Nonconservative Relaxation Scheme

A nonconservative relaxation scheme is derived from a conservative scheme, Eq. (4.15), with the assumption that the Jacobian matrices are locally constant,

$$[\mathbf{A}^+ \delta_x^- + \mathbf{A}^- \delta_x^+](\delta \mathbf{Q}) = -\frac{1}{\sigma_j} \hat{\mathbf{R}}_j. \quad (4.17)$$

This formulation is expected to be a good approximation to Eq. (4.15), except near discontinuities. The corresponding tridiagonal equation is

$$(-\mathbf{A}_j^+, |\mathbf{A}|_j, \mathbf{A}_j^-)(\delta \mathbf{Q}) = -\frac{h}{\sigma_j} \hat{\mathbf{R}}_j. \quad (4.18)$$

Subtracting the two tridiagonal equations, (4.16) and (4.18) yields

$$(-\mathbf{A}_{j-1}^+ + \mathbf{A}_j^+, 0, \mathbf{A}_{j+1}^- - \mathbf{A}_j^-)(\delta \mathbf{Q}) = 0, \quad (4.19)$$

where the coefficients —  $-\mathbf{A}_{j-1}^+ + \mathbf{A}_j^+$  and  $\mathbf{A}_{j+1}^- - \mathbf{A}_j^-$  — are  $O(h)$  small on sufficiently fine meshes in regular flow regions. Near shocks, the nonconservative linearization used in Eq. (4.17) does not satisfy an order property and the nonconservative relaxation scheme would not be effective. These expectations were confirmed through computations with both the conservative and nonconservative relaxation schemes, where update Eqs. (4.16) and (4.18) were solved precisely. Barth [19] analyzed a similar nonconservative relaxation scheme with fixed-point analysis and showed poor performance of the nonconservative scheme for a flow with a shock.

## Distributed Relaxation

Away from boundaries and shocks, the linear update scheme Eq. (4.17) is relaxed with distributed relaxation. Transforming the corrections in Eq. (4.17) to primitive variables  $\mathbf{q} = (u, p, \epsilon)^T$  gives

$$[\bar{\mathbf{A}}^+ \delta_x^- + \bar{\mathbf{A}}^- \delta_x^+](\delta \mathbf{q}) = -\left(\frac{\partial \mathbf{q}}{\partial \mathbf{Q}}\right)_j \frac{1}{\sigma_j} \hat{\mathbf{R}}_j, \quad (4.20)$$

with a right-side vector composed of a combination of the conserved residual terms. The Jacobian matrices are related through a similarity transformation to the conservative Jacobians:

$$\bar{\mathbf{A}} = \frac{\partial \mathbf{q}}{\partial \mathbf{Q}} \mathbf{A} \frac{\partial \mathbf{Q}}{\partial \mathbf{q}}. \quad (4.21)$$

Alternately, by definition, the primitive variable correction equation is

$$\mathbf{L} \delta \mathbf{q} = -\bar{\mathbf{r}}_j, \quad (4.22)$$

where

$$\bar{\mathbf{r}}_j = \left(\frac{\partial \mathbf{q}}{\partial \mathbf{Q}}\right)_j \frac{1}{\sigma_j} \hat{\mathbf{R}}_j. \quad (4.23)$$

The elements of the matrix operator  $\mathbf{L}$  are given in appendix II.

As follows from Appendices I and II, the determinant of  $\mathbf{L}$  for ( $u > 0$ ) in deep subsonic or supersonic regions is

$$\det(\mathbf{L}) = [(u^2 - c^2) \delta_{xx}] u \delta_x^-, \quad (4.24)$$

where

$$\delta_{xx} \equiv \begin{cases} \delta_x^- \delta_x^+ & \text{for fully subsonic flow,} \\ \delta_x^- \delta_x^- & \text{for fully supersonic flow.} \end{cases} \quad (4.25)$$

The first term in Eq. (4.24) represents an approximation to the full-potential operator and the second term represents an upwind approximation for the convection of entropy. In subsonic flow,  $\delta_{xx}$  is a central discretization associated with the ellipticity of the full-potential operator. In fully supersonic flow,  $\delta_{xx}$  is a one-sided or upwind-biased approximation in accordance with the hyperbolic nature of the full-potential operator. For first-order upwind differencing, the full-potential factor is a 3-point operator; for second-order upwind-biased differencing, the factor is a 7-point operator (see appendix II).

In this one-dimensional case, the first two equations in the matrix operator  $\mathbf{L}$  are uncoupled from the third. Thus we need only consider distributed relaxation of the first two equations; the third equation can be solved for  $\delta\epsilon$  in its primitive variable form once  $\delta u$  and  $\delta p$  are found from distributed relaxation. Denoting the upper  $2 \times 2$  block of  $\mathbf{L}$  as  $\bar{\mathbf{L}}$ , an obvious choice for  $\bar{\mathbf{M}}$  is the matrix of cofactors of  $\bar{\mathbf{L}}$ , which gives a diagonal matrix for  $\bar{\mathbf{L}}\bar{\mathbf{M}}$  with the full-potential factors along the diagonal, i.e.,

$$\bar{\mathbf{L}}\bar{\mathbf{M}} = \begin{pmatrix} (u^2 - c^2) \delta_{xx} & 0 \\ 0 & (u^2 - c^2) \delta_{xx} \end{pmatrix}. \quad (4.26)$$

### Boundary Treatment

As pointed out by Sidilkover [15], there is a connection between distributed relaxation and the characteristic variables. For subsonic flow, a Jacobi update to the operators in  $\bar{\mathbf{L}}\bar{\mathbf{M}}$  is equivalent to a Kacmarcz relaxation of the characteristic equations, cast in terms of a characteristic combination of corrections as

$$\begin{pmatrix} (u + c) \delta_x^- (\rho c \delta u + \delta p) \\ (u - c) \delta_x^+ (\rho c \delta u - \delta p) \end{pmatrix} = - \begin{pmatrix} \rho c \bar{r}_1 + \bar{r}_2 \\ \rho c \bar{r}_1 - \bar{r}_2 \end{pmatrix}_j. \quad (4.27)$$

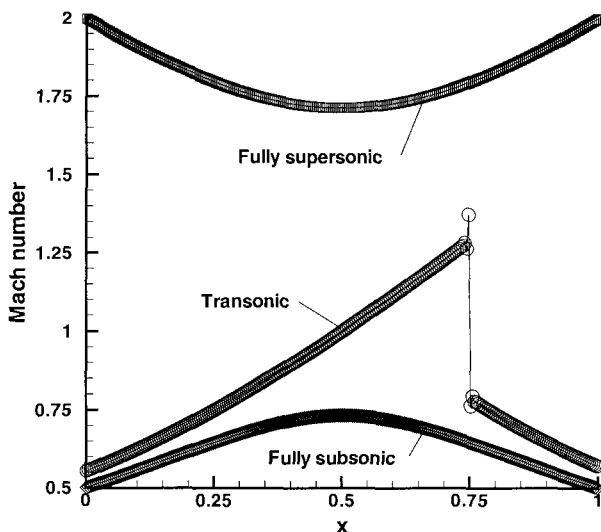
With first-order differencing, an equal and opposite correction to the downstream propagating characteristic variable  $\rho c \delta u + \delta p$  is sent in Kacmarcz relaxation of the first equation to the local cell  $j$  and to the upstream cell  $j - 1$ ; the correction to cell  $j$  is exactly one half of that found from a point Jacobi relaxation of the corresponding characteristic equation. Likewise, an equal

and opposite correction to the upstream propagating characteristic variable  $\rho c \delta u - \delta p$  is sent in Kacmarcz relaxation of the second equation to the local cell  $j$  and to the downstream cell  $j + 1$ . With a local update of the solution at each point, the smoothing rate should be equivalent to Gauss-Seidel relaxation of the standard 3-point Laplacian operator. However, the relaxation at cells adjacent to the boundary is inconsistent with the boundary conditions of the original problem; for example, characteristic boundary conditions would prescribe  $\rho c \delta u + \delta p$  and  $\rho c \delta u - \delta p$  as zero at the upstream and downstream boundaries, respectively.

This example demonstrates the general result mentioned earlier: that distributed relaxation needs to be modified near the boundaries. A simple boundary specification for the ghost variable  $\delta \mathbf{w} = (\delta w_1, \delta w_2)^T$  can be found in this case; enforcing a Neumann condition on  $\delta w_1$  ( $\delta w_2$ ) and a Dirichlet condition on  $\delta w_2$  ( $\delta w_1$ ) at inflow (outflow) in subsonic relaxation is consistent with characteristic boundary conditions in terms of primitive variables. We instead use the more general formulation and apply another local conservative relaxation procedure at the boundary points to reduce the residuals. Distributed relaxation is then applied in the interior with simple Dirichlet conditions for the ghost variables  $\delta \mathbf{w}$ .

## Relaxation Sequence

The general solution procedure was to reduce the local residuals at least two orders of magnitude at the inflow boundary, the outflow boundary, and then at the shock region — both before and after sweeping the entire domain with distributed relaxation. The local conservative relaxation is made over the first two cells adjacent to the boundary. In the shock region, the local conservative relaxation was applied to nine cells centered on the upstream (supersonic) side of the shock. A pseudo-time step corresponding to a Courant number of 100 was used as an underrelaxation factor in the local conservative relaxations; otherwise purely steady-state equations were relaxed. In distributed relaxation, the variables  $(\delta w_1, \delta w_2)^T$  were explicitly used in the implementation; they were first relaxed over the interior cells, then distributions to  $(\delta u, \delta p)^T$  were made, and then the corrections to  $\delta \epsilon$  were computed. As a check, single-grid computations verified that the convergence of the error per relaxation of subsonic equations (4.13) with first-order differencing was identical to that found with a Gauss-Seidel relaxation of the one-dimensional 3-point Laplacian operator with Neumann boundary conditions applied at one end of the domain.



**Figure 1** Mach number distribution computed without limiting on a grid of 257 points for fully subsonic, fully supersonic, and transonic flow with a shock at  $x = 0.75$ .

## Computational Results

Computational results are shown in this section for fully subsonic, fully supersonic, and transonic flow with and without a shock. The Mach number distribution with second-order accurate discretization on a grid of 257 points is shown in Fig. 1 for fully subsonic, fully supersonic, and transonic flow with a shock; the shock location is specified at  $x = 0.75$ . There is no limiting applied in the transonic cases so there are some oscillations at the shock. In this one-dimensional case, the solution can easily be repaired by using an essentially nonoscillatory (ENO) [21] approach (see appendix III) but the emphasis of this work is on the solution procedure and not on the steady-state results.

### Fully Subsonic Flow

The smoothing rate of distributed relaxation for the primitive variable equations is expected to be equal to the worst among the smoothing rates obtained in relaxation of the scalar factors of the determinant of  $\mathbf{L}$ , in our case convection and the full-potential factor. With the full-potential operator

| $h$   | $\ e_d\  : p$           | $\frac{\ e_t\  - \ e_d\ }{\ e_d\ } : p$ |
|-------|-------------------------|-----------------------------------------|
| 1/32  | $0.2654 \times 10^{-3}$ | $< 0.01$                                |
| 1/64  | $0.6341 \times 10^{-4}$ | $< 0.01$                                |
| 1/128 | $0.1547 \times 10^{-4}$ | $< 0.01$                                |
| 1/256 | $0.3817 \times 10^{-5}$ | $< 0.01$                                |

**Table 1** The discretization errors in  $p$  at convergence and the relative  $L_1$ -norm errors after the FMG-1 algorithm for fully subsonic flow.

*relaxed* (Gauss-Seidel) and the convection operator *solved* by downstream marching from the inflow boundary, the convergence rate of the multigrid cycle per relaxation sweep for the first-order accurate discretization should be close to the smoothing factor, 0.447, of Gauss-Seidel relaxation for the one-dimensional 3-point Laplacian operator.

Defect correction is often used in the solution of higher order implicit discretizations to reduce the arithmetic operation count while retaining the target accuracy. One seemingly possible implementation of defect correction is to use first-order accurate discretizations for the convection operators in  $\mathbf{L}$ . The corresponding distribution matrix  $\mathbf{M}$  consists of the first-order operators as well. This implementation, however, is not a good idea for obtaining good *smoothing* rates, because *in terms of high-frequency components*, a low-order operator and a corresponding high-order operator do not necessarily approximate each other well. Indeed, this implementation caused a slowdown in the convergence, and the smoothing rate corresponding to that of the Laplace equation was not attained.

To illustrate this phenomenon, consider the Laplacian operator alone. The elliptic operator  $\delta_{xx}(w_j)$  arising from differencing the primitive variable equations with the second-order upwind-biased operator used here is a 7-point operator (see appendix II). Corrections  $(\delta w)_j$  to the current approximate solutions  $w_j$  computed in Gauss-Seidel defect-correction relaxation with the 3-point Laplacian as a left-side (driver) operator can be found from

$$\frac{1}{h^2}[(\delta w)_{j-1} - 2(\delta w)_j] = -\delta_{xx}(w_j). \quad (4.28)$$

The smoothing factor, computed with local mode analysis, of this defect-correction relaxation (4.28) is  $g_s = 0.525$ , which is substantially smaller than the convergence rate per distributed relaxation of 0.9 observed with a multigrid FV(2,1) cycle applied to Eq. (4.22). The smoothing factor of distributed relaxation for Eq. (4.22) is computed by using local mode analysis as a norm of the matrix  $\hat{\mathbf{G}}$ , defined as



| $h$   | $\ e_d\  : p$           | $\frac{\ e_t\  - \ e_d\ }{\ e_d\ } : p$ |
|-------|-------------------------|-----------------------------------------|
| 1/32  | $0.8890 \times 10^{-4}$ | $< 0.01$                                |
| 1/64  | $0.2151 \times 10^{-4}$ | $0.02$                                  |
| 1/128 | $0.5278 \times 10^{-5}$ | $< 0.01$                                |
| 1/256 | $0.1306 \times 10^{-5}$ | $< 0.01$                                |

**Table 2** The discretization errors in  $p$  at convergence and the relative  $L_1$ -norm errors after the FMG-1 algorithm for fully supersonic flow.

$$\widehat{\mathbf{G}} = I - (1 - g_s)(\widehat{\mathbf{M}}_d)^{-1}\widehat{\mathbf{M}}_t, \quad (4.29)$$

where matrix symbols associated with distribution matrices for target (second-order) and driver (first-order) operators are denoted  $\widehat{\mathbf{M}}_t$  and  $\widehat{\mathbf{M}}_d$ , respectively. The calculated smoothing factor is 0.752, appreciably greater than the expected value of 0.525. The increased smoothing factor is explained by the fact that  $(\widehat{\mathbf{M}}_d)^{-1}\widehat{\mathbf{M}}_t$  is far different from the identity matrix.

With corrections distributed by the second-order distribution matrix operator ( $\mathbf{M}_d = \mathbf{M}_t$ ), the smoothing factor of distributed relaxation for Eq. (4.22) should be identical to that for relaxation of the Laplacian equation. Implementing the defect-correction relaxation (4.28) of the Laplacian equation but distributing corrections with the second-order distribution yielded an FV(2,1) cycle convergence rate per smoothing of 0.51 — close to 0.525 predicted by the mode analysis of the defect-correction relaxation (4.28).

Results for an FMG-1 algorithm, denoting a full multigrid algorithm with one FV(2,1) cycle at each level, are shown for fully subsonic flow in Table 1. All of the norms used in this and the following tables are  $L_1$ -norms. The discretization error in pressure is defined pointwise as  $e_d \equiv \bar{p}_j - p(x_j)^{\text{exact}}$ , where  $p(x)^{\text{exact}}$  is the steady-state solution of Eq. (4.4) and  $\bar{p}_j$  is the discrete solution of Eq. (4.13). A second-order spatial convergence in the  $L_1$ -norm of  $e_d$  is evident. The total error in pressure is defined as  $e_t \equiv p(x_j)^{\text{exact}} - p_j$ , where  $p_j$  is the approximate solution obtained in the FMG-1 algorithm. Small relative difference between  $L_1$ -norms of the total and discretization errors (see Table 1) indicates that optimal efficiency has been attained. The asymptotic convergence rate of the FV(2,1) cycle was 0.50–0.52 per relaxation for any of the four grids.

### Fully Supersonic Flow

In fully supersonic flow, it is generally possible to construct marching methods to solve the full-potential equation. For first-order differencing, the one-sided operator can easily be solved over the domain. For the second-order operator considered here, the defect-correction relaxation with the first-order upwind driver can be defined as

$$\frac{1}{h^2}[(\delta w)_{j-2} - 2(\delta w)_{j-1} + (\delta w)_j] = -\delta_{xx}(w_j). \quad (4.30)$$

This scheme can be solved by marching but, unfortunately, its amplification factor is larger than 1 for some error components. A suitable full-potential driver operator to replace the left-side operator of Eq. (4.30) can be found in the form

$$[\delta_x^- + h \alpha \delta_x^- \delta_x^-]^2, \quad (4.31)$$

where the backward difference operator is defined to be first-order in the above expression. This is a 5-point operator that can be solved by marching. The maximum amplification factor of the supersonic defect-correction relaxation with the driver (4.31) and  $\alpha = 0.23$  is 0.55. With the second-order distribution operator, computations for the fully supersonic flow corresponding to Mach 2 at inflow converged asymptotically at this rate.

Results for the FMG-1 algorithm are shown in Table 2. The  $L_1$ -norm of the discretization error in pressure shows the expected second-order spatial convergence. Optimal efficiency is attained with the FMG-1 algorithm. Here, the performance of the single-grid iteration at each level is the same as an FV(2,1) cycle, because the full-potential factor is being solved (rather than just smoothed) by marching. The asymptotic convergence rate per relaxation was 0.53–0.56 for any of the four grids.

### Transonic Flows

Results for the FMG-1 algorithm are shown for a smooth transonic flow in Table 3. The  $L_1$ -norm of the discretization error in pressure shows the expected second-order spatial convergence. The relative deviations of the  $L_1$ -norm of the total errors from the  $L_1$ -norm of the discretization errors are increased from the fully subsonic or fully supersonic levels but still exhibit optimal performance. The asymptotic convergence rate per relaxation was 0.52–0.56 for any of the four grids.

Results for the FMG-1 algorithm are shown for the transonic flow with a shock in Table 4. The  $L_1$ -norm of the discretization error in pressure shows a

| $h$   | $\ e_d\  : p$           | $\frac{\ e_t\  - \ e_d\ }{\ e_d\ } : p$ |
|-------|-------------------------|-----------------------------------------|
| 1/32  | $0.1075 \times 10^{-3}$ | 0.05                                    |
| 1/64  | $0.2666 \times 10^{-4}$ | 0.09                                    |
| 1/128 | $0.6635 \times 10^{-5}$ | 0.03                                    |
| 1/256 | $0.1655 \times 10^{-5}$ | 0.11                                    |

**Table 3** The discretization errors in  $p$  at convergence and the relative  $L_1$ -norm errors after the FMG-1 algorithm for transonic flow without a shock.

| $h$   | $\ e_d\  : p$           | $\frac{\ e_t\  - \ e_d\ }{\ e_d\ } : p$ |
|-------|-------------------------|-----------------------------------------|
| 1/32  | $0.4396 \times 10^{-2}$ | <0.01                                   |
| 1/64  | $0.2218 \times 10^{-2}$ | 0.015                                   |
| 1/128 | $0.1111 \times 10^{-2}$ | 0.014                                   |
| 1/256 | $0.5558 \times 10^{-3}$ | <0.01                                   |

**Table 4** The discretization errors in  $p$  at convergence and the relative  $L_1$ -norm errors after the FMG-1 algorithm for transonic flow with a shock.

first-order spatial convergence, because there is no limiting of the interpolation at the shock; i.e., maximum local errors occur at the shock and do not diminish with grid refinement. The relative deviations of the  $L_1$ -norm of the total error with the FMG-1 algorithm from the  $L_1$ -norm of the discretization error again exhibit optimal performance. The average convergence rate per relaxation over 10 cycles varied from 0.35 on the coarsest mesh to 0.51 on the finest mesh. Computations for this case with ENO differencing are presented in appendix III and show a similar performance with a monotone solution behavior in the shock region.

### Concluding Remarks

A general multigrid framework for obtaining textbook efficiency to solutions of the compressible Euler and Navier-Stokes equations in conservation law form has been discussed. The general methodology relies on a distributed relaxation procedure to reduce errors in regular (smoothly varying) flow regions; separate and distinct treatments for each of the factors (elliptic and/or hyperbolic) are used to attain optimal reductions of errors. Near the boundaries and near shocks, additional local relaxations of the conservative equations are necessary. Example calculations are made for the quasi-one-dimensional Euler equations

for situations with fully subsonic and fully supersonic flow, as well as transonic flows with and without a shock. All of the calculations showed that the FMG-1 algorithm provides a very accurate approximation of the exact solution.

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## Appendices

### I - Conservative Fluxes

The fluxes are constructed by using the flux-difference splitting of Roe [20] and are given as

$$\mathbf{F}_{j+\frac{1}{2}} = \frac{1}{2}[\mathbf{F}(\mathbf{Q}_R) + \mathbf{F}(\mathbf{Q}_L) - |\tilde{\mathbf{A}}|(\mathbf{Q}_R - \mathbf{Q}_L)] \quad (4.32)$$

where  $\hat{\mathbf{A}} \equiv \partial \mathbf{F} / \partial \mathbf{Q}$  is evaluated with a specific average of the left and right states,  $\mathbf{Q}_L$  and  $\mathbf{Q}_R$ , in order to exactly satisfy the jump conditions for a shock [20]. The left and right states are constructed for first-order accuracy as

$$\mathbf{Q}_L = \mathbf{Q}_j, \quad (4.33)$$

$$\mathbf{Q}_R = \mathbf{Q}_{j+1}, \quad (4.34)$$

and for second-order accuracy with Fromm's discretization as

$$\mathbf{Q}_L = \mathbf{Q}_j + \frac{1}{4}(\mathbf{Q}_{j+1} - \mathbf{Q}_{j-1}), \quad (4.35)$$

$$\mathbf{Q}_R = \mathbf{Q}_{j+1} - \frac{1}{4}(\mathbf{Q}_j - \mathbf{Q}_{j+2}). \quad (4.36)$$

The dissipation matrix  $|\tilde{\mathbf{A}}| \equiv \mathbf{T}|\tilde{\mathbf{\Lambda}}|\mathbf{T}^{-1}$  where  $\mathbf{T}, \mathbf{T}^{-1}$  are diagonalizing matrices, the matrix of eigenvalues is

$$\mathbf{\Lambda} \equiv \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}, \quad (4.37)$$

and

$$(\lambda_1, \lambda_2, \lambda_3)^T = (u + c, u - c, u)^T. \quad (4.38)$$

The tilde superscript denotes that the eigenvalues are limited away from zero as,

$$|\tilde{\lambda}_i| \equiv \begin{cases} |\lambda_i| & \text{if } |\lambda_i| \geq \tilde{\epsilon}, \\ (\lambda_i^2 + (\tilde{\epsilon})^2)/(2\tilde{\epsilon}) & \text{otherwise,} \end{cases} \quad (4.39)$$

mainly to prevent expansion shocks at sonic points for the first-order discretization accuracy, but also to make a smooth transition through the sonic point for the full-potential operator. The value  $\tilde{\epsilon}$  is taken as  $\beta \max\{|\lambda_1|, |\lambda_2|, |\lambda_3|\}$ . This “entropy fix” is not optimal in any sense, as larger  $\beta$  values for sonic points are required for the first order scheme on coarser grids. The value of  $\beta$  was nominally set at 0.1 and was arbitrarily increased to 0.2 for grids of 9 points or less in the implicit first-order Jacobian matrices.

## II - Distribution Matrices

The coefficient matrices  $\bar{\mathbf{A}}^\pm$  are defined as the positive and negative eigenvalue contributions to  $\bar{\mathbf{A}}$ , where

$$\lambda_i^\pm \equiv (\lambda_i \pm |\tilde{\lambda}_i|)/2 \quad (4.40)$$

and  $\bar{\mathbf{A}}$  is written in terms of its eigenvalues,  $\bar{\mathbf{A}}(\lambda_1, \lambda_2, \lambda_3)$ , as

$$\begin{bmatrix} \frac{1}{2}(\lambda_1 + \lambda_2) & \frac{1}{2\rho c}(\lambda_1 - \lambda_2) & 0 \\ \frac{\rho c}{2}(\lambda_1 - \lambda_2) & \frac{1}{2}(\lambda_1 + \lambda_2) & 0 \\ \frac{c}{2\gamma}(\lambda_1 - \lambda_2) & \frac{1}{2\rho\gamma}(\lambda_1 + \lambda_2 - 2\lambda_3) & \lambda_3 \end{bmatrix}. \quad (4.41)$$

The upper  $2 \times 2$  block of  $\mathbf{L} \equiv \bar{\mathbf{A}}^+ \delta_x^- + \bar{\mathbf{A}}^- \delta_x^+$  can be written as

$$\bar{\mathbf{L}} = \begin{bmatrix} t_1 & \frac{1}{\rho c}(t_2) \\ \rho c(t_2) & t_1 \end{bmatrix}. \quad (4.42)$$

and the corresponding matrix of cofactors is

$$\bar{\mathbf{M}} = \begin{bmatrix} t_1 & -\frac{1}{\rho c}(t_2) \\ -\rho c(t_2) & t_1 \end{bmatrix}. \quad (4.43)$$

where

$$t_1 = \frac{1}{2}[(\lambda_1^+ + \lambda_2^+)\delta_x^- + (\lambda_1^- + \lambda_2^-)\delta_x^+], \quad (4.44)$$

$$t_2 = \frac{1}{2}[(\lambda_1^+ - \lambda_2^+)\delta_x^- + (\lambda_1^- - \lambda_2^-)\delta_x^+]. \quad (4.45)$$

The determinant of  $\bar{\mathbf{L}}$  is

$$\begin{aligned} t_1^2 - t_2^2 &= (\lambda_1^+ \lambda_2^+) \delta_x^- \delta_x^- \\ &+ (\lambda_1^+ \lambda_2^- + \lambda_1^- \lambda_2^+) \delta_x^- \delta_x^+ \\ &+ (\lambda_1^- \lambda_2^-) \delta_x^+ \delta_x^+ \end{aligned} \quad (4.46)$$

The operators for second-order upwind-biased differencing are defined as

$$\begin{aligned} \delta_x^- \delta_x^+(w_j) &= \frac{1}{16h^2} (-w_{j-3} + 2w_{j-2} + 17w_{j-1} \\ &\quad - 36w_j + 17w_{j+1} + 2w_{j+2} - w_{j+3}), \end{aligned} \quad (4.47)$$

$$\begin{aligned} \delta_x^- \delta_x^-(w_j) &= \frac{1}{16h^2} (w_{j-4} - 10w_{j-3} + 31w_{j-2} \\ &\quad - 28w_{j-1} - w_j + 6w_{j+1} + w_{j+2}). \end{aligned} \quad (4.48)$$

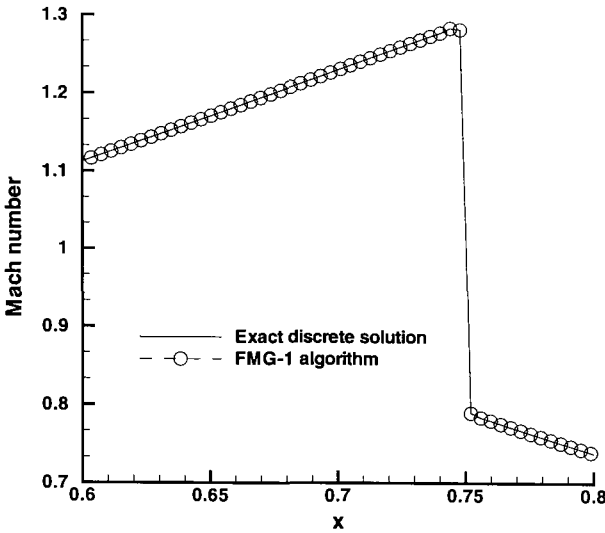
### III - Transonic Shock — ENO Differencing

| $h$   | $\ e_d\  : p$           | $\frac{\ e_t\  - \ e_d\ }{\ e_d\ } : p$ |
|-------|-------------------------|-----------------------------------------|
| 1/32  | $0.3597 \times 10^{-3}$ | $< 0.01$                                |
| 1/64  | $0.9711 \times 10^{-4}$ | 0.01                                    |
| 1/128 | $0.5133 \times 10^{-4}$ | 0.07                                    |
| 1/256 | $0.2665 \times 10^{-5}$ | 0.13                                    |

**Table 5** The discretization errors in  $p$  with ENO differencing at convergence and the relative  $L_1$ -norm errors after the FMG-1 algorithm for transonic flow with a shock.

Uniform application of the upwind-biased interpolations for state variables of appendix I (Fromm's scheme) leads to oscillations at the shock. These oscillations can be eliminated by a limiting procedure to reduce locally the order of approximation in the region of the shock. Here, an ENO approach [21] is used to prevent state-variable interpolations from crossing the shock. At the shock interface, the left (right) state variables,  $\mathbf{Q}_L(\mathbf{Q}_R)$ , are found by one-sided second-order extrapolation; at the first interface upstream (downstream) of the shock, the state variables,  $\mathbf{Q}_L(\mathbf{Q}_R)$ , are found by second-order central averaging.

Fig. 2 presents the Mach number distribution in the region of the shock and shows a nonoscillatory behavior with a one-point representation of the



**Figure 2** Mach number distribution near the shock computed with the FMG-1 algorithm compared to the exact discrete solution ( $h = 1/256$ ).

shock. The discretization errors and relative deviations of the  $L_1$ -norm of the total error with the FMG-1 algorithm are shown in Table 5; comparison with Table 4 shows that the ENO discretization errors are smaller than the discretization errors of the unlimited interpolations, although first-order behavior is still found. In both situations, for this one-dimensional case, second-order accuracy is attained if the  $L_1$ -norm is restricted to regions either far upstream or far downstream of the shock. Although not shown, we note that for this case, uniform second-order accuracy can be found with the ENO procedure above if the entropy fix is dropped at the shock; the shock jump is then recovered identically, and a zero-point shock is recovered. The focus of this investigation is on convergence and both Fig. 2 and Table 5 indicate optimal efficiency has been attained with the FMG-1 algorithm.



# Numerical Solutions of Cauchy-Riemann Equations for Two and Three Dimensional Flows

Mohamed M. Hafez<sup>1</sup> and J. Houseman<sup>1</sup>

## Abstract

For two dimensional flows, the conservation of mass and the definition of vorticity comprise a generalized Cauchy-Riemann system for the velocity components assuming the vorticity is given. If the flow is compressible, the density is a function of the speed and the entropy, and the latter is assumed to be known. Introducing artificial time, a symmetric hyperbolic system can be easily constructed. Artificial viscosity is needed for numerical stability and is obtained from a least squares formulation. The augmented system is solved explicitly with a standard point relaxation algorithm which is highly parallelizable. For an extension to three dimensional flows the continuity equation is combined with the definitions of two vorticity components, and are solved for the three velocity components. Second order accurate results are compared with exact solutions for incompressible, irrotational, and rotational flows around cylinders and spheres. Results for compressible (subsonic) flows are also included.

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## 5.1 Introduction

Numerical flow simulations are usually based on the solutions of Euler and Navier-Stokes equations in terms of primitive or conservative variables. Alternative formulations based on the vector potential or in terms of velocity-vorticity equations are not popular because of the associated difficulties and limitations. Recently there are some efforts to construct numerical schemes based on decoupling the kinematics and the dynamics of the motion. The motivations and the advantages of such formulations are discussed in references [8].

In this work we are interested in calculating the velocity field using the continuity equation and the definition of vorticity. For steady, incompressible, inviscid, irrotational flow the two velocity components can be obtained from the standard Cauchy-Riemann system. The two first-order equations are equivalent to one second-order Laplace equation in terms of the potential or the stream function. Also, a least-squares procedure results in two Laplace equations for the two velocity components.

If the flow is not irrotational and there are also sources in the field, the nonhomogeneous Cauchy-Riemann equations are not reducible to a single second order equation. The velocity vector can be represented as the gradient of a potential plus the curl of another vector. The first component is curl free and the second component is divergence free. This decomposition is always possible under very general conditions according to the Helmholtz theorem. However, for general three dimensional flows, the boundary conditions are complicated. On the other hand, the least squares procedure produces three second order equations for the three velocity components. It is not clear however how to impose easily the entropy condition, in such a formulation, in order to exclude expansion shocks for simulation of transonic flows.

The least squares formulation has other problems as well, even for incompressible flows. First the spurious solution must be excluded via imposing the first order equations on the boundary. Moreover mass may not be conserved at the discrete level. Special arrangements, for example, staggered grids may be necessary to ensure conservation of numerical fluxes. For the vorticity, obtained from the curl of the momentum equations, special treatment is required to guarantee that its divergence vanishes identically at the discrete level.

In the following we will solve generalized Cauchy-Riemann equations embedded in an artificial time process governed by a symmetric hyperbolic system leading to a well posed initial-boundary value problem. Since we are interested only in the steady state solution, the accuracy of the transient behavior is not an issue. Standard convergence acceleration techniques such as multigrid can be employed to improve the efficiency of the calculations (See Appendix A).

To avoid odd and even decoupling and to insure numerical stability, artificial viscosity must be introduced. To guarantee second order accuracy (for incompressible and subsonic flow), artificial viscosity based on the least squares formulation is used (See Hughes et al[3] and Tezduyar and Hughes[9]). The scheme is also similar to Lerat's recent work [5], at least for subsonic flows. The second order terms vanish identically at the continuous level since they are consistent with the Cauchy-Riemann equations. At the discrete level they produce higher order dissipation. Such a construction can be viewed as a compact method to replace the commonly used fourth order dissipation schemes proposed by McCormack[6] and by Jameson[4].

The paper is organized in four sections; governing relations, numerical algorithms, numerical results, and conclusions with some general remarks.

## 5.2 Governing Equations and Boundary Conditions

The equations for flow over a cylinder and over a sphere written in cylindrical and spherical coordinates are given.

### Flow Over a Cylinder

#### Continuity Equation

$$\frac{1}{\rho} \left[ \frac{1}{r} \frac{\partial(r\rho v_r)}{\partial r} + \frac{1}{r} \frac{\partial(\rho v_\theta)}{\partial \theta} \right] = 0 \quad (5.1)$$

#### Vorticity Definition

$$\frac{1}{r} \frac{\partial(rv_\theta)}{\partial r} - \frac{1}{r} \frac{\partial(v_r)}{\partial \theta} = \omega_z \quad (5.2)$$

### Three-Dimensional Flow Around a Sphere

#### Continuity Equation

$$\frac{1}{\rho} \left[ \frac{1}{r^2} \frac{\partial(r^2 \rho v_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta \rho v_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial \rho v_\phi}{\partial \phi} \right] = 0 \quad (5.3)$$

#### Vorticity Definitions

$$\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{1}{r} \frac{\partial(rv_\phi)}{\partial r} = \omega_\theta \quad (5.4)$$

$$\frac{1}{r} \frac{\partial(rv_\theta)}{\partial r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} = \omega_\phi \quad (5.5)$$

and the boundary conditions are:

$$v_r = 0 \quad \text{at } r = r_i \quad (5.6)$$

$$v_\theta \text{ and } v_\phi \text{ are given at } r = r_o, \quad (5.7)$$

where  $r_i$  and  $r_o$  are the radii of inner and outer spheres.

In case of irrotational, isentropic flows the vorticity vanishes identically and the density is related to the speed according to Bernoulli's law:

$$\rho = \left[ 1 - \frac{\gamma - 1}{2} M_\infty^2 (|\vec{q}|^2 - 1) \right]^{\frac{1}{\gamma - 1}}, \quad (5.8)$$

where  $M_\infty$  is the freestream Mach number. For incompressible flows  $M_\infty = 0$ .

### 5.3 Numerical Methods

Incompressible and compressible (subsonic) flows around cylinder and spheres are calculated using a least squares formulation as well as Cauchy-Riemann equations embedded in a symmetric hyperbolic system which is augmented by artificial viscosity. All spatial derivatives are discretized using second order accurate finite volume schemes. The time derivative terms are discretized via first order time differences. The discretized systems are solved using a point Gauss-Seidel relaxation scheme. The source code is implemented in C++ and run on an HP-Visualize C3000 workstation. In the following, the problems we solved are stated. (Transonic flows with shocks are calculated in [7] using similar schemes but with linearized boundary conditions)

The governing equations, denoted

$$L(\vec{w}), \quad (5.9)$$

can be imbedded in the following system

$$\partial_t \vec{w} = L(\vec{w}) + \epsilon L^* L(\vec{w}). \quad (5.10)$$

Where  $L^*$  is the adjoint of the operator  $L$ . For cartesian coordinates and incompressible irrotational flows,  $L^* L(\vec{w})$  are the Laplacians of the  $\vec{w}$  components.

The boundary conditions we imposed are:

*On Solid Surface:* A Dirchlet condition is used for the radial velocity  $v_r$  and a Neumann condition derived from the vorticity equation is used for the angular velocity  $v_\theta$  (and  $v_\phi$ ).

*Far Field:* A Dirchlet condition is used for the angular velocity  $v_\theta$  (and  $v_\phi$ ) and a Neumann condition derived from the continuity equation is used for the radial velocity  $v_r$ .

*Flow Field:* Periodic boundary conditions are used within the flow field for both  $v_r$  and  $v_\theta$  (and  $v_\phi$ ).

## 5.4 Numerical Results

### 5.4.1 Incompressible Flow

We tested the present formulation for cases with analytical solutions, for example, flow with circulation over a cylinder, uniform and shear, as well as a cylinder in a rotating fluid (see Batchelor [1]). We have also calculated flow over a sphere using axisymmetric as well as the full three dimensional equations. In Fig. 1, the streamlines are plotted for incompressible flows over a cylinder and a sphere. In all the above cases, the results are satisfactory in terms of accuracy and convergence. For example the results of flow calculations over a sphere are as follows.

*Example 1:* In this example we assume axisymmetric irrotational flow around a sphere, but we solve for each of the velocity components ( $v_r, v_\theta, v_\phi$ ), as in a fully three-dimensional flow. Numerical solutions are computed in a spherical coordinate system with three space dimensions. The grid spacings used are  $\Delta r = 0.1$ ,  $\Delta\theta = \pi/18$ , and  $\Delta\phi = \pi/18$ . The grid dimensions are  $36 \times 19 \times 37$  and the radius of the sphere is nondimensionalized to 1. The program is run until a residual tolerance of  $10^{-5}$  is met, and the results are shown in Table 1.

### 5.4.2 Compressible Flow

*Example 2:*

We solve for compressible flow over a cylinder with and without density in the Least-Squares formulation using  $\epsilon = 1$  and  $\epsilon = \frac{\Delta r}{2}$  for a cylindrical coordinate system in two space dimensions. The grid spacing is  $\Delta r = 0.1$  and  $\Delta\theta = \pi/18$ . The grid dimensions are  $36 \times 37$ , and the radius of the cylinder is nondimensionalized to 1, the circulation around the cylinder is kept at  $\Gamma = 0$ , the vorticity  $\omega_z = 0.0$ , and *Mach Numbers* of 0.1 and 0.2 are used. The program is run until a residual tolerance of  $10^{-5}$  is met. (See Table 2 and Fig. 2a.)

*Example 3:*

We solve for compressible flow over a sphere using  $\epsilon = 1$  and  $\epsilon = \frac{\Delta r}{2}$  for a spherical coordinate system in two space dimensions. The grid spacing is  $\Delta r = 0.1$  and  $\Delta\theta = \pi/18$ . The grid dimensions are  $36 \times 37$  and the radius of the sphere is nondimensionalized to 1, the vorticity  $\omega_z = 0.0$ , and *Mach Numbers* of 0.1 and 0.2 are used. The program is run until a residual tolerance of  $10^{-5}$  is met. (See Table 3 and Fig. 2b.)

All of the above results are obtained using constant artificial viscosity,  $\epsilon$ . In general, however,  $\epsilon$  can vary from point to point and from iteration to iteration and it can be optimized, depending on local flow conditions, to accelerate the convergence of the calculations.

## 5.5 Concluding Remarks

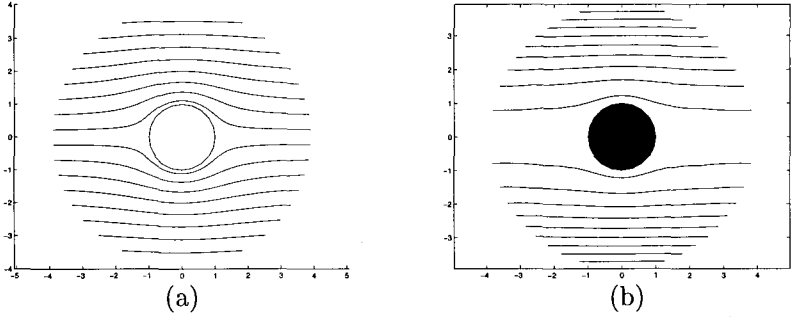
Given the vorticity and the density, the velocity components for two and three dimensional flows can be calculated from a generalized system of Cauchy-Riemann equations. Standard numerical algorithms are applicable to achieve the expected accuracy and efficiency. For a complete flow simulation, the dynamics of the motion must be included to provide the entropy and the total enthalpy and hence the vorticity. The full simulation is still under progress and will be reported separately.

## 5.6 Appendix: Multigrid Convergence Results

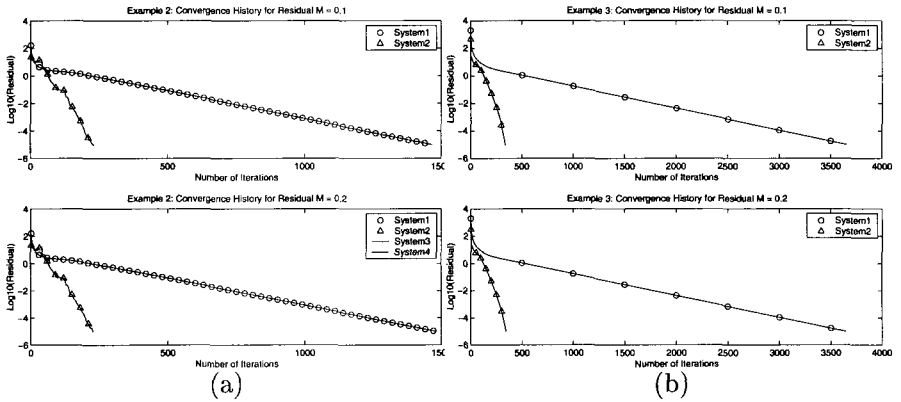
The convergence of the point relaxation schemes can be enhanced by implementing a multigrid V-Cycle scheme. The V-Cycle scheme from Briggs, Henson, and McCormick [2] was modified to compute the incompressible, irrotational flow over a cyclinder using the Cauchy-Riemann equations with  $\epsilon = \frac{\pi}{8}$ . The fine grids used for the computations are  $N = 32^2, 64^2, 128^2, 256^2$ . The coarsest grid used for each run is  $N = 8^2$ . A comparison of point relaxation and V-Cycle( $\nu_1, \nu_2$ ) is shown in Table 4. The parameters  $\nu_1 = 1$  is the number of relaxations used going down the grids and  $\nu_2 = 2$  is the number of relaxations used coming up the grids. The comparison includes the number of iterations until convergence of  $\|r\|_\infty < 10^{-5}$ , the absolute error  $\|e\|_\infty$  between the computed solution and the analytic solution, and the CPU runtime.

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**Figure 1** Streamlines for incompressible flow past (a) cylinder and (b) sphere with  $\omega_\phi = 0$ .



**Figure 2** Convergence histories for the  $\|r\|_\infty$  norm for systems 1 and 2 for (a) Example 2 and (b) Example 3.

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**Table 1** Shown are the total number of iterations and final max norm error of  $(v_r, v_\theta, v_\phi)$  for the computed solution. Where the error is the difference between the exact analytic solution and the numerical results.

|          | $\epsilon$           | Number of Iterations | Absolute Error |
|----------|----------------------|----------------------|----------------|
| System 1 |                      | 1757                 | 0.0084         |
| System 2 | 1                    | 1805                 | 0.0147         |
| System 3 | $\frac{\Delta r}{2}$ | 381                  | 0.0309         |

**Table 2** Shown are the total number of iterations to reach a residual tolerance of  $10^{-5}$  for the computed solutions with  $\epsilon = 1$  and  $\epsilon = \frac{\Delta r}{2}$  and Freestream Mach Numbers 0.1 and 0.2.

|          | $\epsilon$           | Iterations(M = 0.1) | Iterations(M = 0.2) |
|----------|----------------------|---------------------|---------------------|
| System 1 | 1                    | 1467                | 1481                |
| System 2 | $\frac{\Delta r}{2}$ | 231                 | 228                 |

**Table 3** Shown are the total number of iterations to reach a residual tolerance of  $10^{-5}$  for the computed solutions with  $\epsilon = 1$  and  $\epsilon = \frac{\Delta r}{2}$  and Freestream Mach Numbers 0.1 amd 0.2.

|          | $\epsilon$           | Iterations(M = 0.1) | Iterations(M = 0.2) |
|----------|----------------------|---------------------|---------------------|
| System 1 | 1                    | 3657                | 3647                |
| System 2 | $\frac{\Delta r}{2}$ | 341                 | 344                 |

**Table 4** Convergence results

|         | V-Cycle    |                |            | Point Relaxation |                |            |
|---------|------------|----------------|------------|------------------|----------------|------------|
| N       | iterations | $\ e\ _\infty$ | time(sec.) | iterations       | $\ e\ _\infty$ | time(sec.) |
| $32^2$  | 15         | 1.40e-2        | 3          | 532              | 1.40e-2        | 58         |
| $64^2$  | 15         | 3.52e-3        | 12         | 1754             | 3.52e-3        | 823        |
| $128^2$ | 20         | 8.80e-4        | 69         | 6787             | 8.89e-4        | 11810      |
| $256^2$ | 23         | 2.19e-4        | 337        | 26982            | 2.27e-4        | 107622     |



# Efficient High-order Schemes on Non-uniform Meshes for Multi-Dimensional Compressible Flows

A. Lerat, C. Corre and G. Hanss <sup>1</sup>

## 6.1 Introduction

A dissipative scheme of third-order accuracy has been proposed for the compressible Euler and Navier-Stokes equations in Ref. [1]-[2]. This scheme is compact but it does not require the solution of a linear algebraic system (for its explicit form) and it involves a simple and efficient first-order numerical dissipation. These properties are due to a residual-based construction of the scheme. In the present paper, our aim is to study an implementation of the method on a non-uniform mesh that preserves the third-order accuracy rigorously, without any smoothness assumption on the mesh. This study concerns 2-D and 3-D irregular Cartesian meshes, with steps varying according to some stretching or in a random way.

First, the Euler solver on a regular Cartesian mesh is briefly presented in Section 6.2, from two design principles called residual-based compactness and residual-based dissipation. Then, the Euler solver is extended to totally irregular Cartesian meshes in Section 6.3, using a finite-volume formulation, a suitable definition of the accuracy and some residual-based corrections. The generalization to the Navier-Stokes equations is considered in Section 6.4. Finally, numerical experiments are presented in Section 6.5 for three series

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of meshes and five test-cases: a 2-D rotational advection, a 3-D helicoidal advection, a 2-D Poiseuille model problem, a 2-D Burgers problem and a 2-D compressible jet interaction. These test cases allow the comparison of the formal and computed accuracy orders when the solution is smooth and show how the scheme captures discontinuities without limiters.

## 6.2 Euler solver on a regular Cartesian mesh

### 6.2.1 Residual-based compactness

We first consider the two-dimensional compressible Euler equations:

$$w_t + f_x + g_y = 0 \quad (6.1)$$

where  $w$  is the vector of conservative variables and  $f = f(w)$ ,  $g = g(w)$  are the flux components in the  $x$  and  $y$ -directions. The space derivatives in Eq. (6.1) are approximated on a regular Cartesian mesh:  $x_j = j\delta x$ ,  $y_k = k\delta y$ , with  $\delta x \approx \delta y = O(h)$ , using the basic discrete operators:

$$\begin{aligned} (\delta_1 v)_{j+\frac{1}{2},k} &= v_{j+1,k} - v_{j,k} & (\delta_2 v)_{j,k+\frac{1}{2}} &= v_{j,k+1} - v_{j,k} \\ (\mu_1 v)_{j+\frac{1}{2},k} &= \frac{1}{2}(v_{j+1,k} + v_{j,k}) & (\mu_2 v)_{j,k+\frac{1}{2}} &= \frac{1}{2}(v_{j,k+1} + v_{j,k}). \end{aligned}$$

The classical compact approximation for a space derivative is 4th-order accurate on a 3-point stencil. The price to pay for this compact stencil is the numerical solution of a linear algebraic system for each space direction. This extra cost can be avoided when calculating steady solutions by using the *residual-based compactness* (this is also possible for unsteady problems via the introduction of a dual time). The idea is not to discretize each space derivative at high order, but instead to consider the whole residual at steady-state

$$r = f_x + g_y.$$

More precisely,  $f_x$  and  $g_y$  are approximated by special centered difference formulae of second-order accuracy such that the global truncation error can be expressed in terms of derivatives of  $r$  only, namely:

$$\begin{aligned} \frac{1}{\delta x} \delta_1 \mu_1 (I + \frac{1}{6} \delta_2^2) f &= f_x + \frac{\delta x^2}{6} f_{xxx} + \frac{\delta y^2}{6} f_{xyy} + O(h^4) \\ \frac{1}{\delta y} \delta_2 \mu_2 (I + \frac{1}{6} \delta_1^2) g &= g_y + \frac{\delta y^2}{6} g_{yyy} + \frac{\delta x^2}{6} g_{xxy} + O(h^4). \end{aligned}$$

where  $I$  is the identity operator.

Inserting these approximations in Eq. (6.1), we obtain:

$$w_t + \frac{1}{\delta x} \delta_1 \mu_1 (I + \frac{1}{6} \delta_2^2) f + \frac{1}{\delta y} \delta_2 \mu_2 (I + \frac{1}{6} \delta_1^2) g = 0. \quad (6.2)$$

This scheme has a  $3 \times 3$  point stencil and requires no linear algebra. For smooth solutions of Eq. (6.1), its truncation error reads:

$$\epsilon = w_t + r + \frac{\delta x^2}{6} r_{xx} + \frac{\delta y^2}{6} r_{yy} + O(h^4),$$

so that  $\epsilon = O(h^4)$  for an exact steady solution ( $r = 0$ ). The compact scheme (6.2) is thus fourth-order accurate at steady-state.

Residual-based compactness can easily be extended to three space dimensions and also to higher accuracy order (see [1], [2]).

### 6.2.2 Residual-based dissipation

Numerical dissipation is also residual-based, *i.e.* constructed from derivatives of  $r$ . So, it can be first order in the transient phase and therefore can provide robustness in a simple way, while preserving a high accuracy at steady state. More precisely, the dissipative form of the scheme is:

$$w_t + \frac{1}{\delta x} \delta_1 \mu_1 (I + \frac{1}{6} \delta_2^2) f + \frac{1}{\delta y} \delta_2 \mu_2 (I + \frac{1}{6} \delta_1^2) g = \frac{1}{2} \delta_1 (\Phi_1 r_1) + \frac{1}{2} \delta_2 (\Phi_2 r_2) \quad (6.3)$$

where  $r_1$  and  $r_2$  are second-order centered approximations to  $r$  at  $j + \frac{1}{2}, k$  and  $j, k + \frac{1}{2}$  respectively. The coefficients  $\Phi_1$  and  $\Phi_2$  are matrices depending only on  $\delta x$  and  $\delta y$  and on the flux Jacobian matrices  $A = df/dw$  and  $B = dg/dw$ . They have been determined to ensure an optimal multidimensional dissipation (see [1]). For a scalar 2-D hyperbolic equation, they can be written as:

$$\Phi_1 = \text{sgn}(A) \min(1, \frac{1}{\alpha}), \quad \Phi_2 = \text{sgn}(B) \min(1, \alpha) \quad \text{where } \alpha = \frac{\delta x |B|}{\delta y |A|}.$$

For a 2-D hyperbolic system of conservation laws,  $\Phi_1$  and  $\Phi_2$  have rather simple expressions given in [1].

The truncation error of the dissipative scheme (6.3) is:

$$\epsilon = w_t + r + \frac{\delta x^2}{6} r_{xx} + \frac{\delta y^2}{6} r_{yy} - \frac{\delta x}{2} (\Phi_1 r)_x - \frac{\delta y}{2} (\Phi_2 r)_y + O(h^3).$$

For an exact steady solution, we obtain  $\epsilon = O(h^3)$ , *i.e.* a third-order accuracy. Moreover, Scheme (6.3) is compact because its numerical dissipation has been discretized on the same  $3 \times 3$  point stencil as its non-dissipative part.

### 6.2.3 Compact third-order scheme

Being only interested here in steady solutions, the time derivative in (6.3) is simply approximated at first-order. The explicit version of the fully discrete scheme reads:

a) Non-dissipative numerical fluxes:

$$\begin{aligned}\overset{\circ}{F}_{j+\frac{1}{2},k}^n &= [(I + \frac{1}{6}\delta_2^2)\mu_1 f]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= [(I + \frac{1}{6}\delta_1^2)\mu_2 g]_{j,k+\frac{1}{2}}^n\end{aligned}\quad (6.4)$$

b) Numerical dissipation:

$$\begin{aligned}(d_1)_{j+\frac{1}{2},k}^n &= \frac{\delta x}{2} [\Phi_1 (\frac{\delta_1 f}{\delta x} + \frac{\delta_2 \mu_1 \mu_2 g}{\delta y})]_{j+\frac{1}{2},k}^n \\ (d_2)_{j,k+\frac{1}{2}}^n &= \frac{\delta y}{2} [\Phi_2 (\frac{\delta_1 \mu_1 \mu_2 f}{\delta x} + \frac{\delta_2 g}{\delta y})]_{j,k+\frac{1}{2}}^n\end{aligned}\quad (6.5)$$

c) Total numerical fluxes:

$$\begin{aligned}F_{j+\frac{1}{2},k}^n &= (\overset{\circ}{F} - d_1)_{j+\frac{1}{2},k}^n \\ G_{j,k+\frac{1}{2}}^n &= (\overset{\circ}{G} - d_2)_{j,k+\frac{1}{2}}^n\end{aligned}\quad (6.6)$$

d) New cell-values:

$$\Delta w_{j,k}^{expl} = -\Delta t (\frac{\delta_1 F}{\delta x} + \frac{\delta_2 G}{\delta y})_{j,k}^n \quad (6.7)$$

$$w_{j,k}^{n+1} = w_{j,k}^n + \Delta w_{j,k}^{expl},$$

where  $w_{j,k}^n$  denotes the numerical solution at time  $n\Delta t$  and location  $(j\delta x, k\delta y)$ . Note that the above scheme contains no limiter.

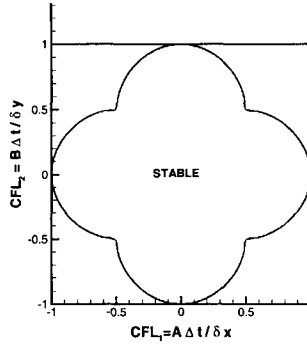
The linear  $L_2$ -stability domain of scheme (6.4)-(6.7) is shown on Fig.1 for a 2-D scalar problem, in terms of the CFL numbers in each space direction.

For increasing the computational efficiency of the solution of steady problems, an implicit version of the scheme has been developed (see [1], [2]). The implicit stage of the method is:

$$\begin{aligned}\Delta w_{j,k} &+ \Delta t [\mu_1 (A \frac{\delta_1 \Delta w}{\delta x}) - \frac{1}{2} \delta_1 (\Phi_1 A \frac{\delta_1 \Delta w}{\delta x})]_{j,k}^n \\ &+ \Delta t [\mu_2 (B \frac{\delta_2 \Delta w}{\delta y}) - \frac{1}{2} \delta_2 (\Phi_2 B \frac{\delta_2 \Delta w}{\delta y})]_{j,k}^n = \Delta w_{j,k}^{expl},\end{aligned}\quad (6.8)$$

$$w_{j,k}^{n+1} = w_{j,k}^n + \Delta w_{j,k}$$

It has been shown that the implicit scheme (6.4)-(6.8) is always linearly stable and offers good damping properties when used with large CFL numbers. In practice, the algebraic linear system (6.8) is solved by alternate-line relaxation of Jacobi or Gauss-Seidel type. Only a few inner iterations of the relaxation process (typically one or two) are necessary to ensure unconditional stability and a damping very close to that of (6.8).



**Figure 1** 2-D stability for the explicit 3rd-order scheme (inviscid problem).

Remark: On a regular mesh, various equivalent expressions of the non-dissipative numerical fluxes (6.4) can be given, that all lead to the same scheme; for instance:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= [\mu_1 f + \frac{1}{6} \delta x \delta_1 \frac{\delta_2 \mu_2 g}{\delta y}]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= [\mu_2 g + \frac{1}{6} \delta y \delta_2 \frac{\delta_1 \mu_1 f}{\delta x}]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.9)$$

combine to give again (6.2). Any linear combination of (6.4) and (6.9) is also equivalent. The following combination will appear in the next Section:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= [(I + \frac{1}{24} \delta_2^2) \mu_1 f + \frac{1}{8} \delta x \delta_1 \frac{\delta_2 \mu_2 g}{\delta y}]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= [(I + \frac{1}{24} \delta_1^2) \mu_2 g + \frac{1}{8} \delta y \delta_2 \frac{\delta_1 \mu_1 f}{\delta x}]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.10)$$

### 6.3 Euler solver on an irregular Cartesian mesh

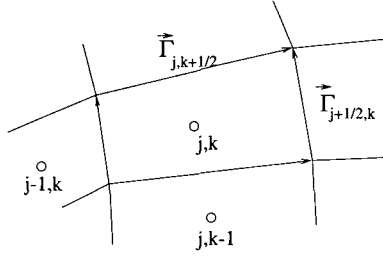
#### 6.3.1 Finite-volume definition of the accuracy order

We now consider the integral form of Eq. (6.1):

$$\frac{d}{dt} \int_{\Omega} w \, dx \, dy + \oint_{\partial\Omega} f \, dy - g \, dx = 0, \quad (6.11)$$

where  $\Omega$  is a bounded domain with boundary  $\partial\Omega$ . Let  $\Omega_{j,k}$  be a quadrangular cell of a structured mesh and  $\vec{\Gamma}_{j+\frac{1}{2},k}$ ,  $\vec{\Gamma}_{j,k+\frac{1}{2}}$  be the cell edges oriented as the increasing mesh indexes (see Fig.2). The cell area is denoted by  $|\Omega_{j,k}|$ , an edge length by  $|\Gamma_{j+\frac{1}{2},k}|$  and the largest edge length in the mesh by  $h$ .

Applied to the cell  $\Omega_{j,k}$ , the exact form (6.11) reads:



**Figure 2** Cell  $\Omega_{j,k}$ .

$$|\Omega_{j,k}|(\overline{w_t})_{j,k} + |\Gamma_{j+\frac{1}{2},k}| \overline{F}_{j+\frac{1}{2},k} - |\Gamma_{j-\frac{1}{2},k}| \overline{F}_{j-\frac{1}{2},k} \\ + |\Gamma_{j,k+\frac{1}{2}}| \overline{G}_{j,k+\frac{1}{2}} - |\Gamma_{j,k-\frac{1}{2}}| \overline{G}_{j,k-\frac{1}{2}} = 0 \quad (6.12)$$

where the bars denote exact averages:

$$(\overline{w_t})_{j,k} = \frac{1}{|\Omega_{j,k}|} \frac{d}{dt} \int_{\Omega_{j,k}} w \, dx dy$$

$$\overline{F}_{j+\frac{1}{2},k} = \frac{1}{|\Gamma_{j+\frac{1}{2},k}|} \int_{\vec{\Gamma}_{j+\frac{1}{2},k}} f \, dy - g \, dx, \quad \overline{G}_{j,k+\frac{1}{2}} = \frac{1}{|\Gamma_{j,k+\frac{1}{2}}|} \int_{\vec{\Gamma}_{j,k+\frac{1}{2}}} g \, dx - f \, dy.$$

For the spatial approximation of (6.12), we consider a finite-volume (FV) scheme of the form:

$$|\Omega_{j,k}|(\overline{w_t})_{j,k} + |\Gamma_{j+\frac{1}{2},k}| F_{j+\frac{1}{2},k} - |\Gamma_{j-\frac{1}{2},k}| F_{j-\frac{1}{2},k} \\ + |\Gamma_{j,k+\frac{1}{2}}| G_{j,k+\frac{1}{2}} - |\Gamma_{j,k-\frac{1}{2}}| G_{j,k-\frac{1}{2}} = 0 \quad (6.13)$$

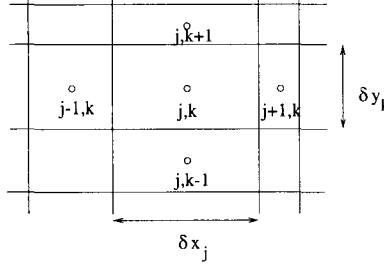
where  $F_{j+\frac{1}{2},k}$  and  $G_{j,k+\frac{1}{2}}$  are numerical fluxes calculated from values located at the centers of neighbouring cells. Let us recall here that the first developments of FV schemes in compressible CFD are due to MacCormack et al. in the early seventies (see [3] for instance).

Following Rezgui [4], Scheme (6.13) will be said to be *accurate at order  $p$  in the FV sense* if

$$F_{j+\frac{1}{2},k} = \overline{F}_{j+\frac{1}{2},k} + O(h^p) \\ G_{j,k+\frac{1}{2}} = \overline{G}_{j,k+\frac{1}{2}} + O(h^p), \quad (6.14)$$

for any smooth exact solution.

It should be noted that, for a general mesh, this accuracy requirement is easier to satisfy than the usual definition based on the truncation error of the whole scheme. A scheme that is accurate at order  $p$  in the FV sense is accurate at order  $p-1$  (at least) in the usual sense. However, the usual definition is not well suited to a FV method. Consider for instance a classical upwind scheme applied to a 1-D hyperbolic equation on an irregular grid. It is well known (see Turkel [5]) that this scheme is inconsistent in the usual sense when using

Figure 3 Cell  $\Omega_{j,k}$ .

a straightforward FV formulation. But it works rather well in practice and Sanders [6] proved its convergence with an appropriate norm.

In the present paper, we will show from various multidimensional numerical experiments that a third-order scheme in the FV sense produces an error (distance of the numerical solution to the exact one) that is really of order three for totally irregular Cartesian meshes.

### 6.3.2 Residual-based correction for the numerical flux

Let us now extend the third-order scheme (6.4)-(6.7) on a general Cartesian mesh. The current cell  $\Omega_{j,k}$  is a rectangle centered at  $(x_j, y_k)$  with edge lengths  $\delta x_j$  and  $\delta y_k$  (see Fig.3) and  $h = \max(\max_j \delta x_j, \max_k \delta y_k)$ .

A straightforward extension (the usual one) consists in using formulae (6.4-6.7) with variable space steps, that is in (6.5) and in the definition of  $\Phi_1, \Phi_2$  :

$$\begin{aligned} (\delta x)_{j+\frac{1}{2},k} &= \frac{1}{2}(\delta x_j + \delta x_{j+1}), & (\delta y)_{j+\frac{1}{2},k} &= \delta y_k \\ (\delta x)_{j,k+\frac{1}{2}} &= \delta x_j, & (\delta y)_{j,k+\frac{1}{2}} &= \frac{1}{2}(\delta y_k + \delta y_{k+1}) \end{aligned}$$

and in (6.7):

$$(\delta x)_{j,k} = \delta x_j, \quad (\delta y)_{j,k} = \delta y_k.$$

This simple extension will be called the *non-weighted* version of the scheme. For an irregular Cartesian mesh, it is only first-order accurate in the FV sense, *i.e.* it satisfies the condition (6.14) with  $p = 1$ .

We now construct numerical fluxes  $F$  and  $G$  satisfying the condition (6.14) with  $p = 3$ . On a general Cartesian mesh, the expressions for the exact fluxes become:

$$\bar{F}_{j+\frac{1}{2},k} = \frac{1}{\delta y_k} \int_{\bar{\Gamma}_{j+\frac{1}{2},k}} f dy, \quad \bar{G}_{j,k+\frac{1}{2}} = \frac{1}{\delta x_j} \int_{\bar{\Gamma}_{j,k+\frac{1}{2}}} g dx. \quad (6.15)$$

Performing a Taylor expansion in the  $y$ -direction, we obtain:

$$\bar{F}_{j+\frac{1}{2},k} = f_{j+\frac{1}{2},k} + \frac{\delta y_k^2}{24} (f_{yy})_{j+\frac{1}{2},k} + O(h^4), \quad (6.16)$$

where  $f_{j+\frac{1}{2},k}$  denotes the exact flux  $f$  at the middle of the edge  $\bar{\Gamma}_{j+\frac{1}{2},k}$ . To complete the discretization, we have to approximate  $f$  and  $f_{yy}$  at  $(j+\frac{1}{2},k)$  using values known at the six cell-centers  $(j,k)$ ,  $(j+1,k)$ ,  $(j,k\pm 1)$ ,  $(j+1,k\pm 1)$ . On a general Cartesian mesh, let us define the 2-point averages and numerical derivative at an edge middle:

$$(\mu_1 f)_{j+\frac{1}{2},k} = \frac{1}{2}(f_{j,k} + f_{j+1,k}), \quad (\tilde{\mu}_1 f)_{j+\frac{1}{2},k} = (1 - \theta_j)f_{j,k} + \theta_j f_{j+1,k}$$

$$(\nabla_1 f)_{j+\frac{1}{2},k} = \frac{f_{j+1,k} - f_{j,k}}{x_{j+1} - x_j} = \frac{2(\delta_1 f)_{j+\frac{1}{2},k}}{\delta x_{j+1} + \delta x_j}$$

where  $\theta_j = (x_{j+\frac{1}{2}} - x_j)/(x_{j+1} - x_j) = \delta x_j/(\delta x_j + \delta x_{j+1})$ . Operators  $\mu_1$  and  $\nabla_1$  are 1st-order accurate and  $\tilde{\mu}_1$  is 2nd-order accurate. In the  $y$ -direction, operators  $\mu_2$ ,  $\tilde{\mu}_2$  and  $\nabla_2$  are similarly defined. Let us also introduce 3-point numerical derivatives at a cell-center, using actual cell-center locations: at  $(j,k)$ ,  $\tilde{\nabla}_1$  is a linear combination of  $(f_{j+p,k})_{p=-1,0,1}$  approximating  $f_x$  at 2nd-order,  $\nabla_1^2$  is a similar combination approximating  $f_{xx}$  at 1st-order and  $\tilde{\nabla}_2$ ,  $\nabla_2^2$  are analogous operators in the  $y$ -direction.

Let us come back to the approximation of  $f_{j+\frac{1}{2},k}$  in (6.16). We first consider the classical average  $\mu_1$ . A Taylor expansion gives:

$$(\mu_1 f)_{j+\frac{1}{2},k} = f_{j+\frac{1}{2},k} + \frac{1}{4}(\delta x_{j+1} - \delta x_j)(f_x)_{j+\frac{1}{2},k} + \frac{1}{16}(\delta x_{j+1}^2 + \delta x_j^2)(f_{xx})_{j+\frac{1}{2},k} + O(h^3). \quad (6.17)$$

In the residual-based approach, the idea is not to cancel the error directly but to add new terms in order to recover derivatives of the residual  $r$  ( $r = f_x + g_y$ ) in the truncation error. So, for the first error term in (6.17), involving  $f_x$ , we add a discretization of a  $g_y$ -term at order 2 and for the second error term in (6.17), involving  $f_{xx}$ , we add a discretization of a  $g_{xy}$ -term at order 1. Thus, we consider the modified approximation to  $f_{j+\frac{1}{2},k}$ :

$$f_{j+\frac{1}{2},k}^* = [\mu_1 f + \frac{1}{4}(\delta x_{j+1} - \delta x_j)\tilde{\mu}_1 \tilde{\nabla}_2 g + \frac{1}{16}(\delta x_{j+1}^2 + \delta x_j^2)\nabla_1 \tilde{\nabla}_2 g]_{j+\frac{1}{2},k}.$$

Clearly, this operator can be expanded as:

$$f_{j+\frac{1}{2},k}^* = f_{j+\frac{1}{2},k} + \frac{1}{4}(\delta x_{j+1} - \delta x_j)r_{j+\frac{1}{2},k} + \frac{1}{16}(\delta x_{j+1}^2 + \delta x_j^2)(r_x)_{j+\frac{1}{2},k} + O(h^3).$$

We now turn to the second term in (6.16). Since first-order accuracy is sufficient here, we approximate  $(f_{yy})_{j+\frac{1}{2},k}$  with the operator  $\nabla_2^2 \mu_1$ . Finally,



the non-dissipative part of the numerical flux can be written as:

$$\overset{\circ}{F}_{j+\frac{1}{2},k} = f_{j+\frac{1}{2},k}^* + \frac{\delta y_k^2}{24} (\nabla_2^2 \mu_1 f)_{j+\frac{1}{2},k},$$

and a similar expression can be found for  $\overset{\circ}{G}$ .

Thus, the non-dissipative numerical fluxes read:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= [(I + \frac{\delta y_k^2}{24} \nabla_2^2) \mu_1 f + \frac{1}{4} (\delta x_{j+1} - \delta x_j) \tilde{\mu}_1 \tilde{\nabla}_2 g \\ &\quad + \frac{1}{16} (\delta x_{j+1}^2 + \delta x_j^2) \nabla_1 \tilde{\nabla}_2 g]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= [(I + \frac{\delta x_j^2}{24} \nabla_1^2) \mu_2 g + \frac{1}{4} (\delta y_{k+1} - \delta y_k) \tilde{\mu}_2 \tilde{\nabla}_1 f \\ &\quad + \frac{1}{16} (\delta y_{k+1}^2 + \delta y_k^2) \nabla_2 \tilde{\nabla}_1 f]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.18)$$

By construction, for a smooth exact steady solution ( $r = 0$ ), we obtain:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= \bar{F}_{j+\frac{1}{2},k}^n + O(h^3) \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= \bar{G}_{j,k+\frac{1}{2}}^n + O(h^3) \end{aligned}$$

where  $\bar{F}$  and  $\bar{G}$  are the exact values defined by (6.15).

Remarks:

- a) For a regular Cartesian mesh, the fluxes (6.18) give back the form (6.10) exactly.
- b) The construction of  $\overset{\circ}{F}$  could also be done from a first evaluation of  $f_{j+\frac{1}{2},k}$  in (6.16) using  $\tilde{\mu}_1$  instead of  $\mu_1$ . Since  $\tilde{\mu}_1$  is second-order accurate, this would save a correction term and lead to:

$$\overset{\circ}{F}_{j+\frac{1}{2},k}^n = [\tilde{\mu}_1 f + \frac{\delta y_k^2}{24} \nabla_2^2 \mu_1 f + \frac{1}{8} \delta x_{j+1} \delta x_j \nabla_1 \tilde{\nabla}_2 g]_{j+\frac{1}{2},k}^n$$

and a similar expression for  $\overset{\circ}{G}$ . We did not choose this approach because it is more difficult to generalize to the Navier-Stokes equations than the one detailed above.

We now consider the discretization of the residual-based dissipation on a general Cartesian mesh. Since this term is of first-order, its derivatives have only to be approximated with a second-order accuracy. This is achieved using again a residual-based correction. More precisely, the numerical dissipation is calculated as follows:

$$\begin{aligned} (d_1)_{j+\frac{1}{2},k}^n &= \frac{\delta x_{j+1} + \delta x_j}{4} [\Phi_1 (\nabla_1 f + \mu_1 \tilde{\nabla}_2 g)]_{j+\frac{1}{2},k}^n \\ (d_2)_{j,k+\frac{1}{2}}^n &= \frac{\delta y_{k+1} + \delta y_k}{4} [\Phi_2 (\mu_2 \tilde{\nabla}_1 f + \nabla_2 g)]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.19)$$

To explain how this works, let us consider for instance  $d_1$ . Of course, the operator  $\nabla_1$  is only first-order accurate:

$$(\nabla_1 f)_{j+\frac{1}{2},k} = (f_x)_{j+\frac{1}{2},k} + \frac{1}{4}(\delta x_{j+1} - \delta x_j)(f_{xx})_{j+\frac{1}{2},k} + O(h^2),$$

but the same is true for  $\mu_1$  and we get:

$$(\mu_1 \tilde{\nabla}_2 g)_{j+\frac{1}{2},k} = (g_y)_{j+\frac{1}{2},k} + \frac{1}{4}(\delta x_{j+1} - \delta x_j)(g_{yx})_{j+\frac{1}{2},k} + O(h^2),$$

so that

$$(d_1)_{j+\frac{1}{2},k}^n = \frac{\delta x_{j+1} + \delta x_j}{4} (\Phi_1)_{j+\frac{1}{2},k}^n \left[ r + \frac{1}{4}(\delta x_{j+1} - \delta x_j)r_x \right]_{j+\frac{1}{2},k} + O(h^3).$$

For a smooth exact steady solution, we obtain:  $(d_1)_{j+\frac{1}{2},k}^n = O(h^3)$ .

The total numerical fluxes  $F$  and  $G$  are still defined by (6.6) and the new cell values are computed according to the finite-volume scheme, *i.e.* for the explicit version:

$$(\delta x_j \delta y_k) \frac{\Delta w_{j,k}^{expl}}{\Delta t} + \delta y_k F_{j+\frac{1}{2},k}^n - \delta y_k F_{j-\frac{1}{2},k}^n + \delta x_j G_{j,k+\frac{1}{2}}^n - \delta x_j G_{j,k-\frac{1}{2}}^n = 0$$

which can also be written as:

$$\frac{\Delta w_{j,k}^{expl}}{\Delta t} = -\frac{(\delta_1 F)_{j,k}^n}{\delta x_j} - \frac{(\delta_2 G)_{j,k}^n}{\delta y_k}. \quad (6.20)$$

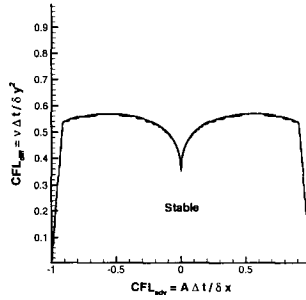
This scheme is third-order accurate at steady-state in the FV sense, even on a totally irregular Cartesian mesh. Its  $3 \times 3$  point stencil is not larger than required when the mesh is regular. In what follows, it will be referred to as *weighted* for comparison with the *non-weighted* version described at the beginning of Section 6.3.2.

## 6.4 Navier-Stokes solver

### 6.4.1 Third-order scheme on a regular Cartesian mesh

Residual-based compactness can be extended to the Navier-Stokes equations by adding the viscous terms in the residual, including the residual in the numerical dissipation. For the sake of brevity, we present here the method for the simplified system:

$$w_t + f_x + g_y = \nu w_{yy} \quad (6.21)$$



**Figure 4** 2-D stability for the explicit 3rd-order scheme (viscous problem).

where  $f = f(w)$  and  $g = g(w)$  are the Euler fluxes and  $\nu$  a positive constant coefficient. On a regular Cartesian mesh, the scheme can be written as:

a) Numerical fluxes without numerical dissipation:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= \left[ \left( I + \frac{1}{6} \delta_2^2 \right) \mu_1 f \right]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= \left[ \left( I + \frac{1}{6} \delta_1^2 \right) (\mu_2 g - \nu \frac{\delta_2 w}{\delta y}) - \frac{1}{12} \nu \frac{\delta_2^3 w}{\delta y} \right]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.22)$$

b) Numerical dissipation:

$$\begin{aligned} (d_1)_{j+\frac{1}{2},k}^n &= \frac{\delta x}{2} \left[ \Phi_1 \left( \frac{\delta_1 f}{\delta x} + \frac{\delta_2 \mu_1 \mu_2 g}{\delta y} - \nu \mu_1 \frac{\delta_2^2 w}{\delta y^2} \right) \right]_{j+\frac{1}{2},k}^n \\ (d_2)_{j,k+\frac{1}{2}}^n &= \frac{\delta y}{2} \left[ \Phi_2 \left( \frac{\delta_1 \mu_1 \mu_2 f}{\delta x} + \frac{\delta_2 g}{\delta y} - \nu \mu_2 \frac{\delta_2^2 w}{\delta y^2} \right) \right]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.23)$$

The total numerical fluxes and new cell values are still defined by (6.6) and (6.7). This scheme involves  $(3 \times 3) + 2 = 11$  points (13 points for the complete Navier-Stokes equations). For smooth solution of Eq. (6.21), the scheme truncation error is:

$$\epsilon = w_t + r + \frac{\delta x^2}{6} r_{xx} + \frac{\delta y^2}{6} r_{yy} - \frac{\delta x}{2} (\Phi_1 r)_x - \frac{\delta y}{2} (\Phi_2 r)_y + O(h^3),$$

where  $r = f_x + g_y - \nu w_{yy}$ . Thus, third-order accuracy is obtained at steady state.

The stability domain is shown on Fig. 4 for the scalar equation  $w_t + Aw_x = \nu w_{yy}$ . An implicit stage can be added to the scheme to obtain unconditional linear stability (see [2]).

### 6.4.2 Third-order scheme on an irregular Cartesian mesh

Proceeding as in Section 6.3.2, we can extend the Navier-Stokes solver to irregular Cartesian meshes. The scheme is now defined from:

a) Numerical fluxes without numerical dissipation:

$$\begin{aligned} \overset{\circ}{F}_{j+\frac{1}{2},k}^n &= [(I + \frac{\delta y_k^2}{24} \nabla_2^2) \mu_1 f + \frac{1}{4} (\delta x_{j+1} - \delta x_j) \tilde{\mu}_1 (\tilde{\nabla}_2 g - \nu \tilde{\nabla}_2^2 w) \\ &\quad + \frac{1}{16} (\delta x_{j+1}^2 + \delta x_j^2) \nabla_1 (\tilde{\nabla}_2 g - \nu \tilde{\nabla}_2^2 w)]_{j+\frac{1}{2},k}^n \\ \overset{\circ}{G}_{j,k+\frac{1}{2}}^n &= [(I + \frac{\delta x_k^2}{24} \nabla_1^2) (\mu_2 g - \nu \nabla_2 w) + \frac{1}{4} (\delta y_{k+1} - \delta y_k) \tilde{\mu}_2 \tilde{\nabla}_1 f \\ &\quad + \frac{1}{16} (\delta y_{k+1}^2 + \delta y_k^2) \nabla_2 \tilde{\nabla}_1 f - \frac{\nu}{48} (\delta y_{k+1} + \delta y_k)^2 \nabla_2^3 w]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.24)$$

where  $(\tilde{\nabla}_2^2 w)_{j,k}$  denotes a second-order approximation of  $w_{yy}$  at  $(j, k)$  computed from values at cell centers  $(j, k \pm 2)$ ,  $(j, k \pm 1)$  and  $(j, k)$ , and  $(\nabla_2^3 w)_{j,k+\frac{1}{2}}$  is the first-order approximation of  $w_{yyy}$  at  $(j, k + \frac{1}{2})$  deduced from  $(j, k + 2)$ ,  $(j, k + 1)$ ,  $(j, k)$  and  $(j, k - 1)$ .

b) Numerical dissipation:

$$\begin{aligned} (d_1)_{j+\frac{1}{2},k}^n &= \frac{\delta x_{j+1} + \delta x_j}{4} [\Phi_1 (\nabla_1 f + \mu_1 \tilde{\nabla}_2 g - \nu \mu_1 \tilde{\nabla}_2^2 w)]_{j+\frac{1}{2},k}^n \\ (d_2)_{j,k+\frac{1}{2}}^n &= \frac{\delta y_{k+1} + \delta y_k}{4} [\Phi_2 (\mu_2 \tilde{\nabla}_1 f + \nabla_2 g - \nu \tilde{\nabla}_2^2 w)]_{j,k+\frac{1}{2}}^n \end{aligned} \quad (6.25)$$

where  $(\tilde{\nabla}_2^2 w)_{j,k+\frac{1}{2}}$  in  $d_2$  denotes a first-order approximation to  $w_{yy}$  at  $(j, k + \frac{1}{2})$  computed from values at cell centers  $(j, k - 1)$ ,  $(j, k)$ ,  $(j, k + 1)$  and  $(j, k + 2)$ , such that its first-order error combines with the one coming from the inviscid terms to yield a y-derivative of the residual, namely:

$$(\tilde{\nabla}_2^2 w)_{j,k+\frac{1}{2}} = (w_{yy})_{j,k+\frac{1}{2}} + \frac{1}{4} (\delta y_{k+1} - \delta y_k) (w_{yyy})_{j,k+\frac{1}{2}} + O(h^2).$$

Note that, when  $\nu = 0$ , formulae (6.24) and (6.25) reduce respectively to (6.18) and (6.19). Besides, when the mesh is regular, formula (6.25) reduces to (6.23).

## 6.5 Numerical experiments

In this section we apply the non-weighted and weighted FV versions of the residual-based scheme defined in section 6.3.2 for the Euler equations and in section 6.4.2 for the Navier-Stokes equations to some test-cases. Inviscid and viscous problems with known analytical solutions are first considered (test-cases 6.5.2 and 6.5.3 are proposed in [7]). They will allow to demonstrate,

through the computation of the  $L_2$ -norm of the error between the exact and the numerical solutions, that the weighted version has a third-order error even on totally irregular Cartesian meshes, while the non-weighted version incurs a severe loss of accuracy on the same meshes. Both versions will also be compared for problems involving discontinuities.

### 6.5.1 The three series of meshes

Three types of Cartesian meshes are used in the following computations: uniform (Fig. 5 (a)), geometrically stretched (Fig. 5 (b)) with a stretching factor of 1.11, and randomly perturbed (Fig. 5 (c)). In this latter case, we start from a uniform grid and we apply to each cell a random scaling factor in the  $x$  and  $y$  direction; the grid is then rescaled to  $[0, 1]^2$  and is such that no general relation exists between neighbouring cells. For each type of mesh, a  $39 \times 39$  grid is first generated, then a  $78 \times 78$  and a  $156 \times 156$  grid are deduced from the coarse one by dividing each cell by 2 in each space direction. Each series of 3 grids allows the computation of an accuracy order. Note that for the series of randomly perturbed meshes, the values of the characteristic mesh size  $h$  is likely to change from one series of computation to the other.

### 6.5.2 2-D rotational advection

This test-case consists of the rotational advection of a smooth Gaussian profile over the square domain  $[0, 1]^2$ . More precisely, we look at the steady solution of the following problem

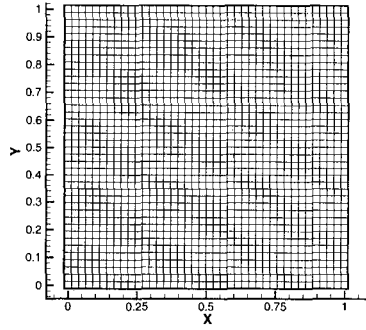
$$\left\{ \begin{array}{ll} \frac{\partial w}{\partial t} + y \frac{\partial w}{\partial x} + (1-x) \frac{\partial w}{\partial y} = 0, & (x, y) \in ]0, 1[^2 \quad t > 0 \\ w(x, y, 0) = 0, & (x, y) \in ]0, 1[^2 \\ w(x, 0, t) = e^{[-50(x-0.5)^2]}, & x \in [0, 1], \quad t \geq 0 \\ w(x, 1, t) = 0, & x \in [0, 1], \quad t \geq 0 \\ w(0, y, t) = 0, & y \in [0, 1], \quad t \geq 0 \end{array} \right.$$

the exact steady-state solution of which is given by

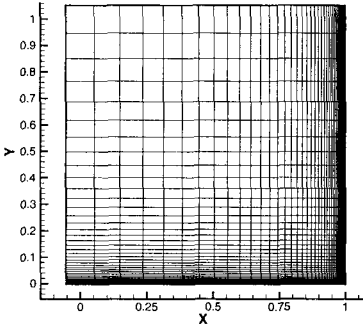
$$w_{exact}(x, y) = e^{[-50(0.5 - (1 - \sqrt{(1-x)^2 + y^2}))^2]}, \quad (x, y) \in [0, 1]^2$$

which means the initial distribution of  $w$  along  $y = 0$  is conserved on any circle of center  $(1, 0)$ . Performing first a series of computations on uniform meshes, we observe the non-weighted and weighted schemes are both third-order accurate and yield actually the same error (see Fig. 6 (a)). This was expected since both versions lead to the same scheme on a uniform grid.

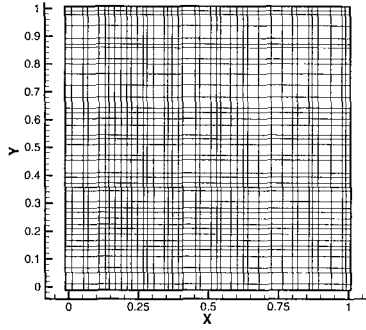
Next, calculations on non-regular grids (either stretched or randomly perturbed) demonstrate the severe loss of accuracy incurred by the non-weighted version, the error order of which drops to about 1.4, while the



(a)



(b)



(c)

**Figure 5** Cartesian mesh with  $39 \times 39$  cells: (a) Uniform (b) Geometrically stretched (c) Randomly perturbed

weighted version, which satisfies the FV accuracy criterion (6.14) with  $p = 3$  has a third-order error (see Fig. 6 (b) and (c)). This is also clearly visible on the solution isovalues (Fig. 7) : while the weighted solution on the random mesh is superimposed on the exact one, the non-weighted solution exhibits strong perturbations caused by mesh irregularities. The convergence history for this case is monitored through the  $L_2$ -norm of  $\Delta w / \Delta t$  over the domain and is plotted on Fig.6 (d); both versions share the same non-weighted implicit stage (6.8). One can notice the weighted scheme yields faster convergence to steady-state than the non-weighted scheme.

### 6.5.3 3-D helicoidal advection

This test-case is a 3-D extension of the previous problem ; we solve  $w_t + z w_x + 0.1w_y - xw_z = 0$  on the domain  $-1 \leq x \leq 0$ ,  $0 \leq y \leq 1$ ,  $0 \leq z \leq 1$ , with appropriate initial and boundary conditions. More precisely, we set a smooth initial condition in the  $z = 0$  plane (2-D Gaussian distribution in  $x$  and  $y$ ), that is both rotated around the  $y$  axis and advected along this same axis (see Fig.8(a)). Computations were performed using series of  $32 \times 32 \times 32$ ,  $64 \times 64 \times 64$  and  $128 \times 128 \times 128$  uniform, geometrically stretched and randomly perturbed Cartesian meshes. However, from now on and for the sake of brevity, we will restrict our presentation of computed orders of accuracy and numerical solutions to the results obtained on the randomly perturbed Cartesian meshes; indeed, it was observed that this latter series made the effects of mesh irregularities even more clearly visible than the geometrically stretched series.

It can be observed on Fig.8(b) that the FV weighted scheme remains truly third-order accurate on randomly perturbed meshes whereas the non-weighted version's accuracy order drops to 1.79. This accuracy loss is made visible on the isovalues of the numerical solutions in the outflow plane  $x = 0$ , when compared with those of the exact solution (see Fig. 9): the weighted solution is free of all the distortions present in the non-weighted one.

### 6.5.4 2-D Poiseuille flow model

We now consider the 2-D viscous problem defined by the following equations :

$$\left\{ \begin{array}{ll} \frac{\partial w}{\partial t} + a \frac{\partial w}{\partial x} = \nu \frac{\partial^2 w}{\partial y^2}, & (x, y) \in ]0, 1[^2, \quad t > 0 \\ w(x, y, 0) = 1, & (x, y) \in ]0, 1[^2 \\ w(x, 0, t) = w(x, L, t) = 0, & x \in ]0, 1[, \quad t \geq 0 \\ w(0, y, t) = \sin(\pi \frac{y}{L}), & y \in ]0, 1[, \quad t \geq 0 \end{array} \right.$$

the exact steady-state solution of which is given by

$$w_{exact}(x, y) = e^{[-\pi^2 \frac{\nu}{a} x]} \sin(\pi y), \quad (x, y) \in [0, 1]^2$$

This problem may be viewed as modelling a Poiseuille flow: an initial profile, prescribed at inflow  $y = 0$ , is advected and diffused between two solid walls on which its value is fixed to zero.

Here again, we observe that the weighted version of the residual-based scheme makes it possible to preserve a genuine third-order error on randomly perturbed Cartesian meshes while the usual non-weighted version is reduced to a first-order accurate method (see Fig. 10(a)). The much better accuracy provided by the weighted version is illustrated on Fig.11, where the isovalues of the numerical solutions are plotted: those of the weighted version are

superimposed on the exact solution, while the non-weighted version displays strong distortions caused by the grid irregularities. Note also that the convergence to steady-state of the implicit weighted version is again faster than that of the implicit non-weighted version (Fig. 10(b)).

### 6.5.5 2-D Burgers problem

This test-case deals with the following non-linear problem:

$$\left\{ \begin{array}{ll} \frac{\partial w}{\partial t} + \frac{\partial(\frac{w^2}{2})}{\partial x} + \frac{\partial w}{\partial y} = 0, & (x, y) \in ]0, 1[^2, \quad t > 0 \\ w(x, y, 0) = (1 - x)w_l + xw_r & (x, y) \in [0, 1]^2 \\ w(x, 0, t) = (1 - x)w_l + xw_r, & x \in [0, 1], \quad t \geq 0 \\ w(0, y, t) = w_l, & y \in [0, 1], \quad t \geq 0 \\ w(1, y, t) = w_r, & y \in [0, 1], \quad t \geq 0 \end{array} \right.$$

where the left and right states  $w_l, w_r$  are chosen so as to produce a compression resolving into a normal (case 1:  $w_l = 1, w_r = -1$ ) or oblique (case 2:  $w_l = 1.5, w_r = -0.5$ ) shock wave.

We observe that, in both cases, the weighted scheme produces straight isolines in the compression region on a randomly disturbed grid while the characteristics provided by the non-weighted version are more perturbed due to the grid irregularities (see Fig. 12 and 13 (a)-(b)-(c)). In case 1, where the shock is aligned with the grid, the weighted solution is very close to the exact one and displays almost no oscillations while the non-weighted solution is slightly oscillatory (see Fig. 12 (d)). In case 2, where the shock is no longer aligned with the grid, both versions display some oscillations but those produced by the weighted scheme are in the present case much weaker than the ones displayed by the non-weighted scheme (see Fig. 13 (d)). Note that the mesh being randomly perturbed the amplitude of these oscillations would not necessarily be the same for another run. However, it was concluded from the many calculations performed on this test-case that the weighted scheme was systematically less oscillatory than the non-weighted one.

### 6.5.6 2-D compressible jet interaction

This problem [8] consists of the interaction of two horizontal, supersonic jets. The upper stream, defined by  $M = 4, \rho = 0.5, p = 0.25$ , and the lower stream defined by  $M = 2.4, \rho = 1, p = 1$  are suddenly brought into contact at  $(x = 0, y = 1/2)$  and their interaction is studied in the domain  $[0, 1]^2$ . This interaction produces an expansion fan and a shock wave propagating respectively in the high pressure and low pressure region, as well as a contact discontinuity resulting of the different densities and velocities behind the two previous waves.



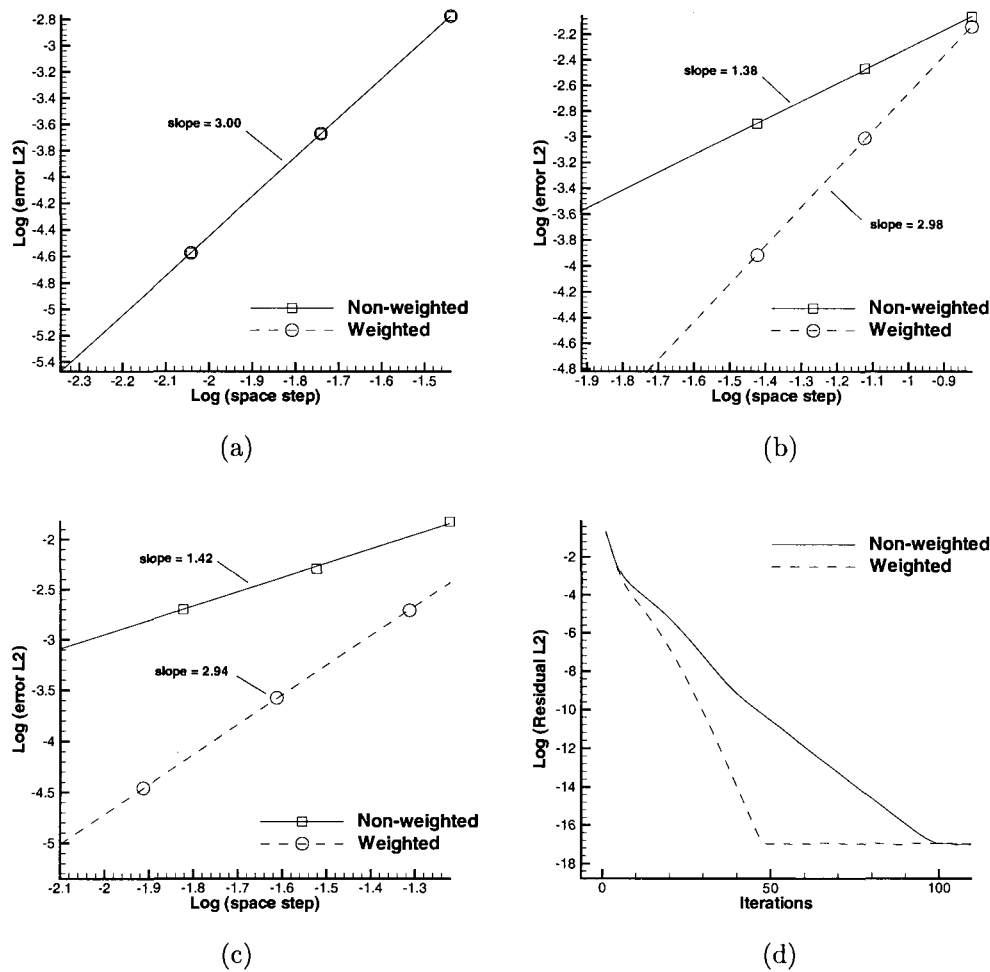
It appears from the Mach contours of the non-weighted and weighted solutions on a randomly perturbed grid that the latter gives consistently a better representation of the waves present in the flow (see Fig.14). It is clear from the Mach number distributions along the outflow boundary (see Fig.15) that the weighted scheme produces sharper shear layer and shock (though slightly oscillatory) as well as an expansion fan closer to the exact solution; the plateaus between the waves are also better represented.

## 6.6 Conclusion

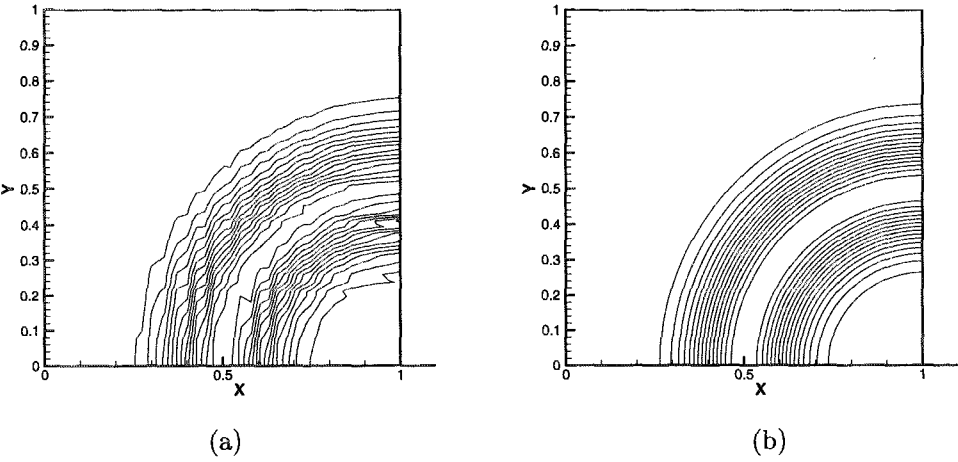
A method has been presented to devise a third-order accurate scheme that retains its order of accuracy without any assumption on the mesh smoothness. Following [4], the idea was, in a FV framework, to approach the exact flux on a face of the control volume at third-order; this has been achieved in a compact way by making use of residual-based corrections, on irregular Cartesian meshes. For inviscid and viscous multidimensional problems, a third-order actual error has been obtained even on totally irregular Cartesian meshes. Ongoing developments of this work include the extension of these ideas to general irregular structured meshes.

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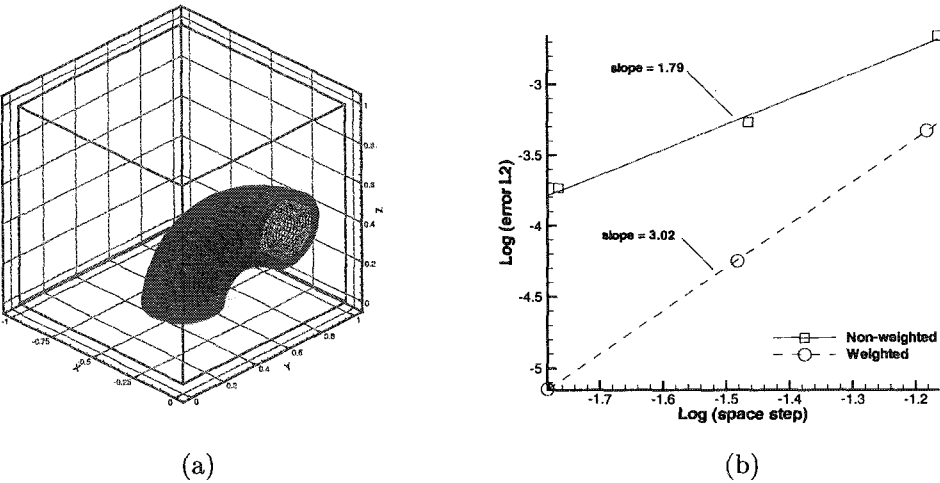
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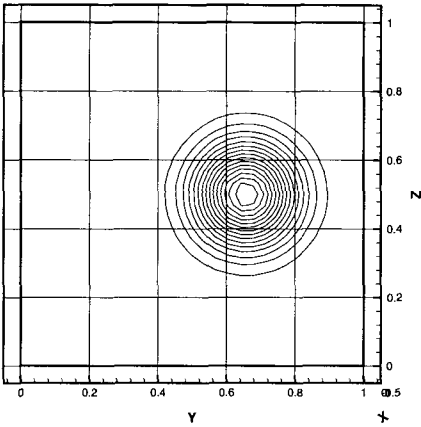
**Figure 6** Rotational advection. Computed error orders on (a) a uniform mesh (b) a stretched mesh (c) a random mesh (d) Convergence to steady-state on a random  $78 \times 78$  grid



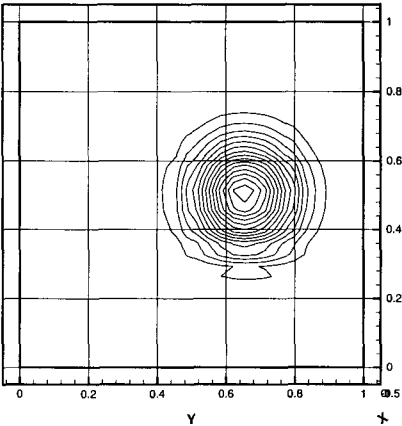
**Figure 7** Isovalues of  $w(x,y)$  on a random  $39 \times 39$  grid (a) non-weighted computation (b) weighted computation



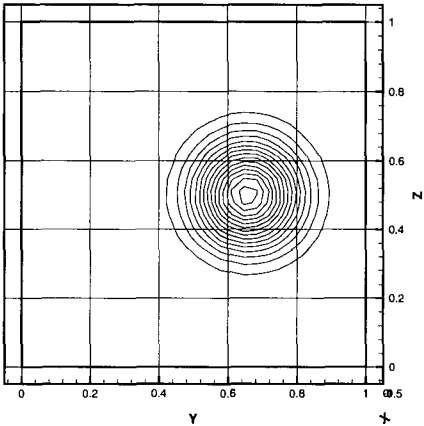
**Figure 8** (a) Exact solution (b) Computed error orders



(a)

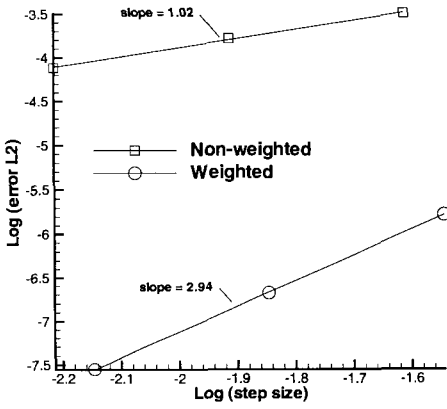


(b)

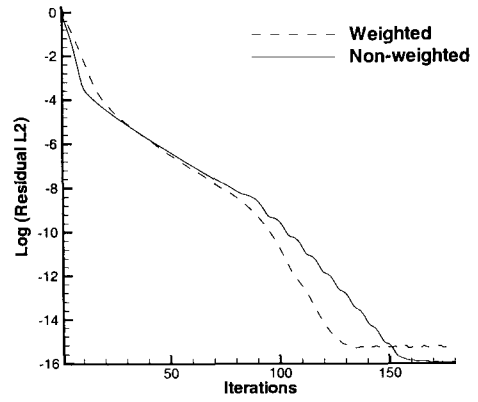


(c)

**Figure 9** Outflow on a random  $32 \times 32 \times 32$  grid (a) Exact solution (b) non-weighted scheme (c) weighted scheme

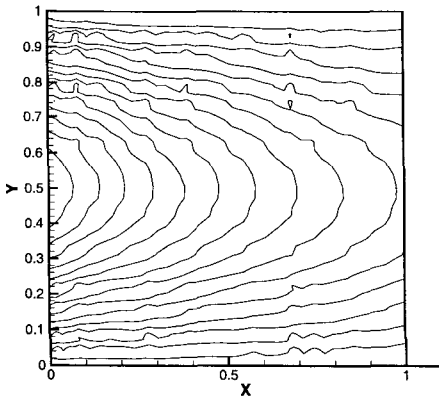


(a)

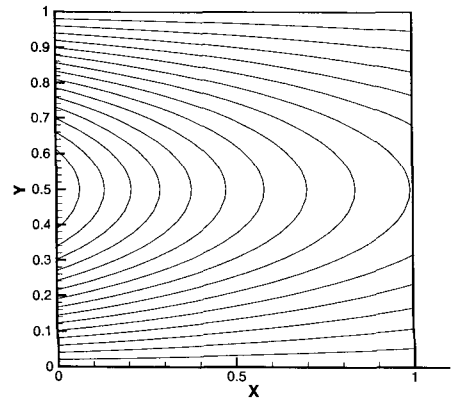


(b)

**Figure 10** 2-D Poiseuille flow model. (a) Computed error orders (b) Convergence to steady-state on a  $78 \times 78$  grid

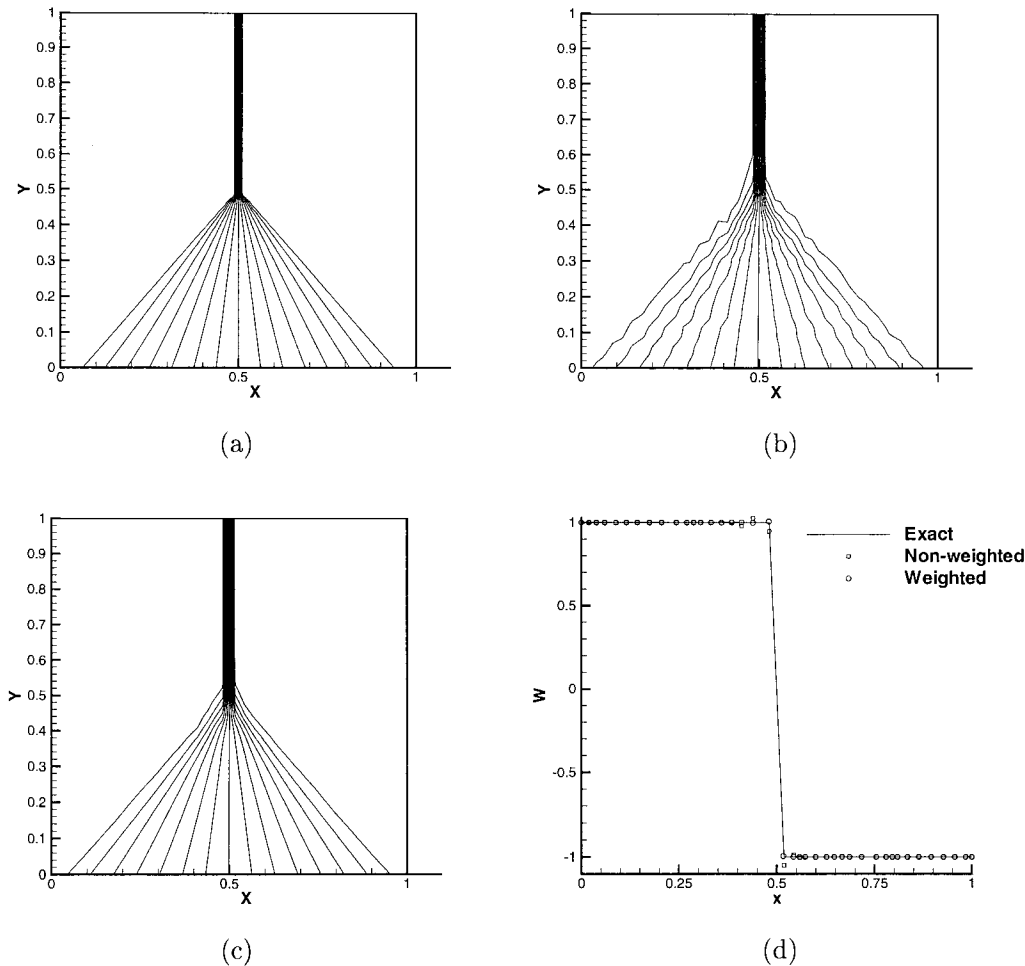


(a)

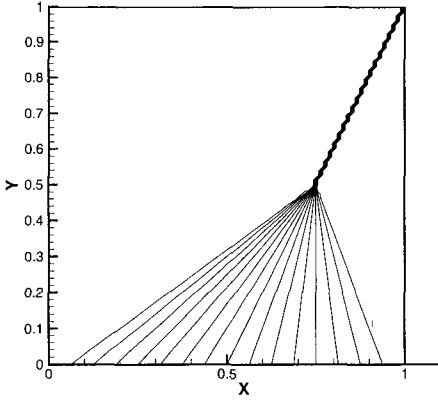


(b)

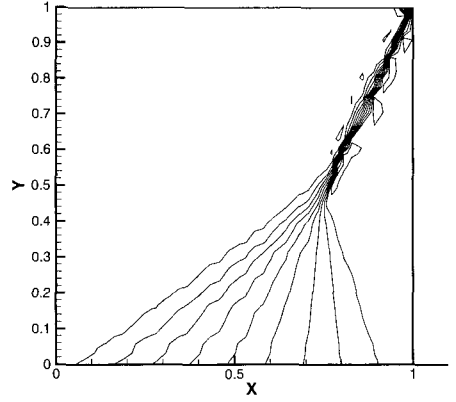
**Figure 11** Isovalues of  $w(x, y)$  on a random  $39 \times 39$  grid (a) non-weighted computation (b) weighted computation



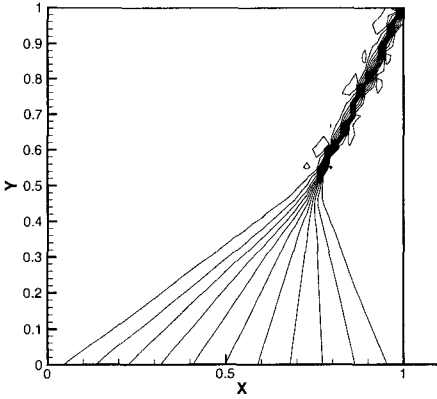
**Figure 12** Isovalues (case1) on a random  $39 \times 39$  grid (a) Exact solution, (b) non-weighted solution (c) weighted solution (d) Solution at  $y = 0.7$



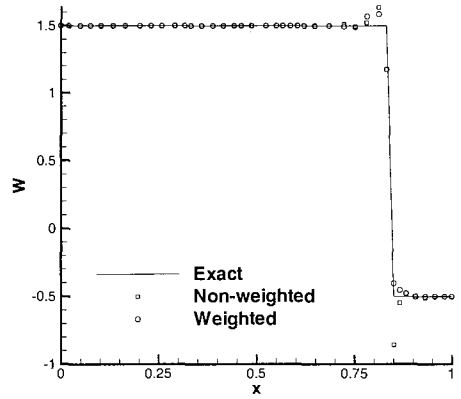
(a)



(b)

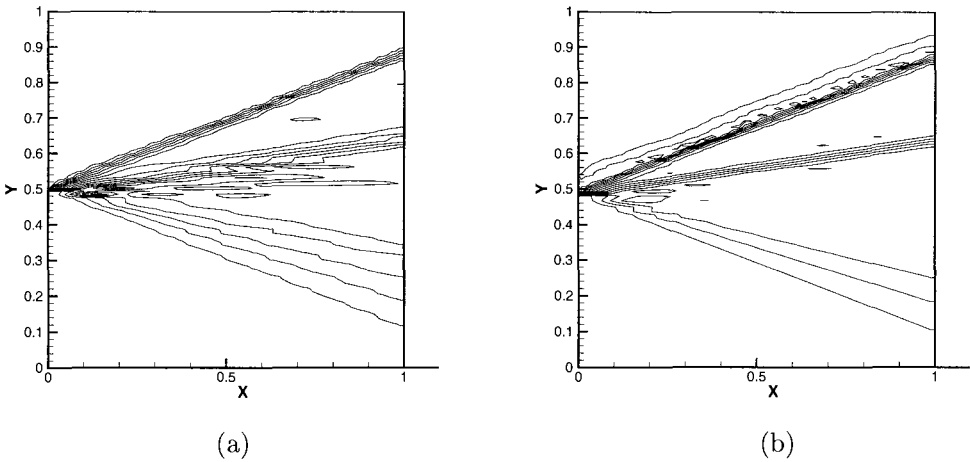


(c)

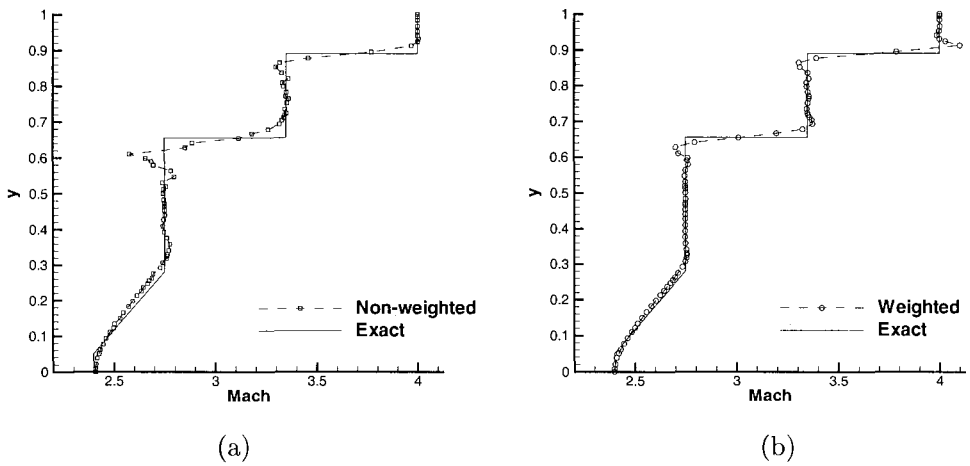


(d)

**Figure 13** Isovalues (case 2) on a random  $39 \times 39$  grid (a) Exact solution, (b) non-weighted solution (c) weighted solution (d) Solution at  $y = 0.7$



**Figure 14** Mach contours on a random  $78 \times 78$  grid (a) non-weighted solution (b) weighted solution



**Figure 15** Mach profile along the outflow boundary  $x = 1$ , on a random  $78 \times 78$  grid (a) non-weighted (b) weighted



# Future directions for computing compressible flows: higher-order centering vs multidimensional upwinding

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## 7.1 Introduction

In the last decades computer performance have improved dramatically with respect to both speed and memory size, so that the relative cost of a given computation has been reduced by approximately a factor of ten every ten years [1]. At the same time, Computational Fluid Dynamics (CFD) has experienced an exponential growth, so that the design and development of modern airplanes, advanced turbomachinery and internal combustion engines have changed dramatically. In fact, it is now possible to compute a very complex flow field (i.e., that around an entire airplane or inside one or more blade passages of a turbomachinery), using millions of computational cells, within hours of CPU time. As a consequence, the still necessary, but very costly, experiments are limited to the final design choices, after performing all preliminary designs by very fast and cheap computer runs. Nowadays, CFD codes for turbomachinery applications are based on numerical methods

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originally derived for aerodynamic applications. They solve the steady-state Reynolds averaged compressible Navier–Stokes equations by means of time marching explicit (Runge–Kutta) [2, 3, 4] or implicit (approximate factorization [5], line relaxation [6]) schemes and their convergence rate is accelerated by various techniques, such as local time stepping, implicit residual smoothing (for the case of explicit schemes), multigrid, etc. [2, 3, 4, 7]. As far as the space discretization is concerned, finite volume (or element) methods are almost universally applied to the conservation-law form of the equations, via a conservative discretization, so as to correctly capture flow discontinuities such as shocks and contact surfaces. The artificial dissipation required to avoid spurious oscillations is either added to the scheme, in case the advection terms in the equations are discretized using centred differences [2], or is “naturally” contained in the scheme itself, if an “upwind” discretization is used for such terms [8, 9, 10, 11, 12]. Recently, these methods have been extended to treat time dependent problems by suitable techniques, such as the dual time stepping [13]. Finally, in order to solve flows of engineering interest, the aforementioned methods employ turbulence models of increasing complexity (algebraic models [14], differential ones [15, 16, 17, 18], large eddy simulation [19]). To date, in fact, it is still unfeasible, for practical problems, to resolve all time and length scales of the unsteady Navier–Stokes equations, namely, to perform a direct numerical simulation [20]. Therefore, CFD is still far from becoming the “unique design tool” in both aerospace and turbomachinery applications, due to serious limitations in the modeling of transition and turbulence, as well as to numerical methods which do not properly account for the multidimensional nature of compressible flows.

In this last respect, the authors have recently contributed to the development of two state-of-the-art methods for solving the compressible Euler and Navier–Stokes equations. The first approach is a finite volume centred scheme which uses a weighed averaged five-point discretization for the inviscid fluxes, to achieve third-order accuracy on uniform and mildly nonuniform grids. The second approach is a multidimensional upwind Fluctuation Splitting (FS) scheme, which is “only” second-order-accurate but allows a more realistic modeling of compressible flow propagation phenomena. Both methods have been conceived specifically for compressible inviscid flows and have been extended to the solution of the Navier–Stokes equations, by using standard second-order-accurate discretizations of the viscous fluxes.

This paper provides a valuable numerical comparison of these two methods, using them to compute well-established test-cases ranging from inviscid smooth flows to viscous flows with shocks. In all cases, three identical grids have been used so as to always achieve grid-convergence as well as to assess grid-sensitivity for both methods.

After a brief description of the two methods, detailed numerical results will be presented for each of the selected test-cases. Finally, a few concluding

remarks will be provided.

## 7.2 High-order centred numerical method

The present method [21] is a high-order-accurate cell-centred finite-volume scheme (HO) for the compressible Euler and Navier–Stokes equations. The main features of the scheme are the introduction of a correction for the second-order dispersion error arising from the spatial discretization of a classical second-order scheme [2] for the Euler equations and the use of weighed averages in the evaluation of the inviscid fluxes so as to retain an effective third-order accuracy (for the steady Euler equations) on mildly irregular curvilinear grids. Here the main features of the scheme will be presented for the case of the one-dimensional Euler equations written in conservation form:

$$U_t + F_x = 0. \quad (7.1)$$

By a standard Taylor series expansion, one can easily show that:

$$U_t|_j + \frac{1}{\delta x} \delta \mu F|_j - \frac{\delta x^2}{6} F_{xxx} = U_t + F_x + O(\delta x^4), \quad (7.2)$$

where

$$(\delta \phi)_{j+1/2} = \phi_{j+1} - \phi_j, \quad (\mu \phi)_{j+1/2} = \frac{1}{2}(\phi_{j+1} + \phi_j).$$

Therefore, if one introduces the second-order-accurate centred five-point discretization of  $F_{xxx}$  in the left-hand side (LHS) of equation (7.2), it provides a fourth-order-accurate non-dissipative discretization of equation (7.1). Then, a dissipative correction is introduced as the standard Jameson artificial dissipation [22], to give:

$$U_t|_j + \frac{1}{\delta x} \delta \left( \mu F - \frac{1}{6} \delta^2 \mu F \right) - \frac{1}{\delta x} \delta \left[ \varepsilon_2 \rho(\bar{A}) \delta U + \varepsilon_4 \rho(\bar{A}) \delta^3 U \right] = 0, \quad (7.3)$$

with

$$\varepsilon_2|_{j+\frac{1}{2}} = k_2 \max\{\nu_j, \nu_{j+1}\}, \quad \varepsilon_4|_{j+\frac{1}{2}} = \max\{0, k_4 - \varepsilon_2|_{j+\frac{1}{2}}\},$$

$$\nu_j = \left| \frac{p_{j+1} - 2p_j + p_{j-1}}{p_{j+1} + 2p_j + p_{j-1}} \right|.$$

In the equations above  $\rho(\bar{A})$  denotes the spectral radius of the average Jacobian matrix  $\bar{A}$ ,  $p$  is the pressure, whereas  $k_2$  and  $k_4$  are constant parameters ( $k_4 = 0.032$  and  $k_2 = 0, 0.5$  for subsonic and transonic flows, respectively). In regions where  $U$  is smooth,  $\varepsilon_2 = O(\delta x^2)$  and  $\varepsilon_4 = O(1)$ ,

so that the third term in the LHS of equation (7.3) is  $O(\delta x^3)$  and the scheme is third-order accurate. The approach described above is extended to multidimensional structured meshes through a cell-centred finite-volume formulation. The conservative variables at the cell centroids are considered as the dependent variables and the numerical fluxes are evaluated using suitably weighed discretization formulas, which take into account the stretching and the skewness of the mesh, see [23] for details. The numerical method is finally extended to the Navier–Stokes equations using a second-order-accurate centred discretization of the viscous fluxes.

For the present steady state computations, a four-stages Runge–Kutta time-integration method [2] is used, with coefficients:  $a_1 = 1/4$ ,  $a_2 = 1/3$ ,  $a_3 = 1/2$  and  $a_4 = 1$ . Implicit residual smoothing [24, 25], local time stepping and a V-cycle multigrid method [7] are implemented to accelerate convergence to steady state.

Standard characteristic boundary conditions are imposed at inflow and outflow points. In the case of inviscid flow calculations, the impermeability condition is imposed at the wall and the pressure is extrapolated from inner cell points, using the cell-averaged pressure gradient evaluated by Green’s theorem. In the case of viscous flows, the pressure at the wall is computed imposing zero pressure gradient and the temperature is evaluated enforcing zero heat-flux.

### 7.3 Fluctuation splitting method

The Euler equations are discretized on a computational domain composed of linear finite elements (triangles). The discrete conservative flux balance over each triangle  $T$ , namely, the fluctuation, can be written in terms of appropriate fluxes through the sides of each triangle (see, e.g., [26, 27]) as:

$$\Phi_{U,T} = \bar{R}\Phi_T, \quad \Phi_T = - \sum_{j=1}^3 \frac{l_j}{2} \bar{\mathbf{A}} \cdot \mathbf{n}_j Q_j = - \sum_{j=1}^3 K_j Q_j. \quad (7.4)$$

In equation (7.4),  $\bar{\mathbf{A}}$  is the Jacobian tensor with respect to the characteristic variables  $Q$ ,  $\bar{R}$  is the cell-averaged projection matrix from  $Q$  to the conservative variables  $U$ ,  $l_j$  is the length of the side of the triangle opposite to node  $j$ ,  $\mathbf{n}_j$  is the inward unit vector normal to  $l_j$ , and

$$K_j = \frac{1}{2} l_j (\bar{A} n_{\xi,j} + \bar{B} n_{\eta,j}), \quad (7.5)$$

$\xi$  and  $\eta$  being natural coordinates. Due to the hyperbolic nature of the system,  $K_j$  can be written as:

$$K_j = (\bar{R}_K \bar{\Lambda}_K \bar{L}_K)_j = (\bar{R}_K \bar{\Lambda}_K^+ \bar{L}_K)_j + (\bar{R}_K \bar{\Lambda}_K^- \bar{L}_K)_j = K_j^+ + K_j^-. \quad (7.6)$$

In equation (7.6),  $\bar{R}_{K,j}$  and  $\bar{L}_{K,j}$  are the right and left eigenvector matrices of  $K_j$ , whereas  $\bar{\Lambda}_{K,j}^+$  and  $\bar{\Lambda}_{K,j}^-$  are the corresponding positive and negative eigenvalue matrices. Introducing the following vectors,

$$Q_{\text{in}} = \left( \sum_{j=1}^3 K_j^- \right)^{-1} \left( \sum_{j=1}^3 K_j^- Q_j \right), \quad Q_{\text{out}} = \left( \sum_{j=1}^3 K_j^+ \right)^{-1} \left( \sum_{j=1}^3 K_j^+ Q_j \right), \quad (7.7)$$

the linear matrix Low Diffusion A (LDA) scheme, which is linearity preserving [28], is obtained as:

$$\Phi_j = -K_j^+ [Q_{\text{out}} - Q_{\text{in}}]. \quad (7.8)$$

The LDA scheme of equation (7.8) is very accurate for subsonic smooth flows and can be considered the optimum compact Fluctuation Splitting (FS) scheme for such conditions. For supersonic flows, a different set of characteristic variables,  $W$ , allows to recast the Euler system into an equivalent set of four scalar advection equations. These are then discretized by the nonlinear Positive Streamwise Invariant (PSI) scheme [28], which can be written as,

$$\hat{\Phi}_j = -K_j^{nl+} (W_j - W_{\text{in}}^{nl}), \quad (7.9)$$

where  $K_j^{nl+}$  is computed using the nonlinear Jacobian  $\bar{\hat{A}}^{nl}$  as defined in [29]. The PSI scheme of equation (7.9) is, to date, the optimum FS scheme for supersonic flows with or without shocks. Finally, in order to compute transonic flows with strong shocks, it is necessary to use locally a monotone scheme, namely, the matrix N scheme of [27], given as:

$$\hat{\Phi}_j = -K_j^{l+} (W_j - W_{\text{in}}^l), \quad (7.10)$$

where  $K_j^l$  and  $W_{\text{in}}^l$  are computed using the Jacobian  $\bar{A}_W$ . In order to pinpoint the “shock cells” where such a lower-order scheme needs to be employed — disregarding those where the transition from supersonic flow conditions to subsonic ones is smooth enough to be properly handled by the LDA scheme — it is necessary to characterize them uniquely. By a careful analysis of the flow properties across a normal shock, it is concluded that “shock cells” are characterized by: i) average cell Mach number,  $M$ , lower than one; ii) at least one supersonic node; iii) at least one subsonic node with local Mach number,  $M_j$ , lower than 0.9. In conclusion, at every step of the computational process, the present hybrid approach flags all cells of the computational domain and distributes each residual using equation (7.9) at supersonic cells, equation (7.10) at shock-cells, and equation (7.8) at the remaining ones.

The residual of the Euler equations at each vertex  $j$  of the computational domain is then evaluated by collecting all contributions coming from the

| <i>grid</i>     | $M_{min}$ |        | $M_{max}$ |        | $L_1(s)(\times 10^{-6})$ |       | $L_\infty(s)(\times 10^{-5})$ |       |
|-----------------|-----------|--------|-----------|--------|--------------------------|-------|-------------------------------|-------|
|                 | HO        | FS     | HO        | FS     | HO                       | FS    | HO                            | FS    |
| $64 \times 16$  | 0.4159    | 0.4189 | 0.9163    | 0.9403 | 53.33                    | 197.8 | 318.0                         | 126.0 |
| $128 \times 32$ | 0.4244    | 0.4181 | 0.9253    | 0.9401 | 8.166                    | 62.55 | 39.80                         | 36.03 |
| $256 \times 64$ | 0.4228    | 0.4177 | 0.9333    | 0.9401 | 1.476                    | 27.58 | 5.220                         | 11.05 |

**Table 1** Channel-flow accuracy study.

surrounding triangles, as:

$$\left(\frac{\partial U}{\partial t}\right)_j = \frac{1}{S_j} \sum_T \Phi_{U,j}^T. \quad (7.11)$$

The numerical method is finally extended to the Navier–Stokes equations using a standard second-order-accurate Galerkin finite-element scheme.

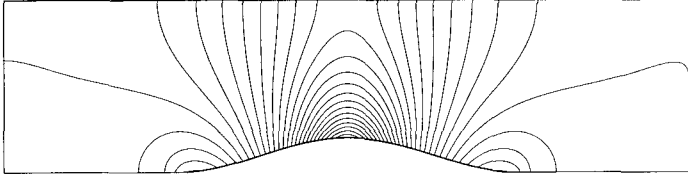
For the present steady state calculations, two different time-integration approaches have been used, namely, the explicit Runge–Kutta scheme of [27] and the implicit Newton-GMRES one of [30].

Standard characteristic boundary conditions are imposed at inflow and outflow points. In the explicit, inviscid code, the wall boundary conditions are enforced using an auxiliary set of ghost cells: isentropic simple radial equilibrium is used together with a characteristic correction to enforce the no-injection condition and evaluate the wall pressure [31]. In the implicit code, the momentum flux is enforced at each wall cell side in the case of inviscid flows, the pressure being approximated as the average of the two nodal values; both velocity components at the wall nodes are set to zero for the case of viscous flows, the zero heat-flux condition being naturally enforced by omitting the boundary heat-flux contributions.

## 7.4 Results and Discussion

Five well-documented test-cases have been considered to evaluate the accuracy-performance of the two numerical methods, using three grids with different resolution.

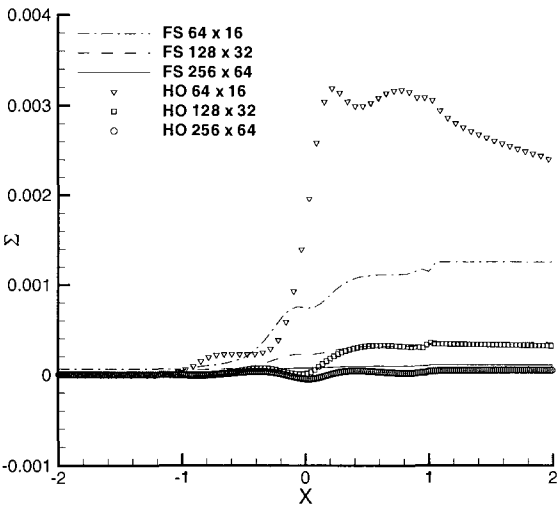
The first test-case is the inviscid subsonic flow through a channel with a cosine shaped wall, 20% restriction and outlet Mach number equal to 0.5. The Mach-number contours computed using the fine ( $256 \times 64$ ) grid with the FS method are provided in figure 1, the corresponding HO solution being substantially coincident. Table 1 shows the minimum and maximum Mach



**Figure 1** Channel-flow Mach-number contours ( $\Delta M = 0.02$ ).

numbers and the  $L_1$  and  $L_\infty$  norms of the entropy error ( $\Sigma = ((p/\rho^\gamma) - (p/\rho^\gamma)_{inlet})/(p/\rho^\gamma)_{inlet}$ ) for the HO method and the FS one, respectively. The  $L_\infty$  norms show that the two methods are almost third-order-accurate and second-order-accurate, respectively, but the  $L_1$  norms decrease less rapidly for both. Moreover, the FS solution appears to be more accurate when using the coarse and medium grids and less sensitive to the grid size, whereas the  $L_1$  norms of the HO method are markedly lower as the mesh is refined. Figure 2 shows the entropy-error distributions along the bottom wall obtained using the three grids: for the coarse grid the FS solution provides the lower error; very close error distributions are obtained on the medium grid; the HO method provides the lower error on the fine grid. Moreover, unlike the HO method, the FS one gives monotone solutions. It is noteworthy that the very good performance of the FS scheme are due to the very accurate wall boundary condition implemented in the explicit code. And in fact the implicit code experiences entropy-error values one order of magnitude higher and provides the following value of  $M_{min}$  and  $M_{max}$ : (0.3955, 0.9306), (0.4059, 0.9370), (0.4119, 0.9392), for the coarse, medium and fine grids, respectively.

The second test-case is the inviscid subsonic flow past an NACA0012 airfoil with  $M_\infty = 0.63$  and  $\alpha = 2^\circ$ . The three grids employed have 96, 192 and 384 cells along the profile, respectively, and the free-stream boundary is located at twenty chords away from the body. Table 2 shows the lift and drag coefficients ( $C_L$ ,  $C_D$ ) together with the  $L_1$  and  $L_\infty$  norms of the entropy error for the two schemes. The  $L_1$  norm indicates that both methods achieve almost their design order of accuracy, whereas the  $L_\infty$  norm still does not show asymptotic behaviour, probably due to the presence of the stagnation-flow region. The lift coefficients computed by the two schemes tend to the same value as the mesh is refined, whereas a lower drag coefficient is provided by the FS scheme, in spite of its higher entropy-error distribution (see figure 3). Finally, the Mach-number distributions along the profile are given in figure 4: all curves are very close, except the coarse-grid HO solution in the front part of the suction side. In conclusion, the results obtained in the present test indicate a comparable accuracy for the two schemes.



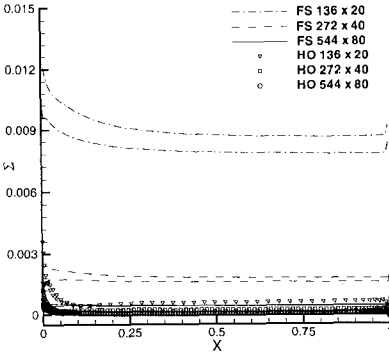
**Figure 2** Channel-flow entropy error distributions along the bottom wall.

The third test-case is the inviscid transonic flow past an NACA0012 airfoil with  $M_\infty = 0.85$  and  $\alpha = 1^\circ$ . The same grids employed in the previous test-case have been used. For such a problem the explicit FS code could only converge on the coarse grid so that the implicit one (using a less accurate wall boundary condition treatment) had to be used. The overall solutions obtained by the two schemes on the fine grid are shown in figure 5 and 6. They are practically identical and show the very good shock-capturing capability of both methods. The distributions of the pressure coefficient ( $C_p = -2(p - p_\infty)/(\rho_\infty u_\infty^2)$ ) along the profile are given in figure 7. Both

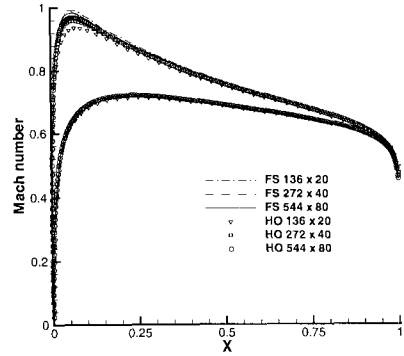
| grid            | $C_L$  |        | $C_D(\times 10^{-4})$ |       | $L_1(s)(\times 10^{-9})$ |       | $L_\infty(s)(\times 10^{-4})$ |       |
|-----------------|--------|--------|-----------------------|-------|--------------------------|-------|-------------------------------|-------|
|                 | HO     | FS     | HO                    | FS    | HO                       | FS    | HO                            | FS    |
| $136 \times 20$ | 0.3176 | 0.3348 | 14.51                 | 12.08 | 154.9                    | 7313. | 38.61                         | 130.5 |
| $272 \times 40$ | 0.3207 | 0.3296 | 5.918                 | 4.274 | 27.54                    | 1872. | 12.51                         | 25.06 |
| $544 \times 80$ | 0.3215 | 0.3279 | 4.060                 | 3.020 | 4.266                    | 1099. | 6.420                         | 9.313 |

**Table 2** Inviscid subsonic flow past an NACA0012 airfoil: accuracy study.





**Figure 3** Entropy error distributions along the profile.



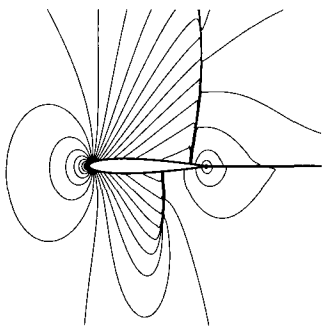
**Figure 4** Mach-number distributions along the profile.

| <i>grid</i>     | $C_L$  |        | $C_D(\times 10^{-2})$ |       |
|-----------------|--------|--------|-----------------------|-------|
|                 | HO     | FS     | HO                    | FS    |
| 136 $\times$ 20 | 0.3841 | 0.3988 | 5.748                 | 6.081 |
| 272 $\times$ 40 | 0.3830 | 0.3844 | 5.793                 | 5.716 |
| 544 $\times$ 80 | 0.3728 | 0.3702 | 5.742                 | 5.646 |

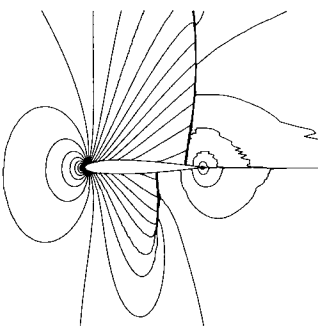
**Table 3** Inviscid transonic flow past an NACA0012 airfoil: accuracy study.

methods provide monotone shocks and very close solutions. Table 3 shows the lift and drag coefficients obtained using the two schemes. Using the fine grid, the difference in the value of the  $C_L$  is about 1% whereas the difference in the value of the  $C_D$  is about 2%. The results agree quite well with the numerical data presented in [32]. For such a case the accuracy of the HO method is clearly superior, as anticipated. To better understand the ill-effect of the less accurate boundary-condition treatment used in the implicit code, the Mach number distributions on the profile are shown in figure 8. The value of the Mach number after the shock is underestimated on all grids, due to excessive entropy production along the body surface. In contrast, the coarse-grid solution of the explicit code is more correct. Notice that such an issue is irrelevant for the following viscous flow calculations.

The next test-case is the well-documented laminar subsonic flow over an NACA0012 airfoil with  $M_\infty = 0.5$ ,  $\alpha = 0$  and Reynolds number, based on



**Figure 5** HO-scheme Mach-number contours ( $\Delta M = 0.05$ ).

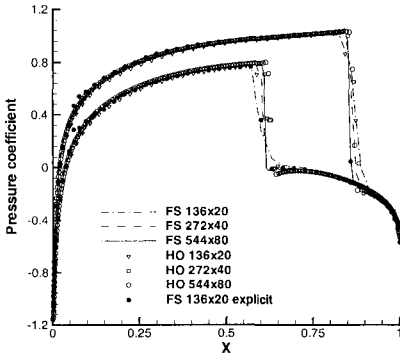


**Figure 6** FS-scheme Mach-number contours ( $\Delta M = 0.05$ ).

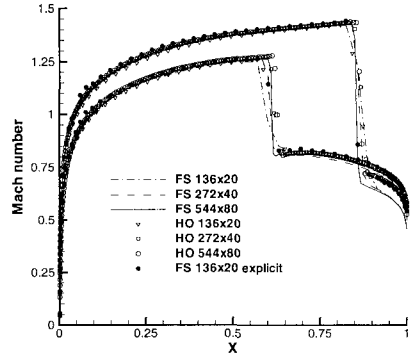
| grid      | $C_D^{inv}$ |            | $C_D^{vis}$ |            | sep. (x/c) |        |
|-----------|-------------|------------|-------------|------------|------------|--------|
|           | HO          | FS         | HO          | FS         | HO         | FS     |
| 132 × 34  | 2.097 (-2)  | 2.251 (-2) | 3.825 (-2)  | 3.502 (-2) | 0.9178     | 0.8642 |
| 266 × 68  | 2.204 (-2)  | 2.244 (-2) | 3.384 (-2)  | 3.313 (-2) | 0.8284     | 0.8255 |
| 532 × 136 | 2.256 (-2)  | 2.237 (-2) | 3.316 (-2)  | 3.277 (-2) | 0.8227     | 0.8186 |

**Table 4** Viscous subsonic flow past an NACA0012 airfoil: accuracy study.

the chord length and free-stream conditions,  $Re_\infty = 5000$ . Three grids have been employed having 112, 224 and 448 cells along the profile, respectively, the free-stream boundary being located at about twenty chords away from the body. The main feature of such flow is the separation occurring close to the trailing edge. The inviscid ( $C_D^{inv}$ ) and viscous ( $C_D^{vis}$ ) drag coefficients, and the separation point computed using the two schemes are provided in table 4. The two sets of results are very close to each other and agree quite well with the data reported in the literature [33]. The FS ones appear to be slightly less grid sensitive and thus possibly more accurate. The distributions along the profile of the pressure coefficient and of the skin-friction coefficient ( $C_f = 2\tau_w/(\rho_\infty u_\infty^2)$ ) are given in figure 9 and 10, respectively. Concerning the pressure coefficient, all curves coincide within plotting accuracy, except the HO solution on the coarse grid. Furthermore, figure 10 again shows a quite good agreement between the two sets of results with minor differences in the



**Figure 7** Pressure-coefficient distributions along the profile.



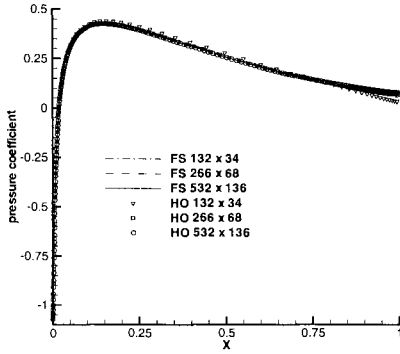
**Figure 8** Mach number distributions along the profile.

| grid             | $C_L$  |        | $C_D$  |        |
|------------------|--------|--------|--------|--------|
|                  | HO     | FS     | HO     | FS     |
| $132 \times 34$  | 0.3396 | 0.3420 | 0.2796 | 0.2761 |
| $266 \times 68$  | 0.3379 | 0.3421 | 0.2758 | 0.2746 |
| $532 \times 136$ | 0.3391 | 0.3397 | 0.2754 | 0.2733 |

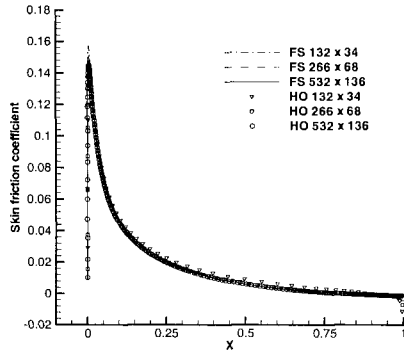
**Table 5** Viscous supersonic flow past an NACA0012 airfoil: accuracy study.

peak value, the HO maximum value being slightly lower (0.1475 vs 0.1480 on the fine grid). For this problem, the results of the HO scheme have been obtained on half-grid, enforcing symmetry, so as to achieve convergence to machine accuracy. Otherwise, on the medium and fine grids the residuals stall after dropping to about  $10^{-3}$ , due to a periodic vortex shedding phenomenon.

The last test-case is the laminar supersonic flow over an NACA0012 airfoil with  $M_\infty = 2$ ,  $\alpha = 10^\circ$ , and  $Re_\infty \approx 1000$ . Table 5 shows the lift and drag coefficients obtained using the two schemes. The two sets of results are comparable and agree quite well with the numerical data provided in [34]. The Mach-number contours computed with the HO and FS schemes using the fine ( $532 \times 136$ ) grid are given in figures 11 and 12, respectively. Both methods capture the shock quite sharply and monotonically. The distributions along the profile of the pressure-coefficient and of the skin-friction-coefficient are given in figure 13 and 14, respectively. All solutions coincide within plotting



**Figure 9** Pressure-coefficient distributions along the profile.



**Figure 10** Skin-friction-coefficient distributions along the profile.

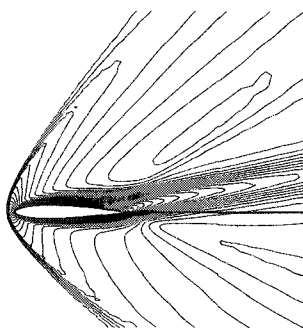
accuracy.

## 7.5 Conclusions

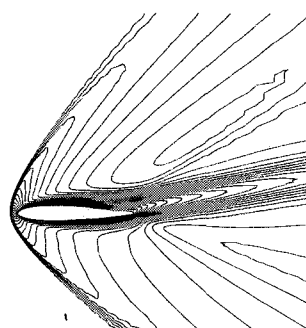
This work provides a very careful one-to-one comparison of the accuracy performance of two state-of-the-art methods for solving the steady-state compressible Euler and Navier–Stokes equations.

The first method is a weighed averaged finite-volume one which approximates the inviscid fluxes with third-order accuracy and the viscous ones with second-order accuracy; the second one is a hybrid multidimensional upwind fluctuation splitting scheme which approximates both inviscid and viscous fluxes with second-order accuracy; both are only first-order-accurate locally at shocks. The lower order FS scheme is seen to perform as well as, if not better than, the HO one for both inviscid- and viscous-flow calculations, mainly due to three reasons: i) the correct modeling of the multidimensional nature of the inviscid fluxes; ii) a very accurate treatment of the inviscid wall boundary conditions; iii) the Galerkin approximation of the viscous fluxes. On the other hand, the HO method is less costly and can be improved with respect to the accuracy of viscous fluxes.

In conclusion, both approaches are worth pursuing towards developing more accurate, robust and efficient CFD tools for advanced aerospace and turbomachinery applications.



**Figure 11** HO-scheme  
Mach-number contours ( $\Delta M = 0.1$ ).



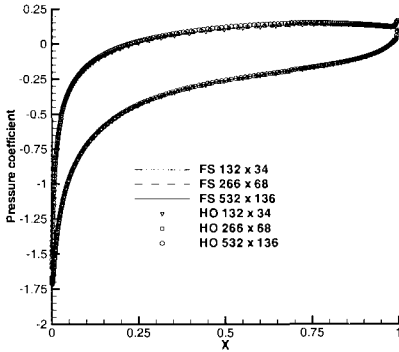
**Figure 12** FS-scheme Mach-number  
contours ( $\Delta M = 0.1$ ).

## 7.6 Acknowledgements

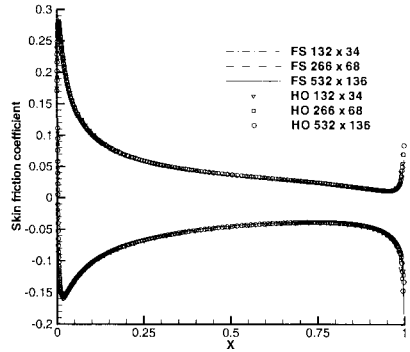
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**Figure 13** Pressure-coefficient distributions along the profile.



**Figure 14** Skin-friction-coefficient distributions along the profile.

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# Extension of Efficient Low Dissipation High Order Schemes for 3-D Curvilinear Moving Grids

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## Abstract

The efficient low dissipative highly parallelizable shock-capturing schemes of essentially fourth-order or higher proposed by Yee et al. [24] is formulated for 3-D curvilinear moving grids in the finite-difference frame work. These schemes consist of high order compact or non-compact non-dissipative base schemes combined with adaptive nonlinear characteristic filters to minimize the use of numerical dissipation away from shock and shear regions. The amount of numerical dissipation is further minimized by applying these schemes to the entropy splitting form of the inviscid flux derivatives. The analysis is given for a thermally perfect gas. The main difficulty in the extension of high order schemes to curvilinear moving grids is the high order numerical evaluation of the geometric terms arising from the coordinate transformation. The numerical evaluation of these terms to insure freestream preservation is done in a coordinate invariant manner. This avoids spurious numerical errors, which would result from previous, noninvariant formulations, when treating axi-symmetric flow.

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## 8.1 Introduction

Most available high order high-resolution shock-capturing numerical schemes are too CPU intensive for practical 3-D complex simulations. In spite of their high-resolution capability for rapidly evolving flows and short term time integrations, these schemes often exhibit undesirable amplitude errors for long time integrations in aeroacoustics, rotorcraft, turbulence and general long wave propagation computations. High order here refers to spatial schemes that are essentially fourth-order or higher away from shock and shear regions. The delicate balance of the numerical dissipation for stability without the expense of excessive smearing of the flow physics after long time integrations poses a severe challenge for unsteady flow simulations of this type. The recently developed high order low-dissipative shock capturing schemes of Yee et al. [24] and their companion papers Yee et al. [25] and Sjogreen & Yee [14] aim at developing methods to overcome some of the difficulties. For efficiency, Yee et al. [24] proposed simple highly parallelizable spatial schemes that consist of a base scheme and nonlinear filters. The base scheme consists of narrow grid stencil high order compact or non-compact centered non-dissipative classical spatial differencings. The filters consist of a product of the dissipative portion of low order total variation diminishing (TVD), essentially non-oscillatory (ENO) or weighted ENO (WENO) schemes and an artificial compression method (ACM) sensor. The role of the ACM sensor is to reduce the amount of numerical dissipation away from shock and shear regions. As an alternative to the ACM sensor, Sjogreen & Yee [14] utilized non-orthogonal wavelet basis functions as multi-resolution sensors to dynamically determine the amount of nonlinear numerical dissipation to be added at each grid point. The resulting wavelet sensors are readily available as more desirable grid adaptation indicators (Gerritsen & Olsson [4]) than the commonly used grid adaptation indicators.

In contrast to hybrid schemes that switch between spectral or spectral-like non-shock-capturing schemes and high order ENO schemes, the high order non-dissipative base scheme is always activated. The final grid stencil of these schemes is five points in each spatial direction if second-order TVD schemes are used as filters, and seven points if second-order ENO schemes are used as filters for a fourth-order base scheme. Studies showed that higher accuracy was achieved with less CPU time and fewer grid points when compared with that of standard high order TVD, positive, ENO or WENO schemes. These schemes are able to accurately simulate a wide range of flow conditions, including long time integrations of wave propagation, computational aeroacoustics, combustion and direct numerical simulation (DNS) of 3-D compressible turbulence. See Yee et al. ([24, 25, 26], Sandham & Yee [13], Sjogreen & Yee [14, 15], Müller & Yee [8] and Polifke et al. [10]. Table 8.1 shows the flow chart of the schemes.

Efficient Low Dissipative High Order Schemes

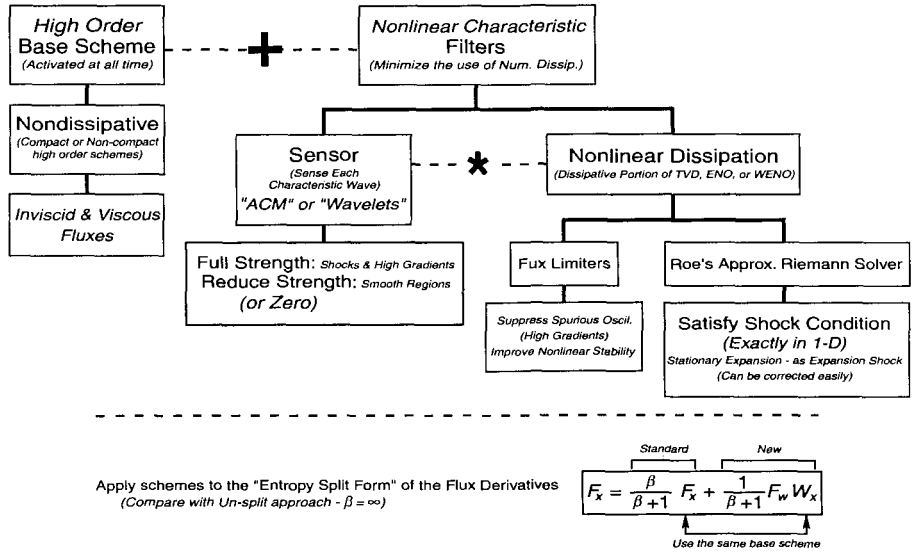


Table 8.1

In Yee et al. [25], these schemes were applied to the entropy splitting form of the inviscid flux derivatives (e.g.,  $F_x$ ; see bottom of Table 8.1.) Studies were conducted to determine to what extent the entropy splitting form of the flux derivative can help in minimizing numerical dissipation, or equivalently, in improving nonlinear stability. They view entropy splitting as a conditioned form of the original conservation laws. Overall, the numerical results of Yee et al. [25], Sandham & Yee [13], Sjogreen & Yee [14] and Polifke et al. [10] indicate a positive benefit from the entropy splitting. The splitting can stabilize spurious noise generated by the non-dissipative or low dissipative spatial discretizations which is a major cause of nonlinear instability. Their study also indicates that entropy splitting **alone** can improve nonlinear stability even when one employs numerical boundary conditions that do not satisfy the Strand [16] condition. This stability property of the entropy splitting is valuable not just for the class of schemes in question, but can also be applied to other schemes commonly used in practice. This emphasizes the fact that one should always try to apply numerical schemes to a more conditioned form of the governing equations.

In Sandham & Yee [13], entropy splitting with a variant of Strand's boundary difference operators that were developed by Carpenter et al. [1] was used in conjunction with the high order non-dissipative central base scheme for a DNS of 3-D shock-free compressible turbulent channel flow.

Very accurate fully developed turbulent statistics were obtained using coarse to moderate grid sizes and without using filters. Their results compared well with the best spectral method of incompressible Navier-Stokes simulations. Modern high-resolution numerical dissipation has been the major factor in improving nonlinear instabilities for short or moderate time integrations (unsteady). Most often, added numerical dissipation is necessary for longer time integration at the expense of excess smearing of the flow physics without resorting to finer grids and extremely small time steps. The use of the entropy splitting form of the flux derivative was shown to be capable of minimizing the use of numerical dissipation.

In Yee et al. [25], the extendibility of the entropy splitting concept to other physical equations of state and evolutionary equation sets was examined. Their study shows that the entropy splitting can be formally extended to a thermally perfect gas, with the internal energy being an arbitrary function of temperature. Although extension of entropy splitting to nonequilibrium flows is not practically feasible, and is not possible for equilibrium real gas and artificial compressibility methods of solving the incompressible Navier-Stokes equations, and magnetohydrodynamic (MHD) flows, the high order numerical schemes in question are applicable to these types of flows. Note that since the Maxwell equations are a linear system of hyperbolic equations that can be easily symmetrized, Strand's numerical boundary operators are still valid, and the numerical schemes in question are also applicable. Entropy splitting is not needed for the Maxwell equations. For nonequilibrium flows, if one solves the species and flow equations separately in a loosely coupled manner, then the flow equations effectively satisfy a locally thermally perfect gas law and a "local" form of entropy splitting is applicable.

In order to apply this scheme in practice, it must be formulated in arbitrary 3-D curvilinear coordinates. A finite-volume formulation is generally preferred over a finite-difference formulation, especially for schemes that are third-order or lower. For fourth-order or higher schemes, finite-volume formulations are very complex. For efficiency, we therefore only consider finite-difference formulations. An important issue then is the treatment of the geometric terms arising from the coordinate transformation. These are the components of the surface area vectors (or metrics) appearing in the transformed fluxes and the volume (or Jacobian) in the definition of the transformed conservative variables. Analytically, these geometric quantities satisfy certain conservation laws. They are the surface area conservation law (or metric identity), and for time-varying grids an additional volume conservation law (sometimes referred to as the geometric conservation law). It would be desirable for these laws to be satisfied numerically. This would result in regions of uniform flow being preserved exactly (within round-off errors). In regions of non-uniform flow, it would hopefully lead to greater accuracy by eliminating errors due to the grid, particularly if it is highly distorted.

In Vinokur [20], it was shown how these two laws can be satisfied numerically by making use of finite-volume concepts. For the surface area vector components, the discretization was found to be identical to the averaging procedure proposed by Pulliam & Steger [11] in the second-order finite-difference frame work. Unfortunately, this use of finite-volume concepts is only valid for second-order accurate central differencing. They cannot be extended to high order compact or non-compact differencings. An alternative method of discretizing the surface area vector components by first rewriting them in an equivalent “conservative” form was proposed by Thomas & Lombard [18]. For second-order accurate central differencing, the surface area conservation law was then numerically satisfied. Gaitonde & Visbal [3] found by numerical experiments on two different curvilinear grids that high order compact and non-compact differencing applied to the Thomas and Lombard form also satisfied that conservation law numerically. An undesirable feature of the expression used is that it is not coordinate invariant. The present coordinate invariant form has been independently proposed by the authors in June 2000 and by Thomas & Neier [19] (P.D. Thomas, private communication, November 2000).

### 8.1.1 Objectives

In this paper we formulate the Yee et al. [24] scheme with entropy splitting (Yee et al. [25]) in 3-D curvilinear coordinates for a thermally perfect gas in a finite-difference approach. The surface area conservation law is satisfied numerically by discretizing a coordinate invariant version of the Thomas and Lombard formulas. The importance of doing this is that it enables us to extend the method to axi-symmetric flow, with an appropriate treatment of the resulting source term. This eliminates spurious numerical errors due to the original Thomas and Lombard expressions. We also provide a rigorous analytic proof that the numerical mixed partial derivatives commute for any grid, using any high order compact or non-compact differencing, with arbitrary numerical boundary conditions. This is necessary to insure that the surface area conservation law is satisfied numerically.

### 8.1.2 Outline

In Section 8.2 we review the method of entropy splitting as applied to the Euler equations for a thermally perfect gas. The equations are formulated for arbitrary 3-D curvilinear coordinates, and specialized to axi-symmetric flow. The section also includes a brief discussion of the Navier-Stokes equations. In Section 8.3 we describe the numerical methods to treat the spatial terms. Particular attention is paid to the treatment of the geometric terms. A section on Roe’s approximate Riemann solver, which is part of the numerical method

for the filters, includes a discussion of some of the issues involved in defining the eigenvectors of the flux Jacobian matrix for generalized 3-D coordinates, as well as the expressions for the Roe-averaged state for a thermally perfect gas. The corresponding forms for nonequilibrium flows are presented in Appendix B. The proof that numerical mixed partial derivatives commute for the high order base scheme in generalized coordinates is presented in Appendix A.

## 8.2 Formulation of Equations

### 8.2.1 Canonical Splitting of Conservation Laws

A system of scalar conservation laws can be written as

$$Q_t + \nabla \cdot \mathbf{F} = 0, \quad (2.1.1)$$

where  $Q$  and  $\mathbf{F}(Q)$  are algebraic *vectors*, but the components of  $\mathbf{F}$  are physical vectors. Letter subscripts indicate partial differentiation. In order to obtain a nonlinearly stable method of solving initial boundary value problems (IBVPs) for the nonlinear system of hyperbolic conservation laws (2.1.1), we transform the governing equations so that the resulting PDEs are nonlinearly stable, including the effect of physical boundary conditions (Olsson & Oliger [9]). We introduce the new *vector*  $W(Q)$  such that  $\mathbf{F}_W$  is symmetric and  $Q_W$  is symmetric and positive definite. Here the matrix  $\mathbf{F}_W$  also has components that are physical vectors. Furthermore,  $W$  is chosen such that both  $\mathbf{F}$  and  $Q$  are homogeneous functions of  $W$  of degree  $\beta$ , i.e., there is a constant  $\beta$  such that for all  $\sigma$

$$Q(\sigma W) = \sigma^\beta Q(W), \quad (2.1.2a)$$

$$\mathbf{F}(\sigma W) = \sigma^\beta \mathbf{F}(W). \quad (2.1.2b)$$

By taking the partial derivative of Eqs. (2.1.2) with respect to  $\sigma$  and then setting  $\sigma = 1$ , we obtain Euler's theorem, namely

$$Q_W W = \beta Q, \quad (2.1.3a)$$

$$\mathbf{F}_W W = \beta \mathbf{F}. \quad (2.1.3b)$$

In order to obtain an energy estimate for stability of the IBVP we introduce a canonical splitting and write  $\nabla \cdot \mathbf{F}$  as

$$\nabla \cdot \mathbf{F} = \frac{\beta}{\beta + 1} \nabla \cdot \mathbf{F} + \frac{1}{\beta + 1} \mathbf{F}_W \cdot \nabla W, \quad \beta \neq -1, \quad (2.1.4)$$

where  $\nabla W$  is a *vector* whose components are the gradients of the components of  $W$ . The “ $\cdot$ ” in the second term in (2.1.4) indicates that when taking the

inner product of a row of  $\mathbf{F}_W$  and the *vector*  $\nabla W$ , the vector dot product is applied when multiplying components. Note that  $\nabla \cdot \mathbf{F}$  has been split into a conservative part and a non-conservative part. If we take the inner product of  $W$  and  $\nabla \cdot \mathbf{F}$ , and use the homogeneity property of  $\mathbf{F}$  and the symmetry property of  $\mathbf{F}_W$ , we obtain

$$W^T \nabla \cdot \mathbf{F} = \frac{1}{\beta + 1} [W^T \beta \nabla \cdot \mathbf{F} + W^T \mathbf{F}_W \cdot \nabla W] \quad (2.1.5a)$$

$$= \frac{1}{\beta + 1} [W^T \nabla \cdot (\mathbf{F}_W W) + \nabla W^T \cdot \mathbf{F}_W W] \quad (2.1.5b)$$

$$= \frac{1}{\beta + 1} \nabla \cdot (W^T \mathbf{F}_W W). \quad (2.1.5c)$$

Applying the canonical splitting to  $Q_t$ , we obtain in a similar manner

$$W^T Q_t = \frac{1}{\beta + 1} (W^T Q_W W)_t. \quad (2.1.6)$$

Taking the inner product of  $W$  with (2.1.1), substituting (2.1.5c) and (2.1.6), integrating over a fixed domain with boundary surface element  $\mathbf{n} dS$ , where  $\mathbf{n}$  is the outward normal, and applying the divergence theorem, results in

$$\frac{d}{dt} \int W^T Q_W W dV = - \oint W^T \mathbf{n} \cdot \mathbf{F}_W W dS. \quad (2.1.7)$$

Here  $\mathbf{n} \cdot \mathbf{F}_W$  is the matrix whose components are the outward normal vector components of  $\mathbf{F}_W$ . From (2.1.7) one can rigorously establish a bound on the rate of growth of the energy norm in terms of the absolute eigenvalues corresponding to the incoming characteristic variables at the boundary of the domain (Olsson & Oliger [9], Gerritsen & Olsson [4]). In doing so, one must make use of the positive definite property of  $Q_W$ .

### 8.2.2 Entropy Splitting of the Euler Equations

In this section we consider the conservation-law form of the 3-D compressible Euler equations. For maximum generality, the analysis is presented for arbitrary three-dimensional, time varying grids. The canonical splitting discussed in Section 2.1 is based on a convex entropy function that satisfies an entropy condition. The splitting is therefore referred to as entropy splitting. While this splitting was originally derived for a perfect gas (Gerritsen & Olsson [4]), it can be extended to flow of a gas that is only thermally perfect, with the internal energy being an arbitrary function of temperature (Yee et al. [25]). This law is valid for a dilute gas consisting of a single chemical species, and is also a very good approximation for air below the temperature when oxygen starts to dissociate (approximately 2000° K).

Let  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  be the unit vectors along the axes of a Cartesian coordinate system. Then the position vector  $\mathbf{r}$  and flow velocity  $\mathbf{u}$  are expressed as

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (2.2.1)$$

and

$$\mathbf{u} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}. \quad (2.2.2)$$

The grid point velocity in a time varying grid is designated by  $\mathbf{v}$ . Let  $\mathbf{u}' = \mathbf{u} - \mathbf{v}$  be the velocity relative to the moving grid. The *vectors*  $\mathbf{Q}$  and  $\mathbf{F}$  for the Euler equations then take the form

$$\mathbf{Q} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ e \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho \mathbf{u}' \\ \rho u \mathbf{u}' + p\mathbf{i} \\ \rho v \mathbf{u}' + p\mathbf{j} \\ \rho w \mathbf{u}' + p\mathbf{k} \\ (e + p)\mathbf{u}' + p\mathbf{v} \end{bmatrix}, \quad (2.2.3)$$

where  $\rho$  is the density,  $p$  is the pressure, and  $e = \rho(\epsilon + q^2/2)$  is the total energy per unit volume.  $\epsilon$  is the specific internal energy and  $q^2 = u^2 + v^2 + w^2$  is the square of the speed. Note that  $\mathbf{F}$  is now defined relative to the moving grid.

The equation of state for a thermally perfect gas is

$$p = \rho \hat{T}, \quad (2.2.4)$$

where  $\hat{T} = RT$  is a normalized temperature and  $R$  is the gas constant.  $\hat{T}$  has the same dimension as  $\epsilon$ . From the first and second laws of thermodynamics we can show that  $\epsilon = \epsilon(\hat{T})$  only. All real gases satisfy the conditions  $\dot{\epsilon} > 0$  and  $\ddot{\epsilon} > 0$ , where  $\dot{\epsilon} = d\epsilon/d\hat{T}$ . The temperature  $\hat{T}(Q)$  is obtained by solving the equation

$$\epsilon(\hat{T}) = \frac{e}{\rho} - \frac{1}{2} \frac{[(\rho u)^2 + (\rho v)^2 + (\rho w)^2]}{\rho^2}. \quad (2.2.5)$$

Equation (2.2.5) has a unique solution since  $\dot{\epsilon} > 0$ .

From the laws of thermodynamics we can relate the dimensionless entropy  $\hat{S} = S/R$  to  $\rho$  and  $\hat{T}$  by

$$\rho f = e^{-\hat{S}}, \quad (2.2.6)$$

where

$$f(\hat{T}) = \exp\left(-\int \frac{\dot{\epsilon}}{\hat{T}} d\hat{T}\right). \quad (2.2.7)$$

The arbitrary constant in the integral of (2.2.7) can be absorbed in the definition of  $\hat{S}$ .

Following Harten [6], we obtain the *vector*  $W(Q)$  from

$$W = \frac{\partial \Gamma}{\partial Q}, \quad (2.2.8)$$



where the convex function  $\Gamma(Q)$  is given by

$$\Gamma = \rho\psi(\hat{S}). \quad (2.2.9)$$

The components of  $W$  are sometimes referred to as “entropy variables”, while  $\Gamma$  is referred to as an “entropy function”. We show in Yee et al. [25] that in order to satisfy the homogeneity and positive definite conditions,  $\psi(\hat{S})$  is given by

$$\psi = \beta e^{-\hat{S}/\beta}, \quad (2.2.10)$$

where  $\beta$  is a constant. This then gives

$$\dot{\psi} = -e^{-\hat{S}/\beta} = -(\rho f)^{1/\beta}, \quad (2.2.11)$$

where we have used (2.2.6) to obtain the second expression. We note from (2.1.4) that the definition of  $W$  allows an arbitrary multiplicative constant. Correspondingly, there is an arbitrary multiplicative constant in the definition of  $\psi$ . We have chosen this constant to obtain the simplest form for (2.2.11). The transformed variable  $W$  can now be written as

$$W = \frac{\dot{\psi}}{p} [e - 2\tilde{\epsilon} - p(1 + \beta) \quad -\rho u \quad -\rho v \quad -\rho w \quad \rho]^T, \quad (2.2.12)$$

where  $\tilde{\epsilon} = \rho\epsilon$  is the internal energy per unit volume. With the aid of (2.2.4), we can combine the components of  $W$  to show that  $\hat{T}$  and  $\epsilon(\hat{T})$  are homogeneous functions of  $W$  of degree 0. It then follows from (2.2.3), (2.2.4), (2.2.11) and (2.2.12) that the *vectors*  $Q$  and  $\mathbf{F}$  are both homogeneous functions of  $W$  of degree  $\beta$ .

While we cannot obtain an explicit formula for  $Q(W)$ , we can derive an explicit expression for the symmetric matrix  $Q_W$  as a function of  $Q$ . The upper triangular part can be written as

$$Q_W = \frac{1}{\dot{\psi}} \begin{bmatrix} a\rho & \rho u & \rho v & \rho w & ae + bp \\ & \rho u^2 - p & \rho uv & \rho uw & u[ae + (b-1)p] \\ & & \rho v^2 - p & \rho vw & v[ae + (b-1)p] \\ & & & \rho w^2 - p & w[ae + (b-1)p] \\ & & & & \frac{ae^2}{\rho} + p(\frac{2be}{\rho} - q^2) + \frac{be^2}{\rho}(1 + \beta) \end{bmatrix}, \quad (2.2.13)$$

where

$$a(\hat{T}, \beta) = \frac{1}{\hat{\epsilon} + 1 + \beta} - 1 \quad (2.2.14)$$

and

$$b(\hat{T}, \beta) = (1 + \beta)a + \beta = \frac{-\hat{\epsilon}}{\hat{\epsilon} + 1 + \beta}. \quad (2.2.15)$$

We prove in Yee et al. [25] that the positive definite condition on  $Q_W$  requires that  $\dot{\psi} < 0$ , a condition already satisfied by (2.2.11), and that

$$\frac{1}{\beta} > \frac{-1}{1 + \dot{\epsilon}}. \quad (2.2.16)$$

Condition (2.2.16) is satisfied if  $\beta > 0$  or  $\beta < -(1 + \dot{\epsilon})$ . Since  $\dot{\epsilon} > 0$ , the maximum value of  $\dot{\epsilon}$  occurs at  $\hat{T}_{max}$ . Therefore, for  $\beta < 0$ ,

$$\beta < -[1 + \dot{\epsilon}(\hat{T}_{max})]. \quad (2.2.17)$$

A sufficiency condition, independent of the flow problem, is obtained by replacing  $\dot{\epsilon}(\hat{T}_{max})$  by  $\dot{\epsilon}(\infty)$ .

**Specialization for a Perfect Gas:** For a perfect gas, with a ratio of specific heats  $\gamma$ , the caloric equation of state becomes

$$\epsilon = \frac{\hat{T}}{(\gamma - 1)} \quad \text{and} \quad \dot{\epsilon} = \frac{1}{\gamma - 1}. \quad (2.2.18)$$

It follows that

$$\rho f = (p\rho^{-\gamma})^{\frac{-1}{(\gamma-1)}}, \quad (2.2.19)$$

$$\dot{\psi} = -(p\rho^{-\gamma})^{\frac{-1}{(\gamma-1)\beta}}, \quad (2.2.20)$$

$$a(\gamma, \beta) = \frac{\gamma - 1}{\gamma + (\gamma - 1)\beta} - 1, \quad (2.2.21)$$

and

$$b(\gamma, \beta) = \frac{-1}{\gamma + (\gamma - 1)\beta}. \quad (2.2.22)$$

The positive definite condition on  $\beta$  then becomes  $\beta > 0$  or  $\beta < \frac{-\gamma}{\gamma-1}$ . See Yee et al. [25] and Sandham & Yee [13] for a study on the beneficial ranges of  $\beta$  for a variety of flows.

### 8.2.3 Formulation in Generalized Curvilinear Coordinates

The equations presented so far can, in principle, be implemented by any numerical method, using any type of grid. Since our interest lies in efficient, high order accurate solutions, we will limit ourselves to finite difference formulation on a structured grid. In this section we therefore consider the formulation of the equations in generalized curvilinear coordinates. We first present the equations for a three-dimensional flow. They will then be specialized to the important case of axis-symmetric flow. The section concludes with a brief discussion of the Navier-Stokes equations.

### 8.2.3.1 Three-Dimensional Flow

An arbitrary, time-dependent transformation from curvilinear coordinates to physical space is written as

$$\mathbf{r} = \mathbf{r}(\xi, \eta, \zeta, \tau), \quad (2.3.1a)$$

$$t = \tau. \quad (2.3.1b)$$

For the computational cell  $d\xi, d\eta$  and  $d\zeta$ , the normalized surface area vectors in the  $\xi, \eta, \zeta$  directions are given by

$$\mathbf{S}^\xi = \mathbf{r}_\eta \times \mathbf{r}_\zeta, \quad \mathbf{S}^\eta = \mathbf{r}_\zeta \times \mathbf{r}_\xi, \quad \mathbf{S}^\zeta = \mathbf{r}_\xi \times \mathbf{r}_\eta. \quad (2.3.2)$$

The normalized cell volume is given by

$$\mathcal{V} = \mathbf{r}_\xi \cdot \mathbf{r}_\eta \times \mathbf{r}_\zeta \quad (2.3.3)$$

and the grid point velocity is given by

$$\mathbf{v} = \mathbf{r}_\tau. \quad (2.3.4)$$

Applying transformation (2.3.1) to the moving grid version of (2.1.1) we obtain

$$\widehat{Q}_\tau + \widehat{E}_\xi + \widehat{F}_\eta + \widehat{G}_\zeta = 0, \quad (2.3.5a)$$

where

$$\widehat{Q} = \mathcal{V}Q, \quad \widehat{E} = \mathbf{S}^\xi \cdot \mathbf{F}, \quad \widehat{F} = \mathbf{S}^\eta \cdot \mathbf{F}, \quad \widehat{G} = \mathbf{S}^\zeta \cdot \mathbf{F}. \quad (2.3.5b)$$

In what follows we will use numerical subscripts to indicate Cartesian components. Thus

$$\mathbf{S}^\xi = S_1^\xi \mathbf{i} + S_2^\xi \mathbf{j} + S_3^\xi \mathbf{k}, \quad (2.3.6)$$

with similar definitions for  $\mathbf{S}^\eta$ ,  $\mathbf{S}^\zeta$ , and  $\mathbf{v}$ . Let

$$\widehat{v}^\xi = \mathbf{S}^\xi \cdot \mathbf{v} = S_1^\xi v_1 + S_2^\xi v_2 + S_3^\xi v_3, \quad (2.3.7)$$

with similar definitions for  $\widehat{v}^\eta$  and  $\widehat{v}^\zeta$ . For the Euler equations, the transformed flux  $\widehat{E}$  is given by

$$\widehat{E} = \begin{bmatrix} \rho \widehat{U} \\ \rho u \widehat{U} + S_1^\xi p \\ \rho v \widehat{U} + S_2^\xi p \\ \rho w \widehat{U} + S_3^\xi p \\ (e + p) \widehat{U} + \widehat{v}^\xi p \end{bmatrix}, \quad (2.3.8)$$

where

$$\widehat{U} = \mathbf{S}^\xi \cdot \mathbf{u}' = S_1^\xi u + S_2^\xi v + S_3^\xi w - \widehat{v}^\xi. \quad (2.3.9)$$

The transformed flux derivative  $\widehat{E}_\xi$  is now split as

$$\widehat{E}_\xi = \frac{\beta}{\beta+1} \widehat{E}_\xi + \frac{1}{\beta+1} E_W^* W_\xi, \quad (2.3.10)$$

where  $E_W^* = \mathbf{S}^\xi \cdot \mathbf{F}_W$  is given by

$$E_W^* = \frac{1}{\psi} \begin{bmatrix} a\rho\widehat{U} & a\rho u\widehat{U} - S_1^\xi p & a\rho v\widehat{U} - S_2^\xi p & a\rho w\widehat{U} - S_3^\xi p & e_{15} \\ (\rho u^2 - p)\widehat{U} - 2uS_1^\xi p & \rho uv\widehat{U} - p(S_2^\xi u + S_1^\xi v) & e_{24} & e_{25} \\ (\rho v^2 - p)\widehat{U} - 2vS_2^\xi p & e_{34} & e_{35} \\ e_{44} & e_{45} \\ e_{55} \end{bmatrix}, \quad (2.3.11)$$

and

$$e_{15} = [ae + (b-1)p]\widehat{U} - \widehat{v}^\xi p, \quad (2.3.12a)$$

$$e_{24} = a\rho w\widehat{U} - p(S_3^\xi u + S_1^\xi w), \quad (2.3.12b)$$

$$e_{25} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}u - \frac{p}{\rho}(e+p)S_1^\xi, \quad (2.3.12c)$$

$$e_{34} = a\rho v\widehat{U} - p(S_3^\xi v + S_2^\xi w), \quad (2.3.12d)$$

$$e_{35} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}v - \frac{p}{\rho}(e+p)S_2^\xi, \quad (2.3.12e)$$

$$e_{44} = (a\rho w^2 - p)\widehat{U} - 2wS_3^\xi p, \quad (2.3.12f)$$

$$e_{45} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}w - \frac{p}{\rho}(e+p)S_3^\xi, \quad (2.3.12g)$$

$$e_{55} = \left[ \frac{ae^2}{\rho} + p\{2(b-1)\frac{e}{\rho} - q^2\} + \frac{p^2}{\rho}\{b(1+\beta) - 2\} \right] \widehat{U} - 2\widehat{v}^\xi \frac{p}{\rho}(e+p). \quad (2.3.12h)$$

The analogous expressions for  $\widehat{F}$  and  $\widehat{F}_\eta$  are obtained from (2.3.8) through (2.3.12) by replacing  $\widehat{U}$  with  $\widehat{V}$  and  $\xi$  with  $\eta$  throughout. Similarly, the expressions for  $\widehat{G}$  and  $\widehat{G}_\zeta$  are obtained by replacing  $\widehat{U}$  with  $\widehat{W}$  and  $\xi$  with  $\zeta$  throughout. Here  $\widehat{V}$  and  $\widehat{W}$  are (2.3.9) with  $\mathbf{S}^\xi$  replaced by  $\mathbf{S}^\eta$  and  $\mathbf{S}^\zeta$  respectively, and  $\widehat{W}$  has no relationship to the entropy splitting vector  $W$  in (2.2.12).

Normally, we need to compute  $\widehat{Q}_W$  for the split form of  $\widehat{Q}_\tau = \frac{\beta}{\beta+1} \widehat{Q}_\tau + \frac{1}{\beta+1} \widehat{Q}_W W_\tau$ . However, we only consider a semi-discrete approach of applying temporal discretizations. Aside from using the split form of the inviscid flux derivatives  $\widehat{E}_\xi$ ,  $\widehat{F}_\eta$  and  $\widehat{G}_\zeta$ , we do not have to use the split form of  $\widehat{Q}_\tau$  for implementation. Thus the final form of the semi-discrete entropy splitting approach still can be expressed in terms of conservative and primitive

variables, making possible easy and efficient implementation in existing computer codes.

From definitions (2.3.2) we can derive the  
Surface Area Conservation Law

$$(\mathbf{S}^\xi)_\xi + (\mathbf{S}^\eta)_\eta + (\mathbf{S}^\zeta)_\zeta = 0, \quad (2.3.13)$$

which is valid for each of the Cartesian components. For time-varying grids, by combining (2.3.13) with (2.3.5), and assuming a uniform flow, we derive the

Volume Conservation Law

$$\mathcal{V}_\tau = (\hat{v}^\xi)_\xi + (\hat{v}^\eta)_\eta + (\hat{v}^\zeta)_\zeta. \quad (2.3.14)$$

Note that we have not written relations (2.3.2) in their component forms. We will show in Section 3.3.1 that in order to satisfy (2.3.13) numerically, these relations must be modified. Finally, we relate our notation to the more familiar one introduced by Steger [17]. These are

$$\mathcal{V} = J^{-1}, \quad \hat{U} = J^{-1}U, \quad S_1^\xi = J^{-1}\xi_x, \quad \hat{v}^\xi = -J^{-1}\xi_t, \quad (2.3.15)$$

with analogous relations for the other quantities, where Steger defines  $J$  as the Jacobian of the transformation and  $U$  is his contravariant velocity in the  $\xi$ -direction.

### 8.2.3.2 Axi-Symmetric Flow

In order to obtain the equations for axi-symmetric flow, we first introduce cylindrical coordinates  $x, r, \zeta$ , where the  $x$ -axis is the polar axis and  $\zeta$  is the polar angle. Introducing the curvilinear coordinates  $\xi, \eta$  in the  $x$ - $r$  plane, we have the transformation equations

$$x = x(\xi, \eta, \tau) \quad \text{and} \quad r = r(\xi, \eta, \tau), \quad (2.3.16)$$

and

$$y = r \cos \zeta \quad \text{and} \quad z = r \sin \zeta. \quad (2.3.17)$$

The surface area vectors can then be obtained from (2.3.2). In particular, the components of  $\mathbf{S}^\zeta$  become

$$S_1^\zeta = 0, \quad S_2^\zeta = -S^\zeta \sin \zeta, \quad S_3^\zeta = S^\zeta \cos \zeta, \quad (2.3.18)$$

where

$$S^\zeta \equiv x_\xi r_\eta - r_\xi x_\eta. \quad (2.3.19)$$

The cell volume is given by (2.3.3) as

$$\mathcal{V} = r S^\zeta. \quad (2.3.20)$$

For axi-symmetric flow, the  $\zeta$ -components of  $\mathbf{u}$  and  $\mathbf{v}$ , as well as the  $\zeta$  derivatives of physical quantities, are all equal to 0. It follows that  $\widehat{v}^\zeta = \widehat{W} = 0$ . The transformed flux  $\widehat{G}$  then becomes

$$\widehat{G} = S^\zeta p [0 \quad 0 \quad -\sin \zeta \quad \cos \zeta \quad 0]^T, \quad (2.3.21)$$

while its  $\zeta$  derivative is

$$\widehat{G}_\zeta = -S^\zeta p [0 \quad 0 \quad \cos \zeta \quad \sin \zeta \quad 0]^T. \quad (2.3.22)$$

The axi-symmetric equations are obtained by setting  $\zeta = 0$ . Then the  $\zeta$ -component of  $\mathbf{u}$  becomes  $w$ , which is set equal to 0. It follows from (2.3.17) that  $z = 0$  and  $r$  can be replaced by  $y$  in (2.3.16), (2.3.19), and (2.3.20).  $\widehat{G}_\zeta$  in (2.3.4) then becomes the source term

$$\widehat{G}_\zeta = -S^\zeta p [0 \quad 0 \quad 1 \quad 0 \quad 0]^T. \quad (2.3.23)$$

Finally, the fourth components of all *vectors*, and fourth rows and columns of all matrices can be eliminated. Note that all the relations for the surface area components must be modified before numerical implementation, as will be indicated in Section 3.3.2. We should also mention that certain limiting forms of some relations must be used for points on the axis of symmetry.

### 8.2.3.3 2.3.3. Navier-Stokes Equations

So far we have treated only the Euler equations, since the canonical splitting is only applied to inviscid terms. For completeness, we make a few remarks about the Navier-Stokes equations, where the additional viscous terms are treated in a standard manner, which are well documented. The system of conservation laws is now written as

$$Q_t + \nabla \cdot (\mathbf{F} - \mathbf{F}_v) = 0. \quad (2.3.24)$$

Relative to a moving grid,  $\mathbf{F}$  is given by (2.2.3), and the viscous flux *vector*  $\mathbf{F}_v$  is given by

$$\mathbf{F}_v = \begin{bmatrix} 0 \\ \mathbf{i} \cdot \boldsymbol{\tau} \\ \mathbf{j} \cdot \boldsymbol{\tau} \\ \mathbf{k} \cdot \boldsymbol{\tau} \\ -q + \boldsymbol{\tau} \cdot \mathbf{u} \end{bmatrix}. \quad (2.3.25)$$

The assumption of a Newtonian fluid yields the relations

$$\boldsymbol{\tau} = \lambda(\nabla \cdot \mathbf{u})\mathbf{I} + \mu[\nabla \mathbf{u} + (\nabla \mathbf{u})^T] \quad (2.3.26)$$

and

$$q = -k\nabla T. \quad (2.3.27)$$

Here  $\mu$  and  $\lambda$  are the first and second viscosity coefficients, and  $k$  is the thermal conductivity.  $\mathbf{I}$  is the identity tensor. In order to express (2.3.25) in generalized coordinates we make use of the relation

$$\nabla = \frac{1}{\mathcal{V}} \left[ \mathbf{S}^\xi \frac{\partial}{\partial \xi} + \mathbf{S}^\eta \frac{\partial}{\partial \eta} + \mathbf{S}^\zeta \frac{\partial}{\partial \zeta} \right]. \quad (2.3.28)$$

The final form of the equations in curvilinear coordinates and the corresponding high order base scheme will not be given here. They may be found in standard references. Here we will only mention the modification of the source term for axi-symmetric flow. The coefficient in (2.3.23) must be modified to account for the normal viscous stress acting in the azimuthal direction. Thus the pressure  $p$  in (2.3.23) must be replaced by

$$p - \lambda \nabla \cdot \mathbf{u} - \frac{2\mu v}{y}. \quad (2.3.29)$$

### 8.3 Numerical Methods

The spatial discretizations proposed in Yee et al. [24] consist of two parts, namely, a base scheme and a filter. When filters are not used, the scheme consists of only the base scheme, which is discussed in Section 8.3.1. The base scheme involves compact or non-compact high order central difference approximations, which are used to evaluate the flux derivatives  $\hat{E}_\xi$ ,  $\hat{F}_\eta$ ,  $\hat{G}_\zeta$ , the surface area components that they contain, and the derivatives  $W_\xi$ ,  $W_\eta$  and  $W_\zeta$ . The form of the filter, which is applied to inviscid terms, is discussed in Section 3.2. The special treatment of the geometric terms necessary to satisfy the surface area conservation law numerically for the base scheme and the filter is described in Sections 3.3 and 3.4. Note that if the filters are of the same spatial order of accuracy as the base scheme, their surface area component discretizations are the same as for the base scheme. If the filters are of lower order of accuracy than the base scheme, for consistency, the corresponding order of discretization for the surface area components need only match the order of the filter.

#### 8.3.1 Spatial Base Schemes

Let  $\varphi_{i,j,k}$  denote the discrete approximation of a function  $\varphi$  at  $(i\Delta\xi, j\Delta\eta, k\Delta\zeta)$ , where  $\Delta\xi$ ,  $\Delta\eta$  and  $\Delta\zeta$  are the grid spacing in the  $\xi$ ,  $\eta$ ,  $\zeta$ -directions, and  $(i, j, k)$  are the corresponding spatial indices. The  $\varphi$  can be a function of  $\hat{Q}$  and/or  $(\xi, \eta, \zeta, \tau)$ . Possible high order central difference approximations for  $\varphi_\xi$  at  $(i\Delta\xi, j\Delta\eta, k\Delta\zeta)$  (with analogous expressions for  $\varphi_\eta$  and  $\varphi_\zeta$ ) can be of the following two types:

Central Differencings: (fourth and sixth-order)

$$\varphi_\xi \approx \frac{-1}{12\Delta\xi} \left( \varphi_{i+2,j,k} - 8\varphi_{i+1,j,k} + 8\varphi_{i-1,j,k} - \varphi_{i-2,j,k} \right), \quad (3.1.1)$$

$$\varphi_\xi \approx \frac{1}{60\Delta\xi} \left( \varphi_{i+3,j,k} - 9\varphi_{i+2,j,k} + 45\varphi_{i+1,j,k} - 45\varphi_{i-1,j,k} + 9\varphi_{i-2,j,k} - \varphi_{i-3,j,k} \right). \quad (3.1.2)$$

Compact Central Differencings: (fourth and sixth-order)

$$\varphi_\xi \approx \frac{1}{\Delta\xi} \left( A_\xi^{-1} B_\xi \varphi \right)_{i,j,k}, \quad (3.1.3a)$$

where for a fourth-order approximation

$$(A_\xi \varphi)_{i,j,k} = \frac{1}{6} \left( \varphi_{i+1,j,k} + 4\varphi_{i,j,k} + \varphi_{i-1,j,k} \right), \quad (3.1.3b)$$

$$(B_\xi \varphi)_{i,j,k} = \frac{1}{2} \left( \varphi_{i+1,j,k} - \varphi_{i-1,j,k} \right), \quad (3.1.3c)$$

and for a sixth-order approximation

$$(A_\xi \varphi)_{i,j,k} = \frac{1}{5} \left( \varphi_{i+1,j,k} + 3\varphi_{i,j,k} + \varphi_{i-1,j,k} \right), \quad (3.1.3d)$$

$$(B_\xi \varphi)_{i,j,k} = \frac{1}{60} \left( \varphi_{i+2,j,k} + 28\varphi_{i+1,j,k} - 28\varphi_{i-1,j,k} - \varphi_{i-2,j,k} \right). \quad (3.1.3e)$$

These central difference approximations are used in applying non-dissipative high order base schemes for terms such as  $\tilde{E}_\xi$ . For the Navier-Stokes equations, the approximations are first used to evaluate the viscous stress tensor  $\tau$  and heat-flux vector  $q$ , and used again to obtain the derivatives of the viscous flux  $\mathbf{F}_v$ . The surface area vector components appearing in all these terms are also evaluated by using these difference approximations, but Eqs. 2.3.2 must first be modified in order to satisfy (2.3.13) exactly. This will be discussed in Section 3.3. For direct numerical simulation (DNS) of compressible turbulent computations, Sandham and Yee use a Laplacian formulation of the viscous and heat conduction terms to ensure that any tendency to odd-even decoupling associated with central schemes can be countered by the fluid viscosity. They also use a special formulation of the continuity equation based on similar arguments.

An alternate method of treating the viscous stress tensor and heat flux in the Navier-Stokes equations is to evaluate them at the half grid points. Using an appropriate high order central difference formula, one would obtain stencils



for the viscous fluxes that are no wider than those for the Euler fluxes. This would require high order interpolation to evaluate the surface area vector components and transport coefficients at the half grid points. The shifted surface area vector components would not satisfy (2.3.13), but this would not affect the preservation of a uniform flow, since the viscous terms give a zero contribution. This alternate method is currently being investigated.

### 8.3.2 Filters

In this section we first review the procedure for applying the characteristic filter to multistage Runge-Kutta type and linear multistep method (LMM) types of time discretizations (Yee et al. [24]). Examples of explicit LMMs are forward Euler and Adams-Bashforth. Examples of implicit LMMs are backward Euler, trapezoidal rule, and three-point backward differentiation. The one-leg formulation of the LMMs of Dahlquist [2] is also applicable. We then discuss forms of the characteristic filter.

#### 8.3.2.1 Procedure to Apply the Filter Step

If a multistage time discretization such as the Runge-Kutta method is desired, the spatial differencing base scheme discussed in the previous section is applied at every stage of the Runge-Kutta method. If viscous terms are present, we use the same order and type of base scheme for the viscous terms as for the inviscid terms.

There are two methods for applying the characteristic filter. Method 1 is to apply the filter at every stage of the Runge-Kutta step. Method 2 is to apply the filter at the end of the full Runge-Kutta step. For inviscid and strong shock interactions, method 1 might be more stable.

If one desires a time discretization that belongs to the class of LMMs, then the filter can be applied as a numerical dissipation vector in conjunction with the base scheme. The filter in this case is evaluated at  $\hat{Q}^n$  for explicit LMMs. For implicit LMMs additional similar filters evaluated at the  $n + 1$  time level are involved. Alternatively, method 2 can be applied to LMMs as well. In this case, we apply the filter after the completion of the implicit time step. One can minimize flux evaluations by using the one-leg formulation of the LMMs of Dahlquist. The only non-dissipative (in time) second-order, two-time level one-leg method is the mid-point implicit method. Note that the noniterative linearized form of the midpoint implicit formula reduces to the regular noniterative linearized trapezoidal formula.

For time marching to steady states using implicit LMMs, certain flow physics only require an explicit dissipation term. Also, the implicit operator can be different from the explicit operator. See Yee [22] and Yee et al. [23] for some efficient conservative linearized implicit forms.

### 8.3.2.2 Nonlinear Characteristic Filters

There are many possible candidates for the filter operator in conjunction with high order base schemes. Here, we prefer using filter operators whose grid stencils have a width similar to that of the base scheme for efficiency and ease of numerical boundary treatment. Higher than third-order filter operators are of course applicable, but they are more CPU intensive and require special treatment near boundary points for stability and accuracy. The filter operator usually consists of the product of a sensor and a nonlinear dissipation. Two possible sensors are considered, the ACM sensor (Harten [5]) and the wavelet sensor (Sjogreen & Yee [14]). See Table 8.1 for the road map. Here we briefly review the ACM sensor and interested readers are referred to Sjogreen & Yee [14] for the wavelet sensor.

Yee et al [24] use nonlinear dissipation terms in conjunction with the Harten ACM sensor applied to each characteristic wave as the filter vector. In essence, the nonlinear dissipation terms act as second or third-order ACM-like operators, unlike Harten's first-order ACM (Harten [5]). The sensor is used to signal the amount of nonlinear dissipation to be added to the high order non-dissipative scheme, one wave at a time. Thus, the current approach is also different in spirit from the original Harten [6] second-order TVD scheme which uses ACM to sharpen the contact discontinuities. Let the filter vector in the  $\xi$ -direction be of the form

$$\tilde{E}_{i+\frac{1}{2},j,k}^* = \frac{1}{2} R_{i+\frac{1}{2}} \Phi_{i+\frac{1}{2}}^* (S^\xi)_{i+\frac{1}{2},j,k}. \quad (3.2.1)$$

$\tilde{E}_{i+\frac{1}{2},j,k}^*$  is the modified form of the nonlinear dissipation portion of the standard numerical flux in curvilinear coordinates.  $(S^\xi)_{i+\frac{1}{2},j,k}$  is the magnitude of the surface area vector  $\mathbf{S}^\xi$  at the  $(i + 1/2, j, k)$  averaged state. The quantity  $R_{i+\frac{1}{2}}$  is the right eigenvector matrix given in Section 3.4.1, using, Roe's approximate solver (Roe's averaged state) with the  $(k, l)$  indices suppressed. We define  $\tilde{F}_{i,j+\frac{1}{2},k}^*$  and  $\tilde{G}_{i,j,k+\frac{1}{2}}^*$  in the same manner. The elements of  $\Phi_{i+\frac{1}{2}}^*$ , denoted by  $(\phi_{i+\frac{1}{2}}^l)^*$  with the  $(k, l)$  indices suppressed, are

$$(\phi_{i+\frac{1}{2}}^l)^* = \kappa \theta_{i+\frac{1}{2}}^l \phi_{i+\frac{1}{2}}^l. \quad (3.2.2)$$

$\phi_{i+\frac{1}{2}}^l$  in (3.2.2) are the elements of the dissipative portion of the high resolution schemes resulting from using a TVD, MUSCL with slope limiters, ENO or WENO scheme. Hereafter, we refer to (3.2.2) as the ACM filter. Formulae for  $\phi_{i+\frac{1}{2}}^l$  involve the eigenvalues, eigenvectors and Roe's averaged state defined in Sections 3.4.1 and 3.4.2, and can be found in the literature (Yee [21, 22], and Yee et al. [23]). See Yee et al. [24, 25] for details and for a discussion of other possible filters.

The function  $\kappa\theta_{i+\frac{1}{2}}^l$  is a mechanism to control the excess numerical dissipation inherent in the dissipative portion of high-resolution shock-capturing schemes. In other words, the elements of  $\Phi_{i+\frac{1}{2}}^*$  are the same as the nonlinear dissipation portion of the TVD, ENO or WENO scheme, multiplied by  $\kappa\theta_{i+\frac{1}{2}}^l$ . The parameter  $\kappa$  is problem dependent. For smooth flows,  $\kappa$  can be very small. It is used to minimize spurious high frequency oscillation producing nonlinear instability. Different physical problems may require different values of  $\kappa$  because of variation in flow properties. The  $\kappa$  value may vary from one characteristic wave to another, and from one region of the flow field to another region with a different flow structure. The range of  $\kappa$  for the numerical experiments in Yee et al. [24, 25] is  $0.0 \leq \kappa \leq 0.7$ . The function  $\theta_{i+\frac{1}{2}}^l$  is the Harten ACM sensor. For a general  $2m + 1$ -point base scheme, Harten recommended

$$\theta_{i+\frac{1}{2}}^l = \max(\widehat{\theta}_{i-m+1}^l, \dots, \widehat{\theta}_{i+m}^l), \quad (3.2.3)$$

$$\widehat{\theta}_i^l = \left| \frac{|\alpha_{i+\frac{1}{2}}^l| - |\alpha_{i-\frac{1}{2}}^l|}{|\alpha_{i+\frac{1}{2}}^l| + |\alpha_{i-\frac{1}{2}}^l|} \right|^p. \quad (3.2.4)$$

The  $\alpha_{i+\frac{1}{2}}^l$  are elements of  $R_{i+\frac{1}{2}}^{-1}(Q_{i+1,j,k} - Q_{i,j,k})$  defined in Section 3.4.1. They are used in (3.2.4), and are needed for  $\phi_{i+\frac{1}{2}}^l$  in (3.2.2) as well. The parameter  $p$  in (3.2.4) is an exponent  $\geq 1$  and is not the “pressure  $p$ ”. The  $\phi_{i+\frac{1}{2}}^l$ ,  $\alpha_{i+\frac{1}{2}}^l$  and  $R_{i+\frac{1}{2}}$  for the corresponding numerical dissipation using the MUSCL formulation are now functions of the left and right states of  $Q$ .

Instead of varying  $\kappa$  for the particular flow problem, one can vary  $p$ . For larger  $p$ , less numerical dissipation is added. Note that by varying  $p \geq 1$  in (3.2.4), one can essentially increase the order of accuracy of the dissipation term. The order of the dissipation depends on the value of  $p$ . One can switch from  $p = 1$  near shock locations to  $p > 1$  at smooth regions. For all of the numerical examples in Yee et al. [24, 25], they use  $p = 1$  and

$$\theta_{i+\frac{1}{2}}^l = \max(\widehat{\theta}_i^l, \widehat{\theta}_{i+1}^l). \quad (3.2.5)$$

The shock-turbulence interaction problems appear to favor this form of  $\theta_{i+\frac{1}{2}}^l$ . To avoid the need to tune the arbitrary parameter  $\kappa$ , Sjogreen & Yee [14] suggest replacing  $\kappa\theta_{i+\frac{1}{2}}^l$  by a multi-resolution non-orthogonal wavelet sensor  $\omega_{i+\frac{1}{2}}^l$ . Their study shows that the wavelet sensor indeed removes the parameter dependence  $\kappa$  and  $p$  with comparable accuracy for the same numerical examples illustrated in Yee et al. [24, 25]. Unlike the ACM sensor, the wavelet sensor can detect most of the distinct flow features including turbulence, leading to an automatic selection of the appropriate distribution of numerical dissipation and a good grid adaptation indicator.

To avoid some conditional statements in the actual computer code and to promote vectorization, several of the functions inside the filter with the potential of dividing by zero are modified. In particular, the sensor (3.2.4), with  $p = 1$ , is replaced by

$$\theta_i^l = \frac{||\alpha_{i+1/2}^l| - |\alpha_{i-1/2}^l||}{|\alpha_{i+1/2}^l| + |\alpha_{i-1/2}^l| + \epsilon}. \quad (3.2.6)$$

In all of the computations, they take  $\epsilon = 10^{-7}$ . (Actually,  $\epsilon$  should have the same dimension as  $\alpha_{i+1/2}^l$ .)

It is emphasized here that neither ACM nor wavelets sensors will be able to improve the accuracy at the shock and shear locations over the inherent shock-capturing capability of the nonlinear dissipation. The accuracy of the shock and shear is dictated by the chosen nonlinear dissipation. The role of the sensors is to allow the full amount of numerical dissipation in shock and shear regions, and to limit the amount of numerical dissipation in regions immediately away from shock and shear locations and the rest of the flow field. Therefore, with a suitable sensor, one does not have to use CPU-intensive high order high-resolution shock-capturing numerical dissipation, since this type of dissipation generally only gives a slightly more accurate solution away from discontinuities while exhibiting the same shock and shear resolution as the high-resolution second or third-order numerical dissipations.

Finally, we note that the ACM filter (3.2.2) or wavelet filter might not be sufficient for (a) time-marching to steady state and (b) spurious high frequency oscillations due to insufficient grid resolution and nonlinear instability away from discontinuities. See Yee et al. [25] for a discussion.

**A Form of the Full Discretization:** Here we illustrate the form of the full discretization using Runge-Kutta methods with the filters applied at the completion of a full Runge-Kutta time step. Let  $\tilde{Q}^{n+1}$  be the solution after one full Runge-Kutta time step using a non-dissipative base scheme. If entropy splitting is employed, the base scheme is applied to the split form of the inviscid flux derivatives. The solution at the next time level  $\hat{Q}^{n+1}$  is

$$\begin{aligned} \hat{Q}_{i,j,k}^{n+1} = \tilde{Q}_{i,j,k}^{n+1} + \frac{\Delta t}{\Delta \xi} \left[ \tilde{E}_{i+\frac{1}{2},j,k}^* - \tilde{E}_{i-\frac{1}{2},j,k}^* \right] &+ \frac{\Delta t}{\Delta \eta} \left[ \tilde{F}_{i,j+\frac{1}{2},k}^* - \tilde{F}_{i,j-\frac{1}{2},k}^* \right] \\ &+ \frac{\Delta t}{\Delta \zeta} \left[ \tilde{G}_{i,j,k+\frac{1}{2}}^* - \tilde{G}_{i,j,k-\frac{1}{2}}^* \right]. \end{aligned} \quad (3.2.7)$$

Here,  $\tilde{E}_{i\pm\frac{1}{2},j,k}^*$ ,  $\tilde{F}_{i,j\pm\frac{1}{2},k}^*$  and  $\tilde{G}_{i,j,k\pm\frac{1}{2}}^*$  are evaluated at  $\tilde{Q}^{n+1}$ .

### 8.3.3 Discretization of Geometric Terms for the Base Scheme

For the base scheme, all first partial derivatives with respect to the transformed variables, such as those in (2.3.10), are approximated by the high order compact and non-compact central difference formulas of Section 3.1. But the surface area vector components in the transformed fluxes and the cell volume in the transformed conservative *vector* are also defined in terms of partial derivatives of the Cartesian coordinates. It would be desirable for the numerical approximations to the surface area vectors to satisfy the surface area conservation law (2.3.13) exactly (within round-off errors). For time-varying grids, we would also like the volume conservation law (2.3.14) to be satisfied numerically. This would eliminate numerical errors in regions of uniform flow, and reduce grid induced errors even for nonuniform flows. Furthermore, in deriving the split formula (2.3.10), (2.3.13) was used in obtaining the expression for the second term.

It was shown by Vinokur [20] that the two geometric conservation laws can be satisfied numerically to second-order accuracy by appealing to finite-volume concepts. Unfortunately, this approach cannot be extended to higher orders of accuracy. Calculating the surface area vectors by applying the high order difference formulas of Section 3.1 to (2.3.2) will result in truncation errors when evaluating (2.3.13). Gaitonde & Visbal [3] showed, by performing some numerical experiments, that (2.3.13) can be satisfied numerically by rewriting the component version of (2.3.2) in an equivalent “conservative” form first proposed by Thomas & Lombard [18]. That form is not coordinate invariant. In Section 3.3.1 we present a coordinate invariant form of the Thomas and Lombard relations for three-dimensional flow. We then derive the corresponding correct relations for axi-symmetric flow in Section 3.3.2. This cannot be accomplished using the original Thomas and Lombard formulas. We will also demonstrate theoretically in Appendix A that applying any arbitrary compact or non-compact difference formulas to the new form of the surface area vector definition (2.3.2) will satisfy the surface area conservation law (2.3.13) exactly.

#### 8.3.3.1 Three-Dimensional Flow

By examining the Thomas and Lombard formulas, one can readily obtain the coordinate invariant form for  $\mathbf{S}^\xi$  as

$$\mathbf{S}^\xi = \frac{1}{2}[(\mathbf{r}_\eta \times \mathbf{r})_\zeta - (\mathbf{r}_\zeta \times \mathbf{r})_\eta], \quad (3.3.1)$$

with the corresponding expressions for  $\mathbf{S}^\eta$  obtained by replacing  $\eta$  and  $\zeta$  by  $\zeta$  and  $\xi$ , respectively, and for  $\mathbf{S}^\zeta$  obtained by replacing  $\eta$  and  $\zeta$  by  $\xi$  and  $\eta$ , respectively. Analytically, (3.3.1) gives the same answer as provided by (2.3.2), noting that since the mixed partial derivatives commute,  $\mathbf{r}_{\eta\zeta} \times \mathbf{r} - \mathbf{r}_{\zeta\eta} \times \mathbf{r} = 0$ .

The commuting property also assures that the answer for  $\mathbf{S}^\xi$  is independent of the choice of origin for  $\mathbf{r}$ , and that (2.3.13) is identically satisfied. In order to show that (2.3.13) is satisfied exactly numerically, we must prove that the mixed partial derivatives commute **numerically**. This we do in Appendix A.

In component form, (3.3.1) is written as

$$S_1^\xi = \frac{1}{2}[(y_\eta z - z_\eta y)_\zeta - (y_\zeta z - z_\zeta y)_\eta], \quad (3.3.2a)$$

$$S_2^\xi = \frac{1}{2}[(z_\eta x - x_\eta z)_\zeta - (z_\zeta x - x_\zeta z)_\eta], \quad (3.3.2b)$$

$$S_3^\xi = \frac{1}{2}[(x_\eta y - y_\eta x)_\zeta - (x_\zeta y - y_\zeta x)_\eta]. \quad (3.3.2c)$$

Expressions for components of  $\mathbf{S}^\eta$  and  $\mathbf{S}^\zeta$  are obtained in an analogous manner. The Thomas and Lombard forms used twice the first term in each parenthesis. While this satisfies (2.3.13), it will not give the correct formulas for axi-symmetric flow, as will be discussed in Section 3.3.2.

From (2.3.7) we see that the transformed grid velocities involve products of surface area components and grid velocity components, which cannot be expressed in “conservative” form. Consequently, we cannot express  $\mathcal{V}$  numerically so as to satisfy (2.3.14) exactly, when the surface area components are given by (3.3.2). We therefore calculate  $\mathcal{V}$  from the invariant expression (2.3.3) as

$$\mathcal{V} = \begin{vmatrix} x_\xi & y_\xi & z_\xi \\ x_\eta & y_\eta & z_\eta \\ x_\zeta & y_\zeta & z_\zeta \end{vmatrix}. \quad (3.3.3)$$

### 8.3.3.2 Axi-Symmetric Flow

The axi-symmetric surface area components are obtained by substituting (2.3.16) and (2.3.17) into (3.3.2), and then setting  $\zeta = 0$ . From (2.3.17) it follows that  $z = 0$  and  $r$  can be replaced by  $y$ . The resulting expressions are

$$S_1^\xi = \frac{1}{2}(y^2)_\eta \quad S_2^\xi = \frac{1}{2}[y_\eta x - x_\eta y - (xy)_\eta] \quad S_3^\xi = 0, \quad (3.3.4a)$$

$$S_1^\eta = -\frac{1}{2}(y^2)_\xi \quad S_2^\eta = -\frac{1}{2}[y_\xi x - x_\xi y - (xy)_\xi] \quad S_3^\eta = 0, \quad (3.3.4b)$$

$$S_1^\zeta = 0 \quad S_2^\zeta = 0 \quad S_3^\zeta = S^\zeta, \quad (3.3.4c)$$

where

$$S^\zeta = \frac{1}{2}[(x_\xi y - y_\xi x)_\eta - (x_\eta y - y_\eta x)_\xi]. \quad (3.3.5)$$

We also have the relations

$$(S_1^\zeta)_\zeta = 0 \quad (S_2^\zeta)_\zeta = -S^\zeta \quad (S_3^\zeta)_\zeta = 0. \quad (3.3.6)$$

Substituting (3.3.4), (3.3.5), and (3.3.6) into (2.3.13), we see that the surface area conservation law is satisfied. The cell volume is given by

$$\mathcal{V} = yS^\zeta. \quad (3.3.7)$$

If we used the Thomas and Lombard form of (3.3.2), the resulting equations (3.3.4) to (3.3.6) would have different expressions. While they would still satisfy (2.3.13), they would lead to numerical errors which violate the requirements of axial symmetry. This is a direct result of the lack of coordinate invariance in their form. Equations (3.3.4c) would still hold, with (3.3.5) replaced by twice the first term in each parenthesis; that is

$$S^\zeta = (x_\xi y)_\eta - (x_\eta y)_\xi. \quad (3.3.8)$$

The expressions for their  $\zeta$  derivatives would not be physically correct. The first two would now be

$$(S_1^\zeta)_\zeta = (yy_\xi)_\eta - (yy_\eta)_\xi \quad (S_2^\zeta)_\zeta = (xy_\xi)_\eta - (xy_\eta)_\xi. \quad (3.3.9)$$

Analytically,  $(S_1^\zeta)_\zeta = 0$  and  $(S_2^\zeta)_\zeta = -S^\zeta$ , corresponding to the fact that the constant  $\zeta$  surfaces are planes parallel to the  $x$ -axis. But numerically, these two conditions are only satisfied to the order of the truncation error of the method. For both these reasons, this would result in small contributions to the source term  $\hat{G}_\zeta$  in both the continuity,  $x$ -momentum and energy equations. By using the coordinate invariant expressions (3.3.2), we satisfy (3.3.6) exactly, and avoid these spurious numerical results.

### 8.3.4 Riemann Solver

#### 8.3.4.1 Flux Jacobians and Eigenvectors

Characteristic based filters require the use of eigenvalues and eigenvectors of flux Jacobian matrices at averaged states. They are best treated using the original conservative *vector*  $Q$  rather than  $\hat{Q}$ . Let  $\mathbf{S}^\xi = S^\xi \mathbf{n}^\xi$ , where  $\mathbf{n}^\xi$  is the unit normal and  $S^\xi$  is the magnitude of the surface area vector  $\mathbf{S}^\xi$ . Similar definitions apply to  $\mathbf{S}^\eta$  and  $\mathbf{S}^\zeta$ . We omit the curvilinear coordinate superscript from now on. At any averaged state, only the direction of the appropriate normal  $\mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k}$  plays a role. Let  $u^n = \mathbf{u} \cdot \mathbf{n}$ ,  $v^n = \mathbf{v} \cdot \mathbf{n}$ , and  $u' = u^n - v^n$ . The transformed flux *vector* can be written as  $\mathbf{S} \cdot \mathbf{F} = SF^n$ , where  $F^n = \mathbf{n} \cdot \mathbf{F}$ . The flux Jacobian matrix is then defined as  $A = (F^n)_Q = \partial F^n / \partial Q$ .

We will first develop relations for a general equilibrium gas. The pressure  $p$  is given by a general equation of state of the form

$$p = p(\rho, \tilde{\epsilon}). \quad (3.4.1)$$

The derivatives will be denoted by

$$\chi = \left( \frac{\partial p}{\partial \rho} \right)_{\tilde{\epsilon}} \quad \text{and} \quad \kappa = \left( \frac{\partial p}{\partial \tilde{\epsilon}} \right)_{\rho}. \quad (3.4.2)$$

If  $h = (\tilde{\epsilon} + p)/\rho$  is the specific enthalpy, the speed of sound  $c$  can then be expressed as

$$c^2 = \chi + \kappa h. \quad (3.4.3)$$

The matrix  $A$  can then be written as

$$A = \begin{bmatrix} -v^n & n_1 & n_2 & n_3 & 0 \\ Kn_1 - u^n u & (1 - \kappa)un_1 + u' & un_2 - \kappa n_1 v & un_3 - \kappa n_1 w & \kappa n_1 \\ Kn_2 - u^n v & vn_1 - \kappa n_2 u & (1 - \kappa)vn_2 + u' & vn_3 - \kappa n_2 w & \kappa n_2 \\ Kn_3 - u^n w & wn_1 - \kappa n_3 u & wn_2 - \kappa n_3 v & (1 - \kappa)wn_3 + u' & \kappa n_3 \\ (K - H)u^n & Hn_1 - \kappa u^n u & Hn_2 - \kappa u^n v & Hn_3 - \kappa u^n w & u' + \kappa u^n \end{bmatrix}, \quad (3.4.4)$$

where  $K = \frac{1}{2}\kappa q^2 + \chi$ , and  $H = h + \frac{1}{2}q^2$  is the total enthalpy per unit mass. The three distinct eigenvalues of  $A$  are

$$\lambda^1 = u', \quad \lambda^2 = u' + c, \quad \text{and} \quad \lambda^3 = u' - c, \quad (3.4.5)$$

where  $\lambda^1$  is a repeated eigenvalue. In order to obtain a corresponding set of linearly independent eigenvectors, we introduce an arbitrary orthonormal basis  $\mathbf{a}^i$  satisfying  $\mathbf{a}^i \cdot \mathbf{a}^j = \delta^{ij}$ , where  $\delta^{ij}$  is the Kronecker delta. One can then define  $a^{ni} = \mathbf{n} \cdot \mathbf{a}^i$  and  $\mathbf{b}^i = \mathbf{n} \times \mathbf{a}^i$ . The right eigenvector matrix  $R$  can then be written as

$$R = \begin{bmatrix} a^{n1} & a^{n2} & a^{n3} & 1 & 1 \\ a^{n1}u + cb_1^1 & a^{n2}u + cb_1^2 & a^{n3}u + cb_1^3 & u + cn_1 & u - cn_1 \\ a^{n1}v + cb_2^1 & a^{n2}v + cb_2^2 & a^{n3}v + cb_2^3 & v + cn_2 & v - cn_2 \\ a^{n1}w + cb_3^1 & a^{n2}w + cb_3^2 & a^{n3}w + cb_3^3 & w + cn_3 & w - cn_3 \\ a^{n1}\tilde{K} + c(\mathbf{b}^1 \cdot \mathbf{u}) & a^{n2}\tilde{K} + c(\mathbf{b}^2 \cdot \mathbf{u}) & a^{n3}\tilde{K} + c(\mathbf{b}^3 \cdot \mathbf{u}) & H + cu^n & H - cu^n \end{bmatrix}, \quad (3.4.6)$$

where  $\tilde{K} = \frac{1}{2}q^2 - \chi/\kappa$ ,  $\mathbf{b}^i = b_1^i \mathbf{i} + b_2^i \mathbf{j} + b_3^i \mathbf{k}$ , and  $\mathbf{b}^i \cdot \mathbf{u} = b_1^i u + b_2^i v + b_3^i w$ . The quantity  $\alpha = R^{-1} \Delta Q$  is expressed most simply as

$$\alpha = \begin{bmatrix} a^{n1}(\Delta \rho - \Delta p/c^2) + \rho(b_1^1 \Delta u + b_2^1 \Delta v + b_3^1 \Delta w)/c \\ a^{n2}(\Delta \rho - \Delta p/c^2) + \rho(b_1^2 \Delta u + b_2^2 \Delta v + b_3^2 \Delta w)/c \\ a^{n3}(\Delta \rho - \Delta p/c^2) + \rho(b_1^3 \Delta u + b_2^3 \Delta v + b_3^3 \Delta w)/c \\ \frac{1}{2}(\Delta p/c^2 + \rho[n_1 \Delta u + n_2 \Delta v + n_3 \Delta w]/c) \\ \frac{1}{2}(\Delta p/c^2 - \rho[n_1 \Delta u + n_2 \Delta v + n_3 \Delta w]/c) \end{bmatrix}. \quad (3.4.7)$$

Note that in computing the filters, we need  $\alpha$  but not  $R^{-1}$  explicitly. For efficiency, one should use 3.4.7 instead of doing the actual product of the matrix  $R^{-1}$  with the vector  $\Delta Q$ , especially for nonequilibrium flows (Yee & Shinn 1989, Yee [22]).



There are three methods of calculating the  $a^{ni}$  and  $\mathbf{b}^i$ . They all require the knowledge of  $\mathbf{n}$  at each averaged state, obtained by calculating  $\mathbf{S} = S\mathbf{n}$  as the vectorial average of the  $\mathbf{S}$  at the two neighboring grid points. This is consistent with the accuracy of the filter procedure. Each method has certain disadvantages. If storage is not an issue, then the third method is preferred. In the first method, the Cartesian unit vectors are used as the basis  $\mathbf{a}^i$ . We thus define

$$\mathbf{a}^1 = \mathbf{i}, \quad \mathbf{a}^2 = \mathbf{j}, \quad \mathbf{a}^3 = \mathbf{k}. \quad (3.4.8)$$

It then follows that

$$a^{n1} = n_1, \quad a^{n2} = n_2, \quad a^{n3} = n_3, \quad (3.4.9)$$

and

$$\begin{aligned} b_1^1 &= 0 & b_2^1 &= n_3 & b_3^1 &= -n_2 \\ b_1^2 &= -n_3 & b_2^2 &= 0 & b_3^2 &= n_1 \\ b_1^3 &= n_2 & b_2^3 &= -n_1 & b_3^3 &= 0. \end{aligned} \quad (3.4.10)$$

The disadvantage of this method is that both  $R$  and  $R^{-1}\Delta Q$  involve many terms to be evaluated. A number of these terms would be absent if the normal  $\mathbf{n}$  was one member of the basis  $\mathbf{a}^i$ . This choice underlies the other two methods. We could then define

$$\mathbf{a}^1 = \mathbf{n}, \quad \mathbf{a}^2 = \mathbf{t}, \quad \mathbf{a}^3 = \mathbf{s}, \quad (3.4.11)$$

where  $\mathbf{n}, \mathbf{t}, \mathbf{s}$  form an orthonormal triad, with  $\mathbf{t}$  as yet undefined, but satisfying  $\mathbf{n} \cdot \mathbf{t} = 0$ , and  $\mathbf{s} = \mathbf{n} \times \mathbf{t}$ . With this choice we then have

$$a^{n1} = 1, \quad a^{n2} = 0, \quad a^{n3} = 0, \quad (3.4.12)$$

and

$$\begin{aligned} b_1^1 &= 0 & b_2^1 &= 0 & b_3^1 &= 0 \\ b_1^2 &= s_1 & b_2^2 &= s_2 & b_3^2 &= s_3 \\ b_1^3 &= -t_1 & b_2^3 &= -t_2 & b_3^3 &= -t_3. \end{aligned} \quad (3.4.13)$$

The best way to define  $\mathbf{t}$  is as

$$\mathbf{t} = \frac{\mathbf{c} \times \mathbf{n}}{|\mathbf{c} \times \mathbf{n}|}, \quad (3.4.14)$$

where  $\mathbf{c}$  is some arbitrary vector. The choice of  $\mathbf{c}$  differentiates the other two methods. In the second method,  $\mathbf{c}$  is taken as a constant. A common choice is to let

$$\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}. \quad (3.4.15)$$

This involves a small number of additional operations, if storage is a problem, and  $\mathbf{t}$  and  $\mathbf{s}$  must be recalculated at each time step. The disadvantage of this method is that it breaks down when  $\mathbf{c}$  and  $\mathbf{n}$  are parallel. To avoid this eventuality, in the third method,  $\mathbf{c}$  is chosen to be one of the vectors  $\mathbf{S}$  at a neighboring grid point. This method involves a few more operations than the second method, but is more robust. As an example, for a  $\xi$  surface at  $(i + 1/2, j, k)$ , we can let  $\mathbf{c}$  be  $\mathbf{S}^c$  at  $(i, j, k)$ .

### 8.3.4.2 Roe Average

Among the various approximate Riemann solvers, the most common one uses the Roe average because of its simplicity and its ability to satisfy the jump conditions across discontinuities exactly. In those solvers based on local linearization, the flux at a point separating two states  $Q^L$  and  $Q^R$  is based on the eigenvalues and eigenvectors of some average  $\bar{A}$ . The optimum choice for  $\bar{A}$  is one satisfying

$$\Delta F^n = \bar{A} \Delta Q, \quad (3.4.16)$$

where  $\Delta(\cdot) = (\cdot)^R - (\cdot)^L$ . This choice of  $\bar{A}$  captures discontinuities exactly. One way of obtaining  $\bar{A}$  is to seek an average state  $\bar{Q}$ , which is a function of  $Q^L$  and  $Q^R$ , such that

$$\bar{A} = A(\bar{Q}). \quad (3.4.17)$$

Such a state is known as a Roe-averaged state [12]. One can easily show that

$$\bar{\mathbf{u}} = \nu \mathbf{u}^L + (1 - \nu) \mathbf{u}^R \quad (3.4.18)$$

and

$$\bar{H} = \nu H^L + (1 - \nu) H^R, \quad (3.4.19)$$

where

$$\nu = \frac{1}{1 + \sqrt{\rho^R / \rho^L}}. \quad (3.4.20)$$

From the definition of  $H$  one then obtains

$$\bar{h} = \bar{H} - \frac{1}{2} \bar{\mathbf{u}} \cdot \bar{\mathbf{u}}. \quad (3.4.21)$$

In (3.4.7) we also need  $\bar{\rho}$ , which is given by

$$\bar{\rho} = \sqrt{\rho^R \rho^L}. \quad (3.4.22)$$

We still need to evaluate  $\bar{\chi}$  and  $\bar{\kappa}$ , which are used to determine  $\bar{c}$  from (3.4.3) and  $\bar{K}$  from its definition. For a general equilibrium gas, various approximate methods of calculating  $\bar{\chi}$  and  $\bar{\kappa}$  have been proposed. Since the entropy splitting is only valid for a thermally perfect gas, we consider only this case, for which

there is an exact solution. The details are found in Yee et al. [25]. The exact formulas are

$$\bar{\kappa} = \frac{\Delta \hat{T}}{\Delta \epsilon} \quad (3.4.23a)$$

and

$$\bar{\chi} = \frac{\epsilon^R \hat{T}^L - \epsilon^L \hat{T}^R}{\Delta \epsilon}. \quad (3.4.23b)$$

Equations (3.4.23a,b) are replaced by

$$\bar{\kappa} = 1/\dot{\epsilon} \quad (3.4.24a)$$

and

$$\bar{\chi} = \hat{T} - \bar{\kappa} \epsilon \quad (3.4.24b)$$

when  $\Delta \epsilon \rightarrow 0$ .

For a perfect gas (3.4.23a,b) reduce to

$$\bar{\chi} = 0 \quad \text{and} \quad \bar{\kappa} = \gamma - 1. \quad (3.4.25)$$

In order to obtain  $R_{i+\frac{1}{2}}$  and  $\Phi_{i+\frac{1}{2}}$  for the filter (3.2.1), the right state and left state (superscripts  $R$  and  $L$  in (3.4.16) - (3.4.23)) are the grid indices  $(i+1, j, k)$  and  $(i, j, k)$ .

#### 8.3.4.3 Non-equilibrium Flow

In Yee et al. [25], we showed that the extension of entropy splitting to fully-coupled non-equilibrium flow is not practically feasible. But the schemes in question are usable, and in addition, one can obtain an exact extension of the Roe's Riemann solver for non-equilibrium flow. This is presented in Appendix B.

## 8.4 Concluding Remarks

In this paper we formulate the Yee et al. [24] scheme with entropy splitting (Yee et al. [25]) in 3-D curvilinear moving grids for a thermally perfect gas. For efficiency, we choose the finite difference formulation. The surface area conservation law is satisfied numerically by discretizing a coordinate invariant version of the Thomas and Lombard formulas. This form was independently proposed by the authors in June 2000, and by Thomas & Neier [19] (P.D. Thomas, private communication, November 2000). Although the formal extension of entropy splitting is limited to a thermally perfect gas, the numerical schemes themselves do not have this restriction. Consequently, the schemes discussed here are applicable to equilibrium real gas, non-equilibrium

and artificial compressibility method of treating incompressible flows, MHD and the Maxwell equations. In addition, the dual purpose wavelet sensors (dynamic numerical dissipation controls and grid adaptation indicators) proposed by Sjogreen and Yee can be a stand alone option for a variety of schemes other than what is discussed here. Numerical experiments with the metric terms in general coordinate transformation that are discretized by the same high order difference operator as the flow variables can be found in Müller & Yee [8] and Polifke et al. [10]. Numerical examples illustrating the performance of the new 3-D metric formulation will be reported in a future paper.

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## Appendix A: The Commutativity of a Class of Numerical Mixed Partial Derivatives

In this appendix we prove that the numerical mixed partial derivatives commute, so that the surface area conservation law is satisfied exactly. We would like to thank Dennis Jespersen of NASA Ames Research Center for providing the essential elements of the proof. We find it convenient to employ a notation which differs from that in the body of the paper. Upper case letters denote a matrix, lower case letters denote an algebraic vector, and Latin subscripts denote their components.

We first introduce the notion of a tensor product (or Kronecker product). Given two arbitrary matrices  $A$  and  $B$ , the tensor product  $A \otimes B$  is defined as a block matrix, each of whose components is the matrix  $B$  multiplied by the corresponding component of the matrix  $A$ . In component form this is written as

$$(A \otimes B)_{ij} = A_{ij} B. \quad (\text{A.1})$$

From the definition (A.1), we can derive an important mixed product rule. If  $A$  and  $C$  are conformable matrices, and  $B$  and  $D$  are also conformable, then

$$(A \otimes B)(C \otimes D) = AC \otimes BD. \quad (\text{A.2})$$

The proof follows immediately by writing (A.2) in component form. Thus

$$\begin{aligned}
 [(A \otimes B)(C \otimes D)]_{ik} &= \sum_j (A \otimes B)_{ij} (C \otimes D)_{jk} \\
 &= \sum_j A_{ij} B C_{jk} D \\
 &= (\sum_j A_{ij} C_{jk}) B D \\
 &= (AC)_{ik} B D \\
 &= (AC \otimes BD)_{ik}.
 \end{aligned} \tag{A.3}$$

Let the  $\xi$ ,  $\eta$ ,  $\zeta$  computational space be discretized with  $l$ ,  $m$ ,  $n$  points in the  $\xi$ ,  $\eta$ ,  $\zeta$  directions, respectively. For a fixed  $\eta$  and  $\zeta$ , the most general finite-difference approximation of the  $\xi$  derivative is

$$A^\xi u_\xi = B^\xi u, \tag{A.4}$$

where  $u$  and  $u_\xi$  are  $l$ -dimensional vectors, and  $A^\xi$  and  $B^\xi$  are  $l$  by  $l$  matrices. Some examples are the central non-compact and compact spatial schemes (3.1.1) - (3.1.3). No restrictions are placed on the nature of  $A^\xi$  and  $B^\xi$ , which incorporate arbitrary boundary conditions on the  $\xi$  boundaries of the computational region. Assume that the discrete unknowns for the whole region are ordered with  $\xi$  values varying first,  $\eta$  values varying next, and  $\zeta$  values varying last. If the **same** finite-difference approximation (A.4) is applied for each  $\eta$  and  $\zeta$  (which implies the same boundary condition along each of the  $\xi$  boundaries), then the approximation to the  $\xi$  derivatives of all the unknowns is

$$\overline{A}^\xi \overline{u}_\xi = \overline{B}^\xi \overline{u}, \tag{A.5}$$

where  $\overline{u}$  and  $\overline{u}_\xi$  are  $l \times m \times n$ -dimensional vectors, and  $\overline{A}^\xi$  and  $\overline{B}^\xi$  are  $l \times m \times n$  by  $l \times m \times n$  matrices given by

$$\overline{A}^\xi = I^n \otimes (I^m \otimes A^\xi), \quad \overline{B}^\xi = I^n \otimes (I^m \otimes B^\xi). \tag{A.6}$$

$I^n$  and  $I^m$  are the  $n$  by  $n$  and  $m$  by  $m$  identity matrices, respectively. Note that the parentheses in (A.6) can be eliminated, since from its definition, tensor multiplication can be shown to be associative.

For a fixed  $\xi$  and  $\zeta$ , the finite-difference approximation of the  $\eta$  derivative can be written as

$$A^\eta v_\eta = B^\eta v, \tag{A.7}$$

where  $v$  and  $v_\eta$  are  $m$ -dimensional vectors, and  $A^\eta$  and  $B^\eta$  are  $m$  by  $m$  matrices. Note that  $A^\eta$  and  $B^\eta$  are again arbitrary, with different boundary conditions on the  $\eta$  boundaries than on the  $\xi$  boundaries being permitted.

If the same finite-difference approximation (A.7) is applied for each  $\xi$  and  $\zeta$ , then the  $l \times m \times n$ -dimensional vectors  $\bar{v}$  and  $\bar{v}_\eta$  of all the unknowns are related by

$$\bar{A}^\eta \bar{v}_\eta = \bar{B}^\eta \bar{v}, \quad (\text{A.8})$$

where the  $l \times m \times n$  by  $l \times m \times n$  matrices  $\bar{A}^\eta$  and  $\bar{B}^\eta$  are given by

$$\bar{A}^\eta = I^n \otimes (A^\eta \otimes I^l), \quad \bar{B}^\eta = I^n \otimes (B^\eta \otimes I^l). \quad (\text{A.9})$$

$I^l$  is the  $l$  by  $l$  identity matrix. Similarly, for a fixed  $\xi$  and  $\eta$ , the finite-difference approximation of the  $\zeta$  derivative can be written as

$$A^\zeta w_\zeta = B^\zeta w, \quad (\text{A.10})$$

where  $w$  and  $w_\zeta$  are  $n$ -dimensional vectors, and  $A^\zeta$  and  $B^\zeta$  are  $n$  by  $n$  matrices.  $A^\zeta$  and  $B^\zeta$  are again arbitrary, with different boundary conditions on the  $\zeta$  boundaries than on the other boundaries being permitted. If the same finite-difference approximation (A.10) is applied for each  $\xi$  and  $\eta$ , then the  $l \times m \times n$ -dimensional vectors  $\bar{w}$  and  $\bar{w}_\zeta$  of all the unknowns are related by

$$\bar{A}^\zeta \bar{w}_\zeta = \bar{B}^\zeta \bar{w}, \quad (\text{A.11})$$

where the  $l \times m \times n$  by  $l \times m \times n$  matrices  $\bar{A}^\zeta$  and  $\bar{B}^\zeta$  are given by

$$\bar{A}^\zeta = A^\zeta \otimes (I^m \otimes I^l), \quad \bar{B}^\zeta = B^\zeta \otimes (I^m \otimes I^l). \quad (\text{A.12})$$

Let the vector  $\bar{u}_{\xi\eta}$  be the finite-difference approximation of all the mixed  $\xi\eta$ -partial derivatives, calculated by first obtaining  $\bar{u}_\xi$  using (A.5), and then using (A.8) to obtain  $\bar{u}_{\xi\eta}$ . If the grid points are equally spaced in the computational space, then the matrices  $A^\xi, B^\xi, A^\eta, B^\eta, A^\zeta$  and  $B^\zeta$  are all **constant**. The final result then becomes

$$\bar{A}^\eta (\bar{A}^\xi \bar{u}_\xi)_\eta = \bar{A}^\eta \bar{A}^\xi \bar{u}_{\xi\eta} = \bar{B}^\eta \bar{B}^\xi \bar{u}. \quad (\text{A.13})$$

If we carry out the operations in the reverse order, we obtain

$$\bar{A}^\xi \bar{A}^\eta \bar{u}_{\eta\xi} = \bar{B}^\xi \bar{B}^\eta \bar{u}. \quad (\text{A.14})$$

Using (A.2), (A.6) and (A.9), it follows that

$$\begin{aligned} \bar{A}^\eta \bar{A}^\xi &= [I^n \otimes (A^\eta \otimes I^l)] [I^n \otimes (I^m \otimes A^\xi)] \\ &= I^n \otimes [(A^\eta \otimes I^l)(I^m \otimes A^\xi)] \\ &= I^n \otimes (A^\eta \otimes A^\xi). \end{aligned} \quad (\text{A.15})$$

In the same manner we show that

$$\begin{aligned}\overline{A}^\xi \overline{A}^\eta &= [I^n \otimes (I^m \otimes A^\xi)][I^n \otimes (A^\eta \otimes I^l)] \\ &= I^n \otimes [(I^m \otimes A^\xi)(A^\eta \otimes I^l)] \\ &= I^n \otimes (A^\eta \otimes A^\xi).\end{aligned}\tag{A.16}$$

Therefore from (A.15) and (A.16) we have established that

$$\overline{A}^\eta \overline{A}^\xi = \overline{A}^\xi \overline{A}^\eta.\tag{A.17}$$

In a similar manner we show that

$$\overline{B}^\eta \overline{B}^\xi = \overline{B}^\xi \overline{B}^\eta.\tag{A.18}$$

From (A.13) and (A.14) it follows that

$$\overline{u}_{\xi\eta} = \overline{u}_{\eta\xi}.\tag{A.19}$$

In a similar manner, using (A.5) and (A.11), the finite-difference approximation of all the mixed  $\xi\zeta$ -partial derivatives can be shown to satisfy

$$\overline{A}^\zeta (\overline{A}^\xi \overline{u}_\xi)_\zeta = \overline{A}^\zeta \overline{A}^\xi \overline{u}_{\xi\zeta} = \overline{B}^\zeta \overline{B}^\xi \overline{u}.\tag{A.20}$$

If we carry out the operations in the reverse order, we obtain

$$\overline{A}^\xi \overline{A}^\zeta \overline{u}_{\zeta\xi} = \overline{B}^\xi \overline{B}^\zeta \overline{u}.\tag{A.21}$$

Using (A.2), (A.6) and (A.12), it follows that

$$\overline{A}^\xi \overline{A}^\zeta = A^\zeta \otimes (I^m \otimes A^\xi) = \overline{A}^\zeta \overline{A}^\xi,\tag{A.22}$$

with an analogous relation involving  $\overline{B}^\xi$  and  $\overline{B}^\zeta$ . From (A.20) and (A.21) it then follows that

$$\overline{u}_{\xi\zeta} = \overline{u}_{\zeta\xi}.\tag{A.23}$$

The same procedure, using (A.8) and (A.11), can be used to show that

$$\overline{u}_{\eta\zeta} = \overline{u}_{\zeta\eta}.\tag{A.24}$$

This completes the proof that all numerical mixed partial derivatives commute.

## Appendix B: Riemann Solver for Non-equilibrium Flow

In this appendix we present the Riemann solver for chemical non-equilibrium flow. The analysis can be readily extended to thermo-chemical nonequilibrium. We consider a mixture of  $ns$  chemical species, each obeying a thermally perfect gas law. Let  $\rho^s$ ,  $R^s$ , and  $\epsilon^s(T)$  be the density, gas constant, and internal energy for species  $s$ , respectively.  $T$  is the temperature of the mixture, assumed to be in thermal equilibrium. Note that we have not introduced a normalized  $\hat{T}$ , since the  $R^s$  are different for each species. All species satisfy the condition  $\dot{\epsilon}^s > 0$ , where  $\dot{\epsilon}^s = d\epsilon^s/dT$ . We can now define the density of the mixture,  $\rho$ , as

$$\rho = \sum_s \rho^s, \quad (\text{B.1})$$

the species mass fraction  $\alpha^s$  as

$$\alpha^s \equiv \frac{\rho^s}{\rho}, \quad (\text{B.2})$$

the pressure  $p$  as

$$p = \rho RT, \quad (\text{B.3})$$

where

$$\rho R = \sum_s \rho^s R^s, \quad (\text{B.4})$$

and the internal energy of the mixture per unit volume as

$$\tilde{\epsilon} = \rho \epsilon = \sum_s \rho^s \epsilon^s. \quad (\text{B.5})$$

The conservation law takes the form

$$Q_t + \nabla \cdot \mathbf{F} = \mathcal{S}^c, \quad (\text{B.6})$$

where  $\mathcal{S}^c$  is a *vector* consisting of source terms for each species. The *vectors*  $Q$  and  $\mathbf{F}$  take the form

$$Q = \begin{bmatrix} (\rho^s) \\ \rho u \\ \rho v \\ \rho w \\ e \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} (\rho^s \mathbf{u}') \\ \rho u \mathbf{u}' + p \mathbf{i} \\ \rho v \mathbf{u}' + p \mathbf{j} \\ \rho w \mathbf{u}' + p \mathbf{k} \\ (e + p) \mathbf{u}' + p \mathbf{v} \end{bmatrix}. \quad (\text{B.7})$$

Note that  $\mathbf{F}$  is again defined relative to a moving grid.  $Q$  and  $\mathbf{F}$  are  $(ns + 4)$ -dimensional *vectors*. We employ compact notations “ $(\rho^s)$ ” and “ $(\rho^s \mathbf{u}')$ ” in which the first term written for each *vector* in (B.7) is a general component of a



$ns$ -dimensional *subvector* corresponding to the  $ns$  species continuity equations. The temperature  $T(Q)$  is obtained by solving implicitly the equation

$$\sum_s \rho^s \epsilon^s(T) = e - \frac{1}{2\rho} [(\rho u)^2 + (\rho v)^2 + (\rho w)^2]. \quad (\text{B.8})$$

Equation (B.8) has a unique solution since  $\rho^s > 0$  and  $\epsilon^s > 0$ .

In order to derive the flux Jacobian matrix, the pressure  $p$  must be expressed in the form

$$p = p(\rho^s, \tilde{\epsilon}). \quad (\text{B.9})$$

The derivatives will be denoted by

$$\chi^s = \left( \frac{\partial p}{\partial \rho^s} \right)_{\tilde{\epsilon}, \rho^r} \quad \text{and} \quad \kappa = \left( \frac{\partial p}{\partial \tilde{\epsilon}} \right)_{\rho^s}. \quad (\text{B.10})$$

With the aid of (B.1) to (B.5) we can show that

$$\kappa = \frac{\sum_s \alpha^s R^s}{\sum_s \alpha^s \tilde{\epsilon}^s} \quad (\text{B.11})$$

and

$$\chi^s = R^s T - \kappa \epsilon^s. \quad (\text{B.12})$$

The matrix  $A$  can then be written as

$$A = \begin{bmatrix} (\delta^{sr} u' - \alpha^s u^n) & (\alpha^s n_1) & (\alpha^s n_2) & (\alpha^s n_3) & 0 \\ (K^r n_1 - u^n u) & (1 - \kappa) u n_1 + u' & u n_2 - \kappa n_1 v & u n_3 - \kappa n_1 w & \kappa n_1 \\ (K^r n_2 - u^n v) & v n_1 - \kappa n_2 u & (1 - \kappa) v n_2 + u' & v n_3 - \kappa n_2 w & \kappa n_2 \\ (K^r n_3 - u^n w) & w n_1 - \kappa n_3 u & w n_2 - \kappa n_3 v & (1 - \kappa) w n_3 + u' & \kappa n_3 \\ ((K^r - H) u^n) & H n_1 - \kappa u^n u & H n_2 - \kappa u^n v & H n_3 - \kappa u^n w & u' + \kappa u^n \end{bmatrix}, \quad (\text{B.13})$$

where  $K^r = \frac{1}{2} \kappa q^2 + \chi^r$  and  $\delta^{sr}$  is the Kronecker delta. All other quantities are defined in Section 3.4. Once again a compact notation is used for the terms written in the first row of  $A$ . The first term written in  $A$  is the  $sr$  component of a  $ns$  by  $ns$  submatrix, and the second to fourth terms are column subvectors of dimension  $ns$ . In addition, the last four terms written in the first column of  $A$  are row subvectors of dimension  $ns$ .

The three distinct eigenvalues of  $A$  are again

$$\lambda^1 = u', \quad \lambda^2 = u' + c, \quad \text{and} \quad \lambda^3 = u' - c, \quad (\text{B.14})$$

where  $c$  is the frozen speed of sound given by

$$c^2 = \sum_s \alpha^s \chi^s + \kappa h. \quad (\text{B.15})$$

$\lambda^1$  is the repeated eigenvalue of multiplicity  $ns+2$ . The choice of Cartesian unit vectors (3.4.8) as the basis  $\mathbf{a}^i$  in order to obtain the set of linearly independent eigenvectors would result in far too many additional terms in  $R$  and  $R^{-1}\Delta Q$  for non-equilibrium flow. We therefore restrict ourselves to the triad  $\mathbf{n}, \mathbf{t}, \mathbf{s}$  as the basis (3.4.11). The right eigenvector matrix  $R$  can then be written as

$$R = \begin{bmatrix} (\delta^{sr}) & 0 & 0 & (\alpha^s) & (\alpha^s) \\ u & cs_1 & -ct_1 & u + cn_1 & u - cn_1 \\ v & cs_2 & -ct_2 & v + cn_2 & v - cn_2 \\ w & cs_3 & -ct_3 & w + cn_3 & w - cn_3 \\ \tilde{K}^r & c(\mathbf{s} \cdot \mathbf{u}) & -c(\mathbf{t} \cdot \mathbf{u}) & H + cu^n & H - cu^n \end{bmatrix}, \quad (\text{B.16})$$

where  $\tilde{K}^r = \frac{1}{2}q^2 - \chi^r/\kappa$ . The quantity  $R^{-1}\Delta Q$  is expressed most simply as

$$R^{-1}\Delta Q = \begin{bmatrix} (\Delta\rho^s - \alpha^s\Delta p/c^2) \\ \rho(s_1\Delta u + s_2\Delta v + s_3\Delta w)/c \\ -\rho(t_1\Delta u + t_2\Delta v + t_3\Delta w)/c \\ \frac{1}{2}(\Delta p/c^2 + \rho[n_1\Delta u + n_2\Delta v + n_3\Delta w]/c) \\ \frac{1}{2}(\Delta p/c^2 - \rho[n_1\Delta u + n_2\Delta v + n_3\Delta w]/c) \end{bmatrix}. \quad (\text{B.17})$$

Equations (3.4.16) to (3.4.22) defining Roe-averaged variables are still valid. In addition, we define

$$\overline{T} = \nu T^L + (1 - \nu)T^R, \quad (\text{B.18})$$

$$\overline{\alpha^s} = \nu \alpha^{sL} + (1 - \nu)\alpha^{sR}, \quad (\text{B.19})$$

and

$$\overline{\epsilon^s} = \nu \epsilon^{sL} + (1 - \nu)\epsilon^{sR}. \quad (\text{B.20})$$

We can then show that (3.4.16) is satisfied if we define

$$\overline{\kappa} = \left( \frac{\sum_s \overline{\alpha^s} R^s}{\sum_s \overline{\alpha^s} \Delta \epsilon^s} \right) \Delta T, \quad (\text{B.21})$$

and

$$\overline{\chi^s} = R^s \overline{T} - \overline{\kappa} \overline{\epsilon^s}. \quad (\text{B.22})$$

Equation (B.21) is replaced by

$$\overline{\kappa} = \frac{\sum_s \overline{\alpha^s} R^s}{\sum_s \overline{\alpha^s} \dot{\epsilon}^s} \quad (\text{B.23})$$

when  $\Delta T \rightarrow 0$ .

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# Fourth Order Methods for the Stokes and Navier-Stokes Equations on Staggered Grids

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*Dedicated to Robert MacCormack on his 60th birthday*

## 9.1 Introduction

In this paper we consider the 2D Stokes equations

$$\begin{aligned}u_t + p_x &= \nu(u_{xx} + u_{yy}), \\v_t + p_y &= \nu(v_{xx} + v_{yy}), \\u_x + v_y &= 0,\end{aligned}$$

and the incompressible 2D Navier-Stokes equations

$$\begin{aligned}u_t + uu_x + vv_y + p_x &= \nu(u_{xx} + u_{yy}), \\v_t + uv_x + vv_y + p_y &= \nu(v_{xx} + v_{yy}), \\u_x + v_y &= 0.\end{aligned}$$

Here  $u$  and  $v$  are the velocity components in the  $x$ - and  $y$ -direction respectively, and  $p$  is the pressure. We develop fourth order compact difference methods of Padé type for both sets of equations. We apply the approximation to the original form with the divergence condition used explicitly as part of

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the system. The boundary conditions are based on the results in [2] for the steady state Stokes equations.

In most applications the velocity field  $\mathbf{v} = (u, v)^T = \mathbf{v}_0$  is given at the whole boundary  $\partial\Omega$ , and almost all analysis has been devoted to this case. However, sometimes it is required to prescribe the pressure, at least on part of the boundary, and we shall consider this case as well.

The divergence condition introduces a restriction on the boundary data

$$\int_{\partial\Omega} \mathbf{v}_0 \cdot \mathbf{n} \, dS = 0, \quad (9.1)$$

where  $\mathbf{n}$  is the outer normal to  $\partial\Omega$ . This restriction can be seen as an effect of the fact that the boundary value problem is singular. The consequence is that the algebraic system to be solved for any consistent approximation, necessarily is singular as well. By formulating the boundary conditions for the Stokes equations in a particular way, such that the boundary value problem becomes non-singular, we are able to construct a non-singular approximation in a straightforward way. The approximation is based on a second order difference approximation on a staggered grid. This type of approximation has been used by others, see for example [3], [6], [5] and [7], and the advantage is that no parasitic solutions are present. In our case, it also allows for an analysis that is very close to the analysis for the continuous case, and we are able to derive the same type of estimates. Some of these results are summarized in Section 9.2.

In Section 9.3 we describe a fourth order compact difference method on a staggered grid for the Stokes equations. By using non-symmetric approximations near the boundaries, the difference operators are well defined at all gridpoints independent of the particular initial-boundary value problem at hand. The algebraic system is solved by an iterative method of Krylov type. In each iteration the intermediate solution is then modified by applying the physical boundary conditions.

It turns out that the results for the steady Stokes equations easily carries over to the time-dependent equations, and we apply exactly the same technique. Furthermore, the results are significant also for the Navier-Stokes equations. For convenience, we write the Stokes equations in the form

$$\begin{aligned} \mathbf{v}_t + \text{Stokes}(\mathbf{v}, p) &= 0, \\ \text{div } \mathbf{v} &= 0, \end{aligned}$$

where  $\text{Stokes}(\mathbf{v}, p)$  denotes the space part of the momentum equations. The time-discretization is obtained by using the backwards differentiation formula

$$\begin{aligned} \frac{3}{2}\mathbf{v}^{n+1} + \Delta t \text{Stokes}(\mathbf{v}^{n+1}, p^{n+1}) &= 2\mathbf{v}^n - \frac{1}{2}\mathbf{v}^{n-1}, \\ \text{div } \mathbf{v}^{n+1} &= 0, \end{aligned} \quad (9.2)$$

where the superscript denotes the time-level.

Considering next the Navier-Stokes equations, we write them in the form

$$\begin{aligned}\mathbf{v}_t + \text{Navier}(\mathbf{v}) + \text{Stokes}(\mathbf{v}, p) &= 0, \\ \text{div } \mathbf{v} &= 0,\end{aligned}$$

where  $\text{Navier}(\mathbf{v})$  denotes the advection terms. For the time-discretization we use a semi-implicit method proposed by Per Lötstedt. It is based on the implicit method above for the Stokes part of the equations:

$$\begin{aligned}\frac{3}{2}\mathbf{v}^{n+1} + \Delta t \text{Stokes}(\mathbf{v}^{n+1}, p^{n+1}) &= 2\mathbf{v}^n - \frac{1}{2}\mathbf{v}^{n-1} \\ -\Delta t(2\text{Navier}(\mathbf{v}^n) - \text{Navier}(\mathbf{v}^{n-1})), & \\ \text{div } \mathbf{v}^{n+1} &= 0.\end{aligned}\tag{9.3}$$

This allows for large time-steps without having to solve the full nonlinear system at each time-level, as is necessary for a fully implicit method. Assuming that we have an efficient and well conditioned solver for the Stokes equations, we have an efficient and well conditioned solver for the Navier-Stokes equations as well.

For direct simulation of turbulence, it is well known that high order methods are required. In Section 9.4, we use the same type of compact high order methods as for the Stokes equations to approximate the Navier-Stokes equations, and we present some preliminary results for a simple model problem.

## 9.2 The Steady Stokes Equations and Staggered Grids

In this section we shall summarize a few of the results in [2], and discuss the particular type of boundary conditions that eliminates the singularity in the system. We consider the steady state equations

$$\begin{aligned}p_x &= \nu(u_{xx} + u_{yy}), \\ p_y &= \nu(v_{xx} + v_{yy}), \\ u_x + v_y &= 0.\end{aligned}\tag{9.4}$$

The theoretical results are obtained for the case with periodic solutions in the  $y$ -direction:  $\mathbf{U}(x, y) = \mathbf{U}(x, y + 2\pi)$ , where  $\mathbf{U} = (u, v, p)^T$ , and the equations are defined in a rectangular domain  $\Omega = \{0 \leq x \leq 1, 0 \leq y \leq 2\pi\}$ .

If the velocity  $u, v$  is prescribed on both sides  $x = 0, 1$ , the restriction (9.1) on the data is

$$\int_0^{2\pi} u(0, y) dy = \int_0^{2\pi} u(1, y) dy,$$

In order to eliminate the singularity that is the basis for this restriction, we formulate the boundary conditions as

$$\begin{aligned} u(0, y) - \frac{1}{2\pi} \int_0^{2\pi} u(0, y) dy &= w_0(y), & u(1, y) &= u_1(y), \\ v(0, y) &= v_0(y), & v(1, y) &= v_1(y), \\ \int_0^{2\pi} p(0, y) dy &= q_0, \end{aligned} \quad (9.5)$$

where it is assumed that  $\int_0^{2\pi} w_0(y) dy = 0$ . However, it should be noted that even if there is a small perturbation in this integral, the problem is still solvable with a small perturbation in the solution as a result. With the notation  $\mathbf{v}_B = (w_0, u_1, v_0, v_1)^T$ , and the norms defined by

$$\begin{aligned} \|\mathbf{v}(\cdot, \cdot)\| &= \left\{ \int_0^{2\pi} \int_0^1 |\mathbf{v}(x, y)|^2 dx dy \right\}^{1/2}, \quad |\mathbf{v}|^2 = u^2 + v^2, \\ \|\mathbf{v}_B(\cdot)\| &= \left\{ \int_0^{2\pi} |\mathbf{v}_B(y)|^2 dy \right\}^{1/2}, \quad |\mathbf{v}_B|^2 = |w_0|^2 + |u_1|^2 + |v_0|^2 + |v_1|^2, \\ \|p(\cdot, \cdot)\| &= \left\{ \int_0^{2\pi} \int_0^1 |p(x, y)|^2 dx dy \right\}^{1/2}, \end{aligned}$$

we have

**Theorem 9.1** *Assume that the boundary data  $w_0, u_1, v_0, v_1$  are  $2\pi$ -periodic in  $y$  and  $\int_0^{2\pi} w_0(y) dy = 0$ , but otherwise arbitrary. Then the system (9.4) with boundary conditions (9.5) has a unique solution, and there is an estimate*

$$\begin{aligned} \|\mathbf{v}(\cdot, \cdot)\|^2 &\leq \text{const}(|q_0|^2 + \|\mathbf{v}_B(\cdot)\|^2), \\ \|p(\cdot, \cdot)\|^2 &\leq \text{const}(|q_0|^2 + \|\mathbf{v}_B(\cdot)\|^2 + \left\| \frac{\partial \mathbf{v}_B}{\partial y}(\cdot) \right\|^2). \end{aligned} \quad (9.6)$$

Instead of prescribing both velocity components, we can substitute one of the components by the pressure. The conditions are

$$\begin{aligned} u(0, y) - \frac{1}{2\pi} \int_0^{2\pi} u(0, y) dy &= w_0(y), & u(1, y) &= u_1(y), \\ p(0, y) - \frac{1}{2\pi} \int_0^{2\pi} p(0, y) dy &= q_0(y), & p(1, y) &= p_1(y), \\ \int_0^{2\pi} v(0, y) dy &= z_0, & \int_0^{2\pi} v(1, y) dy &= z_1. \end{aligned} \quad (9.7)$$

In this case the derivatives of the boundary data are not necessary in the estimate of the solution. We have for  $\|\mathbf{U}\|^2 = \|\mathbf{v}\|^2 + \|p\|^2$



**Theorem 9.2** *Assume that the boundary data  $w_0, u_1, q_0, p_1$  are  $2\pi$ -periodic in  $y$  and  $\int_0^{2\pi} w_0(y)dy = \int_0^{2\pi} q_0(y)dy = 0$ , but otherwise arbitrary. Then the system (9.4) with boundary conditions (9.7) has a unique solution, and there is an estimate*

$$\begin{aligned} \|\mathbf{U}(\cdot, \cdot)\|^2 &\leq \text{const}(|z_0|^2 + |z_1|^2 + \\ &+ \int_0^{2\pi} (|w_0(y)|^2 + |u_1(y)|^2 + |q_0(y)|^2 + |p_1(y)|^2)dy). \end{aligned} \quad (9.8)$$

A third set of boundary conditions is

$$\begin{aligned} v(0, y) &= v_0(y), & v(1, y) &= v_1(y), \\ p(0, y) - \frac{1}{2\pi} \int_0^{2\pi} p(0, y)dy &= q_0(y), & p(1, y) &= p_1(y), \\ \int_0^{2\pi} u(0, y)dy &= w_0. \end{aligned} \quad (9.9)$$

In the same way as above we can prove

**Theorem 9.3** *Assume that the boundary data  $v_0, v_1, q_0, p_1$  are  $2\pi$ -periodic in  $y$  and  $\int_0^{2\pi} q_0(y)dy = 0$ , but otherwise arbitrary. Then the system (9.4) with boundary conditions (9.9) has a unique solution, and there is an estimate*

$$\begin{aligned} \|\mathbf{U}(\cdot, \cdot)\|^2 &\leq \text{const}(|w_0|^2 + \\ &+ \int_0^{2\pi} (|v_0(y)|^2 + |v_1(y)|^2 + |q_0(y)|^2 + |p_1(y)|^2)dy). \end{aligned} \quad (9.10)$$

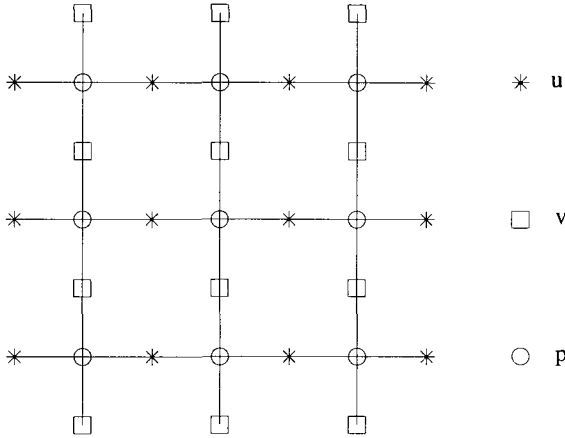
For the discretization in space we use a staggered  $N \times M$  grid according to Figure 1.

The standard second order scheme is for the inner points

$$\begin{aligned} D_{-,x}p_{i,j} &= \nu(D_{+,x}D_{-,x} + D_{+,y}D_{-,y})u_{i-\frac{1}{2},j}, \\ D_{-,y}p_{i,j} &= \nu(D_{+,x}D_{-,x} + D_{+,y}D_{-,y})v_{i,j-\frac{1}{2}}, \\ D_{+,x}u_{i-\frac{1}{2},j} + D_{+,y}v_{i,j-\frac{1}{2}} &= 0, \end{aligned} \quad (9.11)$$

where  $D_{+,x}u_{i,j} = (u_{i+1,j} - u_{i,j})/\Delta x$  and  $D_{-,x}u_{i,j} = (u_{i,j} - u_{i-1,j})/\Delta x$ , and similarly for  $D_{\pm,y}$ .

The discrete boundary conditions are a direct approximation of the continuous ones, except for one velocity component, where we take an average.



**Figure 1** A staggered grid.

The first set of conditions (9.5) are approximated by

$$\begin{aligned}
 \frac{1}{2\pi} \sum_{j=0}^{M-1} \frac{u_{-\frac{1}{2},j} + u_{\frac{1}{2},j}}{2} \Delta y &= w_0(y_j), & \frac{u_{N-\frac{1}{2},j} + u_{N+\frac{1}{2},j}}{2} &= u_1(y_j), \\
 v_{0,j-\frac{1}{2}} &= v_0(y_{j-\frac{1}{2}}), & v_{N,j-\frac{1}{2}} &= v_1(y_{j-\frac{1}{2}}), \\
 \sum_{j=0}^{M-1} p_{0,j} \Delta y &= q_0,
 \end{aligned} \tag{9.12}$$

where  $\sum_{j=0}^{M-1} w_0(y_j) \Delta y = 0$ . For this approximation one can derive an estimate that is completely analogous to the continuous one (9.6):

$$\begin{aligned}
 \|\mathbf{v}\|_h^2 &\leq \text{const} (|q_0|^2 + \|\mathbf{v}_B\|_h^2), \\
 \|p\|_h^2 &\leq \text{const} (|q_0|^2 + \|\mathbf{v}_B\|_h^2 + \|D_{+,y} \mathbf{v}_B\|_h^2),
 \end{aligned} \tag{9.13}$$

where the constants are independent of  $\Delta x, \Delta y$ . Here the discrete norms  $\|\cdot\|_h$  are direct discretizations of the continuous norms  $\|\cdot\|$  defined above.

The assumption of periodic solutions makes it possible to derive the estimates above by using Fourier technique. In most applications the solutions are non-periodic, and we must prescribe boundary conditions also in the  $y$ -direction:

$$\begin{aligned}
 u_{i-\frac{1}{2},0} &= u^0(x_{i-\frac{1}{2}}), & u_{i-\frac{1}{2},M} &= u^1(x_{i-\frac{1}{2}}), \\
 \frac{v_{i,-\frac{1}{2}} + v_{i,\frac{1}{2}}}{2} &= v^0(x_i), & \frac{v_{i,M-\frac{1}{2}} + v_{i,M+\frac{1}{2}}}{2} &= v^1(x_i).
 \end{aligned} \tag{9.14}$$

Here  $u^0$ ,  $v^0$  and  $u^1$ ,  $v^1$  are given data at  $y = 0$  and  $y = 1$  respectively. Numerical experiments in [2] show that the estimates derived in the periodic case also holds for the non-periodic case.

The staggered grid and the special form of the boundary conditions presented here is the same also for the time-dependent problems (9.2),(9.3). Furthermore, it is the basis for the high order approximations to be discussed in the next sections.

### 9.3 A Fourth Order Method for the Stokes Equations

A fourth order compact method can be derived both for regular grids and for staggered grids, see [4] and [1]. For the Stokes equations, all derivatives of first order are centered in the middle between two gridpoints, while all derivatives of second order are centered at a gridpoint. For a function  $f$  and grid spacing  $h$ , we have the formulas

$$\frac{1}{24}f'_{i-1} + \frac{11}{12}f'_i + \frac{1}{24}f'_{i+1} = \frac{1}{h}(f_{i+1/2} - f_{i-1/2}), \quad (9.15)$$

$$\frac{1}{10}f''_{i-1} + f''_i + \frac{1}{10}f''_{i+1} = \frac{6}{5h^2}(f_{i+1} - 2f_i + f_{i-1}). \quad (9.16)$$

In order to solve for the unknowns  $f'$  and  $f''$ , we need closed systems in the form  $Pf' = Qf$ ,  $Rf'' = Sf$ . No boundary conditions are available for the derivatives, and we use one-sided formulas. For the function itself on the right hand side, we could use the physical boundary conditions given above, since the right hand sides correspond exactly to the second order scheme above. However, it is convenient to define  $P, Q, R, S$  independently of the particular problem, and therefore we use one-sided formulas also for  $f$ . There are two alternatives  $P_1, Q_1$  and  $P_2, Q_2$  for the first derivative depending on whether these operators are used for the pressure or for the velocity. The complete formulas are with  $h \rightarrow \Delta x$

$$P_1 f' = Q_1 f:$$

$$P_1 f' = \begin{cases} \frac{1}{552}(f'_{-\frac{1}{2}} + 23f'_{\frac{1}{2}}), \\ \frac{1}{24}(f'_{i-\frac{1}{2}} + 22f'_{i+\frac{1}{2}} + f'_{i+\frac{3}{2}}), & i = 0, \dots, N-1, \\ \frac{1}{552}(23f'_{N-\frac{1}{2}} + f'_{N+\frac{1}{2}}), \end{cases} \quad (9.17)$$

$$Q_1 f = \begin{cases} \frac{1}{552\Delta x}(-25f_0 + 26f_1 - f_2), \\ \frac{1}{\Delta x}(f_{i+1} - f_i), & i = 0, \dots, N-1, \\ \frac{1}{552\Delta x}(f_{N-2} - 26f_{N-1} + 25f_N). \end{cases}$$

$$P_2 f' = Q_2 f:$$

$$P_2 f' = \begin{cases} \frac{1}{24}(-f'_0 + f'_1), \\ \frac{1}{24}(f'_{i-1} + 22f'_i + f'_{i+1}), & i = 1, \dots, N-1, \\ \frac{1}{24}(f'_{N-1} - f'_N), \end{cases} \quad (9.18)$$

$$Q_2 f = \begin{cases} \frac{1}{24\Delta x}(f_{-\frac{1}{2}} - 2f_{\frac{1}{2}} + f_{\frac{3}{2}}), \\ \frac{1}{\Delta x}(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}), & i = 1, \dots, N-1, \\ \frac{1}{24\Delta x}(-f_{N-\frac{3}{2}} + 2f_{N-\frac{1}{2}} - f_{N+\frac{1}{2}}). \end{cases}$$

$$Rf'' = Sf:$$

$$Rf'' = \begin{cases} \frac{1}{100}(f''_0 + 10f''_1), \\ \frac{1}{10}(f''_{i-1} + 10f''_i + f''_{i+1}), & i = 1, \dots, N-1, \\ \frac{1}{100}(10f''_{N-1} + f''_N), \end{cases} \quad (9.19)$$

$$Sf = \begin{cases} \frac{1}{1200\Delta x^2}(145f_0 - 304f_1 + 174f_2 - 16f_3 + f_4), \\ \frac{6}{5\Delta x^2}(f_{i-1} - 2f_i + f_{i+1}), & i = 1, \dots, N-1, \\ \frac{1}{1200\Delta x^2}(f_{N-4} - 16f_{N-3} + 174f_{N-2} - 304f_{N-1} + 145f_N). \end{cases}$$

These formulas are one-dimensional, and are applied for all gridlines in the  $x$ -direction. They are applied in the  $y$ -direction as well, but now  $\Delta x$  and  $N$  are replaced by  $\Delta y$  and  $M$  respectively. We introduce extra indices  $x$  and  $y$  on the operators to indicate the coordinate direction (with space steps  $\Delta x$  and  $\Delta y$  respectively), and write the scheme as

$$\begin{aligned} P_{1x}^{-1}Q_{1x}p - \nu(R_x^{-1}S_x + R_y^{-1}S_y)u &= 0, \\ P_{1y}^{-1}Q_{1y}p - \nu(R_x^{-1}S_x + R_y^{-1}S_y)v &= 0, \\ P_{2x}^{-1}Q_{2x}u + P_{2y}^{-1}Q_{2y}v &= 0. \end{aligned} \quad (9.20)$$

Note that this system denoted by  $A\mathbf{U} = 0$  is not the one that defines the correct solution to our boundary value problem. It is only an intermediate solution to be computed at each iteration of a Krylov type iterative method. In each iteration, we are required to compute  $A\mathbf{U}^{(k)}$ , which in turn requires the solution of the one-dimensional systems  $P_{1x}p_x^{(k)} = Q_{1x}p^{(k)}$  etc. To complete the iteration, the true boundary conditions, for example (9.12), are applied, and  $\mathbf{U}^{(k+1)}$  is obtained. Note that the second order averaging is substituted by a fourth order averaging. Furthermore, the summation formulas are substituted by a forth order approximation of the integrals in (9.5).

To illustrate the accuracy of the scheme, we construct a test problem. We add a forcing function in the second equation, in order to obtain a simple

| N×M             | Errors in $l_2$ -norm, $e_N$ , $N=[10,20,40]$ |                 |                 |
|-----------------|-----------------------------------------------|-----------------|-----------------|
|                 | $  u - u^*  _h$                               | $  v - v^*  _h$ | $  p - p^*  _h$ |
| 10×10           | 2.3e-3                                        | 2.3e-3          | 3.5e-3          |
| 20×20           | 6.7e-5                                        | 6.7e-5          | 2.2e-4          |
| 40×40           | 2.3e-6                                        | 2.3e-6          | 1.2e-5          |
| $e_{10}/e_{20}$ | 34.7                                          | 34.7            | 16.0            |
| $e_{20}/e_{40}$ | 28.7                                          | 28.7            | 18.9            |
| $e_{10}/e_{40}$ | 995                                           | 995             | 302             |

**Table 1** Numerical results for the steady Stokes equation with  $\nu = 1$ .

analytic solution. The problem is

$$\begin{aligned}
 p_x - \nu(u_{xx} + u_{yy}) &= 0, \\
 p_y - \nu(v_{xx} + v_{yy}) &= -4\nu \cos(x) \sin(y), \\
 u_x + v_y &= 0.
 \end{aligned} \tag{9.21}$$

In this case the exact solution is

$$\begin{aligned}
 u^* &= \sin(x) \cos(y), \\
 v^* &= -\cos(x) \sin(y), \\
 p^* &= 2\nu \cos(x) \cos(y).
 \end{aligned} \tag{9.22}$$

The computational domain is  $\Omega = \{0 \leq x, y \leq 6\}$  with  $\nu = 1$ . The algebraic system of equations was solved with an iterative GMRES solver. In Table 1 we display the errors of the calculations in the discrete  $l_2$ -norm, i.e.,

$$||w||_h = \sqrt{\sum_{i,j} |w_{i,j}|^2 \Delta x \Delta y}, \tag{9.23}$$

for each component of the solution. We can see that we achieved fourth-order accuracy (or better) both for the velocity components and the pressure. Note that the accuracy is  $\sim 10^{-3}$  already on the very coarse  $10 \times 10$  grid.

Next we consider the time-dependent Stokes equations, and the time-discretization (9.2). The system corresponding to (9.20) is

$$\begin{aligned}
 \frac{3}{2}u^{n+1} + \Delta t (P_{1x}^{-1}Q_{1x}p^{n+1} - \nu(R_x^{-1}S_x + R_y^{-1}S_y)u^{n+1}) &= 2u^n - \frac{1}{2}u^{n-1}, \\
 \frac{3}{2}v^{n+1} + \Delta t (P_{1y}^{-1}Q_{1y}p^{n+1} - \nu(R_x^{-1}S_x + R_y^{-1}S_y)v^{n+1}) &= 2v^n - \frac{1}{2}v^{n-1}, \\
 P_{2x}^{-1}Q_{2x}u^{n+1} + P_{2y}^{-1}Q_{2y}v^{n+1} &= 0.
 \end{aligned} \tag{9.24}$$

For each timestep, this system is solved by a Krylov type method as described above. Note that the true boundary conditions are applied at each iteration of the Krylov method, not only at the completion of each timestep.

The first test problem was solved in the domain  $\Omega = \{0 \leq x, y \leq 6\}$ . As for the steady state equations, we add forcing functions in the equations to obtain simple analytical solution. The problem is

$$\begin{aligned} u_t + p_x - \nu(u_{xx} + u_{yy}) &= \sin(x) \cos(y) \cos(t), \\ v_t + p_y - \nu(v_{xx} + v_{yy}) &= -\cos(x) \sin(y) \cos(t) \\ &\quad - 4\nu \cos(x) \sin(y) \sin(t), \\ u_x + v_y &= 0, \end{aligned} \quad (9.25)$$

with the exact solution

$$\begin{aligned} u^* &= \sin(x) \cos(y) \sin(t), \\ v^* &= -\cos(x) \sin(y) \sin(t), \\ p^* &= 2\nu \cos(x) \cos(y) \sin(t). \end{aligned} \quad (9.26)$$

The second test problem for the time dependent Stokes equations is flow in a straight channel  $\Omega = \{0 \leq x \leq 1, -1 \leq y \leq 1\}$ :

$$\begin{aligned} u_t + p_x - \nu(u_{xx} + u_{yy}) &= 0, \\ v_t + p_y - \nu(v_{xx} + v_{yy}) &= 0, \\ u_x + v_y &= 0. \end{aligned} \quad (9.27)$$

For this problem we derive an analytical solution with the ansatz

$$\begin{aligned} u^* &= U(y) e^{\alpha x - \omega t}, \\ v^* &= V(y) e^{\alpha x - \omega t}, \\ p^* &= P(y) e^{\alpha x - \omega t}, \end{aligned} \quad (9.28)$$

and the boundary conditions  $U(-1) = U(1) = V(-1) = V(1) = 0$ , by solving a transcendental equation. We obtain

$$\begin{aligned} U(y) &= c_1 \sin(\alpha y) + \frac{2\kappa}{\alpha} c_2 \sin(\kappa y), \\ V(y) &= c_1 \cos(\alpha y) + 2c_2 \cos(\kappa y), \\ P(y) &= \frac{\omega}{\alpha} c_2 \sin(\alpha y), \end{aligned}$$

where

$$\kappa = \frac{\sqrt{\nu^2 + \alpha^2 \nu \omega}}{\nu},$$

for the numerical values  $\nu = 1$ ,  $\alpha = 1$  and

$$\begin{aligned} \omega &= 11.6347883720355431, \\ c_1 &= 0.9229302839450678, \\ c_2 &= 0.2722128679701572. \end{aligned}$$

|                              | Number of time steps |        |        |        |        |
|------------------------------|----------------------|--------|--------|--------|--------|
|                              | 10 000               | 20 000 | 30 000 | 40 000 | 50 000 |
| $eu_{10} =   u_{10} - u^*  $ | 2.3e-4               | 4.5e-4 | 6.8e-4 | 8.9e-4 | 1.1e-3 |
| $ev_{10} =   v_{10} - v^*  $ | 2.3e-4               | 4.5e-4 | 6.8e-4 | 8.9e-4 | 1.1e-3 |
| $ep_{10} =   p_{10} - p^*  $ | 2.0e-3               | 2.2e-3 | 2.5e-3 | 2.7e-3 | 2.9e-3 |
| $eu_{20} =   u_{20} - u^*  $ | 6.4e-6               | 1.3e-5 | 1.9e-5 | 2.5e-5 | 3.1e-5 |
| $ev_{20} =   v_{20} - v^*  $ | 6.4e-6               | 1.3e-5 | 1.9e-5 | 2.5e-5 | 3.1e-5 |
| $ep_{20} =   p_{20} - p^*  $ | 7.7e-5               | 9.5e-5 | 1.1e-4 | 1.3e-4 | 1.5e-4 |
| $eu_{40} =   u_{40} - u^*  $ | 2.2e-7               | 4.5e-7 | 6.7e-7 | 8.9e-7 | 1.1e-6 |
| $ev_{40} =   v_{40} - v^*  $ | 2.2e-7               | 4.5e-7 | 6.7e-7 | 8.9e-7 | 1.1e-6 |
| $ep_{40} =   p_{40} - p^*  $ | 3.2e-6               | 4.2e-6 | 5.1e-6 | 6.0e-6 | 7.0e-6 |
| $eu_{10}/eu_{20}$            | 35.4                 | 35.2   | 35.1   | 35.0   | 35.0   |
| $ev_{10}/ev_{20}$            | 35.4                 | 35.2   | 35.1   | 35.0   | 35.0   |
| $ep_{10}/ep_{20}$            | 26.1                 | 23.5   | 21.7   | 20.5   | 19.6   |
| $eu_{20}/eu_{40}$            | 28.5                 | 28.5   | 28.5   | 28.5   | 28.5   |
| $ev_{20}/ev_{40}$            | 28.5                 | 28.5   | 28.5   | 28.5   | 28.5   |
| $ep_{20}/ep_{40}$            | 24.1                 | 22.5   | 22.1   | 21.6   | 20.7   |
| $eu_{10}/eu_{40}$            | 1009                 | 1003   | 1000   | 999    | 998    |
| $ev_{10}/ev_{40}$            | 1009                 | 1003   | 1000   | 999    | 998    |
| $ep_{10}/ep_{40}$            | 629                  | 527    | 480    | 444    | 405    |

**Table 2** Numerical results for the test problem (9.25) when solving the time dependent Stokes equation with  $\nu = 1$ ,  $\Delta t = 1.0e - 5$  and  $N = M = [10, 20, 40]$ .

Both test problems (9.25) and (9.27) are discretized according to the fourth order scheme (9.24), and the algebraic system of equations was solved with an iterative GMRES solver. A small time step,  $\Delta t = 1.0e - 5$ , was chosen in order to demonstrate the correct order of accuracy in space. In Table 2 and Table 3 the errors are shown. We can see that we achieved better than fourth-order accuracy both for the velocity components and the pressure in both test problems.

### 9.4 A Fourth Order Method for the Navier-Stokes Equations

In this section we present some preliminary results for the incompressible time-dependent Navier-Stokes equations. The analysis of the boundary conditions

|                              | Number of time steps |        |        |        |        |
|------------------------------|----------------------|--------|--------|--------|--------|
|                              | 10 000               | 20 000 | 30 000 | 40 000 | 50 000 |
| $eu_{10} = \ u_{10} - u^*\ $ | 8.5e-4               | 2.7e-4 | 8.3e-5 | 2.6e-5 | 8.1e-6 |
| $ev_{10} = \ v_{10} - v^*\ $ | 3.7e-3               | 1.1e-3 | 3.6e-4 | 1.1e-4 | 3.5e-5 |
| $ep_{10} = \ p_{10} - p^*\ $ | 2.5e-2               | 7.9e-3 | 2.5e-3 | 7.7e-4 | 2.4e-4 |
| $eu_{20} = \ u_{20} - u^*\ $ | 2.3e-5               | 7.1e-6 | 2.2e-6 | 7.0e-7 | 2.2e-7 |
| $ev_{20} = \ v_{20} - v^*\ $ | 1.8e-4               | 5.7e-5 | 1.8e-5 | 5.6e-6 | 1.7e-6 |
| $ep_{20} = \ p_{20} - p^*\ $ | 8.5e-4               | 2.7e-4 | 8.3e-5 | 2.6e-5 | 8.1e-6 |
| $eu_{40} = \ u_{40} - u^*\ $ | 1.4e-6               | 4.4e-7 | 1.4e-7 | 4.3e-8 | 1.3e-8 |
| $ev_{40} = \ v_{40} - v^*\ $ | 9.9e-6               | 3.1e-6 | 9.6e-7 | 3.0e-7 | 9.4e-8 |
| $ep_{40} = \ p_{40} - p^*\ $ | 5.8e-5               | 1.9e-5 | 5.7e-6 | 1.8e-6 | 5.6e-7 |
| $eu_{10}/eu_{20}$            | 37.4                 | 37.4   | 37.4   | 37.4   | 37.4   |
| $ev_{10}/ev_{20}$            | 20.2                 | 20.2   | 20.2   | 20.2   | 20.2   |
| $ep_{10}/ep_{20}$            | 29.8                 | 29.8   | 29.8   | 29.8   | 29.8   |
| $eu_{20}/eu_{40}$            | 16.3                 | 16.3   | 16.3   | 16.3   | 16.3   |
| $ev_{20}/ev_{40}$            | 18.5                 | 18.5   | 18.5   | 18.5   | 18.5   |
| $ep_{20}/ep_{40}$            | 14.6                 | 14.3   | 14.7   | 14.3   | 14.5   |
| $eu_{10}/eu_{40}$            | 610                  | 610    | 610    | 610    | 610    |
| $ev_{10}/ev_{40}$            | 373                  | 373    | 373    | 373    | 373    |
| $ep_{10}/ep_{40}$            | 435                  | 428    | 437    | 428    | 431    |

**Table 3** Numerical results for the test problem (9.27) when solving the time dependent Stokes equation with  $\nu = 1$ ,  $\Delta t = 1.0e - 5$  and  $N = M = [10, 20, 40]$ .

is a direct generalization of the analysis given for the Stokes equations in Section 9.2, and the form of the boundary conditions is exactly the same. The fourth order approximations are also the same, but we need two more ingredients for the advective terms. Since  $u$  and  $v$  are not stored at the same points, fourth order averaging formulas  $Eu$  and  $Ev$  are required. Furthermore, the compact scheme used for derivatives of first order above, must be modified such that it is centered at a gridpoint. We use



$$\tilde{P}f' = \tilde{Q}f:$$

$$\begin{aligned}\tilde{P}f' &= \begin{cases} \frac{1}{12}(f'_0 + 2f'_1), \\ \frac{1}{6}(f'_{i-1} + 4f'_i + f'_{i+1}), & i = 1, \dots, N-1, \\ \frac{1}{12}(2f'_{N-1} + f'_N), \end{cases} \\ \tilde{Q}f &= \begin{cases} \frac{1}{24\Delta x}(-5f_0 + 4f_1 + f_2), \\ \frac{1}{2\Delta x}(f_{i+1} - f_{i-1}), & i = 1, \dots, N-1, \\ \frac{1}{24\Delta x}(-f_{N-2} - 4f_{N-1} + 5f_N). \end{cases}\end{aligned}\quad (9.29)$$

Using  $\tilde{P}_x, \tilde{Q}_x, \tilde{P}_y, \tilde{Q}_y$  to indicate the coordinate direction, we get the system (corresponding to (9.24))

$$\begin{aligned}\frac{3}{2}u^{n+1} + \Delta t (P_{1x}^{-1}Q_{1x}p^{n+1} - \nu(R_x^{-1}S_x + R_y^{-1}S_y)u^{n+1}) &= \\ 2u^n - \frac{1}{2}u^{n-1} - 2\Delta t(u^n\tilde{P}_x^{-1}\tilde{Q}_xu^n + (Ev^n)\tilde{P}_y^{-1}\tilde{Q}_yu^n) + \\ \Delta t(u^{n-1}\tilde{P}_x^{-1}\tilde{Q}_xu^{n-1} + (Ev^{n-1})\tilde{P}_y^{-1}\tilde{Q}_yu^{n-1}), \\ \frac{3}{2}v^{n+1} + \Delta t (P_{1y}^{-1}Q_{1y}p^{n+1} - \nu(R_x^{-1}S_x + R_y^{-1}S_y)v^{n+1}) &= \\ 2v^n - \frac{1}{2}v^{n-1} - 2\Delta t((Eu^n)\tilde{P}_x^{-1}\tilde{Q}_xv^n + v^n\tilde{P}_y^{-1}\tilde{Q}_yv^n) + \\ \Delta t((Eu^{n-1})\tilde{P}_x^{-1}\tilde{Q}_xv^{n-1} + v^{n-1}\tilde{P}_y^{-1}\tilde{Q}_yv^{n-1}), \\ P_{2x}^{-1}Q_{2x}u^{n+1} + P_{2y}^{-1}Q_{2y}v^{n+1} &= 0.\end{aligned}\quad (9.30)$$

Note that the coefficient matrix for the unknown  $\mathbf{U}^{n+1}$  is exactly the same as for the Stokes equations above. The system is complemented by the true boundary conditions in each iteration exactly as described above.

The last test problem demonstrates the ability of the scheme to produce fourth-order accurate solutions also for the time dependent Navier-Stokes equations. We consider the equations

$$\begin{aligned}u_t + uu_x + vv_y + p_x - \nu(u_{xx} + u_{yy}) &= \sin(x)\cos(x)\sin^2(t) \\ &\quad + \sin(x)\cos(y)\cos(t), \\ v_t + uv_x + vv_y + p_y - \nu(v_{xx} + v_{yy}) &= \sin(y)\cos(y)\sin^2(t) \\ &\quad - 4\nu\cos(x)\sin(y)\sin(t) - \cos(x)\sin(y)\cos(t), \\ u_x + v_y &= 0,\end{aligned}\quad (9.31)$$

in the domain  $\Omega = \{0 \leq x, y \leq 6\}$ . The exact solution is the same as for the first test problem for the time dependent Stokes equations (9.26).

The results of the numerical experiment is shown in the Table 4. For this computation the Reynolds number was  $Re = 1/\nu = 2000$  and the time step  $\Delta t = 1.0e-5$ . We can see that we achieved better than fourth-order accuracy

|                              | Number of time steps |        |        |        |        |
|------------------------------|----------------------|--------|--------|--------|--------|
|                              | 10 000               | 20 000 | 30 000 | 40 000 | 50 000 |
| $eu_{10} = \ u_{10} - u^*\ $ | 2.3e-4               | 4.5e-4 | 6.7e-4 | 8.8e-4 | 1.1e-3 |
| $ev_{10} = \ v_{10} - v^*\ $ | 2.3e-4               | 4.5e-4 | 6.7e-4 | 8.8e-4 | 1.1e-3 |
| $ep_{10} = \ p_{10} - p^*\ $ | 1.8e-3               | 1.7e-3 | 1.7e-3 | 1.6e-3 | 1.4e-3 |
| $eu_{20} = \ u_{20} - u^*\ $ | 6.3e-6               | 1.3e-5 | 1.9e-5 | 2.5e-5 | 3.1e-5 |
| $ev_{20} = \ v_{20} - v^*\ $ | 6.3e-6               | 1.3e-5 | 1.9e-5 | 2.5e-5 | 3.0e-5 |
| $ep_{20} = \ p_{20} - p^*\ $ | 5.7e-5               | 5.6e-5 | 5.3e-5 | 5.1e-5 | 4.7e-5 |
| $eu_{40} = \ u_{40} - u^*\ $ | 2.2e-7               | 4.4e-7 | 6.5e-7 | 8.6e-7 | 1.1e-6 |
| $ev_{40} = \ v_{40} - v^*\ $ | 2.2e-7               | 4.4e-7 | 6.5e-7 | 8.6e-7 | 1.1e-6 |
| $ep_{40} = \ p_{40} - p^*\ $ | 2.4e-6               | 2.4e-6 | 2.4e-6 | 2.3e-6 | 2.2e-6 |
| $eu_{10}/eu_{20}$            | 35.8                 | 35.8   | 35.7   | 35.7   | 35.6   |
| $ev_{10}/ev_{20}$            | 35.8                 | 35.8   | 35.8   | 35.8   | 35.8   |
| $ep_{10}/ep_{20}$            | 31.3                 | 31.2   | 31.0   | 30.6   | 30.3   |
| $eu_{20}/eu_{40}$            | 28.7                 | 28.7   | 28.7   | 28.6   | 28.5   |
| $ev_{20}/ev_{40}$            | 28.7                 | 28.7   | 28.7   | 28.7   | 28.6   |
| $ep_{20}/ep_{40}$            | 23.8                 | 23.0   | 22.5   | 22.2   | 21.6   |
| $eu_{10}/eu_{40}$            | 1028                 | 1027   | 1026   | 1022   | 1015   |
| $ev_{10}/ev_{40}$            | 1028                 | 1028   | 1028   | 1026   | 1022   |
| $ep_{10}/ep_{40}$            | 747                  | 717    | 698    | 679    | 654    |

**Table 4** Numerical results for the problem (9.31) when solving the time dependent Navier-Stokes equation with the Reynolds number  $Re = 1/\nu = 2000$ ,  $\Delta t = 1.0e - 5$  and  $N = M = [10, 20, 40]$ .

both for the velocity components and the pressure. As for the previous cases, we note the small error already on the very coarse  $10 \times 10$  grid.

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# Scalable Parallel Implicit Multigrid Solution of Unsteady Incompressible Flows

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## 10.1 Abstract

A scalable parallel iterative implicit multigrid algorithm is presented for complex unsteady incompressible viscous flows containing rotating and moving components, using dynamic relative-motion multiblock structured grids. The algorithm combines a discrete state-variable flux linearization, nonlinear multigrid iteration at each time step, with scalable concurrency introduced by a block-Jacobi LU/SGS scheme at each multigrid level. Semi-empirical performance estimates are developed for parallel CPU, memory and cost efficiencies on existing and hypothetical computing platforms. Scalability is analyzed using these estimates, and results are given in the form of performance landscapes for both memory-constrained sizeup and constant-problem-size scaleup modes. The influence of parameters such as MPI software bandwidth and architecture-specific software tuning is included. These results indicate that the method is scalable in a practical sense for large-scale problems. Subiteration convergence rate and polyalgorithm variants are also discussed, and computed results illustrating a large-scale simulation of a submarine maneuver induced by a ten-degree rudder deflection are given.

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## 10.2 Introduction

Parallel computing can both reduce runtime and provide access to the large but distributed global memory required for large problems in computational fluid dynamics. Scalable parallel solution algorithms have become a key element in the practical solution of large-scale problems and problems with long-term transient evolutions. The present study develops and analyzes a scalable implicit multigrid algorithm. Multigrid methods have been discussed extensively by Brandt [1], and some recent work on parallel multigrid flow solvers for structured grids can be found in [2–5]. One of the first parallel multiblock multigrid schemes for the three-dimensional Euler equations was that of Yadlin and Caughey [2], who introduced the concept of horizontal and vertical multigrid. In horizontal mode, all block interface boundaries are updated after each multigrid level, and in vertical mode, multigrid cycles are completed within each block before updating interface boundaries. An asynchronous vertical implementation was proposed in [2] in which interface boundaries are updated with the most recent data available from adjacent blocks, and this scheme was demonstrated successfully for up to eight processors. The multigrid scheme used here is a horizontal adaptation of the Full Approximation Scheme (FAS) multigrid scheme of Sheng, Taylor and Whitfield [6–7], which was implemented in vertical mode to reduce memory requirements for a single-processor code.

A scalable parallel algorithm without multigrid was developed by Pankajakshan and Briley [8] as a parallel adaptation of the Discretized Newton/Relaxation (DNR) scheme proposed by Taylor and Whitfield [9–10]. Scalable concurrency was introduced in [8] by using Block Jacobi Lower/Upper Symmetric Gauss–Seidel (BJ–LU/SGS) relaxation as the innermost iteration, to solve for Newton iterates. The present work uses nonlinear multigrid iteration cycles at each time step, BJ–LU/SGS at each multigrid level, and extends previous work through further study of parallel performance and scalability. The capabilities of the method for complex unsteady flow applications are illustrated by recent results from a DoD Challenge project [11] on submarine maneuvers induced by a moving control surface.

## 10.3 Basic Unsteady Flow Solver

The present strategy for developing an efficient parallel algorithm is to begin with an effective sequential algorithm and then introduce scalable concurrency modifications that do not significantly degrade the convergence rate or inherent efficiency of the serial algorithm. The basic flow solver is that of Taylor and Whitfield [9–10, 12] and is comprised of an iterative implicit finite-volume scheme, Roe/MUSCL fluxes, numerically computed state-vector flux linearizations, and approximate-Newton iteration solved using LU/SGS relaxation.

### 10.3.1 Upwind Finite-Volume Scheme

The three-dimensional unsteady incompressible Reynolds-averaged Navier-Stokes equations are solved by introducing artificial compressibility [13] to facilitate iterative solution at each physical time step. A cell-centered finite-volume scheme approximating the artificial compressibility formulation for a time-dependent curvilinear coordinate system can be written as

$$\partial \mathbf{q} / \partial \tau = - [\delta_i (\mathbf{f} - \mathbf{f}_v) + \delta_j (\mathbf{g} - \mathbf{g}_v) + \delta_k (\mathbf{h} - \mathbf{h}_v)] = - \mathbf{R}(\mathbf{q}) \quad (1)$$

Here,  $\mathbf{R}(\mathbf{q})$  is the steady residual vector, and  $\mathbf{q} = J(p, u, v, w)^T$  is the solution vector, where  $p$  is pressure,  $u$ ,  $v$ , and  $w$  are Cartesian velocity components, and  $J$  is the Jacobian of the inverse coordinate transformation. The inviscid flux vectors are  $\mathbf{f}$ ,  $\mathbf{g}$ ,  $\mathbf{h}$ , the viscous flux components including modeled turbulent stresses are  $\mathbf{f}_v$ ,  $\mathbf{g}_v$ ,  $\mathbf{h}_v$ , and  $\tau$  is time. The central difference operators are defined as in  $\delta_i(\cdot) \equiv (\cdot)_{i+1/2} - (\cdot)_{i-1/2}$  for each  $i, j, k$  direction, corresponding to the respective curvilinear  $\xi$ ,  $\eta$ , and  $\zeta$  coordinate directions. The artificial compressibility parameter is  $\beta = 5$ . Detailed definitions are given in [9].

### 10.3.2 Numerical Fluxes

The inviscid flux vectors at each cell face are obtained using Roe's [14] approximate Riemann solver and van Leer's MUSCL extrapolation of left and right state vectors,  $\mathbf{q}^R$  and  $\mathbf{q}^L$ , as implemented in the third-order nonlimited form of Anderson, Thomas, and van Leer [15]. The flux approximation can be written for the  $i$  direction as

$$\mathbf{f}_{i+1/2} = \mathbf{f}_{i+1/2}(\mathbf{q}_{i+1/2}^L) + \bar{\mathbf{A}}^-(\mathbf{q}_{i+1/2}^R, \mathbf{q}_{i+1/2}^L) \cdot (\mathbf{q}_{i+1/2}^R - \mathbf{q}_{i+1/2}^L) \quad (2)$$

Here,  $\bar{\mathbf{A}}^-$  is a Roe-averaged evaluation of  $\mathbf{A}^-$ , which is formed from the eigensystem of the inviscid flux Jacobian  $\mathbf{A} \equiv \partial \mathbf{f} / \partial \mathbf{q}$ . The matrix  $\mathbf{A}^-$  is defined by  $\mathbf{A}^- = \mathbf{S} \mathbf{A}^- \mathbf{S}^{-1}$ , where  $\mathbf{S}$  is a matrix whose columns are the right eigenvectors of  $\mathbf{A}$ , and  $\mathbf{A}^-$  is a diagonal matrix containing the negative eigenvalues of  $\mathbf{A}$  as nonzero elements. Analogous definitions for the  $j$  and  $k$  direction fluxes are obtained by replacing  $[i, \mathbf{f}, \mathbf{A}]$  by  $[j, \mathbf{g}, \mathbf{B} \equiv \partial \mathbf{g} / \partial \mathbf{q}]$  or  $[k, \mathbf{h}, \mathbf{C} \equiv \partial \mathbf{h} / \partial \mathbf{q}]$ , and using corresponding eigensystems to define  $\mathbf{B}^-$  and  $\mathbf{C}^-$ . Details and nonsingular eigensystems that use metric information from only one direction are given in [9].

### 10.3.3 Iterative Implicit Unsteady Algorithm

An iterative implicit nonlinear scheme for solving (1) is given by

$$\Delta \mathbf{q}^{n, s+1} / \Delta \tau = - \mathbf{R}(\mathbf{q}^{n+1, s+1}) \quad (3)$$

where  $\Delta(\cdot)^n \equiv (\cdot)^{n+1} - (\cdot)^n$ , and  $s = 0, 1, \dots$  is an iteration index. For the nonlinear spatial residual, a discrete linearization about a previous iteration state  $\mathbf{q}^{n+1, s}$  is written in the form

$$\mathbf{R}(\mathbf{q}^{n+1, s+1}) = \mathbf{R}(\mathbf{q}^{n+1, s}) + \mathcal{L}^{n+1, s}(\Delta_s \mathbf{q}^{n+1, s}) + \mathcal{O}(\Delta_s \mathbf{q}^{n+1, s})^2 \quad (4)$$

where  $\Delta_s(\cdot) \equiv (\cdot)^{s+1} - (\cdot)^s$ , and  $\mathcal{L}^{n+1,s}(\cdot)$  is a linear spatial difference operator made up of flux derivatives to be defined subsequently. This leads to the following iterative linearized implicit scheme:

$$[\Delta\tau^{-1}\mathbf{I} + \mathcal{L}^{n+1,s}(\cdot)](\Delta_s \mathbf{q}^{n+1,s}) = \mathbf{R}_U(\mathbf{q}^{n+1,s}) \quad (5)$$

where a *physical* unsteady residual  $\mathbf{R}_U$  is defined as

$$\mathbf{R}_U(\mathbf{q}^{n+1}) = [\Delta\tau^{-1}\mathbf{I}_p(\mathbf{q}^{n+1} - \mathbf{q}^n) + \mathbf{R}(\mathbf{q}^{n+1})] \quad (6)$$

by replacing the identity matrix  $\mathbf{I}$  by  $\mathbf{I}_p \equiv [0, 1, 1, 1]^T$  in the unsteady residual. The converged solution then satisfies the physical unsteady incompressible approximation  $\mathbf{R}_U = 0$  without an artificial compressibility time derivative in the continuity equation.

### 10.3.4 Numerical State–Vector Flux Linearization

The flux linearization matrices are computed numerically, as proposed by Whitfield and Taylor [9–10]. The authors have found that accurate linearization matrices provide better stability and iterative convergence rates than more approximate flux Jacobians. Whitfield and Taylor [12] have also proposed a new numerical flux linearization in which the numerical fluxes (i.e.,  $f_{i+1/2}$ ) are differentiated with respect to the left and right solution–variable state vectors  $\mathbf{q}^R$  and  $\mathbf{q}^L$ , instead of the nodal values  $\mathbf{q}_i$ , and  $\mathbf{q}_{i+1}$ . This technique is more economical, avoids the issue of whether to omit derivatives with respect to  $\mathbf{q}_{i+2}$  in high–order fluxes, and also seems to perform well in practical calculations. These numerical *state–vector flux linearizations* are defined [12] by

$$\hat{\mathbf{A}}_i^+ = \frac{\partial f_{i+1/2}}{\partial \mathbf{q}_{i+1/2}^L}, \quad \hat{\mathbf{A}}_i^- = \frac{\partial f_{i-1/2}}{\partial \mathbf{q}_{i-1/2}^R} \quad (7a,b)$$

and the  $k$ th column of each matrix is evaluated as in

$$\hat{\mathbf{A}}_i^+ = [\mathbf{f}_{i+1/2}(\mathbf{q}_{i+1/2}^R, \mathbf{q}_{i+1/2}^L + h\mathbf{e}_k) - \mathbf{f}_{i+1/2}(\mathbf{q}_{i+1/2}^R, \mathbf{q}_{i+1/2}^L)]/h \quad (8)$$

where  $\mathbf{e}_k$  is the  $k$ th unit vector and  $h \approx \sqrt{\text{machine zero}}$ . Analogous definitions apply to  $\hat{\mathbf{B}}_j$  and  $\hat{\mathbf{C}}_j$ . The linearized flux derivative operator  $\mathcal{L}^n(\cdot)$  can now be defined as

$$\begin{aligned} \mathcal{L}^n(\cdot) = & -\hat{\mathbf{A}}_{i-1}^+(\cdot)_{i-1} + (\hat{\mathbf{A}}_i^+ - \hat{\mathbf{A}}_i^-)(\cdot)_i + \hat{\mathbf{A}}_{i+1}^-(\cdot)_{i+1} \\ & -\hat{\mathbf{B}}_{j-1}^+(\cdot)_{j-1} + (\hat{\mathbf{B}}_j^+ - \hat{\mathbf{B}}_j^-)(\cdot)_j + \hat{\mathbf{B}}_{j+1}^-(\cdot)_{j+1} \\ & -\hat{\mathbf{C}}_{k-1}^+(\cdot)_{k-1} + (\hat{\mathbf{C}}_k^+ - \hat{\mathbf{C}}_k^-)(\cdot)_k + \hat{\mathbf{C}}_{k+1}^-(\cdot)_{k+1} \end{aligned} \quad (9)$$

### 10.3.5 Solution of the Linearized Scheme using LU/SGS Relaxation

The solution of (5) for  $\Delta_s \mathbf{q}^{n+1,s}$  is obtained by Lower–Upper Symmetric Gauss–Seidel (LU/SGS) relaxation. Introducing a second subiteration index  $m$



and appropriate definitions for the matrix  $\mathbf{D}$  and operators  $\mathcal{L}_1$ , and  $\mathcal{L}_2$ , the LU/SGS scheme can be written as

$$\begin{aligned} [\mathbf{D}^{n+1,s} + \mathcal{L}_1^{n+1,s}(\cdot)](\Delta_s \mathbf{q}^{n+1,s})^* + \mathcal{L}_2^{n+1,s}(\Delta_s \mathbf{q}^{n+1,s})^m &= \mathbf{R}_U(\mathbf{q}^{n+1,s}) \\ [\mathbf{D}^{n+1,s} + \mathcal{L}_2^{n+1,s}(\cdot)](\Delta_s \mathbf{q}^{n+1,s})^{m+1} + \mathcal{L}_1^{n+1,s}(\Delta_s \mathbf{q}^{n+1,s})^* &= \mathbf{R}_U(\mathbf{q}^{n+1,s}) \end{aligned} \quad (10)$$

The definitions of  $\mathbf{D}$ ,  $\mathcal{L}_1$ , and  $\mathcal{L}_2$  for the LU/SGS scheme are

$$\begin{aligned} \mathbf{D}(\cdot) &= [\Delta\tau^{-1}\mathbf{I} + (\hat{\mathbf{A}}_i^+ - \hat{\mathbf{A}}_i^-) + (\hat{\mathbf{B}}_j^+ - \hat{\mathbf{B}}_j^-) + (\hat{\mathbf{C}}_k^+ - \hat{\mathbf{C}}_k^-)](\cdot)_{i,j,k} \\ \mathcal{L}_1(\cdot) &= -\hat{\mathbf{A}}_{i-1}^+(\cdot)_{i-1} - \hat{\mathbf{B}}_{j-1}^+(\cdot)_{j-1} - \hat{\mathbf{C}}_{k-1}^+(\cdot)_{k-1} \\ \mathcal{L}_2(\cdot) &= +\hat{\mathbf{A}}_{i+1}^-(\cdot)_{i+1} + \hat{\mathbf{B}}_{j+1}^-(\cdot)_{j+1} + \hat{\mathbf{C}}_{k+1}^-(\cdot)_{k+1} \end{aligned} \quad (11a,b,c)$$

which satisfy the following consistency relationship:

$$\mathbf{D}(\cdot) + \mathcal{L}_1(\cdot) + \mathcal{L}_2(\cdot) = \Delta\tau^{-1}\mathbf{I}(\cdot) + \mathcal{L}(\cdot) \quad (12)$$

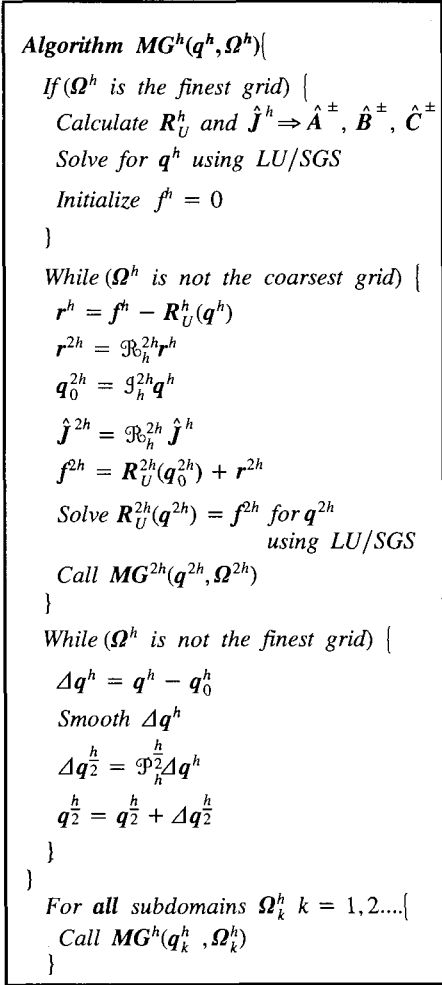
The solution is updated as in  $\mathbf{q}^{n+1,s+1} \leftarrow \mathbf{q}^{n+1,s} + (\Delta_s \mathbf{q}^{n+1,s})$  following a specified number of LU/SGS iterations. The quantity  $\Delta_s \mathbf{q}^{n+1,s}$  approaches zero as the  $s$ -iteration converges, giving a solution to  $\mathbf{R}_U(\mathbf{q}^{n+1}) = 0$ . Further discussion of this algorithm can be found in [16].

## 10.4 Scalable Parallel Implicit Multigrid Algorithm

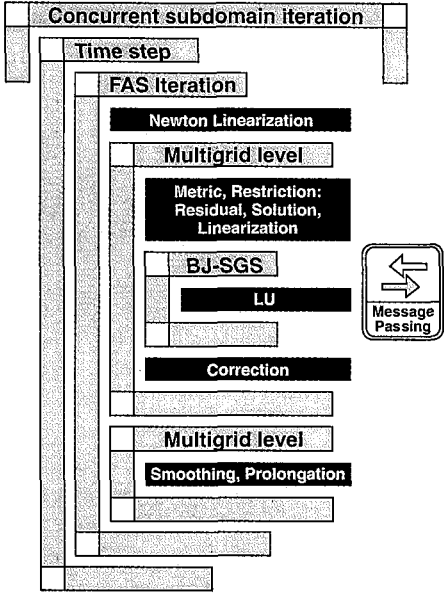
Spatial domain decomposition is a natural approach for introducing concurrency that exploits coarse/medium-grained parallelism among subdomains assigned to multiple processors. The subdomains are equivalent to grid blocks of a multiblock grid, which is either generated or repartitioned to provide load balancing for a specified number of processors. An MPI message-passing approach is used for its advantages in maintaining control, generality (especially for arbitrary subdomain connectivity), and portability across both distributed and shared-memory multiprocessor platforms.

The primary source of sequential data dependency in the basic flow solver is the LU/SGS iteration algorithm, but other parallelization issues arise when using multigrid acceleration. A concurrent *subiteration hierarchy* for implementing nested iterations was established [17] to explore parallel performance issues such as efficient CPU utilization, reduced communications overhead, algorithmic convergence rate, and scalability in the context of both steady and time-accurate solutions. The subiteration hierarchy consists of multiple FAS V-Cycle iterations, Newton iteration at each multigrid level, and LU/SGS iteration, as in the basic multigrid scheme of [6–7]. Concurrency was introduced at the innermost subiteration level by combining block-Jacobi (BJ) updating at subdomain interfaces with LU/SGS iteration within each subdomain. Although increasing the number of subdomains requires more BJ-LU/SGS iterations, the increase in cost is small, and the convergence rate often

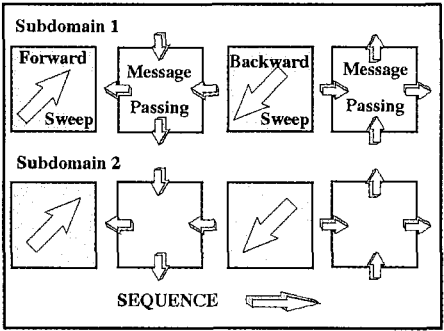
exceeds that obtained by updating interfaces sequentially or at a higher iteration level.



**Figure 1** Pseudocode: Parallel Nonlinear Multigrid Algorithm



**Figure 2** Parallel Subiteration Sequence



**Figure 3** Sequence of Operations for One BJ-LU/SGS Iteration

### 10.4.1 Nonlinear Multigrid Cycles

The nonlinear multigrid scheme is represented by the algorithm given as pseudo-code in Fig. 1. Here,  $q_h^{2h}$  is the restriction operator for the solution vector, while  $\mathcal{R}_h^{2h}$  is the restriction operator for the residual vector and flux linearization matrices. Similarly  $\mathcal{P}_h^{h/2}$  is the prolongation operator for interpolating the corrections from the coarse grid to the next finer grid. Each call to the rou-

time in Fig. 1. represents one V-cycle. The multigrid restriction, prolongation and implicit correction smoothing operators are as defined in [6–7].

#### 10.4.2 Parallel Multigrid Subiteration Sequence

In implementing FAS multigrid schemes, Newton iteration can be used at each multigrid level, and this option is discussed in [6–7, 17]. A more economical algorithm can be constructed by omitting the Newton iteration and using restriction operators to transfer the discrete linearization matrices to coarser grid levels. A schematic diagram of this subiteration sequence and major computational kernels is shown in Fig. 2. Three multigrid levels are generally used, and typical algorithm parameters for unsteady flows are three multigrid cycles at each time step, with eight or more BJ–LU/SGS iterations at each multigrid level. For steady flows, the steady residual  $\mathbf{R}$  is used instead of  $\mathbf{R}_U$ , and local time stepping is used with a single multigrid cycle and five or more BJ–LU/SGS iterations per multigrid level.

#### 10.4.3 BJ-LU/SGS Subiteration

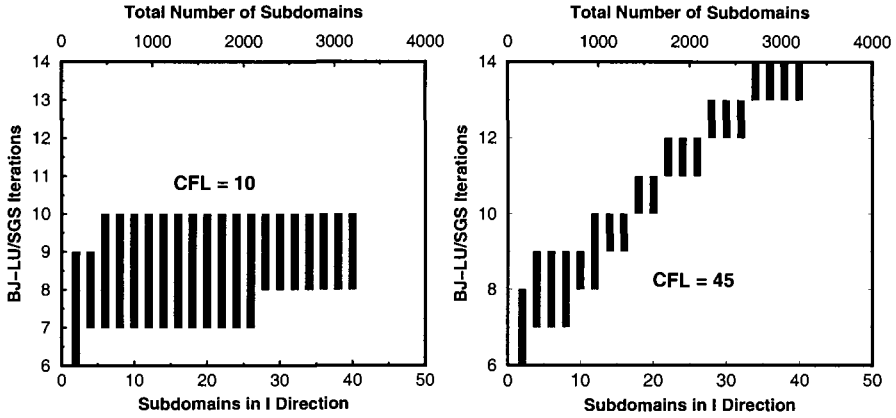
The BJ-LU/SGS subiteration to solve the linearized equations is implemented as a sequence of lower and upper (LU) sweeps through each subdomain, accompanied by message-passing exchanges of solution data at neighboring subdomain interfaces. Three possibilities exist for sequencing the LU sweeps and interfacial data exchanges. The first two variations involve a bi-directional exchange of interfacial data, either following each of the LU sweeps or after the two LU sweeps are both completed. The third method performs unidirectional data transfers after each of the lower and upper sweeps, as illustrated in Fig. 3. This third variation combines rapid global propagation of  $\Delta \mathbf{q}$  with minimal message passing, and is generally the method of choice.

#### 10.4.4 Algorithmic Scalability: BJ-LU/SGS Convergence Rate

Total computational *work* is influenced by the convergence rate of the subiteration and is therefore not a linear function of grid size. Scalability is directly influenced by the convergence rate of the BJ–LU/SGS scheme with an increasing number of subdomains and for different time steps. This was examined empirically in [17], and the results for a linear solution using (10) are illustrated here by a test case for a steady solution having 100,000 grid points, using local time stepping and a near-optimal CFL = 45. Two orders of residual reduction required 5 iterations for a single domain (LU/SGS), and this increases gradually to 8 iterations using BJ–LU/SGS for 200 subdomains, each having 500 points. For comparison, the Red-Black Gauss-Seidel scheme, whose (slower) convergence is independent of the number of subdomains, requires 14 iterations.

Since the innermost iteration need not be carried to complete convergence, the nonlinear solution for  $\mathbf{R}_U = 0$  also depends on the number of BJ–LU/SGS iterations as subdomains are added. Results showing the number of iterations required to reduce the residual to a baseline value obtained with six BJ–LU/

SGS iterations and eight subdomains is shown in Fig. 4 for CFL=10 and CFL=45. The test case was an appended hull at 18.11 degrees incidence with 800,000 points, using three multigrid levels and a single multigrid cycle.



**Figure 4** Effect of Decomposition on BJ-LU/SGS Iterations Required for Specified Residual Reduction

## 10.5 Parallel Performance Estimates and Scalability

The term scalability has numerous definitions and interpretations. One simple definition (useful when runtime is to be minimized) is that scalability implies a linear dependence between effective Mflops and the number of processors. The present goal for scalability on practical CFD simulations is to achieve an acceptable multiprocessor runtime for large-scale practical problems, while maintaining cost-effective use of computing resources such as CPU and memory.

### 10.5.1 Parallel CPU, Memory and Cost Efficiencies

Parallel efficiency and scalability are analyzed here using *a priori* semi-empirical estimates of parallel runtime based on input parameters for a particular flow case and parallel computing platform. The total runtime per time step is expressed as  $T_{Runtime} = T_{CPU} + T_{Comm}$ , where  $T_{CPU}$  is the time spent on floating point operations, and  $T_{Comm}$  is the total time spent on interprocessor communications, including message passing and loading buffer arrays. A concept of *idle or unused memory* is also introduced to represent memory assigned to processors but not actually used during execution (i.e., reserved and unavailable to other users). Unused memory may be available for allocation to other users for shared memory systems but not for distributed memory systems. The parallel CPU efficiency  $\eta_{CPU}$ , communications overhead  $\gamma_{Comm}$ , and memory efficiency  $\eta_{Mem}$  are defined by

$$\eta_{CPU} = \frac{T_{CPU}}{T_{Runtime}}, \quad \gamma_{Comm} = \frac{T_{Comm}}{T_{Runtime}}, \quad \eta_{Mem} = \frac{Mb_{Used}}{Mb_{Reserved}}$$

Finally, a cost efficiency  $\eta_{Cost}$  is defined using rough estimates of the costs of CPU ( $C_{CPU}$ ) and memory ( $C_{Mem}$ ) as a fraction of the total cost of the parallel computer. Here,  $C_{CPU} \doteq 50\%$  and  $C_{Mem} \doteq 30\%$ , are assumed to be typical for current computers, with the remaining 20% attributable to supporting hardware. The following overall cost efficiency

$$\eta_{Cost} = 1 - C_{CPU}(1 - \eta_{CPU}) - C_{Mem}(1 - \eta_{Mem})$$

provides an overall measure of the cost of computing resource utilization.

### 10.5.2 Cost Efficiency and Runtime

It is commonly observed that message-passing overhead is reduced by choosing decompositions with large point volumes and small surface areas, leading to small communication/computation ratios, and by asynchronous overlapping of communications with computation. According to the present performance analysis, the optimal cost efficiency for distributed-memory computers is obtained by choosing the minimum number of processors required to provide the necessary global memory ( $P_{min} = Mb_{Problem}/Mb_{Processor}$ ), since this gives 100% memory efficiency and minimizes message-passing overhead. However, since this runtime may not be acceptable in the user's working environment, the runtime can be reduced by increasing the number of processors used, although this will inherently increase message passing overhead and reduce distributed memory efficiency. The present performance analysis can help guide this tradeoff between faster runtime and less efficient computing resource utilization.

For coarse-grained load balancing on homogeneous processors, the domain should be partitioned into blocks of equal size. If the partitioning cannot be appropriately sized for load balancing due to a complex grid topology and/or nonhomogeneous processors, the load balancing efficiency can be improved by overloading selected processors with a number of the smaller subdomains. The performance analysis also suggests that processors with small memory tend to require exceptional communications speed to be efficient for these simulations.

### 10.5.3 Estimate of CPU Time

To estimate the CPU time, the number of floating point operations ( $OPs$ ) for each of four major algorithm components were first counted using a hardware performance monitor. These are given in operations per grid point as follows:  $OP_{Metric} = 481$  (grid metric evaluation),  $OP_{Res} = 1230$  (residual calculation),  $OP_{Lin} = 5284$  (numerical flux linearizations), and  $OP_{LU} = 330$  (one LU subiteration). Note that the cost of a BJ-LU/SGS iteration is small compared with the residual and Jacobian evaluations. The total CPU time (in seconds per time step) for the finest multigrid level is given by

$$T_{CPU} = W_{MG} [OP_{Metric} + N_{FAS} (OP_{Res} + N_{LU} \times OP_{LU} + OP_{Lin}/N_{Lin})] \frac{N_{Pts}}{P R_{CPU}}$$

where  $W_{MG}$  is a factor representing multigrid work units per multigrid iteration cycle [e. g.,  $(1 + 1/8 + 1/64 \doteq 1.14)$  for three multigrid levels], and the flux linearizations are (optionally) updated every  $N_{Lin}$  multigrid iteration cycles. The other parameters are defined in Table 1.

Since the value of  $R_{CPU}$  achieved in practice is highly dependent on computer architecture, compiler efficiency, and individual programming, the present analysis uses empirical benchmark timings for the target architecture, with message passing suppressed. Effective processor speed can be improved by cache-aware programming (i.e., maximum data reuse). Since the present code is MPI-portable across multiprocessor architectures, it is typically not tuned for any specific machine, although it is *generically* optimized for cache-based architectures.

The present algorithm requires 1.26 Gb of memory per million grid points, using 3 multigrid levels and 64-bit arithmetic. This memory is largely comprised of the following arrays: fluxes (24%), solution variables (18%), grid metrics (22%), 32-bit flux derivative matrices (34%), and buffer arrays (2%). The algorithm is implemented in FORTRAN-90 and MPI with a Single Program Multiple Data (SPMD) structure.

**Table 1** Parameters for Estimating Performance

|                | Computer Parameters                    |           | Algorithm Parameters                             |
|----------------|----------------------------------------|-----------|--------------------------------------------------|
| $P$            | <i>Number of processors</i>            | $N_{Pts}$ | <i>Grid points (millions)</i>                    |
| $R_{CPU}$      | <i>Effective CPU rate (Mflops)</i>     | $N_{MG}$  | <i>Multigrid iteration cycles</i>                |
| $R_{Buf}$      | <i>Effective buffering rate (Mb/s)</i> | $N_{LU}$  | <i>LU/SGS iterations per <math>N_{MG}</math></i> |
| $\beta_{MPI}$  | <i>MPI software bandwidth (Mb/s)</i>   | $L_m$     | <i>Average message length (Mb)</i>               |
| $\sigma_{MPI}$ | <i>MPI software latency (s)</i>        | $N_m$     | <i>Interfaces per subdomain</i>                  |
| $C$            | $1 + \log_2 P$                         | $N_q$     | <i>Number of <math>q^{n+1}</math> updates</i>    |

**10.5.4 Estimate of Communication Time**

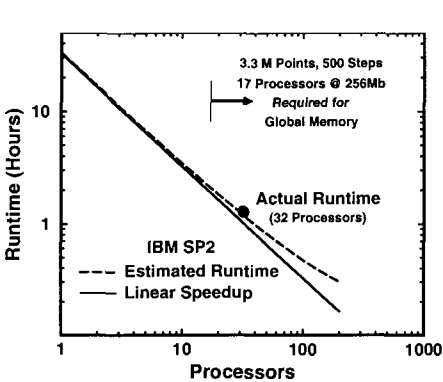
The interprocessor communication estimate is developed from parameters defined in Table 1. and accounts for 1) point-to-point messages exchanging data at subdomain interfaces, 2) loading and unloading of message-passing buffer arrays, and 3) global reduction operations required by algebraic turbulence models. Since the solution process is synchronized at the LU iteration level, all messages are sent at approximately the same time. The possibility of self contention for message bandwidth is included by estimating the bisection bandwidth as  $\sqrt{P}$ , the exact value for a two-dimensional interconnection mesh architecture. Each subiteration involves messages for two updates of  $\Delta q$  in one interface surface, and each time step requires updating  $q$  in two surfaces and one set of global operations. The total communications time per step is estimated as

$$T_{Comm} = W_{MG} N_{MG} N_m (2N_{LU} + N_q) \left[ \sigma_{MPI} + \frac{L_m \sqrt{P}}{\beta_{MPI}} + \frac{L_m}{R_{Buf}} \right] + 4 \left[ C\sigma_{MPI} + \frac{5L_m \sqrt{P}}{8\beta_{MPI}} \right]$$

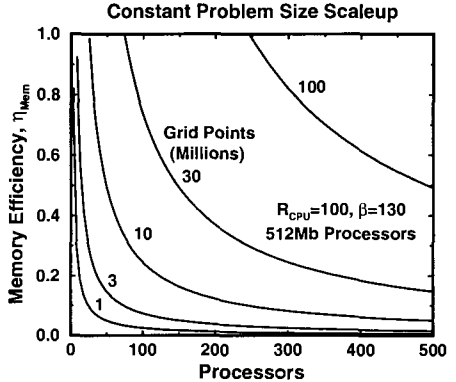
The parameters  $\beta_{MPI}$  and  $\sigma_{MPI}$  are obtained from available literature, and the buffering rate  $R_{Buf}$  is obtained from direct measurement in calibration runs.

### 10.5.5 Comparison of Performance Analysis with Direct Measurements

The estimated runtimes have been compared in [8,17] with actual runtimes for a few cases on an IBM-SP2. The SP2 had 66Mhz POWER2 *wide* nodes with either 128 or 256Kb data caches and memory bandwidth from 1.07Gb/s to 2.13Gbytes/s. The effective processor speed  $R_{CPU}$  measured in [8,17] was 49 Mflops, and the measured MPI software latency and bandwidth for large messages was  $\sigma_{MPI}=62\mu s$  and  $\beta_{MPI}=34Mb/s$ . Some generic cache-based tuning has subsequently increased  $R_{CPU}$  to 72 Mflops for the SP2 and the same code runs at 65–68 Mflops on a Cray-T3E. In Fig. 5, predicted and actual runtimes from [17] are compared for a sample case with 3.3 million points and 500 time steps; the good agreement largely reflects the semi-empirical content of the analysis. The loading of buffer arrays accounts for about 40% of communications cost. Asynchronous messages overlapping communication with computation reduced the message time by 25% but did not save significant runtime since the linear solver was slower, probably due to increased access of nonlocal data. Direct measurements of  $T_{Runtime}$ ,  $T_{CPU}$ , and  $T_{Comm}$  are also routinely monitored and have given  $\eta_{CPU}$  and  $\gamma_{Comm}$  consistent with the present analysis.



**Figure 5** Estimated and Actual Runtime for a Test Problem (3.3M points, 500 time steps)



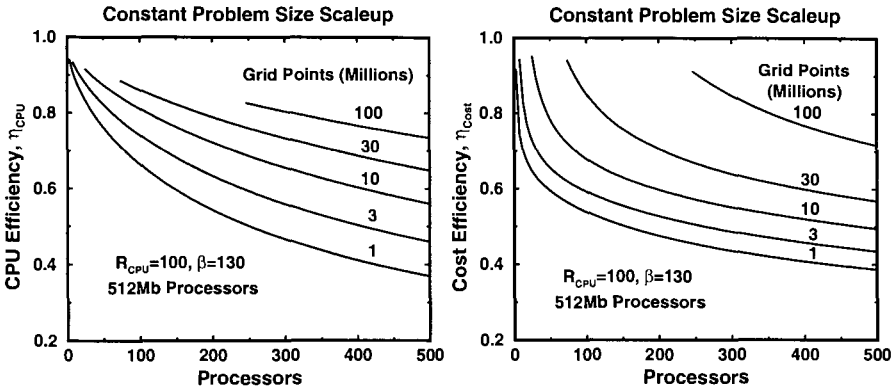
**Figure 6** Memory Efficiency for Constant Problem Size Scaup

### 10.5.6 Scalability: MPI/SPMD Software Implementation

Scalability can be studied by using the present performance analysis to develop a performance landscape for the parallel algorithm on both actual and hypothetical computing platforms. The key computer parameters for scalabil-

ity are  $R_{CPU}$ ,  $\beta_{MPI}$ , and  $R_{Buf}$ ; the MPI latency is negligible since there are a small number of large messages. The present scalability results are given in the form of performance landscapes for the algorithm running on a generic computer having the following parameters:  $R_{CPU}=100\text{Mflops}$ ,  $\beta_{MPI}=130\text{Mb/s}$ ,  $R_{Buf}=30\text{Mb/s}$ ,  $\sigma_{MPI}=15\mu\text{s}$ , and memory of  $512\text{Mb}$  per processor. These parameters are in a range consistent with the Cray T3E and SGI Origin 2000 class of machines. The algorithm parameters are typical of those used for time-accurate solutions: three multigrid levels,  $N_{MG} = 3$ , and  $N_{LU} = 7$ , increasing to 9 for 500 processors.

The performance landscape includes efficiencies for a) a *memory-constrained sizeup* in which the problem size is increased to maintain  $P = P_{\min}$  and  $\eta_{Mem} \approx 100\%$  as processors are added, and b) a *constant-problem-size scaleup* with different numbers of grid points. For constant problem size, reducing the runtime by adding processors beyond the minimum required leads to idle or unused distributed memory. This is illustrated by the rapid decrease in memory efficiency shown in Fig. 6. The CPU and cost efficiencies are shown in Fig. 7, and these also decrease for constant problem size scaleup. Nevertheless, the CPU efficiency remains above 40% of linear speedup except for extreme cases.

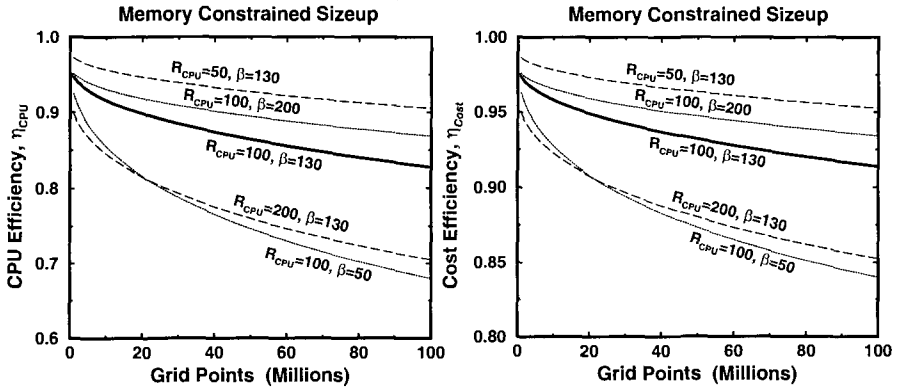


**Figure 7** CPU and Cost Efficiencies for Constant Problem Size Scaleup for Specified Values of Processor Speed and MPI Bandwidth

As mentioned previously, optimal efficiency is maintained for memory-constrained sizeup, and Fig. 8 shows that the CPU and cost efficiencies remain above 80% and 90%, respectively, for up to 100 million points (solid line). The sensitivity to CPU rate and MPI bandwidth are also shown in Fig. 8 for selected values. Although the efficiencies are not overly sensitive to  $R_{CPU}$ , any architecture-specific software tuning to increase  $R_{CPU}$  directly reduces runtime, which satisfies  $T_{\text{runtime}} \propto (PR_{CPU}\eta_{CPU})^{-1}$ . Note that increasing  $R_{CPU}$  lowers efficiency, since  $T_{\text{Comm}}$  is not affected, other than by possibly improving



buffering rate  $R_{Buf}$ . Overall, the present analysis indicates that the method is scalable in a practical sense for large-scale problems.



**Figure 8** CPU and Cost Efficiencies for Memory Constrained Sizeup for Specified Values of Processor Speed and MPI Bandwidth

### 10.5.7 Algorithm Variants

Several algorithm variations have been tested for possible savings in memory or runtime. The flux derivative matrices account for 34% of memory utilization and offer an option for memory reduction by recomputation. If the diagonal matrix  $D$  is computed and saved, and if  $\mathcal{L}_1$  and  $\mathcal{L}_2$  are recomputed for each LU/SGS iteration, then memory is reduced by about 30%, and each subsequent LU/SGS iteration increases runtime by about 16% (SGI PCA/R10000). A second approach to memory reduction is the Frechet variation in which matrix-vector products are calculated directly using

$$(\partial f / \partial q) \Delta q = [f(q + \epsilon \Delta q) - f(q)] / \epsilon$$

without forming the linearization matrices. The runtime breaks even for three LU/SGS iterations but increases by 37% for fifteen iterations.

Another implementation option is to use analytical formulas for the flux derivative matrices, obtained using a symbolic mathematics program (MAPLE 5.0). This option produced a 22% decrease in runtime on an SGI PCA/10000, but gave a 2% increase in runtime on an IBM SP2. A final option is to premultiply Eq. (10) by  $D^{-1}$ , saving  $D^{-1}\mathcal{L}_1$ ,  $D^{-1}\mathcal{L}_2$  and  $D^{-1}R_U$  for reuse in subsequent iterations. The higher cost of the first LU/SGS iteration is amortized over subsequent iterations by avoiding the solution of a (4x4) system. Although the break-even point is processor dependent, in one example it was five iterations, with a 10% savings at fifteen iterations.

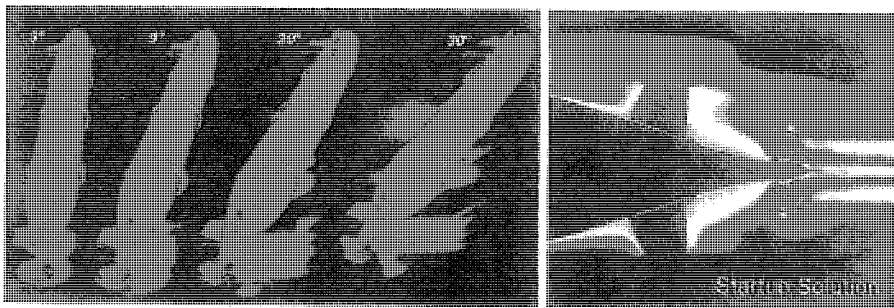
## 10.6 Demonstration: Rudder-Induced Maneuvering Simulation

Maneuvering submarine simulations have been computed [11] using a  $k-\epsilon$  turbulence model and Reynolds number of  $1.2 \times 10^7$ , for a fully configured

SUBOFF/propulsor configuration with realistic control-surface gaps on two sail-plane and four stern-plane surfaces. The vehicle trajectory was determined by coupling the flow solver with a 6-DOF solver for the vehicle motion. The integrated viscous stresses and pressure distribution provides hydrodynamic forces and moments for the 6-DOF equations, whose solution yields the time history of the vehicle's velocity, rotation rate, and trajectory. The multiblock dynamic structured grid topology can have an arbitrary *unstructured* pattern, including possible block surfaces that connect with more than one adjacent grid block. Movable control surfaces are treated by grids that deform locally in blocks near the moving control surface. The grid has 4.5 million points, with sublayer resolution such that grid spacing at the wall corresponds to  $y^+ \leq 1$  everywhere on the surfaces. Details of the technique used for grid motion are given in [18].

A startup solution was first computed for straight-line motion at constant velocity, giving a transient-free solution with periodic motion caused by the rotating propulsor. A solution was then computed for a self-propelled maneuver induced by a rudder deflection of 10 degrees, imposed as a linear motion in time during about one-quarter hull length of travel, after which the rudder is held fixed.

Computed results illustrating this solution are shown in Fig. 9. Axial velocity contours near the stern are shown for the startup solution, and the maneuvering solution is shown at four points in time, corresponding to lateral (horizontal) deflections of 3, 9, 20 and 30 degrees. This notional submarine does not have realistic design values for parameters such as moment of inertia, and consequently, the results demonstrate the capability for predicting maneuvers but do not represent an actual maneuver of a particular submarine. This solution required 350 time steps per propeller revolution and 7000 time steps per hull length traveled. The solution using 50 T3E/256Mb processors required 160 hours (8000 processor hours) for each hull length traveled. The startup solution became periodic after about 2 hull lengths.



**Figure 9** Example: Startup Solution and Submarine Maneuver Induced by Rudder Motion

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# Application of Vorticity Confinement to the Prediction of the Flow over Complex Bodies

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and M. Fan <sup>5</sup>

## Abstract

A technique, “Vorticity Confinement”, is described that represents a very effective, unified way of treating complex, high Reynolds number separated flows with thin convecting vortices, as well as complex solid bodies with thin attached boundary layers.

First, drawbacks of conventional Eulerian computational methods are described and how Vorticity Confinement, which is also Eulerian, ameliorates them. The basic assumptions in Vorticity Confinement are then reviewed. Some details of the method are described.

Following the description, a sequence of results are presented: First, 2-D results for convecting vortices and Cauchy-Riemann flow over a cylinder are presented. These describe the salient features of the method for convecting vortices and for flow over solid surfaces, embedded in a uniform Cartesian grid. Then, 3-D results for flow over complex bodies, including rotorcraft, are presented.

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## 11.1 Introduction

Emphasizing our point of view, most high Reynolds number incompressible flows are characterized by vortical structures which are either fixed, as body conforming boundary layers, or separate and convect, as vortex sheets and filaments. These structures can often be turbulent and are typically approximately modeled by partial differential equations (pde's). This modeling can include a detailed structure, as in eddy viscosity approaches, a zero thickness discontinuity, as implied by inviscid discretized Euler pde's for solid surfaces, or crude models of thin turbulent vortex filament cores implied by inviscid equations with diffusion resulting from numerical discretization error. Unfortunately, these structures are often very thin and these model pde's are then very difficult to solve due to resolution problems. These difficulties result in costly solution strategies involving body fitted grids, even for inviscid treatment, and extensive refinement near the body surface and adaptive grids with extensive refinement within shed vortex sheets and filaments, if any pretense is to be made of accurately resolving the structure of model pde's within these regions<sup>[1]</sup>.

Once we realize that these pde's are only approximate models for these vortical regions, we are led to the idea of modeling them *directly* on the grid using (nonlinear) difference equations, rather than using finite difference equations that approximately resolve the model pde's. This approach allows us to treat these structures as near-singular objects spread over only a few grid cells on an essentially uniform Cartesian computational grid.

This idea is, of course, what is used in shock capturing algorithms as an alternative to computing a detailed Navier Stokes solution for the internal shock structure. However, shocks involve characteristics that point inward, unlike vortical structures, making them easy to capture.

The Vorticity Confinement method, which implements this approach for vortical structures<sup>[2,3]</sup>, has proven to be a particularly effective technique to treat both body conforming boundary layers and shed vortex sheets and filaments. At the current stage of development, very simple structures are modeled for these vortical regions. These are adequate for the problems treated to date, which involve only thin vortical regions and separation from sharp corners. Also, initial applications are currently being developed to treat turbulent boundary layers separating from smooth surfaces.

With the method, solutions are computed on a coarse (typically uniform Cartesian) grid. Even though the method is completely Eulerian, with no Lagrangian marker arrays, shed vortex filaments can be convected indefinitely with no numerical spreading even though they are only a few grid cells in diameter. As such, Vorticity Confinement has the control over vortical structures that Lagrangian "vortex tracking" schemes have. On the other hand, as in conventional Eulerian techniques, the method allows vortex sheets

to be shed or reattach and vortex filaments to merge or reconnect with no logical operations or redistribution of marker arrays, which are typically required in Lagrangian schemes.

In this paper the method is first outlined. Then some representative results are then presented using Vorticity Confinement for flows with the above features involving convecting vortices and flow over cylinders in 2-D and flow over an Ellipsoid and a complex rotorcraft in 3-D.

## 11.2 Conventional Eulerian Methods

Typically, partial differential equations (pde's) are first formulated which express the equations of motion of the fluid. These equations can include explicit models for the turbulent regions. Although the following point may seem trivial, it is important to emphasize: If we consider thin vortical regions at high Reynolds number, either attached (boundary layers) or separated (convecting vortices), for realistic configurations, the goal is always to *model* these regions. This is, of course, because they can be mostly turbulent and a direct Navier-Stokes computation, including the small turbulent scales, is out of the question. This modeling can be explicit, such as, for example, Navier-Stokes-like pde's with an eddy viscosity model and possibly other related transport pde's, or it can be implicit - if the vortical regions are regarded as so thin that the details of the internal structure are not important for determining the overall, "outer" irrotational flows, "Implicit modeling" then implies that a pde is still discretized (for example, the inviscid Euler equations), but the computed results are only taken to be a crude model of the vortical regions. In fact, typically, the goal is then to use  $\leq 10$  grid points for these regions. Because of numerical viscosity, a smooth solution results. Our main point is that, whether the Euler equations (which are first order) or Navier-Stokes-like equations (which are second order) with a model viscosity are used, the result is still a model for the vortical regions at high Reynolds number, (as long as realistic "large scale" problems are computed with only a small number of grid cells devoted to the vortical cross-sections).

Of course, Lagrangian "Vortex Lattice" methods<sup>[4,5,6]</sup> also constitute a model of, for example, the internal structure of a rolling up vortex sheet. Also, the Lagrangian "vortex blob" approach models the internal structure<sup>[7]</sup>. Further, early approaches using analytic methods<sup>[8,9]</sup>, involved treating a vortex sheet as an exact contact discontinuity. These methods can also be thought of as models for the actual internal vortical structure.

The main feature that we want to bring out about conventional Eulerian methods is that, although they involve an efficient discretization and solution of the irrotational or "outer" part of the flow field, they also involve an inefficient model for thin vortical regions: When a pde (or set of pde's) is used

as an approximate model and must be discretized and accurately solved in thin regions, then great computational difficulties can arise. Manifestations include having to use very dense grids for convecting vortices, or fine adaptive grids that require large computational time to “grow” and follow the vortical region so that the discretized pde’s can be resolved<sup>[10]</sup>. Further, the conventional treatment of solid boundaries requires high accuracy at the surface: otherwise, numerical errors, usually in the form of numerically generated vorticity, can be created and diffuse or convect away from the surface, contaminating the solution. The impact on computational requirement is great: Either surface-conforming grids must be used, which are difficult to generate for complex geometries, or non-conforming Cartesian grids can be used with extensive (non-uniform) refinement at the boundary.

Another problem is that it is difficult to see how these models can be discretized and accurately solved (i.e., in a grid independent limit) for separating flow from smooth surfaces, since even the simplest case: laminar 2-D Navier-Stokes equations in the thin boundary layer approximation, have proven to be very difficult to solve<sup>[11]</sup>. Also, since large scale vortical structures are important in the boundary layer, it is difficult to see how an eddy viscosity approach, which is the basis of pde-based models, is appropriate. Thus, current pde models for approximating Reynolds averaged turbulent flow may not be a good approach, both for computational and physical reasons, and perhaps a different type of modeling should be considered.

A final point concerns consistency in accuracy for flows involving nearby separating vorticity, where the flow near the surface is a function of the separated, convecting vorticity as well as the bound, attached vorticity: In conventional methods, the bound and separating vorticity are treated differently (one with conforming grids and one without). This disparity could negate any benefit of using highly accurate conforming grids near a surface, since the final accuracy of the solution cannot be better than that for the nearby convecting vorticity, which does not involve a conforming grid.

## 11.3 Vorticity Confinement

### 11.3.1 - Requirements

The main goal of Vorticity Confinement is to model thin vortical regions using only a few grid points in the cross-section, without requiring them to be aligned with the grid. This seems to be consistent with the fact that we do not have exact equations that are feasible to solve for the structure of these vortical regions and, hence, can only approximately model them and that, further, they may not have to be accurately modeled if they are thin enough. Also, if there are many vortical regions, allocating more than a few



grid cells to their cross sections may not be computationally feasible, even with adaptive grids.

Accordingly, our method must allow vortices to be convected over long distances with no numerical spreading, but must allow a change in shape to be modeled in a controlled way. It should also allow complex solid surfaces, as thin vortex sheets, to be easily embedded in a uniform Cartesian grid with no requirement for generation of body conforming grids. A final requirement for Vorticity Confinement, which has strong implications for the formulation, is that it allow merging of convecting vortices and other changes in topology, such as the ability to convect over objects and reconnect or, for vortex sheets, to separate and reattach.

### 11.3.2 - Basic Concept

The basic idea behind Vorticity Confinement is to develop a set of difference equations on a fixed grid (typically uniform Cartesian) that fulfill the above requirements. This implies that the equations must be an accurate discretization of the Euler pde's in the outer, irrotational regions but reduce to a set of difference equations in the vortical regions where flow quantities vary by  $O(1)$  over a few grid cells.

This vortical region property implies that the equations will not be an accurate discretization of simple pde's there. Instead, they will be (non-linear) *difference* equations that result in the desired model values on the grid nodes. Further, to allow separation, reattachment, merging, etc., the vortical structure cannot be specified. Instead, the structure must *relax* to the desired profile.

The above requirements mean that there should be two basic parameters in the method: a length scale and a time scale. These are directly related to the grid cell size and time step of the computation. i.e., the resulting vortical profile should be a few grid cells wide and the relaxation should take place over a small number of time steps. Of course, if relevant, more complex models (including, for example, boundary layer dynamics or long-term viscous spreading) can be implemented which would involve more parameters. These are currently being formulated for cases with separation from smooth surfaces.

### 11.3.3 - Formulation

The simplest formulation of Vorticity Confinement involves, for incompressible flow, adding two terms to the discretized momentum conservation equations in a primitive variable formulation, which are similar to the diffusion and nonlinear anti-diffusion term for the advecting short pulse discussed in Ref. [12]. These terms are inherently multidimensional and Galilean invariant, depend only on local variables and vanish outside the vortical regions. As

stated, the solution of the equations with these terms is not specified but relaxes to a structure so that, for example, vortices can merge.

For general unsteady incompressible flows, the governing equations with the Vorticity Confinement terms are a discretization of the following equations:

$$\begin{aligned}\nabla \cdot \vec{q} &= 0 \\ \partial_t \vec{q} &= -(\vec{q} \cdot \nabla) \vec{q} + \nabla(p/\rho) + [\mu \nabla^2 \vec{q} - \varepsilon \vec{s}]\end{aligned}$$

where  $\vec{q}$  is the velocity vector,  $p$  pressure, and  $\rho$ , density, and the two terms in brackets are the Confinement terms. The two numerical coefficients,  $\varepsilon$  and  $\mu$ , control the size of the convecting vortical regions or vortical boundary layers and their relaxation time to a quasisteady shape.

There are many possible forms for the second Confinement term. The simplest one seems to be

$$\vec{s} = \hat{n} \times \vec{\omega}$$

where

$$\hat{n} = \frac{\nabla \eta}{|\nabla \eta|}$$

and the vorticity vector is given by

$$\vec{\omega} = \nabla \times \vec{q}$$

The scalar field,  $\eta$ , is defined in two ways:

$$\eta = \begin{cases} |\vec{\omega}| & : \text{Field Confinement} \\ |F| & : \text{Surface Confinement} \end{cases}$$

The simplest implementation of Vorticity Confinement, for convecting vortices, is called "Field Confinement". A simple modification, "Surface Confinement", for boundary layers, will be described in the next section.

For Field Confinement, the unit vector,  $\hat{n}$ , points towards the local centroid of the vortical region, and the Confinement term serves to convect vorticity back towards the centroid as it diffuses away. This convection increases the diffusion term and a steady-state distribution results when the two terms become balanced (for any reasonable values of  $\mu$  and  $\varepsilon$ ). Additional discussions of the formulation can be found in Refs. [13,3,14].

An important feature of the Vorticity Confinement method is that the extra terms are limited to the vortical regions: Both the diffusion term and the Confinement term vanish outside those regions. Another important feature concerns the total change induced by the correction in mass, vorticity and

momentum, integrated over a cross section of a convecting vortex. It is shown in Refs. [13,3,14] that mass and vorticity are explicitly conserved and momentum is almost exactly conserved. A small extension of the method, described in Ref. [15], allows it to also explicitly conserve momentum. This has no observable effect on the results described in this paper, except for cases involving long-term convection of strong 2-D vortices in a weak background velocity field. In those, the momentum conserving extension was used to ensure accurate trajectories.

In general, computed flows do not depend sensitively on the parameters  $\epsilon$  and  $\mu$ , for a range of values. Hence, the issues involved in setting them are similar to those involved in setting numerical parameters in other standard computational fluid dynamics schemes, such as artificial dissipation in many conventional compressible solvers which capture shocks.

The reason for this lack of sensitivity is that, for example, if a vortex core is close to axisymmetric, the velocity outside the core is not sensitive to the vorticity distribution, as long as the radius is kept small and prevented from becoming large due to numerical effects. Similar considerations apply to thin boundary layers. (This is analogous to the artificial shock thickness effects which depend on the dissipation parameter).

In addition to the solitary wave-like features of the vorticity distribution for free convecting vortices in 2D and 3D (convection with fixed shape), two studies, Refs. [3,14], demonstrated the ability of convecting 3D vortex filaments, initially in the form of rings, to merge and re-form. A comparison of these results<sup>[3]</sup> with measurements from an experiment, published in Ref. [14], showed a very close agreement. This demonstrated that the basic computational concept of relaxing to a quasi-steady vortical state through the action of the diffusion and nonlinear terms automatically allows realistic vortex filament reconnection while at the same time preventing spreading due to numerical effects.

It has been shown numerically that vortical solutions to the discretized equations are qualitatively close to those predicted for the continuum ones, even though the vortical regions are only a few cells thick. Roughly speaking, the confinement terms seem to be convecting discretization errors into the vortex center. This point should be addressed by an analysis of the discrete equations themselves, which we are currently carrying out: It must be stressed that we are *not* accurately solving the continuum pde equations within the vortex core or boundary layer, but are solving a set of nonlinear *discrete* equations that relax to the desired structure of a core confined to a few grid cells (even though, these equations become accurate approximations to the Continuum Euler pde's outside the vortical regions).

Finally, it should be mentioned that these solutions should be considered as "zeroth order" solutions, which are very economical but do not take into account dynamics in the vortical cores, such as turbulence effects. The idea

is to include such effects, if they are significant, in a perturbative way using extensions of the Confinement method. In this way, Vorticity Confinement can be regarded as a new type of *frame work* for vorticity dynamical computations.

### 11.3.4 Solid Surface Modeling with Uniform Cartesian Grids

The application of Vorticity Confinement to fixed vortex sheets representing solid surfaces in a non-conforming regular Cartesian grid with no-slip boundary conditions has recently been presented in Refs. [16,12] and [17,18,19]. This represents a very simple, economical way to treat complex bodies since it does not require body conforming or adaptive grid generation and can use a fast Cartesian grid set-up and flow solver. The steps for this method are delineated below:

- The geometry of the body or free surface is specified in a conventional way - such as by the coordinates of a set of points on the surface.
- From this set of points, a smooth function is computed on each point of a regular Cartesian computational grid. The value of this function,  $F(\vec{x})$ , is the (signed) distance of the grid point to the defined surface. Thus, the “level set” of values of  $\vec{x}$  such that  $F(\vec{x}) = 0$  implicitly defines the surface over which the flow is to be solved.
- The flow over the  $F = 0$  surface is computed time-accurately in a sequence of time-steps.

### 11.3.5 Computational Details

For each time-step ( $n$ ), the following computations are executed:

#### **Step a: Velocity Damping in Body**

The velocity,  $\vec{q}^n$  is multiplied by a function of  $F$ ,  $\lambda(F)$ , such that it is reduced for  $F < 0$ . This factor increases to 1 near the surface and no reduction is made in the new velocity at further distances:

$$\vec{q}' = \lambda(F)\vec{q}^n$$

For the ellipsoid presented this represents a reflection condition while for the cylinder and rotorcraft presented and many earlier results this factor was simply set to zero.

#### **Step b: Convection**

A convection-like computation is made to treat part of the momentum equation, as in conventional incompressible “split velocity” methods. (See Ref. [20]). This is a space-discretized version of

$$\vec{q}'' = \vec{q}' - \Delta t \vec{q}' \cdot \vec{\nabla} \vec{q}'$$

#### **Step c: Confinement**

Vorticity Confinement is used to compute a velocity increment such that quasi-steady thin vortical structures are obtained:

$$\vec{q}''' = \vec{q}'' + \Delta t(\epsilon_f \hat{n}_f \times \vec{\omega} - \epsilon_\omega \hat{n}_\omega \times \vec{\omega} + \mu \nabla^2 \vec{q}'') \quad (11.1)$$

Here  $\vec{\omega}$  is vorticity and

$$\hat{n}_f = \frac{\nabla |F|}{|\nabla F|}.$$

$$\hat{n}_\omega = \frac{\nabla |\vec{\omega}|}{|\nabla |\vec{\omega}||}$$

This is a crucial step: it advects vorticity back towards the  $F = 0$  surface and convecting vortical regions – without it vorticity would continually diffuse away leading to, effectively, a highly viscous low Reynolds number solution, since only large regular grid cells are used, rather than very thin body-fitted or adaptively refined cells as in conventional computations. (In earlier studies, the above diffusion was not added explicitly but resulted from discretization of Step b).

In this step  $\epsilon_f(F) = \epsilon$ , a constant, near  $F = 0$  and becomes small for points more than 2 cells away from the body. Also,  $\epsilon_\omega = \epsilon - \epsilon_f$ .

For the 2-D convecting vortex results, as explained above, a simple conservative extension of the added term was used so that momentum was explicitly conserved.

#### **Step d: Pressure Computation**

A pressure is computed such that the velocity at time step  $n + 1$  is divergence-free:

$$\nabla \cdot \vec{q}^{n+1} = 0.$$

This involves solving a Poisson equation

$$\nabla^2 \phi = -\nabla \cdot \vec{q}'''.$$

as in a conventional “split velocity” procedure.

#### **Step f: Velocity Update**

The velocity at the next time step is computed,

$$\vec{q}^{n+1} = \vec{q}''' + \nabla \phi.$$

This agrees with the momentum equation (to first-order in  $\Delta t$ ) where  $\phi$  is related to pressure by:

$$\phi = -\Delta t P / \rho.$$

### 11.3.6 Properties of Converged Solution

At convergence, the discrete approximations to  $\nabla \cdot \vec{q}$  and  $\nabla \times \vec{q}$  are driven to 0 inside the body. Also  $\vec{q}$  is forced to zero inside the body. Since  $\nabla \cdot \vec{q} = 0$  everywhere, there is then no flow through the body surface and vorticity is confined to thin regions on the body surface and thin regions which separate and convect with the flow. These regions are 2-3 grid cells thick, independent of the grid cell size or number of grid points in the overall problem (in the fine-grid limit). Further, since the computed velocity has zero divergence, if the separation locations are accurate, the computed velocity field will be at least first-order accurate in the grid cell size both away from convecting vortices and the body surface. The first-order error is due to the thickness of the computed vortical regions: simple perturbative corrections to bring the solution to second-order accuracy can be formulated within the framework of the method by considering the error as a displacement-thickness effect. This procedure should be simpler for many cases than resorting to the generation of a body conforming grid or adaptive high order refinement. However, for blunt body flows with separation the first order accuracy should be more consistent with the accuracy of the computation of the separating vorticity.

## 11.4 Current Results

### 11.4.1 Convecting Vortices

As a basic test of the ability of Vorticity Confinement to accurately and efficiently convect concentrated vortices, we tested the method for two cases in 2-D: a single vortex convecting in a uniform external field and a pair of vortices of opposite strength convecting in their mutual velocity fields. Both cases were strong tests since the translating velocity of each vortex was much less than that of the maximum velocity of the flow around the vortex (at the edge of the core). Thus, the numerical effects of this “self induced” velocity had to cancel very accurately to obtain accurate translational velocities.

In both cases a  $128 \times 128$  cell grid was used. Also, outer boundary conditions were set based on the computed centroids of the vortices.

#### 11.4.1.1 Single Vortex

In the first case (not shown), a single vortex moved in a uniform velocity field:  $u_\infty = 0.03$ ,  $v_\infty = 0.04$  (normalized by the maximum velocity at the edge of the vortex core). The CFL number based on this same velocity was 0.4. After 5,000 time steps the vortex core was the same size as initially – the contour of  $1/4$  maximum vorticity, (which remained constant) maintained a diameter of  $\sim 3$  grid cells. During the computation the vortex should have traveled 60

cells in the  $x$  direction and 80 in the  $y$  direction (for a total travel of 100 cells). In the computation the point of maximum vorticity in contour plots was less than 1 cell from the predicted position. This was expected since it can be easily shown that the centroid will move at the correct velocity if the method conserves momentum (which it did).

#### 11.4.1.2 *Vortex Pair*

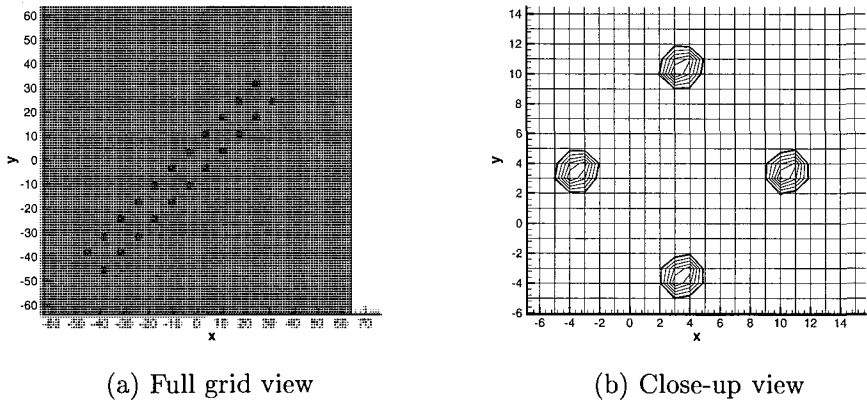
In the second case, the two vortices were at  $45^\circ$ , with centers initially 10 grid cells apart (see Fig. 1, where vorticity contours are plotted). The CFL number was 0.2 and, again, 5,000 time steps were used. Based on their separation (with no external flow) their predicted total translation was 70.7 grid cells in both the  $x$  and  $y$  directions.

The actual translations, again based on the points of maximum vorticity, was within a grid cell of that predicted. Also, as can be seen in Fig.1a, their separation remained constant. Further, as can be seen in Fig.1b, the outer (1/4 maximum vorticity) contour also had a diameter of  $\sim 3$  grid cells. It is important to realize that the unsymmetric appearance of the contours is a figment of the contour plotting software, since the center was not at a cell node or cell center and the vorticity varied by  $O(1)$  in  $1 \sim 2$  cells. At time steps where the center was near a cell center or node, the contours were symmetric about the vortex joining line and the line of travel, and close to axisymmetric.

It should be mentioned that actual vortices, if they obeyed the inviscid Euler equations exactly, would have become deformed due to their mutual induction. The Vorticity Confinement method, as has been emphasized, is intended to represent the effects of small sub-grid scale vortices, which should move like point vortices (in 2-D). The vorticity contours displayed can be thought of as spread images of these small cores plotted on the grid. This is analagous to the philosophy of Large Eddy Simulations.

#### 11.4.2 **Computation of Fuselage Flow**

The next objective is to quantify the model used for treating body surfaces in Cartesian grids: in particular, the ability to accurately predict surface pressure distributions. First, predicted flow is compared to the exact solution for a (simplified) two-dimensional circular cylinder case. Then, predicted surface pressures at various streamwise stations for an ellipsoid are compared to other computed results using complex grids and turbulence models and to wind tunnel data.



**Figure 1** Vorticity contours of translating vortices

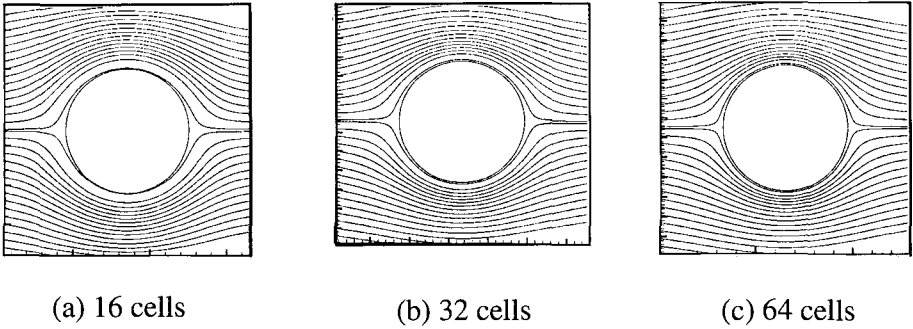
#### 11.4.2.1 Two-dimensional Circular Cylinder

First, results of a basic study of the flow equations: momentum and mass conservation, without the convective term, are presented for flow over a two-dimensional circular cylinder. This is essentially a Cauchy-Riemann solution with attached flow, for which the exact solution is known. This study is important because it validates the basic method of embedding the bound vorticity in a regular Cartesian grid. In these cases, which involved a uniform Cartesian grid, the cylinder was impulsively started from rest. At convergence, the divergence of the velocity field,  $\vec{q}$ , was zero everywhere and the vorticity was zero except for a narrow band near the surface. Also,  $\vec{q}$  was zero inside the cylinder.

Streamlines of the computed solution are presented in Fig. 2 for three different grid resolutions across the cylinder diameter: 16, 32, and 64 cells, respectively. The (numerical) displacement thickness effect of the vortical layer is clearly seen. Computed surface velocity (related to pressure by Bernoulli's relation), together with the exact solution, is shown in Fig. 3 as a function of angle about the center of the cylinder. For these results, an extrapolation, discussed below, was used from outside the vortical layer to the surface. It can be seen that the computed results are very accurate, even for the case with 16 grid cells across the cylinder diameter.

Since the vorticity is concentrated about the cylinder surface ( $F = 0$ ), the velocity is rapidly varying there. Since we use Bernoulli's relation to compute pressure, we must have a method to extrapolate velocity (or computed pressure) from outside the vortical region where the flow is smoothly varying, onto the surface. This extrapolation utilizes Vorticity Confinement and is





**Figure 2** Computed streamlines about circular cylinder for various grid resolutions

described in Refs. [16,21]. The velocity can then be interpolated onto a set of points that lie on the actual body surface and the pressure computed.

Higher order interpolation and extrapolation can be implemented to improve the accuracy of the surface pressure computation. Nevertheless, it can be seen that the predicted results for the two-dimensional circular cylinder are very accurate, even for the case with 16 grid cells across the cylinder diameter.

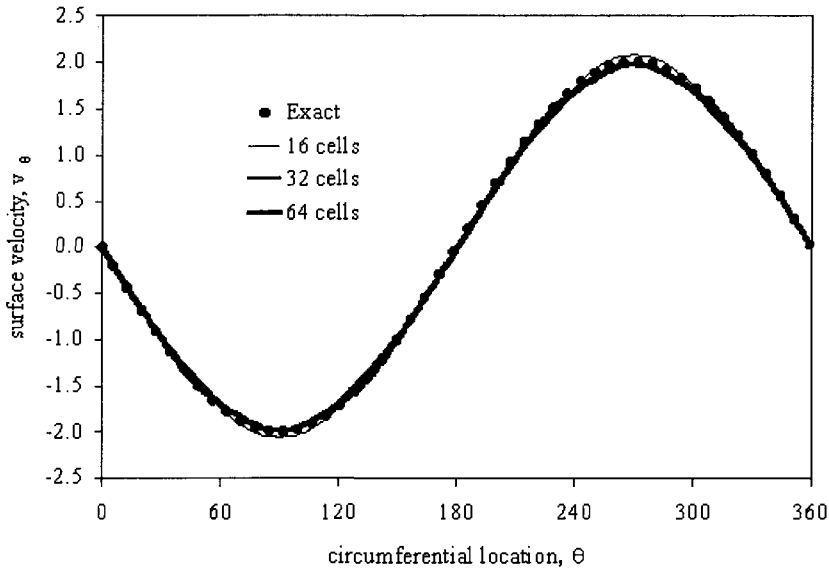
#### 11.4.2.2 Computation of Flow Over Ellipsoid

The method is applied to the computation of the three-dimensional flow over a 6:1 ellipsoid under high angles of attack. This problem is dominated by separation from the smooth surface and large vortical regions and therefore it is a good test case for the method. A quantitative analysis is carried out by comparing the computational results of the scheme with existing measurement data<sup>[22]</sup> and results of a conventional CFD scheme, which is based on a finite volume discretization of the Reynolds Averaged Navier-Stokes Equations and a  $k - \epsilon$  turbulence model<sup>[23]</sup>.

Calculations have been performed for five angles of attack:  $10^\circ$ ,  $15^\circ$ ,  $20^\circ$ ,  $25^\circ$ , and  $30^\circ$ .

- *Computational Grid*

Fig. 4 shows the computational grid employed in the present study. Only one regular, uniform Cartesian grid is used with no refinement near the surface. The number of grid cells in different directions are:  $i_{max} = 188$ ,  $j_{max} = 70$ ,  $k_{max} = 100$ . The maximum number of cells had to be strictly restricted due to limited computational resources. The ellipsoid length is equivalent to 120 grid cells, the diameter equals 20 grid cells. This means, that the mesh only extends one fourth body length ahead and one fourth body length



**Figure 3** Comparison of surface velocity predictions with exact solution for two-dimensional circular cylinder

downstream of the body. The distance to left and right boundary is one diameter, the distance to upper and lower boundary is 2.5 diameters. Some of the differences between results found with Vorticity Confinement and comparison data may be due to the small distances between configuration and outer boundaries. This grid can be compared with that used in the conventional  $k - \epsilon$  turbulence model computation<sup>[23]</sup>, shown in Fig. 5.

- *Vorticity Confinement Parameters*

The Reynolds number of the investigated problem determines the choice of the confinement parameter  $\epsilon$  and the numerical diffusion parameter  $\mu_n$ . Until now, only few investigations of the relation between  $\epsilon$  and  $\mu_n$  and the Reynolds number have been carried out, considering the use of conforming grids. In this study only one particular set of parameters has been used for all calculations. The parameters have been determined by trying, until the pressure distribution for  $\alpha = 25^\circ$  showed best agreement. Only three calculations had to be performed to find an appropriate set. The following set of parameters has been used:  $\epsilon = 0.065$ ,  $\mu_n = 0.0025$ .

- *Time Steps*

300 time steps have been conducted with a time increment of  $\Delta t = 0.4$ . With this time increment, the CFL-condition has been obeyed at every time step. This number of time steps was sufficient to approximately reach a converged solution, which has been checked by conducting additional 150 time steps. The solution after 450 time steps does not show significant differences compared with the solution after 300 steps.

- *Surface Treatment*

The surface boundary algorithm proved to have an influence on the quality of the computational results. Here, we extrapolate external values to nodes just inside the body and reverse sign to enforce a reflect boundary condition. This results in less sensitivity to the pressure extrapolation and separation.

- *General Properties of the Results*

In this section, some results computed with Vorticity Confinement are presented in order to show to what extent the physical properties of the flow problem are resolved quantitatively. In Fig. 6, the configuration is shown with crossflow separation on the lee side for  $\alpha = 25^\circ$ . The free vortex is represented by an isosurface of constant longitudinal vorticity. It consists of a pair of longitudinal vortices which connect with a vortex sheet on the aft end of the body. It is important to notice, that only an isosurface for one value of vorticity can be seen. That is the reason, why the free vortex seems to start at a particular point away from the body surface.

Fig. 7 shows a three-dimensional view of the ellipsoid including secondary streamline pattern at  $x/L = 0.25$ ,  $x/L = 0.33$ ,  $x/L = 0.58$  and  $x/L = 0.8$  for  $\alpha = 30^\circ$ , which have been calculated using Vorticity Confinement. It can be seen, that separation of streamlines occurs earlier with increasing longitudinal location. The streamlines roll up into the primary vortex core. This is conform to the physical behavior of the flow field.

- *Flow Pattern*

Fig. 8 - Fig. 10 show three different views of particle traces indicating cross-flow separation at  $\alpha = 30^\circ$ , computed with Vorticity Confinement and FV-Method, respectively. On the windward side, an attached three-dimensional boundary layer is formed. On the lee side, the flow detaches from the body because of the adverse, circumferential pressure gradient and rolls up into a pair of longitudinal symmetric vortices.

The streamline pattern found with the present method compare fairly well with the pattern found by Tsai and Whitney. A difference can be seen near the aft end of the configuration. This difference is due to the fact, that the solution with FV-Method was found on a mesh, which boundaries have a spherical shape, Fig. 5. Since, probably, only gridpoints of the inner grid have been used for data visualization, streamlines are cut off by the outer boundary of the inner grid.

Separation and reattachment regions are also visible in Fig. 11, which displays the cross flow streamline pattern computed at  $x/L = 0.77$  for

30° angle of attack. The FV-Method solution in Fig. 11 shows primary separation as well as secondary separation. The vortex core of the primary vortex is at a position approximately (0.04, 0.125) in local coordinates. The Vorticity Confinement solution does not resolve secondary separation. One important reason for this might be the grid cell size. The cross-section of the secondary separation corresponds to approximately 3 grid cells of the Cartesian grid. A further difference can be observed concerning location and shape of the vortical region. The location of the vortex core is (0.04, 0.16) and its shape is stretched, compared to FV-Method results. It is also noticeable, that at the separation point streamlines show an eruption-like behavior. The described properties may be caused by the fact that the numerical boundary layers are much thicker than physical boundary layers. This might effect especially the location of the vortex core. The location of the vortex core can also be effected by the fact that inflow boundaries with imposed free stream velocity are very close to the configuration. Although not all details can be resolved, the most important features of the flow are clearly represented by the Vorticity Confinement solution.

- *Surface Pressure*

Using the Vorticity Confinement method on Cartesian grids, the pressure at a point on the surface can only be calculated based on values from surrounding grid points. Since the pressure field is calculated for all gridpoints in the computational domain, regardless, whether they are inside or outside the configuration, no discontinuity on the surface occurs, the distribution is smooth. Hence, no difference can be observed whether extrapolating the pressure from points outside the solid or interpolating it, using points inside and outside. In this study, first the Cartesian coordinates of the investigated point on the surface are calculated, and based on that information, the surrounding grid points are determined. Then, a linear interpolation is applied to compute the pressure for the point on the solid. Fig. 12 - Fig. 16 present a comparison of computed surface pressure, obtained with the presented scheme, with experimental data and results of the FV-method, for angles of attack from 10° to 30° at an axial location 77 percent  $x/L = 0.77$  of the ellipsoid length.  $\phi = 0^\circ$  and  $\phi = 180^\circ$  indicate points on windward side and lee side, respectively. The separation dominated region is from  $\phi = 90^\circ$  up to  $\phi = 180^\circ$ .

The FV-Method results show an excellent agreement for higher angles of attack. Exceptions are visible at 160° azimuth for  $\alpha = 20^\circ$  angle of attack and near 0° azimuth for 30° angle of attack.

The best agreement between Vorticity Confinement results and experimental data can be seen for 20° and 25° angle of attack. The local pressure minimum at approximately 155° azimuth is not resolved for any angle of attack. At lower incidence, especially for  $\alpha = 15^\circ$ , the pressure distribution is slightly corrugated. Part of this phenomena may be attributable to

the linear interpolation of pressure to the investigated point on the surface from surrounding grid points. For  $\alpha = 30^\circ$  differences between Vorticity Confinement results and data increase and also occur in windward regions ( $\phi = 0^\circ - \phi = 90^\circ$ ), where the boundary layer is attached.

Again, the dominating property of the flow, i.e. the adverse, circumferential pressure gradient, which leads to separation, has been predicted correctly and a fairly good agreement to experimental data has been obtained.

### 11.4.3 Computation of Complete Apache Helicopter in Forward Flight

An overset procedure for computing the flow close to the blades, described in Ref. [21] was used for computing flow over the Apache Helicopter in forward flight where the main flow field computation (including fuselage) utilized a uniform Cartesian grid. These calculations were performed for an Apache-A helicopter fuselage with a four bladed rotor. The treatment of the body surface was as described before. Two computational cases, ascending and level forward flight, are presented. The level forward flight solution is also presented in Ref. [16] in order to illustrate the vortex merging property of the basic method. The inner blade fitted C-grid had  $120 \times 26 \times 24$  cells and the background Cartesian grid had  $162 \times 108 \times 54$  cells. The computation required 2.5 hrs/revolution on a CRAY YMP. The tip vortices are visualized by the iso-surface of vorticity magnitude in Figs. 17~18. In these pictures, the contour level of vorticity magnitude is set to 30% of the maximum vorticity magnitude. A short movie which shows the developing of the tip vortices can be viewed at [www.flowanalysis.com](http://www.flowanalysis.com).

## 11.5 Conclusion

The method presented in this paper, based on Vorticity Confinement, has been shown to solve two important incompressible flow problems: the computation of concentrated vortices over long times with no numerical spreading and the computation of flow over complex bodies without body conforming grids or long set-up times. Both computations were completely Eulerian and only required a coarse, regular computational grid.

The new method involves generating solitary wave-like configurations on the computational lattice to represent vortical regions that are direct solutions of nonlinear difference equations, rather than solutions of conventional discrete finite difference approximations to partial differential equation models for these regions. As such, the new method provides a new, efficient framework for efficiently computing a large class of flows.

Of course, as with conventional methods, much work remains to be done for validation and in calibrating models for Reynolds averaged representation of turbulent regions. We are currently developing such models, which are quite different from conventional pde models.

### Acknowledgements

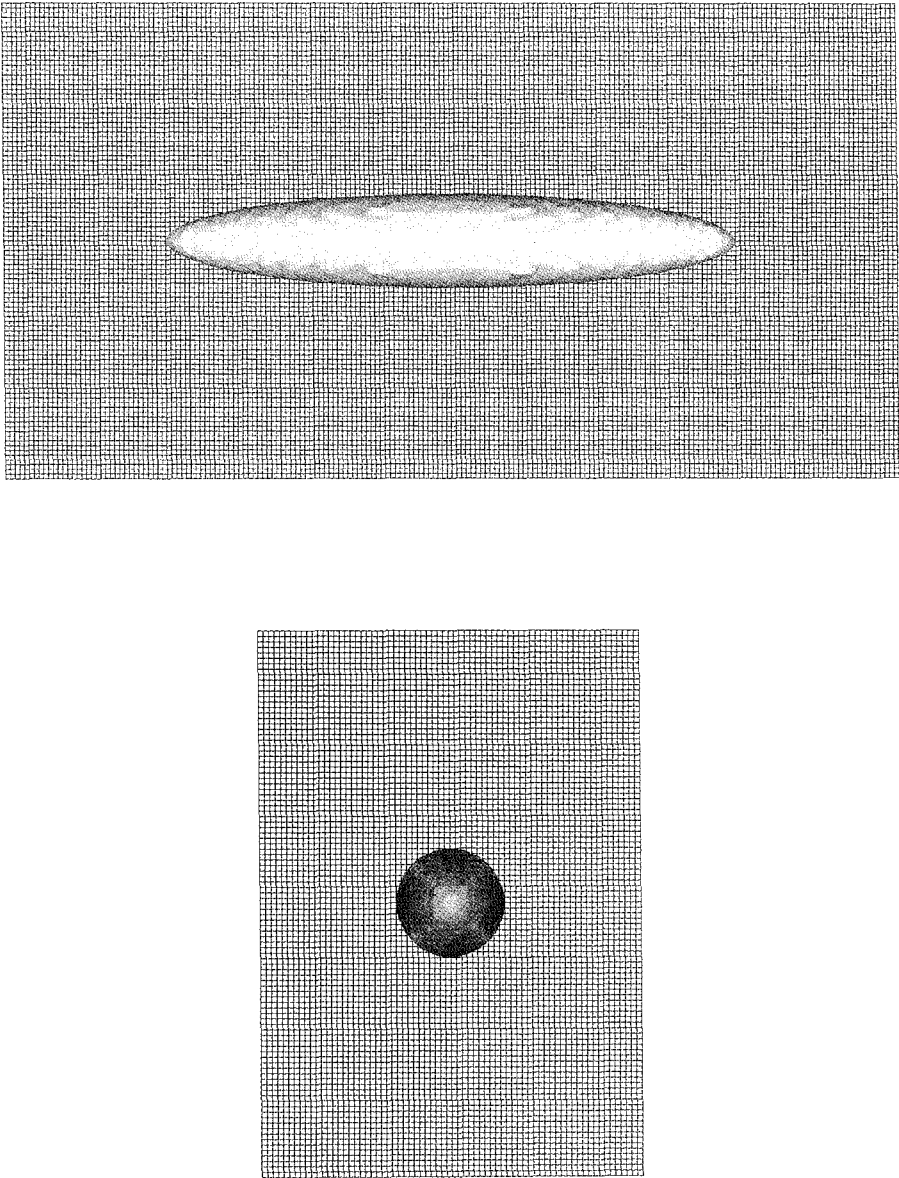
We would like to express our gratitude to Frank Caradonna of the Army Aeroflightdynamics Directorate at Moffett Field for extensive advice on rotorcraft aerodynamic issues.

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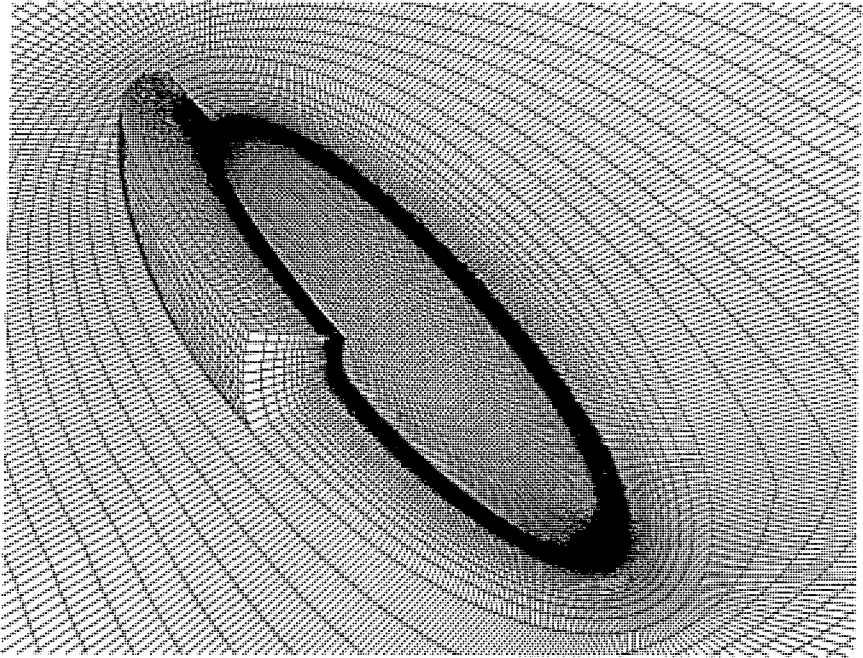
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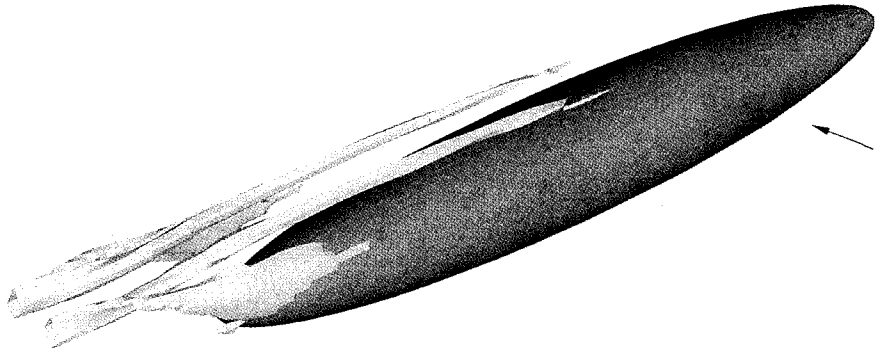


**Figure 4** Side view (top) and front view (bottom) of Computational mesh for present method with embedded ellipsoid

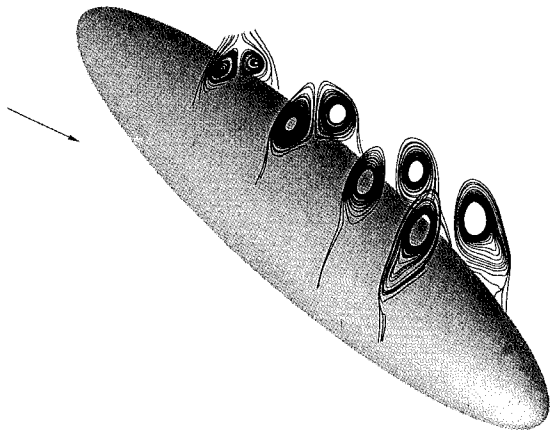




**Figure 5** Computational mesh for the multiblock finite volume method used by Tsai and Whitney



**Figure 6** Separating vortices on lee side of the configuration for  $\alpha = 25^\circ$  calculated with Vorticity Confinement. The arrow indicates the incoming flow.



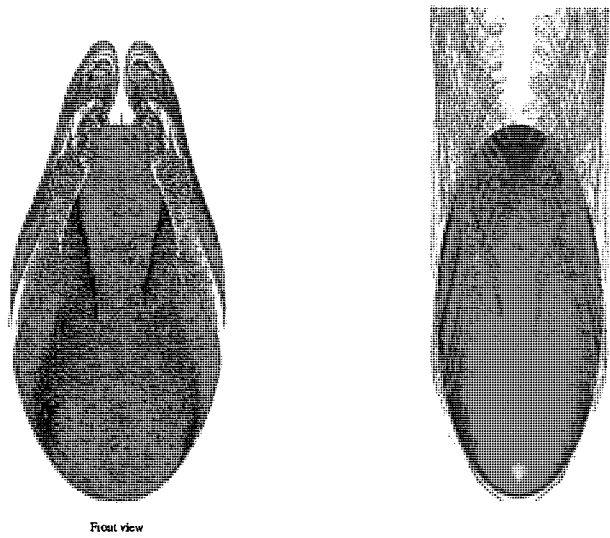
**Figure 7** Threedimensional view on ellipsoid with secondary streamlines at  $x/L = 0.25$ ,  $x/L = 0.33$ ,  $x/L = 0.58$  and  $x/L = 0.8$  for  $\alpha = 30^\circ$  (Vorticity Confinement). The arrow indicates the incoming flow.



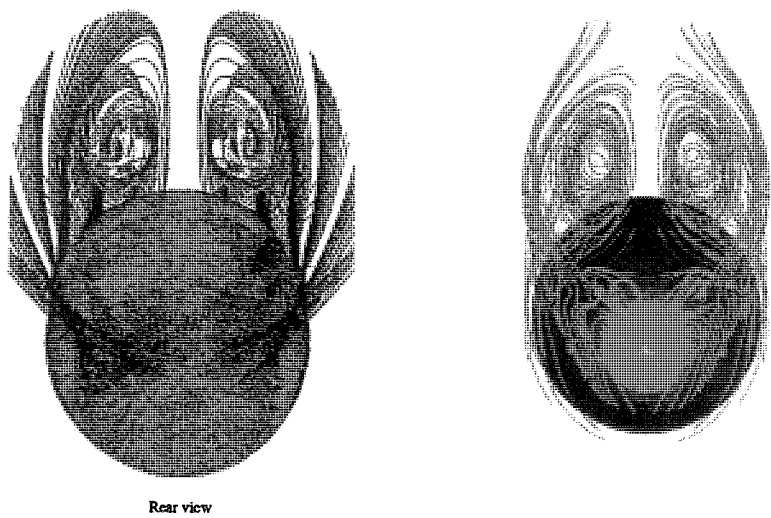
Side view



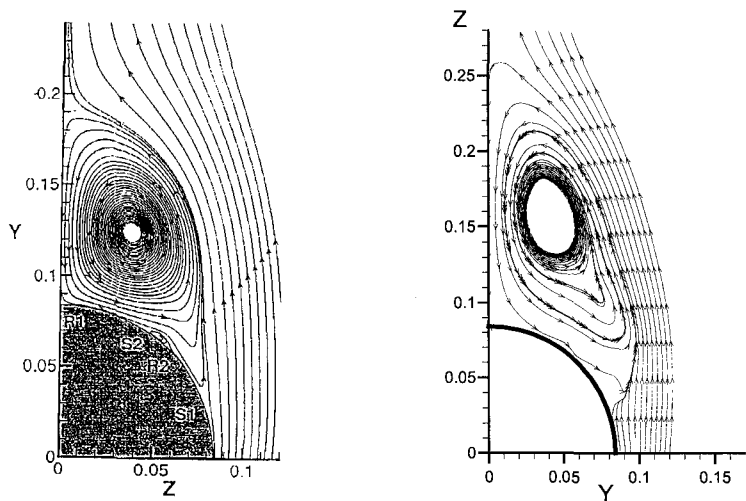
**Figure 8** Side view on streamline pattern of flow over 6:1 ellipsoid at  $30^\circ$  angle of attack. Top: Tsai and Whitney, 1999. Bottom: Vorticity Confinement



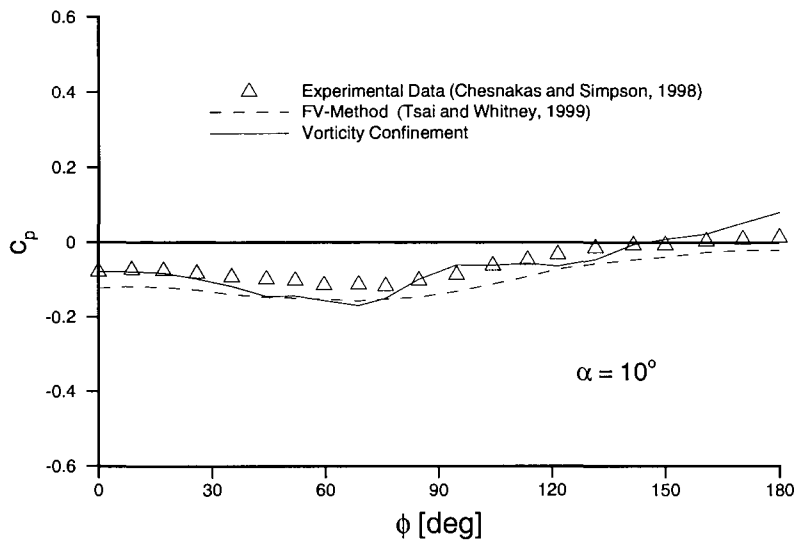
**Figure 9** Front view on streamline pattern of flow over 6:1 ellipsoid at 30° angle of attack. Left: Tsai and Whitney, 1999. Right: Vorticity Confinement



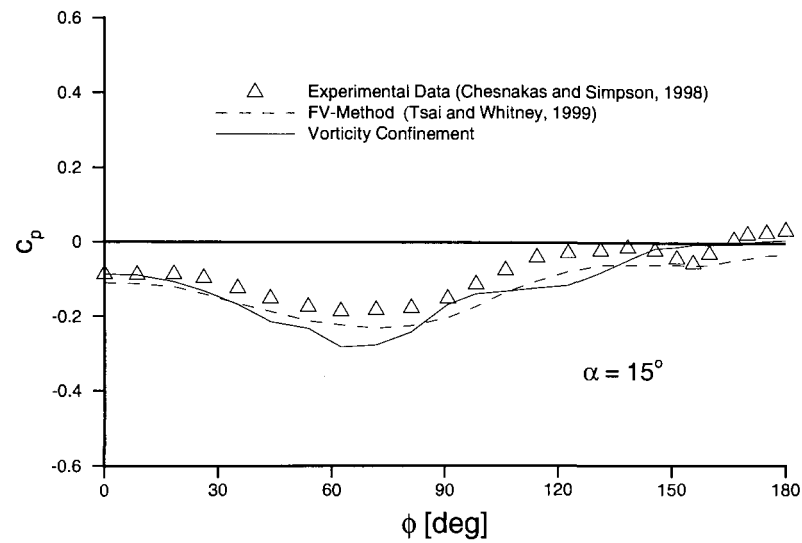
**Figure 10** Rear view on streamline pattern of flow over 6:1 ellipsoid at 30° angle of attack. Top: Tsai and Whitney, 1999. Bottom: Vorticity Confinement



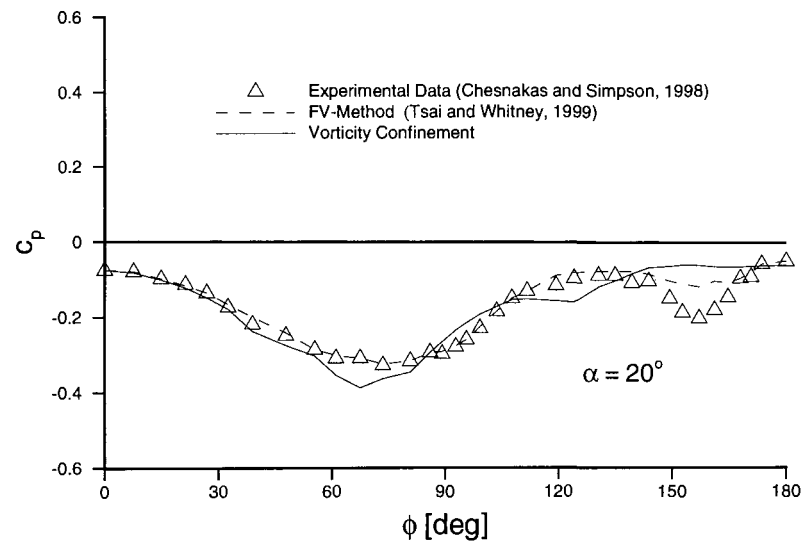
**Figure 11** Cross flow streamline pattern at  $x/L = 0.77$  for  $\alpha = 30^\circ$  : left: Tsai and Whitney, 1999. S1 and S2 indicate location of primary and secondary separation, R1 and R2 indicate location of primary and secondary reattachment. right: Vorticity Confinement



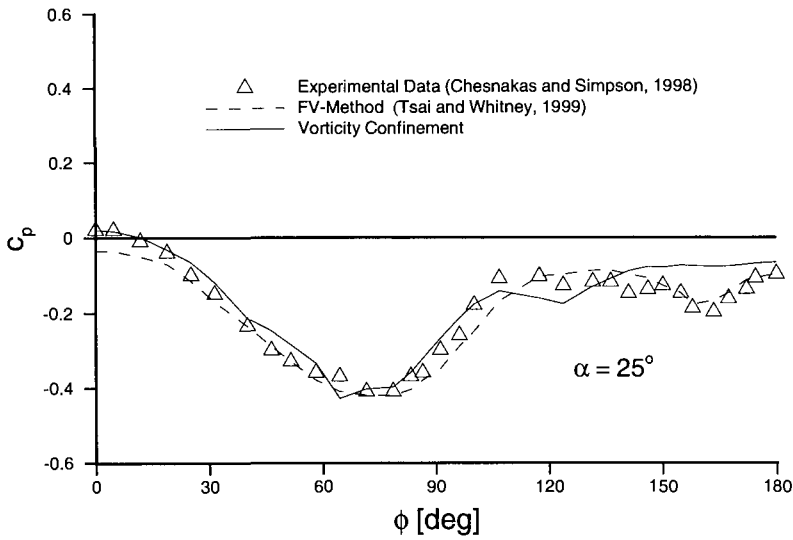
**Figure 12** Pressure coefficient data at  $x/L = 0.77$  and  $\alpha = 10^\circ$



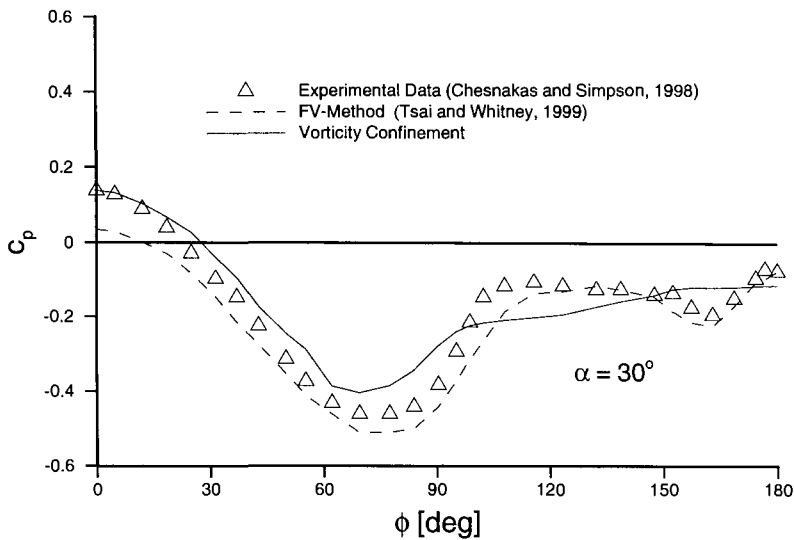
**Figure 13** Pressure coefficient data at  $x/L = 0.77$  and  $\alpha = 15^\circ$



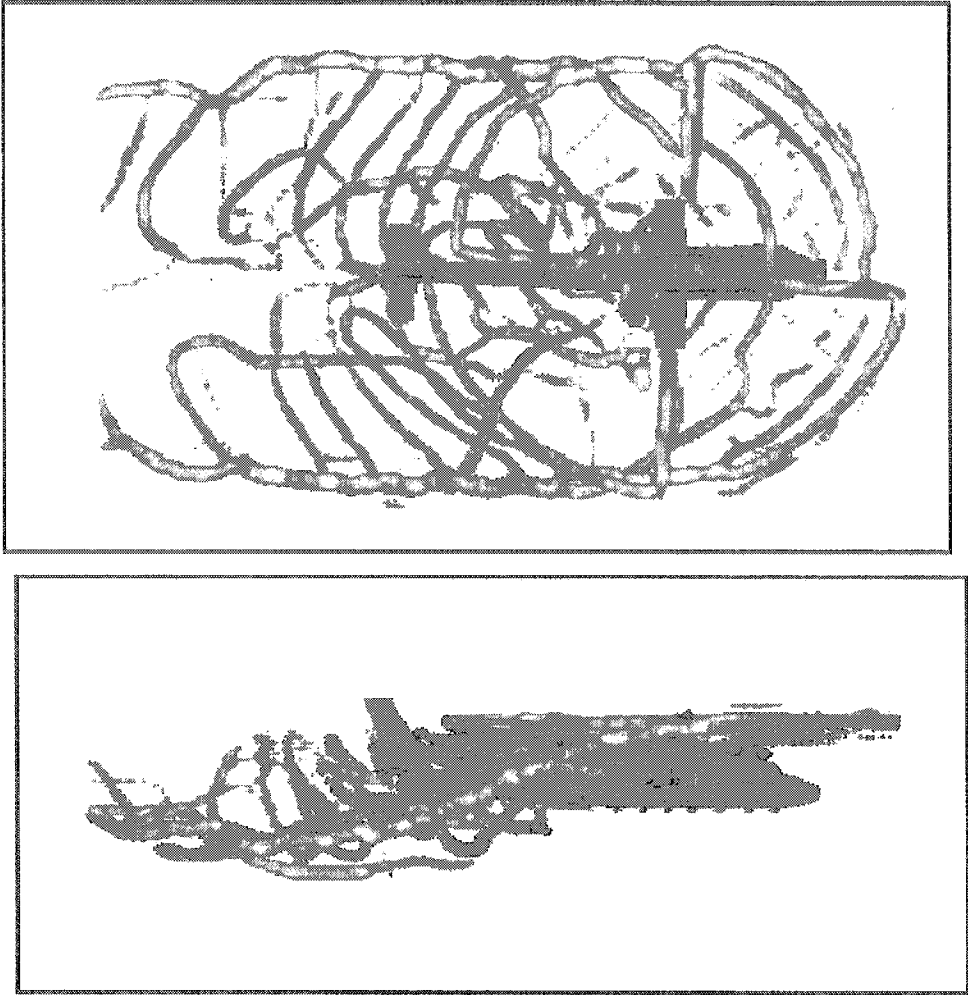
**Figure 14** Pressure coefficient data at  $x/L = 0.77$  and  $\alpha = 20^\circ$



**Figure 15** Pressure coefficient data at  $x/L = 0.77$  and  $\alpha = 25^\circ$

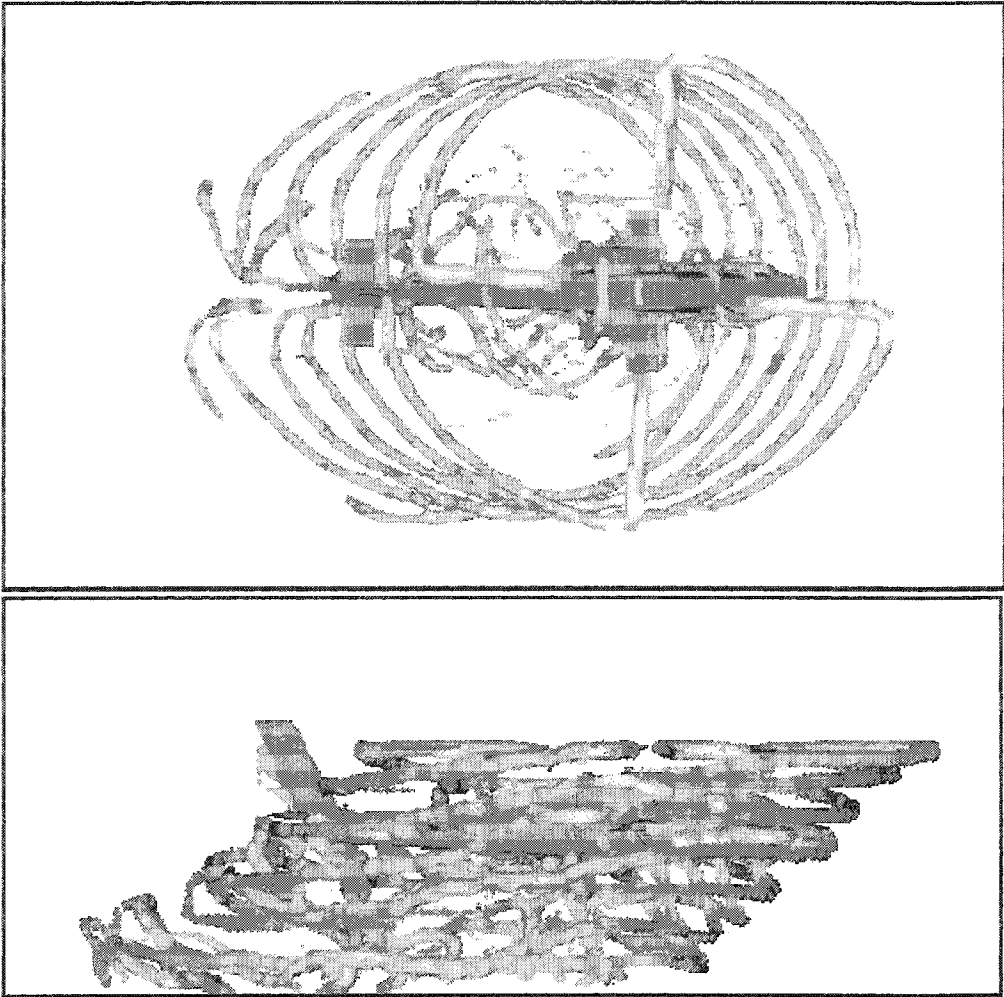


**Figure 16** Pressure coefficient data at  $x/L = 0.77$  and  $\alpha = 30^\circ$



**Figure 17** Computed vorticity iso-surfaces for the Apache helicopter including main rotor blades (High advance ratio) (Top: Top view; Bottom: Side view)





**Figure 18** Computed vorticity iso-surfaces for the Apache helicopter including main rotor blades (Low advance ratio) (Top: Top view; Bottom: Side view)

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# Lattice Boltzmann Simulation of Incompressible Flows

N. Satofuka, M. Ishikura<sup>1</sup>

## 12.1 Introduction

There are two approaches in CFD. Conventional one is to solve the Navier-Stokes equations based on the continuum assumption. In the case of incompressible computations one has to solve the momentum equation together with the Poisson equation to satisfy divergence free condition. The second approach starts from the Boltzmann equation using the Lattice Gas Method. The Boltzmann equation can recover the Navier-Stokes equations by using the Chapman-Enskog expansion. The scheme is ideal for massively parallel computing because the updating of a node only involves its nearest neighbors. Boundary conditions are easy to implement. Therefore the code is simple and can be easily written in the form suitable for parallel processing.

Among the lattice gas methods there exists the Lattice Gas Automata (LGA) [1]. LGA has fundamental difficulty in simulating realistic fluid flows obeying the Navier-Stokes equations. Beside its intrinsic noisy character which makes the computational accuracy difficult to achieve, it contains certain properties even in the fluid limit. The lattice gas fluid momentum equations cannot be reduced to the Navier-Stokes equations because of two fundamental problems. The first is the non-Galilean invariance property due to the density dependence of the convection coefficient. This limits the validity of the LGA method only a strict incompressible region. Second the pressure has an explicit and unphysical velocity dependence. To avoid some of these problems, several

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lattice Boltzmann (LB) models have been proposed. [2]–[4] The main feature of the LB method is to replace the particle occupation variables  $n_i$  (Boolean variables) by the single-particle distribution function (real variables)  $f_i = \langle n_i \rangle$ , where  $\langle \rangle$  denotes a local ensemble average, in the evolution equation, i.e., the lattice Boltzmann equation. The LB model proposed by Chen et al [5] and Qian et al [6] applies the single relaxation time approximation first introduced by Bhatnager, Gross, and Krook in 1954 [7], to greatly simplify the collision operator. This model is called the lattice BGK (LBGK) model.

In the present work, the LBGK method is used to simulate decaying two and three-dimensional homogeneous isotropic turbulence on massively parallel computers (a Hitachi SR2201 and a HP Exemplar V class). Two-dimensional channel flows with/without sudden expansion and three-dimensional duct flows are also calculated. Detailed comparisons of the accuracy, physical fidelity, and efficiency between the LBGK method and traditional Finite Difference Method (FDM) are presented. Speedup and CPU time are presented for a one-dimensional domain decomposition in three-dimensional computation.

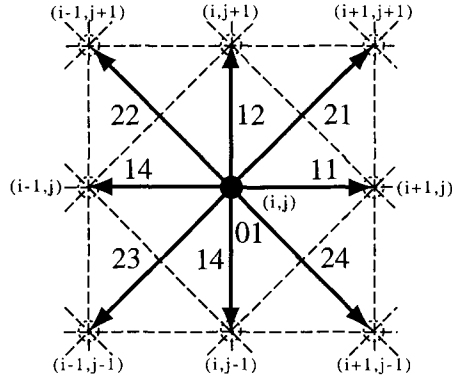
This paper is organized as follows. Section 12.2 presents the lattice Boltzmann methods using square lattice model which is used in this paper. The lattice Boltzmann simulation of two-dimensional homogeneous isotropic turbulence is presented in section 12.3. Accuracy and efficiency of the lattice Boltzmann method in comparison with the conventional higher-order FDM approach are also discussed. Application to two-dimensional channel flows with/without sudden expansion is presented in section 12.4. Extension and application to three-dimensional computation is presented from section 12.5 to 12.7. Section 12.8 presents some results of parallelization. The final section contains concluding remarks.

## 12.2 Lattice Boltzmann Method for Two-dimension

### 12.2.1 Nine-velocity square lattice model

In this section an outline is given of the LB methods with BGK model for the collision operator. A square lattice with unit spacing is used on which each node has eight nearest neighbors connected by eight links as shown in Fig. 1. Particles can only reside on the nodes and move to their nearest neighbors along these links in the unit time. Hence, there are two types of moving particles. Particles of type 1 move along the axes with speed  $|\mathbf{e}_{1i}| = 1$  and particle of type 2 move along the diagonal directions with speed  $|\mathbf{e}_{2i}| = \sqrt{2}$ . Rest particles with speed zero are also allowed at each node.

The occupation of the three types of particles is represented by the single-particle distribution function,  $f_{\sigma i}(\mathbf{x}, t)$ , where subscripts  $\sigma$  and  $i$  indicate the



**Figure 1** Square lattice model.

type of particle and the velocity direction, respectively. When  $\sigma = 0$ , there is only  $f_{01}$ . The distribution function,  $f_{\sigma i}$ , is the probability of finding a particle at node  $\mathbf{x}$  and time  $t$  with velocity  $\mathbf{e}_{\sigma i}$ . The particle distribution function satisfies the lattice Boltzmann equation

$$f_{\sigma i}(\mathbf{x} + \mathbf{e}_{\sigma i}, t + 1) - f_{\sigma i}(\mathbf{x}, t) = \Omega_{\sigma i} \quad (12.1)$$

where  $\Omega_{\sigma i}$  is the collision operator representing the rate of change of the particle distribution due to collisions. According to Bhatnagar, Gross, and Krook (BGK), the collision operator is simplified using the single time relaxation approximation. Hence, the lattice Boltzmann BGK (LBGK) equation (in lattice unit) is

$$f_{\sigma i}(\mathbf{x} + \mathbf{e}_{\sigma i}, t + 1) - f_{\sigma i}(\mathbf{x}, t) = -\frac{1}{\tau} \left[ f_{\sigma i}(\mathbf{x}, t) - f_{\sigma i}^{(0)}(\mathbf{x}, t) \right] \quad (12.2)$$

where  $f_{\sigma i}^{(0)}(\mathbf{x}, t)$  is the equilibrium distribution at  $\mathbf{x}, t$  and  $\tau$  is the single relaxation time which controls the rate of approach to equilibrium. The density per node,  $\rho$ , and the macroscopic velocity,  $\mathbf{u}$ , are defined in terms of the particle distribution functions by

$$\rho = \sum_{\sigma} \sum_i f_{\sigma i}, \quad \rho \mathbf{u} = \sum_{\sigma} \sum_i f_{\sigma i} \mathbf{e}_{\sigma i} \quad (12.3)$$

A suitable equilibrium distribution can be chosen in the following form for particles of each type

$$f_{01}^{(0)} = \rho \alpha - \frac{2}{3} \rho \mathbf{u}^2$$

$$f_{1i}^{(0)} = \rho \beta + \frac{1}{3} \rho (\mathbf{e}_{1i} \cdot \mathbf{u}) + \frac{1}{2} \rho (\mathbf{e}_{1i} \cdot \mathbf{u})^2 - \frac{1}{6} \rho \mathbf{u}^2 \quad (12.4)$$

$$f_{2i}^{(0)} = \rho \frac{(1 - 4\beta - \alpha)}{4} + \frac{1}{12} \rho (\mathbf{e}_{2i} \cdot \mathbf{u}) + \frac{1}{8} \rho (\mathbf{e}_{2i} \cdot \mathbf{u})^2 - \frac{1}{24} \rho \mathbf{u}^2$$

The relaxation time is related to the viscosity by

$$\tau = \frac{6\nu + 1}{2} \quad (12.5)$$

where  $\nu$  is the kinematic viscosity measured in lattice units. Hou et al [8] used the value of  $\alpha = 4/9$  and  $\beta = 1/9$ .

The equilibrium populations are determined by assuming that they can be expressed as a power series in velocity and density of the form;

$$f_{\sigma i}^{(0)} = A_{\sigma i}(\rho) + B_{\sigma i}(\rho)(\mathbf{e}_{\sigma i} \cdot \mathbf{u}) + C_{\sigma i}(\rho)(\mathbf{e}_{\sigma i} \cdot \mathbf{u})^2 + D_{\sigma i}(\rho)\mathbf{u}^2 \quad (12.6)$$

A Chapman-Enskog procedure is then applied to determine the macroscopic behavior of this model. The values of  $A_{\sigma i}$ ,  $B_{\sigma i}$ ,  $C_{\sigma i}$ , and  $D_{\sigma i}$  are chosen so that the macroscopic behavior matches the Navier-Stokes equations to as high an order as possible. The resulting continuity and momentum equations are as follows;

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_\beta}{\partial x_\beta} + O(\epsilon^2) = 0 \quad (12.7)$$

$$\rho \frac{\partial u_\alpha}{\partial t} + \rho u_\beta \frac{\partial u_\alpha}{\partial x_\beta} = \frac{\partial P}{\partial x_\alpha} + \frac{\partial}{\partial x_\beta} \left( \mu \left( \frac{\partial u_\beta}{\partial x_\alpha} + \frac{\partial u_\alpha}{\partial x_\beta} \right) \right) + O(\epsilon^2) + O(M_\alpha^3) \quad (12.8)$$

Characteristic dimensionless parameters are the Mach number,  $M_\alpha = \sqrt{3}U/c$  where  $U$  is a characteristic macroscopic flow speed, the Knudsen number which is proportional to  $\epsilon = c\tau/L$  where  $L$  is a macroscopic flow length, and the Reynolds number,  $Re = \rho UL/\mu$ .

Having chosen the appropriate lattice size and the characteristic velocity,  $\nu$  can be calculated for a given  $Re$  number and then the relaxation time can be determined by using Eq. 12.5. Starting from an initial density and velocity fields, the equilibrium distribution function can be obtained using Eq. 12.4, and  $f_{\sigma i}(\mathbf{x}, t)$  can be initialized as  $f_{\sigma i}^{(0)}(\mathbf{x}, t)$ . For each time step, the updating of the particle distribution can be split into two substeps: collision and advection. It is irrelevant which one is the first for a long time run. The collision process at position  $\mathbf{x}$  occurs according to the right-hand side of the lattice Boltzmann equation given as Eq. 12.2. The resulting particle distribution at  $\mathbf{x}$ , which is the sum of the original distribution and the collision term, is then advected to the nearest neighbor of  $\mathbf{x} + \mathbf{e}_{\sigma i}$ , for particle velocity  $\mathbf{e}_{\sigma i}$ . Then  $\rho, \mathbf{u}$  can be computed from the updated using Eq. 12.3. The updating procedure can be terminated for steady state problems when a certain criterion is satisfied. The method can also be used for transient problems.

## 12.3 Two-dimensional Homogeneous Isotropic Turbulence

### 12.3.1 Initial and boundary conditions

The initial condition of the vorticity  $\omega$  is randomly determined by satisfying the relation,

$$E(k) = \frac{1}{2} \sum_{|\vec{k}-\vec{k}'| \leq 1/2} |\tilde{\omega}(k_1, k_2)|^2 / k'^2 = \frac{2}{3} k \exp\left(-\frac{2}{3}k\right) \quad (12.9)$$

where  $\tilde{\omega}$  denotes the vorticity in the Fourier space,  $k'^2 = k_1^2 + k_2^2$ , and  $k_1$  and  $k_2$  are the wave numbers. The periodic boundary conditions are imposed in the  $x$  and  $y$  directions. The computational domain is a square,  $(0, 0) \leq (x, y) \leq (2\pi, 2\pi)$ .

### 12.3.2 High Reynolds number simulation

As a large-scale direct numerical simulation of high Reynolds number homogeneous isotropic turbulence, simulation for the case with  $\nu = 0.0001$  is carried out. This corresponds to the initial integral scale Reynolds number  $R_L = 25500$ , which is expressed as  $R_L = \Omega/\nu\eta^{1/3}$ .  $\Omega$  and  $\eta$  denote the total energy and the enstrophy dissipation rate, which are defined as

$$\Sigma(t) = \int_0^\infty E(k) dk \quad (12.10)$$

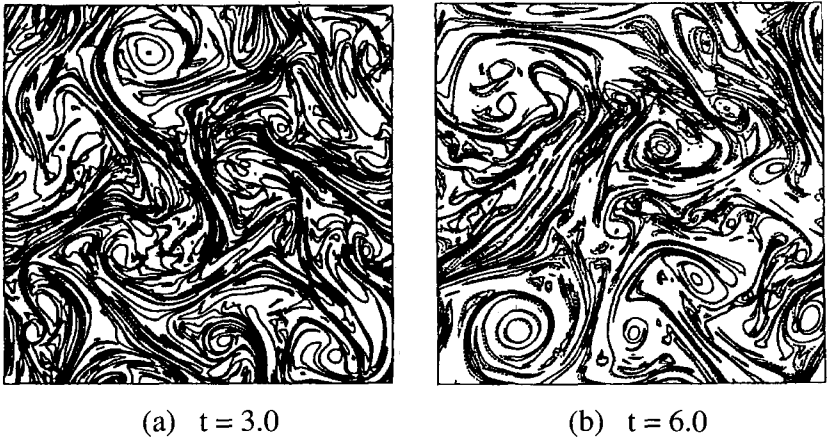
$$\eta(t) = 2\nu \int_0^\infty k^4 E(k) dk \quad (12.11)$$

The number of lattice nodes is  $1025 \times 1025$ . Fig. 2(a) shows comparison of vorticity contour plots at  $t=3.0$  between the square lattice BGK method and the 10th-order FDM, whereas Fig. 2(b) shows comparison at later time  $t=6.0$ . Although slight difference in vorticity contours is noticeable at  $t=6.0$ , strikingly similar features can be found for the LBGK simulation, as compared with the solutions by the 10th-order FDM.

Comparison of the wave number spectra  $k^3 E(k)$  at  $t=3.0$  shows that two methods yield quite a similar answer in terms of the statistical behavior of the flow.

With the present lattice of  $1025 \times 1025$  nodes, the inertial range of two-dimensional turbulence can be resolved. We also found that there is a range of wave number  $k \leq 50$  for which  $k^3 E(k)$  is roughly constant so that  $E(k)$  is proportional to  $k^{-3}$ .

Computational cost of the LBGK method for this high Reynolds number simulation on a SGI POWER ONYX 10000 is compared in Table 1 with that



**Figure 2** Comparison of vorticity cotours between LBGK and FDM (10th-order) at  $\nu = 0.0001$ .

**Table 1** Comparison of computational cost

|                                | LBGK                | FDM(10th-order)             |
|--------------------------------|---------------------|-----------------------------|
| $\sqrt{\langle u^2 \rangle}$   | 0.04                | 1.0                         |
| Time steps                     | 4047                | 1000 ( $\Delta t = 0.001$ ) |
| $T_{NS}$                       | 1.0                 | 1.0                         |
| CPU time                       | 63006               | 136639                      |
| $\langle \text{ratio} \rangle$ | $\langle 1 \rangle$ | $\langle 2.17 \rangle$      |
| CPU time/Time step             | 15.47               | 136.64                      |
| $\langle \text{ratio} \rangle$ | $\langle 1 \rangle$ | $\langle 8.83 \rangle$      |

Computer: Silicon Graphics ONYX 10000  
Number of lattice nodes:  $1025 \times 1025$   
Kinetic viscosity: 0.0001  
Reynolds number: 25500



of the 10th-order FDM. As far as efficiency is concerned, the LBGK method requires less than half CPU time per characteristic time of that of the FDM.

These results show that LBGK method is as accurate as the 10th-order FDM using the same lattice size and can be an alternative to solving the Navier-Stokes equation.

## 12.4 Two-dimensional Channel with Sudden Expansion

### 12.4.1 Descriptions of the problem

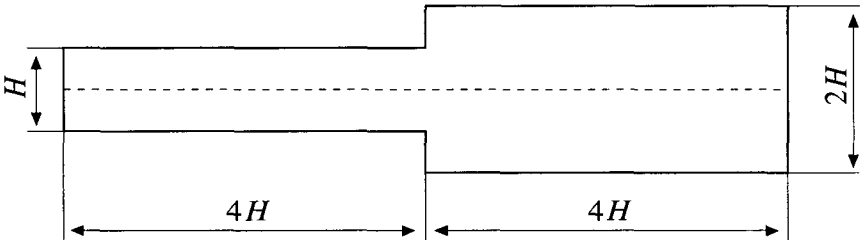
The flow in a symmetric channel with sudden expansion is a hydrodynamic system with a variety of interesting phenomena. The system has been studied both experimentally and numerically.

A two-dimensional channel with an expansion ratio of 1:2 and an aspect ratio of 1:4 for each side of expansion is studied here. At a moderate value of Reynolds number such as  $Re = 10$ , flows in the channel is two-dimensional. The 9 velocity square lattice BGK model for incompressible flow is used to simulate the two-dimensional flow in the channel.

The Reynolds number for the system is defined as

$$Re = \frac{H U_0}{\nu} \tag{12.12}$$

where  $H$  is the height of the entry section,  $U_0$  is the maximum inlet velocity, and  $\nu$  is the kinematic viscosity. The Reynolds number is chosen to be 10 for this simulation. The geometric configuration of two-dimensional channel with sudden expansion is illustrated in Fig. 3.



**Figure 3** Geometric configuration of two-dimensional channel with sudden expansion

### 12.4.2 Boundary and initial conditions

At the entrance ( upstream ), a parabolic profiles of the horizontal component of velocity,  $u$ , with a maximum  $U_0 = 0.1$ , is enforced, and the vertical component of velocity,  $v$ , is set to zero. At the exit ( downstream ), a constant pressure boundary conditions,  $p = 1.0$ , is enforced. At the walls a noslip boundary condition is applied.

The average density,  $\rho_0$ , is set to be 1.0 and the initial value of the velocity field is set to be zero in the interior of the channel.

### 12.4.3 Nonuniform mesh

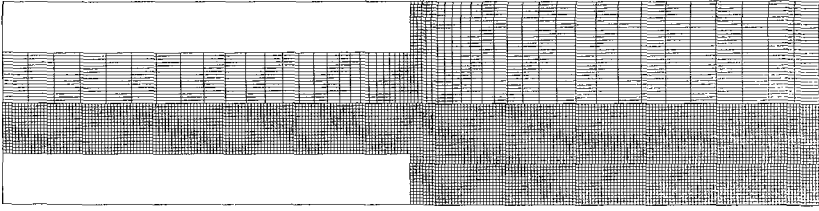
Two types of computational mesh are used in the simulation: uniform mesh shown in the lower half of Fig. 4 ,and nonuniform rectangular mesh shown in the upper half of Fig. 4.

A nonuniform mesh is certainly desirable in many practical applications. What inhibits the use of a nonuniform mesh in convetional LB methods is coupling of the discretization of physical and momentum spaces. The density distribution have to move from one lattice site to another in a single time step to warrant the followed collision process. In the present approach, the computational mesh is uncoupled from the discretization of momentum space and it can have an arbitrary shape. Collisions still take place on the grid points of the computational mesh. After a collision, the density distributions at the lattice sites now may not be exactly determined, they can always be calculated using interpolation. After interpolation collision and advection steps are repeated.

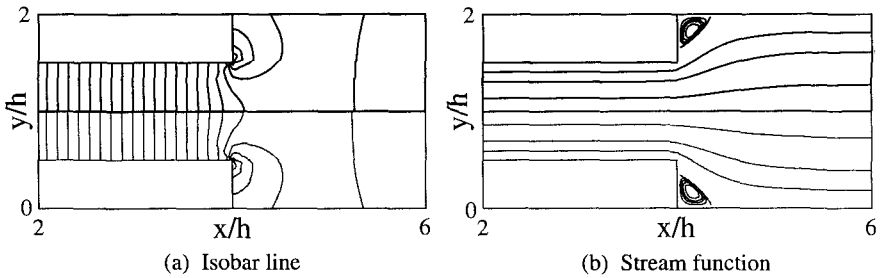
The uniform rectangular mesh of  $128 \times 33 + 129 \times 65$  and the nonuniform one of  $27 \times 33 + 28 \times 65$  are used in the simulation.

### 12.4.4 Results of simulations

Fig. 5 show the isobar line (Fig. 5(a)) and the contour line of the stream function (Fig. 5(b)) for flow in the symmetric channel with sudden expansion. In these figures, upper half corresponds to that simulated by using the nonuniform mesh while lower half indicates one for uniform mesh. It is apparent that the agreement is quite excellent. Computational cost of the simulation with the nonuniform mesh is less than 1/5 of that using the uniform mesh.



**Figure 4** Nonuniform mesh (upper) and uniform mesh (lower)



**Figure 5** Results of simulations

## 12.5 Lattice Boltzmann Method for Three-dimension

### 12.5.1 Fifteen-velocity cubic lattice model

A simple way to extend the square lattice, with vectors to the sides and corners of the square, to three dimensions is to use vectors to the sides and corners of a cube. This defines a body-centered-cubic lattice with  $\mathbf{e}_{1i} \in (\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$ , and  $\mathbf{e}_{2i} \in (\pm 1, \pm 1, \pm 1)$ . The equilibrium distribution functions for these moving populations and a non-moving population are given by

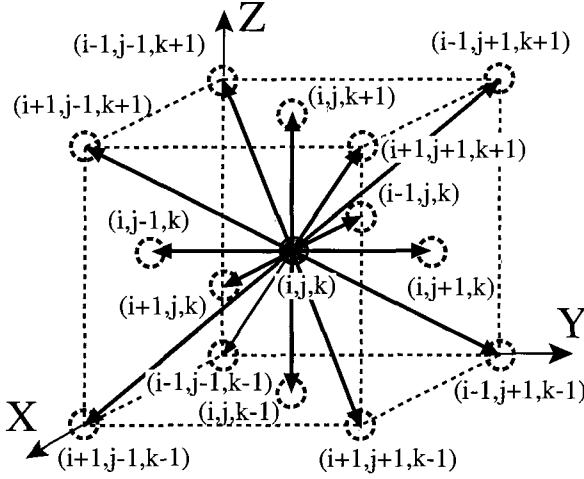
$$f_{0i}^{(0)} = \rho\alpha - \frac{1}{2}\rho\mathbf{u}^2$$

$$f_{1i}^{(0)} = \rho\beta + \frac{1}{3}\rho(\mathbf{e}_{1i} \cdot \mathbf{u}) + \frac{1}{2}\rho(\mathbf{e}_{1i} \cdot \mathbf{u})^2 - \frac{1}{6}\rho\mathbf{u}^2 \quad (12.13)$$

for  $\mathbf{e}_{1i}$  along the lattice axes, and

$$f_{2i}^{(0)} = \rho\frac{(1-6\beta-\alpha)}{8} + \frac{1}{24}\rho(\mathbf{e}_{2i} \cdot \mathbf{u}) + \frac{1}{16}\rho(\mathbf{e}_{2i} \cdot \mathbf{u})^2 - \frac{1}{48}\rho\mathbf{u}^2 \quad (12.14)$$

for  $\mathbf{e}_{2i}$  along the links to the corners of the cube. Values of  $\alpha = 2/9$  and  $\beta = 1/9$  are used.



**Figure 6** Cubic lattice model.

The relaxation time is defined as

$$\tau = \frac{6\nu + 1}{2} \quad (12.15)$$

## 12.6 Three-dimensional Homogeneous Isotropic Turbulence

### 12.6.1 Initial and boundary conditions

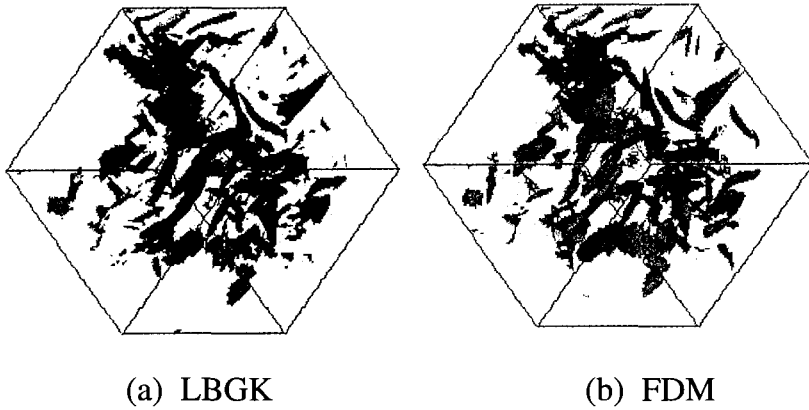
Three-dimensional decaying turbulence is simulated with a random initial condition having the energy spectrum;

$$E(k) = 16\sqrt{2/\pi}\nu_0^2 k_{max}^{-5} k^4 \exp[-2(k/k_{max})^2] \quad (12.16)$$

where we set  $\nu_0 = 1.0$  and the peak wave number  $k_{max} = 2.37841$ . The periodic boundary conditions are imposed in the  $x$ ,  $y$  and  $z$  directions. The computational domain is a cube,  $0 \leq (x, y, z) \leq 2\pi$ .

### 12.6.2 Low Reynolds number simulation

The kinematic viscosity is chosen as  $\nu = 0.0025$ . The initial integral scale and micro scale Reynolds number is 340 which corresponds to medium Reynolds



**Figure 7** Contour surface of enstrophy  
 $(\nu = 0.0025, t = 0.7, 65 \times 65 \times 65, \Omega_{plot} = 100)$ .

number simulation. The number of lattice nodes is  $65 \times 65 \times 65$ .

Contour surface of the enstrophy at  $t=0.7$  for the cubic lattice BGK method and the FDM are shown in Fig. 7(a) and (b) respectively. Plot value is  $\Omega = 100$ . As same as vorticity, contour surface of enstrophy between two methods are similar.

$\Omega$  denotes the enstrophy, which is defined as

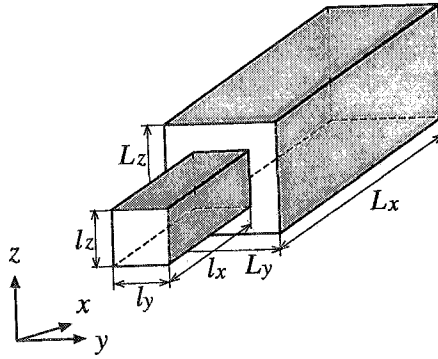
$$\Omega(x, y, z) = \frac{1}{2}(\omega_x^2 + \omega_y^2 + \omega_z^2) \quad (12.17)$$

Comparison between the cubic lattice BGK method and the FDM shows that the cubic lattice BGK method can be an alternative to solving the Navier-Stokes equations.

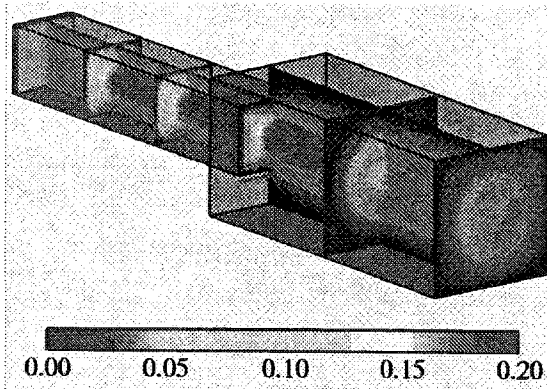
## 12.7 Three-dimensional Duct Flow

### 12.7.1 Description of the problem and Boundary Condition

Numerical simulation of the laminar flow development are carried out in a square duct which undergo a sudden expansion with uniform step height equal to 0.5 times the width of the inlet duct as shown in Fig. 8. The noslip wall boundary conditions,  $u = v = w = 0$ , are used as in the two-dimensional simulations. At the inlet we assume uniform axial velocity  $u = 1.0$ , and the other velocity components are set to zero. The pressure at the outflow boundary is kept constant value of  $p = 1.0$ .



**Figure 8** Square duct with sudden expansion



**Figure 9** Development of axial component of velocity  $Re = 100$

### 12.7.2 Results of simulations

Simulations are carried out by using a uniform cubic lattice for both inlet and outlet ducts. For the inlet duct  $160 \times 33 \times 33$  lattice nodes are used while  $161 \times 65 \times 65$  are used for the outlet duct. The development of axial component of velocity is shown in Fig. 9. In this case the velocity profiles is not yet fully developed due to a shortage of duct length.

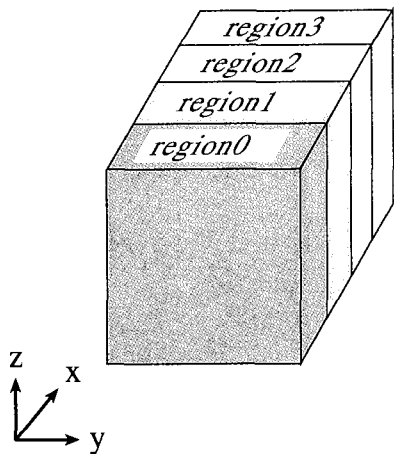


Figure 10 Domain decomposition

12.8 Parallelization

12.8.1 Domain decomposition for three-dimension

Two-dimensional simulation shows that the longer the size of subdomain in horizontal direction, the shorter the CPU time. In other words, the longer the outerloop dimension of data, the shorter the CPU time [9]. Based on this experience in the two-dimension we chose the type of domain decomposition as shown in Fig. 10. Although fifteen distribution functions are placed on each node for the cubic lattice model, only five variables have to be sent across each domain boundary similar to the case of the square lattice model.

12.8.2 Speedup and CPU time

The speedup and CPU time are shown in Fig. 11 and Table 2. In spite of the fact that the number of lattice nodes is small we can get higher speedup as compared with two-dimensional simulation.

Table 2 CPU time and Speedup

| Number of CPU | 1     | 2    | 4    | 8    | 16   |
|---------------|-------|------|------|------|------|
| CPU time      | 18336 | 8921 | 4585 | 2660 | 1522 |
| Speedup       | 1.0   | 2.0  | 4.0  | 6.9  | 11.8 |

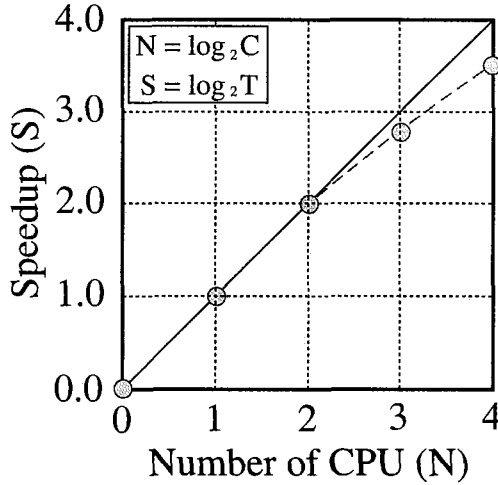


Figure 11 Speedup

## 12.9 Conclusion

Two-dimensional simulations of decaying homogeneous isotropic turbulence and two-dimensional channel flow with sudden expansion using the LBGK method have shown that the method is as accurate as the conventional method using the same lattice size. The LBGK method is able to reproduce the dynamic of incompressible flow and could be an alternative to solving the Navier-Stokes equations. Simulations of three-dimensional flows have also shown accuracy and efficiency of the LBGK method. In parallel computation of the lattice Boltzmann method, it is found that the longer the size of subdomain in horizontal direction the shorter the CPU time. In three dimensional simulation, relatively higher speedup is observed compared with the two-dimensional case.

Further investigation is needed on the accuracy and efficiency of cubic lattice BGK model.

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# Numerical Simulation of MHD Effects on Hypersonic Flow of a Weakly Ionized Gas in an Inlet

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## 13.1 Abstract

In recent years, the possibility of supersonic drag reduction by imposing a magnetic field on a slightly ionized plasma has received a great deal of attention. Some of the results reported in the Russian literature indicate that the shock structure in a slightly ionized gas (plasma) is significantly weaker than that in nonionized gases at the same temperature. This concept and others for shock wave modification/dissipation in supersonic flow are currently being investigated by the U.S. Air Force under the AJAX program. This paper evaluates this concept by numerical simulation. The effect of magnetic field on the weakly ionized flow is studied for a hypersonic inlet using the compressible viscous MHD equations with a bi-temperature model. Two-dimensional MHD equations in generalized coordinates are solved using a modified Runge-Kutta time integration scheme with second-order accurate spatial discretization. A symmetric Davis-Yee Total Variation Diminishing (TVD) flux limiter is employed to damp the oscillations in the shock regions. Numerical simulations are performed for a 2-D scramjet inlet and compared with the 1-D theoretical analysis of Park, Bogdanoff and Mehta; good agreement is obtained. Numerical results also indicate the feasibility of supersonic drag reduction by application of a strong magnetic field on the surface of a body moving at supersonic speed in a weakly ionized gas.

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## 13.2 Nomenclature

|              |                                                                      |
|--------------|----------------------------------------------------------------------|
| $Q$          | field vector                                                         |
| $E$          | flux vector component in the $x$ -direction                          |
| $F$          | flux vector component in the $y$ -direction                          |
| $x$          | Cartesian coordinate                                                 |
| $y$          | Cartesian coordinate                                                 |
| $\rho$       | density                                                              |
| $u$          | velocity component in the $x$ -direction                             |
| $v$          | velocity component in the $y$ -direction                             |
| $w$          | velocity component in the $z$ -direction                             |
| $\mathbf{V}$ | velocity vector                                                      |
| $B_x$        | magnetic field component in the $x$ -direction                       |
| $B_y$        | magnetic field component in the $y$ -direction                       |
| $B_z$        | magnetic field component in the $z$ -direction                       |
| $p$          | static pressure                                                      |
| $p_i$        | ionic pressure                                                       |
| $p_a$        | neutral atom pressure                                                |
| $p_e$        | electronic pressure                                                  |
| $p_o$        | $p_a + p_i$                                                          |
| $\gamma_i$   | specific heat ratio of ionic gas                                     |
| $\gamma_e$   | specific heat ratio of electron gas                                  |
| $\gamma_o$   | specific heat ratio of (ion + neutral atom) gas                      |
| $S_e$        | electron entropy                                                     |
| $e_t$        | total energy per unit mass                                           |
| $c_s$        | speed of sound                                                       |
| $v_{ax}$     | Alfven wave velocity in the $x$ -direction                           |
| $v_{ay}$     | Alfven wave velocity in the $y$ -direction                           |
| $\lambda$    | eigenvalue                                                           |
| $\xi$        | generalized coordinate                                               |
| $\eta$       | generalized coordinate                                               |
| $J$          | Jacobian of transformation                                           |
| $v_{f\xi}$   | fast wave velocity in the $\xi$ -direction                           |
| $v_{f\eta}$  | fast wave velocity in the $\eta$ -direction                          |
| $v_{s\xi}$   | slow wave velocity in the $\xi$ -direction                           |
| $v_{s\eta}$  | slow wave velocity in the $\eta$ -direction                          |
| $L_\xi$      | inverse right eigenvector associated with Jacobian matrix $\bar{A}$  |
| $R_\xi$      | right eigenvector associated with Jacobian matrix $\bar{A}$          |
| $D_\xi$      | diagonal eigenvalue matrix associated with Jacobian matrix $\bar{A}$ |
| $L_\eta$     | inverse right eigenvector associated with Jacobian matrix $\bar{B}$  |
| $R_\eta$     | right eigenvector associated with Jacobian matrix $\bar{B}$          |

|                        |                                                                      |
|------------------------|----------------------------------------------------------------------|
| $D_\eta$               | diagonal eigenvalue matrix associated with Jacobian matrix $\bar{B}$ |
| $\Phi_\xi$             | flux limiter function vector associated with $\bar{A}$               |
| $\Phi_\eta$            | flux limiter function vector associated with $\bar{B}$               |
| $\phi$                 | flux limiter function                                                |
| $\delta, \bar{\delta}$ | entropy correction variables                                         |
| $g$                    | flux limiters                                                        |
| $\mu_f$                | magnetic permeability of the fluid                                   |
| $\sigma$               | electrical conductivity of the fluid                                 |
| $M_\infty$             | free-stream Mach number                                              |
| $Re_\infty$            | Reynolds number                                                      |
| $Re_{m\infty}$         | Magnetic Reynolds number                                             |
| $Pr$                   | Prandtl number                                                       |
| $u_\infty$             | free-stream velocity                                                 |
| $\mu$                  | molecular viscosity                                                  |
| $T_g$                  | gas temperature                                                      |
| $T_e$                  | electron temperature                                                 |
| $n_e$                  | electron number density                                              |
| $n_i$                  | ion density                                                          |
| $n_a$                  | neutral atom density                                                 |
| $n_o$                  | $n_i + n_a$                                                          |
| $K$                    | Boltzmann constant, $1.38 \times 10^{-23}$ Joules/ $^\circ$ K        |
| $D_{12}$               | binary diffusion coefficient                                         |
| $m_e$                  | electron mass                                                        |
| $e$                    | electron charge                                                      |
| $Q_a$                  | collision cross-section                                              |
| $\alpha$               | degree of ionization                                                 |
| $\mathbf{J}$           | electric current density vector                                      |
| $ \mathbf{B} $         | magnitude of magnetic flux vector                                    |
| $\mathbf{E}$           | electric field vector                                                |
| $\rho_e$               | electron density                                                     |
| $\Omega$               | Hall parameter                                                       |
| $\epsilon_o$           | dielectric constant in free space                                    |
| $S_c$                  | source term                                                          |
| $\tau$                 | viscous stress tensor                                                |
| $\mathbf{q}$           | heat flux vector                                                     |
| $k$                    | thermal conductivity                                                 |
| $\alpha$               | degree of ionization                                                 |

### 13.3 Introduction

In recent years, the feasibility of supersonic drag reduction by imposing an electromagnetic field on a weakly ionized plasma has received a great deal of attention. The concept is currently being investigated both numerically and by experiments by the US Air force in collaboration with NASA. The interest in this concept has originated from results reported in the Russian literature wherein the shock structure in a weakly ionized plasma has been shown to be weaker than that in nonionized gases at the same temperature [1-3]. Several hypotheses have been advanced to account for the observed properties of a gas discharge plasma, which can be divided into two groups. The first hypothesis is based on the assumption that weak disturbances propagate in the plasma at a velocity greater than the thermal sound velocity; the second hypothesis (the "energy hypothesis") rests on the assumption that the energy is released in the shock layer behind the shock front (either the release of energy is concentrated in the molecular degrees of freedom or it is a consequence of current flowing in the shock wave). These concepts and others for shock wave modification/dissipation in supersonic flow are currently being investigated by the U.S Air Force under the AJAX program.

In addition, MHD by-pass concept is being evaluated to improve the efficiency and extend the range of conventional scramjet engine. The MHD by-pass scramjet concept (Figure 1) includes an external MHD generator to extract power from the air entering the engine, an ionizer to control the conductivity of the flow entering the engine, a MHD accelerator in series with the internal MHD generator to bypass energy around the engine combustor and return the energy into the exhaust of the engine. The internal MHD generator is used for increasing the pressure, prevent the flow separation and to transform a part of the airflow enthalpy to electric power.

Several experimental and analytical studies [4-7] were performed in the late 1950s and 1960s to study the effects of MHD on hypersonic flow fields. Among them, especially noteworthy is the paper by Kantrowitz [8] wherein he proposed the use of MHD to modify the structure of strong shock waves in a weakly ionized plasma. More recently, Mnatsakanian et al. [9], Bityurin et al. [10] and Cole et al. [11] have studied the MHD effects on aerodynamic and propulsion system flow fields. MHD equations along with Maxwell equations characterize the flow of a conducting fluid in presence of magnetic and electric fields. A theoretical calculation has recently been performed by Park et al. [12] to predict the maximum possible theoretical performance of a MHD-bypass scramjet engine. Bityurin et al. [13] have reported the assessment of the performance potential and scientific feasibility of MHD-bypass hypersonic air-breathing engines by applying fundamental thermodynamic principles. Chase et al. [14] have discussed various technical issues like equilibrium ionization and non-equilibrium ionization related to AJAX system. Kuranov et al. [15] have explored the possibility of increasing scramjet efficiency with proper choice of parameters for the ionizer and other components of a MHD system. Laux et al. [16] have developed a two-temperature kinetic model for air plasmas for the purpose of numerical simulation of a MHD system, which was validated against the experimental data.

The objective of the present work is to study the effect of electromagnetic field on a weakly ionized plasma flow in a scramjet inlet moving at a hypersonic speed using the compressible 2D viscous electroMHD equations with a bi-temperature model and a variable conductivity model. Two-dimensional compressible viscous MHD equations proposed by Powell [17] with a bi-temperature model proposed by Brassier and Gallice [18] are solved using a modified four-stage Runge-Kutta (R-K) time integration scheme with second-order accurate spatial discretization. A non-Monotonic Upwind Scheme for Conservation Laws (non-MUSCL) with Total Variation Diminishing (TVD) flux limiters is used to stabilize the modified R-K scheme. The governing equations are solved in the generalized coordinates. A set of eigenvectors is developed for implementation of the numerical scheme.

The 2-D unsteady compressible viscous MHD code WSUMHD2D reported in Refs. [19, 20] is modified to include the electroMHD model along with a bi-temperature model and a variable electrical conductivity model. The 2-D electroMHD code is employed to compute the hypersonic flow of a weakly ionized gas in a scramjet inlet integrated with a MHD generator for flow conditions employed by Park et al. [12] in their theoretical analysis.

### 13.4 Governing Equations Of Electro-Magnetohydrodynamics

The governing equations of electro-magnetohydrodynamics include the mass, momentum and energy equations of fluid flow and the Maxwell equations of electromagnetics [Ref. 21, 22, 23]. These equations are:

Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \quad (1)$$

Momentum Equations

$$\frac{\partial (\rho \mathbf{V})}{\partial t} + \nabla \cdot [\rho \mathbf{V} \mathbf{V} - \mathbf{B} \mathbf{B} + (p + \mathbf{B} \cdot \mathbf{B} / 2) \mathbf{I}] = \nabla \cdot \boldsymbol{\tau} \quad (2)$$

Energy Equation

$$\frac{\partial (\rho e_t)}{\partial t} + \nabla \cdot (\rho \mathbf{V} e_t) = -\nabla \cdot (\mathbf{V} p) + \nabla \cdot (\mathbf{V} \cdot \boldsymbol{\tau}) + \nabla \cdot \mathbf{q} + \mathbf{E} \cdot \mathbf{J} \quad (3)$$

Maxwell Equations

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (4), \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (5), \quad \nabla \cdot \mathbf{B} = 0 \quad (6), \quad \nabla \cdot \mathbf{E} = \frac{\rho_c}{\epsilon_0} \quad (7)$$

### Pressure-Energy Relation

$$\rho e_t = \frac{p}{\gamma-1} + \frac{\rho(u^2+v^2+w^2)}{2} + |\mathbf{B}|^2 \quad (8)$$

### Heat Flux-Temperature Relation

$$\mathbf{q} = -k\nabla T \quad (9)$$

### Equation of Conservation of Electrical Charge

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad (10)$$

### Ohm's Law of Electric Current Density

$$\mathbf{J} = \sigma (\mathbf{E} + \mathbf{V} \times \mathbf{B}) - \frac{\Omega}{|\mathbf{B}|} (\mathbf{J} \times \mathbf{B} + \nabla p_e) \quad (11)$$

### Electron and Ion Number Density Relations

$$p_e = n_e K T_e \quad (12)$$

$$p_e = \frac{n_e}{(n_i + n_e + n_a)} p_o = \frac{n_e}{(2n_e + n_a)} p_o \quad (13)$$

### Degree of Ionization

$$\alpha = \frac{n_e}{(n_e + n_a)} \approx \frac{n_e}{n_a} \quad (\text{if } n_e \ll n_a) \approx \frac{p_e}{p_o} \quad (14)$$

### Variable Electrical Conductivity Models

The electrical conductivity of plasma is determined by the number of charged particles in the mixture. For weakly ionized gases,  $\sigma \sim \alpha T^{-0.5}$  where  $\alpha < 1e-4$  for weakly ionized gases. Modeling this dependence of  $\sigma$  on  $T$  is critical for accurately capturing the strength of MHD interactions. The following models are employed for this purpose.

The expression for electrical conductivity of a slightly ionized gas ( $\alpha < 1e-4$ ) is given by Chapman and Cowling (Ref. 22) as

$$\sigma_L = \frac{e^2 n_e D_{i2}}{K T_g}, \quad (15)$$



where

$$D_{12} = \frac{3.39(KT_g)^{0.5}}{8n_a Q_a (2\pi m_e)^{0.5}} \quad (16)$$

Substituting the values of various constants in equations (15) and (16), the following equation is obtained.

$$\sigma_L = \frac{c_a \alpha}{Q_a T_g^{0.5}}, \quad (17)$$

where  $Q_a$  is obtained from the collision curve given in Ref. 23 and constant  $c_a$  is evaluated based on the initial values of  $\sigma$  and  $\alpha$ .

The expression for electrical conductivity of a highly ionized gas ( $\alpha \gg 1e-4$ ) is given by the Spitzer-Harm (Ref. 23) relation:

$$\sigma_H = \frac{1.56 \cdot 10^{-4} \cdot T_e^{1.5}}{\ln(12300 T_e^{1.5} / n_e^{0.5})} \quad (18)$$

Using the Kantrowitz's hypothesis (Ref. 8), electrical conductivity for any degree of ionization can be obtained by the expression

$$\frac{1}{\sigma} = \frac{1}{\sigma_L} + \frac{1}{\sigma_H}. \quad (19)$$

### Bi-Temperature Model

The bi-temperature model is included in the MHD equations to simulate ionization effects as ions and electrons by including an advection equation for the electronic entropy. In this model, the conducting fluid is assumed to be made of ions and electrons characterized by their own translational temperatures. The electronic entropy equation is given by

$$\frac{\partial \rho S_e}{\partial t} + \frac{\partial \rho u S_e}{\partial x} + \frac{\partial \rho v S_e}{\partial y} = 0, \quad (20)$$

where

$$S_e = \frac{p_e}{\rho^{1/\gamma_e}}. \quad (21)$$

## 13.5 Governing Equations In Weak Conservation Law Form

The 2-D viscous electroMHD equations with a bi-temperature model in Cartesian coordinate system can be expressed in the following conservation law form:

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} = H_M \left( \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} \right) + \frac{\partial E_v}{\partial x} + \frac{\partial F_v}{\partial y} + S_c \quad (22)$$

$$Q = [\rho \quad \rho u \quad \rho v \quad \rho w \quad B_x \quad B_y \quad B_z \quad \rho e_t \quad \rho S_e]^T \quad (23)$$

$$E = \begin{bmatrix} \rho u \\ \rho u^2 + p + \frac{-B_x^2 + B_y^2 + B_z^2}{8\pi\mu_f} \\ \rho uv - \frac{B_x B_y}{4\pi\mu_f} \\ \rho uw - \frac{B_x B_z}{4\pi\mu_f} \\ 0 \\ uB_y - vB_x \\ uB_z - wB_x \\ \left( \rho e_t + p + \frac{B_x^2 + B_y^2 + B_z^2}{8\pi\mu_f} \right) u - \frac{B_x}{4\pi\mu_f} (uB_x + vB_y + wB_z) \\ \rho u S_e \end{bmatrix} \quad (24)$$

$$F = \begin{bmatrix} \rho v \\ \rho vu - \frac{B_y B_x}{4\pi\mu_f} \\ \rho v^2 + p + \frac{-B_y^2 + B_x^2 + B_z^2}{8\pi\mu_f} \\ \rho vw - \frac{B_y B_z}{4\pi\mu_f} \\ vB_x - uB_y \\ 0 \\ vB_z - wB_y \\ \left( \rho e_t + p + \frac{B_x^2 + B_y^2 + B_z^2}{8\pi\mu_f} \right) v - \frac{B_y}{4\pi\mu_f} (uB_x + vB_y + wB_z) \\ \rho v S_e \end{bmatrix} \quad (25)$$

$$H_M = \begin{bmatrix} 0 & \frac{B_x}{4\pi\mu_f} & \frac{B_y}{4\pi\mu_f} & \frac{B_z}{4\pi\mu_f} & u & v & w & \frac{uB_x + vB_y + wB_z}{4\pi\mu_f} & 0 \end{bmatrix}^T \quad (26)$$

$$E_v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ \tau_{xz} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_x}{\partial x} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_y}{\partial x} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_z}{\partial x} \\ u\tau_{xx} + v\tau_{xy} + w\tau_{xz} - q_x \\ 0 \end{bmatrix} \quad (27),$$

$$F_v = \begin{bmatrix} 0 \\ \tau_{yx} \\ \tau_{yy} \\ \tau_{yz} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_x}{\partial y} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_y}{\partial y} \\ \frac{1}{\mu_f \sigma} \frac{\partial B_z}{\partial y} \\ u\tau_{yx} + v\tau_{yy} + w\tau_{yz} - q_y \\ 0 \end{bmatrix} \quad (28)$$

$$S_c = [0 \ 0 \ 0 \ 0 \ S_5 \ S_6 \ S_7 \ S_8 \ 0]^T \quad (29)$$

$$\text{where } S_5 = \frac{\Omega}{|\mathbf{B}|\mu_f\sigma} \frac{\partial}{\partial y} \left\{ B_y \frac{\partial B_z}{\partial y} + B_x \frac{\partial B_z}{\partial x} \right\}, \quad S_6 = \frac{\Omega}{|\mathbf{B}|\mu_f\sigma} \frac{\partial}{\partial x} \left\{ B_y \frac{\partial B_z}{\partial y} + B_x \frac{\partial B_z}{\partial x} \right\},$$

$$S_7 = \frac{\Omega}{|\mathbf{B}|\mu_f\sigma} \frac{\partial}{\partial x} \left( B_x \frac{\partial B_y}{\partial x} - B_x \frac{\partial B_x}{\partial y} - B_z \frac{\partial B_y}{\partial y} \right), \quad + \frac{\Omega}{|\mathbf{B}|\mu_f\sigma} \frac{\partial}{\partial y} \left( B_x \frac{\partial B_z}{\partial x} + B_y \frac{\partial B_y}{\partial x} - B_y \frac{\partial B_x}{\partial y} \right)$$

$$S_8 = \frac{1}{\mu_f\sigma} \left( \frac{\partial B_y}{\partial x} * \frac{\partial B_y}{\partial x} + \frac{\partial B_z}{\partial y} * \frac{\partial B_z}{\partial y} + \left( \frac{\partial B_x}{\partial y} - \frac{\partial B_y}{\partial x} \right) \left( \frac{\partial B_x}{\partial y} - \frac{\partial B_y}{\partial x} \right) \right),$$

$$\rho e_t = \frac{p}{\gamma-1} + \frac{\rho(u^2 + v^2 + w^2)}{2} + \frac{B_x^2 + B_y^2 + B_z^2}{8\pi\mu_f} \quad (30)$$

In Equations (27) and (28):

$$\tau_{xx} = \frac{2}{3} \mu \left( 2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right), \quad \tau_{xy} = \tau_{yx} = \mu \frac{2}{3} \left( \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

$$\tau_{xz} = \mu \frac{\partial w}{\partial x}, \quad \tau_{yy} = \frac{2}{3} \mu \left( 2 \frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right), \quad \tau_{yz} = \mu \frac{\partial w}{\partial y},$$

$$q_x = -k \frac{\partial T}{\partial x}, \quad \text{and} \quad q_y = -k \frac{\partial T}{\partial y}.$$

Note that

$$p = p_a + p_i + p_e = p_o + p_e, \quad \gamma = \gamma_o(1+b)/(1+a), \quad a = p_e(\gamma_i - 1)/p_o(\gamma_o - 1), \text{ and } b = \alpha\gamma_e/\gamma_o.$$

These equations are nondimensionalized and transformed into a body-fitted coordinate system  $(\xi, \eta)$ . The nondimensionalization is accomplished as follows:

$$t^* = \frac{tu_\infty}{L}, \quad x^* = \frac{x}{L}, \quad y^* = \frac{y}{L}, \quad \rho^* = \frac{\rho}{\rho_\infty}, \quad u^* = \frac{u}{u_\infty}, \quad v^* = \frac{v}{u_\infty}, \quad w^* = \frac{w}{u_\infty}, \quad p^* = \frac{p}{\rho_\infty u_\infty^2},$$

$$T^* = \frac{T}{T_\infty}, \quad \mu^* = \frac{\mu}{\mu_\infty}, \quad B_x^* = \frac{B_x}{\sqrt{\mu_f \rho_\infty u_\infty^2}}, \quad B_y^* = \frac{B_y}{\sqrt{\mu_f \rho_\infty u_\infty^2}}, \quad B_z^* = \frac{B_z}{\sqrt{\mu_f \rho_\infty u_\infty^2}}, \quad Se^* = \frac{Se}{u_\infty^2}$$

The resulting nondimensional parameters are:

$$\text{Reynolds number:} \quad \text{Re}_\infty = \frac{\rho_\infty u_\infty L}{\mu_\infty}$$

$$\text{Prandtl number:} \quad \text{Pr} = \frac{\mu c_p}{k}$$

$$\text{Magnetic Reynolds number:} \quad \text{Re} m_\infty = \sigma \mu_f u_\infty L$$

$$\text{Freestream Mach number:} \quad M_\infty = u_\infty / \sqrt{\gamma_o p_{o_\infty} / \rho_\infty}$$

### 13.6 Governing Equations in Generalized Coordinates

The nondimensional viscous MHD equations in generalized coordinates take the form (source terms are absorbed in viscous terms):

$$\frac{\partial \bar{Q}}{\partial t} + \frac{\partial \bar{E}}{\partial \xi} + \frac{\partial \bar{F}}{\partial \eta} = H_M \left( \frac{\partial \bar{B}_x}{\partial \xi} + \frac{\partial \bar{B}_y}{\partial \eta} \right) + \frac{\partial \bar{E}_v}{\partial \xi} + \frac{\partial \bar{F}_v}{\partial \eta} \quad (31)$$

$$\bar{Q} = \frac{Q}{J}, \quad \bar{E} = \frac{\xi_x E + \xi_y F}{J}, \quad \bar{F} = \frac{\eta_x E + \eta_y F}{J},$$

$$\bar{B}_x = \frac{\xi_x B_x + \xi_y B_y}{J}, \quad \bar{B}_y = \frac{\eta_x B_x + \eta_y B_y}{J}, \quad \bar{E}_v = \frac{\xi_x E_v + \xi_y F_v}{J}, \quad \bar{F}_v = \frac{\eta_x E_v + \eta_y F_v}{J}$$

$$\bar{E}_v = \begin{bmatrix} 0 \\ \frac{\mu}{\text{Re}_\infty J} (a_1 u_\xi + a_3 v_\xi + c_1 u_\eta + c_3 v_\eta) \\ \frac{\mu}{\text{Re}_\infty J} (a_3 u_\xi + a_2 v_\xi + c_3 u_\eta + c_2 v_\eta) \\ \frac{\mu}{\text{Re}_\infty J} (a_4 w_\xi + c_5 w_\eta) \\ \frac{1}{\text{Rem}_\infty J} [a_4 (B_x)_\xi + c_5 (B_x)_\eta] \\ \frac{1}{\text{Rem}_\infty J} [a_4 (B_y)_\xi + c_5 (B_y)_\eta] \\ \frac{1}{\text{Rem}_\infty J} [a_4 (B_z)_\xi + c_5 (B_z)_\eta] \\ \frac{\mu}{\text{Re}_\infty J} \left[ \frac{1}{2} a_1 (u^2)_\xi + \frac{1}{2} a_2 (v^2)_\xi + \frac{1}{2} a_4 (w^2)_\xi + a_3 (uv)_\xi + \frac{1}{\text{Pr}(\gamma-1)M_\infty^2} a_4 T_\xi \right. \\ \left. + \frac{1}{2} c_1 (u^2)_\eta + \frac{1}{2} c_2 (v^2)_\eta + \frac{1}{2} c_3 (w^2)_\eta + c_3 u v_\eta + c_4 v u_\eta + \frac{1}{\text{Pr}(\gamma-1)M_\infty^2} c_5 T_\eta \right] \\ 0 \end{bmatrix} \quad (32)$$

$$\bar{F}_v = \begin{bmatrix} 0 \\ \frac{\mu}{\text{Re}_\infty J} (b_1 u_\eta + b_3 v_\eta + c_1 u_\xi + c_4 v_\xi) \\ \frac{\mu}{\text{Re}_\infty J} (b_3 u_\eta + b_2 v_\eta + c_3 u_\xi + c_2 v_\xi) \\ \frac{\mu}{\text{Re}_\infty J} (b_4 w_\eta + c_5 w_\xi) \\ \frac{1}{\text{Rem}_\infty J} [b_4 (B_x)_\eta + c_5 (B_x)_\xi] \\ \frac{1}{\text{Rem}_\infty J} [b_4 (B_y)_\eta + c_5 (B_y)_\xi] \\ \frac{1}{\text{Rem}_\infty J} [b_4 (B_z)_\eta + c_5 (B_z)_\xi] \\ \frac{\mu}{\text{Re}_\infty J} \left[ \frac{1}{2} b_1 (u^2)_\eta + \frac{1}{2} b_2 (v^2)_\eta + \frac{1}{2} b_4 (w^2)_\eta + \frac{1}{2} b_3 (uv)_\eta + \frac{1}{\text{Pr}(\gamma-1)M_\infty^2} b_4 T_\eta \right. \\ \left. + \frac{1}{2} c_1 (u^2)_\xi + \frac{1}{2} c_2 (v^2)_\xi + \frac{1}{2} c_3 (w^2)_\xi + c_3 v u_\xi + c_4 u v_\xi + \frac{1}{\text{Pr}(\gamma-1)M_\infty^2} c_5 T_\xi \right] \\ 0 \end{bmatrix} \quad (33)$$

In Equations (32) and (33):

$$a_1 = \frac{4}{3}\xi_x^2 + \xi_y^2, \quad a_2 = \xi_x^2 + \frac{4}{3}\xi_y^2, \quad a_3 = \frac{1}{3}\xi_x\xi_y, \quad a_4 = \xi_x^2 + \xi_y^2$$

$$b_1 = \frac{4}{3}\eta_x^2 + \eta_y^2, \quad b_2 = \eta_x^2 + \frac{4}{3}\eta_y^2, \quad b_3 = \frac{1}{3}\eta_x\eta_y, \quad b_4 = \eta_x^2 + \eta_y^2$$

$$c_1 = \frac{4}{3}\xi_x\eta_x + \xi_y\eta_y, \quad c_2 = \xi_x\eta_x + \frac{4}{3}\xi_y\eta_y, \quad c_3 = \xi_y\eta_x - \frac{2}{3}\xi_x\eta_y,$$

$$c_4 = \xi_x\eta_y - \frac{2}{3}\xi_y\eta_x, \quad \text{and} \quad c_5 = \xi_x\eta_x + \xi_y\eta_y.$$

Defining the Jacobian matrices  $\bar{A}$  and  $\bar{B}$  as shown below, the MHD equations can be written as

$$\frac{\partial \bar{Q}}{\partial t} + \bar{A} \frac{\partial \bar{Q}}{\partial \xi} + \bar{B} \frac{\partial \bar{Q}}{\partial \eta} = \frac{\partial \bar{E}_v}{\partial \xi} + \frac{\partial \bar{F}_v}{\partial \eta}, \quad (34)$$

where  $\bar{A} = \frac{\partial \bar{E}}{\partial Q} - H_M \frac{\partial \bar{B}_x}{\partial Q} = (\xi_x A + \xi_y B), \quad \bar{B} = \frac{\partial \bar{F}}{\partial Q} - H_M \frac{\partial \bar{B}_y}{\partial Q} = (\eta_x A + \eta_y B),$

$$A = \frac{\partial E}{\partial Q} - H_M \frac{\partial B_x}{\partial Q}, \quad B = \frac{\partial F}{\partial Q} - H_M \frac{\partial B_y}{\partial Q}$$

## 13.7 Numerical Method

### 13.7.1 Eigenvalues and Eigenvectors

The numerical scheme employed for solving the equation (42) requires the determination of eigenvalues and eigenvectors of the Jacobian matrices. In this section the eigenvalues and eigenvectors are provided. The eigenvalues of the Jacobian matrix  $\bar{A}$  are

$$\lambda_{0\xi} = \xi_x u + \xi_y v, \quad \lambda_{a\xi\pm} = \xi_x (u \pm v_{ax}) + \xi_y (v \pm v_{ay})$$

$$\lambda_{f\xi\pm} = \xi_x u + \xi_y v \pm v_{f\xi}, \quad \lambda_{s\xi\pm} = \xi_x u + \xi_y v \pm v_{s\xi}$$

$$\lambda_{a\xi} = \xi_x u + \xi_y v, \quad \text{and} \quad \lambda_{h\xi} = \xi_x u + \xi_y v.$$

where

$$v_{f\xi}^2 = \frac{1}{2} [a_4 (v_a^2 + c_s^2) + z_\xi], \quad v_{s\xi}^2 = \frac{1}{2} [a_4 (v_a^2 + c_s^2) - z_\xi],$$

$$z_{\xi} = \sqrt{a_4} \sqrt{a_4 (v_a^2 + c_s^2)^2 - \frac{(\xi_x B_x + \xi_y B_y)^2 \gamma p}{\pi \rho^2}}, \quad c_s^2 = \frac{\gamma p}{\rho}$$

$$v_{ax}^2 = \frac{B_x^2}{4\pi\rho}, \quad v_{ay}^2 = \frac{B_y^2}{4\pi\rho}, \quad v_{az}^2 = \frac{B_z^2}{4\pi\rho}, \quad v_a^2 = v_{ax}^2 + v_{ay}^2 + v_{az}^2.$$

The diagonal eigenvalue matrix,  $D_{\xi} = L_{\xi} \bar{A} R_{\xi}$  then becomes

$$D_{\xi} = \begin{bmatrix} \lambda_{0\xi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_{a\xi+} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_{a\xi-} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_{f\xi+} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_{f\xi-} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_{s\xi+} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{s\xi-} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{d\xi} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{h\xi} \end{bmatrix} \quad (35)$$

where  $R_{\xi} = \begin{bmatrix} r_0 & r_{+v_{a\xi}} & r_{-v_{a\xi}} & r_{+v_{f\xi}} & r_{-v_{f\xi}} & r_{+v_{s\xi}} & r_{-v_{s\xi}} & r_{d\xi} & r_{h\xi} \end{bmatrix}$  (36)

and  $L_{\xi} = \begin{bmatrix} l_0 & l_{+v_{a\xi}} & l_{-v_{a\xi}} & l_{+v_{f\xi}} & l_{-v_{f\xi}} & l_{+v_{s\xi}} & l_{-v_{s\xi}} & l_{d\xi} & l_{h\xi} \end{bmatrix}^T$  (37)

are right and left eigenvectors.

Similarly, the eigenvalues of the Jacobian matrix  $\bar{B}$  are

$$\lambda_{0\eta} = \eta_x u + \eta_y v, \quad \lambda_{a\eta\pm} = \eta_x (u \pm v_{ax}) + \eta_y (v \pm v_{ay}),$$

$$\lambda_{f\eta\pm} = \eta_x u + \eta_y v \pm v_{f\eta}, \quad \lambda_{s\eta\pm} = \eta_x u + \eta_y v \pm v_{s\eta},$$

$$\lambda_{d\eta} = \eta_x u + \eta_y v, \quad \text{and} \quad \lambda_{h\eta} = \eta_x u + \eta_y v$$

where  $v_{f\eta}^2 = \frac{1}{2} [b_4 (v_a^2 + c_s^2) + z_{\eta}]$ ,  $v_{s\eta}^2 = \frac{1}{2} [b_4 (v_a^2 + c_s^2) - z_{\eta}]$

and

$$z_\eta = \sqrt{b_4} \sqrt{b_4 (v_a^2 + c_s^2)^2 - \frac{(\eta_x B_x + \eta_y B_y)^2 \gamma p}{\pi \rho^2}}.$$

The diagonal eigenvalue matrix,  $D_\eta = L_\eta \bar{B} R_\eta$  then becomes

$$D_\eta = \begin{bmatrix} \lambda_{v\eta} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_{a\eta+} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_{a\eta-} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_{f\eta+} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda_{f\eta-} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda_{s\eta+} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{s\eta-} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{d\eta} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{v\eta} \end{bmatrix} \quad (38)$$

where  $R_\eta = \begin{bmatrix} r_0 & r_{+v_{a\eta}} & r_{-v_{a\eta}} & r_{+v_{f\eta}} & r_{-v_{f\eta}} & r_{+v_{s\eta}} & r_{-v_{s\eta}} & r_{d\eta} & r_{h\eta} \end{bmatrix} \quad (39)$

and  $L_\eta = \begin{bmatrix} l_0 & l_{+v_{a\eta}} & l_{-v_{a\eta}} & l_{+v_{f\eta}} & l_{-v_{f\eta}} & l_{+v_{s\eta}} & l_{-v_{s\eta}} & l_{d\eta} & l_{h\eta} \end{bmatrix}^T. \quad (40)$

### 13.7.2 The Modified Runge-Kutta (R-K) Time Integration Scheme

The numerical scheme employed for time integration is the modified R-K scheme. The R-K formulation for the 2-D viscous MHD equations in the generalized coordinate system is as shown below:

$$\bar{Q}_{i,j}^{(1)} = \bar{Q}_{i,j}^n - \frac{\Delta t}{4} (L\bar{Q}^n)_{i,j} \quad (41a)$$

$$\bar{Q}_{i,j}^{(2)} = \bar{Q}_{i,j}^n - \frac{\Delta t}{3} (L\bar{Q}^{(1)})_{i,j} \quad (41b)$$

$$\bar{Q}_{i,j}^{(3)} = \bar{Q}_{i,j}^n - \frac{\Delta t}{2} (L\bar{Q}^{(2)})_{i,j} \quad (41c)$$

$$\bar{Q}_{i,j}^{(4)} = \bar{Q}_{i,j}^n - \Delta t (L\bar{Q}^{(3)})_{i,j} \quad (41d)$$

where  $(L\bar{Q})_{i,j} = \frac{\bar{E}_{i+1,j} - \bar{E}_{i-1,j}}{2\Delta\xi} + \frac{\bar{F}_{i,j+1} - \bar{F}_{i,j-1}}{2\Delta\eta}$



$$-H_M \left[ \frac{(\bar{B}_x)_{i+1,j} - (\bar{B}_x)_{i-1,j}}{2\Delta\xi} + \frac{(\bar{B}_y)_{i,j+1} - (\bar{B}_y)_{i,j-1}}{2\Delta\eta} \right] \frac{\partial \bar{E}_v}{\partial \xi} - \frac{\partial \bar{F}_v}{\partial \eta}$$

### 13.7.3 Total Variation Diminishing (TVD) Models

The modified R-K scheme described above becomes unstable for solving Riemann-type problems because it employs central differencing of the convective terms. It is therefore necessary to add an artificial dissipation term to stabilize the scheme. The artificial dissipation term used in this work is based on the Total Variation Diminishing (TVD) model.

The TVD flux limiters are incorporated in the numerical scheme in a post-processor approach. The post-processor approach can be easily applied to many single and multi-step explicit schemes by adding an extra step to the original scheme. This approach becomes convenient because no modification is required to the existing numerical scheme. The addition of TVD model to equation (41d) in the generalized coordinate system can be expressed as:

$$\begin{aligned} \dot{\bar{Q}}_{i,j}^{n+1} &= \bar{Q}_{i,j}^{(4)} \\ &- \frac{\Delta t}{2\Delta\xi} \left[ (R_\xi)_{i+\frac{1}{2},j} (\Phi_\xi)_{i+\frac{1}{2},j} - (R_\xi)_{i-\frac{1}{2},j} (\Phi_\xi)_{i-\frac{1}{2},j} \right] \\ &- \frac{\Delta t}{2\Delta\eta} \left[ (R_\eta)_{i,j+\frac{1}{2}} (\Phi_\eta)_{i,j+\frac{1}{2}} - (R_\eta)_{i,j-\frac{1}{2}} (\Phi_\eta)_{i,j-\frac{1}{2}} \right] \end{aligned} \quad (42)$$

where  $R_\xi$  and  $R_\eta$  are given by equations (36) and (39) respectively. The flux limiter function vectors  $\Phi_\xi$  and  $\Phi_\eta$  are:

$$(\Phi_\xi)_{i+\frac{1}{2},j} = \left[ (\phi_\xi^1)_{i+\frac{1}{2},j} \quad (\phi_\xi^2)_{i+\frac{1}{2},j} \quad (\phi_\xi^3)_{i+\frac{1}{2},j} \quad \dots \right]^T \quad (43)$$

$$(\Phi_\eta)_{i,j+\frac{1}{2}} = \left[ (\phi_\eta^1)_{i,j+\frac{1}{2}} \quad (\phi_\eta^2)_{i,j+\frac{1}{2}} \quad (\phi_\eta^3)_{i,j+\frac{1}{2}} \quad \dots \right]^T \quad (44)$$

The vector elements of (43) and (44) are termed as the flux limiter functions and are given in the following section. Subscripts  $i+\frac{1}{2}, j$  and  $i-\frac{1}{2}, j$  denote that all other related variables are calculated based on the variables  $(\rho, u, v, w, B_x, B_y, B_z, p, \xi_x, \xi_y, \eta_x, \eta_y)$  averaged at grid points  $(i+1, j), (i, j)$  and  $(i, j), (i-1, j)$  respectively.

### 13.7.4 Davis-Yee Second-Order Symmetric TVD Model

As an example of a TVD flux limiter, we describe here the Davis-Yee second-order symmetric TVD model. There are many models that have been proposed in the literature. WSUMHD2D has several TVD models, e.g., Harten-Yee model, Roe-Sweby model, Davis-Yee model, etc. The present calculations have been performed with Davis-Yee model.

For the Davis-Yee symmetric TVD model, the  $l^{\text{th}}$  symmetric flux limiter function is given by:

$$(\phi_{\xi}^l)_{i+\frac{1}{2},j} = -\frac{\Delta t}{\Delta \xi} \left[ (\lambda_{\xi}^l)_{i+\frac{1}{2},j} \right]^2 (g_{\xi}^l)_{i+\frac{1}{2},j} - \phi \left[ (\lambda_{\xi}^l)_{i+\frac{1}{2},j} \right] \cdot \left[ (\alpha^l)_{i+\frac{1}{2},j} - (g_{\xi}^l)_{i+\frac{1}{2},j} \right] \quad (45)$$

$$(\phi_{\eta}^l)_{i,j+\frac{1}{2}} = -\frac{\Delta t}{\Delta \eta} \left[ (\lambda_{\eta}^l)_{i,j+\frac{1}{2}} \right]^2 (g_{\eta}^l)_{i,j+\frac{1}{2}} - \phi \left[ (\lambda_{\eta}^l)_{i,j+\frac{1}{2}} \right] \cdot \left[ (\alpha^l)_{i,j+\frac{1}{2}} - (g_{\eta}^l)_{i,j+\frac{1}{2}} \right] \quad (46)$$

where  $(\lambda_{\xi}^l)_{i+\frac{1}{2},j}$  and  $(\lambda_{\eta}^l)_{i,j+\frac{1}{2}}$  are the  $l^{\text{th}}$  eigenvalues of the diagonal matrices  $D_{\xi}$  and  $D_{\eta}$ . The function  $\phi(z)$  is an entropy correction to  $z$  with

$$\phi(z) = \begin{cases} |z| & |z| \geq \delta \\ (z^2 + \delta^2)/2\delta & |z| < \delta \end{cases} \quad (47)$$

where  $0 < \delta < 0.125$ , and is given by:

$$\delta_{i+\frac{1}{2},j} = \bar{\delta} \left[ \left| U_{i+\frac{1}{2},j} \right| + \left| V_{i+\frac{1}{2},j} \right| + (v_{f_{\xi}})_{i+\frac{1}{2},j} + (v_{f_{\eta}})_{i+\frac{1}{2},j} \right] \quad (48)$$

$$\delta_{i,j+\frac{1}{2}} = \bar{\delta} \left[ \left| U_{i,j+\frac{1}{2}} \right| + \left| V_{i,j+\frac{1}{2}} \right| + (v_{f_{\xi}})_{i,j+\frac{1}{2}} + (v_{f_{\eta}})_{i,j+\frac{1}{2}} \right] \quad (49)$$

with  $U = \xi_x u + \xi_y v$ ,  $V = \eta_x u + \eta_y v$  and  $0.5 < \bar{\delta} < 0.7$ .

The Davis-Yee symmetric limiters are:

$$(g_{\xi}^l)_{i+\frac{1}{2},j} = \min \text{mod} \left\{ 2(\alpha^l)_{i-\frac{1}{2},j}, 2(\alpha^l)_{i+\frac{1}{2},j}, 2(\alpha^l)_{i+\frac{3}{2},j}, \frac{1}{2} \left[ (\alpha^l)_{i-\frac{1}{2},j} + (\alpha^l)_{i+\frac{3}{2},j} \right] \right\} \quad (50)$$

$$(g_{\eta}^l)_{i,j+\frac{1}{2}} = \min \text{mod} \left\{ 2(\alpha^l)_{i,j-\frac{1}{2}}, 2(\alpha^l)_{i,j+\frac{1}{2}}, 2(\alpha^l)_{i,j+\frac{3}{2}}, \frac{1}{2} \left[ (\alpha^l)_{i,j-\frac{1}{2}} + (\alpha^l)_{i,j+\frac{3}{2}} \right] \right\} \quad (51)$$

where  $(\alpha^l)_{i+\frac{1}{2},j}$  and  $(\alpha^l)_{i,j+\frac{1}{2}}$  are the  $l^{\text{th}}$  elements of

$$\alpha_{i+\frac{1}{2},j} = 2(L_\xi)_{i+\frac{1}{2},j} (Q_{i+1,j} - Q_{i,j}) / (J_{i+1,j} + J_{i,j}), \quad (52)$$

$$\alpha_{i,j+\frac{1}{2}} = 2(L_\eta)_{i,j+\frac{1}{2}} (Q_{i,j+1} - Q_{i,j}) / (J_{i,j+1} + J_{i,j}), \quad (53)$$

$$\text{and } L_\xi = R_\xi^{-1}, \quad L_\eta = R_\eta^{-1},$$

$\min\text{mod}(a, b, c, \dots, n) = S \cdot \max[0, \min(a, S \cdot b, S \cdot c, \dots, S \cdot n)]$  and  $S = \text{sgn}(a)$ .

### 13.7.5 Local Time Step (LTS)

For a steady state solution it is advantageous to use the local time stepping to enhance the convergence. The main idea behind the LTS is to march the solution at different time steps at each individual grid point without violation of the stability condition, which is  $\text{CFL} \leq 1.0$ . Each time step is updated at every iteration. The minimum time step in the computational domain is given by:

$$\Delta t = \min \left\{ \frac{\text{CFL} \cdot \Delta \xi}{|\lambda_\xi|_{\max}}, \frac{\text{CFL} \cdot \Delta \eta}{|\lambda_\eta|_{\max}} \right\}.$$

## 13.8 Significant Parameters

The influence of an electromagnetic field on a shock wave can be best characterized in terms of the following parameters:

### Joule Heating

Joule heating ( $J_h$ ) contributes to an entropy increase across the shock. It can be neglected if the plasma has a perfect electrical conductivity. It is associated with the total enthalpy of the flow. The expression for Joule heating is given by

$$J_h = \int \mathbf{J}^2 / \sigma \, dx$$

where  $\mathbf{J}$  is the electrical current density and  $\sigma$  is the variable electrical conductivity

### Hall Parameter

Hall parameter is associated with the Hall current that brings unsteadiness and perturbation in the conducting fluid. The expression for Hall parameter ( $\Omega$ ) is

$$\Omega = \frac{|\mathbf{B}| \sigma}{en_e}$$

MHD Interaction Parameter

MHD interaction parameter determines the magnetic field strength required for achieving meaningful levels of electrical conductivity. It is defined as

$$Q = \frac{\sigma |B|^2 L}{\rho |V|} .$$

13.9 Numerical Simulation of Supersonic Flow in an Inlet.

Park et al. [12] have performed theoretical calculations for a scramjet propulsion system employing the magnetohydrodynamic (MHD) energy bypass concept. The MHD generator upstream of the combustion chamber slows down the flow so that Mach number at the entrance of the combustion chamber is kept below a specified value. The theoretical calculations show that the MHD-bypass scheme can improve specific impulse over that of a conventional scramjet at flight speeds of a rocket engine. The present calculations simulate the theoretical calculations of Park et al. [12] for a scramjet inlet integrated with a MHD generator.

The differences between the physical model employed in the present simulations and the theoretical model used by Park et al. [12] are given in Table 1 below.

Table 1: Physical Models Employed

| Theoretical Calculation of Park et al. [12]    | Present Calculation                             |
|------------------------------------------------|-------------------------------------------------|
| 1-D analysis                                   | 2-D numerical simulation                        |
| Inviscid Flow                                  | Viscous Flow                                    |
| Absence of Shock Waves and Viscous Dissipation | Presence of Shock Waves and Viscous Dissipation |
| Uniform Axial Current Density                  | Non-uniform Axial Current Density               |
| Thermal Equilibrium                            | Thermal Non-Equilibrium                         |

Flow simulation parameters employed in the present study are given in Table 2 below:

Table 2: Flow Simulation Parameters

| Parameter                      | Value                               |
|--------------------------------|-------------------------------------|
| Hydrodynamic Pressure          | $5.372 \times 10^5$ Pascal          |
| Electronic Pressure            | 41.9 Pascal                         |
| Gas Temperature                | 3592 °K                             |
| Electron Temperature           | 3592 °K                             |
| Velocity                       | 3240 m/s                            |
| Mach Number                    | 3.072                               |
| Reynolds Number                | $19.8 \times 10^6$                  |
| Magnetic Reynolds Number       | 0.306                               |
| Prandtl Number                 | 0.72                                |
| Gas Specific Heat Ratio        | 1.4                                 |
| Electron Specific Heat Ratio   | 1.67                                |
| Current Density                | $9.16 \times 10^4$ A/m <sup>2</sup> |
| Voltage Across Electrodes      | 1596 V                              |
| Transverse Voltage Gradient    | 24,650 V/m                          |
| Axial Voltage Gradient         | 5,000 V/m                           |
| Magnetic Field                 | 8.01 T                              |
| Hall Parameter                 | 4.056                               |
| Magnetic Interaction Parameter | 2.865                               |
| Electrical Conductivity        | 75.43 mhos/m                        |
| Degree of Ionization           | $7.8 \times 10^{-5}$                |

Figure 1 shows the contours of various flow variables (total pressure, gas temperature, electron temperature, electrical conductivity, electron number density, Hall parameter, electric current density, axial voltage gradient and transverse voltage gradient) for the scramjet inlet. There are a few common characteristics among the contours of all flow variables. It can be seen from Figure 1 that the shocks are generated from two corners of the inlet at the entrance and they coalesce right after the neck of the inlet. Also these shocks are slightly weaker in flow with ionization compared to that in the flow without ionization.

Figure 2 shows the variation of various flow variables (total pressure, gas temperature, electron temperature, electrical conductivity, electron number density, Hall parameter, electric current density, axial voltage gradient and transverse voltage gradient) at different vertical sections of the scramjet inlet. Figure 2(a) and 2(b) respectively show that the total pressure and gas temperature somewhat decrease in a slightly ionized gas compared to that in a nonionized gas, which is an expected result. The gas temperature is highest in the region near the wall due to heat transfer from the wall to the fluid. The gas temperature in the core region of the inlet is closer to the wall temperature. Electron temperature on the other hand increases in the core region downstream of the neck of the inlet [Figure 2(c)]. As shown in Figure 2(d), electrical conductivity is highest in the wall region near the entrance of the inlet and decreases substantially downstream which is a reflection of the electron number density variation in the flow field as shown in Figure 2(e). Other quantities such as Hall parameter (Figure 2(f)), electric current density

(Figure 2(g)) and axial voltage (Figure 2(h)) show similar behaviour because of their direct dependence upon electrical conductivity.

Figure 3 shows the variation of various flow variables (total pressure, gas temperature, electron temperature, electrical conductivity, electron number density, Hall parameter, electric current density, axial voltage gradient and transverse voltage gradient) along the centerline of the scramjet inlet. There are a few common characteristics among the central-line plots of all the variables except the electron number density. All the variables other than the electron number density reach maximum values, whereas electron number density attains minimum value, at the position where the shocks coalesce. Behind the shocks, values of the variables decrease sharply and eventually become almost constant. It can be observed from the plots of gas temperature ( $T_g$ ) and electron temperature ( $T_e$ ) that thermal non-equilibrium exists ahead of shocks since  $T_e \gg T_g$  and it relaxes toward a thermal equilibrium (not completely achieved) behind the shocks as  $T_e > T_g$ . It is also seen from Figure 3(a) that the total pressure peak is reduced due to the ionization.

Figure 4 shows the comparison of various flow variables (total pressure, gas temperature, flow velocity, Mach number, axial voltage gradient, Hall parameter and electric current density) between the present numerical simulations and the theoretical calculation of Park et al. [12] along the axial direction. The axial variation of all the flow variables in the numerical simulation is obtained by integrating the values along the vertical direction at each section. All flow variables other than Hall parameter decrease with the axial distance, whereas Hall parameter increases with the axial distance in both the numerical simulation and the theoretical calculation. It can be observed from the flow variables that there is some difference between the values in 1-D theoretical analysis and the 2-D numerical predictions.

The axial variation of all the flow variables presented in Figure 4 has been integrated along the axial direction and the integrated values are presented in Table 3. It can be observed from Table 3 that the percentage difference of all parameters between the theoretical analysis and the numerical prediction is within 5% for all the flow variables. This can be considered as a reasonably good agreement between the numerical simulation and the theoretical analysis of Park et al. [12].

Table 3: Comparison Between the 1-D Theoretical Analysis and 2-D Numerical Simulation

| Parameter                                         | Park et al. Calc. [12] | Present Simulation  | % Difference |
|---------------------------------------------------|------------------------|---------------------|--------------|
| Surface Pressure (Pascal)                         | $4.65 \times 10^6$     | $4.48 \times 10^6$  | +3.67        |
| Gas Temp. ( $^{\circ}\text{K}$ )                  | 3560                   | 3619                | -1.65        |
| Flow Velocity (m/s)                               | 2620                   | 2749                | -4.94        |
| Mach Number                                       | 2.60                   | 2.59                | +0.176       |
| Axial Voltage Gradient (V/m)                      | 4600                   | 4375                | +4.89        |
| Hall Parameter                                    | 4.45                   | 4.23                | +4.95        |
| Electric Current Density( $\text{A}/\text{m}^2$ ) | $7.65 \times 10^4$     | $7.303 \times 10^4$ | +4.54        |

### 13.10 Conclusions

Two-dimensional electromagnetohydrodynamics equations with a bitemperature model and a variable electrical conductivity model are employed to compute the influence of magnetic field (generated by the magnets on the body surface) on hypersonic flow of a weakly ionized gas inside a scramjet inlet. Computations are compared with 1-D theoretical analysis; reasonably good agreement is obtained. The computations also show the potential for reducing the supersonic drag of scramjet inlets by the application of electromagnetic field on a weakly ionized plasma.

### 13.11 Acknowledgements

This work has been supported by the Air Force Office of Scientific Research (AFOSR). The authors gratefully acknowledge the support and encouragement of Dr. Len Sakell.

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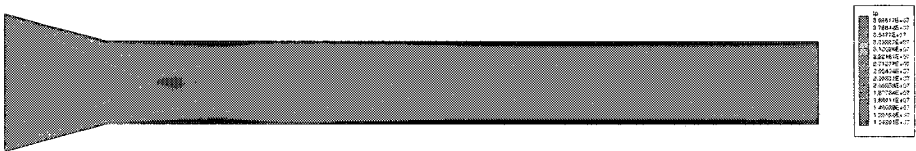


Fig. 1(a) Total Pressure Contours Without Ionization (N/m<sup>2</sup>)

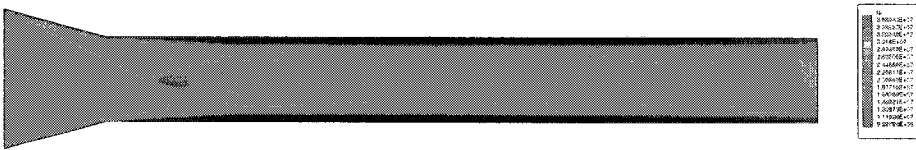


Fig. 1(b) Total Pressure Contours With Ionization (N/m<sup>2</sup>)

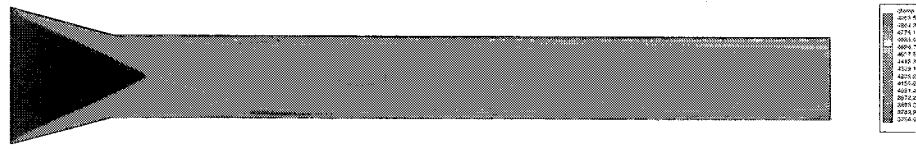


Fig. 1(c) Gas Temperature Contours Without Ionization (°K)

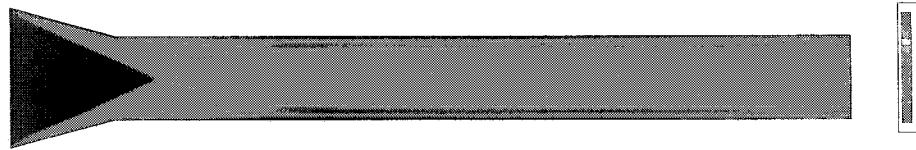


Fig. 1(d) Gas Temperature Contours With Ionization (°K)

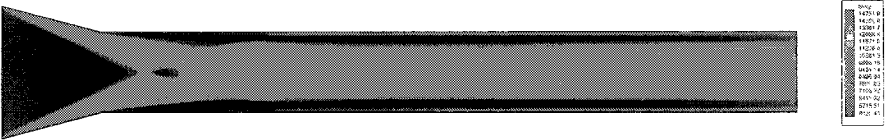


Fig. 1(e) Electron Temperature Contours ( $^{\circ}\text{K}$ )



Fig. 1(f) Electrical Conductivity Contours ( $\text{mhos/m}$ )



Fig. 1(g) Electron Number Density Contours



Fig. 1(h) Hall Parameter Contours



Fig. 1(i) Electrical Current Density Contours (A/m<sup>2</sup>)

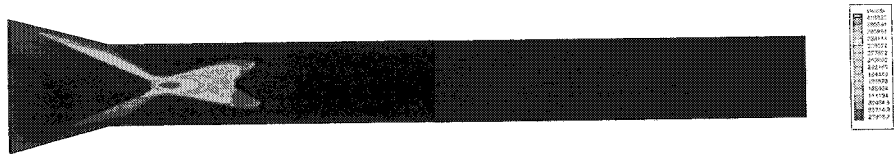


Fig. 1(j) Axial Voltage Gradient Contours (V/m)

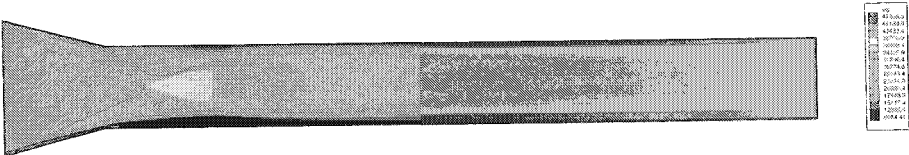


Fig. 1(k) Transverse Voltage Gradient Contours (V/m) Without Ionization

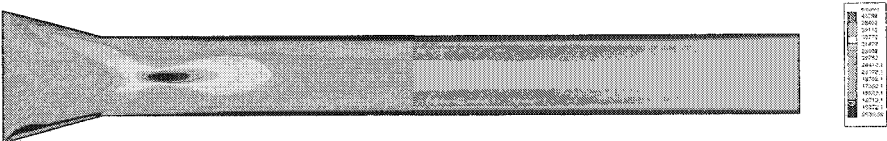


Fig. 1(l) Transverse Voltage Gradient Contours (V/m) With Ionization

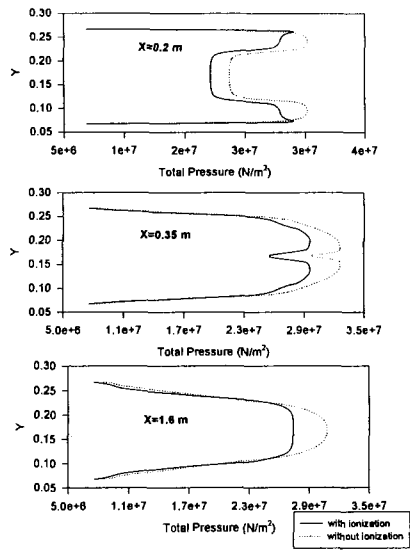


Fig. 2(a) Variation of Total Pressure

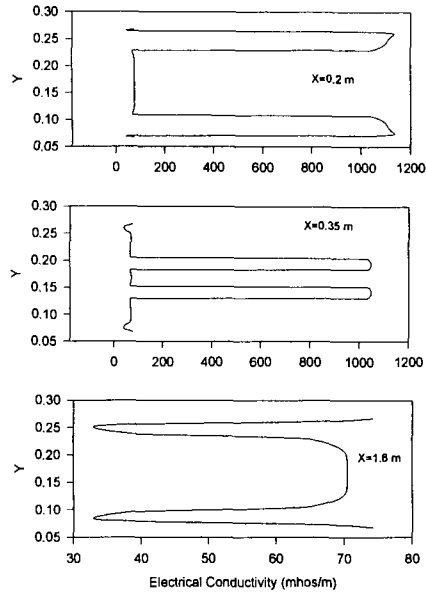


Fig. 2(d) Variation of Electrical Conductivity

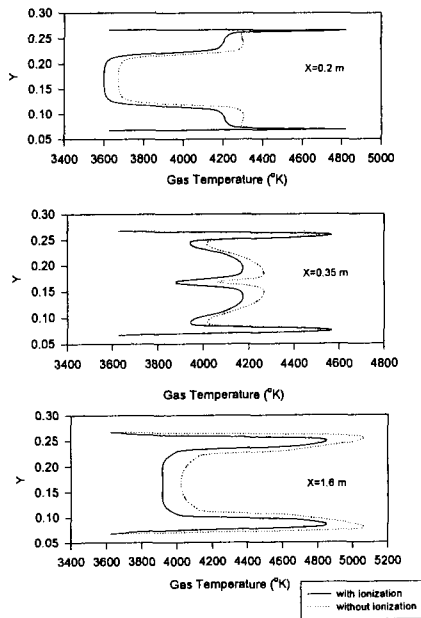


Fig. 2(b) Variation of Gas Temperature

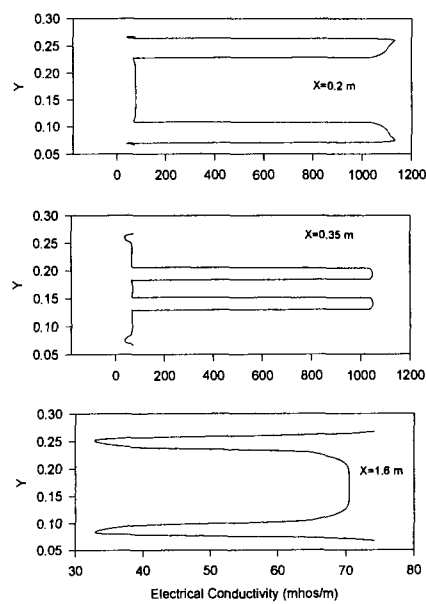


Fig. 2(d) Variation of Electrical Conductivity

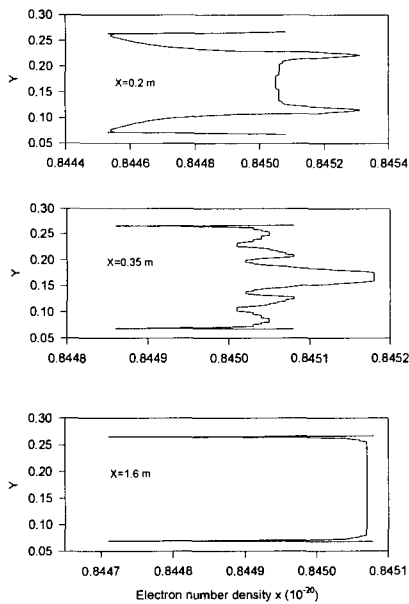


Fig. 2(e) Variation of Electron Number Density

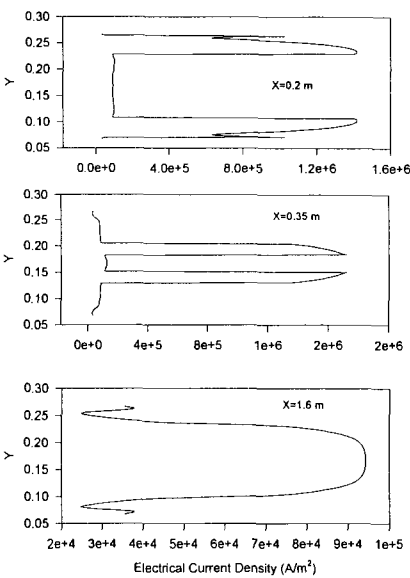


Fig. 2(g) Variation of Electrical Current Density

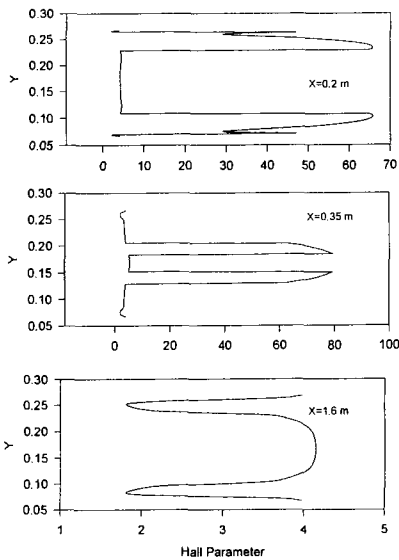


Fig. 2(f) Variation of Hall Parameter

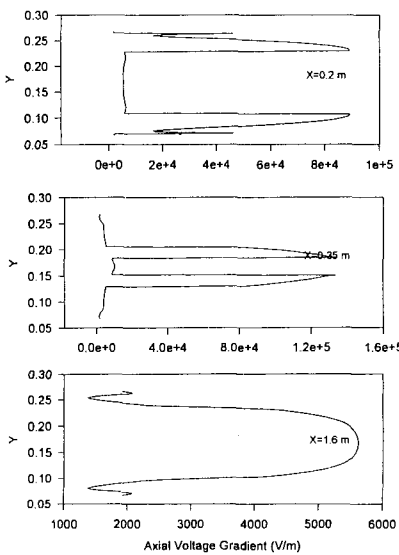


Fig. 2(h) Variation of Axial Voltage Gradient

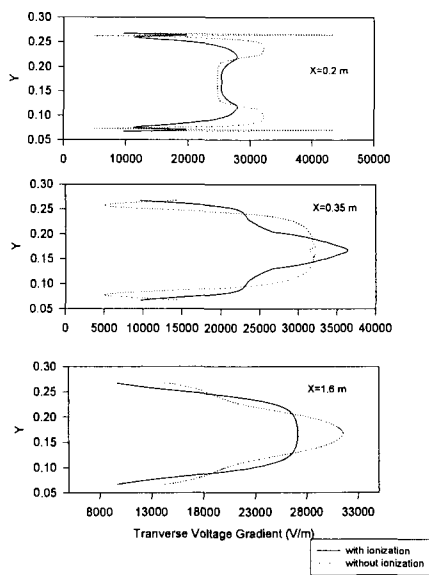


Fig. 2(i) Variation of Transverse Voltage Gradient

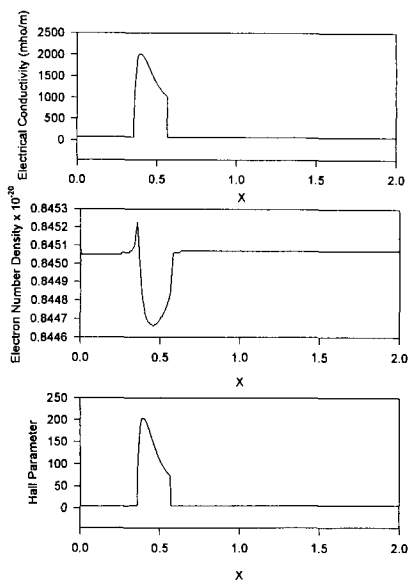


Fig. 3(b) Variation of Electrical Conductivity, Electrical Current Density and Hall Parameter along the Axial Distance

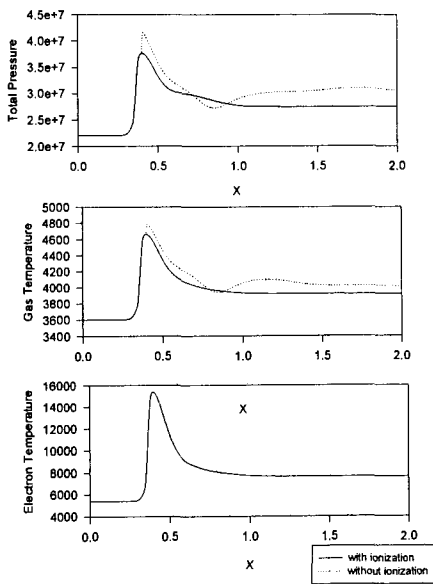


Fig. 3(a) Variation of Total Pressure, Gas Temperature and Electron Temperature along the Axial Distance

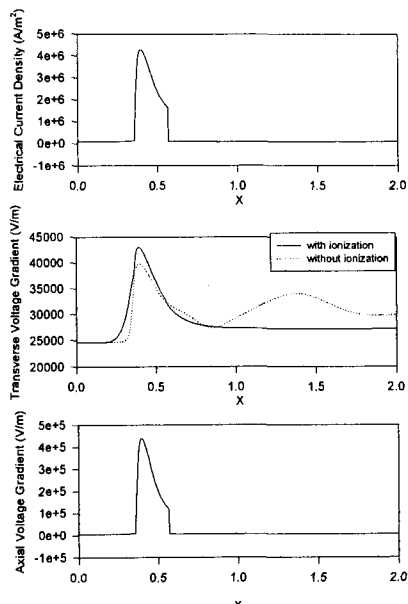


Fig.3(c) Variation of Electrical Current Density, Transverse Voltage Gradient and Axial Voltage Gradient along Axial Distance

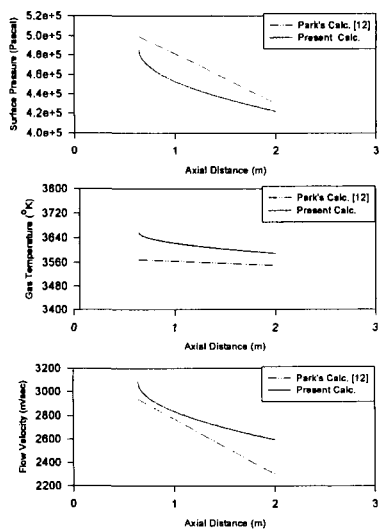


Fig. 4(a) Comparison of Surface Pressure, Gas Temperature and Flow Velocity between Present Simulations and Park's Calculations [12]

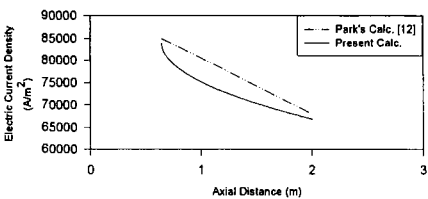


Fig. 4(c) Comparison of Electric Current Density between Present Simulation and Park's Calculations [12]

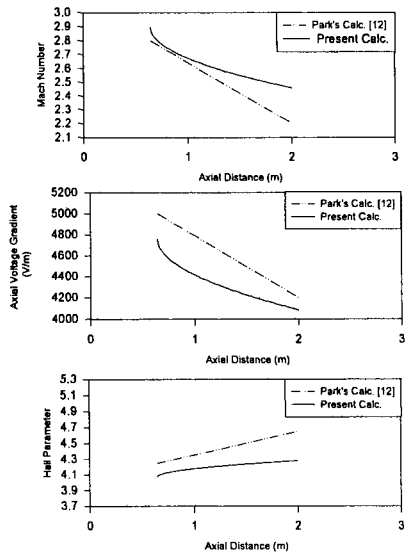


Fig. 4(b) Comparison of Mach Number, Axial Voltage Gradient and Hall Parameter between Present Simulations and Park's Calculations [12].

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# Progress in Computational Magneto-Aerodynamics

Joseph S. Shang<sup>1</sup>  
 Patrick W. Canupp<sup>2</sup>  
 Datta V. Gaitonde<sup>3</sup>

## 14.1 Introduction

In the development of interdisciplinary modeling and simulation technology, electromagnetic phenomena serve as additional flow-field control mechanisms. In fact, four decades ago Resler and Sears recognized the potential application of electromagnetic effects to aerodynamics.<sup>1</sup> Their observations were strongly supported by pioneering research by Bush, Ziemer, and Meyer.<sup>2-4</sup> Magneto-aerodynamics is truly an interdisciplinary endeavor; the interacting physical phenomena require the interplay of aerodynamics, electromagnetics, chemical physics, and quantum physics to describe the ionized gas flow in the presence of electromagnetic fields. This interdisciplinary endeavor not only presents extremely complex science issues, but it also demands a significant knowledge base. However, the prospect for technical breakthrough is too great to be overlooked.

Electromagnetic forces are known to significantly alter the flow fields of electrically conducting media.<sup>5-7</sup> For most hypersonic flights, the air mixture bounded by the shock wave and the vehicle consists of highly

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excited internal energy modes. As the temperature of air exceeds 5000° K, a fraction of dissociating molecules will shed their electrons.<sup>8</sup> The ionized air mixture is then characterized by a finite value of electrical conductivity, which may exceed a value of 100 *mho/m* depending on the flight speed and altitude. The interaction of moving charged particles and an electromagnetic field will generate the Lorentz force as an additional mechanism to relax conventional aerodynamic limitations. Resler and Sears examined a dimensionless parameter that governs the relative magnitude of the Lorentz and dynamics forces. It appears that the Lorentz force becomes dominant at high altitude and high speed.<sup>1</sup> This flow condition, in general, naturally occurs in hypersonic flight.

One of the possible uses of the Lorentz force is for controlling acceleration or deceleration of a gas continuously at subsonic or supersonic speeds without choking, even in constant cross section channels. The coupling of velocity and temperature of electrically conducting media through Joule heating makes possible this flow field manipulation.<sup>1,5,6</sup> The flow orientation of a charged medium can also be manipulated by the intrinsic relationship of the Hall current and the curvature of electron trajectory in a magnetic field. The Hall effect has been used widely as the driving force to divert the orientation of charged fluid particles for combustion enhancement or to reduce the length of inlet compression and distortion.

The Lorentz force always has a component,  $\sigma(U \times B) \times B$ , that will decelerate the flow.<sup>1,4</sup> Across the nonuniform velocity distribution of a shear layer, the decelerating force increases in magnitude with distance away from the solid surface. In general, the net result is a reduction in the velocity and temperature gradients of the shear layer. The flattening of velocity and temperature gradients diminishes both heat transfer and skin friction to the contacting surface.<sup>1,4,5</sup> Under some conditions, the eddy current induced by the turbulent fluid particles crossing lines of magnetic induction interacts with the magnetic field and tends to damp turbulent motion. Even at moderate values of the Hartmann number, the wall shear stress of turbulence will be reduced.<sup>6</sup>

However, shock wave propagation and bifurcation in a plasma field reveals the most pronounced phenomenon in electrically conducting gaseous media. The most striking observation is the drastic increase of the stand-off distance of a bow shock over a blunt body.<sup>2</sup> This behavior stands in stark contrast to all the experimental data and calculations of non-equilibrium hypersonic flows that encounter no electromagnetic field.<sup>8</sup> In hypersonic blunt-body flows, shock wave stand-off distance decreases with increasing Mach number and non-equilibrium effects. In 1959, Ziemer increased the shock wave stand-off distance by applying a magnetic field to a non-equilibrium hypersonic blunt-body flow.<sup>2</sup> His experimental observation received theoretical verification from the research of Bush.<sup>3</sup>

Recent innovations in modifying high-speed flow of an electrically conducting fluid involve aerodynamic and electromagnetic field interactions, as typified in the AJAX concept.<sup>9,10</sup> Research on shock wave propagation in a plasma revived in Russia in the 1980s,<sup>11-13</sup> whereas Ganguly *et al.*<sup>14</sup> have studied the phenomenon most recently. Their collective findings revealed that processes in non-equilibrium, weakly ionized gases substantially modify a traveling shock wave. Specifically, when a shock wave propagates in a plasma, the amplitude decreases and the wave front also disperses. In essence, these findings have all the features of compound waves in motion. If an electrically conducting medium could be introduced upstream of the bow shock wave and the bifurcation quantified, these newly identified physical phenomena would have a revolutionary potential for high-speed flight. Although there has been some debate in identifying the dominant mechanisms controlling plasma modification of compressible flows (e.g., whether thermal or magnetic force effects are responsible), the framework of the magneto-aerodynamic equations describes the first order effect of the electromagnetics on a flow. Studying this effect is then the principal focus of the present effort.

## 14.2 Governing Equations

A unique characteristic of the plasmadynamics of current interest is the existence of a relatively small dielectric constant. Even if the physical phenomena are in the microwave frequency range, the displacement current is still negligible in comparison to the conduction current.<sup>5,6</sup> This simplification reduces the formulation of the generalized Ampère circuit law into a Poisson type. For the present investigation without including the overly complicated non-equilibrium chemical kinetics, the governing system of equations reduces to the following:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho U) = 0 \quad (14.1)$$

$$\frac{\partial B}{\partial t} + \nabla \cdot (UB - BU) = -\nabla \times \left[ \frac{1}{\sigma} \left( \nabla \times \frac{B}{\mu} \right) \right] \quad (14.2)$$

$$\frac{\partial(\rho U)}{\partial t} + \nabla \cdot \left[ \rho U U - \frac{BB}{\mu} + \left( p + \frac{B \cdot B}{2\mu} \right) I \right] = \nabla \cdot \tau \quad (14.3)$$

$$\begin{aligned} \frac{\partial(\rho Z)}{\partial t} + \nabla \cdot \left[ \rho Z U + U \cdot \left( p + \frac{B \cdot B}{2\mu} \right) I - \frac{B}{\mu} (B \cdot U) \right] = \\ \nabla \cdot [q + U \cdot \tau] - \frac{B}{\mu} \cdot \nabla \times \left[ \frac{1}{\sigma} \left( \nabla \times \frac{B}{\mu} \right) \right] + \frac{(\nabla \times B)^2}{\sigma \mu^2} \end{aligned} \quad (14.4)$$

where  $\mu$  denotes the magnetic permeability (which relates the magnetic flux density  $B$ , to the magnetic field intensity,  $H$ ) and

$$\rho Z = \rho e + \frac{B \cdot B}{2\mu} \quad (14.5)$$

The classic ideal magnetohydrodynamics (MHD) governing equations can be deduced from the magneto-aerodynamics system through additional assumptions. First, the concept of infinite electrical conductivity implies that the strength of the motion-induced magnetic field overwhelms that of the applied field.<sup>5,6</sup> Second, in many flows inertial effects greatly outweigh viscous dissipation and heat transfer in the medium. Third, the medium is considered to be isotropic. Then the resulting governing equations are easily obtained by simply setting the right-sides of Eqs. (2)–(4) to zero.

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho U) &= 0 \\ \frac{\partial B}{\partial t} + \nabla \cdot (UB - BU) &= 0 \\ \frac{\partial(\rho U)}{\partial t} + \nabla \cdot [\rho U U - \frac{BB}{\mu} + (p + \frac{B \cdot B}{2\mu})I] &= 0 \\ \frac{\partial(\rho Z)}{\partial t} + \nabla \cdot [\rho Z U + U \cdot (p + \frac{B \cdot B}{2\mu})I - B \frac{B \cdot U}{\mu}] &= 0 \end{aligned} \quad (14.6)$$

The above ideal MHD equations constitute a hyperbolic partial differential system. From the analysis of the governing equations in each one-spatial-time space, eigenvectors and eigenvalues have been found.<sup>15</sup> Because the Lorentz force is perpendicular to both the magnetic field intensity and the electric current, the eigenvalue associated with the normal component of the magnetic field intensity has a null value. The remaining seven eigenvalues of the classic MHD equations can also locally degenerate to coincide with each other, depending on the relative magnitude and orientation of the magnetic field. The seven eigenvalues are

$$u, \quad u \pm C_a, \quad u \pm C_s, \quad \text{and} \quad u \pm C_f$$

These eigenvalues reflect four different wave speeds when a perturbation propagates in a plasma field: the acoustic, Alfvén, slow and fast plasma wave speeds,<sup>5,7</sup>

$$C^2 = \partial p / \partial \rho \quad (14.7)$$

$$C_a^2 = B_n^2 / \rho \mu \quad (14.8)$$

$$2C_s^2 = C^2 + \frac{B^2}{\rho \mu} - \sqrt{\left(C^2 + \frac{B^2}{\rho \mu}\right)^2 - 4C^2 C_a^2} \quad (14.9)$$

$$2C_f^2 = C^2 + \frac{B^2}{\rho \mu} + \sqrt{\left(C^2 + \frac{B^2}{\rho \mu}\right)^2 - 4C^2 C_a^2} \quad (14.10)$$

where  $B_n$  denotes the transverse component of the magnetic flux density with respect to the wave front. Therefore, not all wave speeds propagating in a homogeneous plasma are isotropic; some depend strongly on the polarity of magnetic field.

The graph in Fig. 1 presents the four plasma wave speeds over a phase relationship from  $0^\circ$  to  $180^\circ$  between the wave front and the applied/induced magnetic field. This calculation assumes a homogeneous plasma field and normalizes all wave speeds with respect to the acoustic speed of sound,  $C$ . The parameter  $B^2/(\rho\mu C^2)$  is expressible in terms of the magnetic interaction parameter,  $S = L\sigma B^2/\rho u$  and the magnetic Reynolds number,  $R_m = \sigma\mu uL$  as the following:<sup>5</sup>

$$\frac{B^2}{\rho\mu C^2} = \frac{SM^2}{R_m} \quad (14.11)$$

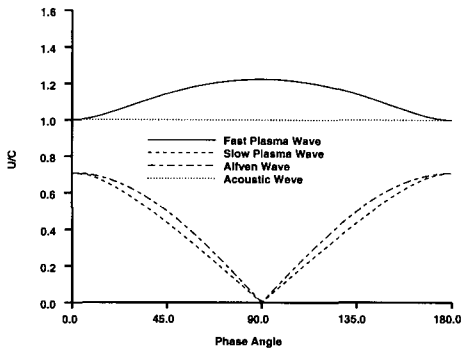
where  $M$  is the Mach number. For an electromagnetic field generated by a glow discharge in a plasma chamber,<sup>6</sup> the parameter  $B^2/(\rho\mu C^2)$  has a value near unity. For the purpose of illustration, the theoretical result is generated by using a value of 0.5. Under this condition, the fast plasma wave can propagate at 1.225 times the speed of sound in a transverse magnetic field. In the full range of phase angles, the fast plasma wave speed is greater than or equal to the speed of sound. The slow plasma wave, on the other hand, propagates at a rate that is lower than the speed of sound over the entire phase angle range. The highest speed of the slow plasma wave is 0.7071 of the sonic speed when the wave motion is parallel to the magnetic field. The slow plasma wave motion ceases when the transverse magnetic component vanishes. The fast and slow plasma wave speeds represent upper and lower bounds for the propagating rates of the Alfvén and acoustic waves. Because the Alfvén wave does not carry energy, there is no entropy producing interaction with other modes of wave motion.<sup>7</sup>

Figure 2 depicts the wave speeds in a glow discharge at a Mach number of 10. Now the fast plasma wave travels at a speed 100 times faster than the speed of sound, and the speed of the slow plasma wave is still bounded by the acoustic speed. In view of the great disparity in wave speed, wave interference is limited to that between the slow plasma wave and the acoustic wave. However, a complete understanding of this phenomenon must be derived from a detailed and systematic experimental and computational research effort.

### 14.3 Numerical Procedures

The governing equations in a Cartesian frame appear in flux vector form as:

$$\frac{\partial V}{\partial t} + \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \frac{\partial F_{x,r}}{\partial x} + \frac{\partial F_{y,r}}{\partial y} + \frac{\partial F_{z,r}}{\partial z} \quad (14.12)$$

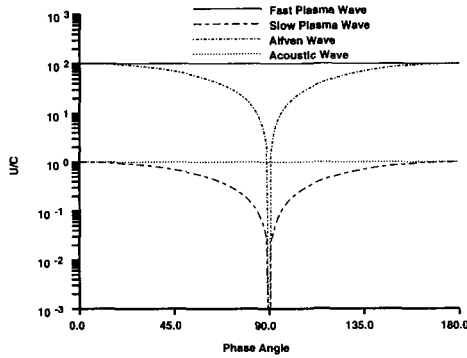


**Figure 1** Wave Propagation Speeds,  $SM^2/R_m = 0.5$

where  $V$  is the vector of conservative dependent variables. The components of flux vectors associated with the ideal MHD equations,  $F_x$ ,  $F_y$ , and  $F_z$ , and the components of the flux vectors associated with the resistivity, viscous dissipation and heat transfer,  $F_{x,r}$ ,  $F_{y,r}$ , and  $F_{z,r}$  can be obtained easily from Eqs. (1)–(4). Only the components in the  $x$  coordinate direction are given here:<sup>16</sup>

$$V = (\rho, \rho u, \rho v, \rho w, \rho Z, B_x, B_y, B_z)^T \tag{14.13}$$

$$F_x = \begin{bmatrix} \rho u \\ \rho u^2 + p + B \cdot B/2\mu - B_x^2/\mu \\ \rho uv - B_x B_y/\mu \\ \rho uw - B_x B_z/\mu \\ (\rho e + p + B \cdot B/\mu) u + (U \cdot B/\mu) B_x \\ 0 \\ uB_y - vB_x \\ uB_z - wB_x \end{bmatrix} \tag{14.14}$$



**Figure 2** Wave Propagation Speed,  $SM^2/R_m = 10^4$

$$F_{x,r} = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ \tau_{xz} \\ \{U \cdot \tau + \kappa \partial T / \partial x \\ + B_y [\partial(B_y/\mu)/\partial x - \partial(B_x/\mu)/\partial y] / \mu \sigma \\ + B_z [\partial(B_z/\mu)/\partial x - \partial(B_x/\mu)/\partial z] / \mu \sigma \} \\ 0 \\ [\partial(B_y/\mu)/\partial x - \partial(B_x/\mu)/\partial y] / \sigma \\ [\partial(B_z/\mu)/\partial x - \partial(B_x/\mu)/\partial z] / \sigma \end{bmatrix} \quad (14.15)$$

The equations of ideal MHD are not a strictly hyperbolic system<sup>17,18</sup> in that the distinct eigenvalues may degenerate in the field to coincide with others. As a consequence, the analytic structure of the weak solution is unknown where the degeneration occurs. Nevertheless, most numerical approaches for solving this system are the characteristic-based, approximated Riemann,<sup>17-21</sup> or TVD schemes.<sup>22</sup> In these approaches, the hyperbolic components (ideal MHD) of the governing equations in conservation form appear as:<sup>16,19,20</sup>

$$\frac{\partial V}{\partial t} + \nabla \cdot F = 0 \quad (14.16)$$

For numerical stability and accuracy enhancements, the ideal MHD flux vectors are further split according to the signs of their respective eigenvalues. The eigenvalues and eigenvectors are derived from the Jacobian coefficient

matrices,  $\partial F_x/\partial W$ ,  $\partial F_y/\partial W$ , and  $\partial F_z/\partial W$  in primitive variables.

$$\frac{\partial V}{\partial t} + \frac{\partial F_x}{\partial W} \frac{\partial W}{\partial x} + \frac{\partial F_y}{\partial W} \frac{\partial W}{\partial y} + \frac{\partial F_z}{\partial W} \frac{\partial W}{\partial z} = 0 \quad (14.17)$$

In order to overcome the numerical difficulty of degenerate eigenvalues, several investigators have modified the Jacobian coefficient matrices. Powell *et al.*<sup>19</sup> successfully imposed Gauss's law,  $\nabla \cdot B = 0$  as a modifier for this purpose. More recently, MacCormack<sup>20</sup> demonstrated that the flux vectors can be expanded to preserve the homogeneous of degree one property. His method takes the following approach. First, the discretized form of the one-dimensional MHD equations is

$$\frac{\delta \mathbf{U}_i}{\Delta t} \mathcal{V}_i + \sum_{sides} \mathbf{F}'_e \cdot \vec{S} = 0 \quad (14.18)$$

where  $\vec{S}$  is the cell interface surface area and  $\mathcal{V}_i$  is the volume of the  $i$ th cell. Introducing transformation Jacobians between primitive and conserved variables,  $S$  and  $S^{-1}$ , this equation can also be written as

$$\frac{\delta \mathbf{U}_i}{\Delta t} \mathcal{V}_i + \sum_{sides} S^{-1} \left( S \frac{\partial \mathbf{F}'_e}{\partial V} \right) S \mathbf{U} \cdot \vec{S} = 0 \quad (14.19)$$

Next, MacCormack splits the redefined ideal MHD flux vector as

$$\mathbf{F}'_e = \mathbf{F}''_e - \mathbf{P} \frac{B_x}{\hat{a}} \quad (14.20)$$

where  $\hat{a}$  is an added conserved variable, taken to be unity, so that Eq. (14.19) takes the form

$$\frac{\delta \mathbf{U}_i}{\Delta t} \mathcal{V}_i + \sum_{sides} S^{-1} \tilde{A}'_e S \mathbf{U} \cdot \vec{S} - \sum_{sides} \mathbf{P} \frac{\partial}{\partial \mathbf{V}} \left( \frac{B_x}{\hat{a}} \right) S \mathbf{U} \cdot \vec{S} = 0 \quad (14.21)$$

The last summation in Eq. (14.21) is identically zero because the quantity  $B_x/\hat{a}$  is homogeneous of degree zero with respect to the conserved variables. The matrix  $\tilde{A}'_e$  contains, as a submatrix, the Jacobian matrix used by Powell. Therefore, this new method uses the same eigensystem as shown in Powell *et al.*<sup>19</sup> to evaluate this part of the flux. An additional flux component arises due to the final column of  $\tilde{A}'_e$ . The numerical method uses a central evaluation of this flux component. Therefore, the discretized equation appears as

$$\frac{\delta \mathbf{U}_i}{\Delta t} \mathcal{V}_i + \sum_{sides} S^{-1} C^{-1} \Lambda C S \mathbf{U} \cdot \vec{S} + \sum_{sides} \mathbf{b} \cdot \vec{S} = 0 \quad (14.22)$$



where the  $\mathbf{b}$  vector is

$$\mathbf{b} = \left( 0, (\beta - 1) \frac{B^2}{2\hat{a}^2}, 0, 0, (\beta - 1) \frac{B^2 u}{2\hat{a}^2}, -\frac{u B_x}{\hat{a}}, -\frac{v B_x}{\hat{a}}, -\frac{w B_x}{\hat{a}} \right)^T \quad (14.23)$$

The flux components in the first summation in Eq. (14.22) split according to the signs of the eigenvalues, which reside on the diagonal of the matrix  $\Lambda$ . Therefore, the split fluxes are

$$\mathbf{F}^\pm = S^{-1} C^{-1} \Lambda^\pm C S U \quad (14.24)$$

The above description assumed a one-dimensional form for the governing equations, but it readily extends to multiple dimensions. MacCormack's original paper actually develops the method in three dimensions, thereby making a significant contribution to numerical techniques for solving the ideal MHD equations.<sup>17-20</sup> One can also write the flux-split method in a form using finite-differences

$$\frac{\delta V}{\delta t} + \frac{\delta(F_x^+ + F_x^-)}{\delta x} + \frac{\delta(F_y^+ + F_y^-)}{\delta y} + \frac{\delta(F_z^+ + F_z^-)}{\delta z} = 0 \quad (14.25)$$

Another innovative numerical procedure for solving the governing equations of magneto-aerodynamics adopts the combination of compact-difference<sup>23</sup> and the classic Runge-Kutta method for the spatial and temporal derivatives, respectively. This approach was successfully applied to solve the time-dependent, three-dimensional Maxwell<sup>24,25</sup> and Navier-Stokes equations.<sup>26,27</sup> Gaitonde successfully implemented this high-order method for solving the magneto-aerodynamic equations.<sup>16</sup> For the compact-differencing scheme, the following formula, given by Lele,<sup>23</sup> approximates the  $\xi$  derivative of a scalar variable,  $\phi$

$$\begin{aligned} \alpha(\phi_\xi)_{i-1} + (\phi_\xi)_i + \alpha(\phi_\xi)_{i+1} &= \frac{(4\alpha-1)}{3}(\phi_{i+2} - \phi_{i-2}) \\ &+ \frac{2(\alpha+2)}{3}(\phi_{i+1} - \phi_{i-1}) \end{aligned} \quad (14.26)$$

The fourth-order, Padé type compact-differencing is recovered by assigning the coefficient  $\alpha$  a value of  $1/4$ , and is the choice of the present numerical approach.

The critically important component of the present approach is a low-pass filter.<sup>24-28</sup> Gaitonde introduced filtering as a key element for practical application of compact-differencing schemes to solve partial differential equations in 1996.<sup>28</sup> He demonstrated that a spatial filter can maintain computational stability on complicated geometries by suppressing the high-frequency, spurious Fourier components that arise through mesh non-uniformity, boundary condition implementation, and the capturing of non-linear phenomena. The general formulation of the filter, based on Lele's

framework, is:

$$\beta\phi'_{i-1} + \phi'_i + \beta\phi'_{i+1} = f_{2n}(\beta, \phi_{i-n}, \dots, \phi_{i+n}) \quad (14.27)$$

where the parameter  $\beta$  must be bounded exclusively between -0.5 and 0.5, and the superscript, ' indicates a filtered variable. The coefficient of the right-side polynomial can be derived in terms of the parameter  $\beta$  by Taylor series and Fourier series analyses.<sup>16</sup> The filter operates on a stencil of  $2n+1$  nodes leading to a  $2n$ -order formula.<sup>23,25</sup> In application, the filter acts as a post processor on a computed solution. For multi-dimensional computations, the applications of the filter are sequential and in alternating coordinate directions.

The formulation of the approximated Riemann or the compact-difference schemes easily extends to a curvilinear frame by introducing the coordinate transformation,  $\xi = \xi(x, y, z)$ ,  $\eta = \eta(x, y, z)$ , and  $\zeta = \zeta(x, y, z)$ . The resulting equations are:

$$\frac{\partial V'}{\partial t} + \frac{\partial F'_\xi}{\partial \xi} + \frac{\partial F'_\eta}{\partial \eta} + \frac{\partial F'_\zeta}{\partial \zeta} = \frac{\partial F'_{\xi,r}}{\partial \xi} + \frac{\partial F'_{\eta,r}}{\partial \eta} + \frac{\partial F'_{\zeta,r}}{\partial \zeta} \quad (14.28)$$

where the transformed dependent variables are defined as  $V' = V/J_c$ , and  $J_c$  is the coordinate transformation Jacobian. The components of flux vectors associated with the ideal MHD equations as well as the resistivity, viscous dissipation and heat transfer terms transform in a similar manner to the curvilinear frame

$$\begin{aligned} F'_\xi &= (\xi_x F_x + \xi_y F_y + \xi_z F_z)/J_c \\ F'_\eta &= (\eta_x F_x + \eta_y F_y + \eta_z F_z)/J_c \\ F'_\zeta &= (\zeta_x F_x + \zeta_y F_y + \zeta_z F_z)/J_c \end{aligned} \quad (14.29)$$

#### 14.4 Rankine-Hugoniot Jump Condition

The effect of electromagnetic force on the shock wave structure can be examined by the modification to the Rankine-Hugoniot condition across a shock. The normal shock jump condition along the  $x$  coordinate in a plasma has been derived from the magneto-aerodynamics equations. The jump condition becomes:<sup>29</sup>

$$\begin{aligned}
[\rho u] &= 0 \\
[p + \rho u^2] &= \int (J_y B_z - J_z B_y) dx \\
[\rho uv] &= \int (J_z B_x - J_x B_z) dx \\
[\rho uw] &= \int (J_x B_y - J_y B_x) dx \\
[\rho uh] &= \int J^2 / \sigma dx \\
[B_x] &= 0 \\
[J_y / \sigma - w B_x + u B_z] &= 0 \\
[J_z / \sigma - u B_y + v B_x] &= 0
\end{aligned} \tag{14.30}$$

where  $h$  denotes the total enthalpy of the flow, and the brackets indicate changes in the end states across the shock wave. The above jump conditions reduce to those derived by Sutton and Sherman<sup>5</sup> if the plasma has a perfect electrical conductivity and the Hall current is negligible. Under these conditions  $J = \vec{k} \frac{d}{dx} \left( \frac{B_y}{\mu} \right) - \vec{j} \frac{d}{dx} \left( \frac{B_z}{\mu} \right)$ , and the resultant jump conditions are obtained by a straight-forward integration with respect to  $x$ .

$$\begin{aligned}
[\rho u] &= 0 \\
[p + \rho u^2] &= -[(B_y^2 + B_z^2)/2\mu] \\
[\rho uv] &= [B_x B_y] \\
[\rho uw] &= [B_x B_z] \\
[\rho uh] &= [-u(B_y^2 + B_z^2)/2\mu + B_x(v B_y + w B_z)/\mu] \\
[B_x] &= 0 \\
[u B_z - w B_x] &= 0 \\
[v B_x - u B_y] &= 0
\end{aligned} \tag{14.31}$$

The above shock jump conditions, regardless of the underlying assumptions, indicate that the presence of an electromagnetic field alters the Rankine-Hugoniot relationship of gasdynamics. The gist of the modification rests on the two additional entropy change mechanisms.<sup>29</sup> One of them is Joule heating, which is positive and will contribute to an entropy increase across the shock. The other mechanism is the work that the electromagnetic forces can perform on the moving gas particles. Depending on the polarity of the induced or applied electromagnetic field, the total entropy for the open system can be reduced. As a direct consequence of entropy reduction, the wave drag of the shock wave will diminish.

To gain some insight into the effects of magnetic fields on the flow of an electrically conducting fluid, the present effort examines the structure of a normal shock wave in a transverse electromagnetic field at a Mach number of 1.5. In order to illustrate the basic idea, the plasma medium is assumed to be a calorically and thermally perfect Helium with a constant electrical conductivity. The free-stream value of the applied transverse magnetic field is  $B_y = 0.03 T$ ,  $B_x = B_z = 0$ , and the electrical conductivity is  $10^5 \text{ mho/m}$ . A pressure of  $10 \text{ Torr}$  and temperature of  $300 K$  completely specify the free-

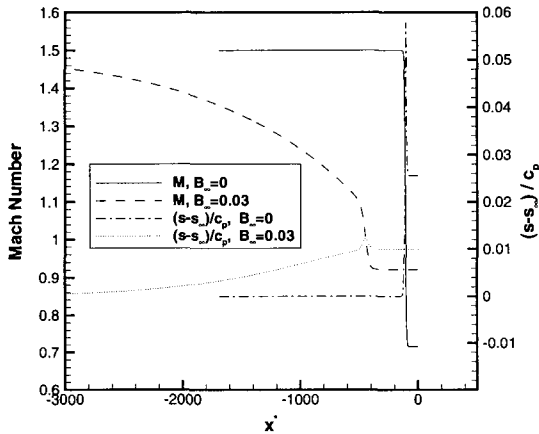
stream state of the gas. Under these conditions, and with the assumption of one-dimensional flow, the governing equations reduce to the following form

$$\begin{aligned}
 \frac{d}{dx} [\rho u] &= 0 \\
 \frac{d}{dx} \left[ \rho u^2 + p + \frac{B_y^2}{2\mu} - \tau \right] &= 0 \\
 \frac{d}{dx} \left[ \rho u h + \frac{u_\infty B_\infty}{\mu} B_y - u\tau - q \right] &= 0 \\
 \frac{d}{dx} \left[ \frac{1}{\sigma\mu} \frac{dB_x}{dx} - uB_y \right] &= 0
 \end{aligned} \tag{14.32}$$

Canupp has applied the Runge-Kutta-Fehlberg method to solve these coupled ordinary differential equations with a variable step size.<sup>30</sup> To ensure numerical stability, the calculation initiates in the subsonic, post-shock region and integrates in the counter flow direction. The fine shock structure is captured by scaling the coordinate with the shock wave Reynolds number,  $x^* = \rho_\infty u_\infty x / \mu_\infty$ . Verification of the solving scheme for a gasdynamic computation (no magnetic field) was accomplished by achieving an excellent agreement with the data of Muntz and Harnett.<sup>31</sup>

To compare the shock structures in the electromagnetic and acoustic fields, Fig. 3 shows results of the numerical calculation for the two cases. The flow is from left to right, and the figure shows both the Mach number and entropy variation through the shock. For each case, both the calculated Mach number and the entropy variations correctly reach the identical upstream asymptotes near  $x^* = -7500$ , although the horizontal axis for this figure only extends to  $x^* = -3000$  to highlight the shock wave internal structure. The compression process of the gasdynamic shock is abrupt and concentrates in a thin shock structure. On the other hand, the signal of the plasma wave propagates far upstream of the gasdynamic shock, as if the compression takes place in two distinct stages. For the same oncoming Mach number, the shock in the plasma moves forward with respect to the gasdynamic shock.

As Fig. 3 shows, the entropy of the gas in the gasdynamic shock rapidly rises to a maximum value at the sonic point then quickly relaxes to the final downstream value. A thin shock structure confines all of the entropy change for this case. The entropy of the shock in plasma exhibits a similar overall behavior, but the changes are much more gradual and less intense. The increase of entropy also initiates farther upstream of the shock in the electromagnetic field than in its gasdynamic counterpart. For both shocks, the entropy rises to a maximum value at the sonic point within the shock and then asymptotically relaxes to different post-shock values. The entropy increment of the shock in plasma is a factor of 2.6 lower than that of the gasdynamic shock for this calculation. Because the entropy variation through a shock in a plasma strongly depends on the polarization of the transverse magnetic field, this



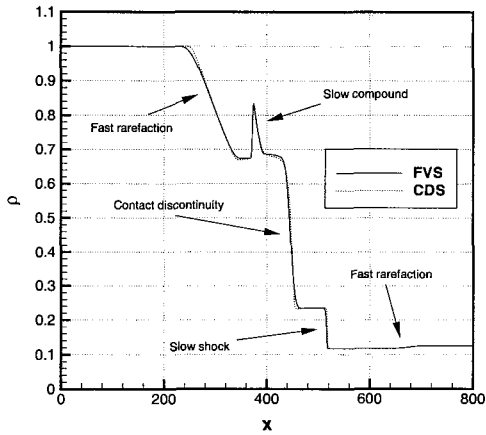
**Figure 3** Acoustic and Plasma Shock Structures

simplified one-dimensional shock calculation does not illustrate all the possible magneto-aerodynamic phenomena. However, its utility lies in its simplicity and ability to demonstrate basic magnetic field effects in gasdynamics.

### 14.5 Ideal MHD Shock Tube Simulation

Although the idealized MHD shock tube is not necessarily reproducible by experiments, it has become a benchmark for developing numerical procedures to solve ideal MHD<sup>17,21,22,34</sup> and magneto-aerodynamic equations.<sup>16</sup> The difficulty of treating degenerate eigenvalues or the admissibility of intermediate (compound) shocks in MHD is still not completely resolved.<sup>17,18</sup> Numerical results generated by the approximated Riemann solver, which is based on eigenvalue and eigenvector analysis, inescapably will receive additional scrutiny in the future. A direct comparison with computations from the characteristics-based and the compact-difference scheme, which does not require specific knowledge of eigenvalues and eigenvectors, is valuable. A side-by-side comparison of gasdynamic and MHD shock tube simulations is even more useful for assessing the salient features of shock waves in plasma.<sup>34</sup>

In order to achieve this goal, calculations by the compact-difference and flux-vector splitting schemes that duplicated the mesh and time-step size of the pioneering effort by Brio and Wu<sup>17</sup> were carried out.<sup>16,34</sup> A total of 800 grid points were used to define the computational domain, the constant spatial and temporal increment are  $\Delta x = 1$  and  $\Delta t = 0.2$ , respectively. For the

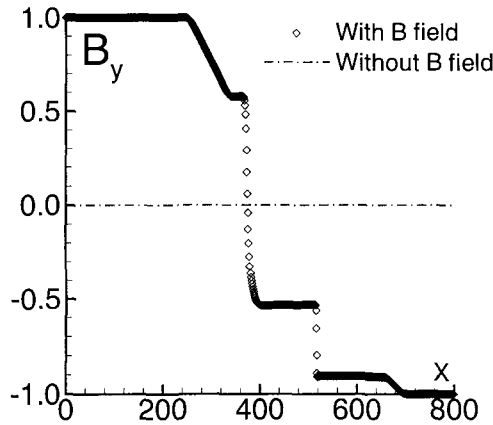


**Figure 4** Comparison of Flux-Vector Splitting (FVS) and Compact Difference (CDS) Methods

compact-differencing solution procedure, a local filter switching procedure is required to change the higher-order filter to a 2nd-order filter locally for shock capturing. The basic shock detection formulation is adopted from the work of Harten and Yee<sup>32,33</sup> and the detailed implementation can be found in Ref. 16. For the characteristic-based calculation, the numerical method is a modified form of Steger-Warming flux-vector splitting using the cell interface values for the flux Jacobian.<sup>34</sup> In regions of strong pressure gradients, the computation reverts to the original Steger-Warming flux-vector splitting scheme.<sup>35</sup>

A comparison of density distributions inside the shock tube appears in Fig. 4. In short, the agreement between solutions of the two entirely different solving schemes is excellent. The results also match those of Brio and Wu.<sup>17</sup> There are small to negligible differences between two numerical results at strong shock jump locations, as one would expect. Again the difference is nearly undetectable in the scale of difference between the gasdynamic and MHD shock waves.

Figure 5 depicts the transverse component of the magnetic field intensity,  $B_y$ . The strenuous variation of the transverse magnetic field component is clearly indicated over the computational domain. The instantaneous and transient phenomena are bounded by the right-running, fast rarefaction precursor of the slow shock toward the low pressure section and the left-running, fast rarefaction wave toward the high pressure section of the shock tube. The  $B_y$  field component undergoes a polarization reversal between the contact discontinuity and the fast rarefaction. Meanwhile, the stream-wise



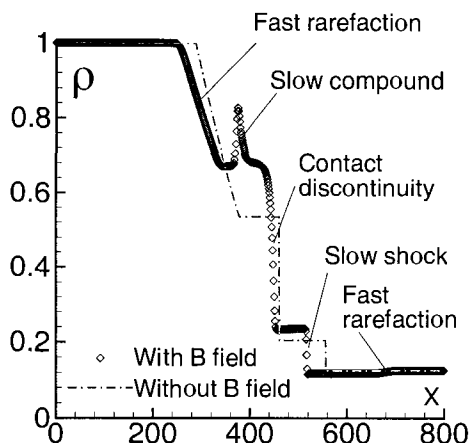
**Figure 5** Transverse Magnetic Field Intensity in MHD Shock Tube

component of the magnetic field intensity,  $B_x$  steadfastly remains at a constant value of 0.75 to ensure the Gauss law of the coplanar magnetic field,  $\nabla \cdot B = 0$  is rigorously enforced.<sup>36</sup>

The density variation within the idealized shock tube, which Fig. 6 presents, best describes the complex wave system. The fast rarefaction waves propagate at speeds higher than the acoustic speed toward both ends of the tube. The precursor of the slow right-running shock, identified as the fast plasma wave, moves into the lower pressure section of the tube. This wave represents one of the unique phenomena of the plasma wave motion.

At the location where the transverse magnetic field reverses its polarity, an intermediate or compound shock emerges. The presence of the compound wave indicates that a slow rarefaction wave can be attached to a slow shock temporally in the plasma field. This unique event occurs when the transverse magnetic field vanishes during the polarity reversal and the distinct plasma waves degenerate. The existence and physical significance of the compound wave are still uncertain.<sup>17,18</sup> At the least, the methods demonstrate the existence of a numerical intermediate shock as a possible weak solution to the nonconvex hyperbolic system. This conclusion is independent of the procedure used to resolve spatial fluxes.<sup>16,17,34</sup>

Figure 7 presents the pressure distribution within the idealized MHD shock tube. The behavior of the pressure variation offers additional confirmation for the different wave speeds of acoustic and plasma fields. The left-running, fast rarefaction wave moves toward the high pressure region followed by a slow shock, which in turn is immediately followed by a rarefaction. Between the



**Figure 6** Density Distribution in MHD Shock Tube

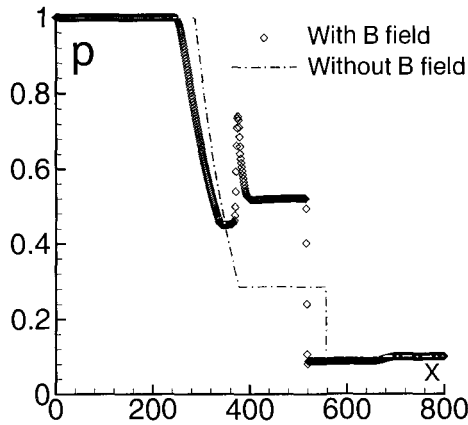
compound wave and the right-running slow shock, the pressure distribution has a constant value across the contact surface. On the compression side of the shock tube, the right-running, slow shock generates a greater pressure jump than the gasdynamic shock when preceded by a right-running fast rarefaction.

Figure 8 shows the streamwise velocity component,  $u$  in the idealized MHD shock tube. Again, the left-running, rarefaction wave rapidly accelerates the flow. The compound wave first decelerates the flow through the shock and immediately accelerates it in the trailing portion of the wave. The maximal stream-wise velocity achieved through the compression-expansion process is lower than in the gasdynamic shock tube calculation. In the highly confined region of large velocity gradient, molecular dissipation processes would not be negligible, as the ideal MHD formulation assumes. Therefore, Myong and Roe's consideration that admissible shocks must have a viscous profile<sup>18</sup> may derive a strong substantiation from solutions of the magneto-aerodynamic equations.

Like the gasdynamic shock tube, the  $u$  velocity component remains a constant value across the contact surface. However, the slow shock deceleration, is greater than its gasdynamic counterpart to recover the added expansion by the right-running, fast rarefaction precursor.

Figure 9 displays the normal velocity component  $v$  in the idealized MHD shock tube. Another unique feature of the plasma shock system is revealed by the additional vorticity generation mechanism in the electromagnetic field. In the region bounded by the right-running, rarefaction precursor and left-





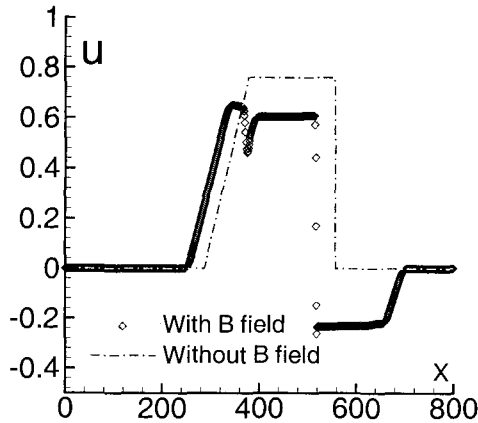
**Figure 7** Pressure Distribution in MHD Shock Tube

running fast expansion waves, the plasma wave system induces a large and negative normal velocity component  $v$ . This feature markedly differs from the corresponding acoustic wave system, which leaves the zero normal velocity component unperturbed,  $v = 0$ . For the acoustic field, it is well-known that the straight shock wave will not produce any vorticity. However, this result shows that the two slow plasma shock waves generate strong counter-rotating vortices with vorticity  $\bar{k}dv/dx$  in the direction normal to the planar shock.

The vorticity generation of a straight plasma shock and wave structure not only provides a clear evidence that the electromagnetic field can alter the entropy production mechanism, but it also points to a new mechanism for flowfield manipulation. The next section will address the role that this mechanism plays in a hypersonic blunt-body flow field.

## 14.6 Hypersonic MHD Blunt Body Simulation

The current interest in applying an electromagnetic field to modify aerodynamic performance focuses on hypersonic flows. Blunt-body computations in acoustic and plasma fields shed light on the basic features of this magneto-aerodynamic flow field. The present analysis computes hypersonic flows past a two-dimensional, cylindrical-nosed body ( $R_n = 0.1 m$ ) in the acoustic and idealized plasma fields at a free-stream Mach number of 5.85. The finite-volume numerical method uses a  $60 \times 50$  elliptically-smoothed mesh system and the MacCormack flux-vector splitting scheme.<sup>20,34</sup> The numerical method is first-



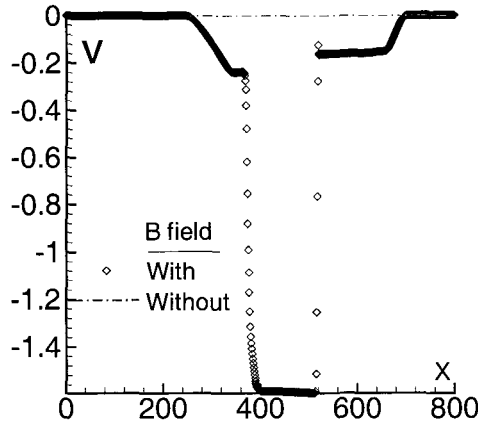
**Figure 8** Streamwise Velocity Component in MHD Shock Tube

order accurate in both space and time. At the free-stream boundary, the flow properties remain constant during each calculation. The body surface behaves as an impenetrable boundary through an inviscid boundary condition. The additional boundary conditions for the MHD computation include constant normal magnetic flux density on the body surface, which establishes a continuous normal component of the magnetic field at the media interface,  $\vec{n} \cdot \delta \vec{B} = 0$ , and an imposed constant value at the far field.

To illustrate the effects of the magnetic field on the flow, Fig. 10 shows a side-by-side comparison of the calculated density contours for two blunt-body flow fields. For the case on the left, no magnetic field exists in the free-stream. For the case on the right, a uniform  $B_\infty = 5\vec{j}$  field exists at the free-stream boundary. The same simulation code calculated the results in both cases. For each flow, a convergence criterion of a five order-of-magnitude drop in solution residual determined when to stop the calculations. The  $L_2$ -norm of the change in solution vector served as a conservative measure of the solution residual after each time step.

The density contours in Fig. 10 show the presence of the detached bow shock wave in both cases. The results reveal that the magnetic field effectively pushes the bow shock wave further upstream of the blunt body. This result is consistent with the finding by Augustinus *et al.*<sup>22</sup> Because the present analysis only intends to illustrate that the idealized electromagnetic field can alter the stand-off distance of the detached shock envelope, it seeks no further detailed description of the effects of various imposed magnetic fields on the flow.

In spite of the difference in the stand-off distance of the bow shock wave

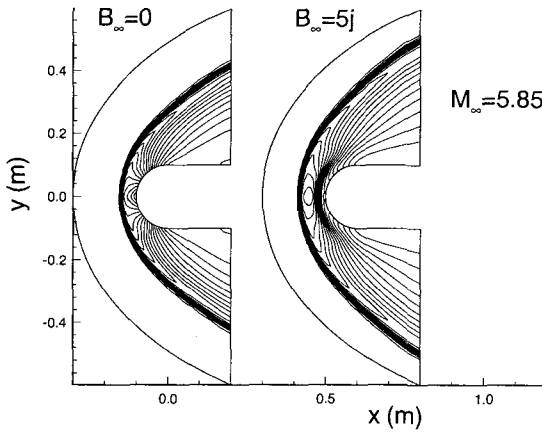


**Figure 9** Normal Velocity Component in MHD Shock Tube

in the two cases, the two wave fronts show some global affinity over the entire flow field. Each flow contains an essentially normal shock wave at the centerline followed by a rapid expansion around the shoulder of the body. The stand-off distance of the gasdynamic calculation agrees well with the correlated experimental results for the acoustic field,  $\Delta/R_n = 0.471$  and  $0.442$ , respectively.<sup>37</sup> From the cases studied, the shock envelope of the plasma field exhibits an outward displacement to the gasdynamic shock by a factor of 1.233 in stand-off distance. In turn, the radius of curvature of the bow shock increases by a factor of 1.103 to that of the acoustic field.

Figure 11 depicts the density distributions along the line of symmetry in each case. The numerical result for the acoustic field ( $B_\infty = 0$ ) shows that the density jumps across the shock wave according to the Rankine-Hugoniot condition and subsequently rises monotonically as the flow further compresses to the stagnation point of the cylindrical nose. In contrast, the post-shock density distribution along the line of symmetry in the plasma field ( $B_\infty = 5\vec{j}$ ) follows a nonmonotonic variation. The lower values of post-shock density in this case provide a clear reason for the increased shock wave stand-off distance. Continuity considerations demand a larger streamtube area to pass the same mass flow through the shock layer.

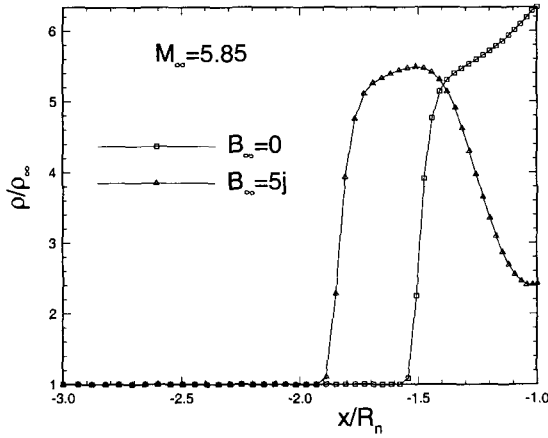
A comparison of Figs. 10 and 11 reveals that the post-shock reduction in density in the plasma field occurs rapidly near the body surface. Across this structure, pressure and velocity also decrease, whereas temperature gradually increases. The source of this structure is uncertain, as it coincides with regions where  $\nabla \cdot B \neq 0$ . Violation of the Gauss law for magnetic field occurs due



**Figure 10** Comparison of Bow Shock Envelopes of a Cylindrical Blunt Body at Mach 5.85

to numerical errors in the fluxes in this region. To alleviate these errors, the calculation employs the relaxation technique described in Ref. 20. Although this algorithm reduces errors in  $\nabla \cdot B$ , it does not completely eliminate them in the shock wave nor in the stagnation region. Future work will continue to resolve this numerical uncertainty.

Figure 12 presents the hydrostatic pressure distributions on the 2-D body surface for both the plasma and acoustic fields. The variable  $s$  measures the distance along the body surface from the stagnation point. Both numerical results display similar behavior in the form of a rapid expansion from the stagnation point toward the downstream afterbody. The present computations also indicate that the surface pressure of the acoustic field is uniformly higher than that of its plasma counterpart over the entire computational domain. However, the large difference between the numerical results does not entirely represent the possible wave drag reduction in the plasma field. For the flow field that contains finite electromagnetic field strength, the Lorentz force must enter into the momentum balance consideration for drag evaluation. In the ideal MHD formulation, the Lorentz force splits into two terms commonly referred to as the Maxwell stress,  $BB$ , and the magnetic pressure,  $B \cdot B/2\mu$ . More importantly, the formulation also neglects the shear stress tensor. Accurate and physically meaningful drag calculations therefore need to employ the complete magneto-aerodynamic equations.



**Figure 11** Variation of Density Along Stagnation Streamline for a Cylindrical Blunt Body at Mach 5.85

## 14.7 Concluding Remarks

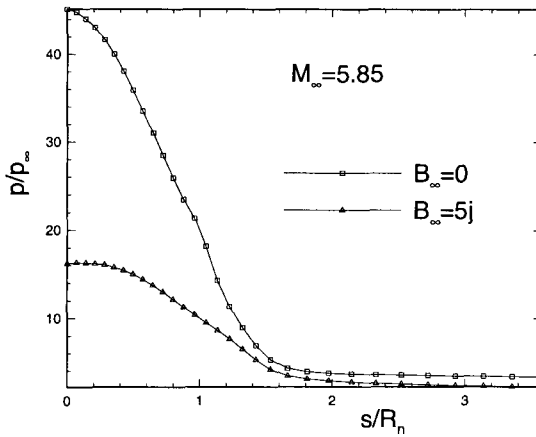
Computational magneto-aerodynamics is recognized as a new frontier for interdisciplinary technology development. A key element of this technical requirement is integrating computational electromagnetics in the time-domain with computational fluid dynamics and computational chemical kinetics. The impact of this interdisciplinary endeavor to high-speed flight may be revolutionary.

The multiple wave speeds that appear as a result of the magnetic field allow for much more complex behavior of an electrically conducting fluid system as compared to non-conducting fluid flows.

The idealized one-dimensional normal shock wave calculation demonstrates a theoretical potential for blunt-body wave drag reduction by showing that the entropy jump across the shock reduces in the presence of a magnetic field. The present results also point out that the transverse electromagnetic field can generate vorticity immediately downstream a straight normal shock.

In addition, the present work has identified the mechanism for changes in bow shock stand-off distance that the magneto-aerodynamic equations predict for blunt-body flows. Specifically, a nonmonotonic variation in density behind the shock leads to larger streamtube area requirements.

Finally, this research found that the flux-vector splitting method introduces large errors in  $\nabla \cdot B$ . Issues that require resolution in the future include understanding the effects that errors in  $\nabla \cdot B$  have on hypersonic blunt-body



**Figure 12** Comparison of Surface Pressure Distributions of a Cylindrical Blunt Body at Mach 5.85

calculations as well as extension of the numerical method to three spatial dimensions and higher-order accuracy.

## 14.8 Acknowledgments

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# Development of 3D DRAGON Grid Method for Complex Geometry

Meng-Sing Liou<sup>1</sup>

Yao Zheng<sup>2</sup>

## 15.1 Introduction

An effective CFD system for routine calculations of engineering problems, usually involving complex geometry, must possess certain features such as: (1) fast turnaround and (2) accurate and reliable solution. The first point, encompassing both the human and machine efforts, entails a short setup time for calculation, minimal memory requirement, and efficient and robust solution algorithm. The second point requires a judicious choice of discretization procedure. Choice of grid methodologies will greatly influence whether the above two criteria are met satisfactorily. In fact, the bulk of human effort, essentially involved in the grid generation, has been seen no significant reduction over the years, while on the contrary the machine effort has been dramatically reduced due to the incredible progress in microchips technology.

A propulsion system is an example of complex geometry, involving many separate geometrical entities with odd shapes and sharp turns. This topology creates challenges to grid generation, especially for viscous flow calculations. For a typical three-dimensional flow calculation, the time spent in generating a grid is about two thirds of the total simulation effort, representing a serious

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bottleneck in the entire analysis cycle [28]. Hence, grid generation continues to be the pacing technology for a practical CFD analysis and is the area where significant payoff can be realized. Furthermore, high quality grids for encompassing viscous regions are essential to yielding an accurate and efficient solution.

To deal with situations in which complex geometry imposes great constraints and difficulties in generating grids, currently composite structured grid schemes and unstructured grid schemes are the two mainstream approaches for solving various CFD problems.

The Chimera grid scheme [26] and similar scheme [1, 6], using overset grids to resolve complex geometries or flow features, are generally classified into the composite structured grid category. Overset grids allow structured grids to be used with good quality, such as orthogonality and smoothness, and with ease to control grid spacing. While being used regularly for flows about complex configuration [4], it has also been used to analyze flows over objects in relative motion [9]. Furthermore, overset grids can be employed as a solution adaptation procedure [18, 2, 22]. However, the nonconservative interpolations to update variables in the overlapped region, without strict satisfaction of the governing equations, can give rise to spurious solution, especially through regions of sharp gradients.

The unstructured grid method is also very flexible to generate grids around complex geometries. In particular, the solution adaptivity is perhaps its greatest strength. However, the unstructured grid method has been shown to be memory and computation intensive [13]. Also, choices of efficient flow solvers are limited, thus further affecting computation efficiency of the method. In practice, it is less amenable than the structured grid to implement a scheme higher than second order accurate in space. In addition, it has been our experiences that generation of 3D viscous grids, for domains such as that around a sharp concave corner, is not as easy as it seems and robustness is the issue.

Clearly, each method has its own strengths and weaknesses. Hence, we have attempted to develop a method that properly combines both the structured and unstructured grids and maximizes their own strengths. In fact some hybrid schemes have already appeared [23, 16]. Today, most of hybrid grid approaches have come from the unstructured grid community, where it is recognized that a structured-like grid (prismatic grid) should be embedded underneath an otherwise unstructured grid in order to better resolve the viscous region [16]. This is done only to address the accuracy issue, but in fact, it presents difficulties for generating this type of grids near a concave corner, thus is short of robustness. Here, a majority of the region is filled with unstructured grids and the grid data remains unstructured—memory issue is still present.

On the contrary, our approach will yield structured grids in the major

portion of the domain and only small regions filled with unstructured grids. Hence, the DRAGON grid [21, 17] is created by means of a Direct Replacement of Arbitrary Grid Overlapping by Nonstructured grid. While the DRAGON grid adapts the thinking of the Chimera grid, it has three important advantages: (1) preserving strengths of the Chimera grid, (2) eliminating difficulties encountered in the Chimera scheme, and (3) enabling grid communication in a fully conservative and consistent manner insofar as the governing equations are concerned. Furthermore, the DRAGON scheme is aimed at achieving the following goals: (1) efficiency, (2) high quality grid, and (3) robustness (generality) for complex multi-components geometry.

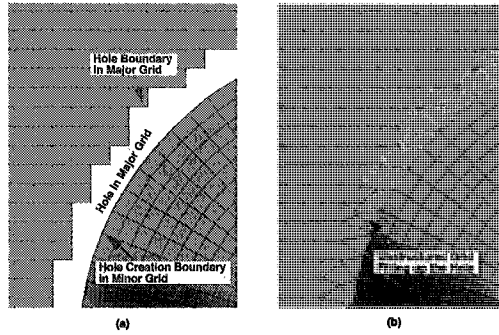
## 15.2 DRAGON Grid

We conclude that (1) the property of maintaining grid flexibility and the quality of the Chimera method are definitely to be preserved, and (2) focusing on improving (choosing) interpolation schemes perhaps only leads to more complication and it does not seem to be a fruitful way to follow.

An alternative method which avoids interpolation altogether and strictly enforces flux conservation is to **solve the region in question on the same basis as the rest of the domain.**

Since the overlapped region is necessarily irregular in shape, the unstructured grid method is most suitable for gridding up this region. Furthermore, this region is in general away from the body where the viscous effect is less important than the inviscid effect, and coarse grid would suffice as far as solution accuracy is concerned. This situation would be amenable to using the unstructured grid, thus minimizing its penalty associated with memory requirements. The combination of both types of grids results in a hybrid grid. Major differences of our approach from other hybrid methods are: (1) We retain the attractive features of the Chimera grid method; (2) We use unstructured grids only in limited regions. In other words, majority of the region is structured hexahedral grids in the DRAGON grid and the unstructured grids cover region where viscous effects are often less important, thus minimizing disadvantages of unstructured grids. To preserve the efficiency of the solution schemes, we opt to integrate two separate solvers – one structured and another unstructured. Although it requires extra efforts in the beginning during the development phase of the solver, the effort is made once for all. Moreover, obtaining two such codes nowadays is no longer a hurdle since many are available and they have been individually validated and used in a production mode. Hence, proper integration of codes essentially is the task to be done.

In what follows we will separately describe the steps taken to generate structured and unstructured grids in the DRAGON grid method. For



**Figure 1** Direct Replacement of Arbitrary Grid Overlapping by Non-structured grid: (a) Chimera grid, (b) DRAGON grid.

essentials of the DRAGON grid technique, we will consider a two-dimensional topology only, but without loss of generality.

### 15.2.1 Structured Grid Region

1. As in the Chimera grid, the entire computational domain is divided up into subdomains. We often designate a major (or background) grid enclosing the complete computational domain and the component grids as minor grids.
2. Create hole regions, referring to Figure 1(a), if overlapped, otherwise the void region is the hole. For example, the outer boundary of a minor grid may be used as the hole creation boundary.
3. The hole boundary points are now forming the new boundaries for the unstructured grid region.

Since the grids need not be overlapped under the DRAGON grid framework, it results in a more flexible procedure than the Chimera grid method.

### 15.2.2 Non-Structured Grid Region

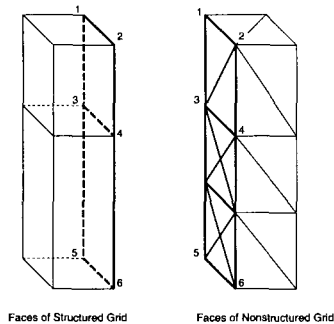
The gap region is inevitably of irregular shape to which triangular cells are especially suitable to adapt. Recall that one important feature in the DRAGON grid is to eliminate any cumbersome interpolation which causes nonconservation of fluxes. Unstructured grids alone are not sufficient to do the task. An additional constraint is imposed to require that the boundary nodes of the structured grid coincide with vertices of boundary triangular cells. In other words, there might be more than one triangle connected to one structured grid cell, but not vice versa. Fortunately, this constraint fits well in unstructured grid generation. The Delaunay triangulation scheme [3, 29, 30]

is applied to generate an unstructured grid in the present work. Figure 1(b) depicts the DRAGON grid with the unstructured grid filling up the hole region. In what follows we give the steps adopting the unstructured cells if the framework of the Chimera grid scheme is used.

1. Boundary nodes provided by the PEGSUS code [27] are reordered according to their geometric coordinates.
2. Delaunay triangulation method is then performed to connect these boundary nodes.
3. Besides from the standard connectivity matrices containing the cell-based as well as edge-based information, we need to introduce additional matrices to describe the connection between the structured and unstructured grids.

### 15.3 Three-Dimensional DRAGON Grid Generation

The extension of the **concept** of DRAGON grid described above to that in three-dimensional space is straightforward. However, more difficulties are anticipated in actual **implementation**. Unique challenges for the case of three dimensions exist in various aspects, such as unstructured grid generation algorithms and code implementation. The development of the three-dimensional DRAGON gridding methodology is emphasized in this work.

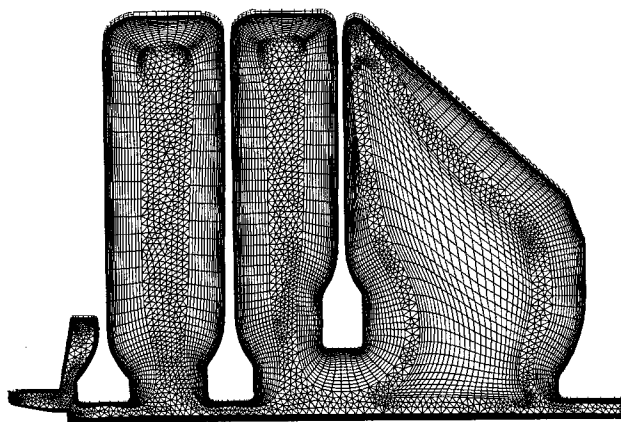


**Figure 2** Interface between structured and nonstructured grids in a three-dimensional DRAGON grid.

With reference to Figure 1, while the boundary edges of the structured grid coincide with edges of boundary triangular cells in two dimensions, the faces of the structured grid in general do not necessarily match the faces of the unstructured grid in the same location for three-dimensional cases as illustrated in Figures 2. This restriction may arise from the grid generation

stage or the requirement to adapt to the physical phenomena. In Figure 2, Points 1, 2, 3, 4, 5 and 6 on the structured grid are made coincident with the corresponding ones on the unstructured grid. Therefore, a scheme can be easily developed to meet flux conservation at common faces without resorting to interpolation of fluxes.

As an example, Figure 3 depicts a DRAGON grid for the compressor drum cavity problem. This grid is created from a base structured grid, where there are both overlapped domains and voids. The structured grid is body-fitted and of high quality for viscous flow simulation. The unstructured grid can be easily generated to fill the overlapped domains and voids. Figure 3 provide a sideview of the DRAGON grid, where the outer rim barely reveals the third dimension.



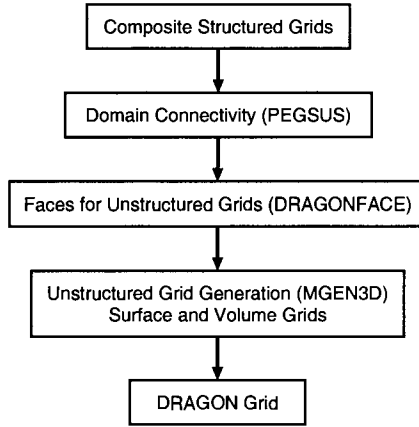
**Figure 3** DRAGON grid for the compressor drum cavity problem (The outer rim barely reveals the third dimension).

### 15.3.1 Programming Aspects

Apart from the PEGSUS code [27], two main modules, called DRAGONFACE and MGEN3D, are created in the three-dimensional DRAGON grid generation, also the flowchart depicted in Figure 4 shows the relationships among the modules involved.

DRAGONFACE, a FORTRAN code, reads in composite structured grids with IBLANK values assigned by the PEGSUS code and writes out topology information of the faces as a surface description for the unstructured grid. Primary steps involved in the DRAGONFACE procedure include: identification of new faces, creation of isolated edges of the surface, and filling





**Figure 4** Program modules involved in the three-dimensional DRAGON grid generation.

openings of the surface to ensure closedness of the surface.

MGEN3D is a C code. It inputs the surface description from the DRAGONFACE procedure and generates volumetric unstructured grids and provides topology information of the corresponding surface grids, giving data communication at the interface between structured and unstructured grids. Main steps involved in the MGEN3D procedure include: spacing source creation, three-dimensional surface triangulation, face reorientation, domain classification, and generation of volumetric grids.

Currently there are two basic types of tetrahedral grid generation schemes: advancing front approach [25] and Delaunay triangulation [19]. Also a coupled approach [10] has been proposed to take the advantages of both schemes. After an investigation into these schemes, Delaunay triangulation has been chosen, partly taking into account the time frame required in the development. The other two approaches, having their own advantages, will be considered in the future.

The algorithm of the MGEN3D code evolves from the framework of previous work [29, 30, 19], and is a Delaunay-type method. The Steiner point [24] creation algorithm has been incorporated into this grid generator.

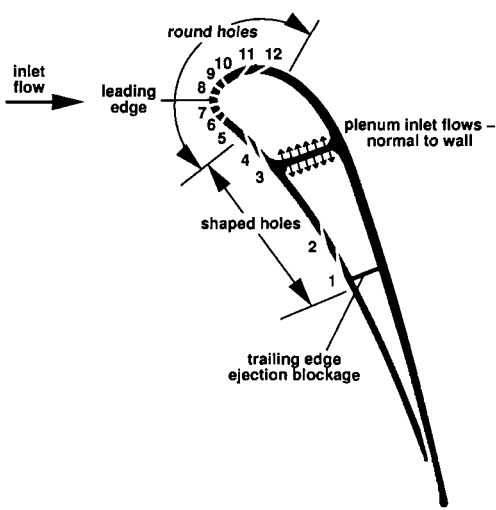
Algorithm involved in the MGEN3D procedure are summarized as follows.

1. Surface description generated via the DRAGONFACE procedure is read in.
2. Surface triangulation is carried out taking into account the baseline grid spacing, point and line source distribution.
3. Domain classification is performed to provide domain information of the

geometry specified by the above surface definition.

- 4. Volumetric grids are generated for the domains. Data of the assembly of individual volumetric grids, including nodal coordinates and connectivities, are finally created.

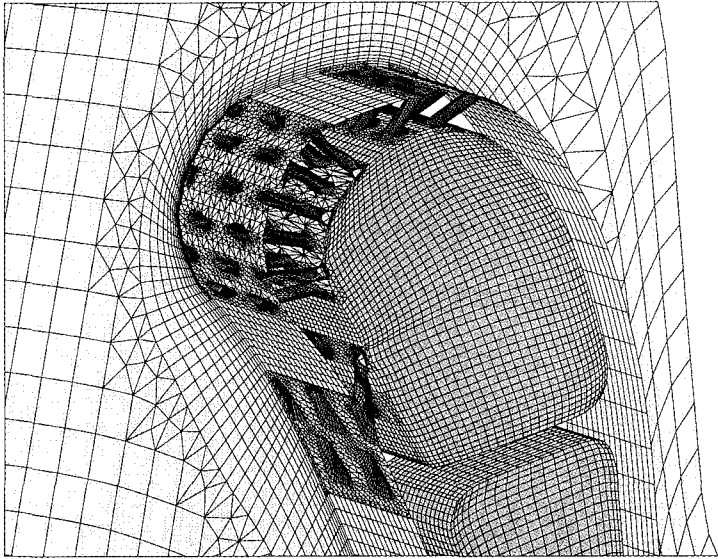
To demonstrate the capability of the DRAGON grid code, we consider a typical film-cooled turbine vane, whose schematic is shown in Figure 5. This geometry includes the vane, coolant plenum, and 33 holes inside of the vane. Figure 6 depicts the resulted DRAGON grid, where the connecting regions between the 33 holes and the flow domain are filled with unstructured grids. It gives a deep insight into this DRAGON grid by means of cutting through the grid, where attention has been paid to the leading edge region.



**Figure 5** Schematic of the vane cross-section and coolant holes.

During the DRAGON grid generation, the 33 individual structured grids for the holes have been created without trying to topologically join them to the grids representing the external flow domain, which would be required in the multiblock grid generation [15]. From this point of view, the DRAGON grid scheme could be considered as a more flexible and easier approach.

However, the user may prefer to have structured grids present in a specific region, due to physical and numerical characteristics. For instance in this example, the user might like to use structured grids to fill the connecting regions between the holes and the flow domain.

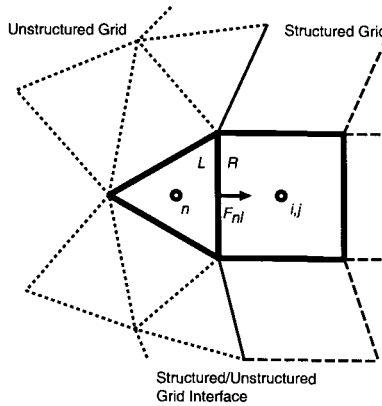


**Figure 6** Cut-through view of the DRAGON grid in the leading edge region of the film-cooled turbine vane.

### 15.3.2 Data Communication through Grid Interfaces

For the DRAGON grid, conservation laws are solved on the same basis on both the structured and unstructured grids. The cell center scheme is used for storage of variables as well as flux evaluations, in which the quadrilateral and triangular cells are used respectively. Figure 7 shows an interface connecting both structured and unstructured grids. As described earlier, the numerical fluxes, evaluated at the cell interface, are based on the conditions of neighboring cells (denoted as L and R cells, respectively). For the unstructured grid, the interface flux  $F_{nl}$ , will be evaluated using the structured-cell value as the right (R) state and the unstructured-cell value as the left (L) state. Consequently, the interface fluxes, which have been evaluated for the unstructured grid, can now be directly applied in computing the cell volume residuals for the structured grid.

Thus, the data communication through grid interfaces in the DRAGON grid guarantees satisfying of the conservation laws and the solution is obtained on the same basis for both structured and unstructured grid regions.



**Figure 7** Fluxes at the cell face connecting the structured and unstructured grids.

## 15.4 Flow Solver

The time-dependent compressible Navier-Stokes equations given by Equation (15.1), in an integral form over an arbitrary control volume  $\Omega$ , are solved:

$$\int_{\Omega} \frac{\partial \mathbf{U}}{\partial t} dv + \int_{\partial \Omega} \vec{\mathbf{F}} \bullet d\vec{\mathbf{S}} = 0, \quad (15.1)$$

where the conservative-variable vector  $\mathbf{U} = (\rho, \rho u, \rho v, \rho w, \rho e_t)^T$ , with the specific total energy is  $e_t = e + |\vec{V}|^2/2 = h_t - p/\rho$ . The flux vector  $\vec{\mathbf{F}}$  includes both the inviscid and viscous fluxes, in which the turbulence variables are also included.

Based on the cell-centered finite volume method, the semi-discretized form, describing the time rate of change of  $\mathbf{U}$  in  $\Omega$  via balance of fluxes through all enclosing faces,  $\vec{S}_l, l = 1, \dots, LX$ , no matter whether they are in the structured or unstructured grid regions, can be cast as

$$\int_{\Omega} \frac{\partial \mathbf{U}}{\partial t} dv + \sum_{l=1}^{LX} \mathbf{F}_{nl} |\vec{S}_l| = 0, \quad (15.2)$$

where  $\mathbf{F}_{nl} = \vec{\mathbf{F}}_l \bullet \vec{n}_l$  and  $\vec{n}_l$  is the unit normal vector of  $\vec{S}_l$ .

The flow code, termed DRAGONFLOW, is made up of two well-validated NASA codes, OVERFLOW [5] and USM3D [11, 12], which are respectively structured- and unstructured-grid codes. A significant effort has been made to add an interface and modify both codes, more so on the OVERFLOW code.

A spatial-factored implicit algorithm is applied in the OVERFLOW code, while a point implicit one in the USM3D code. Since we are only interested in seeking steady state solutions, the differences of convergence rate due to the incompatibility between the structured and unstructured grid regions can be tolerated. However, if unsteady state solutions are of interest, this difficulty can be overcome by a dual time integration approach in which the solution within each physical time step will be iterated to achieve convergence. This subject is left for a future research.

As for the spatial discretization, there is inconsistency in the order of accuracy in that a third-order and a second-order accurate schemes are used respectively in the structured grid-based and unstructured grid-based codes. But a similar limiter function is applied. We use the new flux scheme AUSM<sup>+</sup>, which is described in detail in [20], to express the inviscid flux at the cell faces in both codes. Also, the viscous fluxes are approximated, as usual, by a centered scheme.

## 15.5 Test Cases

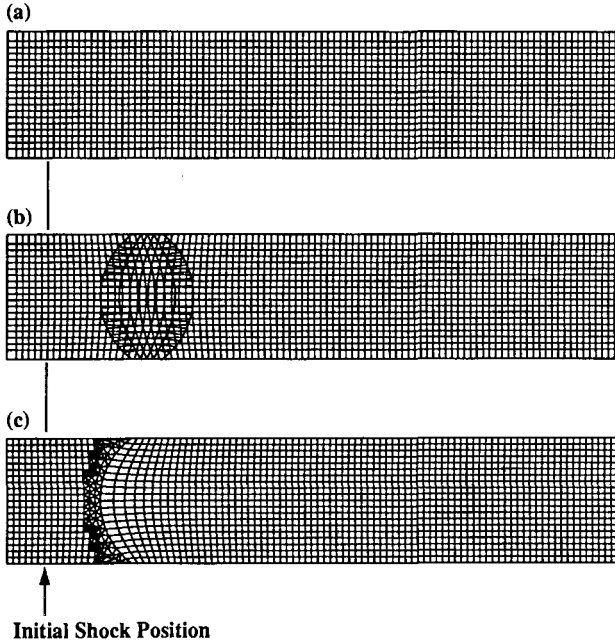
### 15.5.1 Case 1: Shock Tube Problem

First, a 2D shock tube problem was considered. To focus only on the issue of grids, the flow was solved by the basic first-order accurate discretization in both time and space. This case serves to show the effect of interpolation in the Chimera grid and the validity of the DRAGON grid method for a transient problem as a plane shock moves across the embedded-grid region. The shock wave is moving into a quiescent region in a constant-area channel with a designed shock speed  $M_s = 4$ . Solutions were obtained using three grid systems, namely (1) single grid, (2) Chimera grid, and (3) DRAGON grid, as displayed in Figure 8. The single grid solution is used for benchmark comparison.

The pressure distributions along the centerline of the channel, as plotted in Figure 9, shows that the Chimera scheme predicts a faster moving shock in the tube, while the present DRAGON grid and the single grid results coincide, indicating that the shock is accurately captured and conservation property well preserved when going through the region of the embedded DRAGON grid.

### 15.5.2 Case 2: Supersonic Flow in a Converging Duct

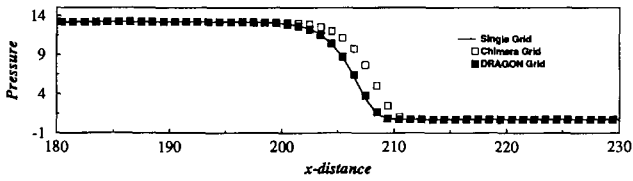
Next we consider a supersonic flow of Mach 3 through a convergent channel. The geometry is sketched in Figure 10, where both the top and front side walls are bent by a wedge angle of 10 degrees, thus creating two wedge



**Figure 8** Grids for moving shock problem ( $M_s = 4$ ): (a) single grid, (b) Chimera grid, and (c) DRAGON grid.

shocks of equal strength and subsequent interactions between them. A strip of unstructured-gridded region, denoted by the shaded region, is placed in the mid-section of the channel. This test case is designed to provide a check on the three-dimensional code against conservation property of the DRAGON grid through a shock. Figure 11 displays the perspective view of the density contours on the top and front side faces, and the exit plane. The two wedge shocks intersect and generate, near the corner edge AD (see Figure 10), a corner shock region manifested by the slip lines emanating from the triple point. As seen in Figure 12, this region progressively becomes larger as the wedge shocks sweep towards the opposite walls. Eventually these two wedge shocks reflect and interact with the flow previously generated by the corner shocks, making the flowfield even more complicated.

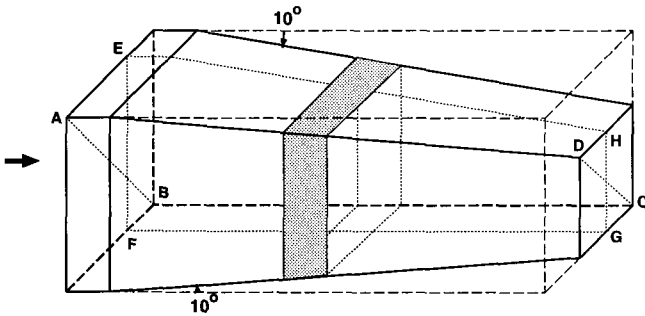
Figure 13 gives an inside view of Mach number contours on a midplane, EFGH (in Figure 10). This reveals quite a rich feature of shock-shock interactions inside the duct, while the shock configuration on the duct walls (Figure 11) is relatively simple. The slip surfaces issuing from the shock triple points are evident in Figure 13 and they subsequently interact with



**Figure 9** Comparison of centerline pressures from the three grids.

the downstream shock.

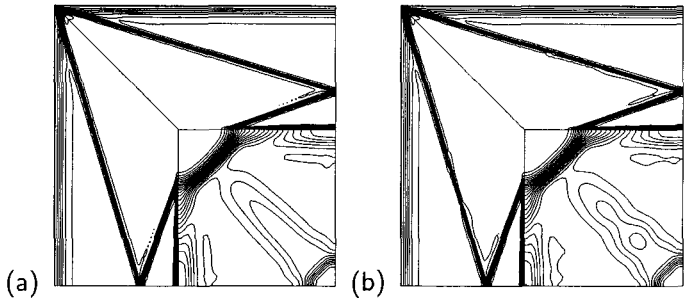
It is evident that the shock profiles pass through the DRAGON grid interfaces seamlessly without creating spurious waves, guaranteeing the conservation property. In each of the above figures, we also include the results based on a single structured grid for validation. We see that both sets of results are essentially the same, except some minute variations in the core region, which are a remnant result of the unstructured grid. This is clearly indicated in Figure 14 where the shock becomes thinner in the unstructured grid region because the grid size is reduced roughly by  $\frac{1}{6}$ . A quantitative measure of the difference between the single grid and DRAGON grid solutions at the exit plane is given in Figure 15, indicating a close agreement of the two solutions.



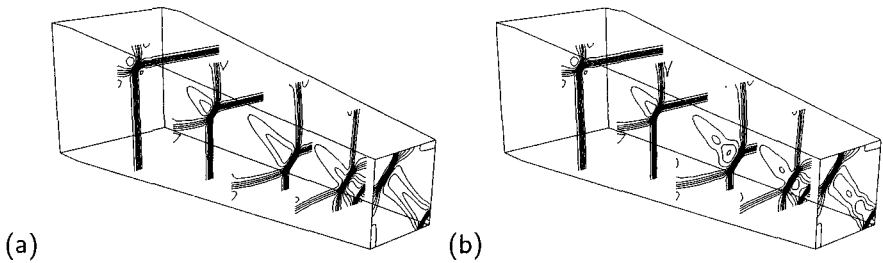
**Figure 10** Sketch of a converging duct where the unstructured grid is in the shaded region for the DRAGON grid.

### 15.5.3 Case 3: Viscous Flow through Annular Cascades

Lastly, we show the results of a flow simulation for an annular cascade of turbine stator vanes developed and tested at NASA Lewis [14]. Figure 16 shows the DRAGON grid, in which a background H-type grid is placed to cover the channel between the vanes, an O-type viscous grid is used to resolve



**Figure 11** Density distributions of the flow on the surface of the converging duct: (a) Single grid; (b) DRAGON grid.



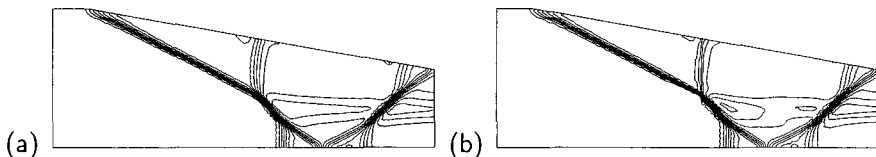
**Figure 12** Mach number distributions of the flow at various stations: (a) Single grid; (b) DRAGON grid.

the region around the vane, and an unstructured grid region between them. Previous calculations have been obtained using a single structured grid [8, 7]. The static pressure distributions of the present DRAGON grid solution are plotted in Figure 17 and they are seen to be in very good agreement with the measured data [14] at three spanwise locations, 13.3%, 50%, and 86.6% respectively.

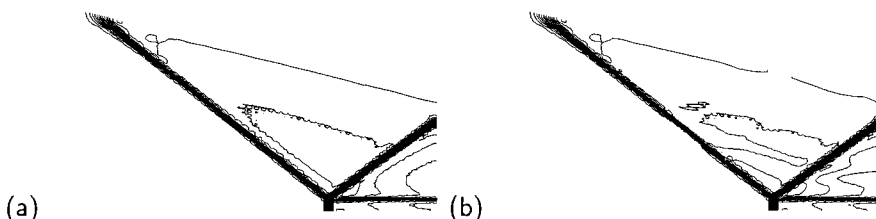
15.6 Concluding Remarks

In the present work, we have concentrated on the extension of the DRAGON grid method into three-dimensional space. Various aspects of the extension, and new challenges for the three-dimensional cases, have been investigated. This method attempts to preserve the advantageous features of both the structured and unstructured grids, while eliminating or minimizing their respective shortcomings. As a result, the method is very amenable to **quickly creating quality** viscous grids for various individual components





**Figure 13** Mach number distributions of the flow at a typical cutting plane EFGH: (a) Single grid; (b) DRAGON grid.



**Figure 14** Density distributions of the flow at the symmetric plane ABCD: (a) Single grid; (b) DRAGON grid.

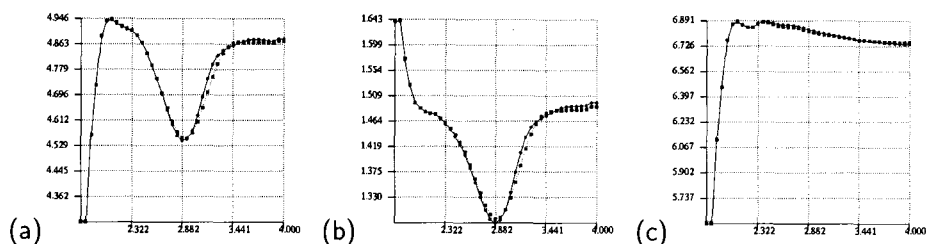
with **complex** shapes found in an engineering system. Computationally, the resulting grid drastically reduces the memory required in the unstructured grid and makes more efficient solvers accessible because the major portion of the DRAGON grid is in the form of the structured grid. Moreover, high quality viscous grids are inherited in the process, unlike in the unstructured grid generation, where structured-grid-like grids (such as prismatic layers) need to be added, but are still based on the unstructured-grid data structure.

Furthermore, the flow solutions confirm the satisfaction of conservation property through the interfaces of structured-unstructured regions, and the results are in very good agreement with the measured data available, thus demonstrating the reliability of the method.

Future plan includes further refinement on unstructured grid generation to improve smoothness and robustness, and further validations and applications to engineering problems with complex geometry.

## Acknowledgments

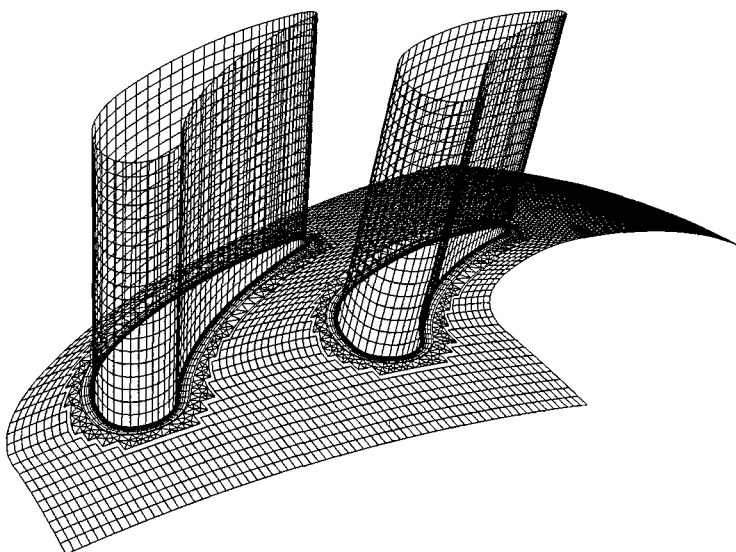
This work is supported under the Turbomachinery and Combustion Technology, managed by Robert Corrigan, NASA Glenn Research Center. We thank J. D. Heidmann and R. V. Chima for providing the vane geometries of the film-cooled turbine and the annular cascade, respectively.



**Figure 15** (a) Density, (b) Mach number and (c) pressure distributions of the flow on line HG. Boxes and circles denote results based on the single grid and the DRAGON grid, respectively.

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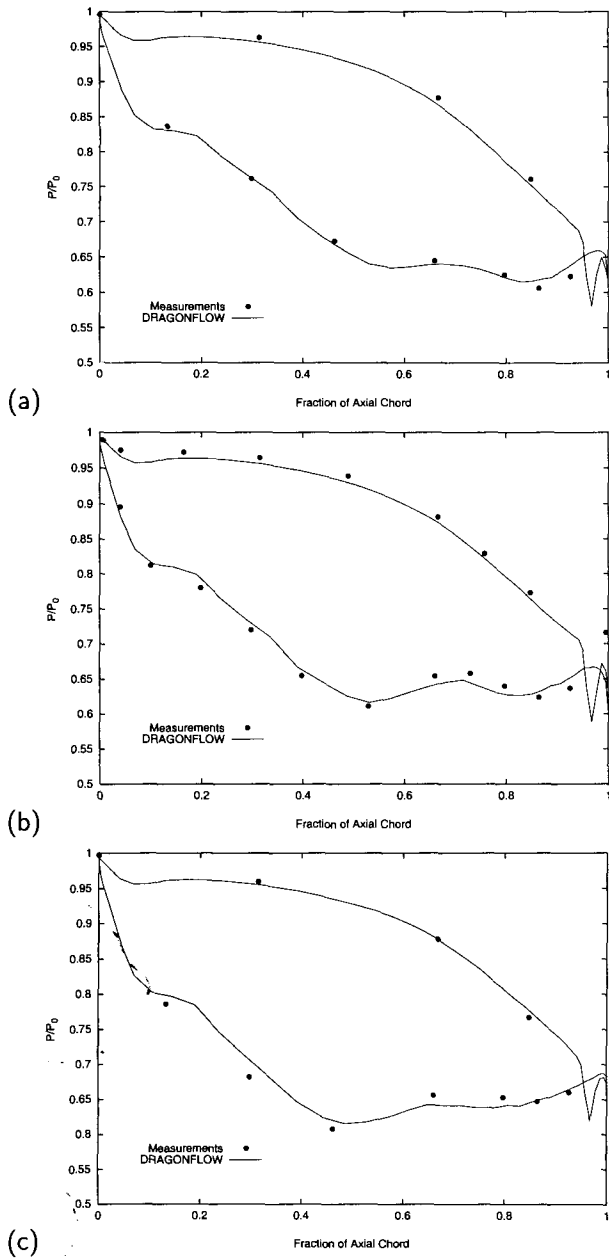
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**Figure 16** DRAGON grid of the annular turbine cascade.

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**Figure 17** Static pressure distributions at (a) 13.3%, (b) 50% and (c) 86.6% of span from hub

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# Application of Multi-Block, Patched Grid Topologies to Navier-Stokes Predictions of the Aerodynamics Of Army Shells

Walter B. Sturek, Sr.<sup>1</sup>  
and  
David J. Haroldsen<sup>2</sup>

## 16.1 Introduction

The Army Research Laboratory is interested in applying state of the art high performance computing tools to predict the aerodynamics of Army shell at angle of attack for a wide range of Mach number from high subsonic to high supersonic. In this paper, the WIND flow solver has been used to study the aerodynamics of two missile configurations. The GridPro grid generation software has been utilized to generate the computational grids. Several aspects of interest concerning the generation of grids for use with WIND and the process of obtaining solutions is discussed. Comparisons are shown for several turbulence models.

Researchers in computational fluid dynamics at the United States Army Research Laboratory are interested in investigating a wide array of complex fluid flow problems. These problems include flow around complex bodies, flow at moderate and high Mach number, and flow at moderate to high angles of attack. A recent study examined the predictive capability of several different Navier-Stokes flow solvers applied to the case of an ogive-cylinder configuration at transonic and supersonic flow velocities at 14-degrees angle of attack[1] . This study extends the previous work by examining the predictive capability of the WIND flow solver to predict flow

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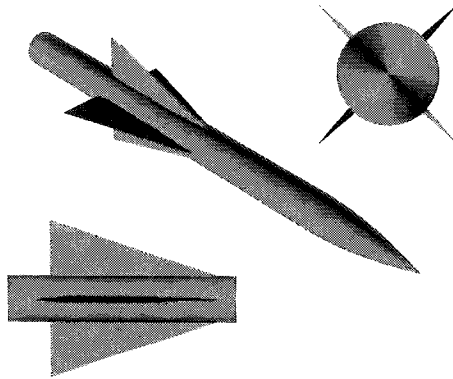
problems for missiles with fins at angle of attack and moderate Mach number.

The WIND package has numerous capabilities that make it potentially attractive for computational researchers. Among these features are the numerous turbulence models, ease of use, portability, parallel processing capability, and the ability to incorporate grids with a generalized topology. This particular feature makes WIND attractive for use with the GridPro grid generation package. GridPro produces structured, multi-block grids with non-overlapping block interfaces.

The focus of this effort is to investigate: 1) the application of WIND for complex flow problems; and 2) the use of multi-block, patched grid topologies. This study considers the application of WIND 1.0 to the study of two different missile configurations at angles of attack of 14 and 40 degrees and at Mach numbers near 2.5. The generation of multi-block, patched grids using the GridPro package is discussed and results for different turbulence models are presented.

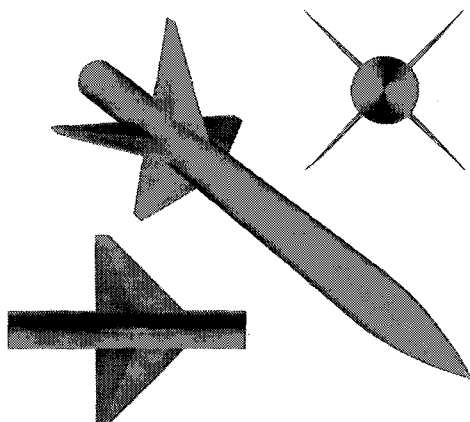
## 16.2 Missile Configurations

Two missile configurations were examined in this study. Both missiles consist of a 3-caliber ogive nose and a 10 caliber cylindrical body. Each missile has four fins, with symmetry about the pitch plane. The specific fin geometry and placement is shown in Figures 1 and 2.



**Figure 1 Missile 1 Configuration.**





**Figure 2 Missile 2 Configuration.**

Missile 1 [2] was studied at roll angles of 0 and 45 degrees, Mach 2.5, and angle of attack 14 degrees with a Reynolds number of  $1.12 \times 10^6$ . Missile 2 [3] was studied at a roll angle of 45 degrees, Mach 1.6 and 2.7, and angle of attack 40 degrees with Reynolds number of 250,000.

### **16.3 Grid Generation**

The grid generation for this investigation was done using the GridPro grid generation software. GridPro is a product of Program Development Corporation in White Plains, NY. This package is of interest because it incorporates a topology-based approach to the generation of grids. This approach emphasizes the underlying topology of the geometric shapes and of any flow features rather than focusing on the geometry of the problem. The package consists of a GUI for topology design, the grid generation software, and utilities for manipulating grids. GridPro produces multi-block, structured grids with the capability to output data in a variety of formats. An important consideration when using GridPro is that the adjacent zones abut but do not overlap. The user can also customize GridPro to output initial boundary data relevant to a particular flow solver.

The user designs the topology by constructing a coarse, unstructured, hexahedral mesh in the region of interest. The hexahedral elements become the individual blocks of the final multi-block grid. The user controls only the topological structure of the grid; the grid generator automatically calculates precise placement of grid lines. Because the gridding process is largely automatic, the user has a significant amount of flexibility in designing the fundamental topology of the grid. This flexibility includes being able to locally define the grid in regions of interest while leaving a coarser grid in regions with insignificant flow variation. After the topology design is complete, the user invokes the grid generation software. The resulting grid for a complex shape may result in hundreds of blocks.

The utilities included with the package allow a variety of operations to the final grid. Two of particular interest are the block merging utility and the clustering utility.

The block merging utility merges the (often) large number of blocks to a more manageable number. The user can control the number of grid nodes that are allowed in each block of the final configuration. Thus, the merging utility together with the initial topology design can be used as an *a priori* “domain decomposition” tool. The other utility of interest clusters grid lines to a particular surface. In practice, the grid generation package is always used to generate Euler grids and the clustering utility is used to obtain a viscous grid. This significantly reduces the time required to obtain a viscous grid.

For the missiles under consideration, the topology design required several days to construct a reasonable topology. The grids were generated using a SGI ONYX with R12000 CPUs. The grid generation generally required on the order of 6 to 8 hours of CPU time. The initial grids in each case had on the order of 375 blocks for one half of the flow volume (symmetry assumed). The number of blocks was reduced by merging to a more tractable number. The final grid configurations are listed in Table 1.

| Missile         | Blocks | Grid Points (in millions) |
|-----------------|--------|---------------------------|
| 1 (0 Deg roll)  | 35     | 4.1                       |
| 1 (45 Deg roll) | 42     | 4.3                       |
| 2               | 26     | 4.2                       |

Table 1 Grid Details.

The topology designed for the missiles did not have rotational symmetry; therefore different topologies and grids had to be generated for missile 1 at different roll angles. Examples of the topologies are shown in Figures 3, 4 and 5. Figure 3 illustrates a close up view of the nose tip topology. A close up view of the fin topology is shown in Figure 4. Figure 5 shows the whole missile topology. Samples of the grids generated are shown in Figures 6 and 7. In these figures, the darker lines indicate the block boundaries and are suggestive of the topology design that was used to create the grid. For the purposes of display, the figures show coarse versions of the final grid before a viscous boundary layer was added to the grid.

16.4 Boundary / Initial Conditions

In all cases, the freestream inflow condition was used on the inflow boundary, the reflection boundary condition was used on the symmetry plane, the freestream outflow condition was used on the outflow plane, and the viscous wall condition was used on the viscous surfaces of the missile body. For missile 1, the total freestream pressure was 20.628 psi and the total freestream temperature was 554.4 degrees Rankine. For missile 2, the static freestream pressure was 0.5637 psi and the static freestream temperature was 248.4 degrees Rankine. The WIND default initialization was used to initialize the flow field variables.

Runs were conducted using the following turbulence models: Baldwin-Lomax (BL), Baldwin-Barth (BB), Spalart-Allmaras (SA), and Shear Stress Transport (SST). The Baldwin-Lomax was run both with and without the option of choosing the maximum number of grid points to search for  $F_{\max}$ . For the former case, maximum grid points 10 and 30 were studied (BL10 and BL30 respectively).

To avoid transient instabilities, the FIXER keyword was used. For missile 1, an initial solution was calculated at a low angle of attack and this solution was used as an initial solution for calculating the solution at a higher angle of attack. For missile 2, the TVD factor was reduced to 1, the CFL crossflow factor was set to 1, and the CFL number was reduced to .4.

## 16.5 Performance / Convergence Criterion

Runs were conducted on Silicon Graphics Origin 2000 or Onyx platforms with multiple processors. Runs were typically conducted using 8 processors and converged solutions could be obtained in 8-12 hours. In each case, the residuals decreased by no more than 3 orders of magnitude over several thousand cycles. To test convergence, solutions were monitored until they were judged to be converged. In the case of missile 2, the loads on the body were calculated using the LOADS keyword in WIND and the solution was considered to be converged when the loads had converged and remained steady for a few hundred cycles.

The parallel performance obtained varied widely depending on the grid used. The grids used for missile 1 were reduced to the final number of blocks while trying to balance the number of nodes in each block. Speedup factors as high as 7.5 were obtained using 8 processors and as high as 14 were obtained for 16 processors. The performance for missile 2 was considerably worse – speedup factors for 8 processors was around 5 – because the block merging process was done so as to minimize the number of blocks, rather than to optimize for parallel performance. An example of the best parallel performance is shown in Figure 8.

## 16.6 Results

Quantities of interest for the study are the pressure coefficient at different stations on the body and fins as well as pitot pressure profiles of the outer flow field at several axial stations. The data presented show examples of the results for missile 1 that were obtained using WIND. The stations and data displayed in the figures were selected with the intention of eventually comparing the computational data with experimental data.

The pitot pressure prediction for missile 1 shown in Figure 9 for roll angle 0 and at axial station  $X/D = 11.5$  show similar results for the SA, BB, and SST models. The BL10 turbulence model predicts a smaller, more intense primary vortex. In addition, this model predicts a more structured solution near the body of the missile. The BL turbulence model predicts a solution that more closely resembles the predictions of

the one- and two-equation models. Comparison of surface pressure predictions on the missile body at various axial stations shows minimal variation between the different turbulence models with the exception of BL10. Sample comparisons are shown in Figures 10 and 11 for roll = 0 degrees.

A reasonable result could not be obtained for missile 2 at the desired angle of attack. Attempts were made to improve the stability of the problem by improving the grid quality and density, reducing the CFL number, and using smoothing options available with WIND. In every instance, the run aborted due to singularities in the flow field. It is most likely that the difficulty was due to the extreme 40-degree angle of attack. Runs using the same grid, the same flow conditions, but at 20 degrees angle of attack converged successfully.

Figures 12 and 13 show three-dimensional views of predicted pitot pressures. Also shown are lines showing the location of the vortex cores for missile 1 at both roll angles. These visualizations were obtained using the PV3 package developed by Dr. Robert Haines of MIT.

## 16.7 Concluding Remarks

The WIND flow solver has been demonstrated to be an efficient tool for increasing and extending the predictive capability of researchers in computational fluid dynamics. WIND has proven to be particularly useful for flow problems with complex geometry although extreme flow conditions caused some difficulties. The ability of WIND to accommodate multi-block, patched grids enabled the use of the GridPro grid generation software. This software package was found to generate high quality structured grids with modest effort and is well suited for use on Army missile configurations. The Spalart-Allmaras turbulence model was found to perform quite well in comparison with the other models used in this study. In particular, no benefit was seen to result from the application of higher order models.

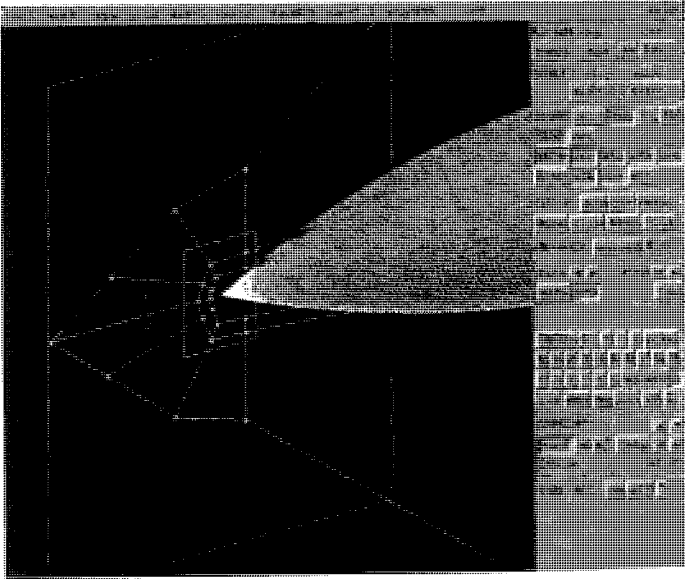
## Acknowledgements

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**Figure 3** Nose tip topology.

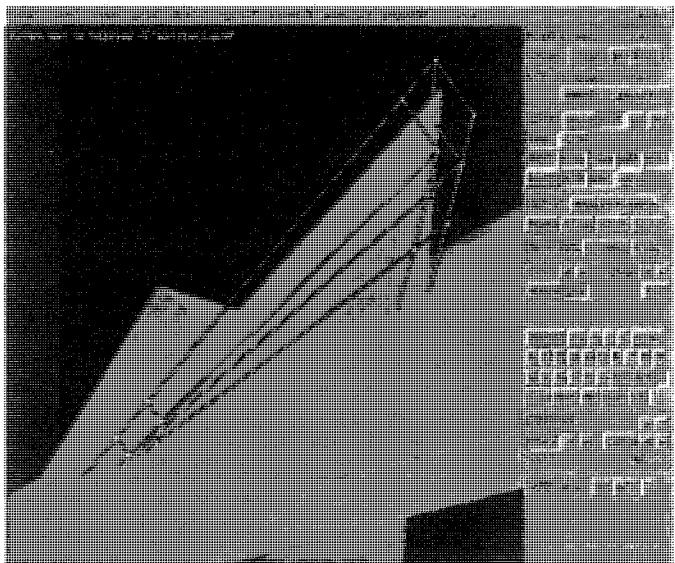


Figure 4 Fin topology.

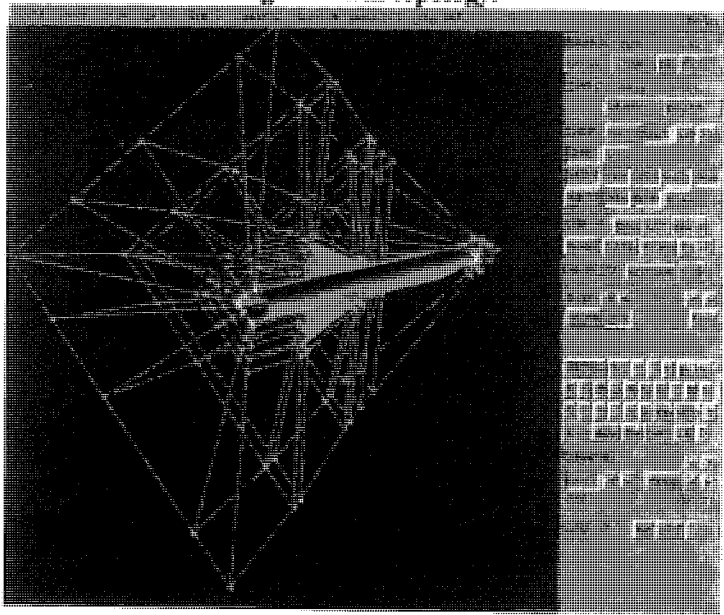
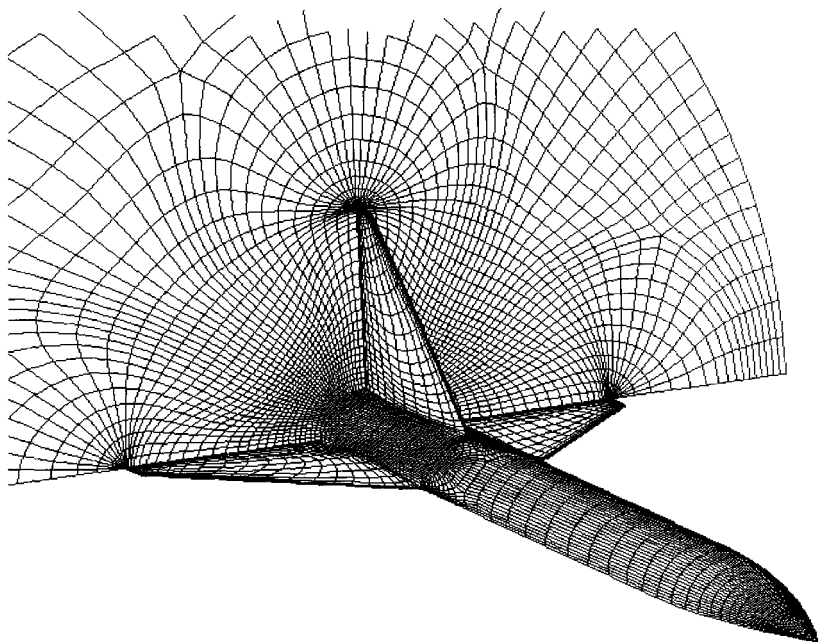
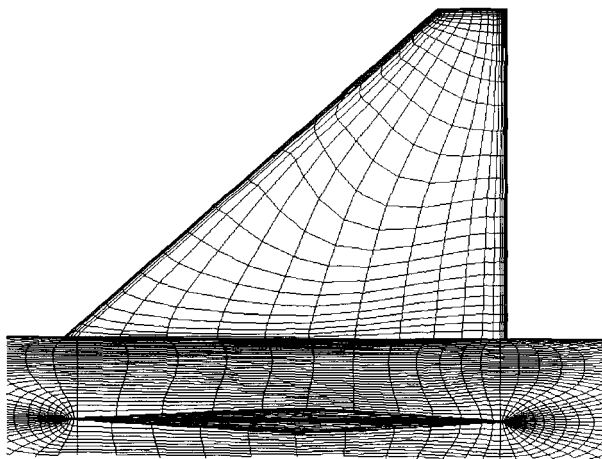


Figure 5 Missile topology.



**Figure 6** Example of a coarse Euler grid for missile 1.



**Figure 7** Fin Grid.

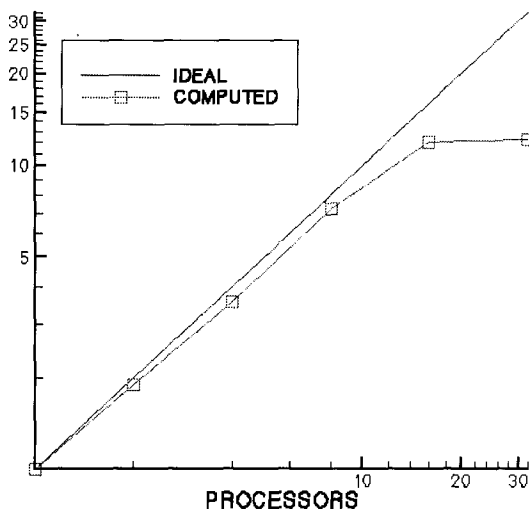


Figure 8 Measured speedup using grid for missile 1, roll angle 0.

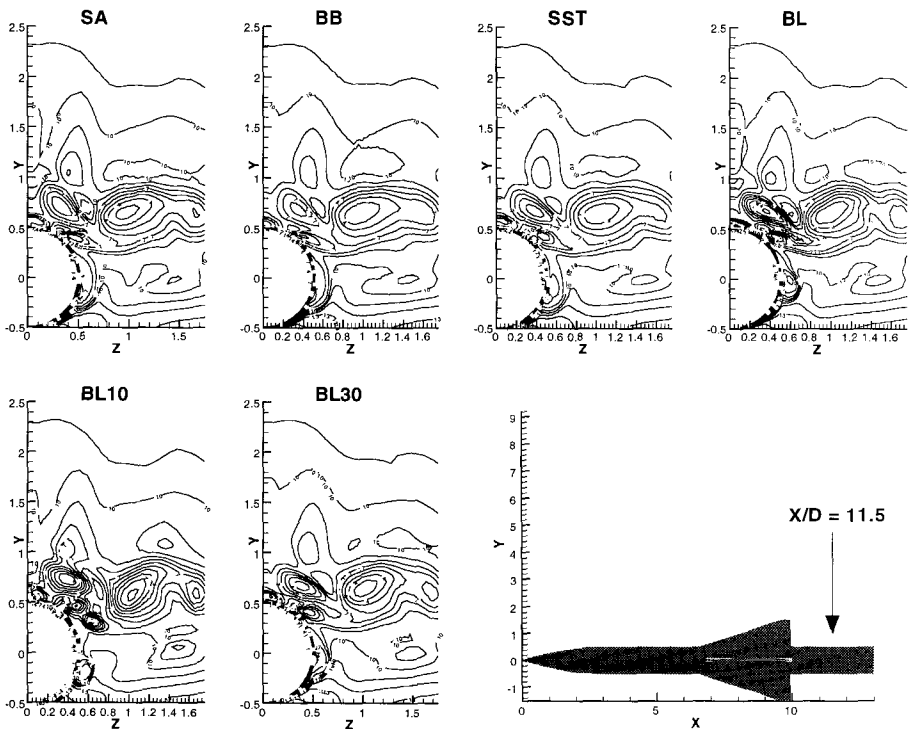


Figure 9 Comparison of pitot pressure predictions at  $X/D = 11.5$  on missile 1.



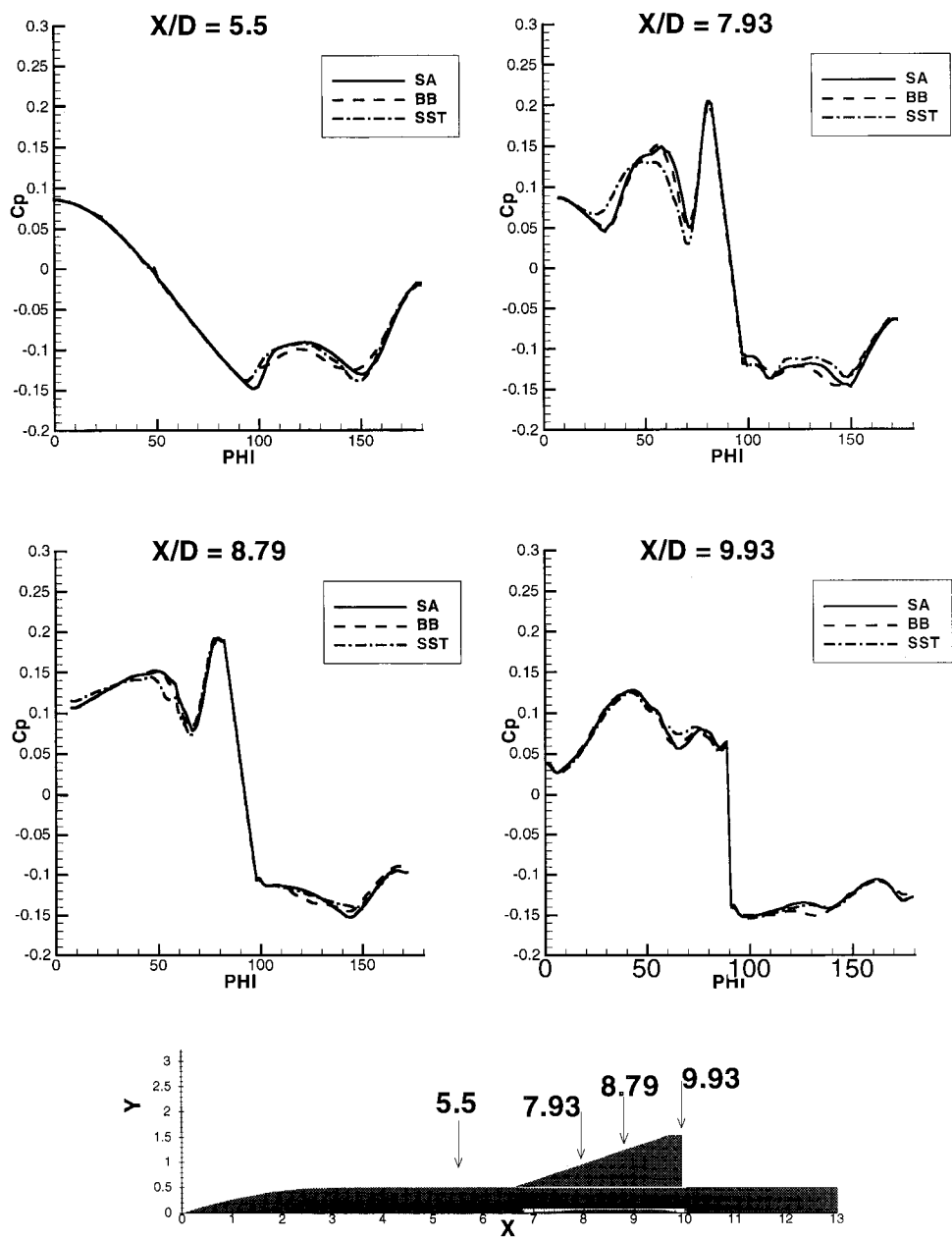


Figure 10 Comparison of surface pressure predictions on the body of missile 1.

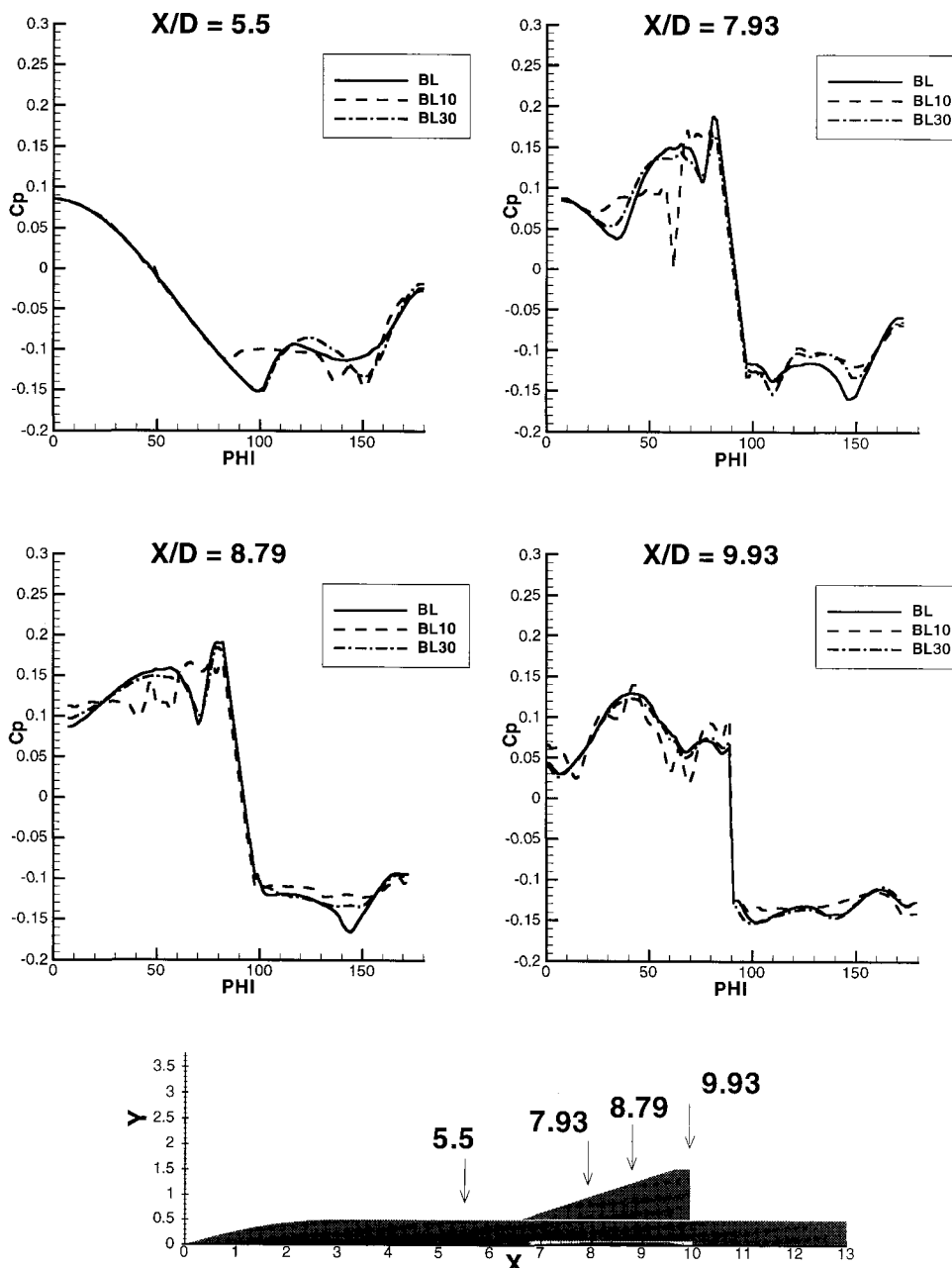
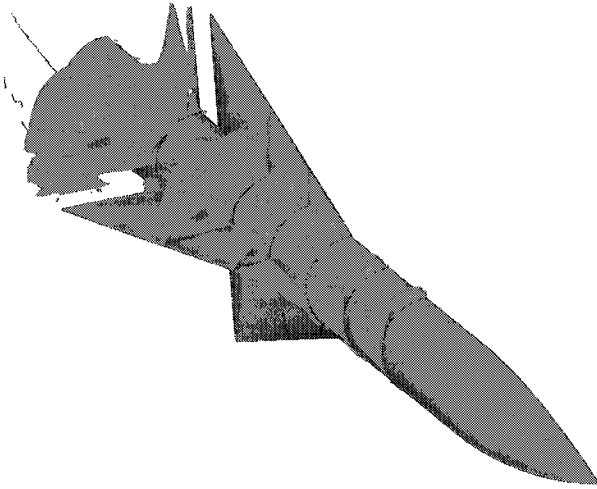
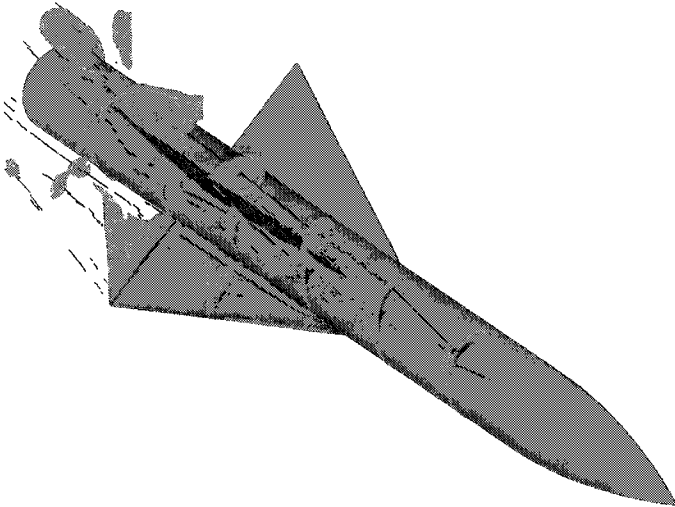


Figure 11 Comparison of surface pressure predictions on the body of missile 1.



**Figure 12 Pitot pressure and vortex core predictions on missile 1 at roll angle 0, SA turbulence model.**



**Figure 13 Pitot pressure and vortex core predictions for missile 1 at roll angle 45, SA turbulence model.**

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# On Aerodynamic Prediction by Solution of the Reynolds-Averaged Navier-Stokes Equations

M. G. Hall<sup>1</sup>

## 17.1 Introduction

An account is given of a numerical experiment on the two-dimensional flow past an aerofoil. The objective is to identify, and reduce, not only the limitations of the turbulence model but also the practical aerodynamic limitations of the Reynolds-averaged approximation itself. The extent that Reynolds averaging limits what is achievable with even an ideal solution of the Reynolds-averaged Navier-Stokes (RANS) equations is considered here. This is of course an important question for the whole range of practical, and generally more complex, configurations, but an aerofoil, or wing section, is probably the simplest from which useful lessons can be learnt. Any serious limitation in the predictability of aerofoil performance must be expected to apply also to winged aircraft.

Here, beforehand, is an outline of the background to the numerical experiment. It is a general view that the range of turbulence scales at the high Reynolds numbers of practical interest is so large that neither direct nor large-eddy simulation is likely to be feasible for some years. For practical aerodynamic prediction it is generally accepted that numerical solution of the RANS equations, with improved turbulence modeling, currently offers the best alternative. The subject has recently been reviewed in depth by Speziale [9]. Improvement in turbulence modeling is much needed but it

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presents a formidable challenge. Even after considerable effort, over several years, satisfactory results have been obtained only for flows with modest departures from the state of equilibrium turbulence. Yet the turbulent flow over a typical aircraft wing, for example, shows pronounced departures from equilibrium under normal operating conditions—at transition, at separation or reattachment, in interactions with shocks, and at the trailing edge. The effort now seems to be directed mainly at developing two-equation and stress-transport models that are better models of the full transport equation for the stress tensor, to improve the simulation of non-equilibrium flows, with an emphasis on minimizing the number of empirical assumptions. Little attention is directed, understandably, at the loss, in the course of RANS averaging, of some features of physical flow.

RANS simulation of the interaction of a shock with a boundary layer provides an illustrative example of an inherent limitation in RANS. The simulated shock will have a thickness related primarily to the cell size of the grid, such that with a fine enough grid the shock pressure rise outside the boundary layer will take place over a distance that is small compared with the boundary layer thickness. In reality, however, the interacting flow is very different [8, 5]. The larger eddies in the boundary layer induce an irregular oscillation of the shock over a distance of the order of the longitudinal dimension of the individual eddies. It follows that the averaged shock pressure rise will be spread over a distance that is also of the order of the large-eddy dimension, and this is typically the same order as the boundary layer thickness. The simulated flow will be qualitatively different from the averaged real flow whatever turbulence model is adopted, and improving the accuracy of the RANS solution by refining the grid will only enhance this difference. There could be serious practical consequences if, for example, the boundary layer is in an adverse pressure gradient and liable to separate. One penalty of the simplification of the Navier-Stokes equations by averaging is the loss of the ability to simulate any specific effect of the large eddies on the adjacent inviscid flow.

Accurate simulation is especially desirable for the flow past the trailing edge of a lifting wing, because changes in the flow at the trailing edge affect the performance of the whole wing. Special attention is given to this in the numerical experiment. The effects of the large eddies are not as easily identified as in shock/boundary-layer interaction. For the present experiment an aerofoil with substantial rear camber, the RAE5225, is chosen; it typifies the wing sections of modern airliners. This aerofoil has already been involved in comparative tests of different RANS codes, and wind-tunnel test results are available for comparison [1]. However, no measurements of turbulence have been made. Rear camber adds to the importance, and complexity, of the flow past the trailing edge. From experience with other aerofoils some, at least, of the main features of this flow can be listed. Firstly, there is a

very pronounced thickening of the boundary layer over the upper surface towards the trailing edge. It might be inferred that normal stresses would be significant in this region. Next, the flow is curved, because in passing the trailing edge it must turn from following the aerofoil surface towards the freestream direction. With rear camber this curvature is pronounced; it is, moreover, preceded upstream of the trailing edge by curvature of the opposite sign. These curvatures are expected to affect the production or damping of turbulence. Again, in the passage of the turbulent flow over the trailing edge the constraint on turbulent fluctuations, by the presence of a solid surface, is abruptly relaxed; there must be a sudden, pronounced change in the structure of the large eddies. This implies a pronounced change in the displacement effect of the shear layer, which would change the pressure field and the lift. Finally, there is an unsteady interaction at the trailing edge between the large eddies and the inviscid flow, which could be pronounced if the averaged flow upstream of the trailing edge is close to separation. For accurate simulation, all of these features should be adequately treated. Treatment of the last two, like the treatment of shock/boundary-layer interaction, seems to be beyond the scope of a strict RANS framework.

The present numerical experiment begins with a set of conventional simulations of transonic flow past the RAE5225 for a Mach number and an incidence at which the lift is moderate and there is no shock. For these simulations an improved version of the vertex-centroid RANS scheme proposed by Hall [2] is coupled to the popular  $k - \omega$  SST turbulence model proposed by Menter [6]. This turbulence model is widely accepted as being one of the best performing models for aerodynamic flows. The results are compared with the corresponding wind-tunnel results. Significant differences are noted. An effort is made to identify the reasons for these. This suggests that the differences might be reduced by a modification of Menter's formula for the eddy viscosity, confined to the region of the trailing edge. Finally, to test the possibility of such an improvement, a simple modification is applied to Menter's formula and the constants in the modifying function are adjusted by trial RANS solutions, at the given Mach number and incidence, to give good agreement with the tunnel results. Then, with the modifying function kept fixed, a simulation is attempted for a higher incidence, where the lift is 60% higher, the extent of supersonic flow is much larger, there is a pronounced shock, and where wind-tunnel results are again available for comparison. The level of agreement obtained for this case is unexpectedly high. The experiment concludes with a discussion of the results.

The computer codes used to generate results for this contribution were written by the author for DERA (formerly RAE), Farnborough. Although the author no longer has any formal links with DERA, informal exchanges have continued and the use of DERA's codes and computing facilities is duly acknowledged. The views expressed here, however, are the author's own.

## 17.2 The RANS Scheme and the Menter Turbulence Model

A summary of the main features of the RANS scheme and the turbulence model follows. There is nothing essentially new in the RANS scheme employed here. It is a finite-volume scheme with a vertex-centroid discretisation, introduced in 1991 by Hall [2] with the aim of combining improved accuracy with improved robustness, relative to the best of the then current cell-centred and cell-vertex schemes. The independent variables are specified at the vertices of the grid cells, but a control volume defined by the cell centroids is used for the integration of both the inviscid and the viscous fluxes. The treatment of artificial dissipation is based on the standard combination of second and fourth differences that has served well for solutions of the Euler equations, except that divided differences are used. The treatment now includes a means of reducing the artificial dissipation where physical dissipation itself serves to damp numerical instability; it is an improved version of the device proposed earlier by Hall [3]. To accelerate convergence, standard multigrid and residual smoothing schemes are used, with Lax-Wendroff time-stepping on the fine grid, for an element of upwinding, and Runge-Kutta time-stepping on the coarse grids.

The numerical accuracy of the RANS scheme is demonstrated here by comparing results obtained by Hall [4], for the RAE5225 aerofoil at a freestream Mach number  $M_\infty = 0.735$  and angle of incidence  $\alpha = 1.57$  degrees, with results obtained on the same C-grid by Benton (BAe) [12], Radespiel (DLR) [12] and Swanson (NASA) [10]. For this RANS validation the simple Baldwin-Lomax turbulence model was adopted. Table 1 below shows the computed lift and drag coefficients from each contributor, for three levels of grid fineness. For the finest grids the different contributors obtain results that are almost identical, with small differences that might be attributable to the differences in their codes. Examination of the change in results with refinement of the grid gives an indication of the accuracy of the numerical method. On this criterion the accuracy of Hall's results on the  $256 \times 64$  grid can be seen to compare favourably with the corresponding accuracies obtained by the others; indeed, the others can match Hall only by computing on a  $512 \times 128$  grid, one level finer.

It should be noted that while the tabled results show that very satisfactory numerical accuracy can generally be achieved in RANS computations, they also show that, at least with the Baldwin-Lomax turbulence model, RANS computations give a very poor simulation of the real flow. The computed pressure distributions show a distinct shock around mid-chord, of which there is no sign in the wind-tunnel measurements. The measured lift and drag coefficients at  $M_\infty = 0.735$  and  $\alpha = 1.59$  (in unpublished supplement to [1]) are 0.4057 and 0.01068 respectively, instead of the computed averages [12], 0.5505 and 0.00978. The computed lift is around 36% too high!



| Contributor | Grid              | $C_L$  | $C_D$    |
|-------------|-------------------|--------|----------|
| Benton      | $128 \times 32$   | —      | —        |
|             | $256 \times 64$   | 0.5538 | 0.010291 |
|             | $512 \times 128$  | 0.5546 | 0.009911 |
|             | $1024 \times 256$ | 0.5521 | 0.009825 |
| Radespiel   | $128 \times 32$   | —      | —        |
|             | $256 \times 64$   | 0.5436 | 0.010460 |
|             | $512 \times 128$  | 0.5499 | 0.009887 |
|             | $1024 \times 256$ | 0.5499 | 0.009799 |
| Swanson     | $128 \times 32$   | 0.5537 | 0.012892 |
|             | $256 \times 64$   | 0.5589 | 0.010069 |
|             | $512 \times 128$  | 0.5550 | 0.009730 |
|             | $1024 \times 256$ | —      | —        |
| Hall        | $128 \times 32$   | 0.5493 | 0.009745 |
|             | $256 \times 64$   | 0.5513 | 0.009771 |
|             | $512 \times 128$  | 0.5504 | 0.009803 |
|             | $1024 \times 256$ | —      | —        |

**Table 1** Computed lift and drag coefficients for the RAE5225 aerofoil, with the Baldwin-Lomax turbulence model.

Menter's turbulence model offers some improvement. This linear model yields an isotropic eddy viscosity for use in the RANS scheme, from two partial differential equations, for the convective transport of turbulent energy and its specific dissipation rate,  $k$  and  $\omega$  respectively. The model resembles the standard  $k - \omega$  turbulence model of Wilcox [11], but there are two important distinguishing features. Menter notes that while the Wilcox model has better numerical stability than the widely used  $k - \epsilon$  model in the inner part of a boundary layer, its results are unduly sensitive to variations in small freestream values of  $\omega$ , a shortcoming from which the  $k - \epsilon$  model is free. He then uses the identity  $\epsilon \equiv k\omega$  to change to  $k, \omega$  variables in the  $k - \epsilon$  model and uses this version of the latter for the outer part of a boundary layer whilst retaining Wilcox's  $k - \omega$  model for the inner part. By introducing a blending function the formulation is reduced to just a pair of equations, for  $k$  and  $\omega$ , as in Wilcox's model, but where Menter's equation for  $\omega$  contains an extra term involving the blending function and cross derivatives. The second feature is a modification to Wilcox's formula for eddy viscosity. Wilcox's formula

$$\mu_T = \frac{\rho k}{\omega}$$

implies that the principal shear stress is directly proportional to the shear strain. It does not provide for shear stress transport (SST), which is a serious defect in the simulation of boundary layers with separation or in adverse pressure gradients, where the shear stress has been observed to be roughly proportional to the turbulent kinetic energy. Menter proposes, as a remedy,

$$\mu_T = \frac{a_1 \rho k}{\max \{a_1 \omega; \Omega F\}}, \quad (17.1)$$

where  $a_1 = 0.31$ ,  $\Omega$  is the magnitude of the vorticity, and  $F$  is a function that changes smoothly from 1.0 inside the boundary layer to zero outside.

For the computations presented here Menter's equations for  $k$  and  $\omega$ , together with his formula (17.1) for the eddy viscosity  $\mu_T$ , are used to update the variables once for each multigrid cycle of the Navier-Stokes solution.<sup>2</sup> In the update the current Navier-Stokes variables are kept fixed while the turbulence variables are advanced a few steps, using a vertex-centroid discretisation and a two-stage time-stepping scheme. Since there is scope for variation, between individual programmers, in the numerical implementation of Menter's model, there will be greater differences between individual results from RANS solutions with the Menter model than with the Baldwin-Lomax model. Results from a standardised validation, such as are presented in Table 1 are not yet available. In the present implementation allowances are made for the fact that near a solid surface the dissipation  $\omega$  varies inversely with the square of the distance from the surface because, as will be demonstrated, this measure improves numerical accuracy.

### 17.3 RANS Results for the Menter Turbulence Model

In this section a comparison of computed results with wind-tunnel measurements is presented for the RAE5225 aerofoil at  $M_\infty = 0.735$ ,  $\alpha = 1.59$  and freestream Reynolds number  $Re_\infty = 6 \times 10^6$ . Table 2 shows computed lift and drag coefficients, for three grid levels, firstly for a simple implementation of the turbulence model with no allowance for the inverse square behaviour of  $\omega$ , and then for implementation with explicit allowances for the behaviour. The measured values are included at the foot of the table.

It can be seen that the simple implementation yields a marked increase in lift, and a marked decrease in drag, as the grid is refined. These changes are far more pronounced than those seen for the Baldwin-Lomax turbulence model in Table 1. On the other hand the results obtained with the explicit allowance for the inverse-square behaviour of  $\omega$  show changes with grid level

<sup>2</sup> The equations adopted here are those presented by Menter and Rumsey [7], which differ in the leading, convective, terms from those presented initially by Menter [6].

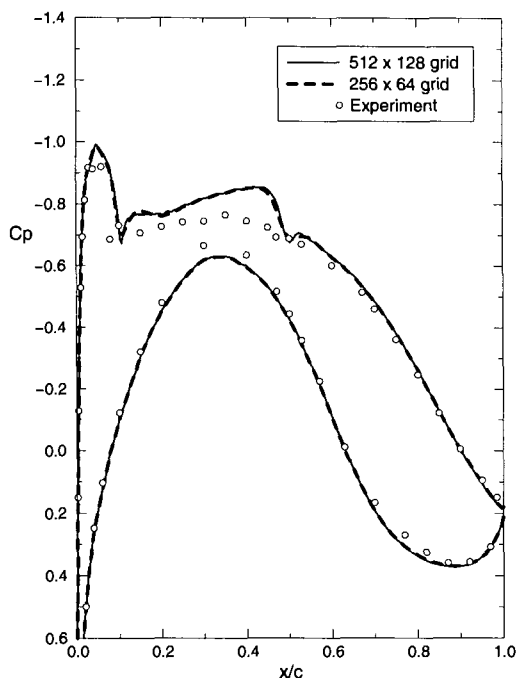
| Case       | Grid             | $C_L$  | $C_D$   |
|------------|------------------|--------|---------|
| simple     | $128 \times 32$  | 0.3822 | 0.01481 |
|            | $256 \times 64$  | 0.4126 | 0.01217 |
|            | $512 \times 128$ | 0.4567 | 0.01099 |
| allowances | $128 \times 32$  | 0.4671 | 0.01180 |
|            | $256 \times 64$  | 0.4675 | 0.01070 |
|            | $512 \times 128$ | 0.4688 | 0.01049 |
| measured   | —                | 0.4057 | 0.01068 |

**Table 2** Computed and measured lift and drag coefficients. Menter turbulence model. RAE5225 aerofoil,  $M_\infty = 0.735$ ,  $\alpha = 1.59$ ,  $Re_\infty = 6 \times 10^6$ .

that approach those in Table 1. In fact the ‘simple’ results tend to the ‘allowances’ results as the grid is refined. For the  $256 \times 64$  grid the ‘simple’ lift does agree satisfactorily with the measured value, but the change with grid refinement and the discrepancy in drag expose this as spurious and misleading; this implementation is not considered any further here. For the above test conditions it therefore appears that, while the Menter turbulence model yields a very satisfactory result for drag, the result for lift is still about 15% too high.

Fig 1 shows, for comparison, computed and measured pressure distributions over the aerofoil surface. It can be seen that the results obtained on the  $256 \times 64$  grid are essentially indistinguishable from those obtained on the  $512 \times 128$  grid. However, the computed and measured pressure distributions over the upper surface differ substantially, with the computed suction pressures being significantly higher than the measured values and the appearance of a weak but distinct computed shock. Only towards the trailing edge is there fair agreement. In fact, the computed pressure distribution is what might be expected for a higher incidence.

Detailed examination of the pressure distributions on the upper surface near the trailing edge provides a possible explanation. Note that the computed pressures maintain a steep gradient through to the trailing edge, whereas the measured pressure gradient is noticeably reduced for  $x/c > 0.95$ . This small difference in computed and measured pressure distributions at the trailing edge is a clue to possible shortcomings of the Menter turbulence model. The reduction in measured pressure gradient on the upper surface is normally associated with a rapid thickening of the boundary layer. Such a thickening would reduce the downward deflection of the flow by the aerofoil, and this could reduce the lift significantly, in the way that upward movement of a trailing-edge flap reduces lift. What, then, are the mechanisms for such a



**Figure 1** Computed and measured surface pressure distributions. Menter turbulence model. RAE5225 aerofoil,  $M_\infty = 0.735$ ,  $\alpha = 1.59$ ,  $Re_\infty = 6 \times 10^6$ .

rapid thickening that are not already included in Menter's model?

It is helpful to recall how a similar question, relating to the same wind-tunnel measurements, has been addressed by Ashill, Wood and Weeks [1]. These authors report on improvements to an interactive viscous-inviscid method for predicting aerofoil flows by the introduction of a higher-order boundary layer model for the viscous part of the method. Their original boundary layer model, which like Menter's turbulence model incorporated an allowance for shear-stress transport to deal with adverse pressure gradients, failed to reproduce the observed reduction of pressure gradient at the trailing edge and yielded a lift coefficient of 0.477, around 17.6% too high. They then made six distinct semi-empirical modifications to their viscous model in an attempt to reproduce the observed results. Two of these were intended to reduce the basic limitations of the boundary layer approximation, to give their method a capability approaching that of a RANS method. The lift is reduced by about 3% to 0.465 which, as might be expected, is close to the result in Table 1 for a RANS solution with the Menter turbulence model. The next three modifications, for the effects of low Reynolds number, normal

stresses and stream curvature, are of the type that researchers incorporate in new non-linear models for the eddy viscosity. Together they reduce the lift by a further 11% to 0.421. Finally there is a special allowance for the exceptional shape of velocity profiles near separation, which reduces the lift by 4% to 0.406, the measured level. Now the authors would be the first to agree that there is a significant degree of uncertainty in each of the modifications, so that the accumulated uncertainty in their sum would be considerable. Moreover there may be effects that are not accounted for at all, for example the sudden change in large-eddy structure at the trailing edge, and the unsteady interaction between the large eddies and the inviscid flow, which were mentioned in the Introduction. Exceptional velocity profiles may be a symptom of the latter unsteady interaction. Simulation of some of the effects would be beyond the scope of a strict RANS formulation, although some allowance for such effects might conceivably be incorporated in the turbulence model. The authors themselves provide evidence of the futility of any attempt to create a universal model with their next test case, the RAE5230, which is a modification of RAE5225 with rear camber increased enough to produce a well-defined trailing-edge separation at the wind-tunnel test conditions. For this, and similar cases, they recommend that for best results the stream-curvature modification, which had provided a third of the required reduction in lift for the RAE5225, be switched off.

The above indicates that any attempt to close the gap between numerical simulations and the physical reality in the usual way by adding non-linear terms to the equations, or resorting to the full transport equations for the stress tensor, is not likely to be an unqualified success. Given sufficient effort and computing capacity the gap could undoubtedly be reduced, but the physical non-linearities and departures from equilibrium are pronounced and there would remain the physical effects that were lost in the Reynolds averaging. In these circumstances a simple modification to the turbulence model, that may be non-physical in the RANS framework, but is focused on the trailing edge and devised with the real, unsteady, large-eddy structured, flow in mind, might be more effective. It would bundle in a simple package all the important trailing edge effects not covered by Menter, some of which might be treated by use of a non-linear turbulence model or by modeling the full transport equations for the stress tensor, and some of which were lost in the Reynolds averaging. None of these effects would be accounted for individually. This possibility is explored here.

## 17.4 A modification to the Menter turbulence model

To restrict the proposed modification of Menter's model to the trailing edge in a simple way the governing convective equations are left unchanged. Instead,

| Case     | Grid             | $C_L$  | $C_D$   |
|----------|------------------|--------|---------|
| Menter   | $128 \times 32$  | 0.4671 | 0.01180 |
|          | $256 \times 64$  | 0.4675 | 0.01070 |
|          | $512 \times 128$ | 0.4688 | 0.01049 |
| modified | $128 \times 32$  | 0.4131 | 0.01175 |
|          | $256 \times 64$  | 0.4070 | 0.01051 |
|          | $512 \times 128$ | 0.4069 | 0.01034 |
| measured | —                | 0.4057 | 0.01068 |

**Table 3** Computed and measured lift and drag coefficients. Menter, and modified, turbulence models. RAE5225 aerofoil,  $M_\infty = 0.735$ ,  $\alpha = 1.59$ ,  $Re_\infty = 6 \times 10^6$ .

a modifying factor  $F_{TE}$  is added to Menter's formula (17.1) for the eddy viscosity to yield

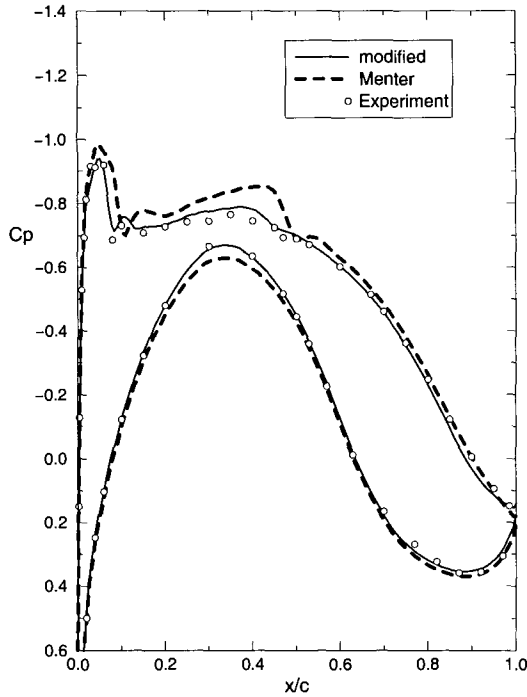
$$\mu_T = \frac{F_{TE} a_1 \rho k}{\max \{a_1 \omega; \Omega F\}}. \quad (17.2)$$

The modifying factor  $F_{TE}$  may take many forms. The only criterion in the choice made is that it should represent an average of the real, complex, highly interactive flow. No special merit is claimed for the form chosen. It is

$$F_{TE} = 1 - A(I) \{1 + G(I, J)\}. \quad (17.3)$$

Here  $I$  and  $J$  are interval counters along the C-lines of the grid and their transversals, respectively. The function  $A(I)$  is a constant  $0 < A < 1$  for  $0.975 \leq x/c \leq 1.0$  over the upper surface, and reduces linearly with  $I$  in both directions, so that  $A = 0$  over most of the aerofoil. It serves to reduce the eddy viscosity over the upper surface in the vicinity of the trailing edge. The function  $G(I, J)$  is added to give a suitable variation in the transverse direction. The magnitude of the change in eddy viscosity and the spatial extent of the region covered can be altered by means of adjustable constants in  $A$  and  $G$ . A number of trials, with the measured level of lift as the target, then yield the results shown in Table 3 for lift and drag.

It can be seen in Table 3 that the computed lift has been satisfactorily matched to the measured value. The drag predicted with Menter's turbulence model was not seriously in error and no attempt has been made to improve it by modification. The corresponding pressure distributions, for the  $256 \times 64$  grid, are shown in Fig 2. As might be expected, the computed pressure distribution is now in satisfactory agreement with its measured counterpart. Closer examination shows that, with the modified turbulence model, a



**Figure 2** Computed and measured surface pressure distributions. Menter, and modified, turbulence models. RAE5225 aerofoil,  $M_\infty = 0.735$ ,  $\alpha = 1.59$ ,  $Re_\infty = 6 \times 10^6$ .

reduction in pressure gradient on the upper surface towards the trailing edge is obtained that is similar to the reduction measured in the tunnel test. This indicates that the modified model has produced the required pronounced thickening of the boundary layer over the trailing edge.

To test the modified turbulence model it is now used for a different flow, without any change to the functions  $A$  and  $G$  in the expression (17.3) for the factor  $F_{TE}$  in the formula (17.2) for the eddy viscosity. The flow past the RAE5225 aerofoil at a higher incidence,  $\alpha = 2.763$ , is calculated. At this incidence the lift is 60% higher and the overall pressure distribution is very different, with a shock wave at around mid-chord. The resulting lift and drag coefficients are compared in Table 4 with the corresponding tunnel-test values, and also with computed values obtained by using Menter's original formula (17.1) for the eddy viscosity.

The corresponding pressure distributions, for the  $256 \times 64$  grid, are presented in Fig 3. The quality of agreement between computed and measured results,

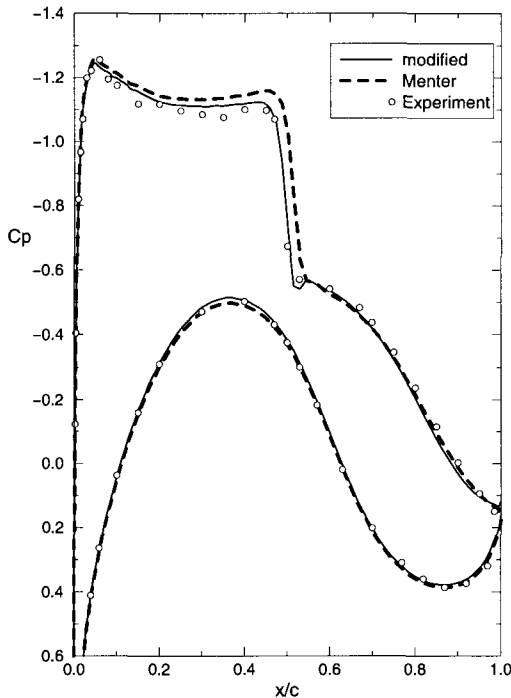
| Case     | Grid             | $C_L$  | $C_D$   |
|----------|------------------|--------|---------|
| Menter   | $128 \times 32$  | 0.6885 | 0.01387 |
|          | $256 \times 64$  | 0.6946 | 0.01396 |
|          | $512 \times 128$ | 0.7015 | 0.01394 |
| modified | $128 \times 32$  | 0.6440 | 0.01280 |
|          | $256 \times 64$  | 0.6572 | 0.01285 |
|          | $512 \times 128$ | 0.6597 | 0.01277 |
| measured | —                | 0.6616 | 0.01293 |

**Table 4** Computed and measured lift and drag coefficients. Menter, and modified, turbulence models. RAE5225 aerofoil,  $M_\infty = 0.737$ ,  $\alpha = 2.763$ ,  $Re_\infty = 6 \times 10^6$ .

as seen in Table 4 and Fig 3, is generally good. The computed lift and drag are only 0.03% and 1.2% in error, respectively. The computed pressure distribution matches the measured distribution well overall, but shows an excessive reduction of gradient at the trailing edge.

It might be argued that this good agreement should be expected because Menter's turbulence model gives a lift, in this case, that is only 6% too high (compared with 15% at  $\alpha = 1.59$ ), so that only a relatively small correction should be needed. On the other hand, Menter's model gives a drag in this case that is 8% too high, which is large compared with the discrepancy of 1.8% at  $\alpha = 1.59$ , so that a relatively large correction to the drag is needed for  $\alpha = 2.763$ . In view of the fact that no effort has been made to match drag in the calibration of the modified formula (17.2) for the eddy viscosity it seems surprising that drag is so well predicted when Menter's formula gives a poor result. This may of course be fortuitous, but there is a possible explanation. At the lower incidence there is no shock wave, or only a very weak shock. With such a flow the drag changes only relatively little for a given change in lift. At the higher incidence there is a shock wave on the upper surface, which contributes its own wave drag to the total drag. For a given change in lift the shock will change its position and its strength; there will be a significant change in the wave drag and, hence, in the total drag. It seems plausible, therefore, that Menter's model gave a poor estimate of drag at the higher incidence because it gave an inaccurate estimate of lift in circumstances where drag was sensitive to lift; once the lift was accurately estimated, by use of the modified eddy viscosity at the trailing edge, RANS together with Menter's model ensured that the shock was set in the correct position and the drag was well predicted.





**Figure 3** Computed and measured surface pressure distributions. Menter, and modified, turbulence models. RAE5225 aerofoil,  $M_\infty = 0.737$ ,  $\alpha = 2.763$ ,  $Re_\infty = 6 \times 10^6$ .

## 17.5 Concluding Remarks

A first test of a simple modification to the SST turbulence model of Menter has yielded good agreement with wind-tunnel measurements. This, however, would be only a first step in any serious development of a turbulence model for practical use. The proposed modification has so far been tested for only one new condition, namely for a new angle of incidence, with the freestream Mach number and the shape of the aerofoil retained at the state for which the modification was fully specified. Tests for a range of conditions, and a range of aerofoils, would be required. These might suggest further modifications. They would certainly expose limitations of the approach.

Obviously, modifications of the present type can be made to any turbulence model of the eddy-viscosity type; they are not restricted to the eddy viscosity model of Menter. Priority should perhaps be given to the development of a two-equation turbulence model in which the specific dissipation rate  $\omega$  is

replaced by a variable that is better suited for numerical computation. It seems perverse to derive the eddy viscosity, which varies with the fourth power of the distance from the surface, by solving a non-linear partial differential equation for a quantity,  $\omega$ , which varies with the inverse square of the distance from the surface.

Finally, the present results indicate that including a simple allowance in a standard turbulence model, for physical effects at the trailing edge that are not covered by the model, could yield a worthwhile improvement in aerodynamic prediction. The allowance bundles together the individual effects, some of which might otherwise be treated by use of a non-linear turbulence model or by modeling the full transport equations for the stress tensor, and some of which were lost in the Reynolds averaging. None of these effects are accounted for individually; their sum is treated as a single trailing-edge effect. The requirement in this approach is, first, to identify their combined physical effect and, then, to devise an appropriate numerical representation.

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# Advances in Algorithms for Computing Aerodynamic Flows

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## 18.1 Introduction

The success achieved in the field of computational fluid dynamics (CFD) over the past thirty years tends to obscure the tremendous challenges faced by the CFD community as the 21st century begins. If we concentrate on the application of CFD to aircraft design, or more specifically on the solution of the Reynolds-averaged Navier-Stokes (RANS) equations in that context, challenges can be identified in the following three areas:

**Computational efficiency.** The computing time required to achieve appropriately resolved solutions must be reduced. This need is particularly pressing as a result of the trend toward an integrated product and process development environment [34] and in the context of aerodynamic and multi-disciplinary design optimization. In order to be fully integrated into the design process, the time required for solution of the RANS equations over three-dimensional configurations must be on the order of a few minutes. This is roughly two orders of magnitude faster than current capabilities. Although increased computer speeds, especially parallel architectures, will undoubtedly help, improvements in algorithms are also needed. Algorithm reliability also has increased importance in a design optimization context. Modern design optimization algorithms, such as adjoint methods, cannot be effective if the flow solver does not converge in relevant areas of the design space.

**Human efficiency.** The need for a reduction in the human effort and

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expertise required for computing flows over complex configurations is perhaps even more urgent, since humans are not governed by Moore's Law. Although it is unwise to expect useful computations to be performed by a user without knowledge of CFD and aerodynamics, the expertise required in the selection of solver and grid parameters must be minimized. At present, the time and knowledge required to generate a computational grid from a complex three-dimensional geometry stored in a format associated with a computer-aided design package is excessive. Although unstructured solution-adaptive grid techniques have great potential in this regard, there are many hurdles to be overcome before their promise can be fulfilled. Estimation of global numerical error is another fundamental issue which would substantially reduce user expertise requirements but remains to be adequately addressed.

**Accuracy of physical modelling.** While numerical errors can be carefully controlled through appropriate grid resolution and other means, the errors resulting from physical models, including turbulence models and prediction of laminar-turbulent transition, are more difficult to estimate and control. The eddy-viscosity turbulence models currently popular for computing aerodynamic flows are generally incapable of accurately predicting subtle phenomena, such as Reynolds number or flap gap effects on high-lift configurations. It appears that Reynolds stress models (or *second-moment closures*) may be the most promising approach. Efforts to incorporate such models into aerodynamic flow solvers should be accelerated, at least to evaluate and guide development of such models, if not for production use at this stage. Further development of turbulence models is dependent upon comprehensive evaluation and testing for a wide range of flows, which in turn requires a reduction in the time needed to compute well-resolved solutions of three-dimensional flows. Furthermore, there is a need for more high-quality experimental datasets which include all of the boundary condition data needed for computations.

In this chapter, we describe and discuss two recent advances aimed at improving the computational efficiency of RANS solvers. The first is an inexact Newton-Krylov algorithm which reduces the computing time needed to achieve a steady-state solution [31]. The second is a higher-order spatial discretization which decreases the grid resolution requirements for a given level of numerical accuracy and thereby also reduces computing expense [12, 51, 13]. The two subjects are covered in separate sections with a comparable format. Each section contains background material and an overview of the algorithm, followed by results and a discussion of the issues raised.

An underlying theme in this chapter is the need for objective measures of algorithm performance. In order to make progress, one must be able to measure it. Although the overall goal, minimization of computing time,

might be straightforward,<sup>2</sup> algorithm assessment is a complicated matter as a result of the dependence on hardware and programming. In any case, the key indicator is computing time, not the number of iterations as is often reported. In the next section, we use a normalized measure of computing time, which, although it does not account for all factors, aids in comparing different algorithms run on distinct computers. Measuring the performance of a spatial discretization is even more difficult, as it requires the ability to calculate numerical error. In the section describing the higher-order spatial discretization, we make extensive use of grid convergence studies for this purpose.

## 18.2 Newton-Krylov Algorithm

### 18.2.1 Background

After discretizing the spatial derivatives in the steady RANS equations, whether through a finite-volume, finite-difference, or a finite-element method, a coupled system of nonlinear algebraic equations is obtained. With  $Q$  representing a vector containing the conservative variables at every node of the grid, we write this system as

$$R(Q) = 0. \quad (18.1)$$

For nonlinear algebraic equations, it is natural to consider Newton's method, which requires the solution of a linear system at each iteration and converges quadratically under certain conditions. This approach can be effective for relatively small problems [3], but the scaling and memory use associated with the direct solution of the linear problem becomes prohibitive as the problem size increases. This motivates the use of inexact-Newton methods in which the linear system which arises at each Newton iteration is solved using an iterative method. The inexact-Newton iteration can be written as

$$\|R(Q_n) + A(Q_n)\Delta Q_n\| \leq \eta_n \|R(Q_n)\|, \quad (18.2)$$

where  $A$  is the Jacobian of  $R$ ,  $\Delta Q_n = Q_{n+1} - Q_n$ ,  $Q_n$  is the current solution, and  $Q_{n+1}$  is the updated solution. The parameter  $\eta_n$  determines the degree of convergence of the iterative solution of the linear system, which controls the convergence of the inexact-Newton method. It is convenient to define the inexact-Newton iterations as *outer iterations* and the iterations required in the solution of the linear system as *inner iterations*. If  $\eta_n$  is equal to zero for all  $n$ , that is, the linear system is solved exactly, then Newton's method is recovered. It is possible to choose nonzero values of  $\eta_n$  such that quadratic

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<sup>2</sup> Even this is an oversimplification, since memory use is an important consideration as well.

convergence is retained, as shown in [11]. Further considerations in choosing a sequence of  $\eta_n$  values are discussed in [14].

If a Krylov subspace method is used to solve the linear system within an inexact-Newton framework, the combination is called a Newton-Krylov algorithm. Although there exist many iterative methods for nonsymmetric linear systems, the generalized minimal residual algorithm (GMRES) [39] is probably the most popular in this context. In order to solve the linear system  $Ax = b$ , GMRES utilizes a subspace given by  $v_1, Av_1, A^2v_1, \dots$ , where the vector  $v_1$  is formed from the the initial guess  $x_0$  as

$$v_1 = \frac{b - Ax_0}{\|b - Ax_0\|}. \quad (18.3)$$

The linear combination of vectors in the subspace which minimizes the residual is found at each GMRES iteration, and if sufficient convergence is not achieved, then an additional vector is added to the subspace. In practice, GMRES is normally restarted once the subspace reaches a specified size, in order to avoid excessive memory use. Wigton et al. [48] and Shakib [40] were the first to use GMRES in CFD. Further noteworthy developments in Newton-Krylov algorithms are reported in [23, 45, 9, 38, 30, 4, 2, 10, 5, 31, 27]. Recent contributions address issues related to parallelization [24] and incorporation of multigrid [25, 19].

Several distinct variations on the Newton-Krylov theme have emerged. We will describe three of them here. The first, associated with [48], applies the Newton-Krylov method to the solution of the nonlinear system of algebraic equations resulting from application of an iterative solver. For example, if an iterative solution technique converges to the solution of  $R(Q) = 0$  through an update formula given by

$$Q_{n+1} = M(Q_n), \quad (18.4)$$

where the operator  $M$  is a function of the iterative technique, then the Newton-Krylov method is applied to

$$Q - M(Q) = 0. \quad (18.5)$$

Effectively, the original solver is used as a preconditioner for GMRES. This approach is easy to add to an existing solver. Furthermore, GMRES requires only matrix-vector products, which can be formed using approximate Fréchet derivatives without explicitly defining the Jacobian matrix. Since the solvers used in [48] do not require the formation and storage of the Jacobian matrix, the resulting algorithm is truly matrix-free and has a modest memory overhead. The primary need for additional memory is associated with the search directions required by GMRES, which can be controlled using restarted GMRES.

In the second approach, known as approximate Newton, the Jacobian matrix is simplified, usually by using the Jacobian associated with a first-order spatial discretization. Rogers [38] gives an example of this strategy. Since a first-order spatial discretization involves only nearest neighbours, the simplified Jacobian requires considerably less storage than the full Jacobian. Furthermore, it is generally better conditioned and possibly diagonally dominant, so convergence of the inner iterations is rapid. However, the possibility of quadratic convergence is lost, and the number of outer iterations required is greatly increased. In terms of computing time requirements, the approximate-Newton approach is inferior to the Jacobian-free inexact-Newton approach which we describe next [31, 5].

In the third approach, the matrix-vector products required by GMRES are again formed using approximate Fréchet derivatives, and thus the full Jacobian need not be formed. However, an approximation to the Jacobian matrix similar to that used in the approximate-Newton framework is used as the basis for the preconditioner for GMRES [31]. Hence this approach is *Jacobian-free* but not *matrix-free*, in contrast to the first approach described. Typically, some form of incomplete lower-upper (ILU) factorization of the approximate Jacobian is used as the preconditioner. Although there is a memory penalty associated with this approach, it can be made very fast, as we shall see below.

### 18.2.2 Algorithm

In this subsection, we present the various components of our Newton-Krylov algorithm, which belongs to the Jacobian-free class described above. While the basic algorithm is relatively simple, several details are critical to its success. In order to provide results which can be easily interpreted, we normalize the computing time by the cost of a single evaluation of the residual. This is an admittedly imperfect attempt to allow for differences in hardware and, to a lesser extent, differing spatial discretizations.

#### *Spatial Discretization*

The spatial discretization is based on finite differences applied through a generalized curvilinear coordinate transformation. Second-order centered differences are used in combination with a blend of second- and fourth-difference scalar artificial dissipation. The second-difference dissipation is significant only near shocks. The algebraic turbulence model of Baldwin and Lomax is used to determine the eddy viscosity. The steady-state solutions obtained are identical to those of the well-established flow solver ARC2D [32]. For all of the examples, the grids have a C topology.

*Initial Phase*

A shortcoming of Newton’s method is that it is not globally convergent. There are several strategies for addressing this issue. One approach is based on the observation that Newton’s method is obtained from the implicit Euler time-marching method with local time linearization applied to the ordinary differential equation

$$\frac{dQ}{dt} = R(Q) \tag{18.6}$$

in the limit as the time step goes to infinity. Hence the implicit Euler method can be used with a finite time step initially, with the time step increasing as the residual is reduced. It is generally more efficient to use an approximate-Newton method until the time step is effectively infinite, at which point the Jacobian-free strategy can be initiated [5].

For the present results, we use the diagonal form of the approximate-factorization algorithm [32] with mesh sequencing to deal with the initial iterations. The residual is first reduced two orders of magnitude on a coarse grid (or a maximum of 150 iterations), followed by 5 iterations on the fine grid. This produces a stable algorithm for all cases studied, and the computing time for the initial phase is very small.

*Inexact Newton Strategy*

Following the initial phase described above, we use the following values of  $\eta_n$ :

- $\eta_n = 0.5$  for the first 10 outer iterations,
- $\eta_n = 0.1$  for the remaining iterations.

Although this approach results in linear, rather than quadratic, convergence, it is very efficient in terms of computing time. Fig. 1 shows the effect of the value of  $\eta_n$  on the computing time, with  $\eta_n$  held constant, for a representative test case. It is clear that a tight tolerance on the linear iterations is not beneficial in terms of computing time, even though it reduces the number of outer iterations. Beyond a certain point, further reduction of the residual of the linear system has no effect on the residual of the nonlinear system. The chosen strategy for  $\eta_n$  avoids this situation, which is known as oversolving the linear system [14].

*Jacobian-Free GMRES*

In order to form the Krylov subspace, GMRES requires the product of the Jacobian matrix  $A(Q_n)$  and an arbitrary vector  $v$ . This can be obtained without actually calculating the Jacobian from

$$A(Q_n)v = \frac{R(Q_n + \epsilon v) - R(Q_n)}{\epsilon}, \tag{18.7}$$



where  $\epsilon$  is calculated from [30]

$$\epsilon \|v\|_2 = \sqrt{\epsilon_m}, \quad (18.8)$$

and  $\epsilon_m$  is the value of machine zero.

### *Preconditioning*

The use of the approximate Fréchet derivative provides an excellent approximation of the product of the Jacobian matrix and an arbitrary vector. The Jacobian matrix associated with the residual function  $R(Q)$  arising from the spatial discretization used is both ill-conditioned and off-diagonal dominant. These properties cause GMRES to stall with very little reduction in the residual. Convergence can be greatly improved by preconditioning the basic system. In fact, the choice of preconditioner can be the most important single factor in determining the performance of the overall algorithm.

With right preconditioning, the system  $Ax = b$  becomes

$$AM^{-1}Mx = b, \quad (18.9)$$

where  $M$  is typically some approximation to  $A$  which is easier to invert than  $A$ . The idea is that the eigenvalues of  $AM^{-1}$  are much more clustered than those of  $A$ , thus improving the performance of GMRES. Note that a preconditioner can be used as an iterative solver, and vice-versa. For example, in the approach of Wigton [48], an iterative solver such as an approximately-factored implicit method or a multistage-multigrid method is used to precondition the system. This entirely avoids the formation and storage of the Jacobian matrix. In contrast, another popular class of preconditioners based on an incomplete lower-upper (ILU) factorization requires the storage of the Jacobian matrix or some reasonable approximation of it. It is popular to base the ILU factorization on an approximate Jacobian matrix associated with a first-order spatial discretization rather than the true Jacobian associated with a higher-order discretization. The original motivation was to reduce the storage and operation count, but Pueyo and Zingg [31] showed that the use of the approximate Jacobian leads to a more effective preconditioner as well. The difficulty with the use of the true Jacobian is that the incomplete  $L$  and  $U$  factors can be very poorly conditioned [16, 6]. Thus, although  $M$  can be a good approximation to  $A$ , the product  $AM^{-1}$  computed numerically can be very far from the identity matrix [31]. The approximate Jacobian formed from a first-order spatial discretization, which is very dissipative, tends to be more diagonally dominant and produces better conditioned incomplete  $L$  and  $U$  factors.

When an upwind spatial discretization is used, it is a straightforward matter to define a first-order discretization and the corresponding Jacobian

for use in forming the preconditioner. With the present spatial scheme, the dependence on next-to-nearest neighbours arises as a result of the fourth-difference artificial dissipation term. In forming the approximate Jacobian, only second-difference dissipation is used, with a coefficient,  $\epsilon_2^l$ , determined from

$$\epsilon_2^l = \epsilon_2^r + \sigma \epsilon_4^r, \quad (18.10)$$

where  $\epsilon_2^r$  and  $\epsilon_4^r$  are the actual second- and fourth-difference dissipation coefficients, respectively, and  $\sigma$  is a user-specified coefficient. Increasing  $\sigma$  makes the approximate Jacobian more diagonally dominant, but possibly an inferior approximation to the true Jacobian. Fig. 2 shows the total number of GMRES iterations required to reduce the outer residual by 12 orders of magnitude as a function of  $\sigma$  for the four test cases to be discussed below. Based on these and other similar results,  $\sigma = 5$  has been selected as a good general-purpose value.

An ILU factorization is formed in the same manner as an LU factorization, but certain nonzero entries are dropped. The simplest such factorization is ILU(0), in which the only entries retained in the L and U factors are those in locations where the matrix being factored has a nonzero entry. Hence the storage required for the L and U factors is identical to that of the matrix. More accurate ILU factorizations can be formed by permitting a certain amount of additional fill, based on either a level-of-fill or a threshold strategy. In the former, only the graph of the matrix is used to determine which entries to retain. With the threshold strategy, the size of the entries is taken into consideration as well. After considerable experimentation, some of which is documented in [31], we have selected the level-of-fill strategy with two levels of fill, i.e., ILU(2). Note that the entries in the approximate Jacobian are 4 by 4 blocks. Rather than using block ILU, we use a scalar version, but if a block contains at least one nonzero entry, we tag all of its entries as nonzero. We call the resulting factorization block-fill ILU(2), or BFILU(2). Fig. 3 shows residual convergence histories plotted against computing time measured in residual function evaluations for a typical test case. BFILU(2) formed from the approximate Jacobian, A1, produces substantially faster convergence than BFILU(0) formed from either A1 or the true Jacobian, A2.

Another factor affecting the performance of the ILU factorization is the ordering of the grid nodes. The reverse Cuthill-McKee (RCM) ordering [8] has become almost ubiquitous, based on studies such as that reported in [31], and is used here. However, we have obtained some preliminary results which indicate that a physically motivated ordering might be even more effective. Finally, we calculate the preconditioner only once, right after the initial phase using the approximate factorization algorithm. This was shown in [31] to reduce the computing time significantly without harming the convergence of GMRES.

### 18.2.3 Results and Discussion

The performance of the Newton-Krylov algorithm described is presented for four turbulent flows, three about the NACA 0012 airfoil and a fourth about the RAE 2822 airfoil. Flow conditions are:

**Case 1** – angle of attack: 0 degrees, free-stream Mach number: 0.3, Reynolds number: 2.88 million

**Case 2** – angle of attack: 6 degrees, free-stream Mach number: 0.3, Reynolds number: 2.88 million

**Case 3** – angle of attack: 1.49 degrees, free-stream Mach number: 0.70, Reynolds number: 9 million

**Case 4** – angle of attack: 2.31 degrees, free-stream Mach number: 0.729, Reynolds number: 6.5 million

The grid about the NACA 0012 airfoil has a C topology with 331 nodes in the streamwise direction and 51 in the normal direction. The distance to the first node from the airfoil surface is  $1 \times 10^{-5}$  chords, leading to a maximum cell aspect ratio in excess of  $1 \times 10^5$ . The grid about the RAE 2822 airfoil has 321 nodes in the streamwise direction and 49 in the normal direction with the same off-wall spacing.

Figures 4 to 7 show the residual histories plotted as a function of the computing time normalized by the time required for a single calculation of the residual. Note that this is simply a normalization in order to provide a means of comparison with other solvers; it does not reflect the actual number of residual evaluations, which is considerably less. In order to provide a reference, results are also presented for an approximately-factored multigrid algorithm (denoted AF-MG), which is typically four to six times faster than the approximate factorization algorithm alone [7]. The multigrid algorithm uses a three-grid sawtooth cycle. The Newton-Krylov algorithm converges twelve orders of magnitude for cases 1, 3, and 4. For case 2, the residual hangs after a reduction of about eight orders. This is caused by the Baldwin-Lomax turbulence model. Freezing the turbulent eddy viscosity leads to complete convergence. In all cases, the Newton-Krylov algorithm converges faster than the approximately-factored multigrid algorithm.

The preconditioner, BFILU(2), can also be used as a solver. Figure 6 shows that it can perform very well, actually converging faster than the approximately-factored multigrid algorithm. However, it is not reliable, diverging for some cases. The residual history of the approximate factorization algorithm without multigrid is also plotted in Fig. 6. It is seen to be much less efficient than BFILU(2). Thus it is clear that BFILU(2) is a key component of the Newton-Krylov algorithm. Furthermore, this indicates that

a fully matrix-free strategy using the approximate factorization algorithm as a preconditioner for GMRES is unlikely to be competitive.

Next we revisit case 2 using a  $265 \times 49$  grid with an off-wall spacing of  $1.2 \times 10^{-6}$ . The reduced off-wall spacing leads to better resolution of boundary layers and reduced numerical error in drag. Residual histories for this case are plotted in Fig. 8. The Newton-Krylov algorithm reduces the residual by twelve orders of magnitude in a computing time equivalent to under 1000 residual function evaluations, actually slightly faster than the convergence on the previous grid shown in Fig. 5. In contrast, the decreased off-wall spacing (and the associated increase in cell aspect ratio) adversely affects the convergence of the approximately-factored multigrid algorithm quite substantially.

For the cases studied, the number of outer iterations for a twelve-order residual reduction ranges from 17 to 22 as a result of the loose inner tolerance ( $\eta_n$ ) used. The total number of inner iterations ranges from 227 to 370, with an average number of inner iterations per outer iteration between 12 and 20. The total computing time on a Pentium Pro 180 processor varies from under six minutes to almost nine minutes. Overall, the Newton-Krylov algorithm converges faster and more reliably than the approximately-factored multigrid algorithm.

## 18.3 Higher-Order Spatial Discretization

### 18.3.1 Background

A complementary means of reducing computing expense is to raise the accuracy of the spatial discretization, thus enabling the use of coarser grids while maintaining the numerical error below an appropriate threshold. Higher-order finite-difference methods have a long history, with [26] and [43] providing much of the initial motivation. Gustafsson et al. [21] provide an interesting discussion, including some early references; further analysis and comparison can be found in [17] and [55].

Much of the analysis demonstrating the efficiency of higher-order methods is based on simple scalar equations with uniform grids and periodic boundary conditions. In the solution of the RANS equations with a nonuniform grid, more general boundary conditions, and added numerical dissipation, the case is not as clear. Consequently, the vast majority of RANS flow solvers in practical use are based on second-order discretizations (sometimes with third-order schemes for the inviscid terms). The use of higher-order schemes is more popular in the solution of problems where high accuracy is needed simply to make the computation feasible, such as simulation of transition and turbulence [33, 50], electromagnetics [54, 41], and aeroacoustics [47, 15].

Application of higher-order methods to practical aerodynamic flows was

initially limited by the fact that many early RANS solvers used the scalar artificial dissipation scheme associated with [22] to provide the numerical dissipation needed for stability. As shown in [1], for example, this scheme is excessively dissipative and is the primary source of error in computations of boundary-layer flows at high Reynolds numbers. Raising the discretization to higher order while retaining the scalar artificial dissipation scheme would be futile. The development of upwind schemes [37] and advanced artificial dissipation schemes [42] was thus critical to the successful implementation of higher-order methods. This illustrates a key aspect of improving the accuracy of a spatial discretization: the largest source of error must be addressed. Unfortunately, this is not necessarily easy to determine.

Since the introduction of sophisticated numerical dissipation schemes, several researchers have applied higher-order methods to practical aerodynamic problems. Examples using structured grids and a curvilinear coordinate transformation are given in [35, 44, 46, 12, 13]. In each case, there are significant benefits in terms of efficiency. In order to achieve these benefits, the following aspects of the discretization must be addressed:

- inviscid fluxes, including artificial dissipation or filtering,
- metrics of the curvilinear coordinate transformation,
- viscous fluxes,
- convective and diffusive fluxes in the turbulence model,
- near-boundary operators,
- extrapolation at boundaries,
- interpolation at zonal interfaces,
- integration for force and moment calculations.

Determining the improvements in efficiency associated with a higher-order spatial discretization requires some means of calculating the numerical error of a solution computed on a given grid. At the present time, the most reliable means to accomplish this is to use a solution computed on a much finer grid as a reference solution. Assuming that the error on the finer grid is much smaller than that on the coarser grid, the error on the coarser grid is simply the difference between the two solutions, thus providing both local and global error estimates. This is currently practical only in two dimensions, and, unfortunately, is not as simple as it sounds. Use of an extremely fine grid can lead to convergence difficulties, both because of increased numerical stiffness and because scales may be resolved which tend to be unsteady, near the trailing edge of an airfoil, for example. One would prefer a systematic grid refinement study, permitting the use of Richardson extrapolation [52], but it can be very difficult (or require excessively fine grids) to achieve suitable asymptotic behavior [36]. For example, some of the grid refinement studies presented in [51] show well-defined asymptotic behavior, while others do not. Furthermore, Zingg et al. [51] show that different spatial discretizations can

produce surprisingly large solution differences even on extremely fine grids.

There are many reasons why the asymptotic convergence behaviour of a flow solver can be difficult to ascertain. One reason is that there are several sources of error of different order. For example, the numerical dissipation used away from shocks and discontinuities generally scales with the grid spacing to third order, while errors in the viscous and inviscid flux derivative approximations are often second order. In addition, there may be local sources of error which are first order, such as the numerical dissipation added near shocks and some approximations used near boundaries. On a sufficiently fine grid, the first-order sources of error must become dominant. However, they may be smaller than the higher-order errors on grids which are practical. On coarse grids, the third-order numerical dissipation may dominate. Further compounding the difficulty is the presence of grid and flow singularities, such as the trailing edge of an airfoil. When these are present, the error behavior obtained in a grid refinement study may not correspond to the local truncation error of the discretization, especially a higher-order discretization. The means by which the grid is refined and the degree of continuity of the underlying representation of the geometry can also affect the order of accuracy realized.

It may appear that the presence of singularities and discontinuities would invalidate the use of higher-order methods, since the higher-order behavior is not achieved. However, the results presented in [35, 44, 46, 12, 13] show that higher-order methods remain beneficial. The error introduced at the singularity is typically local and has little effect on the surrounding solution. Therefore, although a higher-order algorithm may produce second-order convergence in the presence of singularities, it is still likely to produce a smaller error than a second-order algorithm. Hence our emphasis is on the magnitude of the error on a given grid, as opposed to the effective order of accuracy.

### 18.3.2 Algorithm

The present higher-order spatial discretization is implemented using the approximate-factorization algorithm in diagonal form [32] to iterate to steady state. Turbulence is modelled using either the algebraic Baldwin-Lomax model or the one-equation Spalart-Allmaras model. An overview of the spatial discretization is given here; details can be found in [12] and [13].

#### *Inviscid Fluxes, Numerical Dissipation, and Grid Metrics*

Any finite-difference approximation of a first derivative can be written as the sum of a skew-symmetric operator and a symmetric operator. The errors can be classified based on Fourier analysis of the scheme when applied to the linear convection equation. If the amplitude of a harmonic function decays, the error

is classified as dissipative. If the phase speed differs from the actual phase speed, thus becoming a function of the wavenumber, the error is described as dispersive. The dissipative error is associated with the symmetric portion of the operator. If the finite-difference scheme is purely skew-symmetric, i.e., the symmetric portion of the operator is zero, then the semi-discrete scheme is nondissipative. The dispersive error is associated with the skew-symmetric portion, which cannot be zero. Hence all finite-difference schemes produce dispersive error except under idealized conditions, such as specific Courant numbers.

It is widely accepted that a spatial discretization of the Euler equations must include some numerical dissipation, i.e., the symmetric portion of the operator must be nonzero, for convergence and stability.<sup>3</sup> However, it is not known how much dissipation is needed. High-resolution schemes are based on minimizing the first-order dissipation added near discontinuities while satisfying some criterion with respect to total variation, positivity, or monotonicity. A comparable theoretical approach toward minimization of the numerical dissipation in smooth regions of the flow would be highly desirable. Entropy-based approaches have shown some potential [29].

Numerical dissipation is also typically needed in the solution of the RANS equations. Although these equations contain physical dissipation mechanisms, further numerical dissipation can be required depending on the grid resolution. [18] explored the use of the cell Reynolds number as a scaling parameter for the numerical dissipation, while [49] used the vorticity function from the Baldwin-Lomax turbulence model for the same purpose. Although Zingg et al. [51] showed that third-order matrix dissipation is not a major source of error when coupled with second-order centered differencing, such scalings may need to be revisited when higher-order difference schemes are used.

The leading dissipative error term is always of odd order, while the leading dispersive error term is always of even order. For a given symmetric stencil, the dispersive error can always be made one order higher than the dissipative error. The symmetric operator can be multiplied by a small coefficient, however, and hence the dissipative error can be smaller than the dispersive error at finite wavenumbers. This is the strategy followed in [54], using a seven-point stencil. The dissipative error resulting from the fifth-order symmetric operator is smaller than the dispersive error from the sixth-order skew-symmetric operator, except at very small wavenumbers (where the errors are negligible anyway). The similar behavior of the dissipative and dispersive errors as a function of the wavenumber leads to the desirable property that only modes which have a large dispersive error are heavily dissipated.

In the present solver, we use a five-point stencil, leading to a third-order dissipative error and a fourth-order dispersive error. The skew-symmetric

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<sup>3</sup> [28] provides some interesting discussion on this point.

operator is

$$(\delta^a u)_j = \frac{1}{12\Delta x}(-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}). \quad (18.11)$$

The basic symmetric operator is

$$(\delta^s u)_j = \frac{\epsilon_4}{\Delta x}(u_{j+2} - 4u_{j+1} + 6u_j - 4u_{j-1} + u_{j-2}), \quad (18.12)$$

where  $\epsilon_4$  is a coefficient, typically 0.02. Note that  $\epsilon_4 = 1/12$  produces the popular third-order upwind-biased operator. Details of the matrix dissipation scheme used, including the pressure switch used to add first-order dissipation near shocks, can be found in [42] and [12].

The metrics of the curvilinear coordinate transformation are calculated using the skew-symmetric operator given above. It is shown in [51] that the accuracy of the operator for the grid metrics should correspond to that of the skew-symmetric portion of the operator for the inviscid fluxes. If a second-order approximation is used for the metrics, accuracy can be significantly degraded.

### *Viscous Fluxes*

There are a number of possible strategies for discretization of the viscous fluxes, which are in the following form:

$$\partial_x(\alpha \partial_x \beta). \quad (18.13)$$

Some authors [46] simply apply the higher-order first-derivative operator used for the inviscid fluxes twice, with no apparent complications. However, this approach should generally be avoided, as it produces poor damping of high wavenumbers (particularly important with multigrid) and a larger than necessary stencil. A fourth-order approximation can be achieved with a five-point stencil. However, for convenience in programming, we use the following scheme, which leads to a seven-point stencil. The term  $\partial_x \beta_j$  is first approximated at  $j + 1/2$  using the following fourth-order expression:

$$(\delta_x \beta)_{j+1/2} = \frac{1}{24\Delta x}(\beta_{j-1} - 27\beta_j + 27\beta_{j+1} - \beta_{j+2}). \quad (18.14)$$

The coefficient  $\alpha$  is calculated at  $j + 1/2$  using a fourth-order interpolation formula:

$$\alpha_{j+1/2} = \frac{1}{16}(-\alpha_{j-1} + 9\alpha_j + 9\alpha_{j+1} - \alpha_{j+2}). \quad (18.15)$$

The complete operator then becomes:

$$\begin{aligned} \delta_x(\alpha_j \delta_x \beta_j) &= \frac{1}{24\Delta x}[\alpha_{j-3/2}(\delta_x \beta)_{j-3/2} - 27\alpha_{j-1/2}(\delta_x \beta)_{j-1/2} \\ &\quad + 27\alpha_{j+1/2}(\delta_x \beta)_{j+1/2} - \alpha_{j+3/2}(\delta_x \beta)_{j+3/2}]. \end{aligned} \quad (18.16)$$



*Turbulence Models*

In the Baldwin-Lomax model, the vorticity must be calculated at  $j + 1/2$ . This is accomplished using Eq. 18.14. In the Spalart-Allmaras model, the diffusive fluxes are discretized as in Eq. 18.16. The convective terms are discretized using first-order upwinding in order to maintain positivity of the eddy viscosity. Experiments with third-order approximations of the convective terms have shown no improvement in accuracy. Nevertheless, this first-order treatment could eventually become the leading source of error as other sources continue to be addressed.

*Near-Boundary Operators*

One of the problems associated with the use of higher-order methods is the need for near-boundary operators resulting from the large stencil. It can be difficult to find near-boundary operators of suitable accuracy which are stable in conjunction with the interior scheme, because the boundary operator associated with an incoming wave becomes downwind biased [53]. However, reducing the accuracy of the boundary scheme can undermine the benefits of the higher-order interior scheme. In the present algorithm, we have been able to use third-order boundary schemes while maintaining stability.

The following third-order biased operator is used to approximate a first derivative at an interior node which has a boundary node as its  $j - 1$  neighbour:

$$(\delta_x u)_j = \frac{1}{6\Delta x}(-2u_{j-1} - 3u_j + 6u_{j+1} - u_{j+2}). \quad (18.17)$$

The same formula is used to calculate the grid metric at such a node. For the grid metric at a boundary node, the following equation is used:

$$(\delta_x u)_j = \frac{1}{24\Delta x}(-11u_j + 18u_{j+1} - 9u_{j+2} + 2u_{j+3}). \quad (18.18)$$

The dissipation operator for a near-boundary node is

$$\frac{1}{\Delta x}(-u_{j-1} + 3u_j - 3u_{j+1} + u_{j+2}). \quad (18.19)$$

For the viscous fluxes, the near-boundary operators corresponding to Eqs. 18.14, 18.15, and 18.16 are, respectively,

$$(\delta_x \beta)_{j+1/2} = \frac{1}{24\Delta x}(-23\beta_j + 21\beta_{j+1} + 3\beta_{j+2} - \beta_{j+3}), \quad (18.20)$$

$$\alpha_{j+1/2} = \frac{1}{8}(3\alpha_j + 6\alpha_{j+1} - \alpha_{j+2}), \quad (18.21)$$

and

$$\delta_x(\alpha_j \delta_x \beta_j) = \frac{1}{24\Delta x} [-23\alpha_{j-1/2}(\delta_x \beta)_{j-1/2} + 21\alpha_{j+1/2}(\delta_x \beta)_{j+1/2} + 3\alpha_{j+3/2}(\delta_x \beta)_{j+3/2} - \alpha_{j+5/2}(\delta_x \beta)_{j+5/2}]. \quad (18.22)$$

### *Extrapolation at Boundaries*

At a far-field boundary where the flow is subsonic, the solution is determined from a combination of the prescribed free-stream conditions and a state which is extrapolated from the interior, which is dependent on the direction of the flow [51]. The extrapolated state is found using the following formula:

$$u_j = 3u_{j+1} - 3u_{j+2} + u_{j+3}. \quad (18.23)$$

At the airfoil surface, the pressure is calculated from the expression

$$p_1 = \frac{1}{11}(18p_2 - 9p_3 + 2p_4), \quad (18.24)$$

which is derived from a third-order approximation to  $\partial p / \partial n = 0$ . For an adiabatic surface, the density is found using the same formula. We have also used Eq. 18.23 to extrapolate pressure to the airfoil surface, with no change in the results.

### *Interpolation at Zonal Interfaces*

In [35] a higher-order method is applied to overlapping grids, so considerable attention must be paid to interpolation between zones. In the present study, using single-block C grids, a similar issue arises on the wake cut. Although the wake cut can be treated as an interior point, it is currently treated as an interface. The following interpolation formula is used to calculate the solution on the wake cut (denoted by the subscript *wc*) based on the solutions above and below:

$$q_{k_{wc}} = \frac{1}{6}(-q_{k_{wc}+2} + 4q_{k_{wc}+1} + 4q_{k_{wc}-1} - q_{k_{wc}-2}) \quad (18.25)$$

### *Integration for Force and Moment Calculations*

Given a solution computed using a higher-order spatial discretization, a higher-order integration technique should be used to calculate lift and drag. This is accomplished by fitting cubic spline interpolants to the airfoil geometry, the surface pressure, and the skin friction at the surface. A fourth-order one-sided operator is used to calculate the derivative required for the skin

friction. Based on the spline interpolants, an adaptive two-point Gauss-Legendre quadrature with a global error control strategy is used to perform the integration.

### *Linearization*

The diagonal form of the approximate factorization algorithm leads to the solution of scalar banded linear systems. The treatment of the inviscid fluxes is based on a five-point stencil, leading to pentadiagonal systems. Our viscous operator is based on a seven-point stencil, so we use a linearization of the standard second-order viscous operator, since the contribution of the viscous terms to the left-hand side of the diagonal form is quite approximate in any case. Approximate linearizations are also used for the near-boundary operators which lie outside the pentadiagonal structure.

De Rango and Zingg [12, 13] include a number of residual convergence histories comparing the higher-order algorithm with a second-order algorithm also using matrix dissipation. In some cases, convergence is nearly identical, while in others, the higher-order algorithm converges somewhat more slowly. Although this could be associated with the approximations in the linearization of the viscous terms, it is more likely caused by the increased stiffness associated with the improved resolution of the higher-order algorithm.

### **18.3.3 Results and Discussion**

In this section, we present the results of a number of grid convergence studies in order to demonstrate the relative performance of the higher-order spatial discretization. The higher-order discretization (denoted *Higher-order* in the figures) is compared with a discretization using second-order centered differences with matrix dissipation together with lower-order boundary schemes and a second-order integration procedure (denoted *Second-order* in the figures). In some cases, results are also shown for a third-order upwind-biased flux-difference split scheme with second-order viscous terms, grid metrics, and integration (denoted *Third-order upwind* in the figures).

For the first case, we consider a subsonic flow about the NACA 0012 airfoil at an angle of attack of 12 degrees, a Reynolds number based on the chord of 2.88 million, and a free-stream Mach number of 0.16. Laminar-turbulent transition is fixed at 0.05 and 0.8 chords on the upper and lower surfaces, respectively. The flow is characterized by a small region of separated flow on the upper surface near the trailing edge. The experimentally measured lift coefficient for these conditions is 1.240, and the drag coefficient is 0.0180 [20].

Table 1 gives a description of the grids used. Grid A, which was generated using an elliptic grid generator, is a very dense grid providing a reference

| Grid | Dimensions        | Off-Wall Spacing<br>( $\times 10^{-6}$ ) | Leading Edge<br>Clustering<br>( $\times 10^{-3}$ ) | Trailing Edge<br>Clustering<br>( $\times 10^{-3}$ ) |
|------|-------------------|------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| A    | $1057 \times 193$ | 0.23                                     | 0.1                                                | 0.5                                                 |
| B    | $529 \times 97$   | 0.53                                     | 0.2                                                | 1.0                                                 |
| C    | $265 \times 49$   | 1.2                                      | 0.4                                                | 2.0                                                 |

**Table 1** Grids for subsonic case.

solution with little numerical error.<sup>4</sup> Grid B was generated by removing every second node in each coordinate direction from grid A, and grid C was similarly generated from grid B. Consequently, these three grids form a sequence suitable for a grid convergence study. We are primarily interested in the numerical errors in computations on grid C, which is typical of grids used in practice.

Fig. 9 shows the lift, pressure drag, and friction drag coefficients computed on grids A, B, and C using the three discretizations described above with the Baldwin-Lomax turbulence model. Results are plotted versus  $1/N$ , where  $N$  is the total number of grid nodes. The solutions computed on grid A lie very close to one another, confirming that these solutions have very little numerical error. Thus the errors in the solutions computed on grid C can be estimated by comparison with the grid A solutions. All three discretizations produce an error in the lift coefficient of less than one percent on grid C. The errors in the drag components are much larger. On grid C, the second-order scheme produces errors in pressure and friction drag of over 40 and 12 percent, respectively. These errors are of opposite sign, but the overall drag error exceeds 20 percent. The third-order upwind scheme produces somewhat smaller errors than the second-order scheme but much greater errors than the higher-order scheme, indicating that raising the order of the flux extrapolation is not particularly effective unless the accuracy of the grid metrics and other approximations is improved as well. The errors in the drag components computed using the higher-order scheme on grid C are less than 2 percent and are less than the error computed using the other schemes on grid B, which has 4 times as many nodes.

Closer inspection of the solutions reveals that the lower-order schemes lead to an overprediction of the boundary-layer thickness on the upper surface, consistent with overprediction of pressure drag and underprediction of friction

<sup>4</sup> We are hesitant to use the term *grid independent*, since this may imply grid independence to within machine precision, which is certainly not the case. However, the solution computed on grid A can be considered grid independent to within a reasonable tolerance.

drag. This is shown in Fig. 10, which displays the computed boundary-layer velocity profiles on the upper surface at 85 percent chord. The grid A solutions computed using the three discretizations are indistinguishable; the second-order solution is shown. Every fourth point of the grid A solution is plotted, corresponding to the points on grid C. The higher-order solution computed on grid C lies very close to the grid A solution, while the other solutions are significantly in error, consistent with the errors in the drag components.

Next we revisit this case using the Spalart-Allmaras model in order to examine the impact of the discretization of the turbulence model. Results are shown in Fig. 11. The trends are similar to those observed using the Baldwin-Lomax model, demonstrating that the errors associated with the first-order discretization of the convective terms in the Spalart-Allmaras model are not large. Once again, the higher-order results computed on grid C are more accurate than the second-order results on grid B.

As an aside, both turbulence models lead to an overestimation of lift and an underestimation of drag relative to the experimental values, with the Spalart-Allmaras results preferred. It is interesting to note that the grid C results computed using the second-order discretization actually lie closest to the experimental values as a result of cancellation between numerical and physical-model errors. This reinforces the idea that accurate numerical solutions are essential for successful turbulence model evaluation.

Figs. 12 and 13 show analogous grid convergence studies for two transonic flows over the RAE 2822 airfoil. Flow conditions for the two cases are as follows:

**Transonic case I** – angle of attack: 2.31 degrees, free-stream Mach number: 0.729, Reynolds number: 6.5 million

**Transonic case II** – angle of attack: 2.57 degrees, free-stream Mach number: 0.754, Reynolds number: 6.5 million

For both cases, transition is fixed at 0.03 chords on both surfaces. Case II is characterized by a much stronger shock wave on the upper surface and a much larger region of shock-induced boundary-layer separation. A sequence of three grids is again used, with characteristics as given in Table 2. The Spalart-Allmaras turbulence model is used for both cases.

Figs. 12 and 13 show that the higher-order discretization leads to a significant reduction in the error relative to the second-order scheme, generally producing solutions on grid C which are accurate to within 2 percent. The exception is the pressure drag for case 2, for which the higher-order solution has an error of nearly 4 percent, and the solution computed using the second-order scheme has an error just over 5 percent. It is possible that there is insufficient grid resolution in the vicinity of the separation bubble at the shock even for the higher-order scheme. Figs. 14 shows a portion of the computed

| Grid | Dimensions        | Off-Wall Spacing<br>( $\times 10^{-6}$ ) | Leading Edge<br>Clustering<br>( $\times 10^{-3}$ ) | Trailing Edge<br>Clustering<br>( $\times 10^{-3}$ ) |
|------|-------------------|------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| A    | $1025 \times 225$ | 0.23                                     | 0.1                                                | 0.25                                                |
| B    | $513 \times 113$  | 0.53                                     | 0.2                                                | 0.5                                                 |
| C    | $257 \times 57$   | 1.2                                      | 0.4                                                | 1.0                                                 |

**Table 2**   Grids for transonic cases.

pressure coefficient distribution on the upper surface for case II. The solution computed using the higher-order discretization on grid C lies much closer to the grid A solution than that computed using the second-order scheme on grid C. This indicates that, despite the modest reduction in the pressure drag error, the higher-order scheme leads to a substantial reduction in local solution error. This is confirmed by Fig. 15, which shows the computed boundary-layer velocity profiles on the upper surface at 95 percent chord. The error in the velocity profile computed on grid C using the second-order scheme is quite large, while the error in the higher-order results, though visible, is small. In Fig. 15, we also plot a velocity profile computed using the higher-order discretization with the viscous terms discretized using the second-order scheme. This shows that raising the viscous terms to higher order accounts for roughly 10 percent of the overall error reduction.

The results show that the higher-order spatial discretization presented leads to a substantial reduction in the numerical errors obtained on a given grid relative to a similar second-order discretization. This is achieved with an increased computing expense of roughly 6 percent per iteration and a small reduction in convergence rate. In many cases, the higher-order solution computed on a given grid is more accurate than the second-order solution computed on a grid with four times as many nodes. This translates into savings in computing time of at least a factor of four.

**18.4   Concluding Remarks**

We have presented two complementary algorithmic advances which reduce the computing expense of the solution of the RANS equations for the flow over aerodynamic configurations. Based on the results presented for two-dimensional flows, both algorithms show sufficient potential to justify extension to three dimensions and to more complicated physics than the

simple single-element airfoil cases studied to date. Further development and testing is needed to ensure that these algorithms are sufficiently reliable for practical use.

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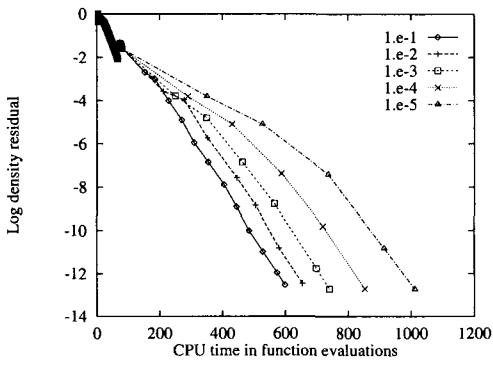
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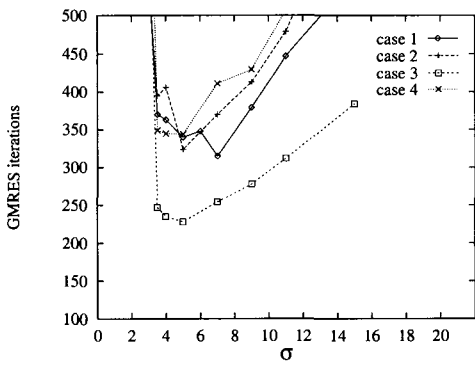
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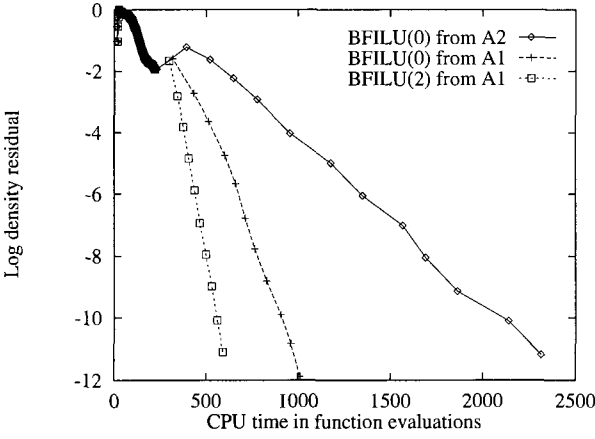
**Figure 1** Effect of  $\eta_n$  on convergence. Values of  $\eta_n$  are given in the legend. Each symbol represents one outer iteration.



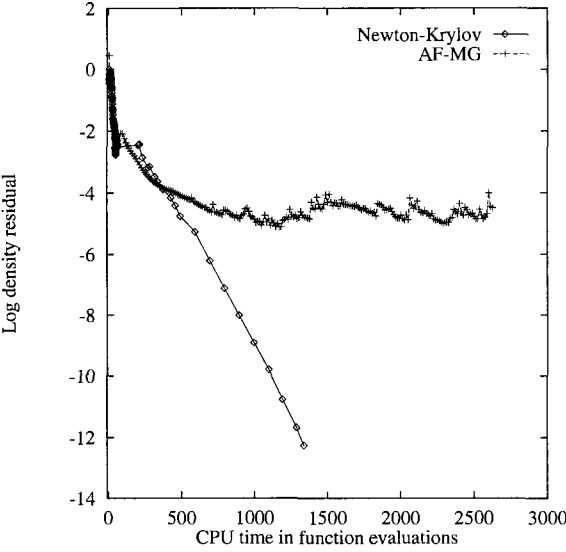
**Figure 2** Effect of  $\sigma$  on convergence.

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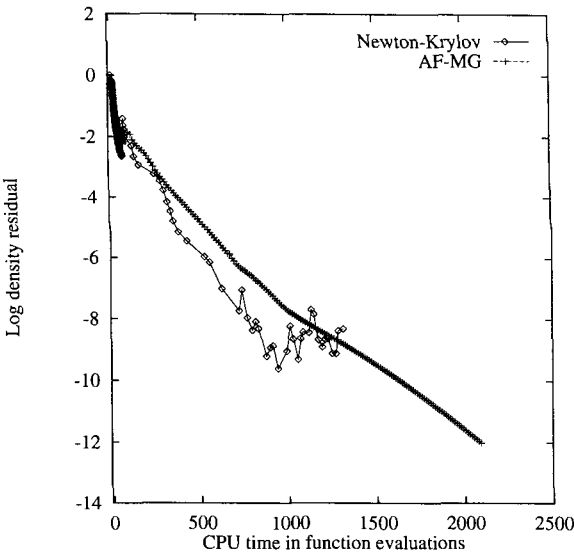
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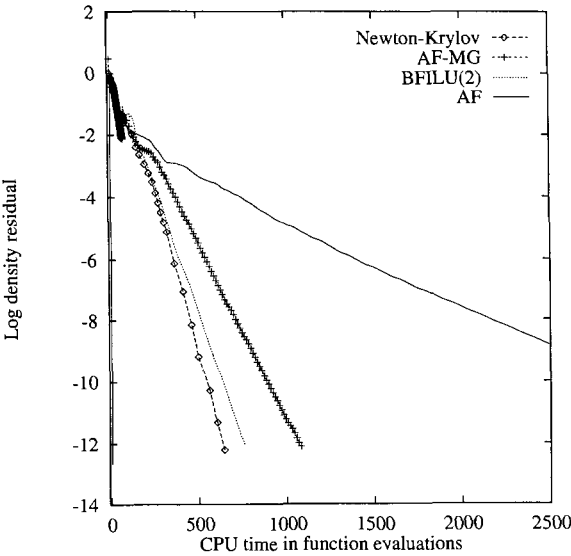
**Figure 3** Performance of various preconditioners.



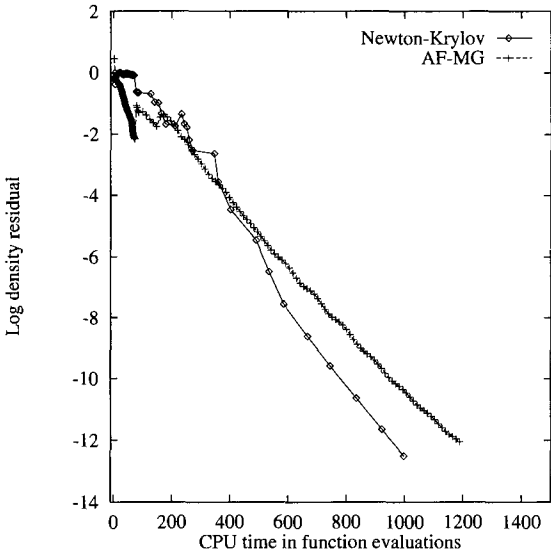
**Figure 4** Residual history for case 1.



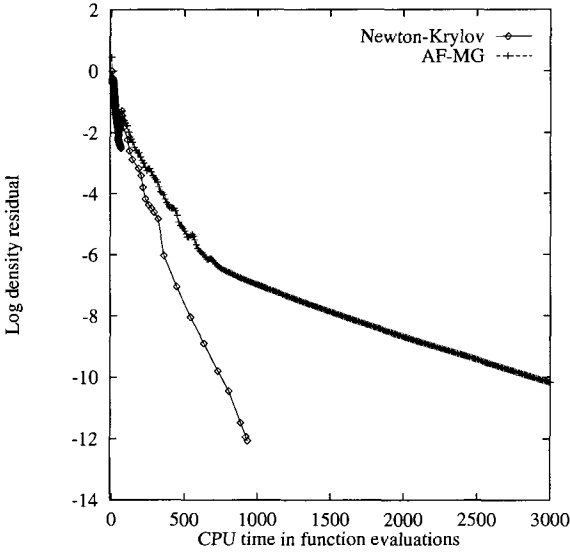
**Figure 5** Residual history for case 2.



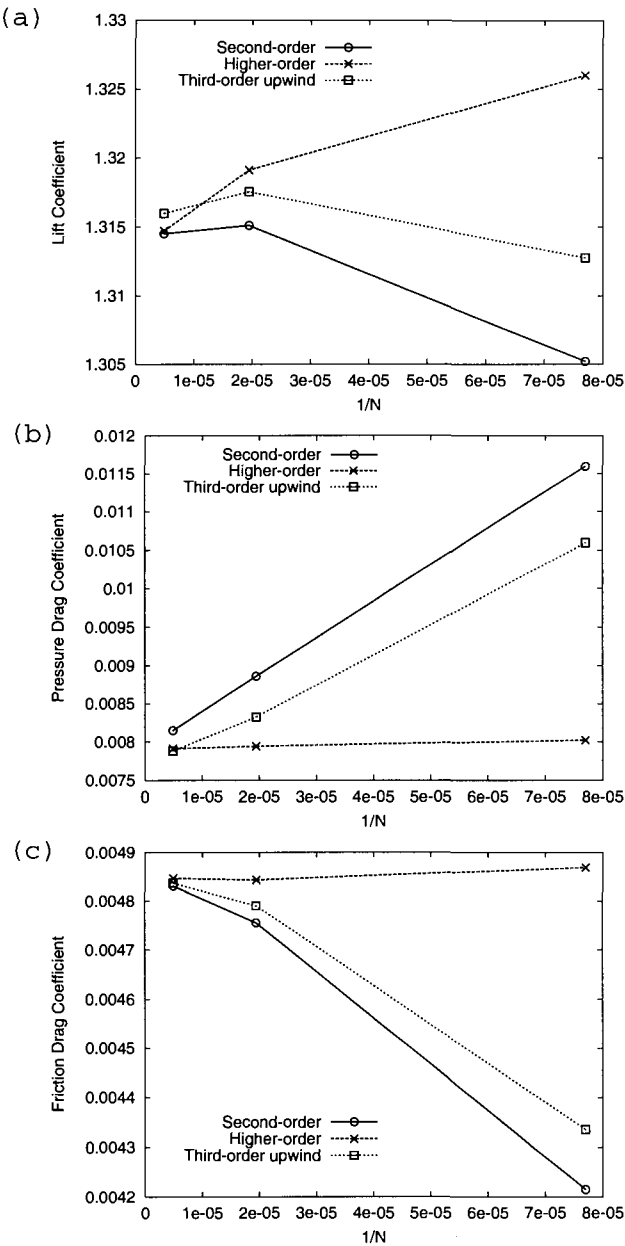
**Figure 6** Residual history for case 3.



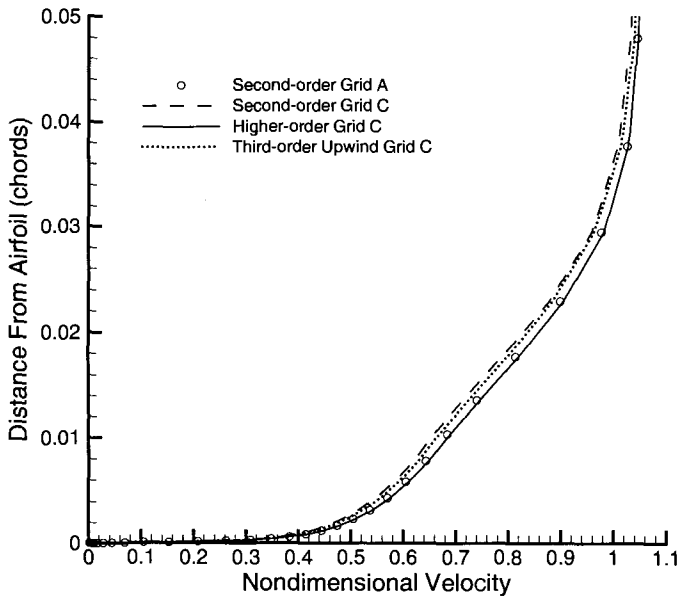
**Figure 7** Residual history for case 4.



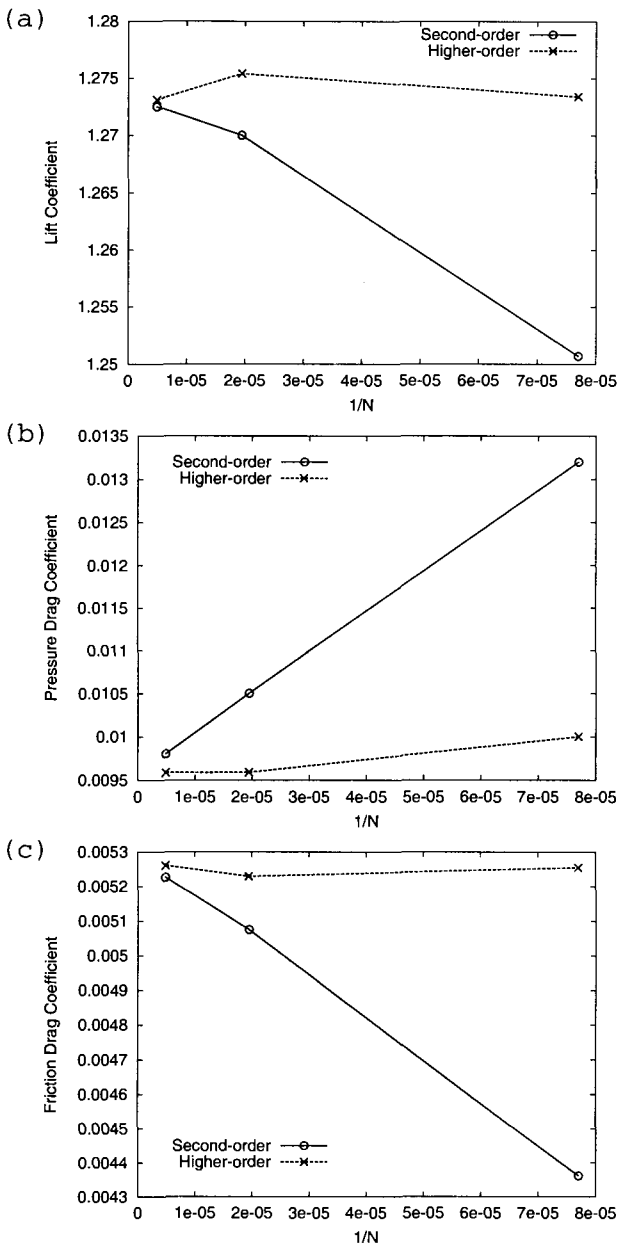
**Figure 8** Residual history for case 2 with reduced off-wall grid spacing.



**Figure 9** Grid convergence study for subsonic case, Baldwin-Lomax model. (a) Lift coefficient. (b) Pressure-drag coefficient. (c) Friction-drag coefficient.

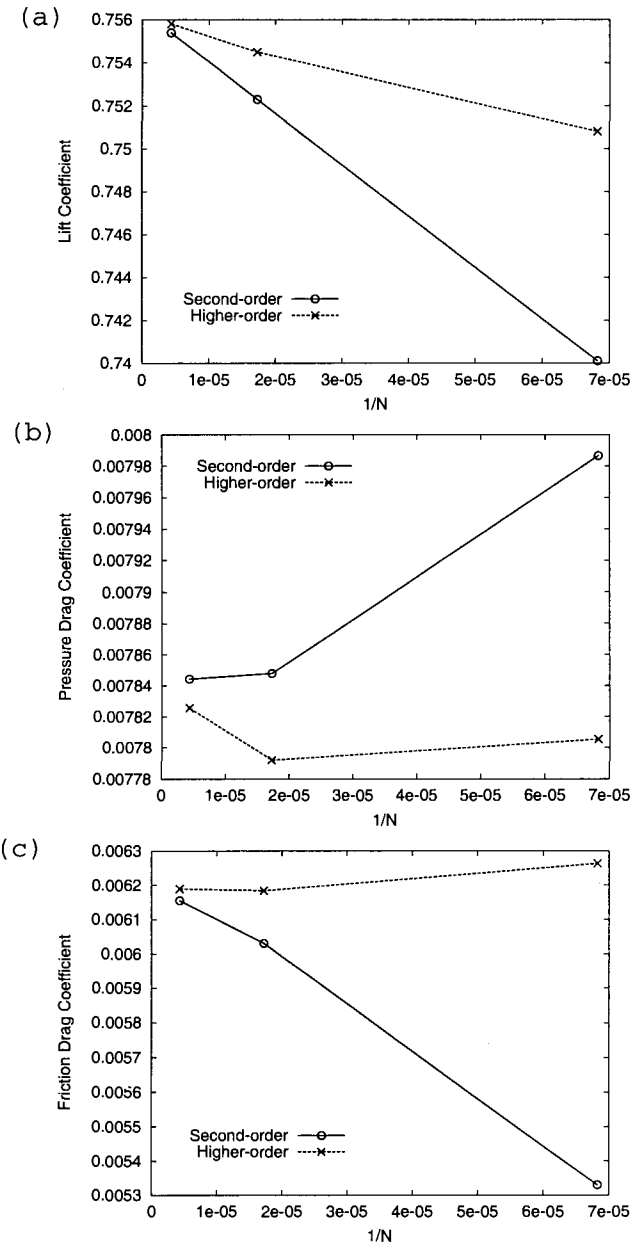


**Figure 10** Boundary-layer profile for subsonic case, upper surface at 85% chord, Baldwin-Lomax model.

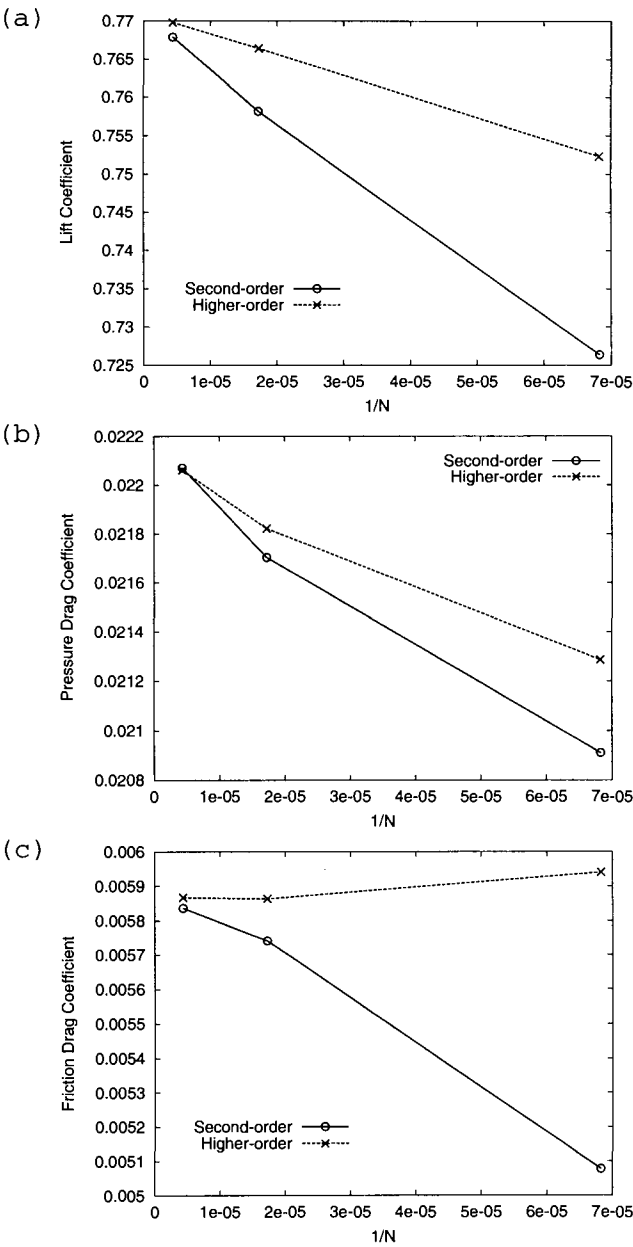


**Figure 11** Grid convergence study for subsonic case, Spalart-Allmaras model.  
(a) Lift coefficient. (b) Pressure-drag coefficient. (c) Friction-drag coefficient.

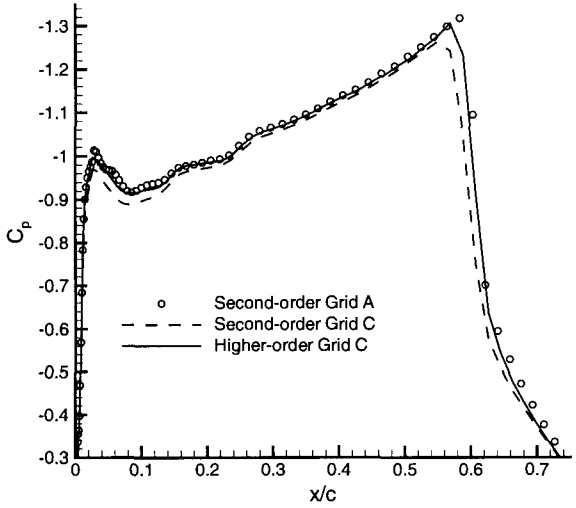




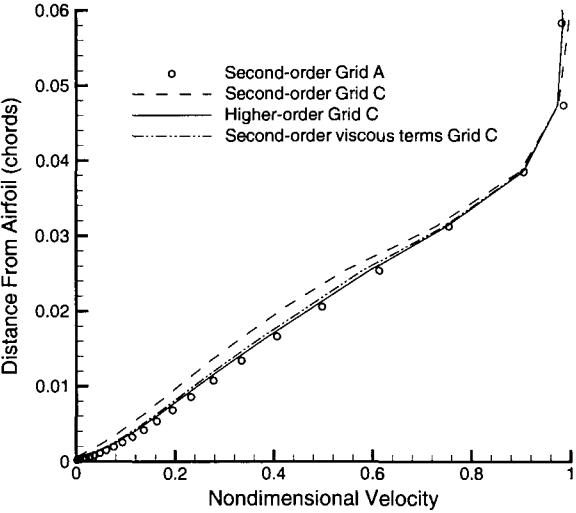
**Figure 12** Grid convergence study for transonic case I, Spalart-Allmaras model.  
(a) Lift coefficient. (b) Pressure-drag coefficient. (c) Friction-drag coefficient.



**Figure 13** Grid convergence study for transonic case II, Spalart-Allmaras model.  
(a) Lift coefficient. (b) Pressure-drag coefficient. (c) Friction-drag coefficient.



**Figure 14** Surface pressure distribution for transonic case II.



**Figure 15** Boundary-layer profile for transonic case II, upper surface at 95% chord.

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# Numerical Simulation of Hypersonic Boundary Layer Stability and Receptivity

Xiaolin Zhong, Chong W. Whang, and Yanbao Ma<sup>1</sup>

## 19.1 Introduction

The prediction of laminar-turbulent transition in hypersonic boundary layers is a critical part of the aerodynamic design and control of hypersonic vehicles[2]. The transition process is a result of the nonlinear response of laminar boundary layers to forcing disturbances, which can originate from many difference sources including free stream disturbances, surface roughness and vibrations. In an environment with weak initial disturbances, the path to transition consists of three stages: 1) receptivity, 2) linear eigenmode growth or transient growth, and 3) nonlinear breakdown to turbulence. The first stage is the receptivity process[7], which converts the environmental disturbances into initial instability waves in the boundary layers. The second stage is the subsequent linear development and growth of boundary-layer instability waves. Relevant instability waves for hypersonic boundary layers include the first mode and higher mode instabilities[5], the Görtler instability over concave surfaces[10], attachment line instability at leading edges, and the cross flow instability in three-dimensional boundary layers. The third stage is the breakdown of linear instability waves, and the transition to turbulence after the linear instability waves reach certain magnitudes[4].

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The stability and transition of supersonic and hypersonic boundary layers was reviewed by [5, 8, 9]. Most of our knowledge of hypersonic boundary layer stability is obtained by the linear stability theory (LST)[5]. Mack[5] found that there are higher acoustic instability modes in addition to the first-mode instability waves in supersonic and hypersonic boundary layers. Among them, the second mode becomes the dominant instability for hypersonic boundary layers at Mach numbers larger than about 4. The existence and dominance of the second mode has been validated by experimental studies[11]. Practical hypersonic vehicles have blunt noses in order to reduce thermal loads. It has been generally recognized that the bow shock in front of a blunt nose has strong effects on the stability and transition of the boundary layer behind it[9].

Due to the complexity of transient hypersonic flow fields involving the instability and receptivity process, an effective approach to studying hypersonic boundary layer stability and receptivity is the direct numerical simulation of the full Navier-Stokes equations. In [13] and [15], we presented and validated a new high-order (fifth and sixth order) upwind finite difference shock fitting method for the direct numerical simulation of hypersonic flows with a strong bow shock and with stiff source terms. The use of the high-order shock-fitting scheme makes it possible to obtain highly accurate mean flow and unsteady solutions, which are free of spurious numerical oscillations behind the bow shock. The method has been subsequently validated and applied to numerical studies of receptivity and stability of two and three-dimensional hypersonic flows over blunt bodies. This paper presents the results of the application of the new method to four test cases involving the receptivity and stability of hypersonic boundary layer flows over 2-D and 3-D blunt bodies. The four test cases are: 1) receptivity of 2-D Mach 15 flow over a parabola, 2) receptivity of 3-D Mach 15 flow over an elliptical cross-section blunt cone, 3) hypersonic Görtler vortices over concave surfaces, and 4) wave propagation in Mach 4.5 flow over a flat plate.

## 19.2 Governing Equations and Numerical Methods

The governing equations and numerical methods are briefly summarized here. Details can be found in previous papers for 2-D flows[13, 14]. The governing equations are the unsteady 3-D Navier-Stokes equations as follows:

$$\frac{\partial U^*}{\partial t^*} + \frac{\partial F_j^*}{\partial x_j^*} + \frac{\partial F_{vj}^*}{\partial x_j^*} = 0 \quad (19.1)$$

where superscript “\*” represents dimensional variables, and  $U^* = \{\rho^*, \rho^*u_1^*, \rho^*u_2^*, \rho^*u_3^*, e^*\}$ . The gas is assumed to be thermally and calorically perfect. The viscosity and heat conductivity coefficients are calculated using

Sutherland's law together with a constant Prandtl number,  $Pr$ . The equations are transformed into body-fitted curvilinear computational coordinates in a computational domain bounded by the bow shock and the body surface. The location and the movement of the bow shock is an unknown to be solved with the flow variables. We nondimensionalize the velocities with respect to the freestream velocity  $U_\infty^*$ , length scales with respect to a reference length  $d^*$  given by the body surface equation, density with respect to  $\rho_\infty^*$ , pressure with respect to  $p_\infty^*$ , temperature with respect to  $T_\infty^*$ , time with respect to  $d^*/U_\infty^*$ , vorticity with respect to  $U_\infty^*/d^*$ , entropy with respect to  $C_p^*$ , wave number with respect to  $1/d^*$ , etc. The dimensionless flow variables are denoted by the same dimensional notation but without the superscript “\*”.

The numerical methods for spatial discretization of the 3-D Navier-Stokes equations are a fifth-order shock-fitting scheme in streamwise and wall-normal directions, and a Fourier collocation method in the periodic spanwise flow direction for the case of a 3-D wedge geometry or in the azimuthal direction for the case of a cone geometry. The spatially discretized equations are advanced in time using a Low-Storage Runge-Kutta scheme of Williamson[12] of up to third order. The numerical accuracy of the computational results is evaluated by grid refinement studies and by comparison with available experimental or theoretical results. Detailed results of the code validation and error assessment can be found in [13, 15].

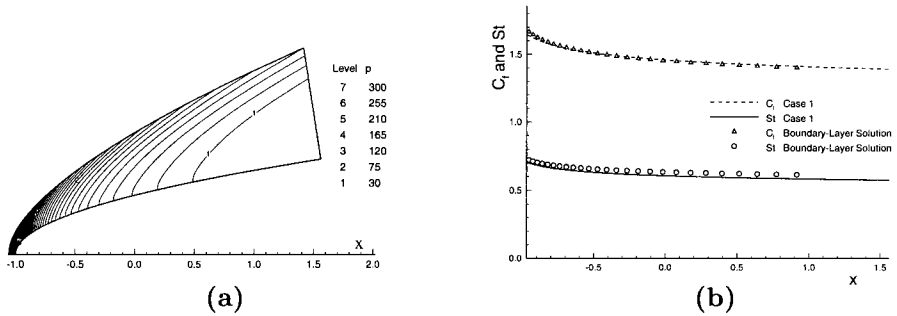
## 19.3 Results and Discussion

### Case 1. Receptivity of Mach 15 flow over a parabola

The first case is the numerical studies of the receptivity of 2-D hypersonic boundary-layer flows to weak free stream acoustic disturbances on a parabolic leading edge. The free stream disturbances are weak monochromatic planar acoustic waves with wave fronts normal to the center line of the body in the following form:

$$q_\infty(x, t)' = |q_\infty'| e^{ik_\infty(x - c_\infty t)} \quad (19.2)$$

where  $q$  represents the perturbation of any flow variables,  $|q_\infty'|$  is the wave amplitude constant,  $k_\infty$  is the wave number, and  $c_\infty$  is the wave speed before reaching the shock. For linear acoustic waves, perturbation amplitudes of nondimensional flow variables satisfy the following relations:  $|s'_\infty| = |v'_\infty| = 0$  and  $|\rho'_\infty| = |p'_\infty|/\gamma = |u'_\infty|M_\infty = \epsilon M_\infty$ , where  $\epsilon$  is a small number, and  $\epsilon M_\infty$  represents the relative amplitude of the free stream acoustic wave. In this paper, only the fast acoustic waves in the free stream are considered. The forcing frequency of the free stream acoustic wave is represented by a dimensionless frequency  $F$  defined by  $F = 10^6 \frac{\omega^* \nu^*}{U_\infty^{*2}}$ .

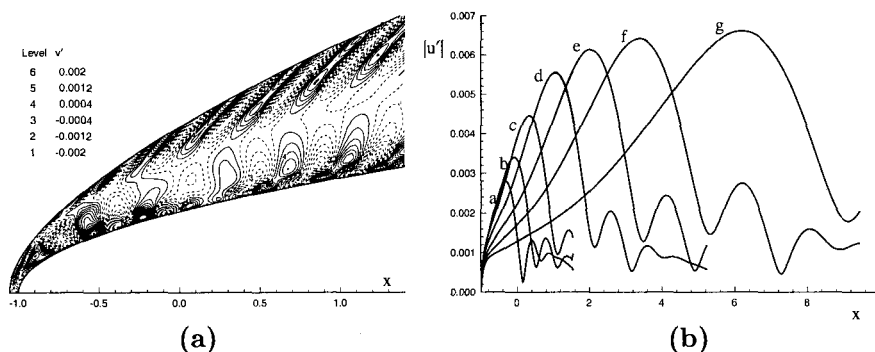


**Figure 1** Steady flow solutions: (a) pressure contours; (b) locally defined skin friction and heat transfer coefficients,  $c_f$  and  $st$ , along the body surface.

The flow conditions for the test case are:  $M_\infty = 15$ ,  $\gamma = 1.4$ ,  $Pr = 0.72$ ,  $Re_\infty = 6026.6$ ,  $T_\infty^* = 193K$ ,  $T_w^* = 1000K$ ,  $r^* = 0.0125\text{ m}$ , and  $d^* = 0.1\text{ m}$ . Figure 1 (a) shows the steady pressure contours. The figure shows that the steady flow over the parabola develops a favorable pressure gradient along the body surface. Due to the effects of nose bluntness, the pressure has a slight variation across the boundary layer in the region near the leading edge. The pressure distribution across the boundary layer approaches constants as the flow develops further down stream. The current numerical solutions of the Navier-Stokes equations are compared with approximate boundary layer equations results, in order to partially validate and to better understand the numerical simulation results. Figure 1 (b) shows a local shear stress coefficient  $c_f$  and Stanton number  $st$  defined by local boundary layer variables. The figure shows that the Navier-Stokes solutions agree very well with the boundary-layer results for both skin friction and heat transfer rates.

Having obtained the steady solutions, boundary-layer receptivity to free-stream acoustic disturbances is studied for two-dimensional hypersonic flow over a parabolic leading edge. The unsteady flow solutions are obtained by imposing acoustic disturbances on the steady flow solutions in the free stream. The subsequent interaction of the disturbances with the shock and the receptivity of the boundary layer over the parabola are computed by using the full Navier-Stokes equations. Seven unsteady cases are considered with nondimensional frequencies  $F$  ranging from 531 to 2655 and  $\epsilon$  ranging from  $5 \times 10^{-4}$  to  $10^{-1}$ . Figure 2 (a) shows the contours of instantaneous perturbation  $v'$ , after the flow field reaches a time periodic state, for the case of  $F = 2655$  at  $\epsilon = 5 \times 10^{-4}$ . The instantaneous contours show the interaction of the free stream disturbances with the bow shock and the development of

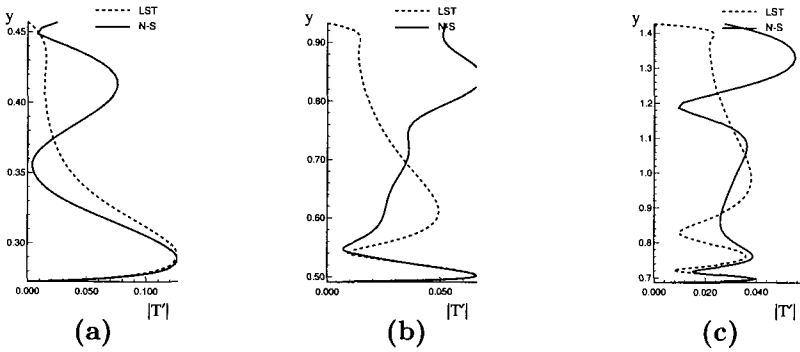




**Figure 2** (a): contours of instantaneous perturbations of velocity components  $v'$  for  $F = 2655$  at  $\epsilon = 5 \times 10^{-4}$ ; (b) profiles of horizontal velocity perturbation amplitudes for seven cases of  $F$ .

instability waves in the boundary layer on the surface. The wave patterns in the region outside the boundary layer are different from those inside the boundary layer. The wave patterns outside the boundary layer are the results of free stream waves passing through the shock and propagating in the flow field. On the other hand, the wave patterns inside the boundary layer show very different structures. These structures are dominantly the boundary waves induced by the forcing waves. The waves in the boundary layer region on the wall, as shown in Fig. 2 (a), contain two separate zones of different wave patterns. The first zone is located in the region of  $x < 0.2$  and the second one is located in the region of  $x > 0.2$ . In the first zone, there is only one peak in the oscillation magnitudes across the boundary layer. On the other hand, the second zone develops oscillations away from the wall with two magnitude peaks. Therefore, these results suggest that the waves developed in the first wave zone correspond to the first mode waves, while those in the second zone are dominated by the second mode waves.

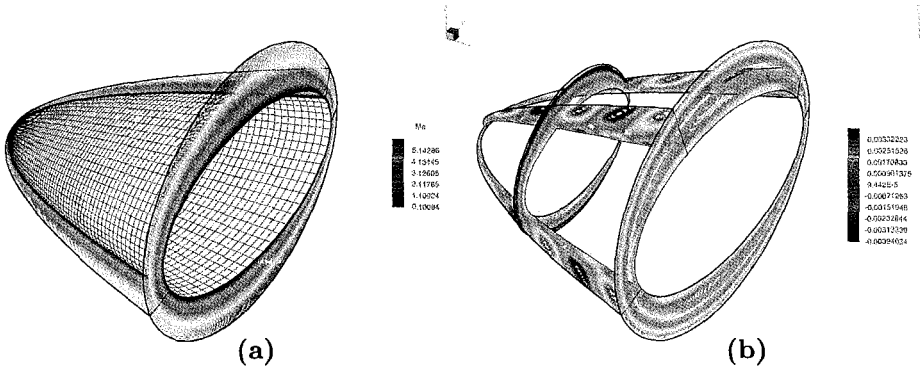
Temporal Fourier analysis is carried out on the numerical solutions of the perturbations of unsteady flow variables after a time periodic state has been reached in a simulation. The identification of the wave modes induced by free stream disturbances is further examined by comparing the linear stability eigenfunctions with Fourier amplitudes of the Navier-Stokes solutions for the receptivity problem. For the purpose of comparison, the eigenfunctions are normalized by their respective first peak values from the wall. Figure 3 shows such comparisons for the temperature amplitudes along grid lines normal to the parabola surface at several  $i$  grid stations for the case of  $F = 2655$ . The figure shows that the numerical solutions at earlier stations in first wave zone agree closer to the linear stability first mode inside of the boundary



**Figure 3** Variation of the temperature amplitudes along grid lines normal to the parabola surface at three grid stations: (a)  $i = 50$  LST: 1st mode, (b)  $i = 100$  LST: 2nd mode, (c)  $i = 140$  LST: 3rd mode ( $F = 2665$  at  $\epsilon = 5 \times 10^{-4}$ ).

layer near the wall. At later stations, the numerical results gradually change to profiles that are closer to the linear stability second mode at  $i = 100$ , the linear stability third mode at  $i = 140$ . The figure also shows that due to the effects of interaction of the bow shock with the flow disturbances, the numerical solutions have very complex flow distributions in the inviscid region outside of the boundary layer. The variations outside the boundary layer are those induced by the forcing disturbances and their interaction with the bow shock. The numerical results and linear stability results agree mainly inside the boundary layer on the wall. Therefore, the results indicate that the instability waves excited in the first region are dominated by the first mode, followed by a gradual transition to the second and third modes downstream.

The nose bluntness in a study of hypersonic boundary layer receptivity can be characterized by a nondimensional Strouhal number defined by  $S = \frac{\omega^* r^*}{U_\infty^*}$ , where  $r^*$  is the nose radius. The increase of forcing frequency at a fixed nose radius is equivalent to the increase of relative nose bluntness. The effects of nose bluntness and forcing frequencies on the receptivity of a hypersonic boundary layer is investigated in this paper by considering seven test cases of different forcing frequencies  $F$  while holding all other flow parameters fixed. The nondimensional frequencies of the seven cases are in the range of  $F = 531$  to 2655 corresponding to the Strouhal numbers in the range of  $S = 0.4$  to 2. Figure 2 (b) shows the distribution of Fourier amplitudes of the fundamental frequency for horizontal velocity perturbations along a grid line near the body surface for the seven test cases of different frequencies, where cases a to g in the figure correspond to increasing frequency from  $F = 531$  to 2655. The results show similar receptivity wave patterns for all frequencies that the first modes



**Figure 4** Steady Mach number contours and unsteady contours of instantaneous entropy perturbations induced by freestream planar acoustic disturbances.

grow and decay, followed by the growth and decay of second and third modes respectively. As frequency decreases, the maximum first mode amplitudes become larger, their locations move downstream, and the first mode instability regions extend to much longer ranges. For the present cases, the maximum first mode amplitudes reach a peak value at a nondimensional frequency around  $F = 500$  to  $600$ . The second mode instability regions appear further downstream to the first mode regions with higher local Reynolds numbers. Similar to the first mode cases, the second modes increase in strength as frequencies decrease, though they are much weaker than the first mode waves for the current flow conditions. Since, the frequency increase corresponds to the increase of the Strouhal number and the relative nose radius, the increase of nose bluntness leads to the decrease in the receptivity coefficient in general.

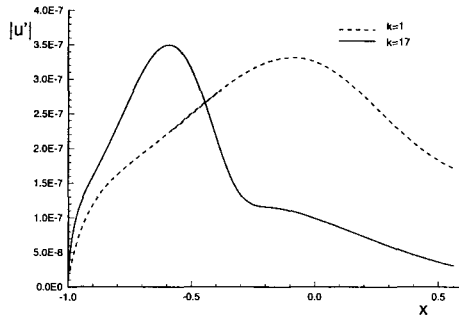
## Case 2. Receptivity of blunt elliptical cone at $M = 15$

The second case is the receptivity of a 2:1 blunt elliptical cone in a Mach 15 freestream at zero angle of attack. The body surface is specified by:

$$x^* = \frac{y^{*2}}{2r_y^*} + \frac{z^{*2}}{2r_z^*} - d^* \quad (19.3)$$

where  $r_y^*$  and  $r_z^*$  are the major and minor nose radius of curvatures. The flow conditions are:  $M_\infty = 15$ ,  $\epsilon = 5 \times 10^{-3}$ ,  $T_w^*/T_{0\infty}^* = 0.167$ , Reference length  $d^* = 0.01m$ ,  $Re_\infty = 0.60266 \times 10^5/m$ , Nose Radius  $r_y^* = .0125m$ , and  $r_y^* : r_z^* = 4 : 1$ . The results presented in this paper are obtained using two sets of grids:  $92 \times 31 \times 64$  and  $92 \times 61 \times 64$  grids.

Figure 4 (a) shows the 3-D steady solutions for Mach number contours. The uneven strength of the bow shock in the major and minor axes creates the

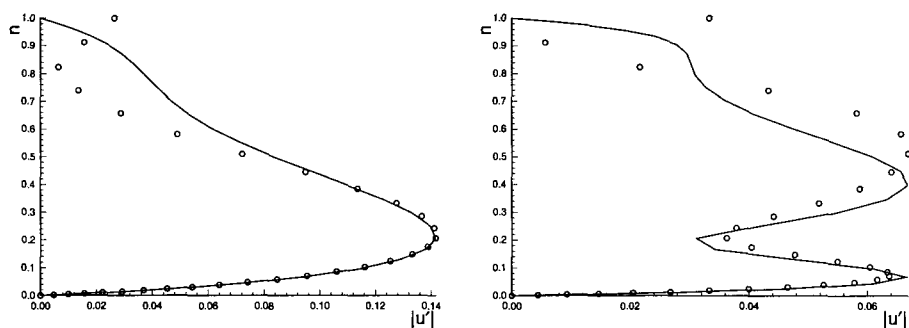


**Figure 5** The distribution of streamwise velocity perturbation amplitudes along the major ( $k = 1$ ) and minor ( $k = 17$ ) axes

circumferential pressure gradient and cross flow velocity. The cross mass flow thickens the boundary layer and enlarges the standoff distance in the minor axis. The numerical results on the heating rate at the stagnation point are compared with those of boundary layer theory obtained by Fay and Riddell [3]. The Navier-Stokes simulation predicts that the stagnation-point heating coefficient is 0.0168, while the Fay and Riddell theory gives 0.0163. The agreement between the computational and theoretical results is good.

Having obtained the steady flow solution, the generation of boundary-layer waves by freestream acoustic disturbances is simulated with a freestream disturbance wave of dimensionless frequency  $F = 4247.9$ . Figure 4 (b) shows 3-D contours for the instantaneous entropy perturbation  $s'$  after the unsteady computations reach a periodic state. The instantaneous contours show the development of wave modes in the boundary layer along the surface. Figure 5 shows the distribution of  $|u'|$  amplitudes along the major and minor axes at a wall-parallel grid line near the wall surface. Both Figs. 4 (b) and 5 show that the velocity perturbations in the boundary layer develop a peak in amplitudes near the leading edge region. The peaks are the maximum induced strength of the first mode in the boundary layer. The freestream forcing waves enter the boundary layer and always generate first modes near the leading edge, which decay rapidly to convert to second modes. The figures also show that the peaks of the first modes are reached earlier in the minor axis ( $k = 17$ ) than the major axis ( $k = 1$ ). This is due to the fact that the minor axis has a thicker boundary layer which develops a peak in  $|u'|$  perturbations faster.

The preceding picture of hypersonic receptivity near the leading edge can be further analyzed by comparisons with the first mode and second mode structure obtained by linear stability analysis. The stability analysis is carried out at each local mean flow section using local parallel assumption. Figure 6 compares the perturbation amplitudes obtained by simulation with the



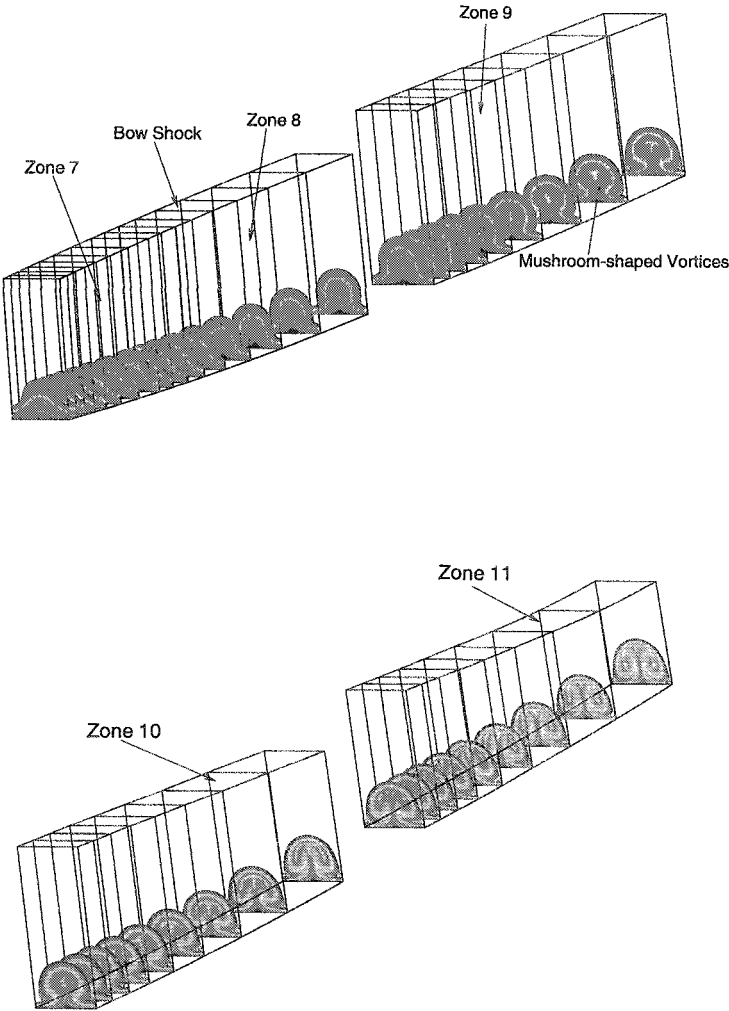
**Figure 6** Distribution of streamwise velocity perturbation amplitudes along wall normal direction for  $k = 1$  (left figure, line: LST first mode, circles: N-S) and  $k = 17$  (right figure, line: LST second mode, circles: N-S) grid lines.

eigenfunctions of the first modes in the major axis and the second modes in the minor axis. The numerical solutions agree well with LST results in the boundary layer. The comparisons show indeed that the induced waves in the boundary layer are dominated by the first modes in the major axis, while the solutions have already developed a second mode structure in the minor axis. The comparison also shows that the numerical solutions capture the mode structure accurately in the boundary layer region.

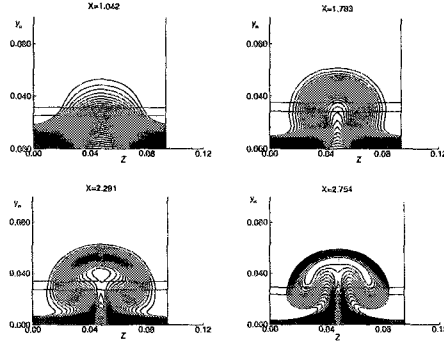
### Case 3. Hypersonic Görtler vortices over concave inlet surfaces

The third case is hypersonic flow over a blunt concave body surface with nonlinear development of 3-D Görtler vortices. Concave compression surfaces in front of hypersonic inlets are susceptible to Görtler counter rotating instability vortices, in addition to other hypersonic instability waves in hypersonic boundary layers[10]. The nonlinear breakdown of the Görtler vortices is a critical step in the laminar turbulent transition process of concave hypersonic boundary layers. Experiments showed that the interaction of nonlinear growth of Görtler vortices and other forms of disturbances plays an important role in the transition process. In the nonlinear growth process, a Görtler vortex develops into a mushroom shaped vortices because the counter-rotating vortices pump fluid with a low streamwise velocity away from the wall. Such mushroom shaped Görtler vortices are characterized by the existence of two regions: the peak region with low velocity and the valley region with high velocity.

The specific flow considered is a  $M_\infty = 15$  flow over a blunt wedge with a concave surface. The freestream Reynolds number based on the nose radius is  $Re_\infty = 15075.3$ . The total length of the flow field is about 40 nose radius. The



**Figure 7** Distributions of iso-contours of streamwise mean velocity along the streamwise direction. The size of grids of each zone is  $161 \times 121 \times 64$ . Mushroom shaped vortices develop as flow moves downstream due to nonlinear effects.



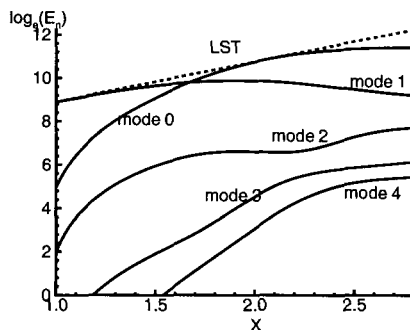
**Figure 8** Sectional streamwise velocity distribution at four different streamwise locations.

steady mean flow solution is 2-D and is computed by a separate simulation using the same shock fitting scheme. Having obtained the steady solution, the linear and nonlinear growth of the Görtler vortices are simulated in a section of the downstream flow field beginning at  $x$  coordinate of 19.4 nose radius. The Görtler vortices are induced by a small perturbations imposed over the steady solution in the inlet of in a localized computational domain. These inlet disturbances are eigenfunctions of a first-mode Görtler vortex obtained from linear stability analysis based on the simulated mean flow. The effect of surface on the Görtler vortex is measured by a nondimensional Görtler number defined by

$$G = Re_{\delta} \sqrt{\frac{\delta}{R}} \quad (19.4)$$

where  $\delta = \sqrt{\frac{\nu_{\infty}^* x^*}{U_{\infty}^*}}$ , and  $R$  is the local radius of curvature of the body surface. In the present test case, the Görtler number at the entrance of the disturbance simulation is 6.7, in which the linear stability analysis predicts that the strength of Görtler vortices is spatially growing as flow moves downstream.

In order to study the nonlinear effects of Görtler instability, we introduce relatively large amplitude disturbances at the inlet of the computational domain. The maximum amplitude of the velocity perturbations in the inlet Görtler mode is  $0.25U_{\infty}$ . Figure 7 shows streamwise velocity as flow moves downstream for the case with inlet perturbations. The figure shows that as the flow moves downstream, the initial linear Görtler vortices develop into mushroom shaped vortices because of nonlinear effects. The bow shock does not have much effect on the flow field because Görtler vortices develop in the viscous layer away from the shock. The spanwise iso-contours of streamwise velocity at four different streamwise locations are shown in figure 8. The peak



**Figure 9** Nonlinear evolution of disturbance energy for various spanwise Fourier modes.

and valley regions are clearly shown in the figure. While the middle region (peak) tends to go up, others become narrower.

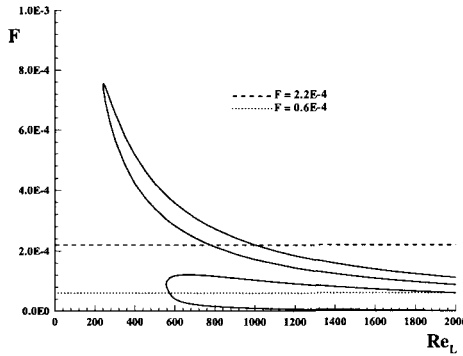
The nonlinear growth of the Görtler vortices can be represented by Fourier spectral energy modes. The streamwise development of the perturbation kinetic energy of each Fourier mode is shown in figure 9. Initially, the Görtler vortex (mode 1) grows linearly while other modes are negligible. As the nonlinear interactions between the modes become more significant downstream, the mean flow correction mode (mode 0) grows considerably, and it dominates mode 1 as the flow moves downstream. In this nonlinear regime, the main interaction is between the fundamental and mean flow correction mode (mode 0). However, effects of higher modes (mode 2, 3, etc) increase as flow moves downstream. At further downstream, these higher modes will increase further and be saturated.

#### **Case 4. Mach 4.5 boundary layer over a flat plate**

The fourth case is to use the fifth-order shock fitting schemes to simulate the linear and nonlinear stability of Mach 4.5 flow over a flat plate. The purpose of this work is to conduct numerical studies of nonlinear breakdown of instability wave in hypersonic boundary layers for both perfect and nonequilibrium real gas flow. Only the perfect gas solutions are presented in this paper. In order to use the shock fitting scheme, the steady flow fields near the leading edge are computed by using a TVD (Total Variation Diminishing) shock capturing scheme. The TVD results near the leading edge are used as inflow conditions for the shock fitting schemes in the down stream flow field. For unsteady flow simulations, the instability wave are induced by weak blowing and suction through a narrow slot on the wall near the leading edge.

The specific flow conditions of the test case is  $M_\infty = 4.5$  flow over a



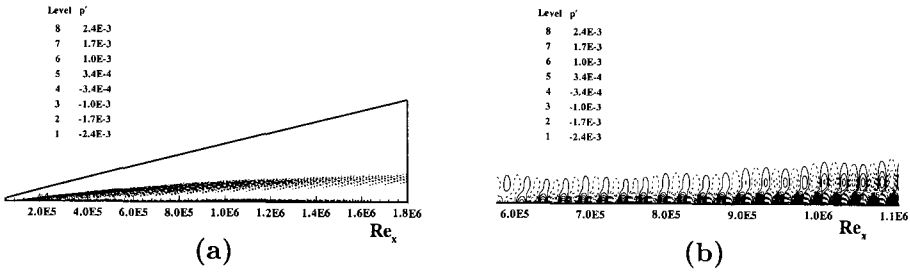


**Figure 10** Neutral curves of stability ( $F$  vs.  $Re_L$ ) for two-dimensional first- and second-mode disturbances at  $M = 4.5$ .

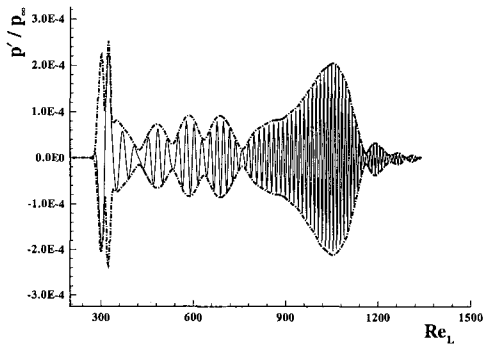
flat plate. The computational domain begins at  $x = 0.006m$  and ends at  $0.25m$ , corresponding to the local Reynolds numbers ranging from  $Re_x = 0.432 \times 10^5$  to  $Re_x = 1.8 \times 10^6$ . In the studies of boundary stability, the Reynolds number based on the length scale of boundary layer thickness,  $Re_L = U_\infty^* L / \nu_\infty^* = \sqrt{Re_x}$ , are often used, where  $L = \sqrt{\frac{\nu_\infty^* x^*}{U_\infty^*}}$ . The values of  $Re_L$  of the computational domain are from 207.9 to 1341.6. The computational domain is divided into 7 zones along streamwise direction with a total of 1221 grid points in the streamwise direction and 121 points in the wall-normal direction. The nondimensional frequency  $F$  of the blow and suction near the inlet of the computational domain is chosen so that the first or the second mode are unstable according to local linear stability theory. The first and second-mode unstable regions for Mach 4.5 flat-plate boundary layers, obtained by using Malik's[6] local LST method, are shown in Fig. 10. According to this figure, two different frequencies are chosen for the blowing and suction perturbations:  $F = 2.2 \times 10^{-4}$  (second mode) and  $F = 0.6 \times 10^{-4}$  (first mode) respectively. The simulation results are compared with the LST predictions for the purpose of code validation.

In this paper, only the results of an unsteady case with forcing wave frequency at  $F = 2.2 \times 10^{-4}$  are presented. Figure 10 shows that as the wave develops downstream, the second mode wave goes through a stable region of decay first, then a narrow unstable region of growth, followed by a stable region again afterward. The region of second mode instability predicted by LST is from  $Re_L = 836$  (branch I neutral stability point) to 1028 (branch II neutral stability point). According LST results, the second mode is the most unstable mode in this region, the wave fields induced by the initial blowing and suction are expected to be dominated by the second mode waves there.

Figure 12 (a) shows the contours of instantaneous pressure perturbation



**Figure 11** Unsteady solutions: (a) instantaneous pressure perturbation contours; (b) a local wave pattern at  $M = 4.5$ ,  $F = 2.2 \times 10^{-4}$ .



**Figure 12** Unsteady solutions of wall pressure fluctuations at  $M = 4.5$ ,  $F = 2.2 \times 10^{-4}$ .

after the flow field reaches a periodic state. It is clear that blowing and suction introduce boundary layer disturbance as well as Mach waves radiation outside the boundary layer. The wave inside the boundary layer on the wall is a combination of several wave modes near the blow and suction region. As the waves develop downstream reaching the branch I neutral stability point of the second mode, the amplitude of the second mode increases because of instability. In this region, the disturbance in the boundary layer is dominated by the second mode wave. The second mode wave decays after passing through the branch II stability point as shown in the contours. Figure 12 (b) shows a local wave patterns in the second growth region. The figure shows a typical second mode wave pattern.

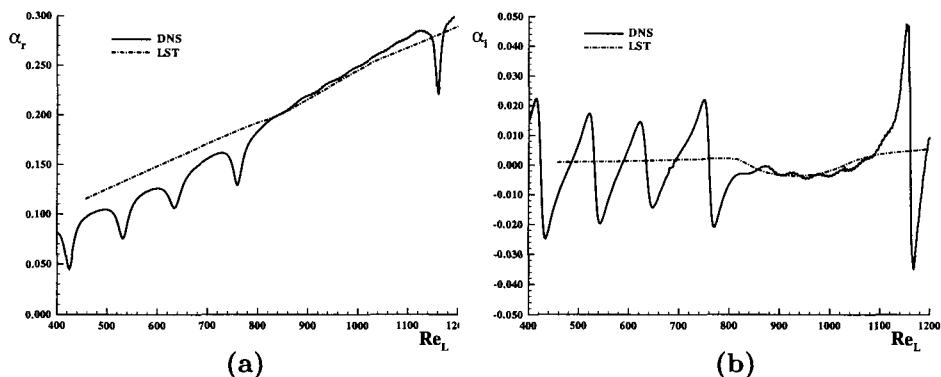
The distribution of pressure perturbations along the surface is plotted in

Fig. 12. The envelope (dash line) of the pressure perturbations shows the change of amplitude as the waves develop downstream. The figure shows that the disturbance enters the second-mode unstable region after going through an initial modulation region. The amplitude of disturbance reaches the maximum at the branch II neutral point of about  $Re_L = \sqrt{Re_x} = 1052$  in the simulation result. After passing the branch II neutral point, the amplitude of disturbance decays rapidly in the second-mode stable region. The region of second mode instability obtained by the numerical simulation as shown in Fig. 12 is approximately in the  $Re_L$  range of 840 to 1052. These values agree very well with the LST prediction of the range from 836 to 1028.

From the numerical results of the unsteady solutions, the streamwise wave numbers and growth rates of the disturbance can be computed. The wave numbers and growth rates from numerical simulation are compared with the eigenvalues obtained from local linear stability analysis based on the same mean flow. Figure 13 shows the comparison of streamwise wave numbers and growth rates between the results from numerical simulation and LST respectively. In the figure, the real part and the imaginary parts of  $\alpha$  are the wave numbers and growth rates respectively. The figures shows that there is a good agreement between the numerical and LST results on  $\alpha_r$  and  $\alpha_i$  in the second-mode's unstable region, located in the region of  $840 < Re_L < 1052$ . The simulation results do not agree with the LST results in the region near the entrance because of the influence of the forcing waves in the blow and suction region. According to Balakumar and Malik's study[1], environmental forcing would generate several discrete modes as well as continuous spectra. The instability waves may be a combination of discrete modes and continuous spectra as well as forcing waves. In the second-mode unstable region, the second-mode waves are dominant. Therefore, it is reasonable that there is good agreement between the results of LST and numerical simulation in this region. We can also compare the eigenfunctions between the results from simulation and LST. Figure 14 shows the structure of velocity disturbance of simulation and LST calculations at the station  $Re_L = 1007.57$ . Despite the difference in the outside boundary layer, the agreement is excellent. At this location, Reynolds number is large, therefore, the nonparallel effect is small. Overall, the numerical simulation can capture the structure of disturbance waves very well.

## 19.4 Concluding Remarks

The stability and receptivity of hypersonic boundary layers are studied by using a fifth-order shock fitting scheme, in order to simulate accurately physical interactions between shock waves and physical waves in the flow fields. Four test cases are discussed in this paper. The results show that the

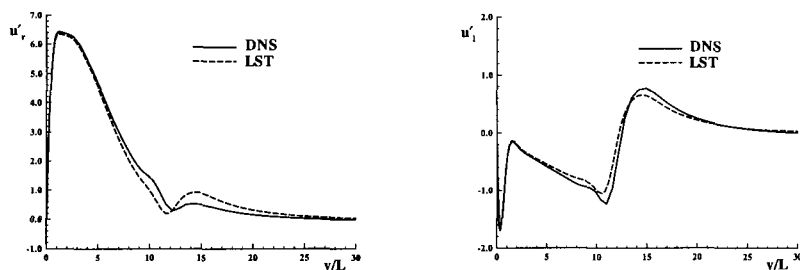


**Figure 13** Comparison of streamwise real and imaginary parts of wave number of obtained by simulation and LST calculation at  $M = 4.5$ ,  $F = 2.2 \times 10^{-4}$ .

fifth-order shock fitting scheme is suitable for conducting detailed numerical studies on supersonic and hypersonic boundary layer receptivity, stability, and transition. For the case of receptivity of Mach 15 flow over a parabola and a blunt elliptical cone, the free stream acoustic disturbances generate first, second, and third mode waves in the boundary layer on the wall. The results indicate that the receptivity coefficients increase when the relative nose radius decreases. For the case of hypersonic Görtler vortices over concave inlet surfaces, the nonlinear development of high and low velocity regions in Görtler vortices produces mushroom shaped vortices. The numerical method is also tested in a Mach 4.5 boundary layer over a flat plate. There is a good agreement between our numerical simulation results and the LST results.

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**Figure 14** Comparison of structure of eigenfunctions at  $M = 4.5$ ,  $Re_L = 1007.57$ , and  $F = 2.2 \times 10^{-4}$ .

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# Time-Dependent Simulation of Incompressible Flow in a Turbopump Using Overset Grid Approach

Cetin Kiris<sup>1</sup> and Dochan Kwak<sup>1</sup>

## ABSTRACT

This paper reports the progress being made towards complete unsteady turbo-pump simulation capability by using overset grid systems. A computational model of a turbo-pump impeller is used as a test case for the performance evaluation of the MPI, hybrid MPI/Open-MP, and MLP versions of the INS3D code. Relative motion of the grid system for rotor-stator interaction was obtained by employing overset grid techniques. Unsteady computations for a turbo-pump, which contains 114 zones with 34.3 Million grid points, are performed on Origin 2000 systems at NASA Ames Research Center. The approach taken for these simulations, and the performance of the parallel versions of the code are presented.

## 20.1 INTRODUCTION

The motivation of this effort is based on two primary elements. First, the entire turbo pump simulation intends to provide a computational framework for the design and analysis for the liquid rocket engine fuel supply system. This effort is part of the High Performance Computing and Communications (HPCC) advanced

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technology application projects. The second objective of this research is to support the design of liquid rocket systems for the space transportation. Since the space launch systems in the near future are likely to rely on liquid rocket engines, increasing the efficiency and reliability of the engine components is an important task. One of the major problems in the liquid rocket engine is to understand the fluid dynamics of fuel and oxidizer flows from the fuel tank to plume. Understanding the flow in the turbo pump through numerical simulation will be of significant value toward finding a better design that is simpler yet more efficient and robust with less manufacturing cost. Until recently, the pump design process was not significantly different from that of decades ago. The current semi-empirical turbomachinery design process does not account for the three-dimensional (3-D) viscous phenomena in the pump flows. Some of these 3-D viscous phenomena include wakes; the boundary layers in the hub, the shroud and the blades; junction flows; and tip clearance flows. Even though computational fluid dynamics (CFD) applications in turbines have been reported widely in the literature, the applications in entire-pump simulations are quite limited. The objective of this paper is to present, and evaluate a parallel computational procedure that simulates the incompressible flow through the entire turbo pump configuration.

A substantial computational time reduction for these 3D unsteady flow simulations is required to reduce design cycle time of the pumps. Part of this speed up will be due to enhancements in computer hardware platforms. The remaining portion of the speed-up will be contributed by advances in algorithms and by efficient parallel implementations. The following section outlines the initial effort and steps taken in order to reach this speed-up.

## 20.2 NUMERICAL METHOD

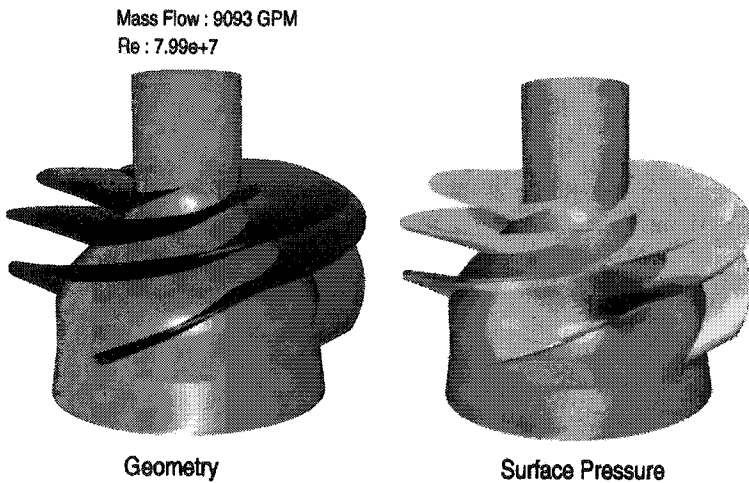
The present computations are performed utilizing the INS3D computer code, which solves the incompressible Navier-Stokes equations for both steady-state and unsteady flows. The numerical solution of the incompressible Navier-Stokes equations requires special attention in order to satisfy the divergence-free constraint on the velocity field because the incompressible formulation does not yield the pressure field explicitly from the equation of state or through the continuity equation. One way to avoid the numerical difficulty originated by the elliptic nature of the problem is to use an artificial compressibility method, developed by Chorin<sup>1</sup>. The artificial compressibility algorithm, which introduces a time-derivative of the pressure term into the continuity equation; the elliptic-parabolic type partial differential equations are transformed into the hyperbolic-parabolic type. A family of flow solvers has been developed<sup>2-3</sup> based on this algorithm. Since the convective terms of the resulting equations are hyperbolic, upwind differencing can be applied to these terms. The current versions, designated as INS3D code, use Roe's flux-difference splitting<sup>4</sup>. The third and fifth-order upwind differencing used here is an implementation of a class of high-accuracy flux-differencing schemes for the



compressible flow equations. To obtain time-accurate solutions, the equations are iterated to convergence in pseudo-time for each physical time step until the divergence of the velocity field has been reduced below a specified tolerance value. The total number of sub-iteration required varies depending on the problem, time step size and the artificial compressibility parameter used, and typically ranges from 10 to 30 sub-iterations. The matrix equation is solved iteratively by using a non-factored Gauss-Seidel type line-relaxation scheme<sup>5</sup>, which maintains stability and allows a large pseudo-time step to be taken.

Details of the numerical method can be found in Refs. 2-3. The GMRES scheme has also been utilized for the solution of the resulting matrix equation<sup>6</sup>. Computer memory requirement for the flow solver INS3D with line-relaxation is 35 times the number of grid points in words, and with GMRES-ILU(0) scheme is 220 times the number of grid points in words. When a fast converging scheme, such as a GMRES-ILU(0) solver, was implemented into the artificial compressibility method, previous computations showed that reasonable agreement was obtained between computed results and experimental data. The line-relaxation scheme in artificial compressibility method could be very expensive for time accurate computations and could lead to erroneous solutions if incompressibility is not enforced in each physical time step.

#### TURBOPUMP INDUCER

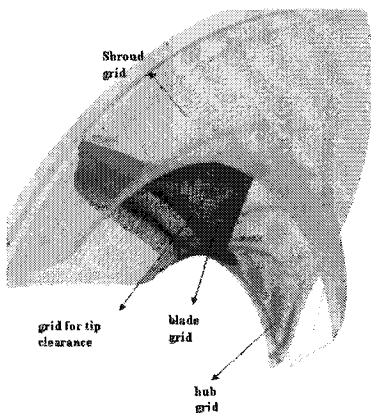


**Figure 1.** Geometry and surface pressure for a pump inducer.

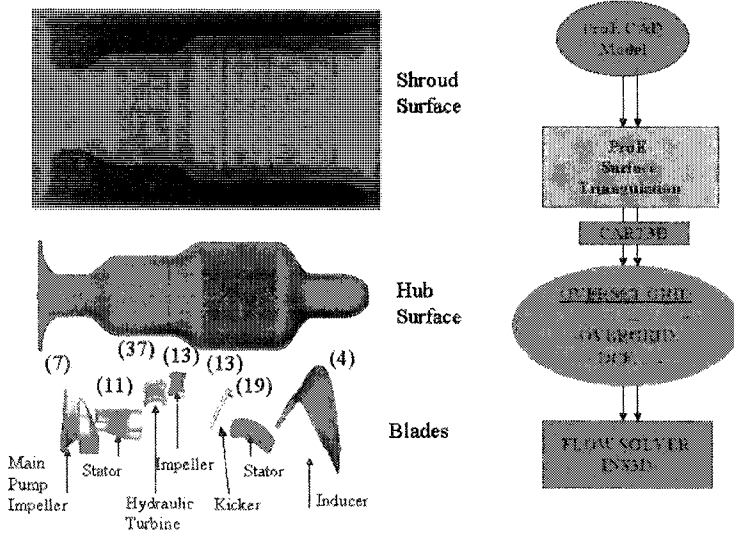
### 20.3 APPROACH AND COMPUTATIONAL MODELS

The geometry for the liquid oxygen pump has various rotating and stationary components, such as inducer, stators, kicker, and hydraulic turbine, where the flow is extremely unsteady. Figure 1 shows the geometry and computed surface pressure of the inducer from steady-state components analysis. When rotating and stationary parts are included, time-dependent simulations need to be carried out due to unsteady interactions.

To handle the geometric complexity, an overset grid approach is used. The overset structured grid approach to flow simulation has been utilized to solve a variety of problems in aerospace, marine, biomedical and meteorological applications. Flow regimes can range from simple steady flows as that of a commercial aircraft, to unsteady three-dimensional flows with bodies in relative motion as in the case of turbopump configurations. A geometrically complex body is decomposed into a number of simple grid components, as shown in figure 2. The freedom to allow neighboring grids to overlap arbitrarily implies these grids can be created independently from each other and each grid is typically of high quality and nearly orthogonal. Connectivity between neighboring grids is established by interpolation at the grid outer boundaries. Addition of new components to the system and simulating arbitrary relative motion between multiple bodies are achieved by establishing new connectivity without disturbing the existing grids. Scalability on parallel compute platforms is naturally accomplished by the already decomposed grid system. For certain problems, it is more efficient to gather the grids into groups of approximately equal sizes for parallel processing.



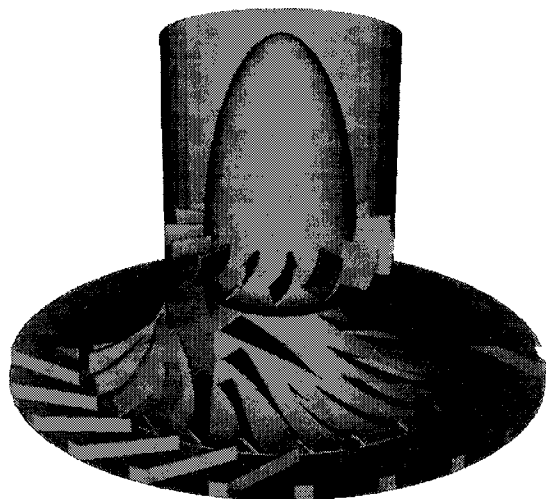
**Figure 2.** Overset grid system for the impeller long blade section with tip clearance



**Figure 3.** A pump model and steps taken in the simulation procedure.

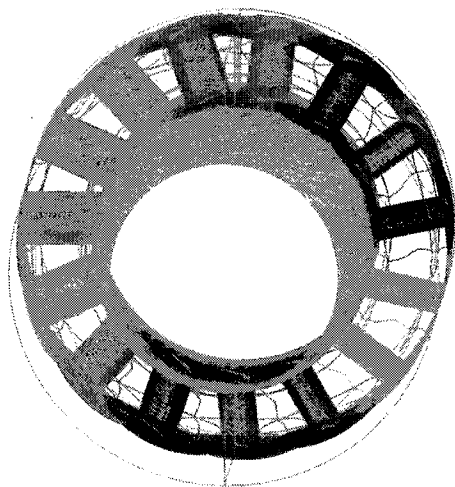
In order to compute the flow on grids with moving boundaries, the overset grid scheme in OVERFLOW-D<sup>7</sup> code is incorporated with the INS3D solver that new connectivity data is obtained at each time step. The overlapped grid scheme allows sub-domains move relative to each other, and provides a general flexibility when the boundary movement creates large displacements. Figure 3 shows the model for boost pump and the steps taken in the simulation procedure. The numbers in figure 3 indicate the number of the blades in each section. The computational grid has been generated by using the OVERGRID<sup>8</sup> software. OVERGRID is a unified graphical interface for performing overset grid generation. The software contains general grid manipulation capabilities as well as modules that are specifically targeted for efficient creation of overlapping grids. General grid utilities include functions for grid transformation, redistribution, smoothing, concatenation, extraction, extrapolation, projection, and many others. Functions especially useful for overset grids include feature curve extraction, hyperbolic and algebraic surface grid generation, hyperbolic volume grid generation, and Cartesian box grid generation. Visualization is achieved using OpenGL while widgets are constructed with Tcl/Tk. The software is portable between various platforms from UNIX workstations to personal computers.

In order to demonstrate the current unsteady solution capability, the SSME shuttle upgrade pump configuration has been selected. Figure 4 shows the geometry of the test rig for this pump being tested at NASA-MSFC facilities. In this particular configuration, the SSME impeller is unshrouded.



**Figure 4.** Geometry of SSME-rig1 shuttle upgrade pump impeller

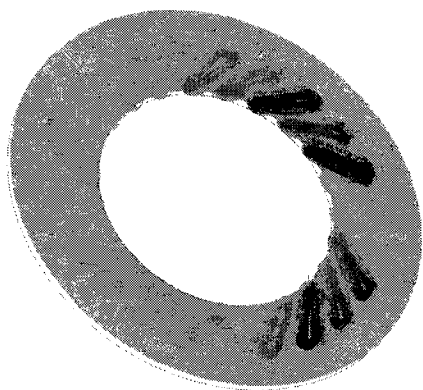
The computational grid for the inlet guide vanes, impeller and diffuser sections of the SSME-rig1 configuration are shown in figures 5,6 and 7, respectively.



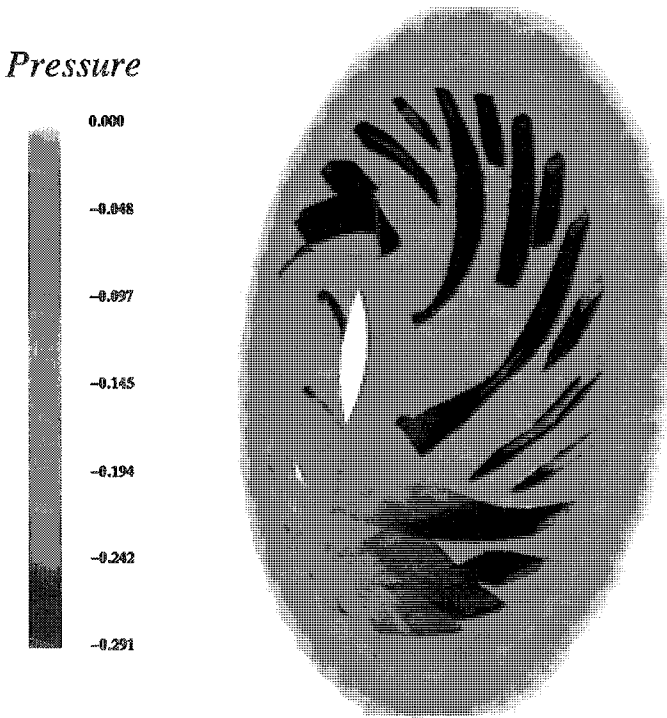
**Figure 5.** Computational grid of SSME-rig1 inlet guide vanes with 17 zones, and 5.5 Million grid points.



**Figure 6.** Computational grid of SSME impeller with 60 zones, and 19.2 Million grid points.



**Figure 7.** Computational grid of SSME-rig1 diffuser with 24 zones, and 6.5 Million grid points.



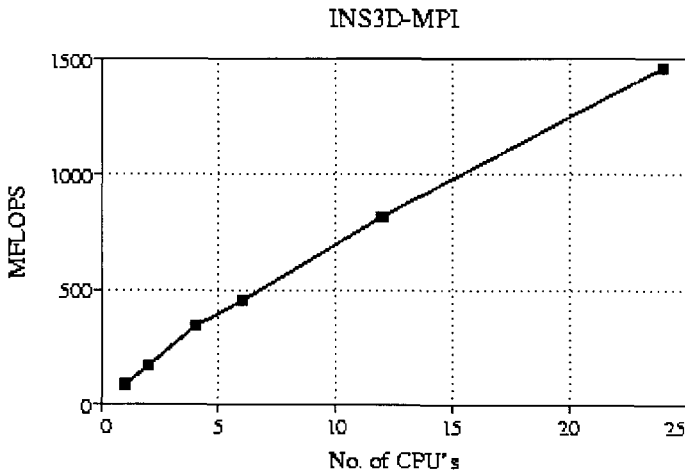
**Figure 8.** Computed surface pressure for SSME-HPFT impeller.

## 20.4 COMPUTED RESULTS

Computed results are obtained for 2.8 Million and 19.2 Million grid points SSME impeller models. Figure 8 shows computed surface pressure of the shrouded SSME impeller. The performance of the two approaches in obtaining multi-level parallelism of the INS3D code is reported in this section. The first approach is hybrid MPI/OpenMP and the second approach is Multi Level Parallelism (MLP) developed at NASA-Ames Research Center. The first approach is obtained by using message-passing interface (MPI) for inter-zone parallelism, and by using OpenMP directives for intra-zone parallelism. INS3D-MPI<sup>9</sup> is based on the explicit message-passing interface across MPI groups and is designed for coarse grain parallelism. The primary strategy is to distribute the zones across a set of processors. During the iteration, all the processors would exchange boundary condition data between processors whose zones shared interfaces with zones on other processors. A simple master-worker architecture was selected because it is relatively simple to implement and it is a common architecture for parallel CFD applications. All I/O was

performed by master MPI process and data was distributed to the workers. After the initialization phase is complete, the program begins its main iteration loop.

The SSME impeller model with 24 zones, and total grid points of 2.8 Million is used as a test case for the coarse grain INS3D-MPI code. For this version of the code, the number of zones in the computational model limits the maximum number of CPU count. Figure 9 shows floating point counts per second for this computation on SGI Origin 2000 platform. The average speedup, as compared to the linear speedup, is about 65% for 24 processors. It should be noted that the number of MPI groups reported in this paper always include the master MPI process. For the first four processors, very good performance is obtained. When the number of CPU count is increased further, the performance of the code is decreased due to the load balancing issues. In order to obtain fine grain parallelism, OpenMP directives are utilized<sup>10</sup>. Figure 10 shows time (in seconds) required per time integration step versus number of processors from the hybrid parallel code. It should be noted that the "time per iteration" reported in this paper includes the time obtaining  $Ax=b$  linear system of equations and the time solving this particular system of equations for the entire grid system. In other words, the iteration term is used for the physical time step, not for the iteration of linear equation solver. It also should be noted that the number of implicit line relaxation sweeps is kept same at each time step. The cases for 4, 12, and 24 MPI groups were plotted in figure 10. For each MPI group, various numbers of threads, such as 1, 2, 4, 8, and 16, were used. The number of CPU count is equal the number of threads multiplied by the number of MPI groups. When number of threads is increased, the performance of the code slows down because the grid size for each zone is relatively small for higher number of threads. This is shown in figure 12.



**Figure 9.** INS3D-MPI performance on the SGI O2K, SSME-HPFT Impeller.

When the problem size is increased from 2.8 Million grid points to 19.2 Million grid points, the SSME impeller has 60 zones where the smallest zone has 74K grid points and the largest zone has 996K grid points. Figure 12 shows the effect of MPI groups on the performance of the code when one OpenMP thread is used. A good load balancing is obtained up to 20 MPI groups. When more than 20 MPI groups are employed, no more improvements in the performance of the code is observed. The number of OpenMP threads is increased to 2, 4, 6, 10, and 15 for the same MPI groups. These cases are plotted in figure 13 and figure 15. Figure 14 shows time per iteration versus total CPU count, and figure 14 shows the cases for various OpenMP threads. The best performance was obtained for 20 MPI groups. In figure 13, the effect of load balancing can be seen for 30 MPI groups. The OpenMP directives show very similar speed-up for 20 and 30 MPI groups (see figure 14).

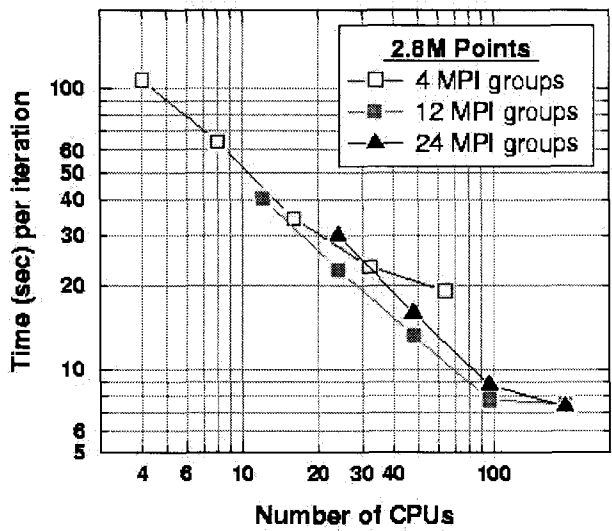


Figure 10. Time per iteration step for various MPI groups (INS3D MPI/OpenMP).



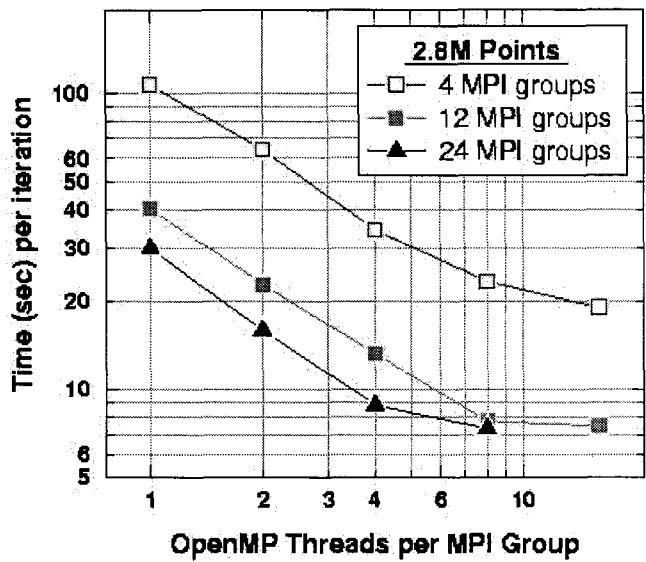


Figure 11. Time per iteration step versus OpenMP threads per MPI group.

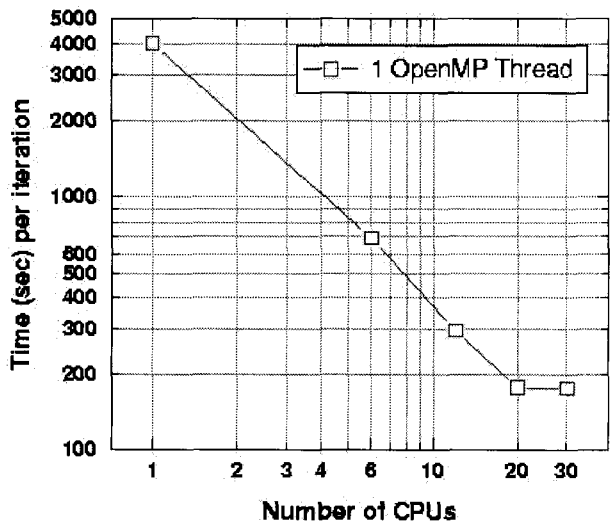


Figure 12. Performance for various MPI groups when one OpenMP thread is used (SSME impeller 19.2 Million grid).

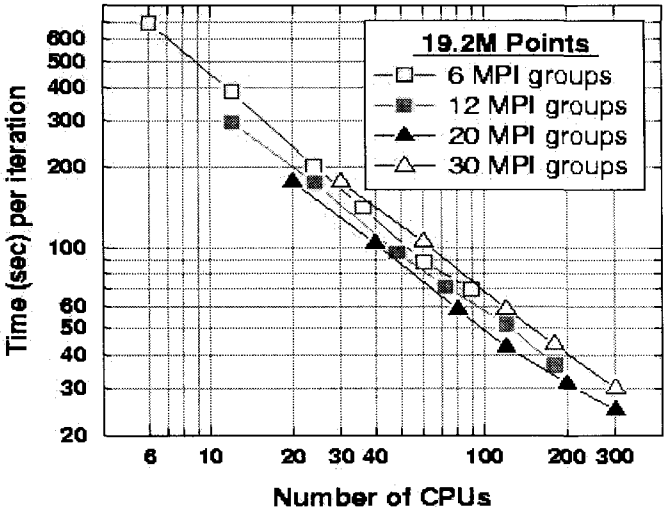


Figure 13. INS3D-MPI/OpenMP performance versus CPU counts for Origin 2000.

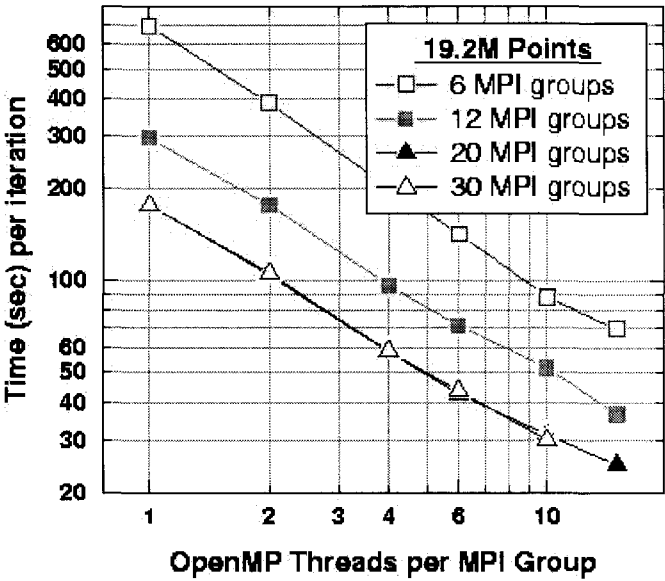


Figure 14. INS3D-MPI/OpenMP performance versus OpenMP threads for various MPI groups.

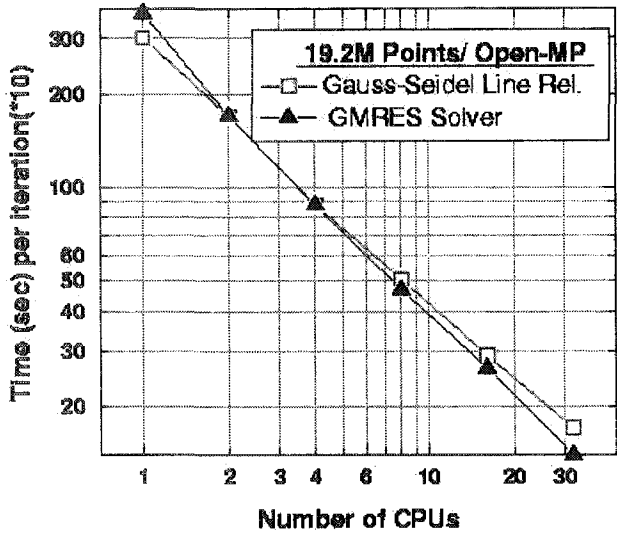


Figure 15. OpenMP performance for two different linear solvers.

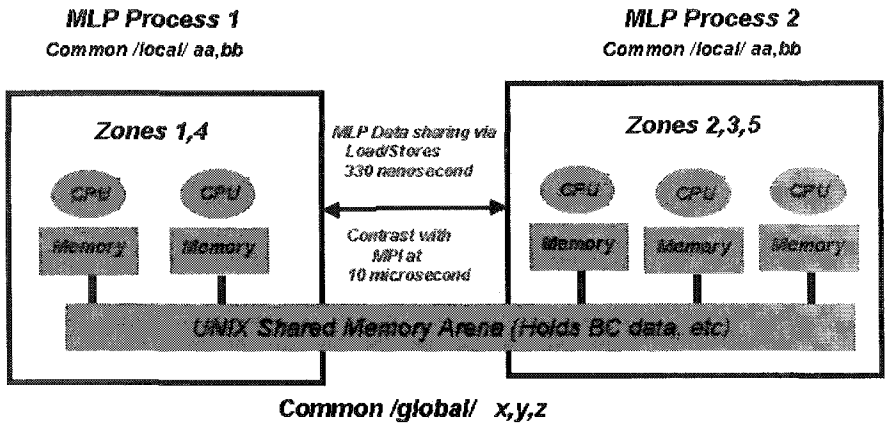
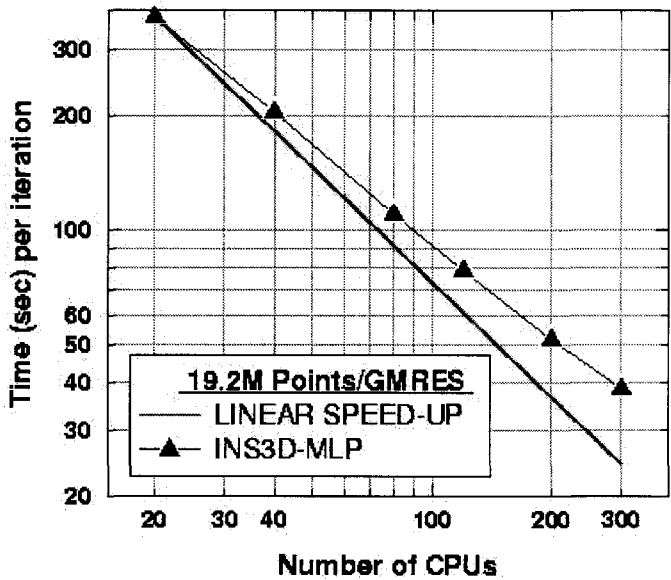


Figure 16. Shared memory MLP organization for NAS-MLP <sup>11</sup>.

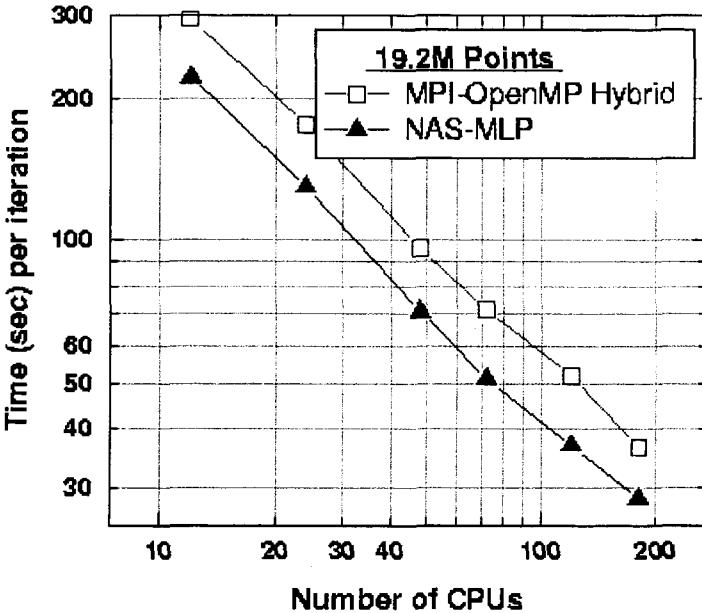


**Figure 17.** INS3D-MLP performance versus CPU counts for Origin 2000.

Figure 15 shows the effect of OpenMP threads for implicit Gauss-Seidel line-relaxation scheme and GMRES-ILU(0) scheme. Four Gauss-Seidel sweeps were performed for line-relaxation scheme, and 20 directional searches were performed for GMRES solver. The OpenMP directives show better speed-up for GMRES solver due to nested loop effect in oblique planes.

The second approach in Multi-Level Parallelism (MLP) is obtained by using NAS-MLP<sup>11</sup> routines developed by Taft for the OVERFLOW code. The shared memory MLP technique developed at NASA Ames Research Center has been incorporated into the INS3D code. This approach differs from the MPI/OpenMP approach in a fundamental way in that it does not use messaging at all. All data communication at the coarsest and finest level is accomplished via direct memory referencing instructions. This approach also provides a simpler mechanism for converting legacy code, such as INS3D, then MPI. For shared memory MLP, the coarsest level parallelism is supplied by spawning of independent processes via the standard UNIX fork. The advantage of the UNIX fork over MPI procedure is that the user does not have to change the initialization section of the large production code. Shared memory MLP is inserted into INS3D in a very similar way that Taft inserts MLP into OVERFLOW code. Library of routines are used to initiate forks, to

establish shared memory arenas, and to provide synchronization primitives. The shared memory organization for INS3D is shown in figure 16. The boundary data for the overset grid system is archived in the shared memory arena by each process. Other processes access the data from the arena as needed. Figure 17 shows INS3D-MLP performance versus CPU count for 19.2 Million-grid points SSME impeller model. GMRES-ILU(0) linear solver was used for these computations. The MLP version of code shows 73% of the linear speed-up performance. When MLP performance is compared with MPI/OpenMP performance (figure 13, 20 MPI groups), 19% more speed up is observed by using MLP version of the code. This comparison can be seen in figure 18. It should be noted that this comparison is preliminary since the further improvements in the fine-grain parallelization of the MLP code are currently underway.



**Figure 18.** Comparison of INS3D-MPI/OpenMP and INS3D-MLP performance.

## 20.5 SUMMARY

An incompressible flow solver in steady and time-accurate formulations has been utilized for parallel turbo-pump computations. Grid systems and numerical procedures are outlined for unsteady turbo pump simulations. Results from 2.8 Million and 19.2 Million grid points SSME impeller models are presented for the performance evaluations of the INS3D-MPI/OpenMP and INS3D-MLP versions of

the code. SSME impeller model with 60 zones showed that up to 20 MPI groups hybrid code showed good scalability. OpenMP directives were more effective for GMRES(ILU) solver than line-relaxation scheme. Shared memory MLP version of the code was developed by using NAS-MLP routines. The SSME impeller computations showed very good scalability for the MLP version.

## 20.6 ACKNOWLEDGEMENTS

Authors are grateful to Tom Faulkner and Jennifer Dacles Mariani for providing the coarse grain INS3D-MPI version of the code, to Henry Jin for his support and valuable ideas and discussions in OpenMP directives, and to James R. Taft for providing NAS-MLP routines.

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# Aspects of the Simulation of Vortex Flows over Delta Wings

Arthur Rizzi, Stefan Görtz and Yann LeMoigne<sup>1</sup>

## 21.1 Introduction

Current and future fighter aircraft are designed to fly transiently at very high angles of attack. This super-maneuverability is now required for dog-fight maneuvers like the “Cobra maneuver” for instance. These new types of maneuvers involve high pitch rates and flight at incidences beyond the static stall angle of attack. The aerodynamics surrounding a maneuvering fighter is very complex, but needs to be understood in order to optimize the airplane design. The superior aerodynamic performance of the fighters in high angle of attack maneuvers are often reached thanks to the use of a delta wing and the benefits of enhanced dynamic stall.

The aerodynamics of these wings at incidence is characterized by the formation of a pair of strong leading-edge vortices on the leeward side of the wing creating an additional lift compared to classical rectangular shapes. At very high incidence, however, the vortex flow breaks down, a zone of recirculation with a turbulent wake appears and the lift decreases. During maneuvers, the same phenomena occur but the flow has to adapt to the moving planform and thus time-lags are observed in the dynamic response. The prediction and the understanding of the hysteresis loops that form a history of the aerodynamic forces, for example, can be very useful for the designer to extend the limit of the flight envelope during maneuvers. The forebody vortices that develop at high angle of attack at the nose of the airplane also

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play a dominant role in the maneuver characteristics. The prediction and possibly control of the asymmetric development of the vortices is needed to enhance the behavior of the aircraft in maneuvers.

Although both experimental and computational studies of vortex flow over delta wings have been conducted over the last five decades in an effort to elucidate the complex fluid physics encountered in the high-angle-of-attack regime, certain details in steady flow are still not resolved, and a good deal of fundamental questions remain unanswered for transient flight, which together justifies the continued research interest and further investigations. For example, the SAAB Gripen has recently undergone significant high-alpha flight tests, and a better understanding of the unsteady aerodynamics underlying the hysteresis effects in the forces and moments acting on the aircraft. This could lead to improved understanding of the flight-test results and to more accurate tuning of the aerodynamic database used in the flight simulators.

### 21.1.1 Stationary Wing

The numerical study of  $70^\circ$  sharp-edged delta wings by Görtz [10] demonstrated that the basic features of the vortical flow as well as the phenomenon of vortex breakdown can be simulated accurately from a qualitative point of view. It was found that both Euler and Navier-Stokes equations are capable of predicting vortex breakdown. The breakdown location was seen to be independent of physical modeling, but it appeared to depend on the different numerical schemes used – Roe's upwind scheme predicted a more upstream burst position than the central scheme. Viscous effects, such as secondary separation, were seen in the Navier-Stokes computations but not in the Euler computations. Vortex trajectories showed good agreement with experimental data in the spanwise direction but the vortex axis was seen to be located too high above the wing. Furthermore, the suction peak on the wing's surface was seen to be too low.

In a follow-up study by Görtz et al.[11] the quantitative accuracy of the modeling was judged on four specific criteria: 1) variation of the integrated aerodynamic coefficients with angle of attack, 2) axial position of the vortex breakdown over the wing, 3) flow topology over the delta wing including surface pressure, and 4) the axial component of velocity along the axis of the vortex core. Good agreement with experimental data could be demonstrated for the aerodynamic coefficients. Reasonable agreement was achieved for the computed breakdown locations, although comparison with wind tunnel measurements proved to be difficult due to scatter in the experimental data. Details of the flow, however, such as the axial component of velocity of the fluid entrained in the vortex core or the surface pressure distribution, were very sensitive to mesh topology, including singular lines, and mesh fineness



at the apex and the leading edges, as well as the size of the mesh cells overall. This result raised the question of how an ‘optimum grid’ should look like, and whether mesh convergence can be achieved. Due to a lack of meaningful experimental data it could at that time not be determined what kind of velocity profile was representing the experimental flow conditions, thus another open question.

The given examples demonstrate that the simulation of delta wings at high  $\alpha$  is not as straight forward as one could be tempted to believe by the numerous published results for low to moderate angles of attack, where the computed flowfield generally shows excellent agreement with experimental data and little mesh sensitivity.

The first part of the present paper aims at evaluating the state of the art of stationary-delta-wing computations and at analyzing as thoroughly as possible the parameters that influence the predicted flowfield. Special attention is being given to the mesh generation process, which appears to have a significant influence on the quality and accuracy of the results. Furthermore, we will attempt to answer the question on how the distribution of the axial component of velocity along the vortex core axis looks like. Therefore, the axial velocity profile computed on grids with different mesh topology and with local grid refinement will be related to experimental data.

### 21.1.2 Pitch-Oscillating Delta

The objective of the second part of the present study is to simulate the unsteady flow around a pitching sharp-edged  $70^\circ$  delta wing (see Figure 1). This is the first, and primary, component in the study of the high- $\alpha$  aerodynamics of a fighter aircraft. A second component could be a slender body that approximates the forebody of a fighter aircraft. Such simulations on relatively simple shapes, either individually or combined together, are useful to better understand the aerodynamics and to predict the forces acting on a maneuvering fighter, with the intention to improve the engineering design. In this study, a delta wing is animated in a sinusoidal oscillatory pitching motion which is a rather simple but fundamental component of a flight maneuver.

This kind of simple motion has been experimentally investigated since the end of the eighties both for the delta wing and the slender body. Brandon [2] documented the evolution of the vortex burst location and the normal force history of a  $70^\circ$  delta wing oscillating in pitch for various mean angles of attack. Soltani et al. [20] results showed the influence of Reynolds number, sideslip angle and reduced oscillatory frequency on the aerodynamic forces of a delta wing pitched from  $0$  to  $55^\circ$ . The influence of Reynolds number and reduced frequency on the vortex breakdown location for a  $30^\circ$ - $40^\circ$  oscillation are reported by Thompson et al. in [21] while in [22], Thompson et al. examined the surface pressure distribution on the pitched delta wing. The

vortex breakdown position and propagation velocity have been studied by LeMay et al. [17] for 29°-39° and 0°-45° oscillations. All these experiment were carried out for a 70° delta wing in subsonic wind tunnels. In general not so many data are available for this geometry and the reported data are not complete: the history of the forces is usually reported, but often the unsteady pressure and the time history of the vortex location, for instance, are missing. On the contrary, the papers that present the latter do not include the forces history.

Numerical simulations of pitching objects, although a quite recent topic for CFD simulations, are also well documented. Concerning delta wings, Fritz [8] simulated a 65° cropped wing in pitch oscillation around a mean angle of attack of 9° and 15° and reported the  $C_l$  history as well as the pressure distribution. In [1], Arthur et al. compare the results (forces and pressure data) of several CFD codes for Euler calculations around a 65° cropped delta wing pitched around 9° and 21°. Kandil & Chuang [14] analysed the pressure distribution on a delta wing pitched around 20.5° mean angle of attack. A double-delta wing was also simulated in pitch oscillations around 22.4° by Ekaterinaris and Schiff [7]. The results of these Navier-Stokes computations are given in the form of  $C_p$  plots and particle tracing pictures. For transonic conditions (Mach=0.85), Navier-Stokes simulations on a 65° cropped delta wing have been made by Kandil and Kandil [15] which resulted in pressure contours plots and the  $C_l$  history. All these papers present important and pioneering results for the case where the amplitude of the oscillations are small and limited to  $\pm 8^\circ$  (see [7]). And the results compare rather well with experimental data which is encouraging for continuing further to study the large amplitude case by numerical simulations.

The computational study here deals with a 70° single-beveled delta wing pitched in sinusoidal oscillations at a low subsonic Mach number. The law of motion for the wing is given by the history of the angle of attack:  $\alpha(t) = \alpha_m + \alpha_0 \sin(\omega t)$ . Low oscillation amplitudes are considered to compare with the simulations presented above, but larger ones (on the order 0°-40°) are also investigated. Indeed, the ability of the code to simulate pitching motions is better assessed for large amplitudes because of the very distinct hysteresis loop, and it is also of more practical interest. The pitching motion and the consequent unsteady flowfield are simulated using a moving grid that rotates with the wing around the pitch axis.

The paper begins with a short description of the numerical method, next are the test cases and generated grids, followed by the discussion of the results and conclusions.

## 21.2 Computational Method

All the test cases are solved using the Navier-Stokes Multi-Block (NSMB) flow solver [25, 9, 26, 5] that is the product of a joint European research project between universities and industry in France, Sweden and Switzerland. The code solves the compressible Navier-Stokes equations on structured multi-block grids using a cell-center finite-volume method to discretize the equations. Several turbulence models have been implemented in NSMB, including the algebraic models of Baldwin-Lomax and Granville, the one-equation model by Spalart and Allmaras, the two-equation  $\kappa - \varepsilon$  models by Chien and Hoffman and the two-equation  $\kappa - \tau$  model.

The pitching-delta problem is solved using a moving grid (the so-called ALE, Arbitrary Lagrangian-Eulerian approach). The whole grid rotates as a rigid body together with the pitched object, and unsteady boundary conditions are implemented for the far field and surface boundaries. The mesh velocity thus has to be taken into account in the governing equations:

$$\frac{\partial}{\partial t} \iiint_{\Omega(t)} U(\mathbf{x}, t) d\Omega + \iint_{\partial\Omega(t)} \mathcal{F} \cdot \mathbf{n} dS = 0 \quad (21.1)$$

where the control volume  $\Omega(t)$  is time-dependent, the fluid state vector  $U$  and flux tensor of inviscid and viscous components  $\mathcal{F} = \mathcal{F}^{inv} + \mathcal{F}^{vis}$  are

$$U = \begin{bmatrix} \rho \\ \rho \mathbf{v} \\ \rho E \end{bmatrix}, \quad \mathcal{F}^{inv} = \begin{bmatrix} \rho(\mathbf{v} - \dot{\mathbf{x}}) \\ \rho \mathbf{v}(\mathbf{v} - \dot{\mathbf{x}}) + p\mathbf{I} \\ \rho E(\mathbf{v} - \dot{\mathbf{x}}) + p\mathbf{v} \end{bmatrix}$$

with  $\rho$ ,  $\mathbf{v}$ ,  $E$  and  $p$  the density, velocity vector, total energy and pressure respectively while  $\mathbf{n}$  denotes the outward normal of the moving boundary  $\partial\Omega(t)$  which has the velocity  $\dot{\mathbf{x}}$ .

The inviscid numerical flux-vectors at cell faces are approximated using Jameson's central scheme or the second and third order Total Variation Diminishing (TVD) version of Roe's upwind scheme applying the Monotone Upwind Scheme for Conservation Laws (MUSCL) extrapolation. The viscous fluxes are calculated at the cell-surface center using the gradient theorem on a shifted control volume. The resulting semi-discrete system of equations,

$$\frac{d}{dt} (U\Omega) + R(U^n) = 0 \quad (21.2)$$

where now  $\Omega$  is the cell volume,  $U$  is the cell state vector and  $R$  is the explicit flux-difference operator (residual), can be integrated in time using, for example, the explicit Runge-Kutta scheme. Alternatively, in NSMB, the implicit system of equations

$$\frac{d}{dt} (U\Omega) + R(U^{n+1}) = 0 \quad (21.3)$$

can be integrated using the Lower-Upper Symmetric Gauss-Seidel (LU-SGS) scheme. All the stationary-wing computations presented here are carried out with Jameson's central-flux scheme and a scalar version of the LU-SGS time-stepping procedure to accelerate the convergence to steady state.

A serial version of the NSMB code was used to perform the calculations on a Fujitsu VX/3 parallel vector computer with a peak performance of 2.2 Gflop/s for each of its three processing elements. The computer is operated by the Center for Parallel Computers (PDC) at the Royal Institute of Technology (KTH).

### 21.2.1 Dual Timestepping

If the problem to solve varies in time, then time accuracy is important, and convergence-accelerating techniques like local time-steps and multigrid cannot be used. Explicit time-stepping becomes impractical, and even implicit schemes are not well-suited because they are only first-order accurate in time due to the linearization that is needed in the solution procedure. Dual timestepping is one way around this dilemma in that it offers higher-order time accuracy while allowing use of efficient convergence procedures like multigrid, but at the cost of an additional iteration loop.

This technique introduces an outer time-stepping loop for a real time-accurate timestep using a fully implicit scheme, and an inner loop with a fictitious timestep to reach a "steady state" at each real timestep. Acceleration techniques like local time-stepping, multigrid or implicit schemes then can be used in the inner loop to iterate the solution to a "steady-state." This implementation follows the technique presented by Jameson [13].

Applying the implicit Backward-Difference-Formula (BDF) scheme of second-order accuracy to Eq.(21.3) yields:

$$\frac{3}{2\Delta t} [U^{n+1} \Omega^{n+1}] - \frac{2}{\Delta t} [U^n \Omega^n] + \frac{1}{2\Delta t} [U^{n-1} \Omega^{n-1}] + R(U^{n+1}) = 0 \quad (21.4)$$

Then define the modified residual,  $R^*(U)$ , for this discretization in time, as

$$R^*(U) = \frac{3}{2\Delta t} [U \Omega^{n+1}] - \frac{2}{\Delta t} [U^n \Omega^n] + \frac{1}{2\Delta t} [U^{n-1} \Omega^{n-1}] + R(U) \quad (21.5)$$

so that the second-order time-accurate solution  $U^{n+1}$  is reached when the iterate  $U$  satisfies the system  $R^*(U) = 0$ .

Thus a new system of ordinary differential equations, similar to Eq(21.2), can be formulated using this modified residual in a fictitious time-stepping procedure to iterate  $U$  forward to the time-accurate level  $U^{n+1}$

$$\frac{d}{dt^*} (U \Omega) + R^*(U) = 0 \quad (21.6)$$

Equation (21.6) is marched to steady-state in the fictitious time  $t^*$ , using any convergence acceleration method implemented in the flow solver, with the slightly modified residual in Eq(21.5).

#### 21.2.1.1 Starting procedure

Since  $U^{n-1}$  is not defined at the very first time step, a special procedure is needed to start the second-order BDF time stepping. This could be done in a number of ways, e.g., by setting  $U^{n-1} = U^n$  first and then use BDF as usual, or by using on just the first step another scheme that only involves  $U^{n+1}$  and  $U^n$  and then continue with BDF. The latter approach is preferable because the former sacrifices global second order accuracy. NSMB uses the first-order backward Euler scheme for the first step because it is easy to implement, and has better damping properties than the second-order trapezoidal scheme, and furthermore it still results in a method that is globally second order accurate.

### 21.3 Test Cases and Grids

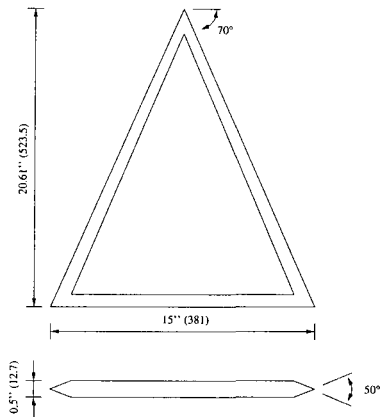
Initially only stand-alone delta-wing planforms that are stationary in attitude have been investigated, i.e., no fuselage is modeled. However, several physical models of the wing have been tested, differing in size and edge beveling.

#### 21.3.1 Model One

The first (wind-tunnel) model treated here is a simple flat plate delta wing of  $70^\circ$  leading-edge sweep, a 20.61 inch (523.5 mm) root chord and a 15 inch (381.0 mm) span at the trailing edge, Fig. 1. The wing is 0.5 inch (12.7 mm) thick and the leading and trailing edges have been sharpened using a  $25^\circ$  bevel on the top and bottom surface.

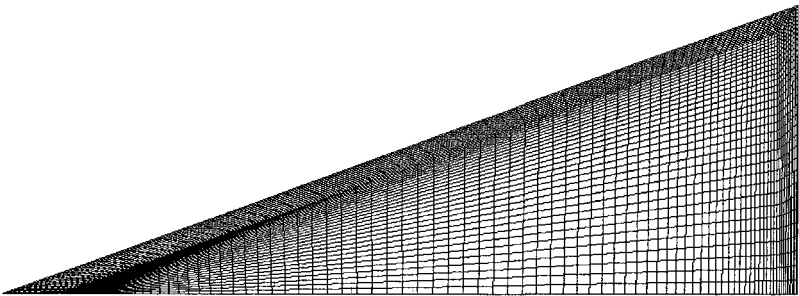
##### 21.3.1.1 H-H Grid: Model 1

A structured H-H type grid consisting of 1,168,512 volumes was generated for this model. The smallest cell size in the wing normal direction is  $0.55 \times 10^{-5}$  root chords. A topology with 18 blocks was considered necessary to fully represent the entire geometry of the wing including all bevels correctly. More than 100 points were placed along the wing in the chordwise direction and up to 46 points in the spanwise direction, Fig. 2. A high number of grid points was placed at the apex, the shape of which has a major influence on the development of the primary vortex and on the onset of vortex breakdown. Due to the bevels, the usual apex grid degeneration is moved aft to the upper flat surface. The far-field boundaries extended 6.5 root chords upstream and 10



**Figure 1** Model 1 (after Soltani & Bragg, 1988)

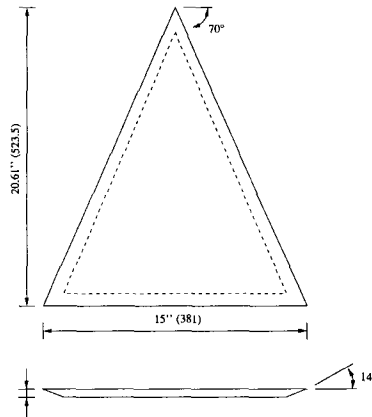
root chords downstream of the wing, and 6.5 root chords in the each direction normal to the wing’s surfaces.



**Figure 2** Upper surface grid, double-bevel model (Model 1)

**21.3.2 Model Two**

The second model is 0.25 inch (6.4 mm) thick with edges sharpened using a 14° bevel from the bottom surface only, Fig. 3. For this model, two grids were generated.



**Figure 3** Model 2 (after after Soltani & Bragg, 1993)

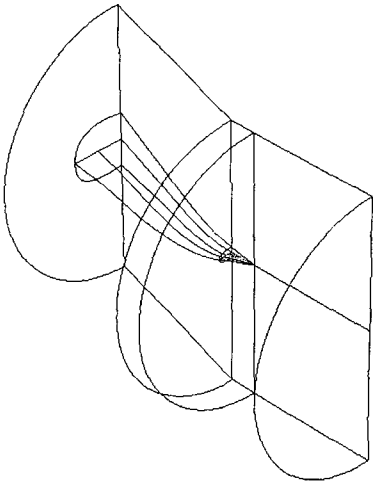
#### 21.3.2.1 E-C Grid: Model 2

The first one is a 7-block embedded-conical (E-C) type grid that was suggested for delta wing computations by Kumar [16]. The wing is enclosed by two blocks (one above and one below) that together form a cone which in turn is embedded in a standard  $H-O$  grid, Fig. 4. The height of the cells adjacent to the wing is  $1.97 \times 10^{-3}$  root chords (this mesh was used for Euler computations only). The block above and below the wing has 65, 65 and 33 points in the streamwise, spanwise and normal direction, respectively, and in total there are 616,448 cells. The grid has a singularity at the apex, which is the location of the cone nose. The farfield boundaries extend 5 root chords away from the wing in all directions and the blocks aft of the wing are angled upward  $22^\circ$  to better resolve the wake flow.

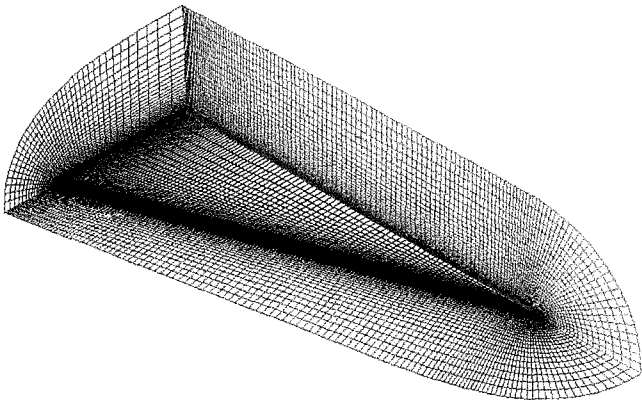
#### 21.3.2.2 C-O Grid: Model 2

The second grid is a 6-block structured C-O mesh of 714,000 cells, see Fig. 5.

The grid approximates the real geometry to make it simpler at the bevel, i.e., the bevel edge is not fully represented at its intersection with the flat lower surface, it is rather a projection of one block face on the surface comprising the bevel and the lower surface. The chosen topology results in a singularity at the apex. 130 points were placed along the wing in the chordwise direction and 40 points in the spanwise direction in an attempt to resolve the flowfield



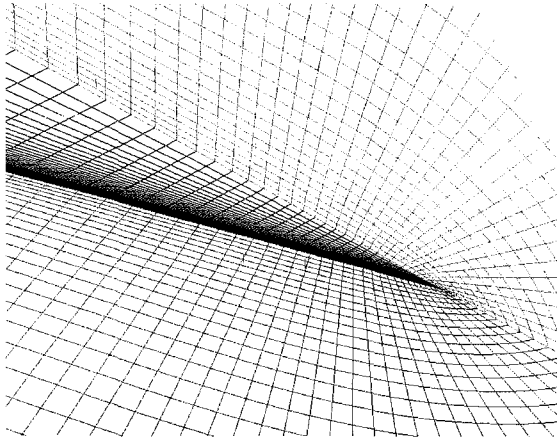
**Figure 4** Domain outline, embedded-conical (E-C) type grid, 7 blocks (Model 2)



**Figure 5** C-O type mesh with polar singularity at the apex (Model 2)



above the wing properly. The grid was refined close to the apex and around the leading edges, Fig. 6. The far-field boundaries extend 1 root chord upstream and 2 root chords downstream of the wing body, and 1 root chord in the direction normal to the body upper surface and 1 root chords in the direction normal to the lower surface.



**Figure 6** Apex detail of C-O type mesh: grid refinement

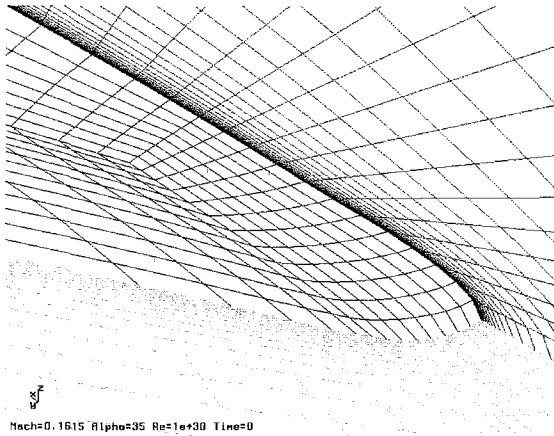
### 21.3.3 Model Three

The third model is a  $65^\circ$  delta wing and is included here because a C-O grid was created without a polar singular line at the apex.

#### 21.3.3.1 Non-singular C-O Grid: Model 3

The singular line at the apex was replaced by a mesh block formed with 4 'fictitious' corners, a block formed with a so-called TV-screen face, Fig. 7. Doing so however required rounding the apex slightly, and then rounding the leading edge. This approach is not possible if the apex is sharp. The resulting 8 block C-O grid with more than 700,000 cells is shown in Fig. 8. The mesh is refined towards the wing's upper and lower surface and at the leading and trailing edge.

All grids were generated using ICEM CFD Hexa. The major disadvantage of this commercial mesh generator is that it offers no effective control over mesh line curvature, i.e., the mesh lines are almost straight. This drawback makes generating a high-quality mesh around the sharp leading edges of delta wings a difficult task to perform. The desired mesh orthogonality near the



**Figure 7** close-up: ‘TV-screen’ solution for Model 3 with rounded apex and rounded leading edge

wing surface cannot always be achieved in critical regions such as the apex or the leading edges, compare Fig. 7.

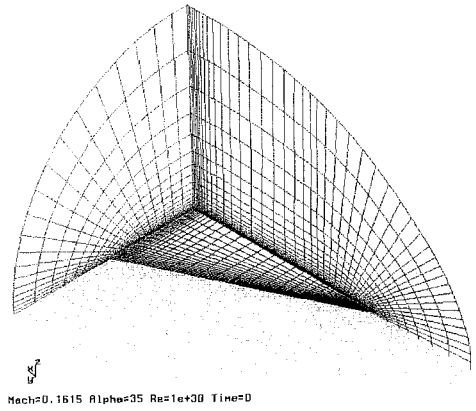
## 21.4 Stationary-Wing Computations and Results

### 21.4.1 Influence of Windtunnel Walls

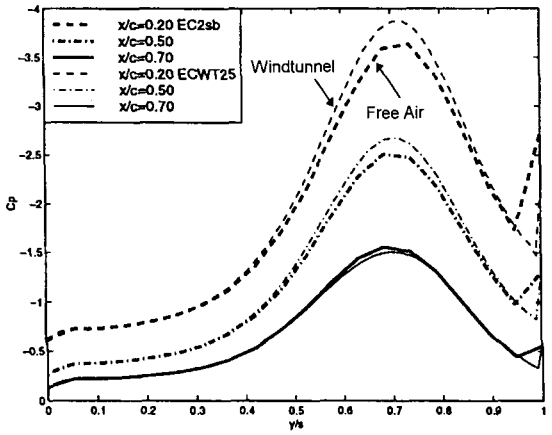
Using the previously described E-C type grid de Try[6] demonstrated that better agreement with experimental data could be achieved for the pressure suction peak when the wind tunnel walls are modeled. Fig. 9 compares the upper surface pressure distribution for free air with that of a case where the wind-tunnel walls were modeled. At the apex of the delta wing the suction peak is significantly lower for the free air solution. This result partly explains the disagreement between computed and measured surface pressure that was found in Ref.[10] and [11]. However, the wind tunnel walls appeared not to have any significant influence on the location of the vortex core axis in the direction normal to the wing’s surface.

### 21.4.2 Influence of Artificial Dissipation

The influence of artificial viscosity on the convergence history has been investigated. Figures 10 and 11 show the lift coefficient versus number of iterations for the Model 3, the delta wing with slightly rounded leading edges, where only the forth order artificial dissipation coefficient  $k$  was varied. It

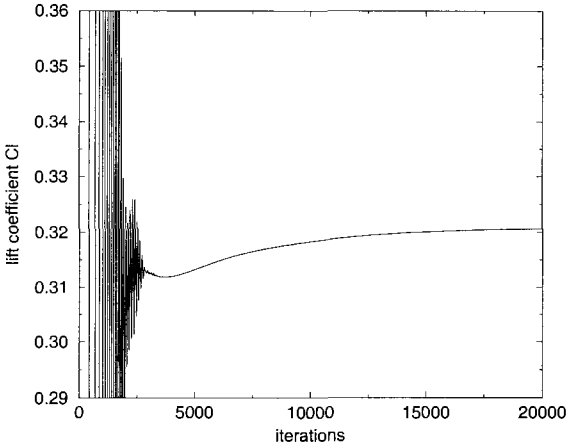


**Figure 8** C-O type mesh with 'TV-screen' block (Model 3)



**Figure 9** Influence of wind-tunnel walls on surface pressure at different chordwise stations, E-C grid, Model 2, Euler results,  $\alpha = 25^\circ$

can be seen that the higher the level of artificial viscosity, the more the vortex flow is damped, leading to smaller gradients of the flow quantities, smaller deviations and consequently a smaller amplitude of the lift-coefficient oscillation.

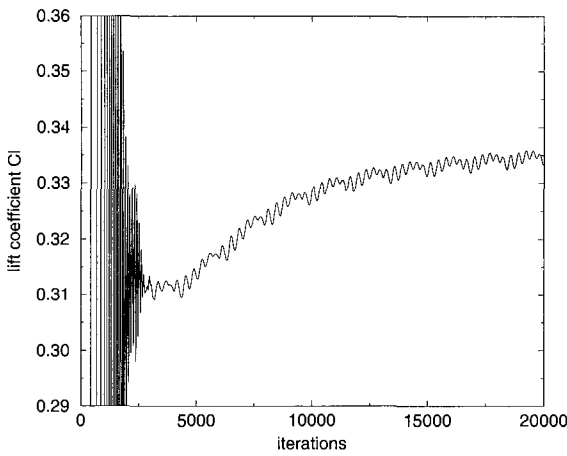


**Figure 10** Lift coeff. versus number of iterations, Model 3, artificial dissipation coeff.  $k = 1/12$

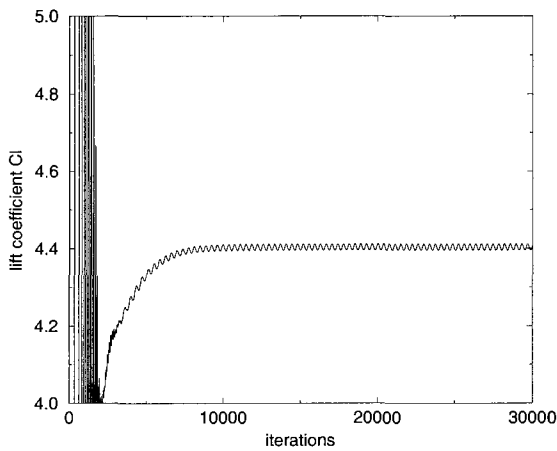
#### 21.4.2.1 *Quasi-steady Flow*

Although we are trying to compute a steady-state solution, in reality the flowfield is better described as quasi-steady. Oscillations appear in the flowfield when a spiral-type vortex breakdown occurs. Downstream of the breakdown point the instantaneous vortex core axis spirals around the mean axis against the sense of rotation of the primary vortex. In addition, the instantaneous vortex axis turns around with respect to time in the sense of the primary vortex with the same frequency as the force and moment oscillations. And this quasi-steadiness can be captured in the simulations if they are not suppressed by the artificial dissipation. Fig. 12 shows the sinusoidal oscillations for Model 2. Flowfield analysis shows that the artificial dissipation also has an influence on the vortex breakdown position.

It is our opinion that this quasi-unsteadiness should not be suppressed, and that the amount of artificial dissipation should only be the minimum needed for a stable computation. Otherwise the computed solution varies as a function of the dissipation level. Alternatively, one could say that the grid



**Figure 11** Lift coeff. versus number of iterations, Model 3, artificial dissipation coeff.  $k = 1/33$



**Figure 12** Lift coefficient versus number of iterations, 6-block CO-type mesh, Model 2, Euler solution, artificial dissipation coefficient  $k = 1/33$ ,  $\alpha = 35^\circ$

must be refined until the effect of artificial dissipation becomes insignificant. This is not only true for Euler computations but for Navier-Stokes simulations using two-equation models as well.

### 21.4.3 Influence of Physical Modeling

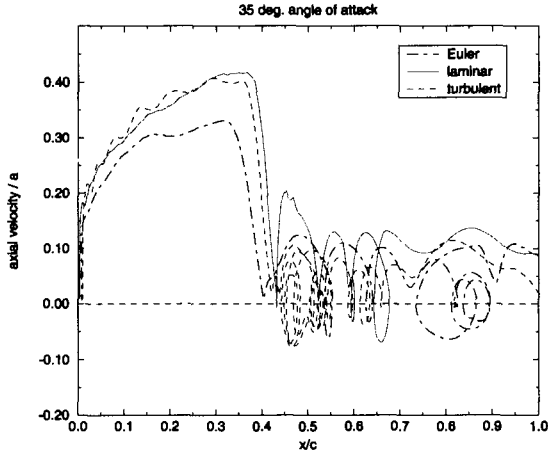
The influence of physical modeling on the axial velocity along the core can be seen for  $35^\circ$  angle of attack in Fig. 13. The peak axial velocity before breakdown is about the same for both the laminar and the turbulent case, but lower for the Euler case. But the axial velocity develops in the same manner along the axis of the vortex core and the fluid is brought to a rest at nearly the same chordwise location in all three cases (at  $x/c = 40\%$  in the Euler case,  $42\%$  in the turbulent case and  $43\%$  in the laminar case). Nevertheless, it can be stated that vortex breakdown occurs slightly earlier in the inviscid case, which could be demonstrated for other angle of attack as well. This is confirmed by theoretical findings by Darmofal[4]. Using the quasi-cylindrical approximation, he demonstrated that at low Reynolds numbers viscosity is responsible for the dissipation of azimuthal vorticity. This viscous effect was found to slow down the inviscid production of negative azimuthal vorticity, which is, according to Brown and Lopez [3], responsible for the axial deceleration of the fluid in the vortex core associated with vortex breakdown. According to the theory, viscosity delays vortex breakdown at low Reynolds numbers. However, Darmofal found that viscous effects are negligible in practical applications, i.e., at practical Reynolds numbers. This was confirmed by a numerical investigation of breakdown in pipes [4] and is generally supported by the present investigation.

The reader is referred to Refs. [10] and [11] for a more detailed investigation of the influence of physical modeling on the aerodynamics of delta wings.

### 21.4.4 Influence of Mesh Topology and Wing Geometry

We investigated the effect of mesh topology, block layout and mesh singularities on convergence rate, surface pressure distribution and axial velocity profile. When avoiding a polar singular line in the computational mesh at the apex by modeling a so-called ‘TV-screen’ block the convergence rate was seen to improve. The best agreement between experimental and computed surface pressure distribution was achieved on the embedded conical (E-C) grid with the drawback that the axial velocity profile was inaccurate. The leading edge geometry (Model 1 vs. Model 2) was seen to have a small effect on the pressure distribution, the aerodynamic coefficients, the breakdown location and the maximum axial velocity on the vortex core axis, whereas it proved to have no influence on the type of axial velocity profile.

When using a H-H type grid the axial component of velocity along the



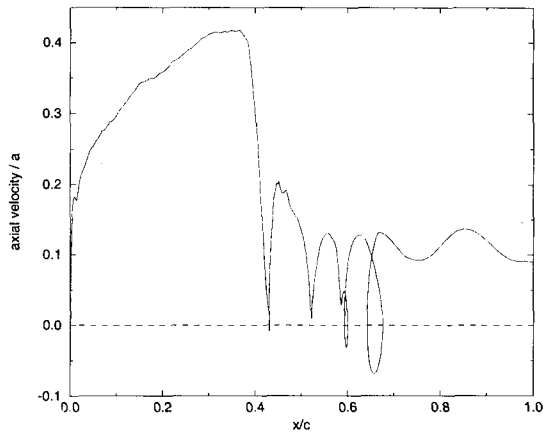
**Figure 13** Influence of physical modeling on axial velocity and breakdown position (H-H type grid,  $\alpha = 35^\circ$ ,  $Re = 1.97 \times 10^6$ ,  $M_\infty = 0.1615$ )

vortex core axis begins with freestream velocity and then accelerates over the apex and along the core as the vortex develops, Fig. 14. The axial velocity continues to increase up to some distance before breakdown. A position is eventually reached where the velocity does not increase further, but reaches some maximum, remaining at that magnitude for some distance, until falling abruptly through the breakdown region (“arc-type” axial velocity profile, also compare Figs. 10 & 12 in Ref. [11]). This kind of velocity profile was found experimentally by Hummel[12] and Verhaagen and Ransbeeck [23] using a 5-hole probe in both cases (which might have influenced the flowfield upstream of the probe in the apex region). Visbal[24] found an “arc-type” velocity profile in one of his computational studies of pitching delta wings.

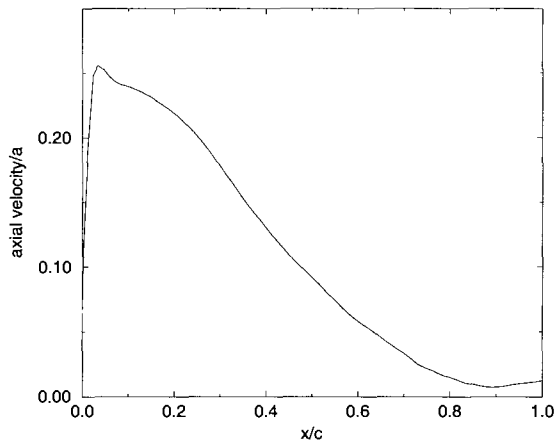
In contrast to that, when using the embedded conical grid suggested by Kumar [16], the axial component of velocity along the core axis increases sharply at the apex and decreases gradually subsequently until zero axial velocity is reached at the point of breakdown (“peaky-type” axial velocity profile, compare Fig. 15, and Fig. 20 in Ref. 2). This result seems to be in conflict with the experimental data by Hummel, Verhaagen and Ransbeeck, and other numerical studies on delta wing vortex flows.

The C-O type grid for Model 2 gave rise to a profile that falls between the first two, Fig. 16. Finally, the third model, the  $65^\circ$  delta wing with rounded apex and leading edges, resulted in a fourth type of velocity profile, Fig. 17.

We believe that the ‘arc-type’ profile, Fig. 14, is correct because of the

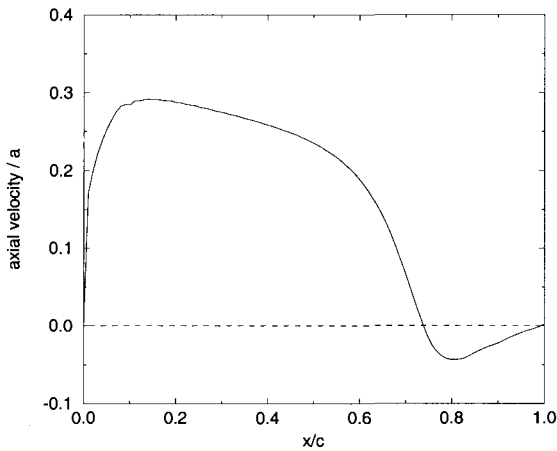


**Figure 14** “Arc-type” axial velocity profile (Model 1, double-bevel model, H-H type grid topology)



**Figure 15** “Peaky-type” axial velocity profile (Model 2, single-bevel model, E-C type grid topology)





**Figure 16** “Flat-type” axial velocity profile (Model 2, single-bevel model, C-O type grid topology)

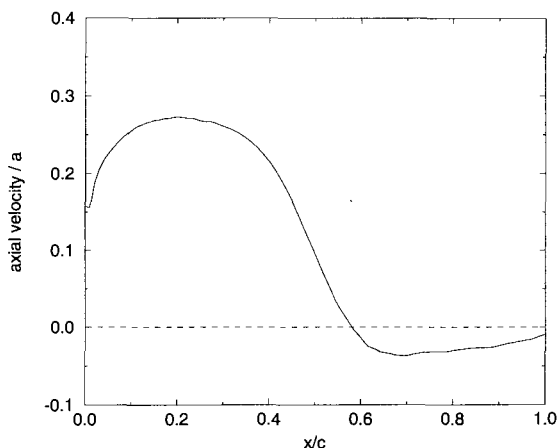
following physical reasoning: upstream of the point of breakdown the free shear layer being emanated from the wing’s leading edge continuously feeds vorticity into the vortex along the entire leading edge, leading to an increase in the axial velocity. When the vorticity-feeding rate from the leading edge exceeds that which can be convected downstream, the vortex filaments tighten up leading to breakdown and a cross-sectional expansion of the vortex core. Furthermore the arc-type profile is the only one featuring the “classical” vortex breakdown phenomenon, namely a sudden decrease in the magnitude of the axial and circumferential velocity components of the core.

The peaky-type profile, Fig. 15, could be explained by a conical singularity, i.e., by the conical mesh block at the apex.

The supposedly correct ‘arc-type’ velocity profile was computed only on one out of seven different meshes, the 18-block H-H type mesh for Model 1. This mesh is superior over all other meshes in this respect because of the high number of grid points at the apex, the refined leading edges, and the fact that the usual apex grid degeneration is moved aft to the upper flat surface.

#### 21.4.5 Influence of Grid Refinement

The C-O type mesh for Model 2 was further refined at the leading edge. On the refined mesh a steeper core velocity gradient was computed in the breakdown region, Fig. 18. A steeper gradient indicates a better resolution of



**Figure 17** “Round-type” axial velocity profile (Model 3, rounded leading edge, C-O type grid with ‘TV-screen’ at apex)

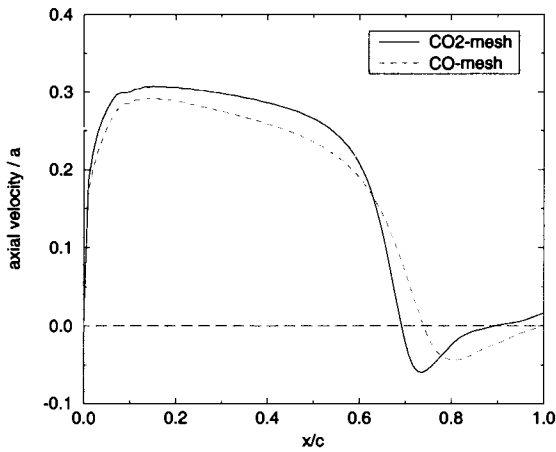
the vortex breakdown phenomenon. Furthermore it leads to a more upstream breakdown position, i.e., the point where the axial component of velocity gets zero is located closer to the apex. The maximum axial velocity is also seen to be somewhat higher on the refined grid, which translates into a stronger vortex and higher suction on the wing surface.

Since recently NSMB can handle patched grids, i.e., blocks with locally refined grids where the non-matching of cells across block boundaries are accounted for. Using patched grids one can increase the grid resolution locally in the regions of interests. This possibility will be used to refine the wing and the wing-body grids in the vortical flow regions in order to approach grid resolution. For cases featuring vortex breakdown the grid will be refined near the point of breakdown in order to resolve the small-scale details of the flow.

A substantially improved resolution of the suction area can be expected, which will lead to a better agreement between computed and experimental surface pressure.

## 21.5 Preliminary results for Pitching Delta

A first test of the mesh and of the different parameters for the computations is made for two static cases at 25 and 35 degrees. These are Euler calculations with a free stream Mach number of 0.1615, an implicit time integration with



**Figure 18** Influence of grid refinement on axial component of velocity along vortex core axis, CO-type mesh, Model 2, Euler solution,  $k = 1/33$ ,  $\alpha = 25^\circ$  (CO  $\Rightarrow$  non-refined mesh; CO2  $\Rightarrow$  mesh refined at leading edge)

the scalar LU-SGS method and a central scheme with artificial dissipation. The results shown in Figure 19 are in very good agreement with the static experimental values from [2] and encourages us to carry on with the dynamic oscillations.

As our initial simulation for the pitching computation, we solve the Euler equations and compare with the experimental testcase from [2]. The delta wing is pitched around its 40% chord location with a reduced pitching frequency (or pitch rate, often denoted  $k$  and defined as  $k = \omega c / 2U_\infty$ ) equal to 0.0376. The free stream Mach number for the computations is 0.2. Two different conditions were computed: one for low amplitude oscillations and the other for large amplitude oscillations.

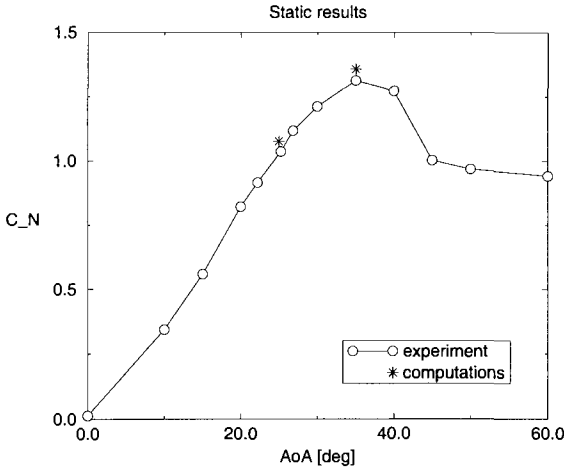
#### *Small amplitude oscillations*

The mean angle of attack is  $12^\circ$  and the wing oscillates in pitch with  $\pm 3^\circ$ .

#### *Large amplitude oscillations*

The wing is oscillated in pitch around  $22^\circ$  with  $\pm 18^\circ$  for the large amplitude oscillations.

All the computations are performed with the implicit dual timestepping method and different Roe upwind schemes: first, second and third order. The



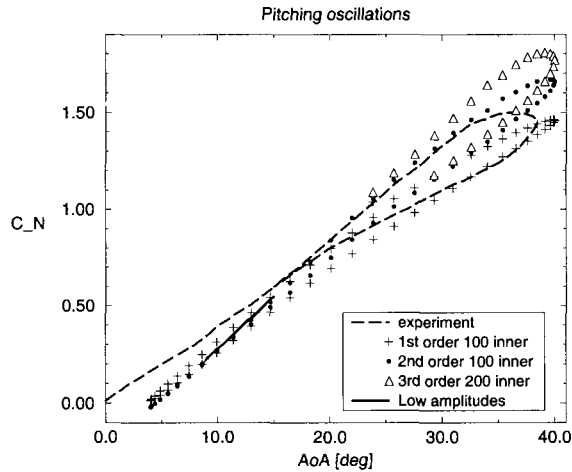
**Figure 19** Normal force  $C_n$  computed at two stationary  $\alpha$  values compared with static measurements

normal force results are presented in Figure 20 along with the experimental results (see [2]).

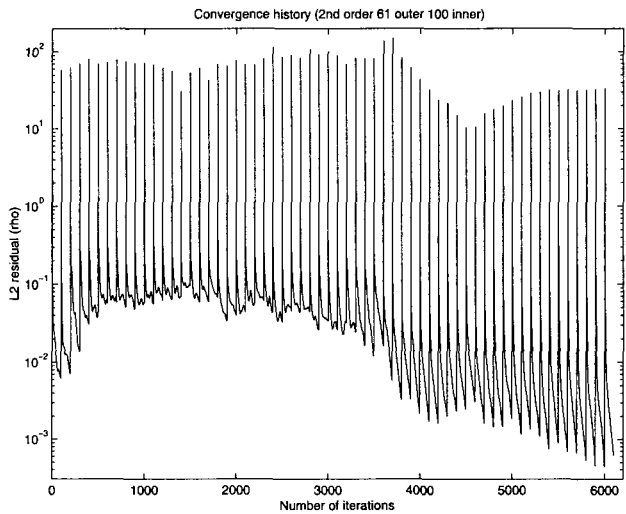
By using fully implicit schemes for the outer time step, large time steps can be made. The problem with the dual time-stepping approach is to find the optimal value of the outer time step, since a too large value of the outer time step requires a large number of inner time steps to converge to a steady state, while with a too small value of the outer time step the gain of the dual time stepping approach compared to the Runge Kutta scheme is small.

One oscillation cycle in the large amplitude case is computed with the dual timestepping method having 60 real timesteps for which 100 subiterations are run to get the “steady state” (in the first and second order cases). In Figure 21, the convergence history of the second order calculation is plotted. This figure clearly shows the principle of the dual timestepping method with the convergence of the inner iterations which is repeated at each outer iteration. Convergence problems were encountered when using the third order Roe upwind scheme. Increasing the number of subiterations to 200 improves the convergence but only a part of the cycle oscillation could be successfully computed.

In Figure 20, the hysteresis loops of the first and second order schemes compare quite well with the experimental values. However, the first order scheme predicts a normal force lower than the experiment and the loop is also thinner. On the other hand, the second order results overshoot the



**Figure 20** Computed normal force  $C_n$  for the pitching oscillations compared to experimental data



**Figure 21** Convergence history for the large amplitude case (second order) showing the inner time step convergence for each of the outer time steps

experimental values at high angle of attack. The first part of the loop computed with the third order scheme is in better agreement with the general shape of the experimental hysteresis loop although in this case too, the values are overpredicted. For all the schemes, the slopes of the curve are very similar and slightly higher than the experimental data. A complete oscillation computed with the third order scheme would probably give more accurate results even if the second order scheme already yields promising results. It is interesting to notice that the higher the order of the scheme is, the higher the normal forces for the large angles of attack are, whereas at smaller angles the different schemes predict quite similar forces.

The low amplitude oscillations are computed using only 24 real timesteps and 80 subiterations as the computation is less demanding. In this case, the results lie under the experimental value but with a slightly higher slope to the line. However, here also the results are encouraging as the points are aligned in a nearly perfect order like the experiment suggests for such angles of attack, indicating that the code NSMB is able to reproduce the vortex flow.

## 21.6 Conclusions and Outlook

### 21.6.1 Stationary Wing

It was demonstrated that the mesh generation process plays an important role in the simulation of the aerodynamics of delta wings at high angle of attack. Details of the flowfield, such as the axial component of velocity along the vortex core axis or the breakdown position were shown to highly depend on mesh topology and mesh fineness at the apex and the leading edge.

The modeling of the wind-tunnel walls could be shown to have a non-negligible influence on the surface pressure distribution close to the apex.

An embedded-conical type grid appeared to be superior over other mesh topologies when the surface pressure is of interest. The axial variation of velocity along the vortex core axis, however, was seen to be unphysical for this mesh type. The supposedly correct axial velocity profile was computed on a fine H-H type grid without singularity at the apex.

The convergence rate was seen to improve when the apex singularity was replaced with a 'TV-screen' block.

Artificial dissipation was shown to influence the damping of the rotating breakdown region. It has to be adapted for each individual test case.

Future work for stationary-delta simulations include ongoing efforts to find an "optimum mesh." Furthermore, it will show time-accurate solutions for a test-case featuring vortex breakdown. The newly available patched grid capability of the flow solver will be used to refine the vortical flow field in regions of interest.

### 21.6.2 Pitching Delta

The preliminary results obtained for the 70° delta wing oscillated in pitch are very promising as the overall hysteresis effect is reproduced in the history of the aerodynamic forces. However, these results are preliminary and can be improved with further development of the code, e.g., self-adapting time steps, and with further experience with the physical characteristics of this unsteady flowfield.

For instance, the influence of the number of real timesteps during an oscillations needs to be further investigated. This is also true for the number of cycles required before getting results that are independent of the cycle number (the first oscillations do not yield exactly the same results). Numerical parameters like the number of subiterations, the space discretization scheme and the CFL number are also very sensitive. The effect of the viscosity, which is known to influence the position of the vortex on the leeward side of the wing and thus modify the lift forces, also must be checked by carrying out the Navier-Stokes computation.

When all these parameters are better understood, the next step will be to modify the pitching conditions. The influence of the pitch rate, the mean angle of attack, the position of the axis of rotation need further investigations. The authors intend to look at the unsteady flow field around the wing and the surface pressure during the oscillations, in addition to the dynamic forces. The final goal is to compute a whole large-amplitude oscillation cycle with flow visualization highlighting the vortex formation and burst in order to better understand the hysteresis phenomenon observed in the experiments.

## 21.7 Acknowledgments

We wish to thank a number of people for their help: Denis Darracq at CERFACS for providing us with the NSMB-ALE code, Jan Vos at EPFL for his assistance in tuning NSMB, Kari Munukka and Peter Johansson at SAAB for grid generation and very helpful advice, and lastly the Center for Parallel Computers (PDC) for their help in running the machines.

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# Selected CFD Capabilities at DLR

W. Kordulla<sup>1</sup>

## 22.1 Introduction

Computational Fluid Dynamics (CFD) experienced dramatic developments since the seventies if one considers the Navier-Stokes solutions in two dimensions as the starting point. One of the first such solutions was that one proposed by MacCormack and based on a variant of the explicit predictor-corrector Lax-Wendroff formulation [1]. The current state of the art is reflected in [2] or in some of the papers cited below. Today nearly any laminar flow can be predicted, in principle. The bottleneck is, as has been and probably will be, the appropriate grid generation for complex 3D geometries. The simulation of stationary turbulent flows on smooth surfaces without massive separation can be handled in a suitable manner with Reynolds- or Favre-averaged Navier-Stokes (RANS) equations, provided that the used code, including the involved turbulence model, has been properly evaluated for the class of problems in question. For turbulent flows with massive separation or those in transition from laminar to turbulent there is still basic research to be done although some achievements can be recognised [10, 11]. Owing to the very different length scales which need to be resolved in such flows, we are still far away from solving directly the laminar Navier-Stokes equations in spite of the algorithmic and grid adaptation efforts undertaken since roughly two decades and in spite of the advances achieved in computer architecture and speed ups. The intermediate steps such as large-eddy simulations profit from the lower number of mesh points needed to resolve only the large turbulence eddies at the expense of having to model physics in the sub-scale layer which seems to

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be easier for turbulent than for transitional flows [5].

The modelling of transitional and turbulent flows is still the major stumbling block for the application of CFD to aeronautical flows at realistic high Reynolds numbers. In the case of high-temperature or reactive flows this difficulty is increased by the modelling of reactive mechanisms in wind-tunnel as well as in free-flight situations [4]. Here, again, substantial work needs to be performed before numerical simulations become truly predictive in situations new to the engineer. This situation provides, in some sense, a life insurance for experimentalists and ground-based facilities. At DLR, the development of numerical methods to integrate the partial differential equations governing various kinds of flows has a long tradition. The considered flows comprise e.g. wakes behind wings [7], three-dimensional viscous flows in turbomachinery [6], aeroelastic phenomena [9] and reactive flows in the combustion chambers of gas turbines. Aerospace is, of course, a central area of application since the corresponding fields of research and technology development represent major foci of work at DLR.

## 22.2 CFD Developments

Roughly 20 years ago various CFD approaches existed at DLR, competing with each other in some sense. In Braunschweig Jameson's explicit Runge-Kutta time-stepping scheme was selected [37], while in Gttingen MacCormack's explicit-implicit approach [38, 39, 40], Beam-and-Warming's implicit scheme [41, 42], and another version of a Runge-Kutta time-stepping method [43, 44] have been used. These approaches were based on structured grids. A little later, a finite-volume method using so-called unstructured meshes was developed in Gttingen, named TAU [45], which abbreviated "triangular, adaptive and unstructured". Over the years with resources decreasing more and more, the approaches which have survived are the evolutions of the Runge-Kutta time-stepping code developed at the Braunschweig site and resulting in the MEGAFLOW project, and a hybrid 3D TAU code. For space flow applications a derivative of an earlier version of the flow solver in MEGAFLOW is being used, called CEVCATS-N.

### 22.2.1 MEGAFLOW Project

Within the framework of the German Aerospace Research Programme DLR led a national CFD Project called MEGAFLOW [16, 20, 21]. This acronym combines the names of the systems for the generation of meshes MegaCads and for the numerical integration of the governing equations of fluid flow FLOWer. The goal was to provide German industry with an efficient numerical tool for predicting the performance of a complete aircraft in cruise as well as

in take-off and landing configurations. In order to fulfil the requirements for industrial implementation the concentrated co-operative effort involved the aircraft industry, DLR and several universities. The first phase of the project, partially funded by the German Federal Ministry for Education, Science, Research and Technology (BMBF), started in mid 1995 and finished at the end of 1998. The total effort of roughly 125 man-years was spent on five activities:

One was the development of the grid generator MegaCads, another was the improvement of the block-structured solver FLOWer, yet another was the validation of the software system for industrial applications. With many partners working on one product, the software quality management of the simulation system was a further activity, and, finally, a small fraction of the overall effort was dedicated to the investigation of alternative methods using unstructured grids based on the TAU code. The system was complemented with a post-processing tool for the computation of aerodynamic forces and an aerodynamic optimisation system.

MegaCads (Multiblock Elliptic grid GenerAtion and CAD System) exhibits a graphical user interface, the graphic module for visualisation and the module for CAD and grid generation techniques [48]. Major efforts were devoted to the improvement of techniques for the generation of surface and volume grids for industrial aircraft applications. These efforts concerned not only the quality of grids [20, 21] but also the speed up of the generation process [16]. This includes the application of advanced techniques for the specification of the control terms for the elliptic grid generation both within grid blocks and across block boundaries (e.g. [47, 48]) and the use of multigrid and parallelization [16]. Due to the parametric concept MegaCads can be used in aerodynamic optimisation loops. Results are shown e.g. in [20].

FLOWer integrates the compressible 3D RANS equations [50] and is based on an explicit cell-vertex finite-volume formulation on block-structured meshes. The baseline scheme uses central discretization in space with Jameson-type artificial viscosity and various explicit multi-stage time-stepping capabilities. Convergence to steady state is accelerated by using local time stepping, implicit residual smoothing and multigrid. Based on previous work in an earlier national project (POPINDA) the code has been extended to a parallel code [16, 20, 21]. It is fully portable for those computers which use the message passing interface MPI. The achievable speed-up relies heavily on appropriate load balancing of the processors. For the numerically accurate simulation of viscous (near-) incompressible flows, e.g. for take-off and landing situations, preconditioning of the compressible flow equations is implemented in FLOWer. The scheme is further enhanced by Chimera-type grid techniques, originally invented by J. Steger and J. Benek to cope with store-separation problems roughly two decades ago. This allows for more flexibility in the description of complex configurations at the expense of the non-trivial task to provide conservative interface boundary conditions for the overlaid or patched

grids, in particular in 3D.

FLOWer simulates unsteady flows as well. The grid motion is represented by additional terms in the governing equations. The severe time-step restriction associated with explicit schemes for viscous-flow simulation is circumvented following Jameson by means of a simple implicit method known as dual time stepping. Combined with multigrid acceleration the resulting scheme was demonstrated to achieve a speed-up of about three orders of magnitude for viscous flows past oscillating wings compared to the basic explicit scheme [16]. Combined with parallelisation this approach is expected to treat flows past most complicated configurations. With the code being able to handle grid deformation the FLOWer system is ready to be used for aeroelastic flow-structure simulations. The use of Chimera techniques enables to simulate the flow past bodies or components in relative motion such as in the case of trains entering tunnels or passing each other [46].

A major point of concern for accurate simulations of viscous flows in view of the demand of aircraft producers with respect to guaranteeing drag counts is the inclusion of physical models for transition as well as turbulence. Therefore, various approaches are followed in the MEGAFLOW project in co-operation with universities to cope with this problem [16, 19, 20, 21].

Owing to the reached high level of quality and efficiency, and because of the extensive validation of the system according to industrial requirements, the MEGAFLOW system is being intensively used by DaimlerChrysler Aerospace in the design processes for new aircraft. However, further development steps are still needed with respect to improved physical modelling, enhanced automation of grid generation, reduced turn-around times, further validation, provision of interfaces for multi-disciplinary simulations and enhanced capabilities of aerodynamic optimisation systems.

For the determination of dynamic coefficients of reentry configurations which are usually required in the low supersonic and transonic flow regime FLOWer is employed [29]. The corresponding results have been the first ones for complete lifting vehicles known to the author.

### 22.2.2 CEVCATS-N

CEVCATS-N with N standing for non-equilibrium combines CEVCATS, the predecessor of FLOWer, but improved and tuned for hypersonic flow simulation, and the physical modelling of the viscous flow solver NSHYP for reacting flows [36, 42, 50]. The code CEVCATS-N integrates the governing equations for viscous flows in finite-volume formulation. The fluid can be a perfect gas or a mixture of gases in thermo-chemical equilibrium or in nonequilibrium. The high-temperature effects of air in thermo-chemical equilibrium are described with the help of vectorizable curvefits based on Tannehill's approximations of thermodynamic and transport properties. For

flows in nonequilibrium a five-species model is usually applied. Appropriate wall boundary conditions allow for the consideration of catalytic effects, radiation and various heat transfer conditions.

The spatial discretisation is based on an explicit second-order finite-volume scheme with the variables stored at the vertices. The thermodynamic and chemical source terms are considered in a point-implicit fashion. The numerical scheme uses a hybrid upwind flux-vector splitting: the van Leer scheme for regions of strong shocks and the AUSM scheme according to Liou and Steffen elsewhere. Second-order accuracy is achieved by MUSCL extrapolation to properly capture strong shocks and contact discontinuities. Local time stepping, implicit residual averaging and full multigrid (with semi-coarsening) are used to accelerate the convergence to steady state similar to the procedures employed in FLOWer. Compared with typical viscous transonic flow simulations, for reentry flow computations the numerical viscous eigenvalues are large in comparison with the acoustic wave speed. At strong shocks large Courant numbers obtained with the help of residual smoothing will result in solution divergence. An adaptive time step is therefore employed which reduces the Courant number to about one at strong shocks.

The computational performance of CEVCATS-N on a NEC SX 4/4 super computer of DLR is approximately 1.1 GFLOPS per single processor. The CPU time per grid point and multigrid cycle is 30  $\mu$ sec for equilibrium and 70  $\mu$ sec for non-equilibrium computations. The use of 4 processors in parallel reduces the latter time to 20  $\mu$ sec.

The code CEVCATS-N has been extensively validated in a variety of test cases organised by ESA and other organisations, in particular by comparison with experimentally obtained data e.g. [24, 25, 26, 27, 28, 31, 32, 34, 35].

### 22.2.3 TAU

The development of TAU, started in the early nineties [45] for unstructured, triangular grids, has been enhanced in the MEGAFLOW project, as well as in the frame work of several EC-funded projects, see e.g. [3]. The involvement in MEGAFLOW enabled especially a detailed comparison of the results predicted for structured and unstructured meshes. The code for unstructured grids must, in particular, reproduce the results for structured grids, although at a larger computational effort owing to the more complex but more flexible code structure. Once the surface of a complex configuration is described sufficiently accurate (water-tight), unstructured grids are known for short generation times including the application of local grid refinement with larger cells where the action is missing. This is, certainly, true if inviscid flows are considered. For viscous-flow simulations at high Reynolds numbers the appropriate resolution of e.g. the boundary layer near walls would require either a tremendous amount of regular-shaped cells or cells with

very acute angles between the corresponding edges. The former leads to vast computation times, the latter results in reduced accuracy or oscillatory distributions of quantities such as the heat transfer. Consequently, hybrid grids are being employed with unstructured tetrahedral meshes away from walls and structured grids, typically layers of prisms, near surfaces [13, 15, 16, 17, 18, 3]. DLR's strategy is to cooperate, in principle, with commercially available 3D grid generation software producers because the in-house work in this field becomes less and less affordable. Features such as adaptation or grid deformation are, however, built in in-house, e.g. [15].

Today's TAU code is a software package for solving either the Euler or RANS equations on the mentioned grids. It is composed of a pre-processing part, the flow solver and the adaptation module. The pre-processing part supports currently tetrahedral, prismatic, pyramidal and hexahedral elements. The generation of meshes for complex geometries, e.g. a wing/body/pylon/engine configuration, is carried out within only a few days, while weeks are required for block-structured grids about the same geometry. Note, that the solver is completely independent of the type of grid cell because it employs an edge-based data structure using a dual-grid approach. The pre-processing does not only generate the dual grid from the initial grid and the corresponding data structure. It also establishes the agglomerated coarser grids for the multigrid approach and the partitioning of the grid, i.e. domain decomposition, for the use on parallel computers. In addition, a colouring of the edges is performed allowing either for vectorization on vector computers or for cache optimisation on cache-based machines.

The flow solver duplicates as much as possible the features present in the FLOWer approach. Thus the solver employs an explicit multi-stage time-stepping Runge-Kutta scheme for the discretization of the temporal gradients. In the case of time-accurate simulations the dual-time stepping method is used. In order to accelerate the convergence to steady state local-time stepping, residual smoothing and the multigrid technique are employed. The solver includes various central and upwind approaches for the calculation of the convective fluxes and a number of turbulence models for aerodynamic flows [14].

The adaptation module of TAU provides the automatic local refinement of the hybrid grids based on the flow features in question using residual- and/or gradient-based sensors. Points in structured sublayers can be redistributed to provide an improved boundary-layer resolution. Since one wanted application of TAU is the multi-disciplinary link with a code for structural dynamics [12], an efficient tool for mesh deformation has been added to the system.

It is noted that developments for DLR's future CFD code are directed towards a combination of the good features of both code systems FLOWer and TAU into an improved TAU+. Also, a transfer of the know how present in CEVCATS-N is currently underway to profit from the main advantage of



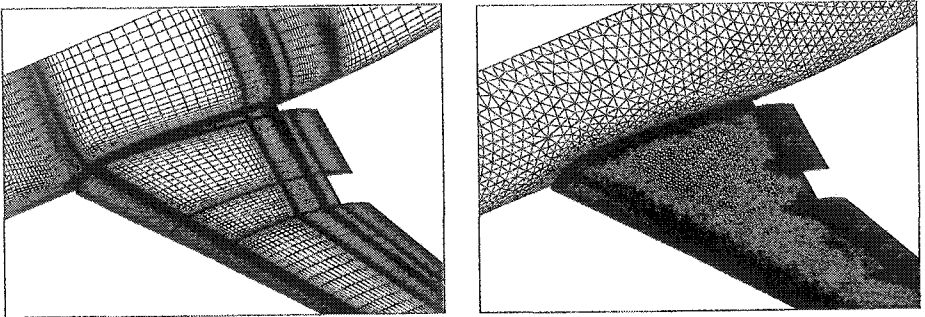
TAU, namely reduced turn-around times whenever new configurations must be considered, and to generate a more flexible code for space flow activities TAU+N(onequilibrium).

Finally, a derivative of the compressible TAU code has been established in recent years to cope with the incompressible flow in the combustion chamber of gas turbines [13]. This work is carried out in the frame work of a DLR project, where the flow solver is being combined with models describing the combustion processes including radiation.

### 22.3 Recent Applications

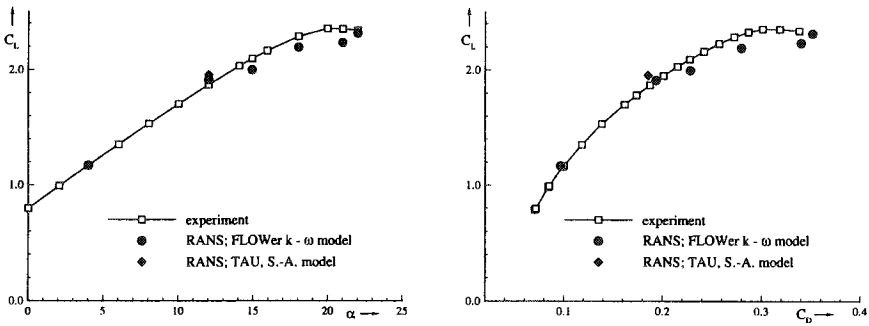
The references selected for the list given below present an overview over a broad range of aerospace applications in three dimensions, from incompressible to reacting hypersonic flows and from steady-state to unsteady flows. A few examples are described here to give an impression on the capabilities present at DLR.

The first example is taken from high-lift aerodynamics, see Figs. 1 and 2 [51]. The configuration is the DLR ALVAST wind-tunnel model, a 1:10 scale model of a twin-jet narrow body commercial aircraft similar to an Airbus A320.



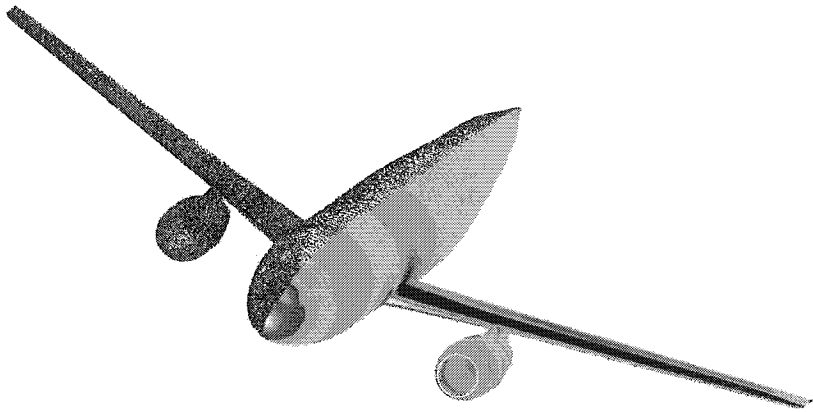
**Figure 1** Surface grid near the wing-fuselage junction: block-structured (left) and hybrid unstructured (right).

The interactive MegaCads module has been used to generate the structured grid with 9.2 million grid points for the complete (half) configuration. An experienced user needed about 3 months to generate the grid, including slat and flap gaps. Fig. 1 shows the surface grid on the left. On the right the surface grid for the adapted unstructured hybrid grid with 5.8 million grid points is observed. The grid was generated with the commercial software CENTAUR [53]. Based on a well-defined CAD description of the model, the grid generation process took just about a week. Figure 2 displays the



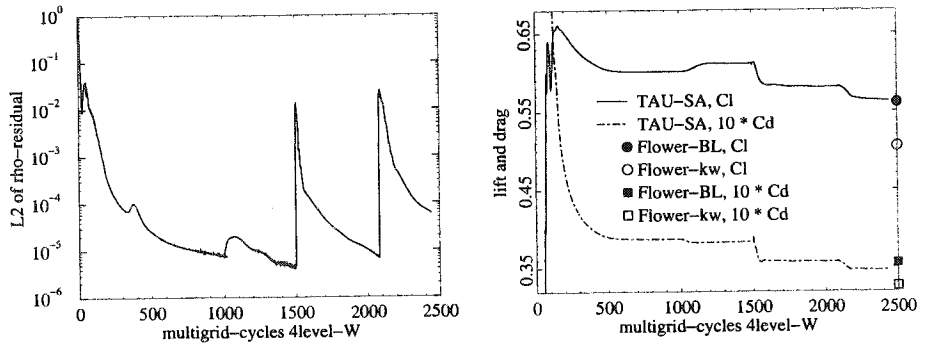
**Figure 2** Comparison of experimentally observed lift/drag curves with predicted ones obtained with structured and hybrid grids.

corresponding lift and drag polars as predicted with FLOWer and TAU, in comparison with experimental data obtained in the large low-speed wind tunnel of DNW. The agreement is good in the linear regime and deteriorates thereafter as is expected, regardless of the turbulence model used. Results of FLOWer for the flow past helicopters are discussed in [22].



**Figure 3** Adapted surface grid and final pressure distribution on the DLR-F6 configuration.

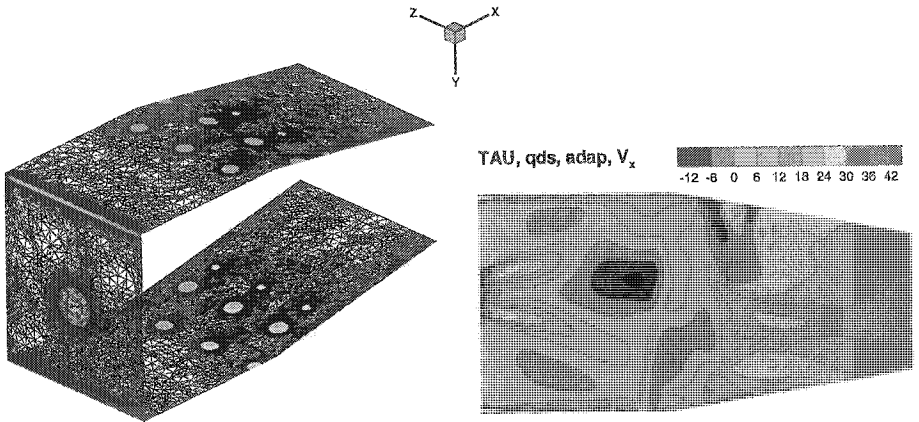
Figures 3 and 4 provide an impression of the results obtained with TAU with an unstructured hybrid mesh [3, 16, 21]. The generic aircraft configuration in cruise conditions is the DLR-F6 wing-body-pylon-nacelle configuration. Starting with an initial grid with prismatic sublayer and tetrahedral elements



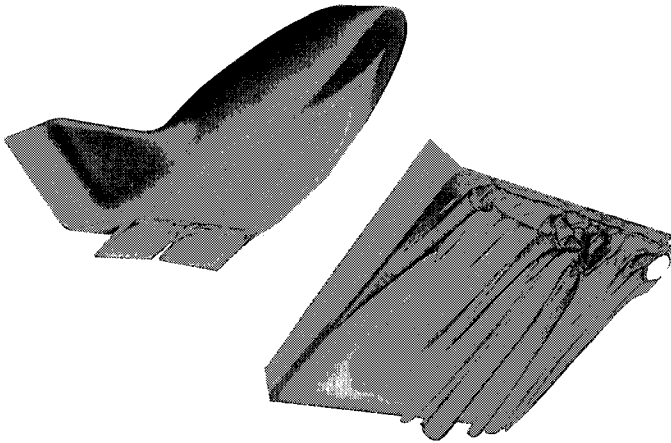
**Figure 4** Convergence history of the complete adaptation cycle, comparison with integral data obtained with a structured grid.

in the outer region with 1.4 million grid points, adapting twice a final grid with 3.5 million points was obtained for half the configuration. Spalart-Allmaras' turbulence model has been used for the simulation. Fig. 4 displays the adapted surface grid and the final pressure distribution, while Fig. 5 reflects the decrease of the residuals and the convergence of lift and drag compared to the results based on a structured grid with 3.8 million points [21].

Figure 5 indicates the capabilities of the incompressible TAU code as applied to inert flow in a generic combustion chamber of a gas turbine [13]. The adaptation feature is neatly observed.

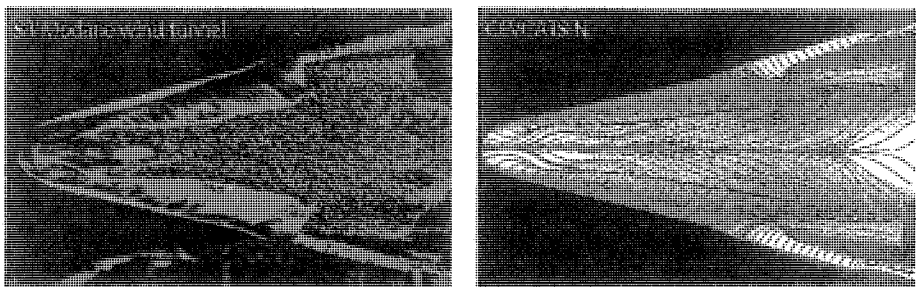


**Figure 5** Adapted grid on partial surface of a generic combustion chamber and axial velocity in a vertical cut.

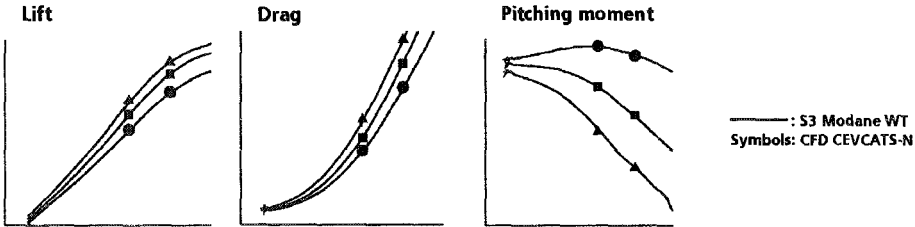


**Figure 6** Heat loads and visualisation of streamwise vortices on the body flaps of X-38 at DNW-RWG wind-tunnel conditions ( $M_\infty = 6$ ,  $Re_\infty = 3.1 \cdot 10^6$ , laminar).

The remaining examples concern the use of CEVCATS-N for the simulation of the aerodynamic and aerothermodynamic performance of lifting reentry vehicles. For the computation of the dynamic coefficients for such vehicles the reader is referred to [29]. Results for the optimization of supersonic transport airplanes and of space vehicles including the consideration of the interaction of forebodies with inlets of airbreathing engines is described e.g. in [30] and [23]. Optimization results for the design of capsule-type vehicles are found in [31]. Figure 6 shows the prediction of the heat load on the body flap of the demonstrator X-38 of NASA's crew rescue vehicle at a condition of the wind tunnel DNW-RWG in Göttingen. The main goal of that experiment carried out in the frame work of the German national technology development



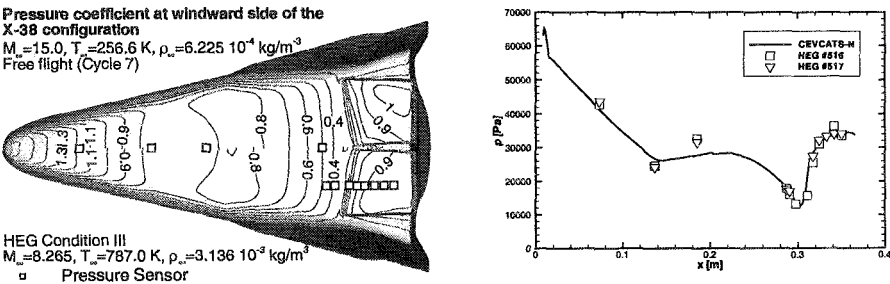
**Figure 7** Laminar flow pattern on the leeward side of X-38 at ONERA S4 Modane wind-tunnel condition ( $M_\infty = 10$ ,  $\alpha = 40^\circ$ ).



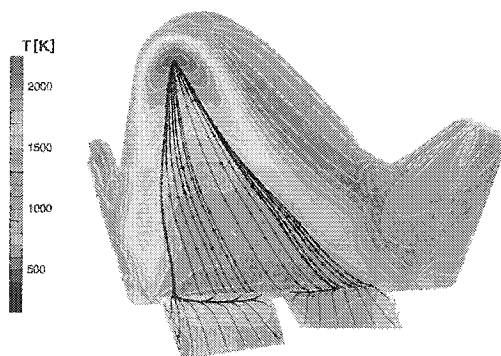
**Figure 8** Aerodynamic coefficients of X-38 at  $M_\infty = 6$  for different angles of attack and body-flap deflections.

programme TETRA was to give evidence of Görtler-type instabilities in the boundary layer on the flaps. In addition to classical pressure and heat-transfer measurements with thermocouples liquid crystals have been used to detect a vortical trace on the flap. In the experiment there was no clear evidence seen. In the numerical simulation, however, with more than 1 million points in the block around the flap a flow structure with streamwise vortices is clearly identified, see Fig. 6 [35]. This is the first time that streamwise counterrotating vortices are predicted on surfaces of a realistic 3D configuration. Further investigations are required to substantiate the findings.

Figures 7 and 8 display results for the X-38 configuration at different wind-tunnel conditions, realised in ONERA's wind tunnels. Figure 8 shows an excellent comparison of predicted and experimentally observed skin-friction lines on the lee-ward side of the demonstrator which is not easy to achieve owing to the low density there and the associated convergence problems. Fig. 8 demonstrates the capability of the code to reproduce experimentally obtained integral quantities and pitching moments for cold hypersonic flows



**Figure 9** Predicted pressure distributions on the surface of X-38 for flight and HEG conditions.



**Figure 10** Temperature distribution with selected skin-friction lines for free-flight condition for  $M = 25$ .

[28, 31, 34]. Figures 9 and 10 show results predicted for reacting hypersonic flow, the former for wind-tunnel conditions in the high-enthalpy facility HEG in Göttingen and for corresponding free-flight condition. Fig. 10 provides the temperature distribution for X-38 in free flight with asymmetric flap setting at  $M = 25$ .

## 22.4 Where to go

The future of predictive CFD depends, on one hand, heavily on the capabilities to model physics appropriately and to make cost-effective models easy to implement in the codes. On the other hand, it depends on enhanced grid generation capabilities with improved flexibility and speed according to the rule "a good mesh is half the solution". Therefore, DLR concentrates on one code in the future which copes with unstructured hybrid meshes, based on the modern data structure of TAU. Aside of this choice the name of the game is multi-disciplinary networks for the various disciplines wrapped in some user-friendly information-technological environment, including the use of meta computing with ultra-fast simulation codes and efficient coupling procedures probably going beyond the currently first steps of loose coupling. At present, DLR supports internal projects producing appropriate shells for multi-disciplinary couplings, e.g. for aeroelastic simulations for transonic aircraft (AEROSTABIL, AMANDA [52]) or flight mechanics (AEROSUM). This also requires the use of CFD in optimisation processes in some quicker and automatic fashion. In space applications a project starts work in the multi-disciplinary direction by coupling hot reacting flows with the heat transfer into structure and the corresponding response as a first step (IMENS).

## 22.5 Acknowledgements

The author could not have written this paper without the support with material of his colleagues in Braunschweig and Göttingen, thanks go to them therefore. Also thanked is, in particular, S. Brück for his support with  $\text{\LaTeX}$ . Thanks are also due to various programmes funded by BMBF, DFG, DLR, ESA/ESTEC and the European Community allowing to generate data.

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# CFD Applications to Space Transportation Systems

Kozo Fujii<sup>1</sup>

## 23.1 Introduction

In early 1970's, CFD in aerospace was first attracts people's attention. The embedded shock wave on a transonic airfoil was automatically captured in the computer simulations and the design process of commercial aircraft was drastically simplified. In the middle of 80's, CFD in aerospace was again attracts people's attention. Innovation of space transportation systems required CFD simulations for the flow environment that is difficult to physically realize on the ground. Recently, we do not see such epoch-making topics in CFD. We would like to understand it optimistically and consider that it indicates that CFD has become in the matured stage and people start using it as one of the simple analysis tools for the fluid dynamic research.

In the present paper, some of the recent applications of CFD to the development and basic research of the space transportation systems at the Institute of Space and Astronautical Science are shown. Three topics are chosen; (1) drag reduction of the boat-tail region of the air breathing engine, (2) flow field analysis of the truncated annular plug nozzles, and (3) analysis of the dynamic instability of the reentry capsule. These examples demonstrate the capability and effective use of the CFD technology for the space applications.

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## 23.2 Numerical Method

### 23.2.1 Basic Equations

The basic equations are the three-dimensional compressible Navier-Stokes equations written in the generalized coordinate system. Since one of the applications requires grid motion, it is necessary to include time metrics terms in the formulation.

These equations are solved timewise, but the basic equations are modified for the information exchange among each grid zone as two of the applications use overset zonal grid systems. The interface method used in the present study is called “Fortified Solution Algorithm (FSA)” where source terms are added to the basic equations as forcing terms to enforce the solutions for the other zones. The details of the algorithm can be found in Ref. 1.

### 23.2.3 Discretizations

The LU-ADI algorithm [2] is used for the time integration with the convective terms discretized by the SHUS scheme developed by Shima and Jonouchi[3]. The SHUS scheme is one of the AUSM-type schemes with the modification for the solution monotonicity. Third-order accuracy is achieved by the MUSCL interpolation [4] using primitive variables. Viscous terms are discretized by the central differencing. The algebraic turbulence model by Baldwin & Lomax [5] is used since the Reynolds number is about on the order of  $10^6$  to  $10^7$  with some modification if necessary.

## 23.3 Results and Discussions

### 23.3.1 Boat-tail drag reduction of an airturbo RAMjet engine

The airturbo RAMjet engine (so called ATREX engine) is under development in the Institute of Space and Astronautical Science (ISAS) for the future use as a booster of TSTO (Two-stage to Orbit) transportation systems[6].

One of the current problems of this RAMjet engine is the amount of drag induced by the boat-tail region. At transonic or low supersonic speeds, the boat-tail drag may go up to more than 20% of the total thrust. Figure 1 shows the schematic picture of the

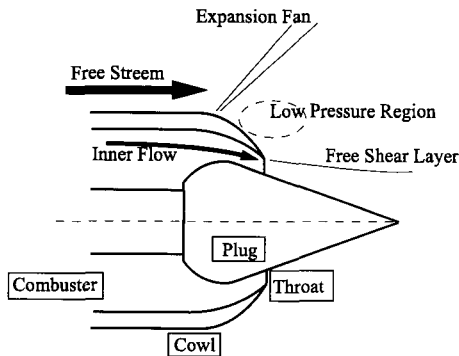


Fig. 1 Boat-tail region of the ATREX plug nozzle

Figure 1 shows the schematic picture of the

boat-tail region. Plug nozzle is used to achieve high thrust for the entire flight path, and there occurs a low-pressure region between strong expansion of the freestream and the plume boundary from the cowl lip. This low-pressure region causes, so called "boat-tail drag".

The basic flow structure is discussed first based on the simulations, and then secondary flow injections are tried for the drag reduction. Three-dimensional Navier-Stokes equations were used for the future computations for the body at angles of attack, although the results here is restricted to axisymmetric flows.

Two grid systems; one for the internal region of the nozzle (101\_17\_81) and the other for the outer flow (347\_17\_151), are used as overset grids. Figures 2 and 3 show these two grids respectively. The arrow in Fig. 3 is the direction and the location of the secondary flow injection. The geometry of the nozzle and the body is taken from the experiment conducted at the ISAS for the ATREX engine development. The design pressure ratio is 9.86 and the nozzle area ratio is 1.92.

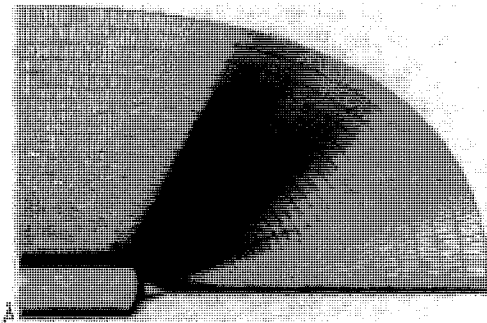


Fig. 2 Computational grids (outer region)

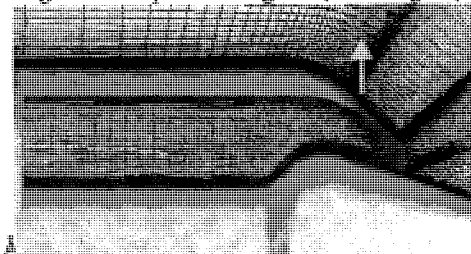


Fig. 3 Computational grids (inner nozzle region)

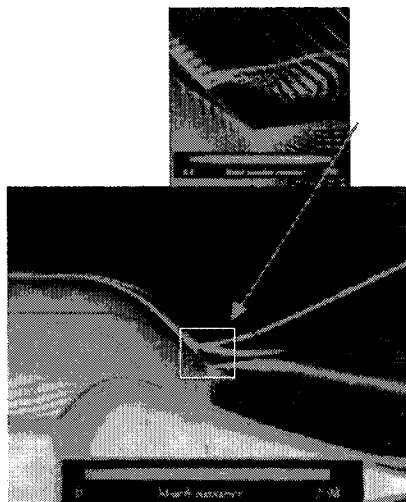


Fig. 4 Mach number contour plots ( $\xi=6.5$ ) with the close-up view of the velocity vector plots

The freestream Mach number is 1.5 and the Reynolds number is  $1.37 \times 10^7$ . The ratios of the chamber total pressure and the freestream static pressure are 4.5, 5.5 and 6.5. The total flow condition for the secondary flow injection is given as that of the internal flows.

The computed Mach number contour plots are shown in Fig. 4 for the pressure ratio of  $\xi=6.5$  with no injection. Strong expansion is observed along the boat-tail region, and as a result, low pressure region is created. Separation shock wave appears in front of the nozzle plume and there occurs a recirculating region between the separation shock wave and the plume. Figure 5 shows the pressure distributions along the boat-tail region and the plug. The pressure at the recirculating region is lower than the freestream pressure and it becomes a cause of the drag. However, much wider low pressure region appears by the expansion in front of the separation shock wave, as observed in Fig. 5, and the reduction of this region turns out to be the key for the reduction of the boat-tail drag. Although not shown here, separation shock wave moves forward and the boat-tail drag reduces when the pressure ratio is increased. The result indicates that the low-pressure region would become small and the drag would reduce by the forward movement of the separation shock wave.

Secondary flow injection was tried for the purpose of reducing the low pressure region. Figure 6 shows the Mach number contour plots. Compared to Fig. 4, separation shock wave moves forward and is located in front of the secondary jet injection. The expansion of the nozzle plume is stronger due to the change of the ambient pressure. In Fig. 7, the pressure distributions for this case are plotted. The low pressure region moves forward to the flat region of the body. The boat-tail drag reduces for all the

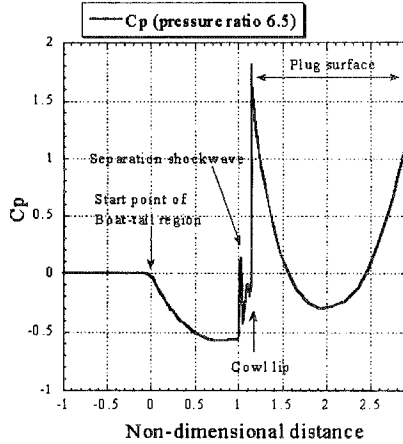


Fig. 5 Pressure distributions over the boat-tail region and the plug surface



Fig. 6 Mach number contour plots ( $\xi=6.5$ ) – with secondary injection

three different pressure ratios by the injection although not shown here. The comparison of the boat-tail drag shows the secondary injection effectively decrease the boat-tail drag at any pressure ratio although it is not shown here.

In summary, the flow characteristics of the boat-tail region were clarified and the reason for the boat-tail drag was identified. The effectiveness of the secondary flow injection for the drag reduction was also demonstrated.

23.3.2 Flow field analysis of the annular truncated plug nozzles

A Single Stage to Orbit (SSTO) vehicle is one of the two launching systems expected in the next century. A highly-efficient propulsion system is one of the key factors to realize a SSTO. Plug nozzles seem to be a good candidate for the system and have been studied in the last several years. Unlike a bell-shaped nozzle, the nozzle flow is not confined by the wall but instead, the exhausted jet is bounded by the external flow. The plug nozzle is considered to have globally better performance as the jet boundary adjusts its configuration to the ambient pressure and therefore the jet expands optimally for the entire conditions (e.g. altitude). In addition, the nozzle performance is not influenced by the truncation (cut-off the rear part) of the nozzle. These favorable features of plug nozzles were confirmed by the previous studies but the discussion on the mechanism was insufficient due to the limited number of experimental and/or computational studies.

In the present research, plug nozzle flow fields are computationally simulated for the investigation of the features above. Two parameters; the chamber to

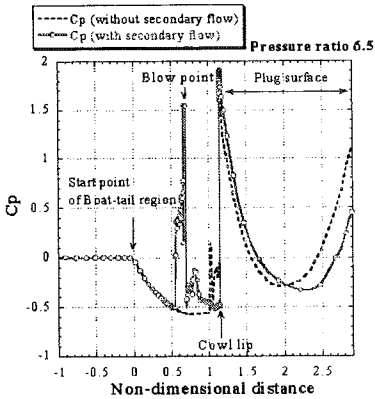


Fig. 7 Pressure distributions along the boat-tail region and plug surface –with injection

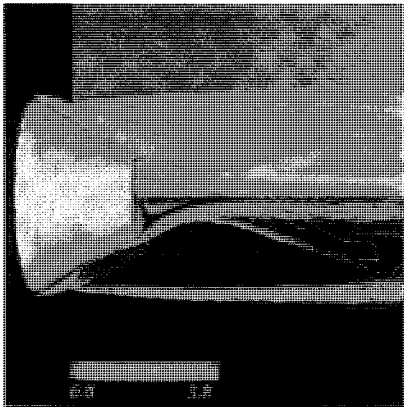


Fig. 8 Computed Mach number distributions (contoured plug nozzle, pressure ratio 71.0 -optimum)

ambient pressure ratio and the length of the nozzle are systematically changed. The flow structures are mainly investigated for the contoured plug nozzle and the thrust performance is evaluated quantitatively with the following discussion of the flow mechanism of the plug nozzles.

An annular-type conical plug having 25 degrees of its half angle is used for the validation of the present computations[7]. Although not shown here because of the length limitation, the computed pressure distributions agree well with the experiment by Tomita et al.[8] for all the pressure ratios and the result showed that the present approach is appropriate for the analysis.

The detailed analysis of the performance is next carried out for the contoured plug nozzle designed by the method of characteristics. Comparison of the truncated nozzles is made for five types of plug nozzles retaining 20, 30, 40, 50 and 100% portion of the total plug length. Optimum expansion is achieved at the pressure ratio 71.0. The Reynolds number is set to  $5 \times 10^5$  based on the length between the plug axis and cowl lip. One of the results is shown in Fig. 8. For this 20% truncated (80% cut-off) nozzle, the flow separates at the base edge and expands. The resultant shear layer at the edge of the base region induces the trailing shock wave when it converges and turns the flow parallel to the axis.

Figure 9 shows the portion of the thrust produced by each component of the plug nozzle at the pressure ratio of 500. The plug truncation results in the shorter nozzle length. The ramp area for the nozzle decreases and the pressure thrust produced at the ramp also decreases by the truncation. On the other hand, the thrust generated by the base pressure increases due to the increase of the base area. It compensates the total thrust loss caused by the decrease of the ramp pressure thrust. Accordingly, the total nozzle thrust becomes almost the same for any nozzle truncation.

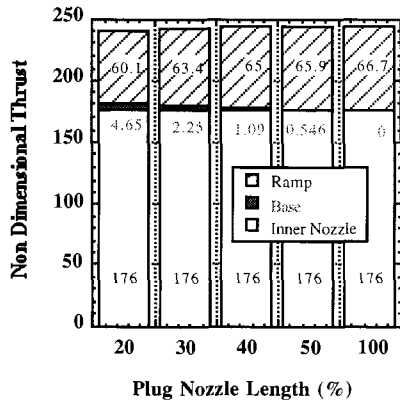


Fig. 9 \_Portions of the thrust for each truncated nozzle at the pressure ratio of 500

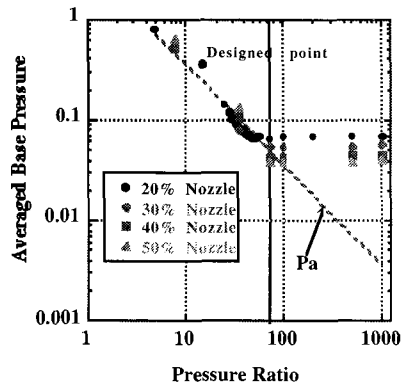


Fig. 10 Averagaed base pressure .vs. pressure ratio (without external flows)



Figure 10 shows the averaged base pressure for the 20, 30, 40 and 50% plug nozzles plotted versus various pressure ratios. The dash line shows the environmental pressure, which decreases as the pressure ratio becomes higher (high altitude). For low-pressure ratios (low altitudes), the base pressure linearly decreases as the environmental pressure decreases. In this region, the jet does not converge in the wake and the base region opens to the external environment. Therefore, the base region is influenced by the external environment and the pressure on the base quantitatively equals to the environmental pressure. The pressure thrust produced as the pressure difference between the environment and the base is small at the low-pressure ratios. On the other hand, at high-pressure ratios the pressure on the base becomes constant despite the variation of the pressure ratios. In the high-pressure ratio region, the jet expands and converges at downstream at the axis. The recirculating region is created behind the base and the base pressure becomes independent from the external flow conditions. As the altitude (pressure ratios) becomes higher, the environmental pressure decreases and the difference between the base pressure and the environment pressure increases. Therefore the base pressure thrust increases. This thrust increase due to the base region makes the plug nozzles to have high efficiency for the entire altitudes.

It has been shown that the base pressure almost equals to the environmental pressure when the pressure ratio is low, and becomes independent from the environmental pressure when the pressure ratio is high. The reason used in the explanation is that the wake region is open for low pressure ratios and is closed and recirculating region is created for high pressure ratios. This is common belief in the past. However, some of the computations showed that the base pressure is not independent from the external pressure even when the wake is closed. What is the mechanism to make a sudden change of the feature of the base pressure? There can be found two references for the mechanism[9,10].

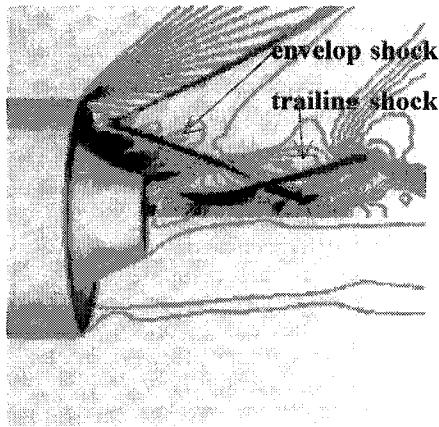


Fig. 11 Pressure contour plots (upper half) and the corresponding subsonic region (pressure ratio: 35.0)

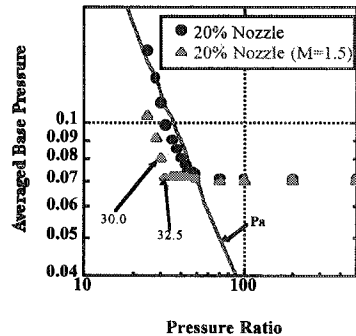


Fig. 12 Averaged base pressure vs. pressure ratio (with external flows)

Both papers indicate that the compression wave/envelop shock emanating from the cowl lip plays an important role. The former defines the criteria based on the location of the wake stagnation at the trailing shock generation, while the latter defines the criteria based on the location of the envelop shock impingement. To discuss the detailed mechanism of the phenomenon, further computations were carried out. As the study in our previous work showed that the external flow does not change the essential flow mechanism except the envelop shock to be strengthened, the flow mechanism is discussed for the simulations with external flows. The result for the 20% truncated nozzle is mainly discussed. As an example of the computed flow fields, the computed pressure contours along with the Mach number contours showing the subsonic region are plotted in Fig. 11 for the pressure ratio 35.0, where two high pressure regions due to the shear-layer impingement and the envelop shock impingement are shown.

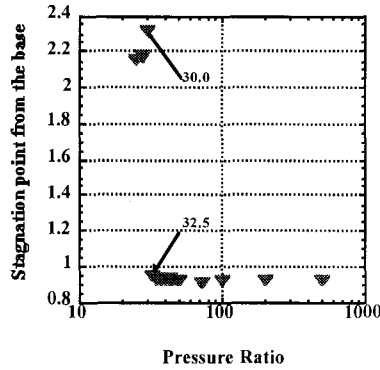


Fig. 13 Location of the stagnation point in the centerline of the wake region

First, relation of the base pressure to the pressure ratio is re-plotted for the cases with external flows in Fig. 12. The flow feature changes at the pressure ratio 32.5. At the pressure ratio 30.0 the base pressure is influenced by the external flows but is not so for the value more than 32.5. At the pressure ratio 30.0, the trailing shock wave impinges on the subsonic part of the base region. At pressure ratio 50.0, the trailing shock wave impinges on the supersonic part of the base region. At pressure ratio 35.0, the trailing shock wave impinges on the subsonic region (See Fig. 11), but the base pressure is independent from the external flows according to Fig. 12. Whether the shock impinges to the subsonic or supersonic part of the base region does not necessarily corresponds to the feature change of the base pressure[11]. Before looking into the flow-field details near the pressure ratio 32.5, locations of the stagnation points in the centerline of the wake are plotted against the pressure ratio in Fig. 13. The stagnation location changes its position at the pressure ratio 32.5, the value where the

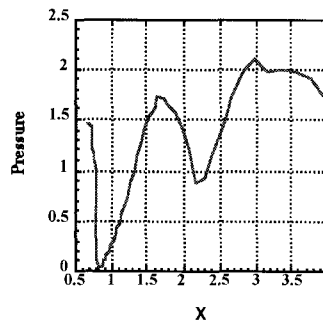


Fig. 14 Computed pressure contour plots and the pressure distributions along the centerline (pressure ratio 30.0)

strong change occurs on the base pressure. It indicates that the sudden movement of the stagnation point may be the key for the sudden change of the base pressure.

The pressure distributions along the wake centerline are plotted in Fig. 14 for the pressure ratio 30.0. There are two high-pressure regions in the wake. First peak occurs due to the shear layer impingement and the second peak occurs due to the envelop shock impingement. The second peak is much higher. The flow stagnates there and strong reverse flow occurs. The result for higher pressure ratio (35.0) is plotted in Fig. 15. Front peak is now higher than the second peak. Another pressure peak appears as the forward flow from the front stagnation and the backward flow from the second stagnation impinge each other. The results for still higher pressure ratios all show that the front stagnation pressure peak is much higher than the peaks downstream. The flow near the base never feels the pressure downstream as the flow downstream does not go into the base region due to the existence of the high pressure region created by the shear layer impingement.

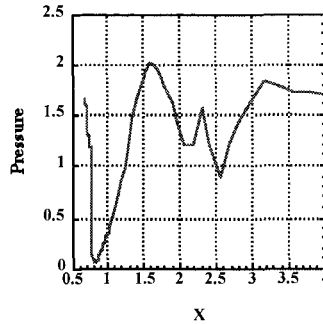


Fig. 15 Computed pressure contour plots and the pressure distributions along the centerline (pressure ratio 35.0)

The study here clearly indicates that the two high pressure regions induced by the shear layer and the envelop shock wave in the wake rule the flow mechanism. Whether the base pressure is influenced by the environmental pressure or not depends on the reverse flow from the stagnation region induced either by the envelop shock wave and the shear layer impingement. The analysis shows that the base pressure characteristics are governed not by the pressure wave (speed of sound) but by the convective velocity phenomenon in the wake region[11].

### 23.3.3 Analysis of the dynamic instability of a reentry capsule

A blunt reentry capsule tends to be dynamically unstable at transonic speeds. The instability phenomenon has been known since 1960's and has been mainly studied with the experiment. The mechanism of the instability, however, has not been discussed until recently. The Institute of Space and Astronautical Science has a sample return project, where the capsule with the sample of an asteroid experiences high-speed reentry. The parachute decelerating the capsule will be opened at low speed, and therefore the capsule faces this instability at transonic speeds.

Recently, Hiraki at the ISAS studied the flow field by the experiment and showed that the instability is caused by the phase delay of the base pressure. The following com-

putational research by the present authors also supported this result[12].

In the present paper, a series of the computational studies[12-14] on this subject is summarized. A new post-processing method which was really inevitable for the good understanding of the flow structure was developed during the course of the study.

The flow field over the capsule in the forced pitching oscillation is first simulated. The maximum pitch angle is  $\pm 20$  degrees and the frequency of the oscillation is 20 Hz which is the observed value in the wind tunnel experiment by Hiraki, et al. The maximum diameter is 100 mm and the body length is 50 mm. The corresponding Reynolds number is  $2.5 \times 10^6$  and the freestream Mach number for the computation is 1.3. The total number of the grid points used in the study is roughly 1 million. Figure 16 shows the structure (schlieren image) of the flow field over the capsule in the forced pitching oscillation. There occurs a bow shock wave, shear layer, recompression shock wave, and these make the flow field to be complicated. As the Hiraki's experiment showed, there is a time delay in the base pressure. Therefore, the time history of the pressures on the representative points in the front and base are plotted to confirm the experimental observation. The result shows that there is no pressure delay on the front part, but there is a time delay of roughly 3 msec (15 degrees) in the base pressure. The frequency of the pitching oscillation observed in the experiment is 20 Hz and the reduced frequency becomes order of  $10^{-2}$ . Therefore, we may consider the flow field is quasi-steady although we have to consider the time delay of the base pressure.



Fig. 16 Schlieren image of the flow field over an oscillating capsule (M=1.3, pitch=-20 to 20 degrees)

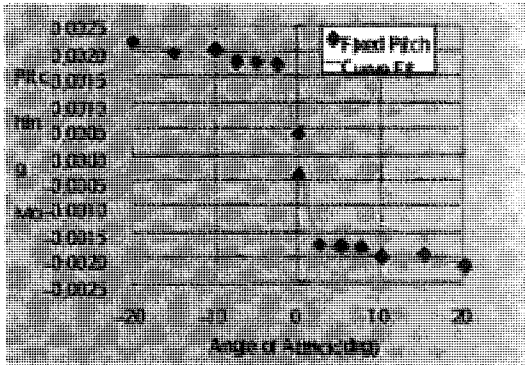


Fig. 17 Base pitching moment for each pitch angle (result for the fixed pitch)

The flow field over the capsule at the fixed pitch angle is therefore studied in detail. The pitching moment created only by the base pressure distributions is plotted in Fig. 17 for several pitch angles. Two important features are observed in this plot. First, pitching moment is positive for negative pitch angles, and negative for positive pitch angles. The pitching moment created by the base region contributes to the static stability. Second, the pitching moment is almost constant at high pitch angles (absolute values), and the pitching moment abruptly changes its sign near zero pitch angle. The result indicates that the base flow field is similar for most of the pitch angles except near zero degrees. The instantaneous streamlines plotted in Fig. 18 clearly prove it. There appears a pair of vortices; one large vortex in the bottom and one small counter-rotating vortex above.

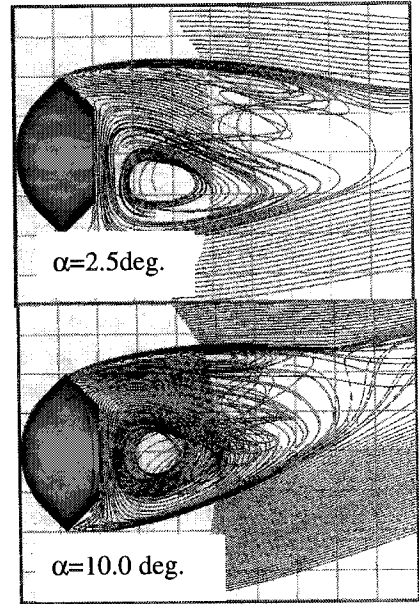


Fig. 18 Instantaneous streamlines for the capsule at fixed pitch

The pitching moment for the capsule now in the pitching oscillation for several cycles is plotted in Fig. 19. Also plotted are the curves that correspond to 'constant delay model'. The 'constant delay model' means that the flow field for the capsule in pitch oscillation is interpreted as quasi-steady solution with the time delay of the base pressure. In the 'constant delay model', there occurs certain time delay of the (base) pitching moment and hysteresis appears in a cycle of the plots. Figure 19 indicates, although some data scatter is observed, that the 'constant delay model' is acceptable. A few msec-delay exists in the base flow field for the capsule in pitching oscillation and we may discuss the flow structure based on the fixed pitch simula-

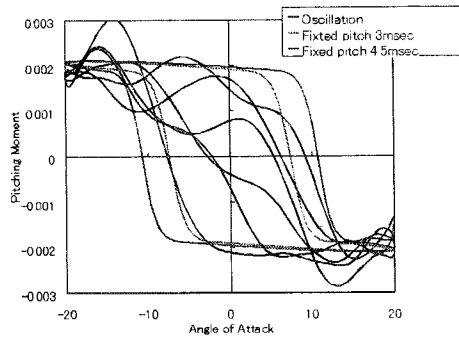


Fig. 19 base pitching moment for the capsule in pitch oscillation - hysteresis loop

tions with certain time-delay in mind.

Time-averaged base-pressure distributions along the centerline from the bottom to the top are plotted in Fig. 20 for the simulation result at several fixed pitch angles. The pressure level becomes lower as the pitch angle increases. There is a relatively high-pressure region on the top part (right side in the figure) of the base for all the pitch angles, and the difference between this region and the other lower region becomes larger as the pitch angle increases. Although not shown here, this high pressure region is created by the convection of strong reverse flows in the recirculating region in the wake (see Fig. 18). It should also be noted that the high pressure region appears only for the capsule that has a tendency of dynamic instability (See Ref. 13 for the details ). As there is a time delay of the base pressure for the capsule in pitch motion, the pressure difference for the higher and lower parts of the capsule base becomes smaller than the capsule at fixed pitch angle. It would contribute to the dynamic instability as the high pressure region does not appear immediately to the pitch angle change in the motion.

To confirm that this high pressure region exists in the flow field for the capsule in pitch motion and that there is a time delay in the base flow field, time-accurate particle traces are plotted for the capsule in pitch motion. However, the flow field does not show clear structure. The reason is that the amplitude of the vortex shedding from the edge of the capsule is almost the same as the amplitude of the flow structure change by the pitch motion and they cannot be separated from the animation. A new concept of the digital filter is defined. For the time sequence of the flow variables at each grid point, frequency filter is operated. In this case, low-pass filter (LPF) with the cut-off frequency of 150 Hz is operated for the high-frequency flow structure to be removed. The result clearly indicates that the high pressure region exists in the flow field for the capsule in pitch motion and there is a time delay in the base flow field. The flow structure changes after a certain pitch angle delay. Figure 21 shows an instantaneous shot of the particle traces. The details of the flow change can be found in Ref. 14. This filtering technique is useful for many types of the flow fields. The method is discussed in Ref. 15 with other examples.

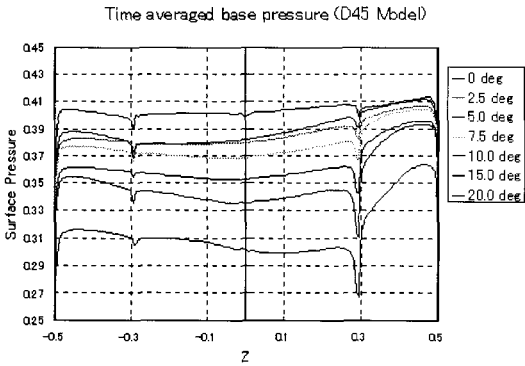


Fig. 20 Pressure distributions on the vertical centerline of the base –fixed pitch

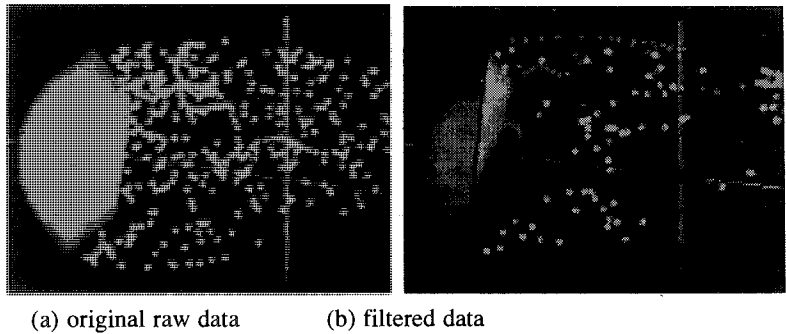


Fig. 21 Time-accurate particle traces in the base region

The vortex-identifying technique developed by Sawada [16] is used to clarify the vortex structure in the wake region. The result is shown in Fig. 22. The two counter-rotating vortices that appeared in Fig. 18 are not isolated each other, but they are a combined one vortex ring connected as the figure shows. Lower part (when positive pitch angle) is located forward and this strong ring creates large recirculating region near the base (see Fig. 18). The other part of the ring is rather weak and located away from the base. Because of the difference of each side of the vortex, there occurs a pair of the two longitudinal vortex streets toward downstream. This longitudinal vortices are similar to the wing tip vortex. The strength of the tip vortex reflects the static lift coefficient of the wing, and if the same is true for the capsule, the static lift of the capsule is directly connected to the formation of the pair of the longitudinal vortices. Dynamic instability occurs due to the time delay of the high-pressure region on the upper part of the base. This high-pressure region occurs due to the vortex structure behind the capsule and the vortex structure is associated with the pair of the longitudinal vortices. Based on such understanding of the flow mechanism, we may discuss the dynamic instability of this type of capsule based on the static lift characteristics, as the strength of the longitudinal vortices is determined by the strength of the lift (just like a tip vortex). The discussion has just started, and no conclusions have been obtained yet, but some of the results indicate the positive reason for this assumption.

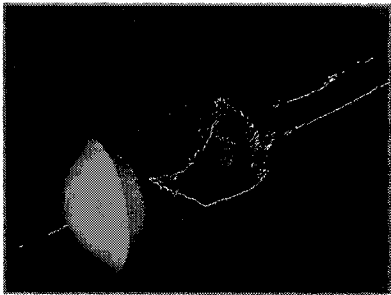


Fig. 22 Vortex core distributions

23.4 Conclusions

Some of the recent applications of CFD to the development and basic research of the space transportation systems at the Institute of Space and Astronautical Science were presented. Three topics were chosen; (1) drag reduction of the boat tail region of the air breathing en-

gine, (2) flow field analysis of the truncated annular plug nozzles, and (3) analysis of the dynamic instability of the reentry capsule. The flow-field and the boat-tail drag characteristics were clarified and the effect of the secondary flow injection was demonstrated. The flow analysis of the truncated plug nozzle showed the characteristics of the plug nozzles, and the key flow features behind these characteristics was revealed. The analysis of the dynamic instability of a reentry capsule showed that the delay time for the base pressure was the key for the instability, and the flow mechanism was clarified. These examples demonstrated a recent capability of the CFD technology for the space applications.

### 23.5 Acknowledgement

The research presented here is the collaborative result of the author's laboratory in the Institute of Space and Astronautical Science. The study on the boat-tail drag reduction was conducted by Kazuhiro Imai, (currently MHI) the graduate student of Aerospace Dept. University of Tokyo. The study on the annular truncated plug nozzle for the RLV was mainly conducted by Takashi Ito (currently graduate student, Aerospace Dept. University of Tokyo), the graduate student of Aoyama-Gakuin University. The study on the dynamic stability of the capsule was mainly conducted by Susumu Teramoto, the research associate in the laboratory.

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# Multipoint Optimal Design of Supersonic Wing Using Evolutionary Algorithms

Shigeru Obayashi, Yukihiro Takeguchi,  
and Daisuke Sasaki<sup>1</sup>

## 24.1 INTRODUCTION

Demand for developing a new Supersonic Transport (SST) is expected to become larger, because people travel overseas more frequently. Last ten years, several efforts have been made to develop a new SST in the States, Europe and Japan. In Japan, National Aerospace Laboratory (NAL) is studying and designing a SST to launch the small supersonic experimental airplane<sup>1</sup> in 2002.

Considering a new SST design, there exist many technical difficulties to overcome.  $L/D$  must be improved, and the sonic boom should be prevented. However, there is a severe tradeoff between lowering the drag and boom. As a result, a new SST is expected to cruise at a supersonic speed only over the sea and to cruise at a transonic speed over the land. This means that the important design objectives are not only to improve a supersonic cruise performance but also to improve a transonic one. For example, a large sweep angle can reduce the wave drag, but it limits the allowable aspect ratio due to structural problems. Therefore, there are many tradeoffs to be addressed in designing a SST.

To identify global tradeoffs, the problem can be treated as multiobjective (MO) optimization. MO optimization seeks to optimize the components of a vector-valued objective function. In general, the solution to this problem is not a single point unlike single objective optimization, but a family of points known as the Pareto-optimal set. Pareto solutions, which are members of the Pareto-optimal set, represent tradeoffs among multiple objectives. Multiobjective Genetic Algorithms (MOGAs) are unique optimization methods to sample multiple Pareto solutions efficiently and effectively.<sup>2</sup> GAs are the optimization methods that imitate the natural evolution. Since GAs seek optimal solutions in parallel using a population of design candidates, MOGAs can identify multiple Pareto solutions at the same

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time without specifying weights between objectives.

This paper considers the multipoint aerodynamic optimization of a wing shape for a SST at both supersonic and transonic cruise conditions. Aerodynamic drags will be minimized at both conditions under lift constraints. Bending moment at the root will also be minimized so as to prevent all the Pareto solutions having impractically large aspect ratios. In the aerodynamic optimization, design variables specify planform shapes, camber, thickness distributions and twist distributions. In the previous study, the same MO optimization was performed using a potential solver and an Euler solver under the inviscid flow assumption.<sup>3</sup> To consider more realistic flow fields such as a possible flow separation, the viscous effect is considered in the present optimization. Thus, a Navier-Stokes solver is used to evaluate the wing performance at both cruise conditions. Finally, the resulting Pareto solutions are analyzed and compared with NAL's design and the previous results.

## 24.2 OPTIMIZATION METHOD

Application of GAs to MO optimization has many advantages. Their advantages originate in the algorithms themselves, which imitate the mechanism of the natural evolution, where a biological population evolves over generations to adapt to an environment by selection, crossover and mutation. In design optimization problems, fitness, individual and genes correspond to an objective function, design candidate and design variables, respectively.

GAs search from multiple points in the design space simultaneously and stochastically, instead of moving from a single point deterministically like gradient-based methods. This feature prevents design candidates from settling in local optimum. Moreover, GAs do not require computing gradients of the objective function. These characteristics lead to following three advantages of GAs: 1, GAs have capability of finding global optimal solutions. 2, GAs can be processed in parallel. 3, high fidelity CFD codes can easily be adapted to GAs without any modification because GAs use only objective function values.

GAs have been extended to solve MO problems successfully.<sup>2</sup> GAs use a population to seek optimal solutions in parallel. This feature can be extended to seek Pareto solutions in parallel without specifying weights between the objective functions. The resultant Pareto solutions represent global tradeoffs. Therefore, MOGAs are quite unique and attractive methods to solve MO problems.

Figure 1 shows the flowchart of MOGAs in the present study. The following describes genetic operators employed here in brief. Traditionally, GAs use binary numbers to represent design parameter values. For real function optimizations like the present aerodynamic optimization, however, it is more straightforward to use real numbers. Thus, the floating-point representation is used here. Selection is based on the Pareto ranking method and fitness sharing.<sup>2</sup> Each individual is assigned to its rank according to the number of individuals that dominate it. A standard fitness sharing function is used to maintain the diversity of the population. To find the extreme Pareto solutions more effectively, the so-called best- $N$  selection<sup>4</sup> is also

coupled with. Blended crossover<sup>5</sup> (BLX- $\alpha$ ) described below is adopted. This operator generates children on a segment defined by two parents and a user specified parameter  $\alpha$ . In this optimization, a weighted average of new design variables is used as

$$\begin{aligned} \text{Child1} &= \gamma\_Parent1 + (1-\gamma)\_Parent2 \\ \text{Child2} &= (1-\gamma)\_Parent1 + \gamma\_Parent2 \\ \gamma &= (1 + 2\alpha)\_ran1 - \alpha \end{aligned} \quad (1.1)$$

where Child1,2 and Parent1,2 denote encoded design variables of the children (members of the new population) and parents (a mated pair of the old generation), respectively. The random number shown here *ran1* is uniform in [0,1]. Parameter  $\alpha$  is set to 0.5 except for the six planform design variables. Since the planform has a large impact on aerodynamic performance, its design parameters have to be given conservatively. Otherwise, the computation diverges and many children cannot be evaluated. Therefore, parameter  $\alpha$  is set to 0.0 for those six design variables. Mutation takes place at a probability of 20%. If the mutation occurs, then Eqs. (1) will be replaced by

$$\begin{aligned} \text{Child1} &= \gamma\_Parent1 + (1-\gamma)\_Parent2 + m\_(\text{ran2}-0.5) \\ \text{Child2} &= (1-\gamma)\_Parent1 + \gamma\_Parent2 + m\_(\text{ran2}-0.5) \end{aligned} \quad (1.2)$$

where *ran2* are also uniform number in [0,1] and *m* is set to 10% of the given range of each design variable.

Objective functions for each individual are to be evaluated using a CFD solver. In order to evaluate the viscous effect, the three-dimensional Navier-Stokes equations should be solved. In this study, the three-dimensional, compressible, thin-layer Navier-Stokes solver is used to evaluate aerodynamic performances in both transonic and supersonic cruise conditions. This Navier-Stokes code employs total-variation-diminishing type upwind differencing and the lower-upper factored symmetric Gauss-Seidel scheme.<sup>6</sup> The multigrid method<sup>7</sup> is also used to accelerate the convergence. The turbulence model in this code adopts an algebraic mixing length model by Baldwin and Lomax.<sup>8</sup>

## 24.3 FORMULATION OF THE PRESENT OPTIMIZATION PROBLEM

The present optimization problem can be stated as follows.

[Objective functions]

1. Drag coefficient for transonic cruise,  $C_{D,t}$
2. Drag coefficient for supersonic cruise,  $C_{D,s}$
3. Bending moment at the wing root for supersonic cruise condition,  $M_{\text{root}}$

[Constraints]

1. Lift coefficients,  $C_{L,t} = 0.15$  and  $C_{L,s} = 0.10$  at cruise conditions
2. Wing area,  $S = 60$
3. Maximum airfoil thickness,  $t/c \geq 0.03$

[Flow conditions]

1. Transonic cruising Mach number: 0.9
2. Supersonic cruising Mach number: 2.0
3. Reynolds number based on the root chord length at both conditions:  $1.0 \times 10^7$

In the present optimization, all the three objective functions are to be minimized. Both the supersonic and transonic drag coefficients are evaluated by using a Navier-Stokes flow solver. The bending moment is computed by directly integrating the pressure load at the supersonic cruise condition. To maintain lift coefficients constant, the angle of attack is predicted by using  $C_{L\alpha}$  obtained from the finite difference. Thus, three Navier-Stokes computations are performed per evaluation. During the aerodynamic optimization, wing area is frozen at a constant value. The wing thickness is also constrained for structural strength.

Design variables are categorized to planform, airfoil shapes and the wing twist. The wing planform is determined by six design variables as shown in Fig. 2 and their ranges are written in Table 1. A chord length at the wing tip is determined accordingly because of the fixed wing area. Airfoil shapes are composed of its thickness distribution and camber line. The thickness distribution is represented by a Bezier curve defined by nine polygons<sup>9</sup> as shown in Fig. 3. Table 1 also shows their design ranges. The thickness distributions are defined at the wing root, kink, and tip and then linearly interpolated in the spanwise direction. The total number of polygons is 27 for the entire thickness distribution. The camber surfaces composed of the airfoil camber lines are defined at the inboard and outboard of the wing separately. Each surface is represented by the Bezier surface defined by four polygons in the chordwise direction and three in the spanwise direction. For instance, Figure 4 shows the camber line with its control points at the root. It is concave only at the root and it becomes convex at the other spanwise locations similar to the warp design based on the linearized theory. The number of polygons that defines two camber surfaces is 20 polygons in total. Finally, the wing twist is represented by a B-spline curve with six polygons as shown in Fig. 5. As a result, 66 design variables are used to define a wing shape.

The present optimization was performed on FUJITSU VPP700E supercomputer system at The Institute of Physical and Chemical Research. The system has 160 PE's with 384 GFLOPS and 320 GB. The master PE manages MOGA, while the slave PE's compute the Navier-Stokes code. The population size was set to 64 so that the process was parallelized with 8-64 PE's depending on the availability. It should be noted that the parallelization was almost 100% because of the Navier-Stokes computations dominated the CPU time.

## 24.4 OPTIMIZATION OF A SUPERSONIC TRANSPORT WING

### 24.4.1 Overview of Pareto solutions

The evolution was computed for 30 generations. After the present optimization by MOGA, all the solutions evaluated were sorted again to find Pareto solutions as much as possible. As a result, the final Pareto solutions were obtained in the three-

dimensional objective function space as shown in Fig. 6. The tradeoff surface with the objective functions is exhibited in the figure. It also shows four typical planform shapes;  $C_{D,t}$  minimum,  $C_{D,s}$  minimum, bending moment minimum and a certain Pareto solution. The extreme Pareto solutions, three planform shapes that minimize the respective objective functions appear physically reasonable.

To present tradeoffs between the objectives more clearly, Pareto solutions are projected into the two-dimensional plane as shown in Figs. 7-9. Figures 7 and 8 present the tradeoffs between transonic and supersonic drag coefficients. The solutions are labeled by the aspect ratio, and the taper ratio using different symbols in Figs. 7 and 8, respectively. In Fig. 7, wings with larger aspect ratios achieve lower drag coefficients as expected in the aerodynamic theory. Figure 8 shows that the wings that have the taper ratios smaller than 0.4 have good aerodynamic performances, but further decrease of the taper ratio does not correspond to the reduction of cruising drag directly. On the other hand, the wings with the taper ratios larger than 0.4 have the lower bending moments and poor aerodynamic performances as shown in Fig. 9.

#### 24.4.2 Comparison with NAL's second design

To examine the quality of the present Pareto solutions, two Pareto solutions are compared with NAL's second design. NAL SST Design Team already finished the fourth aerodynamic design for the experimental supersonic airplane to be launched in 2002. To summarize their design concepts briefly, the first design determined the planform shapes among 99 candidates, and then the second design was performed by the warp optimization based on the linearized theory. The third design aimed a natural-laminar-flow (NLF) wing by an inverse method using a Navier-Stokes code. Finally, the fourth design was performed for a wing-fuselage configuration. Because a fully developed turbulence is assumed in the present Navier-Stokes computations, it is improper to compare the present Pareto solutions to NAL's NLF wing design. Therefore, the NAL second design is chosen for a comparison.

Table 2 summarizes comparisons of two Pareto solutions with NAL's second design. The aerodynamic calculation of NAL's second design is performed here by using the same Navier-Stokes solver. Pareto solutions A and B presented here are superior to NAL's second design in all three objectives. Figure 10 shows the wing planforms of the three, indicating a large difference of planform shapes between the present solutions and NAL's design. The present planforms are similar to the "arrow wing" planform and the NAL's planform is similar to the conventional "delta wing" planform.

The thickness distributions of the three wings are shown in Fig. 11. The trend of thickness distributions of Pareto solutions A and B are quite similar, having a blunt leading edge and a thin trailing edge. The thickness distribution of NAL's design is simply taken from an existing NLF airfoil. In contrast, the present optimization is performed under a fully turbulent flow with the thickness constrained. Therefore, the maximum thickness appears near the leading edge. Then, the thickness is reduced toward the trailing edge to prevent the rapid growth of the boundary layer.

#### 24.4.3 Difference between the viscous and inviscid calculations

The present viscous designs are compared with the inviscid designs computed previously.<sup>3</sup> By comparing the two optimization results, the difference of the wing shapes due to the viscous effect becomes clear. The Pareto solutions, which are found to outperform NAL's design in all three objectives at both cases, are selected for the comparison.

A comparison of the planform shapes is shown in Fig. 12. Both planform shapes are similar to the "arrow wing" planform, but the shapes are slightly different. The present wing has a less sweep angle and a less taper ratio than the optimized wing under the inviscid flows. A highly swept wing tends to have a flow separation near the wing tip. The present viscous design appears better than the inviscid design to prevent the tip separation.

Figure 13 shows a comparison of the thickness distributions at the root. It shows the quite different distributions. In the viscous case, the wing is thicker near the leading edge and thinner near the trailing edge. However, in the inviscid case, the wing is very thick. The  $C_p$  distributions shown in Fig. 14 explain their difference clearly. In the inviscid case, the  $C_p$  distribution has a discontinuity at the trailing edge, and therefore it generates the lift even at the trailing edge. However, such a thick airfoil probably causes a flow separation. On the other hand, there is no discontinuity at the trailing edge for the viscous flow case. It is important to consider the viscous effect for designing thickness distributions.

### 24.5 CONCLUSION

The multipoint design optimization of a wing for a SST has been performed by using MOGA. Three objective functions are used to minimize the supersonic drag, the transonic drag and the bending moment at the wing root. The complete wing shape is represented by in total of 66 design variables. The Navier-Stokes solver is used to evaluate those aerodynamic drags.

Successful optimization results are obtained. The planforms of the extreme Pareto solutions appear physically reasonable. Global tradeoffs between the objectives are presented. Two Pareto solutions have better performance in all three objective functions compared with NAL's second design. The comparison of the present Pareto solution with the optimal wing designed previously under the inviscid flow is also carried out to examine the viscous effect. The viscous effect is found to have a large influence on the thickness distribution. The present result is found better to prevent the possible boundary layer separation. The analysis of the Pareto solutions suggests that a desirable planform shape is a new type of the arrow wing with a relatively large taper ratio and a relatively small aspect ratio similar to the previous inviscid results.

### ACKNOWLEDGMENTS

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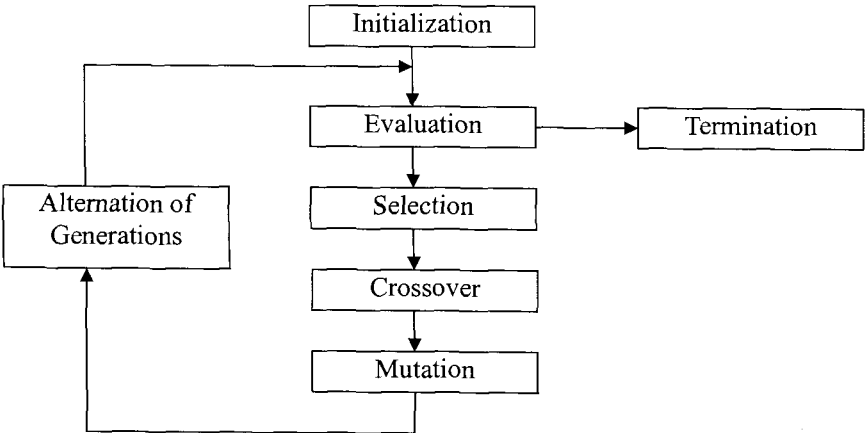


Figure 1: Flowchart of GAs

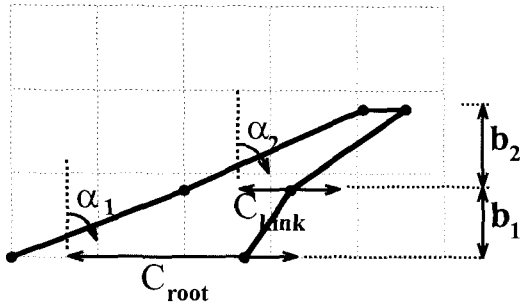


Figure 2: Wing planform definition

| Planform shape             |            |       |
|----------------------------|------------|-------|
| Chord length               | $C_{root}$ | 10-20 |
|                            | $C_{kink}$ | 3-15  |
| Span length                | $b_1$      | 2-7   |
|                            | $b_2$      | 2-7   |
| Sweep angle (deg)          | $\alpha_1$ | 35-70 |
|                            | $\alpha_2$ | 35-70 |
| Thickness distribution     |            |       |
| Max thickness (%)          | $Zp_4$     | 3-4   |
| Max thickness location (%) | $Xp_4$     | 15-70 |

Table 1: Ranges of planform and thickness distribution design variables

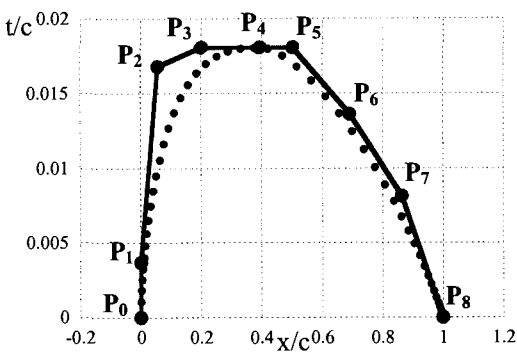
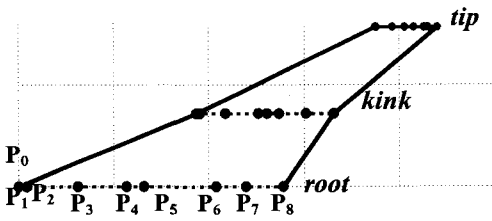


Figure 3: Wing thickness definition

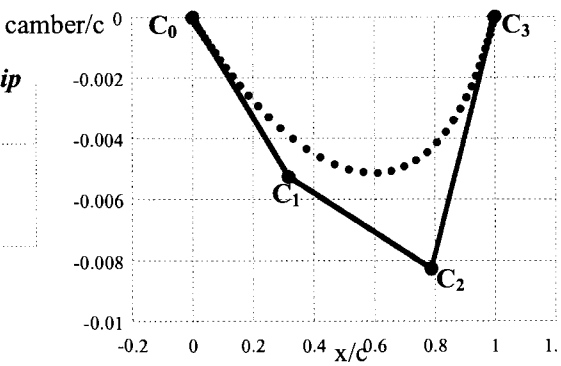
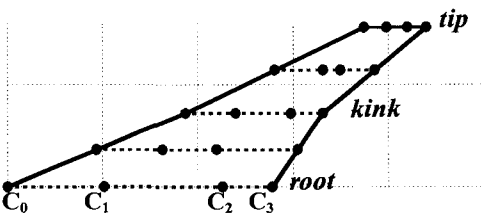


Figure 4: Wing camber definition

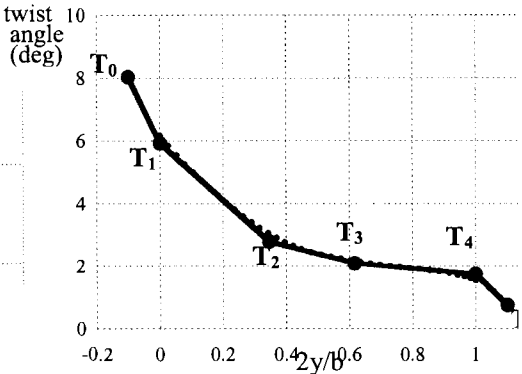
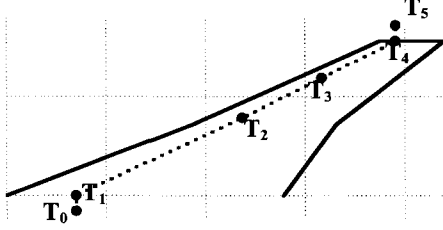


Figure 5: Wing twist definition

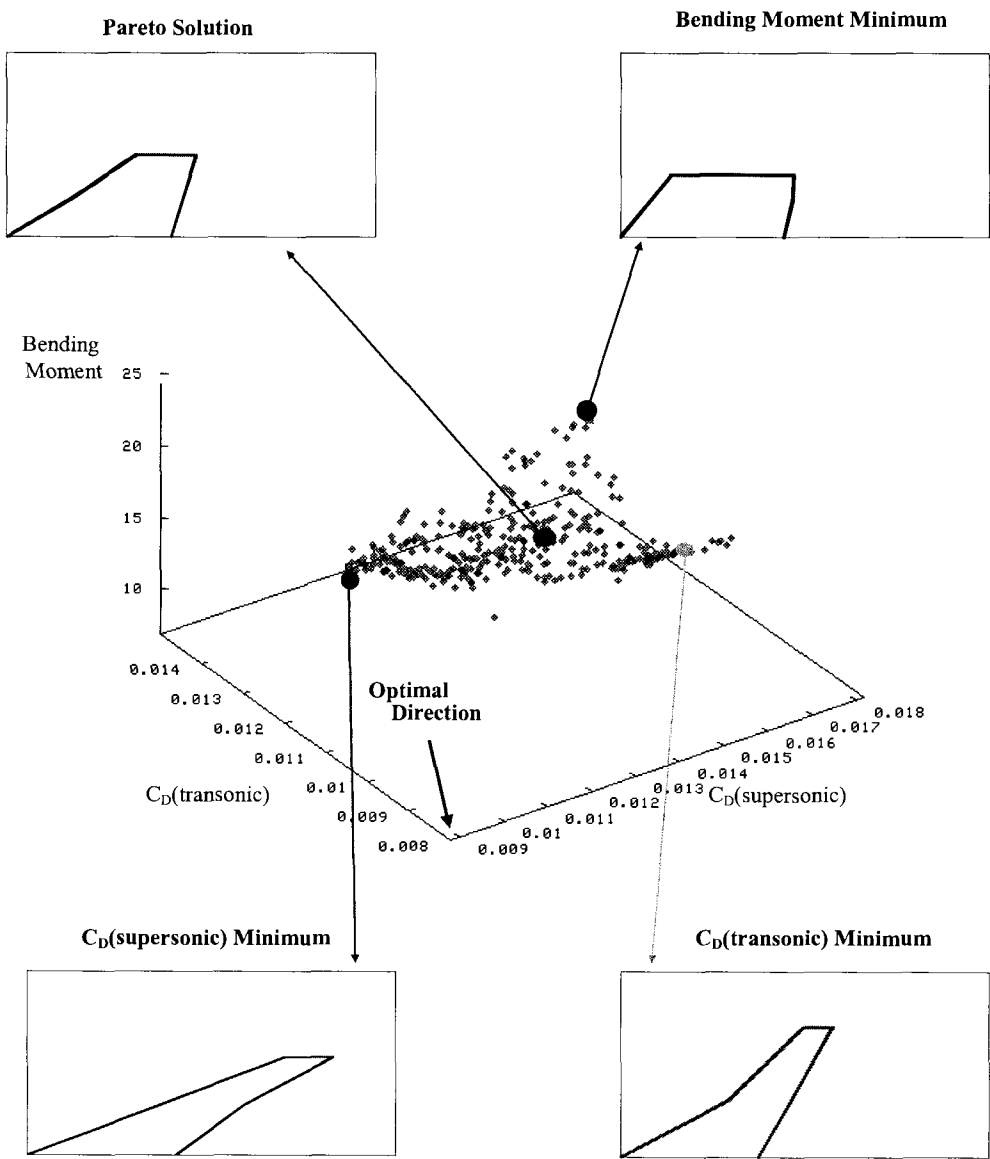


Figure 6: Pareto front in the objective function space and typical planform shapes

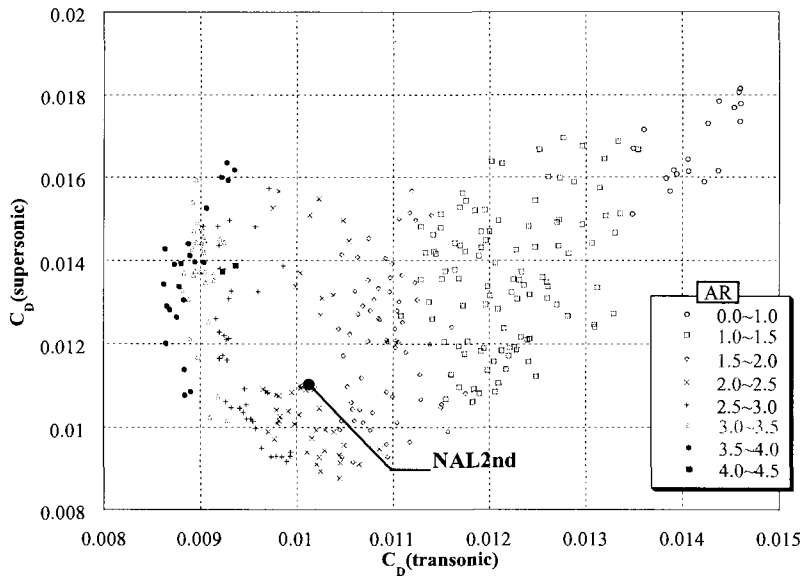


Figure 7: Projection of Pareto front to supersonic and transonic drag tradeoffs labeled according to aspect ratios. NAL's design is plotted here for a comparison although it is not Pareto optimal.

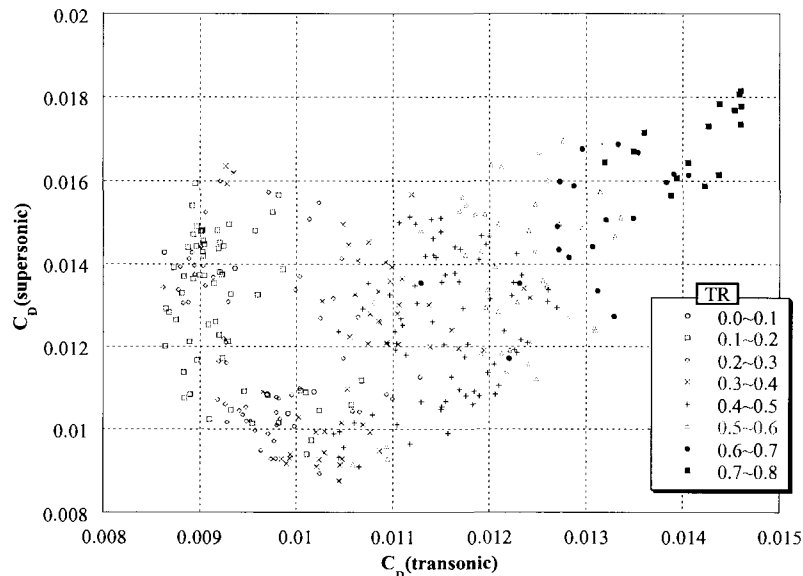


Figure 8: Projection of Pareto front to supersonic and transonic drag tradeoffs labeled according to taper ratios

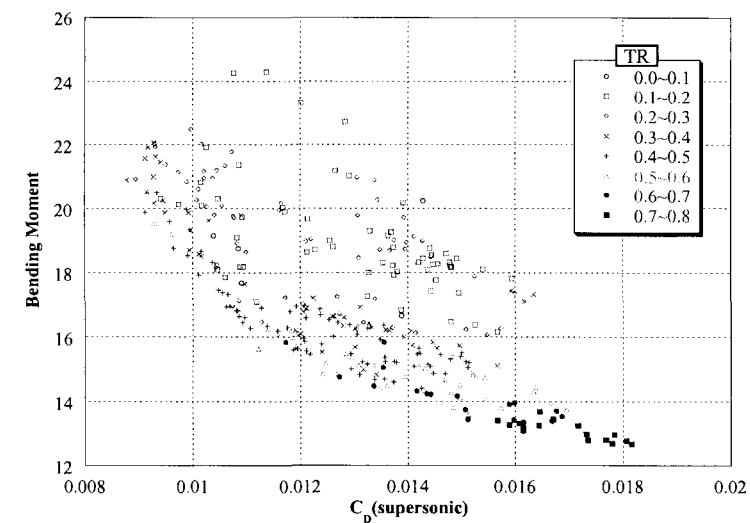


Figure 9: Projection of Pareto front to bending moment and supersonic drag tradeoffs labeled according to taper ratios

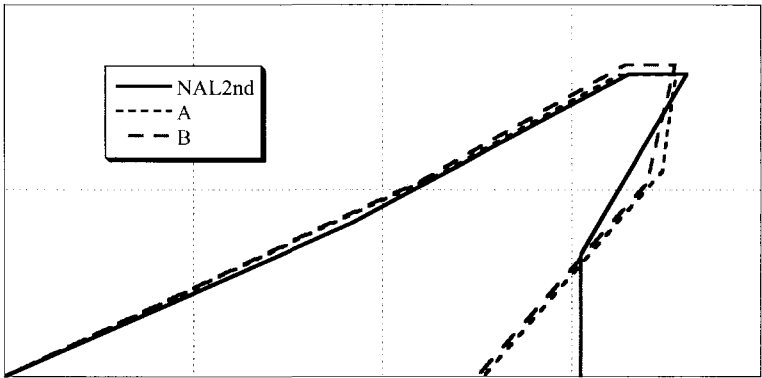
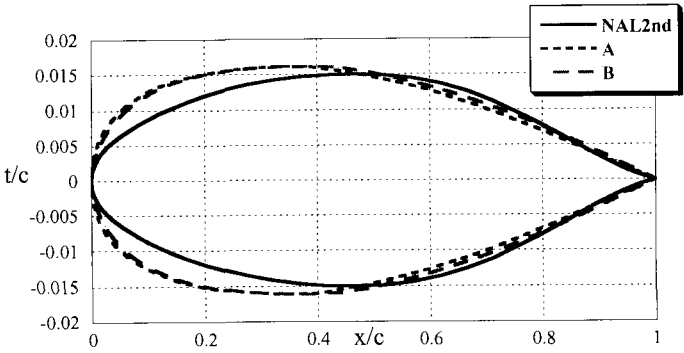


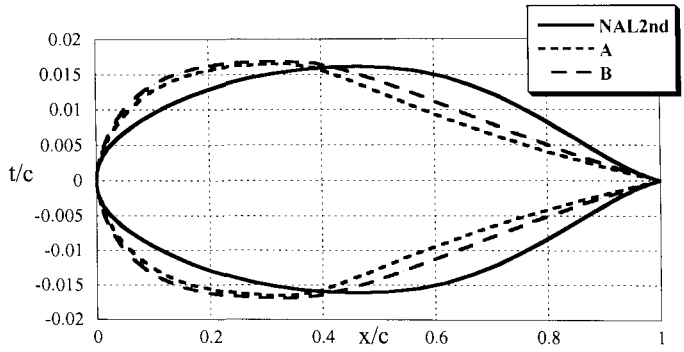
Figure 10: Comparison of planform shapes between selected Pareto solutions and NAL's design

|        | Aspect Ratio | Taper Ratio | $C_D(\text{transonic})$<br>( $\times 10^{-4}$ ) | $C_D(\text{supersonic})$<br>( $\times 10^{-4}$ ) | Bending Moment |
|--------|--------------|-------------|-------------------------------------------------|--------------------------------------------------|----------------|
| A      | 2.19         | 0.12        | 100.40                                          | 109.38                                           | 18.18          |
| B      | 2.34         | 0.11        | 100.96                                          | 108.89                                           | 18.18          |
| NAL2nd | 2.20         | 0.20        | 100.99                                          | 110.92                                           | 18.52          |

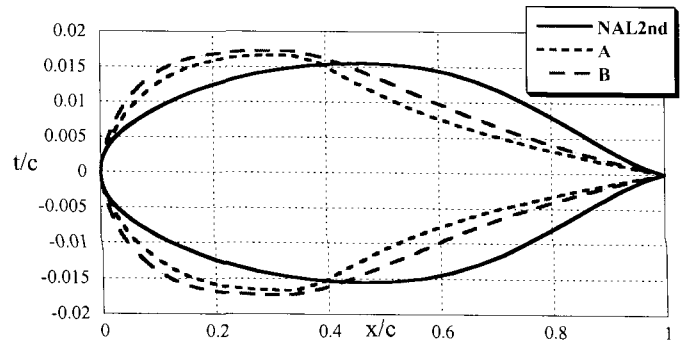
Table 2: Performance comparison among selected Pareto solutions and NAL's design



c) 66% spanwise location



b) 33% spanwise location



a) 0% spanwise location

Figure 11: Comparison of thickness distributions between selected Pareto solutions and NAL's design

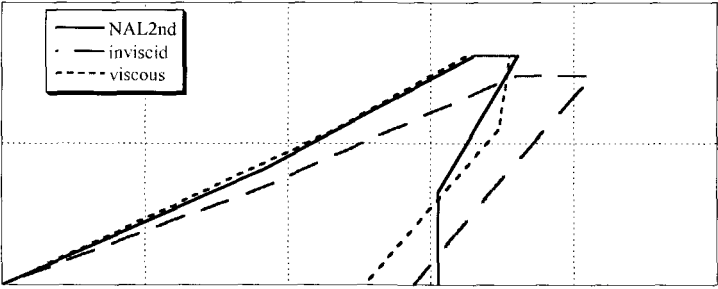


Figure 12: Comparison of planform shapes of the viscous and inviscid designs with NAL's design

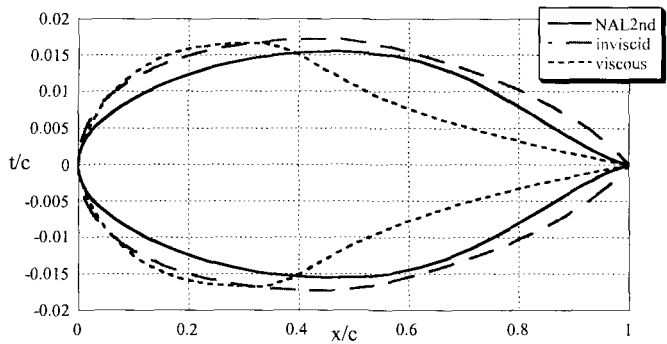


Figure 13: Comparison of thickness distributions of the viscous and inviscid designs with NAL's design

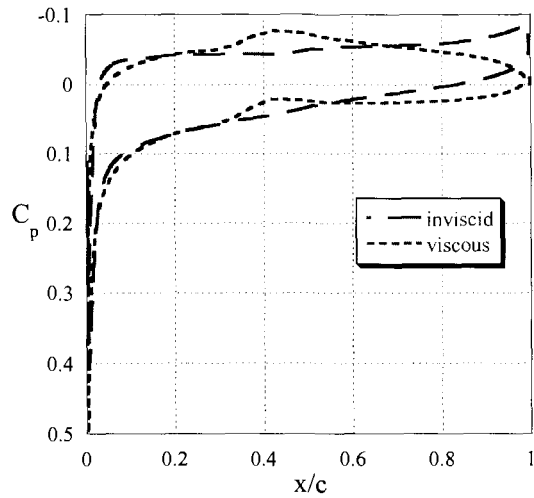


Figure 14: Comparison of  $C_p$  distributions of the viscous and inviscid designs with NAL's design



# Information Science – A New Frontier of CFD

Koichi Oshima<sup>1</sup> and Yuko Oshima<sup>2</sup>

## 25.1 Out of Deterministic Systems Into Complex Systems

Since the Renaissance in Europe, and in the Orient also, the purpose of science has been to find out the “Rule of Nature” and to obtain the “Result”. “Causality”, that is, “Deterministic Relation between Cause and Effect” has been the fundamental presumption of modern science. Scientists treat only such “Deterministic” phenomena. They, as the “Subject”, only observe the “Objects”, and never interact with their “Object”.

The fundamental method of “Exact Science”, to which Fluid Dynamics and CFD also belong, is:

1. Scientists define the parameters which are most influential to the objects.
2. They develop models, which describe nature as a function of these parameters, through known formula and/or experimental observation of the phenomena.
3. They solve these model equations mathematically, often numerically, and verify the solution procedure which they have used. This is “Verification”.
4. They compare the solutions with the real world through Experiments or Observation. This is “Validation”.

Among the many fields of modern science, CFD has been one of the most successful, and has come to be used for practical design of aircraft, for example. Probably, this is because that the fundamental model equations – the Navier-Stokes equations – were well established, and the governing parameters were

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clearly defined. That is, aircraft design problem concerns to the geometrically simple shapes and the physical properties of flows such as density, viscosity, etc. are clearly determined.

Turbulence is still an unsolved problem, since it contains macro- and micro-phenomena simultaneously. Similarly since the middle of the 20 century, we are forced to face complex phenomena, which contain different scales of time and/or of space. Such systems are necessarily "Non-deterministic". They involve the cases where we treat the human society. Thus the new science to treat the "Complex System" of the real world emerged and Social Science merged into Natural Science

Fundamental characters of the complex systems are:

1. Mutual Interference of Cause and Effect, where the feed back effects are observed and are often positive, and "Subject" and "Object" are considered to belong in the same class.
2. Mutual Interference between the micro- and macro-Systems. Chaos in micro-scale phenomena make "Emergent Evolution", and grow into macro-scale phenomena. During such process "Fractal characters" are often recognized.
3. Self-Organizing character. Random motion often grows into organized motion, which is, in some sense, contrary to the second law of thermodynamics..

## 25.2 Computers vs Human Brain

In the 1950's, immediately after the emergence of digital technology, its application to the experimental fluid dynamics began and continuously grows with its progress. Today it becomes indispensable part of experimental fluid dynamics.

Numerical computation came in a little late and its growth has been fully depending on the technological progress of computers. In fact, fluid dynamicists did not contribute its hardware, except in the beginning stage of 1940 - 50's. Throughout the eras of Computing Centers, Supercomputers, and Massively Parallel Machines, fluid dynamicists use these machines of their institutions and produce the endless data flow, solving the same equation - Navier-Stokes equation. Today's only problems left are how to use these data? We simply do not have memory space to save those data!

Remedy is in the flow visualization technology, which can produce virtual world of the computed flow field in real time on 3-dimensional space. Massive data flows from the computers are used directly on real time, and do not need to be saved in the memory. So-called post-processing of the computed data is now fully automated, and the necessary design data, for example, various

force data are directly sent to the design office.

This situation is quite similar in the laboratory experiment as well as field observation. Today's wind tunnel experiments are fully automated, and measured data are directly sent to design office, where the design softwares take care of everything. (It is almost ironical that those who do not know fluid dynamics design an aircraft.) In weather forecasting bureau, various kinds of observation tools, ranging from classical weather post to satellite data, produce huge amount of data flow continuously. They are transferred to the central station, processed on real time, and distributed also in time. Thus accuracy of weather forecasting has been dramatically improved recently. As experimental tool, information technology plays far more important role than as computational tool does.

Here in order to understand the role of computers, we compare them with human brain. Computer has a so-called Processor-base architecture, and produces "Hard" output. Data are fed to input and von Neumann processor calculates, then the data output flows out, which have to be stored somewhere before used.

Brain is Memory-base architecture, and produces "Soft" answer. The Data input is used for Index-search of the already stored associated memory. Then the output is the Referenced Memory. Its function is some similarity with the so-called Object Oriented Database (OODB).

Information technologies may continuously grow to:

1. Massively Parallel System, the projects of Earth Simulator and TERA Observatory have already starts.
2. Non-von Neumann Computers, such as Neuro- , Quantum - computers will be a reality soon.
3. Simulation / Visualization technology using PLD (Programmable Logic Device) is under progress.

### 25.3 Information Science

Now we can call this new science based on information science and technology as "Information". Its major function is:

1. Data Acquisition at global observation station, including satellite and local laboratory, and Data Transfer on line in real time to the processors.
2. Indexing those data with the input condition, and Processing, which includes CFD calculation, and Storage them as OODB.
3. When requested, call these Associated Database and provide the results as Virtual Reality.

Information Science can provide to society

1. Direct advice for risk management, for example, Usu Volcano eruption,

Earthquake prediction, Environmental crisis. Widely distributed data acquisition system and real time data processing will be key factor for management of such disaster.

2. We can solve the complex fluid dynamic flows without modeling the micro-scale phenomena. Instead, on-line, real time data acquisition at the real site, combined with macro-flow analysis by CFD, will be useful to directly predict the flow.

3. Risk control of severe accident of large systems, such as nuclear power plants, rocket launching, aircraft crash, etc. could be effectively managed using these on-line, real time data transfer systems, combined with computer analysis, such as PSA (Probabilistic Safety Assessment), FTA (Fault Tree Analysis), or FMPA (Failure Mode and Effect Analysis).

Above all, the Information Technology is now partly replacing Human Cleverness in various aspects of life. And, indeed, information science is a new frontier of CFD.

# Integration of CFD Into Aerodynamics Education

Earll M. Murman<sup>1</sup> and Arthur Rizzi<sup>2</sup>

## 26.1 Introduction

Robert MacCormack's career in Computational Fluid Dynamics started at NASA's Ames Research Center where he produced the 1969 seminal paper on "MacCormack's Method". In 1981 he moved to academia to introduce CFD into aerospace engineering curriculum and academic research projects, first at the University of Washington and later at Stanford University. Only a few schools, notably Iowa State, had a strong aeronautical CFD program prior to 1981. In response to this deficiency, NASA funded the "Graduate Student Training Programs in Computational Fluid Dynamics" at seven universities, including Stanford (Kutler, [1]). By the mid 1980s, graduate student CFD curriculum was established and the first wave of PhDs emerged, many to become academic, industry and government leaders of CFD. Bob's research, teaching and personal leadership style were major contributors to this new generation of CFDers.

This Symposium honoring Bob MacCormack offers an opportunity to reflect on the state of CFD curriculum in aerospace engineering given the many changes since 1981. CFD has evolved to become the mainstay of aerodynamic analysis and design. External factors which affect aerospace engineering and CFD have vastly changed. In the spirit of the Symposium theme, we challenge our colleagues to think boldly about the role of CFD in aerodynamics education. We ask questions, offer thoughts and pose a hypothesis about the future directions of integrating CFD into

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aerodynamics education. We present the results of an informal survey of colleagues addressing elements of this topic, and offer some brief comments about our own current pedagogy. Although the same questions might be asked for CFD applied to heat and mass transfer, production processes, and other applications, we limit our scope to our own areas of expertise. The many changes also impact research directions and opportunities, but we leave these topics to other papers in the Symposium.

## 26.2 Changes from 1981 to 2000

Nineteen years is an educational generation. The students entering college the fall of 2000 were born the year Bob started at University of Washington. The field of CFD is a generation older, information technology has undergone a revolution, and the aerospace industry has changed substantially. We give a brief review of these changes before we consider educational topics.

Table 1 contrasts a number of CFD attributes from 1981 to 2000. For a 1981 referenced point, we take the Proceedings of the AIAA 5<sup>th</sup> CFD Conference [1]. For 2000, we use the present Symposium proceedings. To each of these we add our own knowledge. In 1981, the potential of CFD as a revolutionary force in aerodynamics was recognized (Lomax, Hall [1]). By 2000, the full impact has been realized. CFD has diffused from the activity of the specialist to the main tool of the practicing aerodynamicist. Three-dimensional steady or unsteady, inviscid Euler or Reynolds averaged Navier Stokes solvers using structured or unstructured grids for actual flight configurations are the norm. Indeed, the June 2000 issue of *Aerospace America* has full page color advertisements from three vendors of such capability [2]. Methods based upon linear panel and vortex lattice models, fully coupled inviscid/viscous solvers, and 2D and 3D non-linear inviscid solvers are on the desktop. Extension of basic CFD methods to include other physics is common. Only a direct solution of turbulent flows for other than model problems is beyond reach. Truly, there has been a revolution in aerodynamic design and analysis.

Table 2 contrasts a number of Information Technology attributes from these two sample years. Again, a complete revolution has occurred. In 1981, the IBM PC and Apple MacIntosh hadn't been introduced. Mainframe supercomputers were accessed by hardwired "dumb" alphanumeric or CRT graphic terminals. The ARPA Net was used for file transfer or remote user access to limited, time shared supercomputers. Programming was FORTRAN oriented. Memory was a precious resource. Today's information technology rich environment is well known and needs no elaboration.

| 1981                                                                                                          | 2000                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2D/3D linear methods well developed<br>Mainstay of applied aero analysis                                      | 2D/3D linear methods on the desktop                                                                                                                                                                                       |
| 2D/3D transonic full potential<br>methods used for airfoil and wing-body<br>configuration analysis            | 2D/3D transonic full potential<br>methods on desk top. Utilized for<br>realistic design optimization                                                                                                                      |
| 2D viscous/inviscid interaction based<br>on inverse iterative coupling                                        | 2D viscous/inviscid interaction by<br>direct Newton solvers on the desktop                                                                                                                                                |
| Limited application of Euler and<br>Navier Stokes methods to 3D<br>configurations.                            | Euler and Navier Stokes methods<br>widely used for realistically complex<br>3D configurations.                                                                                                                            |
| Unsteady time accurate Euler and<br>Navier-Stokes for 2D configurations                                       | Unsteady time accurate Euler and<br>Navier Stokes for complex 3D<br>configurations.                                                                                                                                       |
| Parabolized Navier-Stokes used for 3D<br>hypersonic analysis                                                  | Full 3D Euler and Navier-Stokes used<br>for hypersonic analysis                                                                                                                                                           |
| "Global" turbulence models.                                                                                   | "Local" turbulence models (wall<br>functions) and some LES                                                                                                                                                                |
| Multigrid methods emerging                                                                                    | Considerable use of multigrid methods,<br>robustness a challenge                                                                                                                                                          |
| Limited Finite Element methods                                                                                | Finite Element methods widely used                                                                                                                                                                                        |
| Grid generation for complex bodies is a<br>significant barrier. Single or block<br>structured grids the norm. | Robust grid generation methods for<br>complex bodies using unstructured<br>quadrilateral/triangular or<br>hexagonal/tetrahedral grids plus<br>variety of overset and block structured<br>grids. Grid quality is an issue. |
| Grid adaptation limited to mesh point<br>redistribution                                                       | Robust fully adaptive grids with cell<br>division for local features                                                                                                                                                      |
| Monochrome graphic terminals with<br>"pen plotter" style output                                               | Visually rich, interactive, color, pixel<br>oriented graphic output                                                                                                                                                       |
| Limited coupling to non-aerodynamic<br>simulations and with other physics                                     | Frequent coupling to structures, control<br>and with other physics                                                                                                                                                        |
| Major goal is to execute simulations in<br>reasonable time for fully 3D<br>configurations                     | Major goal is to execute simulations<br>with known accuracy, reliability within<br>design cycles of real products                                                                                                         |
| CFD executed by inhouse experts                                                                               | CFD executed by novices and<br>purchasable from vendors                                                                                                                                                                   |

Table 1 - Representative CFD attributes in 1981 and 1990

Aerospace has also gone through major changes as illustrated in Table 3. An interesting article published in the early 1990s, *The Airplane as A Computer Peripheral*, captured in a popular way just how much the design, function and operation of the airplane has evolved with the

| 1981                                                                                         | 2000                                                                         |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Mainframe - terminal architecture                                                            | Client-server architecture                                                   |
| Commercial supercomputers and specialized number crunchers (e.g Illiac IV, array processors) | Commercial lap top and desk-top workstations with parallel processor servers |
| Limited ARPA Net access                                                                      | Unlimited Internet access                                                    |
| CRT "pen plotter" graphics                                                                   | Interactive, 3D color graphics workstations                                  |
| RAM a precious commodity                                                                     | RAM rich                                                                     |
| FORTTRAN oriented                                                                            | C , Object Oriented, MATLAB,.....                                            |
| Word processing                                                                              | Integrated lap/desktop presentations                                         |

Table 2 - Representative information technology attributes in 1981 & 1990

advent of information technology. The post Cold War era has seen major restructuring of the industry and government laboratories. New military aircraft starts have greatly decreased as shown in Figure 1. The information economy with all its "dot coms" has emerged, setting investor expectations at levels that devalue aerospace business. The cumulative market value of the top ten U.S. aerospace companies<sup>3</sup> at the end of 1999 was about that of Dell Computer. National and global alliances are becoming the norm, for example with partnerships on the Joint Strike Fighter program or the Star airline alliance. The mission of aerospace vehicles has expanded from moving mass (payloads) to being nodes in the global information infrastructure.

| 1981                                                        | 2000                                                              |
|-------------------------------------------------------------|-------------------------------------------------------------------|
| Reagan Cold War buildup                                     | "Last" fighter" (JSF) in competition                              |
| <i>Higher, Faster, Farther</i> mantra                       | <i>Better, Faster, Cheaper</i> mantra                             |
| 15 major U.S. aerospace companies                           | 9 major U.S. aerospace companies                                  |
| Shuttle just flown                                          | Space station being assembled                                     |
| 10 <sup>9</sup> Revenue Passenger Kms<br>7.8% annual growth | 3 × 10 <sup>9</sup> Revenue Passenger Kms<br>5% annual growth     |
| Aerospace largest sector in exports                         | \$37B Export of royalties & licensing<br>\$25B Export of aircraft |
| The "old economy" dominates                                 | The "new economy" dominates                                       |
| Few alliances or partnerships                               | National, regional and global alliances and partnerships          |

Table 3 - Representative aerospace industry attributes in 1981 and 1990

<sup>3</sup> Boeing, Honeywell, UTC, General Dynamics, Textron, Lockheed Martin, Raytheon, TRW, Northrop Grumman and Litton Industries.



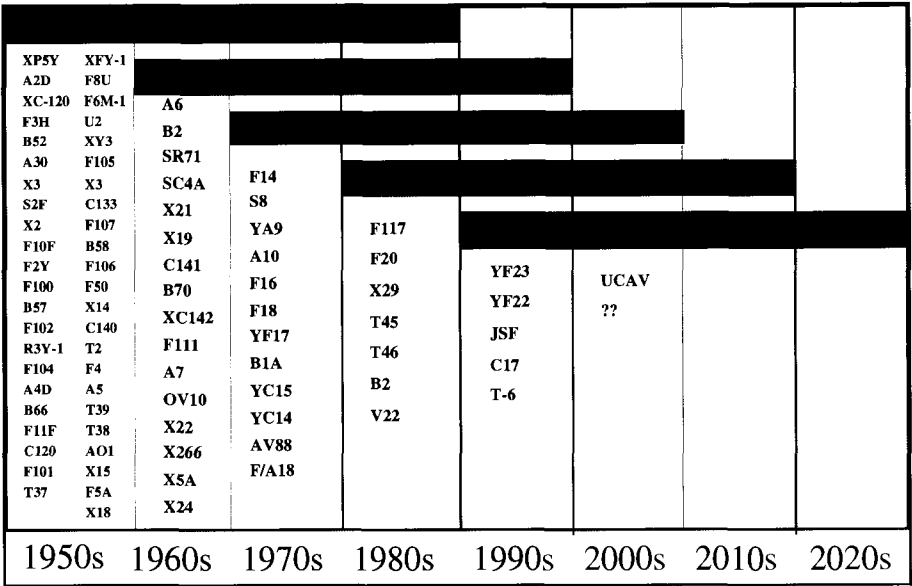


Figure 1 - New US military aircraft programs by decade and career lengths of a typical engineer. Hernandez [3].

With all these changes in CFD, information technology, and the aerospace field, should one expect that university curriculum can retain its 1981 structure? We certainly don't think so, and neither do others (e.g. [4], [5], [6]). Realistically, aerodynamics will play a lesser role in the future aerospace curriculum. Yet students will need to achieve solid working level abilities in the subject. Much of the content of the present aerodynamics curriculum was established in the years between World War II and the mid 1960s. It has served well the generation that was developing many new planes since then, as shown in Figure 1 for the US military sector. But the outlook in Figure 1 as well as similar trends in large commercial aircraft suggest a different future for today's students.

26.3 Educational Considerations and Questions

The trends in industry are away from specialization needed for engineering science innovation and towards more holistic thinkers needed for product development innovation, e.g [5], [7]. Technical competence is expected and rewarded. But narrow specialization representative of the engineering science based educational paradigm of the Cold War years is of less value for a future career than system expertise, group skills, and multifunctional capabilities. Whether we wish it or not, the demand for aerodynamicists has lessened while the demand for system and information engineers has grown. Although still a necessary discipline in the design of aircraft,

aerodynamics is no longer the driving technology in aerospace. These are factors the aerodynamics community needs to accept as they are beyond our control.

In addition, the learning styles have changed as the generation of students entering college has grown up in an information rich, media oriented world. The students of today seek and respond better to hands-on experiential learning rather than the sequential basics-to-application approach which characterizes the vast majority of college curriculum. Actively involving students in the learning process through in-class exercises and out of class projects is more effective than the traditional lecture-listener mode. In many ways our educational system mirrors a mass production model, where knowledge is transmitted to the students "just-in-case" they ever will need it. Pedagogy more attuned to today's student might be the "just-in-time" approach where knowledge is sought and acquired just as the student needs to apply it.

On top of these factors the impact of information technology on educational delivery is just beginning [6], [8], [9]. Desktop computing and web based access to information provide numerous opportunities for new delivery of material. This capability can make obsolete or significantly alter traditional teaching materials such as textbooks and chalk-board lectures. With information readily available, the role of the instructor might change substantially. There are even predictions that universities as we know them will become obsolete. Whether we agree or not with such visions, it behooves us to avoid reactive thinking and consciously consider future opportunities.

With these framing thoughts, we pose the following questions:

As the demand for other skills increase for aerospace engineers and the need for aerodynamic specialists lessen, how should we adjust the aerodynamics curriculum in undergraduate and graduate education?

How will we reduce the number of aerodynamics subjects a typical aerospace engineer will take (1-2 for undergraduate and 3-4 for graduate student), while simultaneously imparting a level of understanding sufficient for their career?

With CFD being the primary analysis and design tool of aerodynamicists, do we have the best balance of classical theory, experiment, and computational approaches to prepare our students for their future careers?

Can we capitalize on CFD methods to address the experiential learning styles of today's students to provide a solid education in aerodynamics?

We leave these questions for the reader to consider. But we thought it might be time to once again, as in the early 1980s, to pause and consider the integration of CFD into aerospace curriculum. At that time, the need was to bring CFD into graduate level education "to meet the critical manpower need of the national aerospace effort by increasing the number of competent specialists in the area of CFD" (Kutler [1]). We speculate that much of the current CFD curriculum is a legacy of this previous need. With the many changes articulated above, we surmise today's needs are different and offer the following:

*Hypothesis: "Today's aerodynamics engineer needs to be fluent in modern CFD methods and tools, and must know how to utilize them in conjunction with theory and experiment for aerodynamic analysis and design. Therefore, we see an increasing need for fully integrating CFD into mainstream aerodynamics pedagogy."*

## 26.4 Findings from an Informal Survey

We thought it would be interesting to sample the opinions of colleagues through an informal survey. During June 2000 we e-mailed a survey to a number of academic colleagues teaching aerodynamics in the U.S. and Europe. We also distributed the survey to the Symposium attendees. In all, we received 40 responses from faculty and non-faculty colleagues in the US, Europe and Asia. We emphasize the results of this survey present below do not represent a scientific or statistically valid sample, nor are the questions necessarily posed in a way to avoid confounding factors.

Our first question addressed the degree of current usage of CFD in aerodynamics subjects. The requested information covered subject level, Mach number regime, CFD methods, codes and texts used. An aggregated summary of responses is shown in Table 4. Very little CFD is used at the underclass undergraduate level, representative of a first subject in aerodynamics. CFD methods begin to enter at the upperclass undergraduate level, with more usage in the US than Europe. Typical methods are 2D panel, 3D vortex lattice, and boundary layer methods. At the graduate level, there is considerable use of CFD methods reported. The European educators use somewhat more advanced methods than their US colleagues. As might be expected, 2D methods dominate over 3D

| Enter in the table below information regarding CFD methods you use in aerodynamic subjects for teaching fundamental understanding, and/or application of aerodynamic principles. |                                                                                                                                                                                           |                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Student Yr                                                                                                                                                                       | United States                                                                                                                                                                             | Europe                                                                                                                |
| <b>Undergrad<br/>1<sup>st</sup> and 2<sup>nd</sup> yr</b>                                                                                                                        | Limited use of subsonic 2D panel, vortex lattice and boundary layer methods. One response on supersonic methods                                                                           | No usage                                                                                                              |
| <b>Undergrad<br/>3<sup>rd</sup> and 4<sup>th</sup> yr</b>                                                                                                                        | Considerable use of subsonic 2D panel and viscous methods. Some use of vortex lattice methods. Some use of transonic/supersonic 2D methods. One use of 3D Euler                           | Limited use of 2D panel methods. Limited use of 2D and 3D Euler and Reynolds averaged Navier Stokes methods           |
| <b>Graduate<br/>Entry Level</b>                                                                                                                                                  | Considerable use of 2D Euler and Navier-Stokes methods, as well as some use of 2D full potential, panel, boundary layer and 3D vortex lattice methods                                     | Variety of 2D Euler, Navier-Stokes, Panel, and Boundary layer methods. Some 3D panel and vortex lattice methods used. |
| <b>Graduate<br/>Advanced<br/>Level</b>                                                                                                                                           | Considerable use of 2D Euler and Navier-Stokes methods, as well as 3D Euler and Navier-Stokes methods. Some use of 2D full potential, panel, boundary layer and 3D vortex lattice methods | Considerable use of 2D and 3D Euler and Navier-Stokes methods, including unsteady methods.                            |

Table 4 - Reported usage of CFD methods in mainstream aerodynamics subjects.

methods, but there is a reasonable representation of the latter.

Beyond the information reported in Table 4, responders indicated many different CFD software packages are used, most being locally developed, and few being used at more than one school. XFOIL and MSES represent the only real exceptions. A variety of textbooks and notes are in use. Among the more frequently cited are Anderson [10], Anderson, Tannehill and Pletcher [11], and Hirsch [12].

The second question directly addressed our hypothesis. Responses in Table 5 show strong agreement for integrating CFD into graduate level aerodynamics subjects, with lesser agreement for undergraduate subjects -

the latter being stronger in the US than Europe. Other viewpoints, including teaching CFD separately from aerodynamics, had average responses below a “3”. Europeans felt stronger than the US responders that CFD should be approached on a “need-to-know” basis in student research, independent, and design projects.

| What is your viewpoint regarding the integration of CFD methods into mainstream aerodynamics subjects compared to other pedagogical approaches for educating aerodynamic engineers on use of CFD tools and methods? |        |         |         |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|---------|---------|
| 1 = strongly disagree      3 = somewhat agree      5 = strongly agree                                                                                                                                               |        |         |         |         |
| Viewpoint                                                                                                                                                                                                           | Avg US | Avg Eur | Avg All | STD All |
| Number of responses                                                                                                                                                                                                 | 24     | 15      | 40      |         |
| CFD approaches should be integrated into graduate level mainstream aerodynamics subjects.                                                                                                                           | 4.3    | 4.4     | 4.4     | 0.8     |
| CFD approaches should be integrated into undergraduate level mainstream aerodynamics subjects.                                                                                                                      | 3.7    | 3.1     | 3.5     | 1.2     |
| CFD approaches should be taught on a “need-to-know” basis as part of student research or independent projects.                                                                                                      | 2.4    | 3.4     | 2.8     | 1.3     |
| CFD approaches should be taught separately from mainstream aerodynamics subjects.                                                                                                                                   | 2.9    | 2.3     | 2.7     | 1.3     |
| CFD approaches should be taught on a “need-to-know” basis as part of a design subject.                                                                                                                              | 2.0    | 3.0     | 2.3     | 1.1     |
| CFD approaches are best learned through “on-the-job” experience after graduation.                                                                                                                                   | 1.9    | 2.4     | 2.1     | 1.2     |
| CFD approaches are best learned through summer internships in industry or government.                                                                                                                               | 1.9    | 1.9     | 2.0     | 1.1     |

Table 5 - Viewpoints on where CFD should be integrated into the curriculum sorted by the overall average response.

Survey responders offered a number of comments on the integration of CFD into aerodynamics curriculum. A summary of these reveal there are multiple viewpoints ranging from strong integration of CFD and aerodynamics to a more traditional serial approach of underlying aerodynamics and numerical analysis subjects preceding a CFD subject.

Our next question focused on the difficult choices of what to emphasize in an integrated approach. Table 6 reports strong agreement that courses should have a deep coverage of the underlying aerodynamic theory as well as interpretation of the output aerodynamic parameters. Less depth of coverage is desired of the user controllable parameters and underlying numerical methods. Little emphasis is recommended for the programming

implementation. Added comments by survey responders focused on the depth of treatment of numerical methods, with a spread in views between depth and light coverage.

| <b>Assuming that CFD methods are integrated into mainstream aerodynamics subjects, in how much depth should the following topics be covered in order for the student to acquire the necessary understanding for effective application of the related CFD tool or method?</b> |        |         |         |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|---------|---------|
| 1 = cover lightly      3 = cover in moderate depth      5 = cover deeply                                                                                                                                                                                                     |        |         |         |         |
| Topic                                                                                                                                                                                                                                                                        | Avg US | Avg Eur | Avg All | STD All |
| Number of responses                                                                                                                                                                                                                                                          | 24     | 13      | 39      |         |
| Theoretical formulation of aerodynamic problem (e.g. fluids assumptions, governing equations, boundary conditions)                                                                                                                                                           | 4.4    | 4.6     | 4.5     | 0.8     |
| Interpretation of output (e.g. force and moment coefficients, pressure distributions, flow field features)                                                                                                                                                                   | 4.2    | 4.5     | 4.3     | 0.8     |
| User controlled parameters (e.g., grid resolution/quality, degree of convergence, numerical constants)                                                                                                                                                                       | 3.6    | 4.2     | 3.8     | 1.1     |
| Formulation of numerical solution (e.g. discretization method, error sources, solution of discrete equations)                                                                                                                                                                | 3.7    | 4.0     | 3.8     | 1.0     |
| Underlying numerical methods (e.g. dispersion & dissipation errors , matrix solution methods, stability, convergence)                                                                                                                                                        | 3.4    | 3.6     | 3.5     | 1.3     |
| Programming implementation (e.g. programming language, program structure, memory utilization and speed of execution, parallelzation)                                                                                                                                         | 2.4    | 2.5     | 2.5     | 1.4     |

Table 6 - Priorities for content of aerodynamic subjects with CFD integrated into the curriculum sorted by overall average response

The last question addressed barriers to integrating CFD into aerodynamics curriculum. Only responses from the academics are shown in Table 7. It is interesting to see that none of the choices offered on the survey were judged to be “significant” barriers. All the choices were in the range of being “somewhat of a barrier”. In general, the pedagogical issues were ranked somewhat higher than the technology factors. However, given the standard deviation, any statistical significance in these differences is doubtful. Interestingly, comments added by the responders focused on the difficulty of keeping software current, licensing issues, professor and teaching assistant proficiency with software, and obsolescence of routines or packages.

**Based upon your first hand knowledge, please rate the significance of the following possible barriers to integrating CFD methods into the teaching of aerodynamics.**

1 = not a significant barrier; 3 = somewhat a barrier; 5 = a significant barrier

| Possible Barrier                                         | Avg US | Avg Eur | Avg All | STD All |
|----------------------------------------------------------|--------|---------|---------|---------|
| Number of responses                                      | 21     | 13      | 34      |         |
| Student preparation/prerequisites                        | 3.5    | 3.6     | 3.6     | 1.2     |
| Technical support (teaching assistants or staff)         | 3.3    | 3.5     | 3.3     | 1.2     |
| Revising curriculum to include CFD into aerodynamics     | 3.4    | 2.8     | 3.2     | 1.4     |
| Knowing the best pedagogical approach                    | 2.9    | 3.2     | 3.0     | 1.1     |
| Textbook coverage of CFD methods                         | 2.9    | 2.8     | 2.9     | 1.1     |
| Available preprocessing software                         | 2.8    | 2.8     | 2.9     | 1.2     |
| Software documentation                                   | 2.8    | 3.3     | 2.9     | 1.0     |
| Availability of example problems and exercises           | 2.4    | 3.1     | 2.7     | 1.4     |
| Robustness of CFD solvers                                | 2.5    | 2.8     | 2.7     | 1.2     |
| Available CFD application packages/programs              | 2.4    | 2.7     | 2.6     | 1.2     |
| Support of department chair or equivalent                | 2.4    | 2.9     | 2.5     | 1.4     |
| Computer speed/memory resources                          | 2.1    | 2.7     | 2.4     | 1.0     |
| Your experience in teaching CFD in aerodynamics subjects | 2.1    | 2.8     | 2.3     | 1.4     |
| Available post processing software                       | 2.5    | 2.2     | 2.3     | 1.2     |
| Computer graphic resources                               | 2.1    | 2.6     | 2.3     | 1.0     |
| Measuring student achievement                            | 1.9    | 2.5     | 2.1     | 1.0     |

Table 7 - Barriers to integrating CFD into aerodynamics curriculum, sorted by the over all average response.

## 26.5 Examples of Integration

The authors have been experimenting with integrating CFD into undergraduate aerodynamics subjects. We close the paper with brief accounts of our experiences.

This past academic year, the first author participated in a new approach to teaching the junior-senior level aerodynamics subject at MIT [13]. The class was centered around a Lockheed-Martin contributed case study related to modeling F-16 wing-body aerodynamics. An integrated theory, experiment, and CFD approach was adopted with "goodness" measures for the methods based upon their ability to predict relevant aircraft performance parameters (range, specific excess power, dash time to supersonic speed, take-off distance). CFD methods were introduced as tools, with the background algorithms covered only at a conceptual level. The coverage reflected the priorities suggested in Table 6, but time pressures resulted in less emphasis on the numerical solution and methods

than recommended by the survey responders. CFD methods introduced were 2D panel (linearly varying vortex), 3D vortex lattice (AVL), 3D Euler (Unstructured grid, Runge-Kutta time integration, FELISA), integral boundary layer methods, and coupled panel/boundary layer methods (XFOIL). Overall, the students achieved a entry-level understanding of these CFD tools and how to combine them with other aerodynamic analysis methods. They constantly desired more depth on CFD and theoretical methods, but time pressures relegated such excursions to more advanced subjects. In terms of barriers, they seemed to align well with those in Table 7. We didn't encounter any major barriers, but certainly the top five in the table were all met to some degree. Overall, we were pleased with the experiment, and plan to continue refining it in the upcoming academic year.

As part of the requirements for the M.Sc. degree in Aeronautics at the Royal Institute of Technology (KTH) in Stockholm, students must take two courses in aerodynamics and may elect a third, advanced course. Since 1993 the second author has been teaching the second required course (fourth year in the M.Sc. program). As might be expected from an instructor with a CFD background, the content of the course the first time it ran emphasized the high-end methods, namely full potential and Euler. At the end of the course, one student, who had had previous experience working in the aerospace industry, commented that the impression given was, "when confronted with an aerodynamic problem, we should immediately run to a big computer". His comment was taken to heart, and in subsequent years the lectures and course content evolved to emphasize understanding of the basics together with use of some practical computational methods. And this aligns reasonably well with the content priorities listed in Table 6. There is, in any case, no escaping the computer. Along with the wind tunnel it is a fundamental tool of aerodynamics, but it need not be big. Rather it should fit on the desktop and be at the students finger tips, like paper and pencil, and used effortlessly in the problem-solving process. The utility of the computer in aerodynamics is huge, ranging from digitized semi-empirical methods like Digital Datcom on up to Euler and Navier-Stokes solvers. Moreover suitable software is becoming more readily available. For example Public Domain Aeronautical Software (see <http://www.pdas.com>) offers a collection of programs in this range, some of which may be suitable for course work.

In its present form the aerodynamics course at KTH (see <http://www.flyg.kth.se/edu-dir/ugrad-dir/prog-dir/4E1211prog.html> for the course prospectus) focuses on the analysis and design of airfoils and wings using coupled panel-integral- boundary-layer theory in 2D and the vortex-lattice method in 3D as implemented in the industrial-strength tools XFOIL and AVL (Athena Vortex Lattice method). In addition the airfoil in transonic (inviscid) flow is studied with a TSP method. Going well beyond a black-box approach, substantial time and effort are spent in understanding



how and what results these tools produce. In fact the course contains the first five items in Table 6, with the greatest emphasis on item 1 and then descending to the least on item 5. Several years experience in the classroom however revealed one substantial barrier to this goal, namely software documentation. XFOIL, in particular, is a very sophisticated piece of software, and one simply cannot teach all the details of how it is put together. Pedagogical aids therefore are being developed in MATLAB at KTH to act as stepping stones in the process of understanding codes like XFOIL and AVL. Two such are PABLO (panel/boundary-layer) and TORNADO (vortex lattice) programmed more or less verbatim from their text-book description. Experience thus far indicates that this stepping-stone approach is helpful.

## 26.6 Summary

In the time that Bob MacCormack has been in academia, a new generation of students has been born and will be entering college this fall. In their lifetimes, CFD has evolved from an "impending revolution" to become the standard analysis and design tool for aerodynamic practitioners. Adding to this are many changes in information technology, the state of the aerospace industry, and student learning styles since 1981. As educators we believe that these factors are calling for substantial changes in the way we teach aerodynamics and in the specific course content. We therefore offer a hypothesis that CFD (in the broadest meaning of the term) should be fully integrated into mainstream aerodynamics subjects.

We have undertaken an informal survey of colleagues in the U.S., Europe, and at this Symposium to provide a cross-section of views relevant to this hypothesis, and a number of trends can be discerned. CFD is beginning to be integrated into the traditional aerodynamic courses at the higher-level undergraduate and graduate levels, with simpler methods entering the undergraduate courses and more advanced ones in the graduate courses. The emphasis in teaching these methods is most on basic understanding and less on the actual programming implementation. And while a number of barriers to integrating these methods into courses do indeed exist, none are considered to be significant. We also report on our own recent experiments in integrating CFD into undergraduate aerodynamic subjects, and how our personal experiences compare to the views of our colleagues.

We foresee that as aerospace curriculum broadens to meet the educational needs of this next generation of students, we will be challenged to rethink our subject content and its delivery. It is apparent to us that a major gap has developed between the traditional aerodynamics curriculum and what is required now and in the future. In our view, the real challenge is "how" do we fully integrate CFD into the mainstream aerodynamics curriculum. In the light of this, we strongly suggest that educators give serious thought

to the structure of aerodynamics curriculum and undertake experiments to try new approaches. Further, we recommend that the results of these experiments be shared widely through printed publications and the Internet.

## 26.7 Acknowledgements

We would like to acknowledge the many people who took time to respond to our survey, who are too numerous to mention by name.

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# Frontiers of Computational Fluid Dynamics 2002

This series of volumes on the "Frontiers of Computational Fluid Dynamics" was introduced to honor contributors who have made a major impact on the field. The first volume was published in 1994 and was dedicated to Prof Antony Jameson; the second was published in 1998 and was dedicated to Prof Earl Murman. This volume is dedicated to Prof Robert MacCormack.

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